

Merged research report for the Surreal project

Omnific-Preserving Automorphisms

Convex-scale stabilizers, definable constants,
nondefinable monomials, and algebraic-parameter rigidity

Built from fifteen manuscripts dated September 23, 2026

Abstract

Which symmetries of a Hahn field preserve its omnific-type integer part, and what does that integer part remember? Let k have characteristic zero, let $\Gamma \neq 0$ be a divisible ordered abelian group, write C_δ for the principal convex subgroup generated by $\delta > 0$, and let $\mathfrak{o} \subseteq k$ be a unital subring. A strong coefficient-fixing 1-automorphism σ of $k((t^\Gamma))$ preserves $\mathfrak{o} \oplus \Pi$, where Π is the purely infinite ideal, exactly when, for every $\delta > 0$, the shift- δ coefficient of its logarithmic character $\log(t^{-g}\sigma(t^g))$ vanishes on C_δ . The proof uses no exponential–logarithm correspondence and applies verbatim to the surreal field and $\mathbf{Oz}$. Where that correspondence is available, the same condition for the operator logarithm $\text{Log } \sigma$ is equivalent; the two coefficient functionals are different.

Consequences: the stabilizer is trivial exactly in Archimedean rank one; its common fixed field is the Hahn field on the bottom convex subgroup, and for $(\mathbf{No}, \mathbf{Oz})$ it is $\mathbb{R}$; in finite rank r its derived length is at most $r - 1$, with equality for $\mathbb{Q}^r$; explicit successive-scale flows generate a faithful $\mathbb{Z} \wr \mathbb{Z}$; for every set of surreal parameters the pointwise stabilizer is nonsolvable and carries a faithful action of $\mathbb{R}^\kappa$ for every set cardinal κ . Consequently no set of parameters makes the Conway monomials, the omega-map, any valuation-representative rule, the simplicity relation or the Gonshor exponential definable in $(\mathbf{No}, +, \cdot, <, \mathbf{Oz})$, although the pair defines the three zero-cut projections and the floor map; a monomial can even be sent to a two-term omnific integer. An equivalent form of the criterion bounds the support of the principal unit $t^{-g}\sigma(t^g)$ itself, and every element of the stabilizer keeps each monomial in the signed Archimedean block of its exponent. A second part proves algebraic-parameter rigidity of $\mathbf{Oz}$: embeddings into polynomial, Laurent and finite-type domains land in the coefficients, strong cancellation holds, and $\text{ML}(\mathbf{Oz}[T_1, \dots, T_n]) = \mathbf{Oz}$, while explicit commuting derivations integrate to formal flows that no polynomial family realizes. A third part proves that every ring automorphism of $\mathbf{Oz}$ extends to a strong automorphism of $\mathbf{No}$, as constant-term tests detect Hahn summability, and classifies the strong monomial-preserving embeddings that preserve the integer parts; proper self-embeddings of $\mathbf{No}$ preserving and reflecting $\mathbf{Oz}$ can move a real number. A fourth part adds a third proof of the constant-term detection, by induced matchings, detection inside subfields containing enough all-ones series, and equivariance of formal power-series evaluation. A fifth part shows that the Hahn summation cannot be topologized: for a densely ordered exponent group every ring topology realizing all Hahn sums as limits is pulled back from the constants, so on $\mathbf{Oz}$ it sees only $\mathbb{Z}$ and on $\mathbf{No}$ it is indiscrete, and a set-sized full Hahn field has a Hausdorff one exactly when its exponent group is 0 or $\mathbb{Z}$. A sixth part recovers the coefficients after adjoining external monomials of any rank: $\exists y (y^2 = 1 + x^4)$ defines the coefficient field of $\text{Frac}(F[G])$ for $F = \mathbf{No}, \mathbf{No}(i)$, a sparse power bound gives coefficient rigidity and cancellation at every finite rank, cancellation fails at infinite rank, and every field automorphism is monomial exactly when $\text{Hom}(G, \mathbb{Z}) = 0$ (with a published defect theorem). Three more parts compute the full formal orbit field of the constant-term Euler flow, show that the set groups acting faithfully on $\mathbf{No}$ or $\mathbf{Oz}$ are exactly the left-orderable ones, with prescribed fixed fields, and prove that every derivation of $\mathbf{Oz}$ integrates to a unique formal Hasse–Schmidt flow. A tenth part re-derives that classification, adds independence of the orbit of ω and a proper-class relative fixed field, and solves difference equations for these automorphisms: $\sigma x - x = b$ exactly when $\text{ct}(b) = 0$, $\sigma x/x = a$ exactly when $\text{lc}(a) = 1$, every scalar first-order equation, and cocycles of groups with a central element.

This report merges fifteen independently written manuscripts, prints shared results once with every source named, and keeps every source’s limitations. It is an AI-assisted draft: it has not been refereed, and nothing in it has been formalized in Lean.

Sources: manuscripts 09 (the base text), 04 and 02, pinned to repository commit [fb5c4b5](#), 08, pinned to [befe739](#), the later addition 10, pinned to [58cd8e1](#), the batch-30 additions 11, 12 and 13, pinned to [cf56b89](#), [a6c68ac](#) and [0865f04](#), the batch-31 addition 14, pinned to [3d40856](#), and the batch-32 additions 15, which names blobs of this report’s source at [b3fa9e2](#) and [20c4c9f](#), and 16, pinned to [1ab41af](#), and the batch-33 additions 17 and 18, pinned to [efc5446](#), and 19, pinned to [889b87c](#), and the batch-34 addition 20, pinned to [e93a06d](#); their titles are in [sections 1.2, 20.1, 33.1, 38.1, 43.1, 53.1, 60.1, 67.1 and 76.1](#). Results already printed in the omnific Diophantine report [docs/surreal/omnific-diophantine-geometry](#) are cited, not reprinted. The provenance of every section is recorded in [section A](#); non-claims are collected in [section B](#). Finite checks accompany the report and check finite identities only ([section C](#)).

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1 Scope, results, and conventions

A Conway normal form carries three kinds of information: coefficients, exponents, and the monomials that represent the exponents. The omnific integers keep the purely infinite part of a normal form and an ordinary integer constant coefficient. This report asks how much of the normal-form presentation is determined by the surreal field together with this distinguished ring, and which automorphisms preserve it.

The answer is asymmetric. The coefficient field, the cut between negative and nonnegative Hahn exponents, the constant coefficient and the floor map are recoverable from the pair $(\mathbf{No}, \mathbf{Oz})$; much of this was already proved in the omnific Diophantine report (section 8). The monomial cross-section is not recoverable. Even an arbitrary *set* of named parameters leaves enough automorphisms to move a monomial to an infinite series and to defeat every definable choice of valuation representatives. The mechanism is a family of derivations that change a higher Archimedean scale by a lower one without ever moving a purely infinite term across the constant-term boundary. Part I classifies these automorphisms and derives the consequences. Part II, from a separate manuscript, shows the complementary rigidity: $\mathbf{Oz}$ admits no nonconstant *algebraic* family of embeddings, although it admits formal flows of all orders. Part III, from three later manuscripts, removes the strongness hypothesis from the classification for $(\mathbf{No}, \mathbf{Oz})$ and shows that the rigidity of automorphisms does not pass to proper embeddings. Part IV, from a ninth manuscript, derives the same automatic strongness a third time and adds a third detection proof and detection in subfields. Part V, from a tenth manuscript, contrasts this algebraic preservation of all Hahn sums with their topological realization: every jointly continuous ring topology that realizes them forgets Π . Part VI, from an eleventh, carries the coefficient rigidity of part II to monomial parameters of arbitrary rank. Three more complete the picture of part II's formal flows and of the symmetry groups: part VII computes the full formal orbit field of one Euler flow, of which part II had only the monomial part; part VIII identifies the set groups of automorphisms preserving $\mathbf{Oz}$ as the left-orderable groups, with prescribed fixed fields; and part IX shows that every derivation of $\mathbf{Oz}$ integrates formally, answering question 19.2. Part X, from a fifteenth manuscript, derives the left-orderable classification a second time and solves the difference equations of these automorphisms.

1.1 Principal results

Bracketed numbers name the source manuscripts (section 1.2); [merge] marks statements assembled or generalized by the merge, each with a complete proof. The Hahn conventions are fixed in section 1.5: $K = k((t^\Gamma))$ with well-ordered supports, $t^\gamma = \omega^{-\gamma}$ on the surreals, Π the series supported in $\Gamma^{<0}$, $\mathcal{A}_k = k \oplus \Pi$ and $\mathcal{R}_\mathfrak{o} = \mathfrak{o} \oplus \Pi$, so that $\mathcal{R}_\mathbb{Z} = \mathbf{Oz}$ when $k = \mathbb{R}$ and $\Gamma = \mathbf{No}$.

- (1) **The convex-support criterion** [04, 09][merge]. For every characteristic-zero k , every unital $\mathfrak{o} \subseteq k$, every nonzero divisible Γ (set-sized, or $\Gamma = \mathbf{No}$), and every strong k -linear 1-automorphism σ : $\sigma(\mathcal{R}_\mathfrak{o}) = \mathcal{R}_\mathfrak{o}$ if and only if the logarithmic-character coefficients $a_{\sigma, \delta}$ vanish on C_δ for every $\delta > 0$, if and only if σ commutes with the three zero-cut projections (theorem 3.3). The proof is 04's and imports no exponential–logarithm correspondence. For contracting strong derivations the analogous criterion holds with 09's coefficients c_δ (theorem 3.4); where the correspondence is available the operator form follows (corollary 3.5). The functionals $a_{\sigma, \delta}$ and c_δ differ; their vanishing conditions are equivalent (example 3.2). For single fixed-shift flows 08 proves the criterion without divisibility (theorem 4.10). Source 10 proves it again in an equivalent form, on the support of the principal unit $t^{-g}\sigma(t^g)$ rather than its logarithm, with a least-forbidden-shift proof

kept as a second route; the inclusion $\sigma(\mathcal{R}_o) \subseteq \mathcal{R}_o$ already suffices, every signed Archimedean block of exponents is preserved, and the necessity half needs no strongness ([theorem 3.8](#), [proposition 3.10](#), and [remark 3.9](#)).

- (2) **Rank and fixed fields** [04, 08, 09, 10]. The stabilizer $\mathcal{U}_{\mathcal{R}_o}(\Gamma)$ is trivial exactly when Γ is Archimedean ([theorem 4.13](#)); for set-sized Γ its common fixed field is the Hahn field on the bottom convex subgroup ([theorem 4.15](#)); and $\text{Fix}(\mathcal{U}_{\mathbf{Oz}}) = \mathbb{R}$, also for all field automorphisms of **No** preserving **Oz** ([theorem 7.2](#)). For every set A of parameters the elementary shears fixing A have common fixed field $\mathbb{R}((t^{V_A}))$, V_A the rational span of the supports ([theorem 7.4](#)). Binomial twists $t^g \mapsto t^g F(t^\delta)^{\theta(g)}$ with an arbitrary profile F have fixed field $k((t^{\ker \theta}))$, no nontrivial finite orbits and a relatively algebraically closed fixed field ([theorems 4.18 and 4.20](#) and [corollary 4.21](#)), and in set-sized fields definable elements obey a support bound ([proposition 7.8](#)) [10].
- (3) **Group structure** [04, 08, 09]. In finite ordered rank r the stabilizer and its Lie algebra have derived length at most $r - 1$, with equality for $\mathbb{Q}^r$ ([theorems 6.3 and 6.6](#)). Two successive-scale flows generate a faithful $\mathbb{Z} \wr \mathbb{Z}$ with exact commutator leading terms ([theorems 6.8 and 6.10](#)). For every set P of surreal parameters, the pointwise stabilizer of P in $\mathcal{U}_{\mathbf{Oz}}$ carries a faithful action of the full product $\mathbb{R}^\kappa$ for every set cardinal κ and is nonsolvable ([theorems 7.6 and 7.7](#)).
- (4) **What the pair remembers** [08, 09]. Beyond the reconstruction package of the Diophantine report, the three zero-cut projections P_-, P_0, P_+ on all of **No**, the floor map and truncation at a named monomial are parameter-free definable in $(\mathbf{No}, \mathbf{Oz})$, and every field automorphism preserving **Oz** commutes with them, without strongness ([theorem 8.2](#) and [corollary 5.2](#)).
- (5) **What it forgets** [04, 08, 09, 10]. For every set P of parameters, some strong 1-automorphism preserving **Oz** fixes P pointwise and moves a Conway monomial to an infinite series; so neither the monomial class nor the omega-map is P -definable ([theorem 9.1](#)), and no P -definable rule selects a representative of each natural-valuation class ([theorem 9.4](#)). Source 10 moves ω^{ω^β} to the two-term omnific integer $\omega^{\omega^\beta} + \omega^{\omega^\beta - \delta}$ and shows that neither the simplicity relation nor the Gonshor exponential is P -definable, even with all reals and the zero-cut projections named ([theorems 9.8 and 9.10](#)).
- (6) **Surcomplex consequences** [04, 08, 09, 10]. The criterion holds for $\mathbf{Oz}[i]$; real shears commute with conjugation, and 10's two-term witnesses also preserve the order of the real axis and the projections ([proposition 10.5](#)); an imaginary-time shear fixing any set moves the real axis inside the leading-term kernel ([theorem 10.2](#)); a wild coefficient lift moves an ordinary real ([proposition 10.3](#)); and the common fixed field of all automorphisms preserving $\mathbf{Oz}[i]$ is $\mathbb{Q}$ ([corollary 10.4](#)).
- (7) **Algebraic-parameter rigidity** [02]. Every nonzero element of **Oz** or $\mathbf{Oz}[i]$ divides a monomial whose roots of all orders lie in the ring ([corollary 13.3](#)). Consequently injective homomorphisms into polynomial and Laurent rings over a domain land in the coefficients ([theorems 15.1 and 15.3](#)), polynomial cylinders over **Oz** recover **Oz** under every isomorphism ([theorem 15.4](#)), $\text{ML}(\mathbf{Oz}[\mathbf{T}]) = \mathbf{Oz}$ ([theorem 16.2](#)), and embeddings of set-sized root-covered rings into finite-type domains land in a finite field of algebraic constants ([theorem 14.4](#)). Commuting derivations of **Oz** integrate to formal flows of all orders that no polynomial family realizes, with an exact transcendence degree for their monomial orbit fields ([theorems 17.3 and 17.5](#)).
- (8) **Automatic strongness and proper embeddings** [11, 12, 13; 14]. A set-indexed family in a full Hahn field is summable exactly when, for every y , only finitely many $\text{ct}(x_j y)$ are

nonzero, and countable $\{0, 1\}$ -valued tests suffice; so a field isomorphism of full Hahn fields commuting with ct up to a coefficient isomorphism is strong (theorems 21.4 and 22.1). Every ring automorphism of $\mathbf{Oz}$ extends uniquely to a strongly $\mathbb{R}$ -linear automorphism of $\mathbf{No}$ (theorem 23.2), which answers question 11.1; every automorphism of $(\mathbf{No}, \mathbf{Oz})$, or of a full real Hahn pair with divisible exponent group, is $M_{\chi, \tau} \circ u$ with u as in theorem 3.3 (theorem 24.1). Strong exact-monomial embeddings preserving the integer parts correspond to coefficient sections into the Hahn field on the bottom gap, and move reals exactly when the positive value image is not coinital (theorem 26.3 and corollary 26.7); explicit proper embeddings preserve and reflect $\mathbf{Oz}$ and send a real b to $b + \omega^{-1}$ (theorem 27.3). Further: constant-term adjoints at set size and a proper-class obstruction, closed images and a continuity dichotomy, nonelementarity, continuum many nonconjugate copies, parameter-fixed copies and surcomplex versions (sections 25 and 28 to 30).

- (9) **A fourth derivation** [14]. Source 14 proves the automatic strongness of the real pair again, in less generality, and is credited beside 12 and 13. It adds a third proof of detection: an induced matching yields a detector with all coefficients 1, so monomials and all-ones series on increasing sequences suffice (lemma 34.1 and corollary 34.2). Detection and transport of sums hold in every subfield containing those series (proposition 35.2), but not in the Puiseux field (example 35.3, [merge]). Pair automorphisms commute with formal power-series evaluation at infinitesimals (theorem 36.1), the adjoint is a sign-reversed transpose (proposition 36.2), and an extension of a ring automorphism of $\mathcal{R}_\Gamma$ is unique (proposition 36.5, [merge]).
- (10) **Topological collapse** [15]. For a densely ordered Γ , the ring topologies on $\mathfrak{o} \oplus \Pi$ with jointly continuous multiplication in which every Hahn sum is the limit of its finite partial sums are exactly the pullbacks along ct of the ring topologies on $\mathfrak{o}$ (theorem 39.4); so on $\mathbf{Oz}$ and $\mathbf{Oz}[i]$ they see only $\mathbb{Z}$ and $\mathbb{Z}[i]$, countable monomial sums already force this, and on $\mathbf{No}$ and $\mathbf{No}(i)$ only the indiscrete topology qualifies (theorem 39.7, proposition 39.8, and corollary 39.9). A set-sized full Hahn field has a Hausdorff such topology exactly when $\Gamma = 0$ or $\Gamma \cong \mathbb{Z}$ (theorem 39.12). The congruence topologies already in the collection are instances (remark 39.6, [merge]). Source 15 also re-derives automatic strongness a fourth time, credited in part III; describes the strong $\mathbb{R}$ -linear dual of $\mathbf{No}$ by class coefficient functions (proposition 40.5); and gives a pairing topology realizing exactly the Hahn sums, with separately but not jointly continuous multiplication (theorems 41.1 and 41.3).
- (11) **Coefficient recovery at arbitrary rank** [16]. For a torsion-free abelian group G and F real closed or algebraically closed of characteristic zero, $\exists y (y^2 = 1 + x^4)$ defines F in $F(U^G) = \text{Frac}(F[U^G])$, and a denominator-cleared form defines $\mathbf{Oz}$ and $\mathbf{Oz}[i]$ in their monoid algebras (theorem 45.4). A nonmonomial P/Q that is an n th power in any monomial extension has $n \leq \max(\text{wt } P, \text{wt } Q) - 1$, sharply (theorem 46.2). Consequently theorems 15.1 and 15.3 to 15.5 extend to monoid algebras of arbitrary rank, cancellation of finite rank (theorems 47.1, 48.1 and 50.1), with exact failure for $\mathbb{Q}^{(\kappa)}$, κ infinite (theorem 50.4); and, granted a defect theorem of Lin and Wang, every field automorphism of $F(U^G)$ is monomial exactly when $\text{Hom}(G, \mathbb{Z}) = 0$ (theorem 49.2). These are finite-support extensions, not Hahn fields (example 51.1).
- (12) **Full formal orbit fields** [17]. For the Euler derivation D_0 of theorem 17.6 and its constant field $\mathbf{C}_0 = \ker D_0$, the Taylor image $\mathcal{E}_{0,s}(\mathbf{No})$ and $\mathbf{No}(s)$ are linearly disjoint over $\mathbf{C}_0$, all polynomial relations descend from $\mathbf{C}_0$, a trajectory is algebraic only if it is constant, and s is transcendental over the full orbit field (theorem 53.1). Explicit countably supported purely infinite omnific integers, one for each ordinal, are independent over $\mathbf{C}_0(\omega)$, so the orbit field has no set-sized transcendence basis (theorem 53.2); after $s = \log(1 + v)$ the rational and algebraic trajectories are classified by $\mathbf{C}_0(\omega)$ and its relative algebraic closure

(theorem 53.3). This answers, for D_0 , the lower bound left after corollary 17.9.

- (13) **Left-orderable symmetry groups** [18]. Every nontrivial left-orderable set group acts faithfully on **No** by strongly additive, real-fixing, **Oz**-preserving double Hahn lifts (monomial maps M_{1,ι_φ} of part III) whose nonidentity elements all fix exactly a prescribed two-level support field containing any given set of parameters, and freely outside $\mathbb{R}$ and $\mathbb{Z}$; set groups acting faithfully on **No** or **Oz** are left-orderable (theorem 60.1 and corollaries 60.2 and 60.3). With conjugation, the admissible groups are G_0 and $G_0 \times C_2$, G_0 left-orderable (theorem 60.4).
- (14) **All derivations integrate** [19]. Every derivation of **Oz** or **Oz** $[i]$ maps the ring into Π , and $\delta^n/n!$ is its unique iterative integral Hasse–Schmidt extension (theorem 67.1), which answers question 19.2; all formal automorphisms tangent to the identity are exponentials of integral Lie series, finite jets always lift, and the jet extensions do not split (theorem 67.2); the quartic formula of part VI becomes universal in **Oz** $[[s]]$ and no set of parameters defines the coefficient copy there (theorem 67.3).
- (15) **Universal symmetries and difference equations** [20]. An independent derivation of the left-orderable classification, adding that the orbit of ω is algebraically independent (theorem 76.1) and a relative action whose fixed field is a proper-class two-level field (theorem 77.5). For a nonidentity element, $\sigma x - x = b$ is solvable exactly when $\text{ct}(b) = 0$, $\sigma x/x = a$ exactly when $\text{lc}(a) = 1$, every nonzero Laurent polynomial in σ is bijective off the constants (theorem 76.2), with a three-condition retraction for the relative action (theorem 79.7); every scalar first-order equation is classified (theorem 80.1), and cocycles of groups with a central element are characters plus coboundaries (theorem 80.2). The difference core re-derives the single-dilation report’s resolvent (`dsup:thm:resolvent`).

These statements are proved here, but proving a statement here is not a claim that nobody proved it earlier. The general theory of strong Hahn-field automorphisms and of the exponential–derivation correspondence is prior work (section 1.3); each source names its most distinctive contributions, and no exhaustive priority claim is made (section B).

1.2 Provenance of this report

The report is built from fifteen manuscripts written independently: five for parts I and II, described here, three for part III, described in section 20.1, one for part IV, described in section 33.1, and one each for parts V to X, described in sections 38.1, 43.1, 53.1, 60.1, 67.1 and 76.1. Four were placed together in commit `f4c9504` (batch 26, items 02, 04, 08 and 09; they keep these numbers here, as their shipped file prefixes do). Manuscripts 02, 04 and 09 pin repository commit `fb5c4b53` (12 commits before the placement); manuscript 08 pins `befe739b` (10 commits before). Both pins contain the batch-24 placement `be06fc8` but precede the batch-24 and batch-25 writes, so no source saw the reconstruction package now in the omnific Diophantine report (its section `odg:def:sec:reconstruction`, placed after both pins). The fifth, manuscript 10, is item 05 of batch 28, placed as an addition in commit `c6359e4` under the file prefix `10-support-cut-`. It pins `58cd8e1`, a commit on a line that does not contain `f4c9504`: its author never saw this report, and its repository comparison predates it. Its material carries the label sub-prefix `opa:sc:`.

Source	Manuscript	Principal contribution to this report
09	<i>Omnific-Preserving Automorphisms: convex-scale stabilizers, definable constants, and nondefinable monomials</i>	Base text and structure; the criterion for any characteristic-zero k and unital $\mathfrak{o}$; the derivation form with c_δ ; the Archimedean dichotomy; the strong factorization; the finite-rank derived-length bound and its sharpness for the abstract group; set-parameter nondefinability and the absence of any definable representative rule; $\mathbb{R}^\kappa$ actions and nonsolvable stabilizers; projections and truncation.
04	<i>Automorphisms Preserving the Omnific Integers: a convex-support criterion, fixed fields, and nondefinability of Conway monomials</i>	The import-free proof of the criterion through the logarithmic character (the proof of record here); exact displacement and fixed fields; the bottom subgroup and the relative Hahn hull; the common-positive-shift exponential–logarithm correspondence; the explicit commutator formula; the one-term class and the RV sort; the real axis inside the leading-term kernel.
08	<i>What the Omnific Integer Part Remembers: coefficient reconstruction, convex-scale automorphisms, nondefinable monomials, and an explicit wreath product</i>	The divisibility-free threshold for single fixed-shift flows; phase shears; the faithful $\mathbb{Z} \wr \mathbb{Z}$ and exact commutator leading terms; the floor map and the order formula on $\mathbf{Oz}$; the wild coefficient lift and the Gaussian fixed field $\mathbb{Q}$.
02	<i>Algebraic-Parameter Rigidity of the Omnific Integers: strong cancellation, finite-type embeddings, and nonalgebraizable formal flows</i>	All of part II .
10	<i>Omnific Integers Do Not Determine Surreal Monomials: exact support-cut stabilizers, fixed fields, and parameterwise nondefinability</i>	A fifth proof of the criterion, in the support-cut form, with the least-forbidden-shift argument as a second route; necessity without strongness; Archimedean blocks; twists with a general profile, their fixed fields, finite orbits and relative algebraic closedness; the set-sized definable-closure bound; the two-term witness; nondefinability of simplicity and of the Gonshor exponential; the question on other integer parts.
11	<i>Coefficient Gaps and Proper Self-Embeddings of the Surreal Field: omnific preservation, topological dichotomies, and nonconjugate copies</i>	Base of the embedding sections of part III : the bottom-gap classification, Taylor admissibility, closed images and the continuity dichotomy, nonelementarity, nonconjugate copies, parameter-fixed copies, conjugation-compatible surcomplex embeddings.

Source	Manuscript	Principal contribution to this report
12	<i>Automatic Strongness and Proper Embeddings of the Omnific Integers: constant-term duality, full stabilizers, and the surreal-surcomplex boundary</i>	Base of the automatic-strongness sections of part III : scalar detection, the constant-term isomorphism theorem, automatic strongness of $\text{Aut}(\mathbf{Oz})$, the factorization at set size and the Archimedean corollary, adjoints and retractions, the proper-class functional, regular cardinals.
13	<i>Automatic Hahn Linearity of Omnific Isomorphisms: residue duality, full stabilizers, and coefficient-drifting self-embeddings</i>	The same main theorems as 12, printed once; its countable binary witness, isomorphisms between different groups, the coefficient-matrix formula, the example that constant-term preservation does not preserve $\mathbf{Oz}$, the small-integer-part example and the failure of pair homogeneity.
14	<i>Automatic Summability from Omnific Arithmetic: constant-term duality, the full omnific stabilizer, and a sharp set-class distinction</i>	A third derivation of 12's and 13's main theorems, printed once with them; all of part IV : the induced-matching route to detection, detection in subfields with the countable detector property, formal evaluation, the general adjoint matrix, the case $\Gamma = \mathbb{R}$ and one new question.
15	<i>Algebraic Rigidity and Topological Collapse of Surreal Arithmetic: Hahn summation, omnific integer topologies, and the discrete-exponent boundary</i>	All of part V : the classification of Hahn-compatible ring topologies on $\mathfrak{o} \oplus \Pi$ for dense Γ , the omnific and field specializations, the cyclic boundary for full Hahn fields, a disjoint-row detector with interpolation, the class description of the strong dual of $\mathbf{No}$, the pairing topology and six new questions; a fourth derivation of 12's and 13's main theorems, printed once with them.
16	<i>Recovering Surreal Coefficients from Monomial Extensions: quartic definitions, arbitrary-rank rigidity, and sharp cancellation</i>	All of part VI : the quartic coefficient definitions, the sparse power bound, arbitrary-rank coefficient rigidity and automorphism formulas, the $\text{Hom}(G, \mathbb{Z}) = 0$ criterion, the finite/infinite cancellation threshold and ten new questions; it generalizes theorems 15.3 to 15.5 .
17	<i>Full Formal Orbit Fields of Surreal Arithmetic: proper-class independence, exact algebraic trajectories, and the effect of changing the formal parameter</i>	All of part VII : exact full-image descent for D_0 over $\ker D_0$, the additive parameter excluded, ordinal families of omnific witnesses, the multiplicative clock and its exact algebraic core, the monomial test, the surcomplex case, the abstract theorem and eight new questions; it answers, for D_0 , the lower bound left in part II .

Source	Manuscript	Principal contribution to this report
18	<i>Relative Universality and Exact Fixed Fields of Strong Omnific Symmetries: group realization, two-level Hahn supports, and conjugation on the surcomplex field</i>	All of part VIII : realization of every left-orderable set group with one prescribed fixed field, free equivariant cut filling, the converse, the field–ring equivalence, the fraction gap, explicit and Klein-bottle examples, the conjugation-compatible classification, the parity tower, the phase-twist fixed fields and eight new questions; it extends theorems 7.2 and 7.6 .
19	<i>All Derivations Integrate: Formal Symmetry and Coefficient Definability of Omnific Arithmetic</i>	All of part IX : automatic integration of every derivation, answering question 19.2 ; the integral exponential–logarithm correspondence, the nonsplit jet tower, residue sections and their common core, the polynomial–formal definability boundary (its quartic formula is part VI ’s, printed once) and seven new questions.
20	<i>Universal Symmetries and Exact Difference Equations over the Surreals: left-orderable groups, omnific stabilizers, fixed fields, and surcomplex descent</i>	All of part X : an independent derivation of part VIII ’s classification with the orbit of ω independent, a second cut filling, the translation family, a proper-class relative fixed field with an independent interval of double exponentials, the weighted Green operator, constant-coefficient operators and the operator field, the inner difference lemma, global and relative multiplicative criteria, the first-order criterion, central and free-group cocycle normal forms, set-sized models and eight new questions; its difference core re-derives the single-dilation report.

Manuscripts 04, 08 and 09 prove one classification at three generalities ([section 3](#)) and the same consequences: rank triviality, fixed field $\mathbb{R}$, set-parameter nondefinability and a nonnilpotent stabilizer. They are printed once. 09 is the base because its hypotheses on $(k, \mathfrak{o})$ are the weakest and its group-theoretic results the widest; 04’s proof is used for the criterion itself because it needs no imported correspondence, and 08’s single-flow version is kept as a marked case because it needs no divisibility. Manuscript 10, added later, is a fifth derivation of the same classification under the same hypotheses ([section 3.3](#)). Their reconstruction and definability sections re-derive results now in the Diophantine report; those are cited, and only the remaining extras are printed ([section 8](#)). Manuscript 02 has a different subject, algebraic families rather than automorphisms preserving a support condition, and forms [part II](#) with the sub-prefix `opa:par:`; its relation to [part I](#) is recorded in [section 19](#). None of the four contradicts another or any report in the collection. [Section A](#) records, section by section, what each source proved, where the merge had to choose, and every symbol renamed from a source.

Manuscript 10 re-derives the criterion, the rank dichotomy, the strong factorization, the fixed field of a shear, set-parameter nondefinability of the monomials and the Gaussian witnesses.

It has the hypotheses of [theorem 3.3](#) exactly: characteristic-zero k , an arbitrary unital $\mathfrak{o}$, divisible Γ , a set or **No**. None of this is a strengthening. These results are printed once, with 10 added to their sources. Its genuinely new material is appended at the end of the sections it belongs to, so that no existing number changes: [section 3.3](#) (support-cut form, second proof, blocks), [section 4.7](#) (general twists and their fixed fields), [example 6.13](#), [section 7.4](#), [section 9.2](#) (two-term witnesses, simplicity and the exponential), [proposition 10.5](#) and [question 11.7](#). 10 contradicts neither this report nor any other; its novelty statement, written before this report existed, is corrected in [section 11.1](#).

Manuscripts 11, 12 and 13, items 03, 04 and 05 of batch 30, were placed as an addition in commit 21375f8. They form [part III](#), placed after [part II](#) and before the appendices, so that no existing theorem, section or equation number changes; their labels carry the sub-prefix `opa:as:`. 12 and 13 prove the same main theorems and are printed once, with 12 as the base; 11 is the base of the embedding sections, which contain the embeddings of 12 and 13 as instances. Where the merge chose is recorded in [section 20.1](#). None of the three contradicts this report or another; they answer [question 11.1](#), and the passages of [part I](#) that recorded it as open now carry status notes.

Manuscript 14, item 07 of batch 31, was placed as an addition in commit 9d28e28. It pins 3d40856, which contains the batch-26 text of this report but none of the material of sources 10–13, and it proves the main theorems of 12 and 13 again, for $\Gamma = \Delta$ divisible. Those results are printed once in [part III](#), with 14 added to their sources; its genuinely new material forms [part IV](#), placed after [part III](#) and before the appendices, so that no existing number changes, with the sub-prefix `opa:cm:`. It contradicts neither this report nor any other; its statement that it removes the strongness hypothesis was accurate at its pin and is stale at the merge ([section 37.2](#)).

Manuscripts 15 and 16, items 03 and 08 of batch 32, were placed as additions in commit 7d04483. They form [parts V](#) and [VI](#), placed after [part IV](#) and before the appendices, so that no existing number changes, with the sub-prefixes `opa:tc:` and `opa:cr:`. 15 names no commit; it read this report's source as written in b3fa9e2 and then in 20c4c9f, so it saw [part III](#) but not [part IV](#). Its topological results are new; its algebraic half re-derives [part III](#) and is printed there once, with 15 added to the sources ([section 38.2](#)). 16 pins 1ab41af, which contains [part III](#) but not [part IV](#); its prior manuscript is source 02, and it generalizes [part II](#) to arbitrary rank ([section 43.2](#)). Neither contradicts this report or another report.

Manuscripts 17, 18 and 19, items 03, 05 and 08 of batch 33, were placed as additions in commit aa9c891. They form [parts VII](#) to [IX](#), one part each, placed after [part VI](#) and before the appendices, so that no existing number changes, with the sub-prefixes `opa:of:`, `opa:es:` and `opa:hs:`. 17 and 18 pin efc5446 ([parts I](#) to [III](#)); 19 pins 889b87c (also [part IV](#)); none saw [parts V](#) and [VI](#). 17 answers, for one Euler flow, the lower bound left after [corollary 17.9](#); 18 answers no named question and generalizes [theorem 7.6](#) to every left-orderable group, in the monomial factor of [theorem 24.1](#); 19 answers [question 19.2](#), and its quartic formula re-derives [theorem 45.4](#) and is printed once there. None contradicts this report or another report.

Manuscript 20, item 01 of batch 34, was placed as an addition in commit a7a435f under the file prefix 20-universal-symmetries-. It forms [part X](#), placed after [part IX](#) and before the appendices, so that no existing number changes, with the sub-prefix `opa:us:`. It pins e93a06d, which contains [parts I](#) to [IV](#) but not [parts V](#) and [VI](#) nor [parts VII](#) to [IX](#); so its universality theorem is an independent second derivation of [part VIII](#)'s, printed once there. Its difference-equation core re-derives the single-dilation report, which was in the tree at its pin but which it did not read ([section 76.2](#)). It answers no named question, contradicts neither this report nor another, and adds a new instance to [question 66.2](#).

1.3 Imported results

Conway normal forms, real closedness of **No** and the omnific integers are classical [7, 11]; factorization theory of **Oz** is due to L’Innocente and Mantova [19]. The monomial/character/kernel decomposition of valued Hahn automorphisms is due to Kuhlmann and Serra [17], and its surreal form, with the class-size conventions used here, to Kaplan, Krapp and Serra [15]; their rigidity of order together with simplicity [15, Proposition 5.1] is used in [theorem 9.10](#). Ehrlich and Kaplan’s initial subdomains and canonical integer part [8] are a different notion from a truncation-closed subring or a subring stabilized by an automorphism, and none is substituted for another here [10]. The collection’s surcomplex automorphism report (`docs/surcomplex/surcomplex-field-automorphisms`) already constructs the fixed-shift derivations and flows used below, in greater generality (its `saut:thm:shiftflow`), and the four-layer decomposition of valued automorphisms (`saut:thm:decomp`).

The correspondence between contracting strong derivations and strong 1-automorphisms is due to Bagayoko, Krapp, Kuhlmann, Panazzolo and Serra.

Cited theorem 1.1 ([3, Theorem 3.13]; [15, Fact 4.2]). *Let k have characteristic zero and Γ be an ordered abelian group. If ∂ is a contracting strongly k -linear derivation of $k((t^\Gamma))$, then*

$$\text{Exp}(\partial)(f) = \sum_{n \geq 0} \frac{\partial^n(f)}{n!}$$

is a strongly k -linear 1-automorphism, and $\partial \mapsto \text{Exp}(\partial)$ is a bijection between contracting strongly k -linear derivations and strongly k -linear 1-automorphisms, with inverse $\text{Log}(\sigma) = \sum_{n \geq 1} (-1)^{n+1} (\sigma - \text{id})^n / n$. Composition corresponds to the Baker–Campbell–Hausdorff operation.

Remark 1.2 (What is proved where). [3] proves the correspondence for set-sized Γ . Source 09 attributes the surreal case to [15, Fact 4.2]; the placement dossier, reading an HTML rendering, doubted that this Fact states the bijection. The merge re-read the PDF of [15] (version 3): Fact 4.2 does state the bijection, cites [3, Theorem 3.13], and adds that the correspondence “was established in the case that G is a set, but it also holds for G a proper class”, without a separate proof. So 09’s citation is accurate, and for $\Gamma = \mathbf{No}$ the operator-logarithm statements rest on that assertion. The criterion itself does not: its proof ([theorem 3.3](#)) and 04’s correspondence on the common-positive-shift class ([theorem 4.8](#)) are complete and elementwise, and the explicit constructions on **No** use one set-sized well-ordered set of shifts ([proposition 4.4](#)). Every statement below that uses [cited theorem 1.1](#) says so.

1.4 Class size and foundations

Unless $\Gamma = \mathbf{No}$ is explicitly under discussion, exponent groups are sets, so fields and automorphism groups are ordinary sets. For $\Gamma = \mathbf{No}$ we work in NBG with global choice [04, 08, 09]; every individual Hahn support and every summation family is a set. An automorphism of **No** is an individual class map; no set or NBG class whose members are all such proper-class graphs is formed. A statement about $\mathcal{U}_{\mathbf{Oz}}$ is a statement about individual maps, their compositions, or a set-indexed action encoded by one class relation; this is the convention of [15]. A common-fixed-field statement is pointwise; nonsolvability means that no ordinary finite derived length annihilates all iterated commutators; valuation representatives are treated through an equivalence relation on nonzero elements, not through proper-class cosets [09]. A quotient notation G/H in a full-class application abbreviates an additive map on G vanishing on H [08]. Bases are chosen only for set-sized vector spaces ([lemma 7.3](#)).

1.5 Notation and conventions

Part I uses the small- t Hahn convention of sources 04 and 09. **Part II** keeps its source's large-monomial convention, which is the convention of the other omnific reports; see [section 12.1](#). **Part III** uses the t -convention; its further symbols are fixed in [section 20.3](#), and those of **parts IV to VI** in [sections 33.3, 38.3 and 43.3](#); the monomials U^g of **part VI** are external indeterminates, not Hahn monomials. **Parts VII to X** use the large-monomial convention of **part II**; their symbols are fixed in [sections 53.3, 60.3, 67.3 and 76.3](#), and their formal variables s and v are external indeterminates. For actual surreals

$$t^\gamma = \omega^{-\gamma}. \quad (1.1)$$

Thus negative t -exponents are purely infinite terms and positive ones are infinitesimal. The tempting false reading is the opposite: in this Part, II is supported at *negative* t -exponents, and a “positive shift” $\delta > 0$ moves a term towards the infinitesimal side. Source 08 wrote in the ω -convention with decreasing supports; its statements have been translated, and its derivation $D_{\varphi,\delta}$ with $D_{\varphi,\delta}(\omega^g) = \varphi(g)\omega^{g-\delta}$ is $D_{\delta,\theta}$ below with $\theta(\gamma) = \varphi(-\gamma) = -\varphi(\gamma)$.

Notation	Meaning in part I , and the sources' symbols
$K = k((t^\Gamma))$	Hahn field, k of characteristic zero; $v(f) = \min \text{supp } f$, $v(0) = \infty$; $\text{lc}(f)$ the coefficient at $v(f)$.
Π	Series supported in $\Gamma^{<0}$, zero included (the purely infinite ideal). Sources: $\mathcal{J}$ (08, 09), $\mathcal{I}$ (04).
$\mathcal{A}_k = k \oplus \Pi$	Series supported in $\Gamma^{\leq 0}$, the boundary ring. Sources: B (09), $\mathcal{A}$ (04), $\mathcal{B}_k$ (08); the Diophantine report's $\mathcal{A}_k(\Gamma)$.
$\mathcal{R}_o = o \oplus \Pi$	The integer-part ring for a unital subring $o \subseteq k$; $\mathcal{R}_{\mathbb{Z}} = \mathbf{O}_{\mathbb{Z}}$ for $(\mathbb{R}, \mathbf{N}o)$ and $\mathcal{R}_{\mathbb{Z}[i]} = \mathbf{O}_{\mathbb{Z}[i]}$ for $(\mathbb{C}, \mathbf{N}o)$. Sources: A_D (09), O_D (04), A_k (08); the Diophantine report's $\mathcal{R}_o(\mathbb{K}, \Gamma)$. The letter D of the sources is freed for derivations.
$\mathcal{O}, \mathfrak{m}$	Series supported in $\Gamma^{\geq 0}$ and $\Gamma^{>0}$: the valuation ring and its maximal ideal.
P_-, P_0, P_+	Projections onto Π , k and $\mathfrak{m}$ in $K = \Pi \oplus k \oplus \mathfrak{m}$; P_0 restricted to $\mathcal{A}_k$ is the retraction ct .
C_δ	Principal convex subgroup $\{\gamma : \gamma \leq n\delta \text{ for some } n \geq 1\}$ of $\delta > 0$. Sources: C_δ (04, 09), H_δ (08), $H(a)$ (02); the transcendence report's $C(a)$.
$\delta \ll \epsilon$	$n\delta < \epsilon$ for every ordinary $n \geq 1$.
H_g	$\{h : n h < g \text{ for all } n \geq 1\}$ for $g \neq 0$ [04]; the same convex subgroup as the quotient report's H_a with $a = g $.
$a_{\sigma,\delta}$	Logarithmic-character coefficient $[t^\delta] \log(t^{-g}\sigma(t^g))$ of a 1-automorphism [04].
c_δ	Operator coefficient $[t^{\gamma+\delta}]\partial(t^\gamma)$ of a derivation ∂ [09]; for $\partial = \text{Log } \sigma$ it differs from $a_{\sigma,\delta}$ (example 3.2).
$D_{\delta,\theta}$	The derivation $t^\gamma \mapsto \theta(\gamma)t^{\gamma+\delta}$ for additive $\theta : \Gamma \rightarrow k$ and $\delta > 0$. Sources: $D_{\delta,\theta}$ (09), $D_{\delta,a}$ (04), $D_{\varphi,\delta}$ (08, sign as above).
$\mathcal{U}_{\mathcal{R}_o}(\Gamma), \mathcal{U}_{\mathbf{O}_{\mathbb{Z}}}$	Strong k -linear 1-automorphisms preserving $\mathcal{R}_o$ (independent of o by theorem 3.3). Sources: $\mathcal{U}_{A_D}$ (09), $\mathcal{U}_D(G)$ (04).

Notation	Meaning in part I , and the sources' symbols
$q_\beta(\gamma)$	The coefficient of ω^β in the Conway normal form of the <i>exponent</i> $\gamma \in \mathbf{No}$. Sources: q_η (09), $[\omega^\beta]g$ (04), c_a (08), χ_b (02), ℓ_η (quotient report).
$\exp(u)$, Exp, Log	The Hahn series $\sum u^n/n!$ for $v(u) > 0$ (E in 08); operator exponential and logarithm.

Two further collisions are resolved in the new material. Source 09's parameter monomial $H = \omega^\eta$ and its family H_j are written $\eta = \omega^\beta$ and η_j here, leaving H_g to 04's convex subgroup; and 08's automorphisms A_1, B_1 are written in 04's normalization A_s, B_u , with $A_1^{(08)} = A_{-1}$ and $B_1^{(08)} = B_{-1}$ ([section 6](#)). A 1-automorphism is a field automorphism that preserves the valuation and the leading coefficient of every nonzero series. *Strong* means that every Hahn-summable family is sent to a summable family with the same sum; a strong automorphism is one whose inverse is also strong (04's convention); *k-linear* includes fixing k pointwise.

Source 10's symbols. Source 10 writes in the large-monomial convention $X^g = \omega^g$ with reverse-well-ordered supports, that is $X^g = t^{-g}$; its statements are translated into the t -convention, and a t -exponent $\gamma < 0$ is its exponent $g = -\gamma > 0$. Two of its letters collide with this report's, and the tempting false readings are wrong:

- 10's H_h , the principal convex subgroup generated by h , is C_δ here, while 10's C_g , the elements infinitesimal relative to g , is H_g here. The letters C and H are *swapped*: 10's C_g is not a principal convex subgroup, and H_δ here is not 10's H_h .
- 10's projection pr_+ onto the positive ω -exponents, the purely infinite part, is P_- here, and its pr_- is P_+ ; its ct is P_0 . The subscripts are reversed by the sign convention.

Further renamings: 10's positive-exponent ideal $\mathcal{I}$ is Π ; its $\mathfrak{m}$ is $\mathfrak{m}$, the same set; its $R_D = D + \mathcal{I}$ is $\mathcal{R}_o$ with $D = o$; its multiplier ring B is $\mathcal{A}_k$; its stabilizer $\mathcal{U}_D$ is $\mathcal{U}_{\mathcal{R}_o}(\Gamma)$; its shift h is δ , and its condition " $h \ll g$ " is $\delta \ll |\gamma|$; its principal unit $u_g = 1 + \varepsilon_g$ is $1 + \varepsilon_\sigma(-g)$ ([section 3.3](#)); its exponent functional $\lambda_b = [\omega^b] \cdot$ is q_b ; its additive profile P and its formal variable T are Θ and ξ , since P and T are taken here; its twist $\sigma_{\lambda, F, h}$ is $\sigma_{h, -\lambda}^F$ of (4.4), the sign coming from $g = -\gamma$; its class twist $\Sigma_{b, h}$ is $\Sigma_{b, \delta}$ of (9.2); its expanded structure $\mathcal{N}^\dagger$ is $\mathfrak{S}^\dagger$ of [theorem 9.10](#).

Part I

Automorphisms preserving the omnific integers

2 Hahn support, the boundary ring, and strong maps

2.1 Summability

An element of $K = k((t^\Gamma))$ is a formal sum $f = \sum_{\gamma \in S} a_\gamma t^\gamma$ with $S \subseteq \Gamma$ well ordered. A set-indexed family $(f_i)_{i \in I}$ is *Hahn summable* if the union of its supports is well ordered and

each exponent lies in only finitely many of the supports; its sum is formed coefficientwise with finite sums in k . No convergence in the fine topology of the surreals is involved; a strong sum is a support-controlled algebraic construction [04, 08, 09].

We use the Neumann support principle in its word form: if S is well ordered and $T \subseteq \Gamma^{>0}$ is well ordered, then $S + \langle T \rangle_{\mathbb{N}}$ is well ordered, and each exponent receives only finitely many contributions from a choice of an element of S and a finite word in T , counted across all word lengths [3]. The one-shift case is proved directly.

Lemma 2.1 (One-shift summation [04, 08, 10]). *If $T \subseteq \Gamma$ is well ordered and $\delta > 0$, then $T + \mathbb{N}\delta$ is well ordered, and each $\gamma \in \Gamma$ has only finitely many representations $\gamma = g + n\delta$ with $g \in T$ and $n \in \mathbb{N}$.*

Proof. An infinite strictly decreasing sequence $g_j + n_j\delta$ has an infinite subsequence on which the n_j are nondecreasing; on it the g_j strictly decrease, contradicting well-ordering of T . Infinitely many representations of one γ would have unbounded, hence eventually increasing, values of n , and the corresponding $g = \gamma - n\delta$ would strictly decrease in T . $\square$

For $h \in \mathfrak{m}$ the series $\exp(h) = \sum_{n \geq 0} h^n/n!$ and $\log(1+h) = \sum_{n \geq 1} (-1)^{n+1} h^n/n$ are Hahn summable, and they are inverse group isomorphisms between $(\mathfrak{m}, +)$ and $(1 + \mathfrak{m}, \cdot)$. This local exponential is the only exponential used in the criterion; neither $\exp$ nor Exp asserts anything about the Gonshor exponential of $\mathbf{No}$.

Lemma 2.2 (Coefficient polynomial [04]). *For $h \in \mathfrak{m}$ and $\delta > 0$ the expression $P_{h,\delta}(X) = \sum_{n \geq 1} \frac{X^n}{n!} [t^\delta] h^n$ is a polynomial over k . For every $q \in \mathbb{Q}$ one has $P_{h,\delta}(q) = [t^\delta] \exp(qh)$, and its coefficient of X is $[t^\delta] h$.*

Proof. The support principle gives only finitely many contributions to the exponent δ across all powers h^n , not merely within each power. So only finitely many displayed terms are nonzero. The rest is coefficientwise. $\square$

2.2 The rings and the zero cut

The additive decompositions

$$K = \Pi \oplus k \oplus \mathfrak{m}, \quad \mathcal{A}_k = k \oplus \Pi, \quad \mathcal{R}_\mathfrak{o} = \mathfrak{o} \oplus \Pi, \quad \mathcal{O} = k \oplus \mathfrak{m} \quad (2.1)$$

define the projections P_-, P_0, P_+ [04, 09]. Membership in Π means that *all* exponents are negative, which is stronger than $v(f) < 0$: the series $t^{-1} + t$ has negative valuation and is not in Π [04]. Π is a nonunital ideal of $\mathcal{A}_k$ and of $\mathcal{R}_\mathfrak{o}$; $\mathcal{A}_k$ is not a valuation ring; P_0 is a ring retraction $\mathcal{A}_k \rightarrow k$ restricting to $\mathcal{R}_\mathfrak{o} \rightarrow \mathfrak{o}$, but not a ring map on K : $P_0(t^{-1})P_0(t) = 0 \neq 1 = P_0(1)$. For $k = \mathbb{R}$, $\mathfrak{o} = \mathbb{Z}$ the ring $\mathcal{R}_\mathbb{Z}$ is an integer part of K ; for $\Gamma = \mathbf{No}$ it is $\mathbf{Oz}$ and its floor function is the Diophantine report's `odg:thm:floor`.

The units $\mathcal{A}_k^\times = k^\times$, and more generally the fact that a product of two elements of $\mathcal{A}_k$ lying in $k^\times$ has constant factors, are the Diophantine report's constant-product lemma (`odg:lem:constantproduct`, `odg:def:lem:units`); sources 04, 08 and 09 reprove it, and it is not reprinted.

Lemma 2.3 (The divisible part [04, 08]). *If $\bigcap_{n \geq 1} n\mathfrak{o} = 0$ (for example $\mathfrak{o} = \mathbb{Z}$ or $\mathbb{Z}[i]$), then $\Pi = \bigcap_{n \geq 1} n\mathcal{R}_\mathfrak{o}$, the largest divisible subgroup of $\mathcal{R}_\mathfrak{o}$; it is preserved by every ring automorphism of $\mathcal{R}_\mathfrak{o}$. This holds for every ordered abelian Γ , divisible or not.*

Proof. A series in Π is n times a series with the same support. Conversely the constant coefficient of an element of the intersection lies in $\bigcap_n n\mathfrak{o} = 0$. Every divisible subgroup lies in the intersection. $\square$

For **Oz** this is the identity `odg:eq:divkernel` of the Diophantine report; it is printed because the threshold theorem uses it for non-divisible Γ .

Definition 2.4 (Principal convex scale). For $\delta > 0$ let $C_\delta = \{\gamma \in \Gamma : |\gamma| \leq n\delta \text{ for some } n \geq 1\}$, and write $\delta \ll \epsilon$ if $n\delta < \epsilon$ for all $n \geq 1$. Then $\gamma \notin C_\delta$ if and only if $\delta \ll |\gamma|$; $C_\delta \subsetneq C_\epsilon$ if and only if $\delta \ll \epsilon$; and C_δ is divisible when Γ is. For $g \neq 0$ the set $H_g = \{h : n|h| < |g| \text{ for all } n \geq 1\}$ is a convex subgroup, and a finite sum of positive elements of H_g is still smaller than $|g|$ [04].

3 The convex-support criterion

3.1 Two coefficient functionals

Let σ be a strong k -linear 1-automorphism. Since $t^{-g}\sigma(t^g)$ has leading term 1, its local logarithm is defined [04]:

$$h_\sigma(g) = \log(t^{-g}\sigma(t^g)) \in \mathfrak{m}, \quad a_{\sigma,\delta}(g) = [t^\delta]h_\sigma(g) \quad (\delta > 0), \quad (3.1)$$

so that $\sigma(t^g) = t^g \exp(h_\sigma(g))$. For a contracting strong k -derivation ∂ (that is, $v(\partial f) > v(f)$ for $f \neq 0$) put [09]

$$c_\delta(\gamma) = [t^{\gamma+\delta}]\partial(t^\gamma) \quad (\delta > 0), \quad \text{so} \quad \partial(t^\gamma) = t^\gamma \sum_{\delta>0} c_\delta(\gamma)t^\delta. \quad (3.2)$$

For fixed γ the nonzero terms have well-ordered set support; no assumption is made that all shifts together form one well-ordered set.

Lemma 3.1 (Additivity [04, 09]). *The maps $h_\sigma : \Gamma \rightarrow (\mathfrak{m}, +)$, $a_{\sigma,\delta}$ and c_δ are additive, hence $\mathbb{Q}$ -linear when Γ is divisible. If Γ is divisible and an additive $\theta : \Gamma \rightarrow k$ vanishes on $(-\delta, 0)$, it vanishes on C_δ .*

Proof. The quotient at $g+g'$ is the product of the quotients at g and g' , and $\log$ takes products to sums. For c_δ the product rule gives $t^{-\gamma-\eta}\partial(t^{\gamma+\eta}) = t^{-\gamma}\partial(t^\gamma) + t^{-\eta}\partial(t^\eta)$. For the last claim, given $0 \neq h \in C_\delta$ choose N with $|h|/N < \delta$; one of $\pm h/N$ lies in $(-\delta, 0)$, and additivity gives $\theta(h) = 0$. $\square$

Example 3.2 (The functionals differ [merge]). Let $\Gamma = \mathbf{No}$, $k = \mathbb{R}$, and $\partial = D_{1,q_1} + D_{\omega,q_2}$ (notation of section 1.5; the t -exponent γ has $q_1(\gamma)$ and $q_2(\gamma)$ as its coefficients at ω and ω^2). Take $\sigma = \text{Exp}(\partial)$, an element of $\mathcal{U}_{\mathbf{Oz}}$ by proposition 4.4, and $g = \omega^2$. Then $\partial(t^g) = t^{g+\omega}$ and $\partial^2(t^g) = t^{g+\omega+1} + t^{g+2\omega}$, and the only word in the shifts $\{1, \omega\}$ reaching the exponent $\omega+1$ applies the shift ω first; $\log(1+u) = u - u^2/2 + \dots$ adds nothing at $\omega+1$, since u^2 starts at 2ω . Hence

$$t^{-g}\sigma(t^g) = 1 + t^\omega + \frac{1}{2}t^{\omega+1} + (\text{terms at exponents } > \omega+1), \quad a_{\sigma,\omega+1}(\omega^2) = \frac{1}{2},$$

whereas $c_{\omega+1}(\omega^2) = 0$: the operator coefficients of ∂ at g are $c_\omega(g) = 1$ and zero at every other shift. So $a_{\sigma,\delta} \neq c_\delta$ in general; 04's condition concerns the logarithm of the monomial image, 09's the logarithm of the operator. Both vanishing conditions hold here, and by theorem 3.3 and corollary 3.5 each is equivalent to preservation of **Oz**, hence to the other. The placement dossier checked the example by exact computation.

3.2 The criterion

The following statement combines 09's generality with 04's proof. 04 proved it for $k = \mathbb{R}, \mathbb{C}$ and $\mathfrak{o} = \mathbb{Z}, \mathbb{Z}[i]$, passing through [lemma 2.3](#); 09 proved it for every characteristic-zero k and unital $\mathfrak{o}$, but through the operator logarithm ([cited theorem 1.1](#)). The first step below is 09's and removes the need for $\bigcap n\mathfrak{o} = 0$; the rest is 04's.

Theorem 3.3 (Convex-support criterion [04, 09, 10][merge]). *Let k have characteristic zero, let $\mathfrak{o} \subseteq k$ be a unital subring, let $\Gamma \neq 0$ be divisible (a set, or $\Gamma = \mathbf{No}$), and let σ be a strong k -linear 1-automorphism of $K = k((t^\Gamma))$. The following are equivalent:*

- (i) $\sigma(\mathcal{R}_{\mathfrak{o}}) = \mathcal{R}_{\mathfrak{o}}$;
- (ii) $\sigma(\mathcal{A}_k) \subseteq \mathcal{A}_k$;
- (iii) $\sigma(\Pi) \subseteq \Pi$;
- (iv) $a_{\sigma, \delta}(C_\delta) = 0$ for every $\delta > 0$;
- (v) for every $g \neq 0$, every exponent $\delta \in \text{supp } h_\sigma(g)$ satisfies $\delta \ll |g|$;
- (vi) σ commutes with P_- , P_0 and P_+ .

When they hold, $\sigma(\Pi) = \Pi$, $\sigma(\mathcal{A}_k) = \mathcal{A}_k$, $\sigma(\mathfrak{m}) = \mathfrak{m}$, the constant coefficient of every series is preserved, and every purely infinite monomial stays purely infinite, though not necessarily a monomial. In particular the group $\mathcal{U}_{\mathcal{R}_{\mathfrak{o}}}(\Gamma)$ does not depend on $\mathfrak{o}$.

Proof. (i) $\Rightarrow$ (ii) [09]. For $\gamma < 0$, $t^\gamma \in \mathcal{R}_{\mathfrak{o}}$, so $\sigma(t^\gamma) \in \mathcal{R}_{\mathfrak{o}} \subseteq \mathcal{A}_k$. An element of $\mathcal{A}_k$ is $c + \sum_{\gamma < 0} a_\gamma t^\gamma$ with $c \in k$; strong k -linearity writes its image as $c + \sum a_\gamma \sigma(t^\gamma)$, a summable family of series supported in $\Gamma^{\leq 0}$, whose sum is supported there.

(ii) $\Rightarrow$ (iv) [04]. Fix $\delta > 0$ and $0 \neq g \in C_\delta$; put $x = -|g| < 0$, so $h_\sigma(x) = \pm h_\sigma(g)$, and choose $N \geq 1$ with $|g| \leq N\delta$. By [lemma 2.2](#), $P(X) = [t^\delta] \exp(Xh_\sigma(x))$ is a polynomial. For rational $0 < q < 1/N$, divisibility gives $qx \in \Gamma^{< 0}$ and additivity gives $\sigma(t^{qx}) = t^{qx} \exp(qh_\sigma(x))$, whose coefficient at $qx + \delta$ is $P(q)$. Since $q|g| < \delta$, this exponent is strictly positive, while $\sigma(t^{qx}) \in \mathcal{A}_k$ has no positive exponents. So P has infinitely many zeros and vanishes; its linear coefficient is $a_{\sigma, \delta}(x)$, so $a_{\sigma, \delta}(g) = 0$. This tests every coefficient of the logarithm; testing only the first nonzero one would not suffice.

(iv) $\Leftrightarrow$ (v) [04]. $g \notin C_\delta$ means $\delta \ll |g|$; consider each nonzero coefficient.

(v) $\Rightarrow$ (iii) [04]. For $g < 0$, $\text{supp } h_\sigma(g) \subseteq H_g^{> 0}$, so every exponent of $\exp(h_\sigma(g))$ is 0 or a finite sum of elements of $H_g^{> 0}$, hence smaller than $|g|$; thus $\sigma(t^g) = t^g \exp(h_\sigma(g)) \in \Pi$. For $f \in \Pi$ write $\sigma(f) = \sum c_g \sigma(t^g)$ by strongness; all supports are negative.

(iii) $\Rightarrow$ (vi) [04]. A 1-automorphism maps $\mathfrak{m}$ into $\mathfrak{m}$, and σ fixes k ; so each summand of (2.1) is mapped into itself and σ commutes with the projections.

(vi) $\Rightarrow$ (i) [04]. If $\sigma(y) \in \Pi$ with $y = y_- + c + y_+$, then $0 = P_0 \sigma(y) = c$ and $0 = P_+ \sigma(y) = \sigma(y_+)$, so $y \in \Pi$. With (iii) this gives $\sigma(\Pi) = \Pi$, and since σ fixes $\mathfrak{o} \subseteq k$, $\sigma(\mathcal{R}_{\mathfrak{o}}) = \mathcal{R}_{\mathfrak{o}}$. The same argument gives $\sigma(\mathcal{A}_k) = \mathcal{A}_k$ and $\sigma(\mathfrak{m}) = \mathfrak{m}$, and (vi) preserves constant coefficients. Condition (ii) does not mention $\mathfrak{o}$, which proves the last assertion. $\square$

Only (ii) $\Rightarrow$ (iv) uses divisibility; it divides an *exponent*, while characteristic zero separately permits rational scalars [09]. The argument is elementwise with set-sized supports, so it applies verbatim to $\Gamma = \mathbf{No}$, $k = \mathbb{R}$, $\mathfrak{o} = \mathbb{Z}$, that is to $\mathbf{Oz}$, and to $\mathbf{Oz}[i]$ with $(k, \mathfrak{o}) = (\mathbb{C}, \mathbb{Z}[i])$ [04]. The theorem classifies *existing* strong maps: an arbitrary additive $h : \Gamma \rightarrow \mathfrak{m}$ need not define a strong surjective field map by $t^g \mapsto t^g \exp h(g)$ [04] ([question 11.2](#)).

Theorem 3.4 (Boundary-preserving derivations [09]). *Let $\Gamma \neq 0$ be divisible and let ∂ be a contracting strong k -derivation of K . Then $\partial(\mathcal{A}_k) \subseteq \mathcal{A}_k$ if and only if $c_\delta(C_\delta) = 0$ for every $\delta > 0$. In that case $\partial(\mathcal{A}_k) \subseteq \Pi$, $\partial(\Pi) \subseteq \Pi$, and ∂ commutes with P_-, P_0, P_+ .*

Proof. If $\partial(\mathcal{A}_k) \subseteq \mathcal{A}_k$ and $-\delta < \gamma < 0$, then $t^\gamma \in \mathcal{A}_k$ and $\gamma + \delta > 0$, so $c_\delta(\gamma) = 0$; lemma 3.1 extends this to C_δ . Conversely, if $\gamma < 0$ and $c_\delta(\gamma) \neq 0$, then $\gamma \notin C_\delta$, so $-\gamma > n\delta$ for all n and in particular $\gamma + \delta < 0$. Every term of $\partial(t^\gamma)$ is negative; constants are killed; strongness sums over any negative support. Positive exponents stay positive under positive shifts, which gives the projection identities. $\square$

The condition excludes a boundary term as well as a crossing term: testing $-\delta/2$ gives $c_\delta(-\delta) = 0$. This matters when constant coefficients are restricted to $\mathfrak{o}$ rather than k [09]. Source 04 proves the same criterion for derivations with common positive shifts (theorem 4.7), testing t^{-u} for $0 < u \leq \delta$.

Corollary 3.5 (Operator form [09]). *Let σ be a strong k -linear 1-automorphism of the form $\sigma = \text{Exp}(\partial)$ for a contracting strong k -derivation ∂ with $\text{Log}(\sigma) = \partial$. Then the conditions of theorem 3.3 are equivalent to $\partial(\mathcal{A}_k) \subseteq \mathcal{A}_k$ and to $c_\delta(C_\delta) = 0$ for all $\delta > 0$. For set-sized Γ every strong k -linear 1-automorphism has this form, so*

$$\mathcal{U}_{\mathcal{R}_\mathfrak{o}}(\Gamma) = \text{Exp}(\mathfrak{L}_A), \quad \mathfrak{L}_A = \{\partial \text{ contracting strong } k\text{-derivation} : c_\delta(C_\delta) = 0 \text{ for all } \delta > 0\}.$$

For $\Gamma = \mathbf{No}$ the same equality holds if the correspondence of cited theorem 1.1 is granted for proper-class Γ , as [15, Fact 4.2] asserts (remark 1.2); on the common-positive-shift class it is proved in theorem 4.8.

Proof. If the conditions hold, $\sigma(\mathcal{A}_k) = \mathcal{A}_k$; then each $(\sigma - \text{id})^n$ maps $\mathcal{A}_k$ into $\mathcal{A}_k$, and $\text{Log}(\sigma)$ is a strongly summable family of series supported in $\Gamma^{\leq 0}$, so $\partial(\mathcal{A}_k) \subseteq \mathcal{A}_k$. This step is taken in the k -algebra $\mathcal{A}_k$, not in $\mathcal{R}_\mathfrak{o}$, so no division by integers inside $\mathfrak{o}$ is needed. Theorem 3.4 converts this into the coefficient condition. Conversely, under the coefficient condition every iterate of ∂ maps $\mathcal{A}_k$ into Π and kills k , so $\text{Exp}(\pm\partial)$ preserve $\mathcal{R}_\mathfrak{o}$. The last assertions are cited theorem 1.1. $\square$

The two routes are genuinely different. 04's functional is read off the monomial images and needs no correspondence; 09's is read off the generator and gives the Lie-theoretic structure of section 6. Neither condition is stated in terms of the other: each is equivalent to preservation of $\mathcal{R}_\mathfrak{o}$ (example 3.2).

Remark 3.6 (What has and has not been classified [04, 09]). The classification is inside the strong, coefficient-fixing, leading-term-preserving automorphisms. It does not assert that every abstract field automorphism preserving $\mathcal{R}_\mathfrak{o}$ is strong; for the real pair, fixing of $\mathbb{R}$ is automatic (corollary 5.2), strongness is not known to be. The coefficient functionals come from an existing strong map; arbitrary functions satisfying the vanishing condition cannot be assembled into an operator without checking summability and invertibility.

Status (batch 30). For the real pair, strongness is now known to be automatic as well: every field automorphism of $\mathbf{No}$ preserving $\mathbf{Oz}$, and every automorphism of a full real Hahn pair with divisible exponent group, is strong (theorem 23.2 and corollary 23.4), so the classification covers them all (theorem 24.1). The remark stands as written for a general $(k, \mathfrak{o})$ (question 32.13), and its last sentence is unaffected (remark 24.2).

3.3 The support-cut form and a second route [10]

Source 10 proves [theorem 3.3](#) a fifth time, without knowledge of this report and under the same hypotheses. Its condition bounds the support of the principal unit itself rather than of its logarithm, and its necessity proof isolates the least forbidden shift instead of testing a polynomial in a rational parameter. The two forms are equivalent, so the statement is printed once, as further equivalent conditions, and 10's proof is kept as a marked second route. For a strong k -linear 1-automorphism σ put

$$\varepsilon_\sigma(\gamma) = t^{-\gamma}\sigma(t^\gamma) - 1 \in \mathfrak{m}, \quad 1 + \varepsilon_\sigma(\gamma) = \exp(h_\sigma(\gamma)). \quad (3.3)$$

Multiplicativity makes $\gamma \mapsto 1 + \varepsilon_\sigma(\gamma)$ a character $\Gamma \rightarrow (1 + \mathfrak{m}, \cdot)$ [17]. For $u \in 1 + \mathfrak{m}$ and $q \in \mathbb{Q}$ let $u^q = \sum_{r \geq 0} \binom{q}{r} (u - 1)^r = \exp(q \log u)$; the sum is Hahn summable by the support principle, and $(u^q)^n = u^{qn}$ for integers n is a formal identity of binomial series [10].

Lemma 3.7 (Leading-one roots [10]). *If $u, w \in 1 + \mathfrak{m}$ and $u^n = w^n$ for some integer $n \geq 1$, then $u = w$. Hence, for a character $c : \Gamma \rightarrow 1 + \mathfrak{m}$ on a divisible Γ , $c(q\gamma) = c(\gamma)^q$ for all $q \in \mathbb{Q}$ and $\gamma \in \Gamma$.*

Proof. Otherwise $u/w = 1 + \eta$ with $0 \neq \eta \in \mathfrak{m}$, and the term of least exponent of $(1 + \eta)^n - 1$ is n times that of η , which is nonzero in characteristic zero. For $q = a/b$ with $b \geq 1$, both $c(q\gamma)$ and $c(\gamma)^q$ have b th power $c(a\gamma) = c(\gamma)^a$. $\square$

Theorem 3.8 (Support-cut form [10]). *Under the hypotheses of [theorem 3.3](#), each of the following conditions is equivalent to the conditions (i)–(vi) there:*

- (vii) $\sigma(\mathcal{R}_0) \subseteq \mathcal{R}_0$;
- (viii) for every $\gamma < 0$, every $\delta \in \text{supp } \varepsilon_\sigma(\gamma)$ satisfies $\delta \ll |\gamma|$, that is $\text{supp } \varepsilon_\sigma(\gamma) \subseteq H_\gamma$;
- (ix) $\text{supp } \varepsilon_\sigma(\gamma) \subseteq H_\gamma$ for every $\gamma \neq 0$.

In source 10's notation, (viii) reads: every shift $h > 0$ occurring in ε_g , for $g > 0$, satisfies $nh < g$ for every $n \geq 1$.

First route, from the proof of [theorem 3.3](#). (i) $\Rightarrow$ (vii) is trivial, and (vii) $\Rightarrow$ (ii) is the proof of (i) $\Rightarrow$ (ii) there, which uses only $\sigma(t^\gamma) \in \mathcal{R}_0$ for $\gamma < 0$. For fixed $\gamma \neq 0$ the series $\varepsilon_\sigma(\gamma) = \exp(h_\sigma(\gamma)) - 1$ has its support among the finite sums of elements of $\text{supp } h_\sigma(\gamma)$, and $h_\sigma(\gamma) = \log(1 + \varepsilon_\sigma(\gamma))$ has its support among the finite sums of elements of $\text{supp } \varepsilon_\sigma(\gamma)$. Since H_γ is a subgroup, (ix) is equivalent to condition (v) of [theorem 3.3](#); this is 10's logarithmic corollary. Finally (ix) $\Rightarrow$ (viii) is trivial, and (viii) $\Rightarrow$ (ix) because, for $\gamma > 0$, $1 + \varepsilon_\sigma(\gamma) = (1 + \varepsilon_\sigma(-\gamma))^{-1} = \sum_{n \geq 0} (-\varepsilon_\sigma(-\gamma))^n$ has its support among the finite sums of elements of $\text{supp } \varepsilon_\sigma(-\gamma) \subseteq H_{-\gamma} = H_\gamma$. $\square$

Second route: least forbidden shift [10]. We prove (vii) $\Rightarrow$ (viii) directly. Fix $\gamma < 0$ and write $\varepsilon_\sigma(\gamma) = \sum_{\delta \in \mathcal{S}} b_\delta t^\delta$ with $\mathcal{S} \subseteq \Gamma^{>0}$ well ordered and all $b_\delta \neq 0$. Suppose some $\delta \in \mathcal{S}$ lies outside H_γ , and let δ be the least such element; the elements of $\mathcal{S}$ below δ lie in H_γ . As $\delta \notin H_\gamma$ there is $N \geq 1$ with $|\gamma| \leq N\delta$. Choose a rational $0 < q < 1/N$; divisibility gives $q\gamma \in \Gamma$ with $0 < q|\gamma| < \delta$, and [lemma 3.7](#) gives $t^{-q\gamma}\sigma(t^{q\gamma}) = (1 + \varepsilon_\sigma(\gamma))^q$. In this binomial series the linear term contributes qb_δ at the exponent δ . A term of degree $r \geq 2$ contributes there only through $\delta = \delta_1 + \dots + \delta_r$ with $\delta_j \in \mathcal{S}$; each δ_j is then smaller than δ , hence in H_γ , and so is their sum, which is impossible. So the coefficient of $\sigma(t^{q\gamma})$ at the exponent $q\gamma + \delta > 0$ is exactly $qb_\delta \neq 0$, whereas $t^{q\gamma} \in \Pi \subseteq \mathcal{R}_0$ and no element of $\mathcal{R}_0$ has a positive exponent. This contradicts (vii). $\square$

Conversely, under (viii) every exponent of $\sigma(t^\gamma)$, $\gamma < 0$, is γ or $\gamma + \delta$ with $\delta \ll |\gamma|$, hence negative; strongness gives $\sigma(\Pi) \subseteq \Pi$, which is (iii) of [theorem 3.3](#), and 10 concludes by the same injectivity and surjectivity argument as (iii) $\Rightarrow$ (vi) $\Rightarrow$ (i) there. $\square$

The two necessity proofs are genuinely different. That of [theorem 3.3](#) (from 04) shows, for each shift, that the coefficient of $\exp(qh_\sigma(x))$ is a polynomial in q with infinitely many zeros, and so tests every coefficient of the logarithm. 10's proof needs one rational q : at the least shift outside H_γ no nonlinear term of the binomial series can contribute. Both use divisibility of Γ only to form $q\gamma$, and neither uses $\mathfrak{o}$ beyond $\mathfrak{o} \subseteq k$. The inclusion form (vii) is implicit in the proof of [theorem 3.3](#) but not in its statement; 10 states it.

Remark 3.9 (Necessity without strongness [10]). The second route applies σ only to monomials, and uses only that σ is a field automorphism with $\sigma(t^\gamma) \in t^\gamma(1 + \mathfrak{m})$ for every γ : multiplicativity on monomials, [lemma 3.7](#), and $\sigma(t^{q\gamma}) \in \mathcal{R}_\mathfrak{o}$. So every k -fixing 1-automorphism σ of K with $\sigma(\mathcal{R}_\mathfrak{o}) \subseteq \mathcal{R}_\mathfrak{o}$, strong or not, satisfies (viii), and (ix) by inversion of units. What fails without strongness is the passage from monomials to arbitrary series, which the converse and all consequences for nonmonomial elements use. For **(No, Oz)** fixing of $\mathbb{R}$ is automatic ([corollary 5.2](#)), so every field automorphism of **No** that preserves **Oz** and the leading terms of monomials satisfies the monomial condition [10]. The merge notes that the necessity half of 04's route is equally monomial: it evaluates σ only at the monomials t^{qx} . This is partial information on [question 11.1](#), not an answer.

Status (batch 30). The remark is subsumed: [question 11.1](#) is now answered ([theorem 23.2](#)), every such automorphism of **(No, Oz)** is strong, and the passage from monomials to arbitrary series is automatic ([corollary 24.4](#)).

Proposition 3.10 (Signed Archimedean blocks [10]). *Call nonzero $\gamma, \gamma' \in \Gamma$ block-equivalent if they have the same sign and $|\gamma| \leq n|\gamma'|$, $|\gamma'| \leq n|\gamma|$ for some $n \geq 1$; the exponent 0 forms a block of its own. If σ satisfies the conditions of [theorem 3.3](#), every exponent of $\sigma(t^\gamma)$ lies in the block of γ . Consequently σ commutes with the projection onto the terms whose exponents lie in any prescribed union of blocks, and $\sigma(k((t^\Delta))) = k((t^\Delta))$ for every convex subgroup $\Delta \subseteq \Gamma$.*

Proof. By (ix) of [theorem 3.8](#), the exponents of $\sigma(t^\gamma)$ are $\gamma + \delta$ with $\delta = 0$ or $\delta \in H_\gamma^{>0}$, and then $2\delta < |\gamma|$. For $\gamma < 0$ this gives $|\gamma|/2 < |\gamma + \delta| \leq |\gamma|$, and for $\gamma > 0$ it gives $\gamma \leq \gamma + \delta < 3\gamma/2$; either way $\gamma + \delta$ is in the block of γ . If Λ is a union of blocks and $f = f_\Lambda + f'$ splits the terms of f according to Λ , strongness puts the support of $\sigma(f_\Lambda)$ inside Λ and that of $\sigma(f')$ outside it, so the projection of $\sigma(f)$ is $\sigma(f_\Lambda)$. A convex subgroup is a union of blocks, so σ maps $k((t^\Delta))$ into itself; σ^{-1} satisfies the same conditions, which gives equality. $\square$

The zero cut is a union of blocks, so this recovers commutation with P_-, P_0, P_+ ; [theorem 4.10](#) fixes $k((t^{C_\delta}))$ pointwise for single flows, whereas here the whole stabilizer preserves every convex Hahn subfield setwise. This is not preservation of every truncation: a twist may send t^γ to $t^\gamma + t^{\gamma+\delta}$ ([theorem 9.8](#)), and a cut strictly between the two exponents separates the two terms. The preserved blocks are determined by Archimedean scale, not by individual exponents [10].

4 Shears, flows, and the rank dichotomy

4.1 Homogeneous shears

Let $\theta : \Gamma \rightarrow (k, +)$ be additive and $\delta > 0$. The formula

$$D_{\delta, \theta} \left(\sum_{\gamma} a_{\gamma} t^{\gamma} \right) = \sum_{\gamma} a_{\gamma} \theta(\gamma) t^{\gamma + \delta} \quad (4.1)$$

defines a contracting strong derivation: translation keeps supports well ordered and families summable, and additivity of θ gives the product rule on monomials, hence on all series. This is the fixed-shift derivation of the surcomplex automorphism report, `saut:thm:shiftflow`, which constructs its exponential for every additive θ and every $\delta > 0$; its n th iterate has coefficient $\prod_{j < n} \theta(\gamma + j\delta)$. Sources 04, 08 and 09 re-derive it; only 04 credited it. The integer-part condition is the additional restriction $\theta(C_\delta) = 0$, not merely $\theta(\delta) = 0$.

Theorem 4.1 (Homogeneous shears [04, 09]; after `saut:thm:shiftflow`). *Suppose $\theta(C_\delta) = 0$. Then $D_{\delta,\theta}^n(t^\gamma) = \theta(\gamma)^n t^{\gamma+n\delta}$ and*

$$\sigma_{\delta,\theta}(t^\gamma) := \text{Exp}(D_{\delta,\theta})(t^\gamma) = t^\gamma \exp(\theta(\gamma)t^\delta). \quad (4.2)$$

The strong extension $\sigma_{\delta,\theta}$ lies in $\mathcal{U}_{\mathcal{R}_\circ}(\Gamma)$ for every $\circ$, with inverse $\sigma_{\delta,-\theta}$, and $\sigma_{\delta,\theta}\sigma_{\delta,\psi} = \sigma_{\delta,\theta+\psi}$ when θ, ψ both vanish on C_δ . With $\Phi_s = \sigma_{\delta,s\theta}$ ($s \in k$), $s \mapsto \Phi_s$ is a homomorphism, injective if $\theta \neq 0$. Conversely $\text{Exp}(D_{\delta,\theta})$ preserves $\mathcal{R}_\circ$ only if $\theta(C_\delta) = 0$.

Proof. $\theta(\delta) = 0$ gives $\theta(\gamma + n\delta) = \theta(\gamma)$ and the iterate formula. For a series supported on S , all terms of the image lie in $S + \mathbb{N}\delta$; lemma 2.1 gives well-ordering and finite contributions across all n , for single inputs and for summable families. On monomials the prescription is multiplicative by additivity of θ and the exponential law; strongness extends this to all series. The map fixes t^δ and every coefficient of $\exp(\cdot)$, so applying it termwise gives $\sigma_{\delta,\theta}\sigma_{\delta,\psi}(t^\gamma) = t^\gamma \exp((\theta + \psi)(\gamma)t^\delta)$, which gives the group law and the inverse. Every correction has larger exponent, so leading terms are fixed. Here $h_\sigma(\gamma) = \theta(\gamma)t^\delta$, whose only coefficient $a_{\sigma,\delta} = \theta$ annihilates C_δ ; apply theorem 3.3. The coefficient at $\gamma + \delta$ of $\Phi_s(t^\gamma)$ is $s\theta(\gamma)$, which gives injectivity. The converse is the same criterion, since $a_{\sigma,\delta} = \theta$ is the only functional. $\square$

The summability above is proved directly; no appeal to cited theorem 1.1 is made. For $\gamma < 0$ with $\theta(\gamma) \neq 0$, every $\gamma + n\delta$ stays negative, because $-\gamma > n\delta$ for all ordinary n : the whole exponential, not only a Taylor polynomial, stays on the correct side of the boundary [09].

Theorem 4.2 (Exact displacement and fixed field [04, 08, 09, 10]). *Let $\theta(C_\delta) = 0$, $s \neq 0$, and $f = \sum c_\gamma t^\gamma$. If $T_\theta(f) = \{\gamma \in \text{supp } f : \theta(\gamma) \neq 0\}$ is empty, $\Phi_s(f) = f$. Otherwise let $\gamma_0 = \min T_\theta(f)$; then*

$$v(\Phi_s(f) - f) = \gamma_0 + \delta, \quad [t^{\gamma_0+\delta}](\Phi_s(f) - f) = s\theta(\gamma_0)c_{\gamma_0}.$$

In particular $\text{Fix}(\Phi_s) = k((t^{\ker \theta}))$.

Proof. Only exponents with $\theta(\gamma) \neq 0$ and powers $n \geq 1$ contribute to the difference, at $\gamma + n\delta \geq \gamma_0 + \delta$, with equality only for $\gamma = \gamma_0$, $n = 1$. $\ker \theta$ is a divisible subgroup, so the Hahn field on it is a subfield. $\square$

This first-displacement computation appears three times in the sources: as 04's exact displacement, as 08's leading term of $T(x) - x$, and inside 09's fixed-field proof. It is printed once.

Example 4.3 (A nonmonomial image [04, 09]). Let $\Gamma = \mathbf{No}$, $\delta = 1$, $\theta = q_1$, the coefficient of ω in an exponent. Every member of C_1 is finite, so $q_1(C_1) = 0$. The flow fixes $\mathbb{R}$ and $\omega = t^{-1}$, and

$$\Phi_s(\omega^\omega) = \omega^\omega \exp(-s\omega^{-1}) = \sum_{n \geq 0} \frac{(-s)^n}{n!} \omega^{\omega-n}.$$

For $s \neq 0$ this is not a monomial; every exponent $\omega - n$ is positive, so it is an omnific integer. The sum is a Hahn sum with strictly decreasing exponents, not a limit in the fine topology.

4.2 Families of shifts and a proved correspondence

Proposition 4.4 (Uniformly supported shifts [09]). *Let $S \subseteq \Gamma^{>0}$ be a set-sized well-ordered set and $(\theta_s)_{s \in S}$ additive maps $\Gamma \rightarrow k$. Then $D(f) = \sum_{\gamma} \sum_{s \in S} a_{\gamma} \theta_s(\gamma) t^{\gamma+s}$ is a contracting strong derivation, the terms of $\text{Exp}(D)(f)$ are supported in $\text{supp } f + \langle S \rangle_{\mathbb{N}}$ and form a summable family, and if $\theta_s(C_s) = 0$ for all s then $\text{Exp}(D) \in \mathcal{U}_{\mathcal{R}_0}(\Gamma)$.*

Proof. The support principle applies to $\text{supp } f + S$ and to all finite words in S ; every coefficient of D , its iterates and products is a finite sum. Additivity of each θ_s gives the product rule; replacing one input support by the union of the supports of a summable input family gives strongness; each output exponent exceeds its input exponent. Preservation of $\mathcal{R}_0$ follows from the no-crossing argument of [theorem 3.4](#) applied at each step of a word in S , which shows that $\text{Exp}(\pm D)$ preserve Π and fix k . $\square$

All explicit constructions on **No** below use one set-sized well-ordered set of shifts, so they do not depend on the proper-class case of [cited theorem 1.1](#) [09]. Source 04 proves the correspondence itself on a class of this kind.

Definition 4.5 (Common positive shifts [04]). A strong k -linear operator L has *common positive shifts* if there is a well-ordered set $S \subseteq \Gamma^{>0}$ with $\text{supp } L(t^g) \subseteq g + S$ for every g . Write S^* for the additive monoid generated by S .

Lemma 4.6 (Iterated operator summability [04]). *If L has common positive shifts, then $(L^n f)_{n \geq 0}$ is Hahn summable for every f , formal power series in L define strong operators, and coefficientwise formal identities between such series hold under composition.*

Proof. $\text{supp } L^n f \subseteq \text{supp } f + nS$. Translate $\text{supp } f$ by its minimum and apply the support principle to the positive well-ordered set $S \cup ((\text{supp } f - \min \text{supp } f) \setminus \{0\})$; a base exponent is used once and later shifts come from S , so every exponent receives finitely many contributions across all paths and all n . For a summable input family use the union of supports; regrouping is justified because every coefficient has finitely many contributions. $\square$

Theorem 4.7 (Derivations with common shifts [04]). *Let $S \subseteq \Gamma^{>0}$ be well ordered and $a_{\delta} : \Gamma \rightarrow k$ additive for $\delta \in S$. Then $D = \sum_{\delta \in S} D_{\delta, a_{\delta}}$ is a strong k -linear derivation with common positive shifts, and every such derivation has a unique expression of this form. When $\bigcap_n n\mathfrak{o} = 0$, D maps $\mathcal{R}_0$ into itself if and only if $a_{\delta}(C_{\delta}) = 0$ for all $\delta \in S$.*

Proof. Summability and the Leibniz rule are termwise. Conversely put $a_{\delta}(g) = [t^{g+\delta}]D(t^g)$; the Leibniz rule for t^{g+h} gives additivity, and strong linearity the expression. If $D(\mathcal{R}_0) \subseteq \mathcal{R}_0$ and $x \in \Pi$, then $D(x) = nD(x/n) \in n\mathcal{R}_0$ for all n , so $D(x) \in \Pi$ by [lemma 2.3](#). For $0 < u \leq \delta$ the coefficient of $D(t^{-u})$ at the nonnegative exponent $\delta - u$ is $a_{\delta}(-u)$, with no cancellation between distinct shifts, so it vanishes; additivity and divisibility extend this to C_{δ} . The converse is the no-crossing argument. $\square$

In this theorem D is a derivation; the sources' coefficient ring D is $\mathfrak{o}$ here, which removes the collision 04 flagged.

Theorem 4.8 (Exponential and logarithm on common shifts [04]). *On operators with common positive shifts, $D \mapsto \text{Exp}(D)$ and $\sigma \mapsto \text{Log}(\sigma) = \sum_{n \geq 1} (-1)^{n+1} (\sigma - \text{id})^n / n$ are inverse bijections between strong k -linear derivations and strong k -linear field automorphisms whose difference from the identity has common positive shifts; the shift set may enlarge to $S^* \setminus \{0\}$. They restrict to a bijection between derivations preserving $\mathcal{R}_0$ and automorphisms preserving $\mathcal{R}_0$ (for $\bigcap_n n\mathfrak{o} = 0$).*

Proof. All series are justified by [lemma 4.6](#). For a derivation D , the iterated Leibniz rule gives multiplicativity of $\text{Exp } D$ after regrouping, and $\text{Exp } D \text{Exp } (-D) = \text{id}$ gives bijectivity; leading terms are fixed since all other shifts are positive. Conversely let $L = \sigma - \text{id}$ and $\sigma^{[X]} = \sum_{n \geq 0} \binom{X}{n} L^n$. For each ordinary $m \geq 0$ this is the binomial expansion of σ^m , so for every exponent γ the t^γ coefficient of $\sigma^{[X]}(fg) - \sigma^{[X]}(f)\sigma^{[X]}(g)$, a polynomial in X by [lemma 4.6](#), vanishes at all $m \geq 0$ and is zero. Differentiating at $X = 0$, with $\frac{d}{dX} \binom{X}{n} \big|_{X=0} = (-1)^{n-1}/n$, shows that $\text{Log } \sigma$ is a derivation; one-variable formal identities give the inverse correspondence. If D preserves $\mathcal{R}_0$ it maps Π into Π , so $\text{Exp}(\pm D)$ do; if $\sigma(\Pi) = \Pi$, then $(\sigma - \text{id})(\Pi) \subseteq \Pi$ and the strong logarithmic sum has negative support. $\square$

4.3 Single-shift flows without divisibility

Source 08 treats one flow at a time but allows any ordered abelian group and does not require $\theta(\delta) = 0$. Its statements are translated from the ω -convention ([section 1.5](#)). For an ordered abelian group G , $(k, \mathfrak{o}) = (\mathbb{R}, \mathbb{Z})$ or $(\mathbb{C}, \mathbb{Z}[i])$, let Π_G be the series supported in $G^{<0}$ and $\mathcal{R}_G = \mathfrak{o} \oplus \Pi_G$.

Proposition 4.9 (Falling-factorial iterates [08]). *For any additive $\theta : G \rightarrow k$ and $\delta > 0$, $D_{\delta, \theta}^n(t^\gamma) = \prod_{j=0}^{n-1} (\theta(\gamma) + j\theta(\delta)) t^{\gamma+n\delta}$, and $T^s = \sum_{n \geq 0} s^n D_{\delta, \theta}^n / n!$ ($s \in k$) is a strong k -linear 1-automorphism with $T^s T^u = T^{s+u}$. If $\theta(\delta) = 0$, $T^s(t^\gamma) = t^\gamma \exp(s\theta(\gamma)t^\delta)$; if $c = \theta(\delta) \neq 0$, $T^s(t^\gamma) = t^\gamma (1 + sct^\delta)^{\theta(\gamma)/c}$ as a formal binomial series. In either case $T^s(t^{-\delta}) = t^{-\delta} - s\theta(\delta)$.*

Proof. The iterate formula is induction. Summability, strongness and the group law follow from [lemma 2.1](#) as in [theorem 4.1](#), and the pairs (n, m) with $n + m = r$ are finite in number; multiplicativity follows from the iterated Leibniz rule. The two monomial formulas are the substitutions of the iterate. For $\gamma = -\delta$ every product of length at least two has the factor $\theta(-\delta) + \theta(\delta) = 0$. $\square$

This is `saut:thm:shiftflow` once more, over an arbitrary ordered abelian G ; 08 reproves the summability for that reason.

Theorem 4.10 (Threshold for single flows [08]). *Let G be any ordered abelian group, $\delta > 0$, $\theta : G \rightarrow k$ additive and $s \in k \setminus \{0\}$. The following are equivalent: (a) $T^s(\mathcal{R}_G) \subseteq \mathcal{R}_G$; (b) $T^s(\mathcal{R}_G) = \mathcal{R}_G$; (c) $\theta(C_\delta) = 0$. When they hold, T^s fixes every series supported in C_δ and preserves the constant coefficient of every series.*

Proof. Assume (a). By [lemma 2.3](#), $\Pi_G = \bigcap_n n\mathcal{R}_G$, so $T^s(\Pi_G) \subseteq \Pi_G$. The constant term of $T^s(t^{-\delta})$ is $-s\theta(\delta)$, so $\theta(\delta) = 0$. If $0 > \gamma \in C_\delta$ and $\theta(\gamma) \neq 0$, choose n with $\gamma + n\delta > 0$; the coefficient of $T^s(t^\gamma)$ there is $(s\theta(\gamma))^n / n! \neq 0$, and the exponents $\gamma + m\delta$ are distinct, contradicting $T^s(t^\gamma) \in \Pi_G$. So θ vanishes on the negative part of C_δ , hence on C_δ . Conversely, under (c), monomials with exponents in C_δ are fixed; if $\gamma < 0$ lies outside C_δ , then $\gamma < -n\delta$ for all n , so all $\gamma + n\delta$ stay negative; positive exponents outside C_δ stay positive. So no term enters or leaves exponent 0, $T^{\pm s}(\mathcal{R}_G) \subseteq \mathcal{R}_G$, and (b) follows. (b) $\Rightarrow$ (a) is trivial. $\square$

Only divisibility of *coefficients* by ordinary integers is used, so G need not be divisible. The theorem covers the elementary family only: even at rank one, exponent rescalings and characters preserve the integer part, and it does not assert that every leading-term-preserving automorphism is the exponential of one fixed-shift derivation [08].

Remark 4.11 (Divisibility and the ring $\mathcal{A}_k$ [09]; corrected). Source 09 remarks that divisibility of Γ is essential, because with $\Gamma = \mathbb{Z}$ the translation below preserves $\mathcal{A}_k(\mathbb{Z}) = k[t^{-1}]$. That is right for the boundary ring $\mathcal{A}_k$ and wrong if read for the integer-part ring: the translation

$t^{-1} \mapsto t^{-1} - 1$ sends $\frac{1}{2}t^{-1} \in \mathcal{R}_{\mathbb{Z}}(\mathbb{Z}) = \mathbb{Z} + t^{-1}\mathbb{R}[t^{-1}]$ to $\frac{1}{2}t^{-1} - \frac{1}{2} \notin \mathcal{R}_{\mathbb{Z}}(\mathbb{Z})$, as [theorem 4.10](#) predicts. So for $\Gamma = \mathbb{Z}$ the stabilizers of $\mathcal{A}_k$ and of $\mathcal{R}_{\mathbb{Z}}$ differ, and the independence of $\mathfrak{o}$ in [theorem 3.3](#) genuinely needs divisibility, whereas the $\mathcal{R}_{\mathbb{Z}}$ -form of the criterion holds without divisibility for single flows. Whether the general criterion for $\mathcal{R}_{\mathbb{Z}}$ holds for every non-divisible Γ is left open ([question 11.6](#)).

4.4 Why translations fail

Example 4.12 (Translations [04, 08, 09]; the saut example). Let $\Gamma = \mathbb{Q}$, $\delta = 1$, $\theta(\gamma) = \gamma$. Then $D^n(t^\gamma) = \gamma(\gamma+1) \cdots (\gamma+n-1)t^{\gamma+n}$ and $\text{Exp}(D)(t^\gamma) = t^\gamma(1-t)^{-\gamma}$; on t this is $t \mapsto t/(1-t)$, and on t^{-1} it is $t^{-1} \mapsto t^{-1} - 1$, which preserves $k[t^{-1}]$. But

$$\text{Exp}(D)(t^{-1/2}) = t^{-1/2}(1-t)^{1/2} = t^{-1/2} - \frac{1}{2}t^{1/2} - \frac{1}{8}t^{3/2} - \dots$$

has positive support, so neither $\mathcal{A}_k$ nor $\mathcal{R}_{\mathfrak{o}}$ is preserved: $\theta(C_1) \neq 0$. The inverse $t \mapsto t/(1+t)$ is 04's example and $X \mapsto X + s$, $X = t^{-1}$, is 08's. This is exactly the transparent example $T_1(t) = t/(1-t)$ of the surcomplex automorphism report (its $\lambda = \ell$, the constant-coefficient functional, with $\delta = 1$, after `saut:thm:shiftflow`); on the surreals $\ell(1) = 1 \neq 0$, so T_1 does not preserve **Oz**: it sends $\omega^{1/2}$ to $\omega^{1/2} - \frac{1}{2}\omega^{-1/2} - \dots \notin \mathbf{Oz}$. Preservation of a few familiar infinite integers is not enough; all rational monomial powers must be tested.

The Taylor coefficient flows of that report (`saut:thm:coeffflow`) move real constants and never preserve **Oz**: this follows from [corollary 5.2](#), and directly, for $d(b) = 1$ they send $bt^{-\delta/2}$ to $bt^{-\delta/2} + t^{\delta/2}$. Composed with a proper exponent embedding whose positive values dominate every multiple of δ , they do preserve and reflect **Oz**, as embeddings ([theorems 26.6](#) and [27.3](#)).

4.5 The rank dichotomy and the bottom subgroup

Theorem 4.13 (Rank dichotomy [04, 09, 10]; [08] for single flows). *For a nonzero divisible ordered abelian group Γ , any characteristic-zero k and unital $\mathfrak{o} \subseteq k$: $\mathcal{U}_{\mathcal{R}_{\mathfrak{o}}}(\Gamma) = \{\text{id}\}$ if and only if Γ is Archimedean. For every ordered abelian G , the single flows of [theorem 4.10](#) preserving $\mathcal{R}_G$ are trivial if G is Archimedean, and some are nontrivial if G is a set with a proper C_δ .*

Proof. If Γ is Archimedean, $C_\delta = \Gamma$ for all δ , so by [theorem 3.3](#) every $h_\sigma(g)$ vanishes; σ fixes all monomials, hence everything. If some C_δ is proper, choose (by a basis of the set-sized $\mathbb{Q}$ -space Γ/C_δ , or of the rationalization of the torsion-free G/C_δ for 08's case) a nonzero $\mathbb{Q}$ -linear θ into $\mathbb{Q} \subseteq k$ vanishing on C_δ ; [theorem 4.1](#) or [theorem 4.10](#) gives a nonidentity element. For $\Gamma = \mathbf{No}$ explicit functionals q_β replace the choice ([section 7](#)). $\square$

For $(k, \mathfrak{o}) = (\mathbb{R}, \mathbb{Z})$ and a set-sized Archimedean Γ , every field automorphism of $\mathbb{R}((t^\Gamma))$ preserving $\mathcal{R}_{\mathbb{Z}}$, strong or not, is therefore a monomial map $M_{\chi, \tau}$ ([corollary 24.3](#)).

Definition 4.14 (Bottom convex subgroup [04]). $B(\Gamma) = \bigcap_{\delta > 0} C_\delta$, possibly zero.

Theorem 4.15 (Common fixed field at arbitrary set-sized rank [04]). *For a set-sized nonzero divisible Γ , $\text{Fix}(\mathcal{U}_{\mathcal{R}_{\mathfrak{o}}}(\Gamma)) = k((t^{B(\Gamma)}))$.*

Proof. If $g \in B(\Gamma)$, every $a_{\sigma, \delta}(g)$ vanishes by [theorem 3.3](#), so t^g is fixed, and strong linearity fixes the Hahn field on $B(\Gamma)$. If $g \in \text{supp } f$ lies outside $B(\Gamma)$, choose δ with $g \notin C_\delta$ and a $\mathbb{Q}$ -linear θ with $\theta(C_\delta) = 0$ and $\theta(g) = 1$; [theorem 4.2](#) shows Φ_1 moves f . $\square$

Example 4.16 ([04, 09]). Let $\Gamma = \mathbb{Q}^r$, ordered so that the *last* nonzero coordinate decides the sign (09's reverse-lexicographic order; 04 orders by the first coordinate, which is the same group with coordinates reversed). Then the coordinate vectors satisfy $0 < e_1 \ll e_2 \ll \dots \ll e_r$, $C_{e_j} = \mathbb{Q}e_1 \oplus \dots \oplus \mathbb{Q}e_j$, and $B(\Gamma) = \mathbb{Q}e_1$. For $r = 1$ the stabilizer is trivial; for $r \geq 2$ it is nontrivial and fixes exactly $k((t^{\mathbb{Q}e_1}))$. The presence of a smallest Archimedean layer, not the number of layers, controls the fixed field. For $\Gamma = \mathbf{No}$, $B(\mathbf{No}) = 0$: for $g \neq 0$ take $\delta = |g|/\omega$.

4.6 Phase shears

Theorem 4.17 (Finite-dimensional phase shears [08]). *Let H be a convex subgroup of an ordered abelian group G , let $\mathfrak{m}_H$ be the series supported in $H^{>0}$, and let L be an additive map on G vanishing on H , with values in $\mathfrak{m}_H$ and image in a finite-dimensional k -subspace. Then $T_L(t^g) = t^g \exp(L(g))$ extends strongly to a k -linear 1-automorphism of the Hahn field that fixes $k((t^H))$ pointwise, preserves $\mathcal{R}_G$ and constant coefficients, and satisfies $T_L T_M = T_{L+M}$, $T_L^{-1} = T_{-L}$, and $T_L = \text{id} \Leftrightarrow L = 0$.*

Proof. Let $u_1, \dots, u_d$ span the image of L and $U = \bigcup_j \text{supp } u_j$, a well-ordered subset of $H^{>0}$. All supports of all powers in all $\exp(L(g))$ lie in the monoid generated by U , which is well ordered with finite representations; with the input support this gives one summable bound, hence the strong extension. Additivity of L gives multiplicativity. Exponents in H have zero phase, so $k((t^H))$ is fixed; in particular each $M(g)$ and its exponential are fixed, which gives $T_L T_M = T_{L+M}$ on monomials and then strongly. Injectivity follows from injectivity of $\exp$ on $\mathfrak{m}$. If $g \notin H$, every exponent of $t^g \exp(L(g))$ lies in $g + H$, which has constant strict sign by convexity; so positive and negative supports are preserved and no term reaches exponent 0. $\square$

The finite-dimensionality is a sufficient uniform support condition and must not be dropped silently: independent phases for infinitely many exponents need not form a summable family [08]. Taking $H = C_\delta$ and $L(g) = s\theta(g)t^\delta$ recovers the flows of [theorem 4.10](#).

4.7 Twists with a general profile [10]

Source 10 builds its explicit automorphisms directly, without derivations. Fix $\delta > 0$, a formal variable ξ , and an additive map $\Theta : \Gamma \rightarrow \xi k[[\xi]]$ into the power series without constant term. Define

$$\sigma_{\delta, \Theta} \left(\sum_{\gamma} a_{\gamma} t^{\gamma} \right) = \sum_{\gamma} a_{\gamma} t^{\gamma} \exp(\Theta(\gamma)(t^{\delta})), \quad (4.3)$$

where $\Theta(\gamma)(t^{\delta})$ is $\Theta(\gamma)$ evaluated at $\xi = t^{\delta}$. For $\Theta(\gamma) = \theta(\gamma)\xi$ this is the shear $\sigma_{\delta, \theta}$ of (4.2). Writing $\Theta(\gamma) = \sum_{n \geq 1} \theta_n(\gamma)\xi^n$ with additive θ_n , the map is Exp of $\sum_n D_{n\delta, \theta_n}$, a derivation with the one well-ordered shift set $\mathbb{N}_{>0}\delta$ as in [proposition 4.4](#) (an identification by the merge); 10's proof below is direct.

Theorem 4.18 (Twists [10]). *If $\Theta(\delta) = 0$, then $\sigma_{\delta, \Theta}$ is a strong k -linear 1-automorphism of K with strong inverse $\sigma_{\delta, -\Theta}$, and $\sigma_{\delta, \Theta} \sigma_{\delta, \Psi} = \sigma_{\delta, \Theta + \Psi}$ for every second such Ψ ; the map $\Theta \mapsto \sigma_{\delta, \Theta}$ is injective. If $\Theta(C_\delta) = 0$, then $\sigma_{\delta, \Theta} \in \mathcal{U}_{\mathcal{R}_o}(\Gamma)$ for every unital $\mathfrak{o} \subseteq k$; conversely [merge] $\sigma_{\delta, \Theta} \in \mathcal{U}_{\mathcal{R}_o}(\Gamma)$ forces $\Theta(C_\delta) = 0$. For set-sized Γ , the additive group $\text{Hom}_{\mathbb{Q}}(\Gamma/C_\delta, \xi k[[\xi]])$ thus embeds in $\mathcal{U}_{\mathcal{R}_o}(\Gamma)$.*

Proof. The support of $t^{\gamma} \exp(\Theta(\gamma)(t^{\delta}))$ lies in $\gamma + \mathbb{N}\delta$. For an input supported in S , [lemma 2.1](#) makes $S + \mathbb{N}\delta$ well ordered with finitely many pairs (γ, n) at each exponent, which defines (4.3); for a summable family of inputs use the union of their supports. So the map is strong. Additivity

of Θ gives multiplicativity on monomials, and strongness extends it to products of series, which are summable sums of monomial products; k is fixed. Since $\Theta(\delta) = 0$, the map fixes t^δ and hence every series in t^δ with coefficients in k , so $\sigma_{\delta, \Theta} \sigma_{\delta, \Psi}(t^\gamma) = t^\gamma \exp(\Theta(\gamma)(t^\delta)) \exp(\Psi(\gamma)(t^\delta)) = \sigma_{\delta, \Theta + \Psi}(t^\gamma)$, and strongness extends the identity; $\Psi = -\Theta$ gives the inverse. Every correction lies at a larger exponent, and no term arising from a larger input exponent reaches the least one, so leading terms are fixed. Evaluation at $\xi = t^\delta$ and $\exp$ are injective, which gives injectivity. If $\Theta(C_\delta) = 0$ and $\Theta(\gamma) \neq 0$, then $\gamma \notin C_\delta$, so every shift $n\delta$ satisfies $n\delta \ll |\gamma|$; thus $\text{supp } h_\sigma(\gamma) \subseteq H_\gamma$ for every $\gamma \neq 0$, and [theorem 3.3](#) applies. Conversely, if $0 \neq \gamma \in C_\delta$, say $|\gamma| \leq N\delta$, no shift $n\delta$ lies in H_γ , so condition (v) there forces $h_\sigma(\gamma) = \Theta(\gamma)(t^\delta) = 0$. $\square$

The field-automorphism part needs only $\Theta(\delta) = 0$; the stronger $\Theta(C_\delta) = 0$ is exactly the integer-part condition [10]. (For a single linear profile, [proposition 4.9](#) even allows $\theta(\delta) \neq 0$.) For additive $\theta : \Gamma \rightarrow k$ with $\theta(\delta) = 0$ and a profile $F \in 1 + \xi k[[\xi]]$, the choice $\Theta = \theta \log F$ gives the *binomial twist*

$$\sigma_{\delta, \theta}^F \left(\sum_{\gamma} a_{\gamma} t^{\gamma} \right) = \sum_{\gamma} a_{\gamma} t^{\gamma} F(t^{\delta})^{\theta(\gamma)}, \quad F^a = \exp(a \log F) = \sum_{n \geq 0} \binom{a}{n} (F - 1)^n. \quad (4.4)$$

Its inverse is $\sigma_{\delta, -\theta}^F = \sigma_{\delta, \theta}^{1/F}$, composition adds θ for fixed F, δ , and $\sigma_{\delta, \theta}^{\exp \xi} = \sigma_{\delta, \theta}$. The most economical profile is $F = 1 + \xi$:

$$\sigma_{\delta, \theta}^{1+\xi}(t^{\gamma}) = \sum_{n \geq 0} \binom{\theta(\gamma)}{n} t^{\gamma + n\delta}, \quad (4.5)$$

whose coefficients are polynomials over $\mathbb{Q}$ in $\theta(\gamma)$, with no choice of branch even for $k = \mathbb{C}$ [10]. Source 10's $\sigma_{\lambda, F, h}$ is $\sigma_{h, -\lambda}^F$ ([section 1.5](#)).

Example 4.19 (Two scales [10]). Let $\Gamma = \mathbb{Q}^2$ as in [example 4.16](#), so $e_1 \ll e_2$, and let θ be minus the e_2 -coordinate, which vanishes on $C_{e_1} = \mathbb{Q}e_1$. Then $\sigma = \sigma_{e_1, \theta}^{1+\xi}$ sends $t^{xe_1 + ye_2}$ to $t^{xe_1 + ye_2} (1 + t^{e_1})^{-y}$; for instance $\sigma(t^{-e_2}) = t^{-e_2} + t^{-e_2 + e_1}$ and

$$\sigma(t^{-e_2/2}) = t^{-e_2/2} + \frac{1}{2} t^{-e_2/2 + e_1} - \frac{1}{8} t^{-e_2/2 + 2e_1} + \frac{1}{16} t^{-e_2/2 + 3e_1} - \dots$$

Every exponent $-e_2/2 + ne_1$ is negative, however large n is. With $\Gamma = \mathbb{Q}$ the prescription $t^{-1} \mapsto t^{-1}(1+t)$ would force a positive exponent at $t^{-1/2}$ (compare [example 4.12](#)).

Theorem 4.20 (Fixed field and first displacement [10]). *Let $\theta(C_\delta) = 0$, let $F = 1 + c_m \xi^m + c_{m+1} \xi^{m+1} + \dots$ with $m \geq 1$ and $c_m \neq 0$, and put $\sigma = \sigma_{\delta, \theta}^F$. Then $\text{Fix}(\sigma) = k((t^{\ker \theta}))$. More precisely, if $f = \sum a_{\gamma} t^{\gamma}$ is not supported in $\ker \theta$ and $\gamma_* = \min\{\gamma \in \text{supp } f : \theta(\gamma) \neq 0\}$, then*

$$v(\sigma(f) - f) = \gamma_* + m\delta, \quad [t^{\gamma_* + m\delta}](\sigma(f) - f) = a_{\gamma_*} \theta(\gamma_*) c_m.$$

For every integer $n \neq 0$, $\text{Fix}(\sigma^n) = \text{Fix}(\sigma)$.

Proof. Monomials with $\theta(\gamma) = 0$ are fixed, hence, by strongness, all of $k((t^{\ker \theta}))$. For $a \neq 0$ the first nonconstant term of F^a is $ac_m \xi^m$, so for $\theta(\gamma) \neq 0$ the difference $\sigma(t^{\gamma}) - t^{\gamma}$ has least term $\theta(\gamma) c_m t^{\gamma + m\delta}$ and all other terms at exponents at least $\gamma + (m+1)\delta$. Exponents $\gamma < \gamma_*$ of f contribute no difference, and exponents $\gamma > \gamma_*$ contribute only beyond $\gamma + m\delta > \gamma_* + m\delta$. So nothing cancels the term at $\gamma_* + m\delta$. The composition law gives $\sigma^n = \sigma_{\delta, n\theta}^F$, and $\ker(n\theta) = \ker \theta$ in characteristic zero. $\square$

For $F = \exp \xi$ this is [theorem 4.2](#); the case of a general profile and the statement for powers are 10's.

Corollary 4.21 (No finite motion [10]). *Every finite orbit of σ in [theorem 4.20](#) is a fixed point, and $\text{Fix}(\sigma)$ is relatively algebraically closed in K .*

Proof. A finite orbit gives $\sigma^n(f) = f$ for some $n \geq 1$, hence $\sigma(f) = f$. If f is algebraic over $\text{Fix}(\sigma)$, all $\sigma^n(f)$ are roots of its minimal polynomial, whose coefficients are fixed; so the orbit is finite. $\square$

This is relative algebraic closedness only: with real coefficients the fixed field need not contain a square root of -1 [10].

Proposition 4.22 (Independent exponent cosets [10]). *Let $V = \ker \theta$ and let $(\gamma_j)_{j \in J}$ have $\mathbb{Q}$ -linearly independent images in Γ/V . Then the monomials t^{γ_j} are algebraically independent over $k((t^V))$.*

Proof. A relation involves finitely many indices and has the form $\sum_{\nu \in A} b_\nu t^{\nu_1 \gamma_1 + \dots + \nu_r \gamma_r} = 0$ with $b_\nu \in k((t^V))$, distinct ν and some $b_\nu \neq 0$. For $\nu \neq \mu$, $\sum_j (\nu_j - \mu_j) \gamma_j \notin V$, so the nonzero summands have supports in distinct cosets of V and cannot cancel. $\square$

Remark 4.23 (Rank, divisibility and what is classified [10]). Source 10 proves [theorem 4.13](#) with a binomial witness: if $\delta \ll |\gamma_0|$ with $\gamma_0 < 0$ and θ is $\mathbb{Q}$ -linear with $\theta(C_\delta) = 0$ and $\theta(\gamma_0) = 1$, then $\sigma_{\delta, \theta}^{1+\xi}(t^{\gamma_0}) = t^{\gamma_0} + t^{\gamma_0 + \delta}$, and the map fixes $k((t^{C_\delta}))$ pointwise. Its divisibility example is stated for $\mathfrak{o} = k$: $t^{-1} \mapsto t^{-1} + 1$ preserves $\mathcal{A}_k(\mathbb{Z}) = k[t^{-1}]$ for $\Gamma = \mathbb{Z}$, and 10 says explicitly that it does not preserve $\mathbb{Z} + t^{-1}\mathbb{R}[t^{-1}]$. This agrees with [remark 4.11](#) as corrected there; since 10 assumes divisibility throughout, it does not bear on [question 11.6](#). At Archimedean rank the monomial maps $t^\gamma \mapsto t^{2^\gamma}$ and the character rescalings still preserve $\mathcal{R}_\mathfrak{o}$; they lie outside the leading-term-fixing factor and are no counterexample to [theorem 4.13](#).

5 The strong stabilizer and all pair automorphisms

Call σ *valued* if $v(\sigma f) = \tau(v(f))$ for an order automorphism τ of Γ . For a character $\chi : (\Gamma, +) \rightarrow (k^\times, \cdot)$ put

$$M_{\chi, \tau} \left(\sum a_\gamma t^\gamma \right) = \sum a_\gamma \chi(\gamma) t^{\tau(\gamma)}. \quad (5.1)$$

It is a strong field automorphism preserving $\mathcal{R}_\mathfrak{o}$, since τ preserves signs of exponents and $\chi(0) = 1$; $M_{\chi, \tau} M_{\chi', \tau'} = M_{\chi' \cdot (\chi \circ \tau'), \tau \tau'}$ [04, 09].

Theorem 5.1 (Strong valued factorization [04, 09, 10]). *Every strong valued k -fixing automorphism σ with $\sigma(\mathcal{R}_\mathfrak{o}) = \mathcal{R}_\mathfrak{o}$ factors uniquely as $\sigma = M_{\chi, \tau} \circ u$ with $u \in \mathcal{U}_{\mathcal{R}_\mathfrak{o}}(\Gamma)$, where τ is the induced exponent automorphism and $\chi(\gamma) = \text{lc}(\sigma(t^\gamma))$. The monomial maps form a subgroup normalizing $\mathcal{U}_{\mathcal{R}_\mathfrak{o}}(\Gamma)$, and*

$$M_{\chi, \tau} D_{\delta, \theta} M_{\chi, \tau}^{-1} = D_{\tau(\delta), \chi(\delta) \theta \circ \tau^{-1}}, \quad (5.2)$$

so the convex condition is invariant, since $\tau(C_\delta) = C_{\tau(\delta)}$.

Proof. Multiplicativity of leading coefficients gives that χ is a character. $u = M_{\chi, \tau}^{-1} \sigma$ fixes the leading term of each monomial; since the images of distinct monomial terms have strictly ordered valuations, strongness shows that u fixes the leading term of every series, and u preserves $\mathcal{R}_\mathfrak{o}$. The data τ, χ are forced by σ , hence so is u . Conjugating a 1-automorphism by a monomial map gives a 1-automorphism preserving $\mathcal{R}_\mathfrak{o}$. Formula (5.2) is checked on monomials. $\square$

The ambient decomposition is due to Kuhlmann–Serra [17], and its surcomplex form is the four-layer decomposition `saut:thm:decomp` $U \rtimes \text{Hom}(\Gamma, \mathbb{C}^\times) \rtimes (\text{Aut } \mathbb{C} \times \text{Aut } \Gamma)$ of the surcomplex automorphism report; [theorem 5.1](#) is its restriction to the $\mathcal{R}_o$ -stabilizer with the residue layer trivial. Sources 04, 09 and 10 credit [17]; the new information is the description of u by [theorem 3.3](#). Source 10 states the factorization as $\text{Stab}(\mathcal{R}_o) = \mathcal{U}_{\mathcal{R}_o}(\Gamma) \rtimes \{M_{\chi, \tau}\}$, restricting to positive characters when k is ordered and only order-preserving maps are considered; it is not a new decomposition theorem for $\text{Aut}(\mathbf{No})$, and coefficient-moving or nonstrong automorphisms lie outside it [10]. For the real pair neither kind exists: coefficients are fixed by [corollary 5.2](#) and strongness is automatic by [theorem 23.2](#), so the splitting covers every automorphism ([theorem 24.1](#)). For $k = \mathbb{R}$ and divisible Γ every character is positive, $\chi(\gamma) = \chi(\gamma/2)^2$; for $\Gamma = \mathbb{Q}$ the factor u is absent and every such real automorphism is $t^q \mapsto c^q t^{\lambda q}$ with $c > 0$, $\lambda \in \mathbb{Q}_{>0}$, so the higher-rank shears are a new phenomenon, not complicated one-variable substitutions [09]. The monomial maps include the dilations S_a of the Diophantine report (`odg:def:ex:dilation`) and the character twists ([remark 9.5](#)).

Corollary 5.2 (All pair automorphisms [08, 09, 10]). *Every field automorphism of $\mathbf{No}$ preserving $\mathbf{Oz}$ setwise fixes $\mathbb{R}$ pointwise, preserves order and the natural valuation ring, and commutes with P_- , P_0 , P_+ and with the floor map of $\mathbf{Oz}$; this needs no strongness. Consequently, among strong automorphisms, the $\mathbf{Oz}$ -stabilizer is exactly $\{M_{\chi, \tau}\} \rtimes \mathcal{U}_{\mathbf{Oz}}$, with χ positive; by [theorem 23.2](#) every automorphism preserving $\mathbf{Oz}$ is strong, so this is the whole stabilizer ([theorem 24.1](#)).*

Proof. By `odg:def:thm:autreal` of the Diophantine report, every automorphism of $\mathbf{Oz}$ extends uniquely to $\mathbf{No}$, preserves order, $\mathbb{R}$ and the finite ring, and fixes $\mathbb{R}$ pointwise; restriction and extension are inverse correspondences [08, 09]. The sets and graphs of [theorem 8.2](#) are parameter-free definable, so they are preserved. Fixing $\mathbb{R}$ pointwise makes a strong such map k -fixing, and preserving the valuation ring makes it valued, so [theorem 5.1](#) applies. The last clause is [theorem 23.2](#), proved in [part III](#) from the first sentence of this corollary and [theorem 22.1](#), with no circularity. $\square$

6 Convex rank, solvability, and a wreath product

6.1 The bracket law

Lemma 6.1 (Homogeneous brackets [04, 08, 09]). *For additive $\theta, \psi : \Gamma \rightarrow k$ and $\delta, \epsilon > 0$,*

$$[D_{\delta, \theta}, D_{\epsilon, \psi}] = D_{\delta + \epsilon, \theta(\epsilon)\psi - \psi(\delta)\theta}. \quad (6.1)$$

If $\theta(C_\delta) = 0$ and $\psi(C_\epsilon) = 0$, the bracket is zero when $C_\delta = C_\epsilon$; if $C_\delta \subsetneq C_\epsilon$ it is a multiple of $D_{\delta + \epsilon, \psi}$, whose shift lies in the C_ϵ scale and whose functional vanishes on $C_{\delta + \epsilon} = C_\epsilon$.

Proof. On t^γ the two compositions differ at $\gamma + \delta + \epsilon$ by $\psi(\gamma)\theta(\gamma + \epsilon) - \theta(\gamma)\psi(\gamma + \delta) = \theta(\epsilon)\psi(\gamma) - \psi(\delta)\theta(\gamma)$; strongness extends the identity. Equal scales: each functional kills the other's shift. If $C_\delta \subsetneq C_\epsilon$, then $\psi(\delta) = 0$. $\square$

Source 08 records the same identity in its own normalization (its bracket $[D_{\varphi, \delta}, D_{\psi, \epsilon}] = D_{\psi(\delta)\varphi - \varphi(\epsilon)\psi, \delta + \epsilon}$, which is (6.1) after the sign change of [section 1.5](#)). Elementary derivations with shifts in the same Archimedean class commute; noncommutativity is an interaction between layers [04]. For general strong derivations the computation applies after expansion into shift coefficients: the two compositions are strongly summable and every output coefficient is a finite sum [09].

6.2 A filtration by convex scales

Say that Γ has *finite ordered rank* r if its distinct nonzero principal convex subgroups are exactly $0 < C_1 < \cdots < C_r = \Gamma$. The layer of a shift δ is the j with $C_\delta = C_j$. A shift in layer r contributes nothing to $\mathfrak{L}_A$ (corollary 3.5), since its functional vanishes on Γ . Let $\mathfrak{F}_j \subseteq \mathfrak{L}_A$ consist of the derivations whose nonzero shifts have layers at least j , so $\mathfrak{L}_A = \mathfrak{F}_1 \supseteq \cdots \supseteq \mathfrak{F}_r = 0$. Selecting the terms of prescribed layers from the images of monomials keeps a summable subfamily, and the new coefficient functions stay additive, so the $\mathfrak{F}_j$ are strongly closed Lie ideals [09].

Lemma 6.2 ([09]). $[\mathfrak{F}_j, \mathfrak{F}_j] \subseteq \mathfrak{F}_{j+1}$ and $[\mathfrak{L}_A, \mathfrak{F}_j] \subseteq \mathfrak{F}_j$.

Proof. Apply lemma 6.1 to pairs of shifts: equal layers cancel and unequal layers give a term in the larger layer. $\square$

Theorem 6.3 (Finite-rank bound [09]). *If the set-sized group Γ has finite ordered rank r , then $\mathfrak{L}_A$ and $\mathcal{U}_{\mathcal{R}_o}(\Gamma)$ have derived length at most $r - 1$.*

Proof. Inductively $\mathfrak{L}_A^{(n)} \subseteq \mathfrak{F}_{n+1}$. For the group, put $G_j = \text{Exp}(\mathfrak{F}_j)$; by the Baker–Campbell–Hausdorff part of cited theorem 1.1 each G_j is a subgroup, and the logarithm of a commutator of two elements of G_j is a strongly summable Lie series each of whose terms contains a bracket of two elements of $\mathfrak{F}_j$, hence lies in $\mathfrak{F}_{j+1}$. So $[G_j, G_j] \subseteq G_{j+1}$, and $G_r = \{\text{id}\}$. $\square$

The Lie bound is elementary; the group bound uses the imported correspondence, which [3] proves for set-sized Γ .

6.3 Sharpness, including the abstract group

In $\mathbb{Q}^r$ ordered as in example 4.16, let θ_i extract the i th coordinate, $X_{j,i} = D_{e_j, \theta_i}$ for $j < i$, and

$$a_0 = 0, \quad a_k = \sum_{h=1}^k 2^{k-h} e_h, \quad Y_{k;j,i} = D_{a_k + e_j, \theta_i} \quad (k < j < i \leq r).$$

These satisfy the criterion, since $C_{e_j} = \mathbb{Q}e_1 \oplus \cdots \oplus \mathbb{Q}e_j$.

Lemma 6.4 (Balanced brackets [09]). $Y_{k;j,i} \in \mathfrak{L}_A^{(k)}$, and $[Y_{k-1;k,j}, Y_{k-1;j,i}] = Y_{k;j,i}$ for $k \geq 1$.

Proof. At $k = 0$ this is the definition. In the bracket, $\theta_j(a_{k-1} + e_j) = 1$ and $\theta_i(a_{k-1} + e_k) = 0$, and the shift is $2a_{k-1} + e_k + e_j = a_k + e_j$; apply (6.1) and induct. $\square$

Lemma 6.5 (Mixed-parameter detection [09]). *Take finitely many homogeneous positive-shift derivations and an iterated Lie bracket W of them. Replace each leaf occurrence D_ℓ by $\text{Exp}(s_\ell D_\ell)$ with independent scalar parameters and each bracket by the group word $ghg^{-1}h^{-1}$, giving $\mathcal{W}(\mathbf{s})$. In its action on a monomial, the coefficient of the product of all leaf parameters is the action of W . If $W \neq 0$, some specialization $\mathbf{s} \in k^m$ gives a nonidentity group word.*

Proof. Formally, $(1+sD+\cdots)(1+uE+\cdots)(1-sD+\cdots)(1-uE+\cdots) = 1+su[D, E]+\text{terms of other multidegrees}$; induct on the bracket tree, using that the subtrees have disjoint parameter sets and that a group word is the identity when a required leaf parameter is zero. Choose a monomial and an output exponent on which W acts nontrivially. Only a finite set of shifts occurs, so by the support principle the coefficient at that output is a polynomial in the parameters, nonzero by the first part. An infinite field has a point where it does not vanish. Repeated leaves are handled by their separate parameters. $\square$

Theorem 6.6 (Sharp derived length [09]). *For $\Gamma = \mathbb{Q}^r$ as above, every characteristic-zero k and every unital $\mathfrak{o} \subseteq k$, the group $\mathcal{U}_{\mathfrak{o}}(\Gamma)$ and its Lie algebra $\mathfrak{L}_A$ have derived length exactly $r - 1$.*

Proof. For $r = 1$ use [theorem 4.13](#). For $r \geq 2$, $0 \neq Y_{r-2;r-1,r} \in \mathfrak{L}_A^{(r-2)}$ by [lemma 6.4](#), and [lemma 6.5](#) turns its balanced bracket into a nonidentity element of the $(r - 2)$ nd derived subgroup. The upper bound is [theorem 6.3](#). $\square$

The lower bound does not identify the abstract derived subgroup with the exponential of the derived Lie algebra; it uses independent parameters and a nonzero mixed coefficient. The upper bound uses BCH and strongly closed ideals [09].

6.4 Metabelian and not nilpotent: one group, printed once

On $\Gamma = \mathbf{No}$ put $D = D_{1,q_1}$, $E = D_{\omega,q_2}$ and $E_n = D_{\omega+n,q_2}$. Both D and E satisfy the criterion: q_1 kills the finite exponents and q_2 kills every exponent bounded by an ordinary multiple of ω .

Proposition 6.7 ([04, 09]). *$[D, E_n] = E_{n+1}$ and $[E_m, E_n] = 0$; the E_n are nonzero and linearly independent over $\mathbb{R}$. So the Lie algebra spanned by $D, E_0, E_1, \dots$ is metabelian and not nilpotent. The same holds in rank three: in $\mathbb{Q}^3$ with $X = D_{e_1,\theta_2}$, $Y = D_{e_2,\theta_3}$ one has $(\text{ad } X)^n Y = D_{e_2+n e_1,\theta_3} \neq 0$.*

Proof. $q_1(\omega+n) = 1$, $q_2(1) = q_2(\omega+n) = 0$; apply (6.1). $E_n(t^{\omega^2}) = t^{\omega^2+\omega+n}$ gives independence. The rank-three case is the same computation [09]. $\square$

Put $A_s = \text{Exp}(sD)$ and $B_u = \text{Exp}(uE)$ for $s, u \in \mathbb{R}$, and $[F, G] = FGF^{-1}G^{-1}$. Here e^{st} denotes the Hahn series $\exp(st)$ in the monomial $t = \omega^{-1}$.

Theorem 6.8 (Exact commutators [04, 08]). *$A_s B_u A_s^{-1} = \text{Exp}(u e^{st} E)$ and $[A_s, B_u] = \text{Exp}(u(e^{st} - 1)E)$. With $C_0 = B_u$ and $C_{n+1} = [A_s, C_n]$,*

$$C_n = \text{Exp}(u(e^{st} - 1)^n E), \quad C_n(t^{-\omega^2}) = t^{-\omega^2} \exp(-u(e^{st} - 1)^n t^\omega),$$

so for $s, u \neq 0$, $C_n(t^{-\omega^2}) - t^{-\omega^2}$ has leading term $-us^n t^{-\omega^2+\omega+n}$. Hence $\langle A_s, B_u \rangle$ is metabelian and not nilpotent. At $s = u = -1$, the leading term is $(-1)^n \omega^{\omega^2-\omega-n}$.

Proof. D fixes t (as $q_1(1) = 0$) and $A_s(t^\omega) = t^\omega e^{st}$. On monomials, $A_s E A_s^{-1}(t^\gamma) = q_2(\gamma) t^{\gamma+\omega} e^{st}$, because E kills every series in t and $q_1(\gamma + \omega) - q_1(\gamma) = 1$; so $A_s E A_s^{-1} = e^{st} E$, and exponentiating gives the first formula. E kills $\mathbb{R}[[t]]$, so the derivations $h(t)E$, $h \in \mathbb{R}[[t]]$, commute and their exponentials add coefficients, which gives the second formula and, by induction, C_n . All shifts lie in the monoid generated by the well-ordered set $\{1\} \cup (\omega + \mathbb{N})$, so [theorem 4.8](#) justifies all sums. The maps $\text{Exp}(h(t)E)$ form an abelian group normalized by A_s , which acts by $h \mapsto e^{st}h$; so the generated group is abelian-by-abelian. Since $q_2(-\omega^2) = -1$, $C_n(t^{-\omega^2})$ is as displayed, and $(e^{st} - 1)^n$ has leading term $s^n t^n$; the leading term of the difference is therefore nonzero at every depth, so the lower central series never terminates. $\square$

Source 08 works with $A_1^{(08)} = A_{-1}$ and $B_1^{(08)} = B_{-1}$ and proves faithfulness, which identifies the group. Write $v = t = \omega^{-1}$, $u_0 = t^\omega = \omega^{-\omega}$, $P = t^{-\omega} = \omega^\omega$, $M = t^{-\omega^2} = \omega^{\omega^2}$, and for h supported in $C_\omega^{>0}$ let $N_h(t^\gamma) = t^\gamma \exp(-q_2(\gamma)h)$, a phase shear ([theorem 4.17](#)) with $N_h N_{h'} = N_{h+h'}$, fixing v, u_0, P and with $N_h(M) = M \exp(h)$. Then $B_{-1} = N_{u_0}$, $A_{-1}(u_0) = u_0 e^{-v}$, $A_{-1}(P) = P e^v$, and $A_{-1} N_h A_{-1}^{-1} = N_{A_{-1}(h)}$.

Lemma 6.9 (Finite exponential independence [08]). *For a Laurent polynomial $p \in \mathbb{Z}[X, X^{-1}]$ put $h_p = u_0 p(e^{-v})$ and $\mathcal{N}_p = N_{h_p}$. Then $p \mapsto p(e^{-v})$ is injective, $p \mapsto \mathcal{N}_p$ is an injective homomorphism, and $A_{-1}\mathcal{N}_p A_{-1}^{-1} = \mathcal{N}_{Xp}$.*

Proof. If $\sum_{n \in F} k_n e^{-nv} = 0$ with $|F| = r$ distinct integers, comparing the coefficients of v^j , $0 \leq j < r$, gives a Vandermonde system with distinct nodes $-n$, so all $k_n = 0$. Injectivity of $\mathcal{N}$ follows from $\mathcal{N}_p(M) = M \exp(h_p)$, and $A_{-1}(h_p) = u_0 e^{-v} p(e^{-v}) = h_{Xp}$. $\square$

Theorem 6.10 (A faithful $\mathbb{Z} \wr \mathbb{Z}$ [08]). *The map $\mathbb{Z}[X, X^{-1}] \rtimes \mathbb{Z} \rightarrow \mathcal{U}_{\mathbf{Oz}}$, $(p, m) \mapsto \mathcal{N}_p A_{-1}^m$, with $(p, m)(q, n) = (p + X^m q, m + n)$, is an injective homomorphism onto $\langle A_{-1}, B_{-1} \rangle$. So this group is the restricted wreath product $\mathbb{Z} \wr \mathbb{Z}$, of derived length exactly two. It extends $\mathbb{C}$ -linearly to $\mathbf{No}(i)$ preserving $\mathbf{Oz}[i]$, and it already lives on the set-sized exponent group $\mathbb{Q}\omega^2 + \mathbb{Q}\omega + \mathbb{Q}$.*

Proof. Lemma 6.9 gives the multiplication law, and $\mathcal{N}_{X^n} = A_{-1}^n B_{-1} A_{-1}^{-n}$ shows that the image is generated by A_{-1}, B_{-1} . If $\mathcal{N}_p A_{-1}^m = \text{id}$, applying it to P gives $P e^{mv} = P$, so $m = 0$, and then to M gives $h_p = 0$, so $p = 0$. Finite Laurent polynomials under multiplication by X are $\bigoplus_{\mathbb{Z}} \mathbb{Z}$ with the shift. The group is nonabelian and its normal subgroup $\mathcal{N}$ is abelian with cyclic quotient. $\square$

Remark 6.11 (All nonzero times [merge]). The same proof gives $\langle A_s, B_u \rangle \cong \mathbb{Z} \wr \mathbb{Z}$ for all real $s, u \neq 0$: here $B_u = N_{-ut\omega}$, $A_s N_h A_s^{-1} = N_{A_s(h)}$ and $A_s(t^\omega) = t^\omega e^{st}$, so with $h_p = -ut^\omega p(e^{st})$ the map $(p, m) \mapsto N_{h_p} A_s^m$ obeys the same law; the Vandermonde nodes ns are distinct, and $A_s^m(P) = P e^{-mst}$ forces $m = 0$.

Corollary 6.12 (Every set stabilizer is not nilpotent [04, 08]). *For every set $P \subseteq \mathbf{No}$, the pointwise stabilizer of P in $\mathcal{U}_{\mathbf{Oz}}$ contains a subgroup isomorphic to $\mathbb{Z} \wr \mathbb{Z}$, metabelian and not nilpotent.*

Proof. Choose ordinals $0 < \beta < \beta'$ above every inner exponent of every outer exponent of every member of P (lemma 7.1), and replace ω, ω^2 by $\epsilon = \omega^\beta, \omega^{\beta'}$, using the functionals $q_\beta, q_{\beta'}$ and the shifts $1, \epsilon$. Both functionals vanish on the exponents of P and satisfy the criterion, and $q_\beta(\epsilon) = 1, q_\beta(1) = q_{\beta'}(\epsilon) = q_{\beta'}(1) = 0$; the proofs are unchanged. $\square$

These examples live in rank three, consistent with theorem 6.3 (derived length at most two there) [08, 09].

Example 6.13 (Binomial twists at three scales [10]). Source 10 shows noncommutativity with binomial twists at the scales of $x = \omega, y = \omega^\omega$ and $z = \omega^{\omega^2}$. Let $\sigma = \sigma_{1, -q_1}^{1+\xi}$ and $\tau = \sigma_{\omega, -q_2}^{1+\xi}$, that is $\sigma(\omega^g) = \omega^g(1 + \omega^{-1})^{q_1(g)}$ and $\tau(\omega^g) = \omega^g(1 + \omega^{-\omega})^{q_2(g)}$; both lie in $\mathcal{U}_{\mathbf{Oz}}$ by theorem 4.18, since $q_1(C_1) = 0$ and $q_2(C_\omega) = 0$. With $u = 1 + x^{-1}$,

$$\sigma(x) = x, \sigma(y) = yu, \sigma(z) = z, \quad \tau(x) = x, \tau(y) = y, \tau(z) = z(1 + y^{-1}),$$

so $\sigma\tau(z) - \tau\sigma(z) = -z/(xy(1 + x^{-1})) \neq 0$ and $[\sigma, \tau](z) = z(1 + 1/(yu))/(1 + 1/y) \neq z$, with $[\sigma, \tau] = \sigma\tau\sigma^{-1}\tau^{-1}$. The rational expressions denote the corresponding Hahn series. The example proves only that the integer-part stabilizer is not abelian and gives no presentation of it; the exponential shears at the same scales have the exact structure of theorems 6.8 and 6.10.

7 Fixed fields and arbitrarily large stabilizers

7.1 Exponent coefficient functionals

For $\beta \in \mathbf{No}$ let $q_\beta(\gamma) = [\omega^\beta]\gamma$ be the coefficient of ω^β in the Conway normal form of the exponent γ (the *inner* normal form; the outer support of $f = \sum c_\gamma t^\gamma$ consists of the exponents γ). Normal-form addition makes $q_\beta : \mathbf{No} \rightarrow \mathbb{R}$ additive; it need not commute with infinite sums inside the exponent group, and only finite additivity is used [04]. If $0 < \delta \ll \omega^\beta$ then $q_\beta(C_\delta) = 0$: a nonzero γ with $|\gamma| \leq n\delta$ has leading Conway exponent at most that of δ , which is below β [04, 08, 09].

Lemma 7.1 (Parameter avoidance [04, 08, 09, 10]). *Let $P \subseteq \mathbf{No}(i)$ be a set, E the union of the outer supports of the real and imaginary parts of its members, and T the union of the inner supports of the members of E . Then E and T are sets, and for every ordinal $\beta > 0$ above all members of T , q_β vanishes on E .*

Proof. Replacement and union; every set of surreals has a strict upper bound, for example an ordinal above all their birthdays. This bounds a set of supports; it does not choose a bound for all exponents at once. $\square$

Theorem 7.2 (The common fixed field [04, 08, 09]). *$\text{Fix}(\mathcal{U}_{\mathbf{Oz}}) = \mathbb{R}$, and already the elementary shears σ_{δ, q_β} have common fixed field $\mathbb{R}$. The same holds for all field automorphisms of $\mathbf{No}$ preserving $\mathbf{Oz}$. The common fixed elements in $\mathbf{Oz}$ are $\mathbb{Z}$, and the common fixed subring of all ring automorphisms of $\mathbf{Oz}$ is $\mathbb{Z}$.*

Proof. All these maps fix $\mathbb{R}$ (corollary 5.2). If $f \notin \mathbb{R}$, some nonzero $\gamma \in \text{supp } f$ has $q_\beta(\gamma) \neq 0$ for some β , for instance its leading inner exponent [08]. Put $\theta = q_\beta$ and $\delta = \omega^{\beta-1}$; then $\theta(C_\delta) = 0$ and $\theta(\delta) = 0$, and theorem 4.2 shows that $\sigma_{\delta, \theta}$ moves f : its first displacement is $a_\lambda \theta(\lambda) t^{\lambda+\delta}$, λ the least exponent of f with $\theta(\lambda) \neq 0$. Ring automorphisms of $\mathbf{Oz}$ are restrictions of field automorphisms preserving $\mathbf{Oz}$ (odg: def: thm: autreal). $\square$

Source 04 reaches the same conclusion through lemma 7.3 with $V = 0$; the explicit choice above is 08's and 09's.

7.2 Relative fixed fields and the Hahn hull

Lemma 7.3 (Set-sized coefficient separation [04]). *Let $V \subseteq \mathbf{No}$ be a set-sized $\mathbb{Q}$ -subspace and $g \in \mathbf{No} \setminus V$. There are $\delta > 0$ and an additive class map $a : \mathbf{No} \rightarrow \mathbb{R}$ with $a(V) = 0$, $a(C_\delta) = 0$ and $a(g) = 1$.*

Proof. Let $T = \bigcup_{x \in V \cup \{g\}} \text{supp}_\omega(x)$, a nonempty set of inner exponents, and $p_T(x) = ([\omega^\beta]x)_{\beta \in T} \in \mathbb{R}^T$. It is injective on the set-sized space $W = V + \mathbb{Q}g$, so $\ell_0(p_T(v + qg)) = q$ is a well-defined $\mathbb{Q}$ -linear functional on $p_T(W)$; extend it through a basis of the set $\mathbb{R}^T$ to ℓ and put $a = \ell \circ p_T$. Choose β_0 below every member of T and $\delta = \omega^{\beta_0}$; then $p_T(C_\delta) = 0$. $\square$

For a set $A \subseteq k((t^{\mathbf{No}}))$ ($k = \mathbb{R}$ or $\mathbb{C}$) let S_A be the union of the outer supports, $V_A = \text{span}_{\mathbb{Q}}(S_A)$ and $\mathcal{H}(A) = k((t^{V_A}))$; let $\mathcal{E}_A$ be the group generated by the elementary shears Φ_s (with real s) that fix A pointwise, in the sense of section 1.4.

Theorem 7.4 (Exact elementary hull [04]). *For every set $A \subseteq k((t^{\mathbf{No}}))$, $\text{Fix}(\mathcal{E}_A) = \mathcal{H}(A) = k((t^{V_A}))$.*

Proof. By [theorem 4.2](#) a nonidentity shear with functional a fixes all of A iff a vanishes on S_A , iff $a(V_A) = 0$; such shears fix every series supported in V_A . If $f \notin \mathcal{H}(A)$, take $g \in \text{supp } f \setminus V_A$ and apply [lemma 7.3](#). $\square$

Corollary 7.5 (Monomial parameters [04]). *For a set $S \subseteq \mathbf{No}$ with $V = \text{span}_{\mathbb{Q}}(S)$, the pointwise stabilizer in $\mathcal{U}_{\mathbf{Oz}}$ of $\{t^g : g \in S\}$ has common fixed field exactly $\mathbb{R}((t^V))$.*

Proof. The elementary subgroup already has this fixed field. Conversely an ordered field automorphism fixing t^g fixes its positive rational powers, hence all t^v , $v \in V$, and strong linearity fixes every series supported on V . $\square$

Naming $\omega = t^{-1}$ gives the fixed field $\mathbb{R}((t^{\mathbb{Q}}))$, much larger than $\mathbb{R}(\omega)$; naming all monomials with real exponents gives $\mathbb{R}((t^{\mathbb{R}}))$ [04]. On sets, $A \mapsto \mathcal{H}(A)$ is extensive, monotone and idempotent, and $\mathcal{H}(A)$ is the smallest Hahn subfield with divisible exponent group containing every monomial of the normal forms of A [04]. For nonmonomial parameters [theorem 7.4](#) computes the fixed field of $\mathcal{E}_A$ only: a general strong automorphism fixing a series may do so by cancellation, and $\mathcal{H}(A)$ is not asserted to be the model-theoretic definable closure [04] ([question 11.3](#)).

7.3 Product actions and nonsolvable stabilizers

Theorem 7.6 (Faithful $\mathbb{R}^\kappa$ actions [09]). *For every set $P \subseteq \mathbf{No}$ and every set cardinal κ there is a faithful action of the full product $\mathbb{R}^\kappa$ by strong 1-automorphisms preserving $\mathbf{Oz}$ and fixing P pointwise.*

Proof. Let E, T be as in [lemma 7.1](#), and choose an ordinal β above κ and all of T . For $j < \kappa$ put $\delta_j = \omega^j$, $\eta_j = \omega^{\beta+j}$ (ordinal addition in the index) and $\theta_j = q_{\beta+j}$. Then $\theta_j|_E = 0$, $\theta_j(C_{\delta_j}) = 0$, $\theta_j(\delta_\ell) = 0$ and $\theta_j(\eta_\ell)$ is the Kronecker delta. For $\mathbf{a} \in \mathbb{R}^\kappa$ let $D_{\mathbf{a}}(t^\gamma) = t^\gamma \sum_{j < \kappa} a_j \theta_j(\gamma) t^{\delta_j}$; the shifts form one well-ordered set, so [proposition 4.4](#) applies with no finiteness condition on $\mathbf{a}$. $D_{\mathbf{a}}$ kills the exponents of P . The $D_{\mathbf{a}}$ commute by (6.1) and add, so $\mathbf{a} \mapsto \text{Exp}(D_{\mathbf{a}})$ is a homomorphism, and $\text{Exp}(D_{\mathbf{a}})(t^{\eta_j}) = t^{\eta_j} \exp(a_j t^{\delta_j})$ detects every a_j . The result is one class relation encoding a set-indexed action. $\square$

Batch 33. Source 18 realizes every nontrivial left-orderable set group, $\mathbb{R}^\kappa$ included, faithfully by strongly additive $\mathbf{Oz}$ -preserving automorphisms fixing a given set P , with the exact fixed field $\mathbb{K}_{\Sigma_2(P)}$ for every nonidentity element, and shows that set groups acting faithfully are left-orderable ([theorem 60.1](#) and [corollary 63.4](#)); its maps lie in the monomial factor $\{M_{\chi, \tau}\}$, not in $\mathcal{U}_{\mathbf{Oz}}$. It also gives single automorphisms whose fixed field is exactly $\mathbb{R}$ and fixed ring exactly $\mathbb{Z}$, sharpening [theorem 7.2](#) ([corollary 60.3](#) and [remark 63.5](#)).

Theorem 7.7 (Nonsolvable stabilizers [09]). *For every set $P \subseteq \mathbf{No}$ the pointwise stabilizer of P in $\mathcal{U}_{\mathbf{Oz}}$ is nonsolvable; for every $r \geq 2$ it contains a subgroup whose derived length is at least $r - 1$, and exactly $r - 1$ granted [cited theorem 1.1](#) for $\Gamma = \mathbf{No}$.*

Proof. Choose ordinals $\beta \leq \eta_1 < \dots < \eta_r$ above T , and put $e_j = \omega^{\eta_j}$, $\theta_i = q_{\eta_i}$. Then $e_1 \ll \dots \ll e_r$, $\theta_i(e_j)$ is the Kronecker delta, and each D_{e_j, θ_i} ($j < i$) satisfies the criterion and kills P . The balanced brackets of [lemma 6.4](#) are unchanged, so [lemma 6.5](#) shows that the group generated by all $\text{Exp}(s D_{e_j, \theta_i})$, $s \in \mathbb{R}$, has derived length at least $r - 1$; this already gives nonsolvability. The upper bound applies the filtration by the $r - 1$ scales $C_{e_1}, \dots, C_{e_{r-1}}$ as in [theorem 6.3](#); its group step uses BCH. $\square$

7.4 A definable-closure bound in set-sized Hahn fields [10]

For a set-sized Γ consider the structure $K^\dagger = (K; 0, 1, +, -, \cdot, \mathcal{R}_o, (c)_{c \in k}, P_-, P_0, P_+)$, with the Hahn order added when k is ordered. For a set $A \subseteq K$ of parameters let S_A be the union of the supports of its members, as in [theorem 7.4](#), and put $V_\delta(A) = \text{span}_{\mathbb{Q}}(S_A) + C_\delta$ for $\delta > 0$ (10's S , A_S and $V_h(S)$).

Proposition 7.8 (Definable-closure support bound [10]). *Every element of K definable over A in $K^\dagger$ is supported in $V_\delta(A)$ for every $\delta > 0$; that is, $\text{dcl}(A) \subseteq \bigcap_{\delta > 0} k((t^{V_\delta(A)}))$.*

Proof. Let $g \in \text{supp } f$ lie outside the $\mathbb{Q}$ -subspace $V_\delta(A)$. A basis of the set-sized $\mathbb{Q}$ -space $\Gamma/V_\delta(A)$ gives a $\mathbb{Q}$ -linear $\theta : \Gamma \rightarrow \mathbb{Q} \subseteq k$ with $\theta(V_\delta(A)) = 0$ and $\theta(g) = 1$. The twist $\sigma_{\delta, \theta}^{1+\xi}$ lies in $\mathcal{U}_{\mathcal{R}_o}(\Gamma)$ ([theorem 4.18](#)); it fixes k , preserves $\mathcal{R}_o$, commutes with P_-, P_0, P_+ ([theorem 3.3](#)), preserves lc and hence the order, and fixes every monomial with exponent in S_A , hence every member of A . By [theorem 4.20](#) it moves f . An element definable over A is fixed by every automorphism of $K^\dagger$ fixing A . $\square$

The bound may be strict. It is not a quantifier-elimination result and does not describe the whole definable closure; it says only that support excluded by a separating exponent functional cannot be introduced by a definition [10]. For $\Gamma = \mathbf{No}$ see [corollary 9.11](#).

8 What the pair (No, Oz) remembers

Sources 04, 08 and 09 each reconstruct the coefficient field from **Oz**; 08 and 09 add first-order definitions of the ordinary integers and of Π . All of this is now in the omnific Diophantine report, placed after the three pins, and is cited rather than reprinted:

- Π is defined in **Oz** and **Oz**[i] by $\exists y (x^2 = 2y^2)$ (`odg:def:thm:ideal`); the first-order definitions of Π and of the graph of ct through ordinary integers are `odg:cor:definablect`, and a positive-existential definition of $\mathbb{Z}$ is `odg:thm:guard`; Pell rigidity is `odg:cor:pell`;
- the multiplier identity $(\Pi : \Pi) = \mathcal{A}_k$, $k = \{0\} \cup \mathcal{A}_k^\times$ and its internal first-order form (`odg:def:thm:multiplier`, `odg:def:cor:internal`); $\text{Frac } \mathbf{Oz} = \mathbf{No}$ (`odg:thm:fractions`);
- the pure ring **Oz** interprets **No** with $\mathbb{R}$, the order, the finite ring, the infinitesimals and the standard part (`odg:def:thm:realrecovery`); the value group (`odg:def:cor:valuegroup`); every automorphism of **Oz** extends uniquely to **No** and fixes $\mathbb{R}$ pointwise (`odg:def:thm:autoreal`);
- for **Oz**[i]: $\mathbb{C}$ is definable (`odg:def:cor:internal`), every field automorphism of $\mathbb{C}$ lifts coefficientwise and restriction to $\mathbb{C}$ splits (`odg:def:prop:coeffaut`); a phase twist shows that neither **No** nor conjugation is definable from **Oz**[i] (`odg:def:lem:twist`, `odg:def:thm:norealaxis`).

In the sources: 04's intrinsic coefficient invariance, 08's characteristic reconstruction, its restriction correspondence and its split restriction to $\text{Aut } \mathbb{C}$, and 09's corollary on all pair automorphisms re-derive `odg:def:thm:multiplier`, `odg:def:thm:autoreal` and `odg:def:prop:coeffaut`; source 10's coefficient-rigidity proposition re-derives the first two, with credit ([section 9.2](#)). 08's formula $J(x) :\Leftrightarrow \forall n (\text{Int}(n) \wedge n \neq 0 \Rightarrow \exists y x = ny)$ is the definition `odg:cor:definablect` in structure (present at 08's pin); 09's order-free formula $\Phi_J(a) :\Leftrightarrow a \in \mathbf{Oz} \wedge \forall u, v \in \mathbf{Oz} [u^2 - 2v^2 = 1 \Rightarrow \exists b \in \mathbf{Oz} a = ub]$ is a variant using all Pell first coordinates; both are dominated by the one-witness existential formula `odg:def:thm:ideal`,

and 08's existential definition of $\mathbb{Z}$ (six witnesses, two inequations) by `odg:thm:guard`. 09 observes that the order symbol can be omitted, positivity being definable by nonzero squares. The following extras are not in the Diophantine report.

Lemma 8.1 (Order on $\mathbf{Oz}$ through fractions [08]). *For $z \in \mathbf{Oz}$, $z \geq 0$ if and only if $\exists a, b \in \mathbf{Oz} (b \neq 0 \wedge a^2 = zb^2)$.*

Proof. If $a^2 = zb^2$ with $b \neq 0$ then $z = (a/b)^2 \geq 0$. Conversely a square root of $z \geq 0$ exists in the real closed field $\mathbf{No}$ and is a quotient a/b of omnific integers (`odg:thm:fractions`). $\square$

The formula is existential with one inequation, consistent with the Diophantine report's result that the nonnegative cone is not positive-existentially definable (`odg:prop:notdiophantine`).

Theorem 8.2 (Zero-cut projections and floor [08, 09]). *In $(\mathbf{No}, +, \cdot, <, \mathbf{Oz})$, equivalently in the fraction field interpreted in the pure ring $\mathbf{Oz}$, the graphs of the projections P_-, P_0, P_+ on all of $\mathbf{No}$ (so $P_0 = \text{ct}$ extended to $\mathbf{No}$) and of the floor map $x \mapsto \lfloor x \rfloor_{\mathbf{Oz}}$ are parameter-free definable. For every monomial $m = t^\alpha$, truncation below α , $T_{<\alpha}(f) = m P_-(f/m)$, is definable with the single parameter m . The same definitions work in $(\mathbb{R}((t^\Gamma)), \mathbb{R}_\mathbb{Z})$ for every nonzero divisible ordered abelian Γ .*

Proof. $\Pi, \mathbb{R}, \mathcal{O}$ and $\mathfrak{m}$ are definable by the results cited above (in the t -convention $\mathcal{O}$ is the finite ring and $\mathfrak{m}$ the infinitesimals). For $x \in \mathbf{No}$, $P_-(x)$ is the unique $j \in \Pi$ with $x - j \in \mathcal{O}$: existence by splitting the support at 0, uniqueness from $\Pi \cap \mathcal{O} = 0$. Then $P_0(x)$ is the unique $c \in \mathbb{R}$ with $x - j - c \in \mathfrak{m}$, since $\mathbb{R} \cap \mathfrak{m} = 0$, and $P_+ = \text{id} - P_- - P_0$ [09]. The floor graph is $y \in \mathbf{Oz} \wedge y \leq x < y + 1$, unique because distinct omnific integers differ by at least one [08]; its value is `odg:thm:floor`. Dividing by m translates exponents by $-\alpha$, P_- keeps those below 0, and multiplying by m restores them [09]. For the set-sized field the proof uses only real coefficients, the Hahn order (the field is real closed since Γ is divisible), and the definitions of Π and $\mathbb{R}$: the formula of `odg:def:thm:ideal` works for every Γ , and the multiplier identity `odg:def:thm:multiplier` holds inside the whole Hahn field by the same one-monomial proof [09]. $\square$

Every pair automorphism therefore preserves $\Pi \oplus \mathbb{R} \oplus \mathfrak{m}$ summandwise, fixes the real summand, and satisfies $\text{ct}(\sigma x) = \text{ct}(x)$ and $\sigma(\lfloor x \rfloor) = \lfloor \sigma x \rfloor$ ([corollary 5.2](#)).

Remark 8.3 (Zero truncation is not the normal form [08, 09]). P_0 is definable on all of $\mathbf{No}$ but is not multiplicative there. Definability of P_- does not define the set of monomials, and truncation at a named monomial is not a uniform definable choice of such a boundary at every valuation; the distinction matters in comparison with work on truncation [5]. Recovering every coefficient by $[\omega^g]x = \text{ct}(\omega^{-g}x)$ would need the monomial map $g \mapsto \omega^g$, and [section 9](#) shows that this is a genuine obstruction.

Source 08's Question 2, whether the reconstruction of $\mathbb{C}$ from the pure ring $\mathbf{Oz}[i]$ is first-order, is answered positively by `odg:def:cor:internal`, whose formula $\exists y (x^2 = 2y^2)$ defines $\Pi_\mathbb{C}$ in $\mathbf{Oz}[i]$ and whose multiplier formula then defines $\mathbb{C}$; 09 gives an independent route, its Pell-divisibility formula with the Gaussian Pell solutions (such as (i, i)) handled, followed by the idealizer. The question is not repeated here.

9 What the pair forgets: monomials and representatives

Theorem 9.1 (Set-parameter invisibility [04, 08, 09, 10]). *For every set $P \subseteq \mathbf{No}$ there is a nonidentity strong real-linear 1-automorphism σ of $\mathbf{No}$ with $\sigma(\mathbf{Oz}) = \mathbf{Oz}$ that fixes $P \cup \mathbb{R} \cup \{\omega\}$*

pointwise and sends a positive Conway monomial in $\mathbf{Oz}$ to a nonmonomial in $\mathbf{Oz}$. Explicitly, with β as in [lemma 7.1](#) and $\eta = \omega^\beta$, one may take $\sigma = \sigma_{1,q_\beta}$, and then

$$\sigma(t^{-\eta}) = t^{-\eta} \exp(-t) = \sum_{n \geq 0} \frac{(-1)^n}{n!} t^{-\eta+n}, \quad \text{that is} \quad \sigma(\omega^{\omega^\beta}) = \sum_{n \geq 0} \frac{(-1)^n}{n!} \omega^{\omega^\beta-n}. \quad (9.1)$$

Consequently, in $\mathfrak{S} = (\mathbf{No}, 0, 1, +, \cdot, <, \mathbf{Oz})$ neither the monomial class $\mathfrak{M} = \{\omega^x : x \in \mathbf{No}\}$ nor the graph of $x \mapsto \omega^x$ is definable with parameters from any set, even after naming $\mathbb{R}$ as a predicate, every ordinary real, ω , and any further set of surreal constants.

Proof. q_β kills C_1 since $\beta > 0$, so $\sigma \in \mathcal{U}_{\mathbf{Oz}}$ by [theorem 4.1](#). It fixes each monomial with exponent in E , hence every member of P by strongness, and it fixes $\mathbb{R}$ and $t^{-1} = \omega$ because $q_\beta(-1) = 0$. Since $q_\beta(-\eta) = -1$, (9.1) follows; all exponents $-\eta + n$ are negative because η is infinite, and the terms at $-\eta$ and $-\eta + 1$ show that the image is not a monomial. A class definable from parameters fixed pointwise by an automorphism of $\mathfrak{S}$ is invariant under it; σ preserves every symbol and moves an element of $\mathfrak{M}$ out of $\mathfrak{M}$. A definable graph of the omega-map would make its range definable by an existential quantifier. $\square$

The three sources give the same construction: 04 with $a = [\omega^\beta]$ and explicit fixing of $\mathbb{R} \cup \{\omega\}$, 08 with the witness $\omega^{\omega^\alpha} \mapsto \sum_n \omega^{\omega^\alpha-n}/n!$ (its sign convention), 09 with q_η ; 10 uses a binomial twist instead of the exponential shear, and its image has exactly two terms ([theorem 9.8](#)). An individual formula uses finitely many parameters; the theorem is the stronger statement that one automorphism fixes an entire prescribed set. Nothing is claimed after naming a proper class of parameters: naming every surreal would remove the symmetry [04, 08]. Expanding the structure by the already definable order, floor or zero-truncation maps cannot remove the obstruction [08].

Corollary 9.2 ([04, 09, 10]). *The monomial class is not determined, as an automorphism-invariant subclass, by $\mathbf{Oz}$, its ordered fraction field and any set of parameters; the standard normal-form section of the natural valuation is not first-order recoverable from these data. The Conway monomials lying in $\mathbf{Oz}$ are not definable in the pure ring $\mathbf{Oz}$ with parameters from any set.*

Proof. The range of the section is $\mathfrak{M}$. For the pure ring, restrict the witness of [theorem 9.1](#) to $\mathbf{Oz}$, with parameters in $\mathbf{Oz}$; automorphisms of $\mathbf{Oz}$ and field automorphisms of $\mathbf{No}$ preserving $\mathbf{Oz}$ correspond (`odg:def:thm:autreal`). $\square$

9.1 No definable representative rule

Write $x \sim_v y$ if $v(x) = v(y)$ on $\mathbf{No}^\times$. A *valuation representative rule* is a function $s : \mathbf{No}^\times \rightarrow \mathbf{No}^\times$ with $v(s(x)) = v(x)$ and $s(x) = s(y)$ whenever $x \sim_v y$. For a set-sized field this is a section of the quotient by $\mathcal{O}^\times$; the formulation also makes sense on the class $\mathbf{No}$ without treating proper-class equivalence classes as elements. No multiplicativity, positivity or monomial-image assumption is imposed [09].

Lemma 9.3 (A whole fiber moves [09]). *For σ and η as in [theorem 9.1](#), every f with $v(f) = -\eta$ satisfies $v(\sigma f - f) = -\eta + 1$ and $\text{lc}(\sigma f - f) = -\text{lc}(f)$.*

Proof. This is [theorem 4.2](#) with $\gamma_0 = -\eta$ and $\theta(-\eta) = -1$: later input terms contribute beyond $-\eta + 1$, higher iterates at $-\eta + n$, $n \geq 2$. $\square$

Theorem 9.4 (No definable section [09]). *For every set $P \subseteq \mathbf{No}$, no P -definable function on $\mathbf{No}^\times$ is a valuation representative rule.*

Proof. Let $x = t^{-\eta}$. Since σ fixes valuations, $\sigma x \sim_v x$ and $s(\sigma x) = s(x)$, while P -definability gives $s(\sigma x) = \sigma(s(x))$. So $s(x)$ would be a fixed point of σ with valuation $-\eta$, contradicting lemma 9.3. $\square$

This is the strongest nondefinability statement of the sources; it covers any representative rule, including one-term ones, and does not follow from character twists. The nondefinability is relative to the field-and-integer-part language; Conway's normal forms remain set-theoretically defined in the ambient foundation [09].

Remark 9.5 (Character twists, one-term series and the RV sort [04, 02]). Bare nondefinability of the coefficient-one monomials already follows from character twists: for a nonzero additive $a : \mathbf{No} \rightarrow \mathbb{R}$ with $a(V_A) = 0$ (lemma 7.3), $\sum c_g t^g \mapsto \sum c_g e^{a(g)} t^g$ preserves $\mathbf{Oz}$, fixes A and sends some monomial to a non-unit multiple of itself [04]. These twists are the $M_{\chi, \text{id}}$ of theorem 5.1; they are 02's $\tau_{\chi, t}$ (remark 17.10), the real form of the Diophantine report's phase twist (`odg: def: lem: twist`), and the collection's character twists (`saut: thm: decomp`); they are established and not claimed new. What the shears add is motion inside the leading-term kernel with more than one nonzero term. In particular the class of nonzero one-term series $\{c\omega^g : c \in \mathbb{R}^\times, g \in \mathbf{No}\}$ is not definable from the ordered field, $\mathbf{Oz}$ and set parameters: the shear sends a one-term series to a multiterm one, which rescaling cannot do [04]. Nondefinability also survives adding the sort $\text{RV}(K) = K^\times / (1 + \mathfrak{m})$ with its multiplication, the quotient map and every element of that sort named, since a 1-automorphism acts trivially on it (for the class $\mathbf{No}$ this is a metatheoretic expansion, read formula by formula). This must be distinguished from naming a representative map $\text{RV}(K) \rightarrow K$, which the Conway monomials partly constitute and which would destroy the symmetry by design: $\Phi_s(t^{-\omega})$ has leading term $t^{-\omega}$, yet Φ_s moves $t^{-\omega}$ [04].

Remark 9.6 (Contrasts in the collection). The single dilation $S_2(\sum a_g t^g) = \sum a_g t^{2g}$ defines the canonical monomials, every coefficient and truncation in the language of fields with one automorphism (`dsup: thm: definability` in `docs/surcomplex/single-dilation-hahn-support`); ($\mathbf{No}, \mathbf{Oz}$) does not, even with set parameters. The two expansions differ exactly on the monomial section, and there is no conflict: the shears do not commute with S_2 , in line with that report's centralizer theorem (`dsup: thm: centralizer`). The shears are not exponential: by `cor: question54` of the rigidity report `docs/surreal/exponential-automorphism-rigidity` (from its `thm: no`), every exponential 1-automorphism of $\mathbf{No}$ is the identity, which is stronger than [15, Proposition 5.2] (strongly linear exponential 1-automorphisms are trivial), the result cited by 09. The shears are compatible with that report's displacement lemma: their displacement $-t^{-\eta+1}$ is infinite. Nor do they preserve the omega-map, whose compatibility questions [15, Questions 5.6, 5.7] are untouched here.

9.2 Two-term witnesses, simplicity and the exponential [10]

For a surreal $b > 0$ and a surreal δ with $0 < \delta < 1$, the functional q_b vanishes on C_δ : every member of C_δ is finite, and a finite surreal has no positive exponent in its normal form. Source 10's class twist is the binomial twist $\Sigma_{b, \delta} = \sigma_{\delta, -q_b}^{1+\xi}$; in normal forms,

$$\Sigma_{b, \delta} \left(\sum_g a_g \omega^g \right) = \sum_g a_g \omega^g (1 + \omega^{-\delta})^{q_b(g)}, \quad (9.2)$$

with the real binomial series. Each image has a set-sized support inside $\{g - n\delta\}$, and the proof of theorem 4.18 is elementwise, so $\Sigma_{b, \delta}$ is a strong real-linear 1-automorphism of $\mathbf{No}$ with inverse given by $-q_b$; it fixes $\mathbb{R}$, preserves the order and $\mathbf{Oz}$, and commutes with P_-, P_0, P_+ [10].

Remark 9.7 (An exponent index is not a fixed element [10]). $\Sigma_{b,\delta}$ fixes the monomial $\omega^{-\delta}$ but need not fix the surreal δ : the equality $q_b(-\delta) = 0$ controls the monomial with exponent $-\delta$, not the exponents inside the normal form of δ . (For $\delta = \omega^{-\omega^b}$ one has $\Sigma_{b,\delta}(\delta) \neq \delta$.) This is the inner/outer distinction of [section 7](#), and it is why an automorphism can fix all valuations without commuting with the omega-map.

Theorem 9.8 (Two-term witnesses [10]). *For every set $S \subseteq \mathbf{No}$ there is an ordinal $\beta > 0$ such that, for every surreal δ with $0 < \delta < 1$, $\Sigma_{\beta,\delta}$ fixes S pointwise and*

$$\Sigma_{\beta,\delta}(\omega^{\omega^\beta}) = \omega^{\omega^\beta} + \omega^{\omega^\beta - \delta}, \quad (9.3)$$

a two-term omnific integer and not a Conway monomial. These images are pairwise distinct as δ ranges over $(0, 1)$, and they form a proper class.

Proof. Take β as in [lemma 7.1](#), above the inner supports of the outer supports of the members of S . Then q_β vanishes on every exponent of every member of S , so (9.2) fixes their monomials and, by strongness, the members themselves. Since $q_\beta(\omega^\beta) = 1$, the binomial exponent at ω^{ω^β} is one, which gives (9.3) with no further terms. Both ω^β and $\omega^\beta - \delta$ are positive, since ω^β is infinite, and they differ, so by uniqueness of normal forms the image is an omnific integer and not a monomial. Distinct δ give distinct second exponents. The class $(0, 1)$ is proper, since $\alpha \mapsto \omega^{-(\alpha+1)}$ injects the ordinals into it, and δ is recovered from the image, so the images form a proper class. $\square$

The witness of [theorem 9.1](#) is an infinite series; here two terms suffice. The theorem gives, for each set of parameters, one uniform formula for individual automorphisms indexed by the set parameters β, δ and a proper class of values at one input; it forms no proper class of automorphisms [10].

Corollary 9.9 (Bounded birthdays [10]). *For every ordinal α there is a nonidentity strong ordered-field automorphism of $\mathbf{No}$ that preserves $\mathbf{Oz}$ and fixes every surreal of birthday less than α .*

Proof. The surreals of birthday less than α form a set; apply [theorem 9.8](#). $\square$

The automorphism is not asserted to preserve the birthdays of other elements: the statement is pointwise fixing of a stage, not preservation of the simplicity hierarchy [10], which by [theorem 9.10](#) no such automorphism preserves.

The proof of [theorem 4.20](#) uses only the set-sized support of one input, so the fixed class of a class twist is explicit [10]:

$$\text{Fix}(\Sigma_{b,\delta}) = \left\{ \sum_g a_g \omega^g : q_b(g) = 0 \text{ for every } g \in \text{supp} \right\}, \quad (9.4)$$

and for a nonfixed f , with g_* its greatest exponent having $q_b(g_*) \neq 0$, the leading term of $\Sigma_{b,\delta}(f) - f$ is $a_{g_*} q_b(g_*) \omega^{g_* - \delta}$. The fixed class field is relatively algebraically closed in $\mathbf{No}$, in the finite-polynomial sense, by the proof of [corollary 4.21](#). If $(r_j)_{j \in J}$ is a $\mathbb{Q}$ -linearly independent family of reals with $|J| = 2^{\aleph_0}$, the monomials $\omega^{r_j \omega^b}$ are algebraically independent over (9.4) by [proposition 4.22](#). No exact transcendence degree of the class extension is claimed [10].

Let $\mathfrak{S}^\dagger = (\mathbf{No}; 0, 1, +, -, \cdot, <, \mathbf{Oz}, (r)_{r \in \mathbb{R}}, P_-, P_0, P_+)$, the structure $\mathfrak{S}$ of [theorem 9.1](#) with every real and the three zero-cut projections named; naming all reals uses a set-sized language. Definability has the usual finite-formula meaning, formula by formula in the class structure; no truth predicate for classes is assumed. Simplicity is the relation of being a proper initial segment of a sign sequence, not comparison of birthdays [10].

Theorem 9.10 (What remains undefinable [10]). *For every set $S \subseteq \mathbf{No}$ none of the following is first-order definable over S in $\mathfrak{S}^\dagger$: (i) the monomial class $\mathfrak{M}$; (ii) the graph of $x \mapsto \omega^x$; (iii) the simplicity relation; (iv) the graph of the Gonshor exponential. The class $\mathfrak{M} \cap \mathbf{Oz}$ is not definable over any set of parameters in the ordered ring $(\mathbf{Oz}; 0, 1, +, -, \cdot, <)$.*

Proof. Take $\Sigma = \Sigma_{\beta, \delta}$ from [theorem 9.8](#). It fixes S and every named real and preserves every symbol of $\mathfrak{S}^\dagger$. Items (i) and (ii) follow as in [theorem 9.1](#), since Σ moves the monomial $Z = \omega^{\omega^\beta}$ out of $\mathfrak{M}$.

(iii) A bijection of $\mathbf{No}$ preserving order and simplicity is the identity [[15](#), Proposition 5.1]: the simplicity tree has one root, each node has two immediate successors distinguished by the order, and each limit sign sequence has a unique continuation, so induction on length fixes every node. A definition of simplicity would make Σ such a bijection, but $\Sigma \neq \text{id}$.

(iv) Put $d = \Sigma(Z) - Z = \omega^{\omega^\beta - \delta}$, which is positive and infinite. The Gonshor exponential is strictly increasing, extends the real exponential and satisfies $\exp(x + y) = \exp(x)\exp(y)$, so $\exp(d)$ is positive infinite and $v(\exp d) \neq 0$. As Σ fixes valuations, $v(\Sigma(\exp Z)) = v(\exp Z)$, whereas $v(\exp(\Sigma Z)) = v(\exp Z) + v(\exp d) \neq v(\exp Z)$. So Σ does not commute with the exponential, while the graph of a definable function is invariant.

For the ordered ring take $S \subseteq \mathbf{Oz}$: the restriction of Σ is an automorphism of $(\mathbf{Oz}; 0, 1, +, -, \cdot, <)$ fixing S , and Z and $\Sigma(Z)$ both lie in $\mathbf{Oz}$, but only Z lies in $\mathfrak{M}$. $\square$

Items (i), (ii) and the ordered-ring clause are [theorem 9.1](#) and [corollary 9.2](#), the order of $\mathbf{Oz}$ being definable in its pure ring ([lemma 8.1](#)), and the projections being definable in the pair ([theorem 8.2](#)); 10 re-derives them. Items (iii) and (iv) are new here. They need only some nonidentity 1-automorphism of $\mathfrak{S}^\dagger$ fixing S : the merge notes that for (iv) the rigidity report's `cor:question54` (every exponential 1-automorphism of $\mathbf{No}$ is the identity, from `thm:no`, whose proof is the same valuation comparison) would equally serve with the witness of [theorem 9.1](#). The formal exp of principal units is not the Gonshor exponential, and the class twists are not claimed to preserve the latter; they provably do not. The displacement mechanism is related to the exponential rigidity arguments of [[15](#)] and of that report; what is added is that the witness preserves $\mathbf{Oz}$ and fixes an arbitrary set of parameters [10].

Corollary 9.11 (Support bounds for definable elements [10]; outer bound [merge]). *Let $A \subseteq \mathbf{No}$ be a set and f definable over A in $\mathfrak{S}^\dagger$. Then $f \in \mathbb{R}((t^{V_A}))$, with $V_A = \text{span}_{\mathbb{Q}}(S_A)$ as in [theorem 7.4](#). In particular $B_+(\{f\}) \subseteq B_+(A)$, where $B_+(A) = \bigcup_{s \in A} \bigcup_{g \in \text{supp } s} (\text{supp } g \cap \mathbf{No}^{>0})$ is the positive second-level support: definability creates no new positive inner support position.*

Proof. Every elementary shear fixing A is an automorphism of $\mathfrak{S}^\dagger$ fixing A , so $f \in \text{Fix}(\mathcal{E}_A) = \mathbb{R}((t^{V_A}))$ by [theorem 7.4](#). Each exponent of f is then a finite rational combination of exponents of members of A , and the normal-form support of a finite sum lies in the union of the supports. Source 10's direct proof of the inner bound: if $b > 0$ is in the inner support of an exponent of f but not in $B_+(A)$, then $\Sigma_{b, \delta}$ fixes A and moves f by (9.4); b need not be an ordinal. $\square$

Both are upper bounds only. They do not say that every element obeying them is definable, nor that every coefficient is recoverable from the named projections [10]; compare [question 11.3](#).

Floor, fractions and coefficient rigidity [10]. Source 10 also proves that every ordered-field automorphism stabilizing $\mathbf{Oz}$ commutes with the floor map, that $\text{Frac } \mathbf{Oz} = \mathbf{No}$ makes any automorphism fixing $\mathbf{Oz}$ pointwise the identity, and that every ring automorphism of $\mathbf{Oz}$ extends uniquely to $\mathbf{No}$ and fixes $\mathbb{R}$ pointwise, through $\Pi = \bigcap_n n\mathbf{Oz}$, the multiplier ring $\mathcal{A}_k$ of Π and its units $\mathbb{R}^\times$, with no strongness assumption. These are [corollary 5.2](#), `odg:thm:fractions`,

`odg:def:thm:multiplicier` and `odg:def:thm:autreal`, which 10 credits as prior repository results; they are not reprinted. There is no conflict between $\text{Frac } \mathbf{No} = \mathbf{No}$ and the moving of monomials: a ring determines its fraction field up to isomorphism, not a distinguished monomial section inside it; the automorphisms preserve $\mathbf{Oz}$ setwise and move some of its elements [10].

10 Surcomplex and Gaussian consequences

Identify $\mathbf{No}(i)$ with $\mathbb{C}((t^{\mathbf{No}}))$ by taking coefficientwise real and imaginary parts; the union of the two supports is well ordered. Then $\mathbf{Oz}[i] = \mathbb{Z}[i] \oplus \Pi_{\mathbb{C}} = \mathbf{Oz} + i\mathbf{Oz}$, which is not an ordered ring; conjugation is coefficientwise [04, 08, 09].

Theorem 10.1 (Gaussian stabilizer and conjugation-compatible witnesses [04, 09, 10]). *Theorem 3.3 applies to $(k, \mathfrak{o}) = (\mathbb{C}, \mathbb{Z}[i])$ with the same condition. The real-coefficient shears and actions of sections 7 and 9 extend $\mathbb{C}$ -linearly, commute with conjugation, preserve $\mathbf{No}$ and $\mathbf{Oz}[i]$; for every set of surcomplex parameters they fix the parameters and give nondefinability of the monomials and of any valuation representative rule, even with the real axis, conjugation and every ordinary complex constant named. Their common fixed field is $\mathbb{C}$, and for every set $A \subseteq \mathbf{No}(i)$ the real elementary shears fixing A have common fixed field $\mathbb{C}((t^{V_A}))$.*

Proof. The criterion was proved for every characteristic-zero k and unital $\mathfrak{o}$, and its proof uses no order on coefficients. The functionals q_β are real, so the derivations and their exponentials commute with conjugation and restrict to the real automorphisms on $\mathbf{No}$. Use the union of the real and imaginary supports in lemma 7.1; the moved monomial is real, and lemma 9.3 holds with any nonzero complex leading coefficient. For the fixed field, the first displacement of theorem 4.2 is a product of a nonzero real functional value and a nonzero complex coefficient. The hull statement is theorem 7.4 with $k = \mathbb{C}$ [04]. $\square$

For these real-axis-preserving maps the surreal modulus is equivariant, $|\sigma(z)| = \sigma(|z|)$, since $|z|$ is the nonnegative square root of $z\bar{z}$; this is not a statement that moduli are fixed [09].

Theorem 10.2 (The real axis inside the leading-term kernel [04]). *For every set $A \subseteq \mathbf{No}(i)$ there is a strong $\mathbb{C}$ -linear 1-automorphism of $\mathbf{No}(i)$ that preserves $\mathbf{Oz}[i]$, fixes A pointwise and does not preserve $\mathbf{No}$. Hence $\mathbf{No}$ is not definable in $(\mathbf{No}(i), 0, 1, +, \cdot, \mathbf{Oz}[i])$ with parameters from any set, even with all ordinary complex constants named and the Hahn valuation ring as a predicate.*

Proof. Take β above the inner exponents of the outer supports of A , and the flow Φ_s of theorem 4.1 with $\theta = q_\beta$, $\delta = 1$ and the imaginary time $s = i$. It fixes A and is strong, $\mathbb{C}$ -linear, leading-term preserving and $\mathbf{Oz}[i]$ -preserving. With $\eta = \omega^\beta$, $\Phi_i(t^{-\eta}) = t^{-\eta} \exp(-it) = t^{-\eta} - it^{-\eta+1} - \frac{1}{2}t^{-\eta+2} + \dots$ has a nonreal coefficient, although $t^{-\eta}$ is real; all its exponents are negative. A 1-automorphism preserves every valuation, hence the valuation ring. $\square$

Phase twists preserving $\mathbf{Oz}[i]$ and moving the real axis are already in the collection (`saut:thm:phase`; `odg:def:lem:twist`, `odg:def:thm:norealaxis`); the new points are arbitrary set-parameter fixing *inside the complete leading-term kernel* and multiterm images. The real shears commute with conjugation and the imaginary-time shear does not; the two conclusions must not be combined into a claim that one automorphism commutes with conjugation and moves its fixed field [04].

Proposition 10.3 (A wild coefficient lift [08]). *There is a strong automorphism of $\mathbf{No}(i)$ that preserves $\mathbf{Oz}[i]$, fixes $\mathbb{Z}[i]$ and every Conway monomial, and sends an ordinary real number outside $\mathbb{R}$. So neither $\mathbb{R}$ nor $\mathbf{No}$ is invariant under all field automorphisms preserving $\mathbf{Oz}[i]$.*

Proof. Choose a real r transcendental over $\mathbb{Q}$ and a transcendence basis of $\mathbb{C}$ over $\overline{\mathbb{Q}}$ containing it. On the rational function field it generates, fix $\overline{\mathbb{Q}}$ and the other basis elements and send $r \mapsto r + i$ (inverse $r \mapsto r - i$), and extend to an automorphism ρ of $\mathbb{C}$ by the set-sized algebraic-closure extension theorem; the extension is onto, since its image is algebraically closed and $\mathbb{C}$ is algebraic over the base. The coefficientwise lift $\hat{\rho}$ (`odg:def:prop:coeffaut`) preserves $\mathbf{Oz}[i]$ and monomials and sends r to $r + i$. $\square$

Corollary 10.4 (Gaussian fixed field [08, 09]). *The common fixed elements in $\mathbf{No}(i)$ of all field automorphisms preserving $\mathbf{Oz}[i]$ are exactly $\mathbb{Q}$; in $\mathbf{Oz}[i]$ they are $\mathbb{Z}$.*

Proof. The shears of [theorem 10.1](#) reduce a common fixed point to $\mathbb{C}$. Every field automorphism of $\mathbb{C}$ lifts coefficientwise and preserves $\mathbf{Oz}[i]$ (`odg:def:prop:coeffaut`). A nonrational algebraic number is moved by a conjugate embedding extended to $\mathbb{C}$; a transcendental one by a translation of a transcendence basis element; rationals are fixed. Finally $\mathbb{Q} \cap \mathbf{Oz}[i] = \mathbb{Z}$. $\square$

This agrees with `saut:prop:fixedfields` of the surcomplex automorphism report, $\text{Fix}(\text{Aut } \mathbf{No}(i)) = \mathbb{Q}$, for the larger collection of all field automorphisms. The real and complex answers differ because a real automorphism preserving $\mathbf{Oz}$ fixes every ordinary real, $\mathbb{R}$ being parameter-free definable in the pair ([corollary 5.2](#), `odg:def:thm:autreal`), whereas a complex one preserving $\mathbf{Oz}[i]$ can move reals unless real structure is required; the explicit witnesses of [theorem 10.1](#) fix all of $\mathbb{C}$ [09]. *Correction* (batch 30): this sentence formerly gave as the reason that a real automorphism “preserves order and cannot move an ordinary real”. Read literally that is false: the Taylor automorphism of `saut:thm:coeffflow` is a strong order-preserving automorphism of $\mathbf{No}$ sending a real b to $b + t^\delta$. It does not preserve $\mathbf{Oz}$ ([example 4.12](#)); preservation of $\mathbf{Oz}$, not of the order, is what fixes $\mathbb{R}$. The comparison and the fixed fields above are unchanged, so this corrects the stated reason and retracts no result. Three mechanisms move the real axis while preserving $\mathbf{Oz}[i]$: the phase twist of the Diophantine report (fixes $\mathbb{C}$, preserves supports), the imaginary-time shear of [theorem 10.2](#) (fixes $\mathbb{C}$ and all leading terms), and the wild lift of [proposition 10.3](#) (fixes monomials, moves $\mathbb{R}$). The real shears of [theorem 10.1](#) preserve the real axis.

Proposition 10.5 (Two-term Gaussian witnesses [10]). *For every set $S \subseteq \mathbf{No}(i)$ there is an ordinal $\beta > 0$ such that for every surreal $0 < \delta < 1$ the $\mathbb{C}$ -linear extension $\tilde{\Sigma}_{\beta,\delta}(x + iy) = \Sigma_{\beta,\delta}(x) + i \Sigma_{\beta,\delta}(y)$ is a strong $\mathbb{C}$ -linear 1-automorphism of $\mathbf{No}(i)$ which (i) fixes S and $\mathbb{C}$ pointwise; (ii) preserves $\mathbf{No}$ and $\mathbf{Oz}[i]$ and commutes with conjugation; (iii) preserves the order of $\mathbf{No}$ and commutes with P_-, P_0, P_+ ; (iv) sends ω^{ω^β} to the two-term element (9.3). Consequently, with parameters from any set, neither the monomial class nor, on the real axis, the graph of the omega-map, the simplicity relation or the graph of the Gonshor exponential is definable in $\mathbf{No}(i)$ expanded by conjugation, the predicate $\mathbf{No}$, the order of $\mathbf{No}$, the predicate $\mathbf{Oz}[i]$, every complex constant and P_-, P_0, P_+ .*

Proof. Apply [theorem 9.8](#) to the set of real and imaginary parts of the members of S . The binomial coefficients in (9.2) are real, so $\tilde{\Sigma}_{\beta,\delta}$ is the same formula on complex-coefficient normal forms; it commutes with conjugation because it acts by one real map on both components and fixes i . [Theorem 3.3](#) with $(k, \mathfrak{o}) = (\mathbb{C}, \mathbb{Z}[i])$ gives (ii) and (iii) for $\mathbf{Oz}[i]$ and the projections. The witness is real. A definition of the monomial class would be preserved, contradicting (iv); for the real-axis statements restrict a putative definition to $\mathbf{No}$ and argue as in [theorem 9.10](#). No global surcomplex exponential is introduced. $\square$

Its fixed class field is the complex-coefficient analogue of (9.4), with the same first displacement, no nontrivial finite orbits and relative algebraic closedness [10]. Beyond [theorem 10.1](#), the proposition adds the two-term witness, the order of the real axis, the projections, and simplicity and the exponential. It differs from the phase twists of the Diophantine report, which move the real axis (`odg:def:lem:twist`): here the real axis and conjugation are preserved, and the obstruction lies inside the fixed real axis [10].

11 Scope, answered questions, and further questions

11.1 What is settled, and status in other reports

For a strong infinitesimal automorphism, preservation of the integer part is exactly a convex-scale annihilation condition; the size of the residual symmetry is measured by a sharp solvability bound in finite rank and by arbitrarily large parameter-fixing actions on $\mathbf{No}$; and the omnific predicate recovers the coefficient fields and zero-cut truncation but no definable valuation cross-section [09]. An attempt to reconstruct Conway monomials by a first-order formula in the pair cannot succeed with any set of constants; a classification of strong $\mathbf{Oz}$ -automorphisms by coefficient and exponent actions alone misses the higher-rank factor $\mathcal{U}_{\mathbf{Oz}}$; and an alleged rank-one nontrivial 1-automorphism preserving a divisible-exponent integer part must violate a support hypothesis, as [example 4.12](#) shows [09].

The results bear on four questions of other reports. They are recorded here; the other reports are not edited by this merge.

- `osq:q:invisible` of `docs/surreal/set-sized-quotients-of-omnific-integers` asks, in its second clause, how much an automorphism of $\mathbf{Oz}$ can do inside Π while fixing ct . *Partly answered*: every automorphism of $\mathbf{Oz}$ fixes $\mathbb{R}$ and ct (`odg:def:thm:autreal`, [corollary 5.2](#)); the *strong* ones are exactly $\{M_{\chi,\tau}\} \ltimes \mathcal{U}_{\mathbf{Oz}}$, with $\mathcal{U}_{\mathbf{Oz}}$ classified by [theorem 3.3](#); they move monomials to infinite series, fix any set of parameters, carry $\mathbb{R}^\kappa$ and have nonsolvable stabilizers ([theorems 7.6, 7.7 and 9.1](#)). Nonstrong automorphisms, and the first clause of that question, remain open. *Status* (batch 30): there are no nonstrong automorphisms of $\mathbf{Oz}$ ([theorem 23.2](#)), so the second clause is *answered* ([corollary 24.4](#) and [section 32.1](#)); the first clause remains open.
- `odg:def:q:realform` of the Diophantine report asks which natural structure on $\mathbf{Oz}[i]$ rules out the coefficient-fixing twists and determines the real omnific subring. *Negative information, not an answer*: naming the valuation ring, every complex constant, the RV sort, or requiring leading-term fixing does not suffice ([theorem 10.2](#)); and in $(\mathbf{No}(i), \mathbf{Oz}[i])$ even $\mathbb{R} \subseteq \mathbb{C}$ is not invariant ([proposition 10.3](#)).
- The surcomplex automorphism report’s research direction “effective descriptions inside the leading-term kernel” (its section `saut:sec:questions`) is *not* answered. Its $\mathbf{Oz}$ -stabilizing part is now described by coefficient functionals ([theorem 3.3](#) and [corollary 3.5](#)), with finite support certificates on the common-positive-shift class ([theorem 4.8](#)); no exhaustion or generation theorem is proved ([question 11.4](#)). Its omega-map compatibility questions are untouched.
- `dsn:q:languages` of `docs/foundations-and-computation/definable-surreals-and-omnific-integers` asks about languages between the ordered field with $\mathbf{Oz}$ and set theory. *Further information, not an answer* [10]: in $(\mathbf{No}, +, \cdot, <, \mathbf{Oz})$, even with every real, the zero-cut projections and any set of parameters named, neither the simplicity relation nor the graph of the Gonshor exponential is definable ([theorem 9.10](#)), whereas

order and simplicity together already define ω (`dsn:prop:simplicity-omega`); definable elements obey the support bound of [corollary 9.11](#).

The novelty statements of the batch-26 sources were checked against the tree at their merge: no report contains a classification of $\mathbf{Oz}$ -preserving automorphisms, the rank dichotomy or set-parameter nondefinability of monomials, and `osq:q:invisible` was recorded as open. Their reconstruction and definability claims were stale ([section 8](#)). Source 10's statement that none of the report guides it inspected states the support-cut criterion, and its proposal of the criterion, the rank boundary, the fixed fields and the parameter witnesses as its contribution, were accurate at its pin `58cd8e1`, which does not contain this report. They are stale at the merge: this report was placed in `f4c9504` and written in `9b80a30`, and those results are [theorems 3.3, 4.2, 4.13](#) and [9.1](#). 10 is recorded as an independent re-derivation of them, and its new material is the list in [section 1.2](#). Its other repository statements (the fraction field and the multiplier and coefficient results of the Diophantine report, the phase twists of its reconstruction audit) are accurate.

11.2 Further questions

These questions arise from the proofs; they are not presented as named historical problems [04, 08, 09, 10].

Question 11.1 (Strongness [04, 09, 10]). Which ring automorphisms of $\mathbf{Oz}$ extend to strong field automorphisms of $\mathbf{No}$? Does preservation of $\mathbf{Oz}$ force strongness, or is there a nonstrong $\mathbf{Oz}$ -preserving example? Equivalently for 09: characterize the nonstrong automorphisms of $(\mathbf{No}, \mathbf{Oz})$ that commute with all set-indexed normal-form sums.

[Corollary 5.2](#) removes coefficient motion from this question but does not settle preservation of infinite sums; nothing is inferred from the existence of nonstrong automorphisms of the ambient field [15]. Source 10 asks what characterizes the nonstrong $\mathbf{Oz}$ -preserving 1-automorphisms, and whether every one is strong. *Partial information* [10]: the necessity half of the criterion holds without strongness on every monomial ([remark 3.9](#)); the passage to arbitrary series is what remains. The question stays open. *Status* (batch 30): *answered* by sources 12 and 13 (and independently by source 14, batch 31, [part IV](#)). Every ring automorphism of $\mathbf{Oz}$ extends uniquely to a strongly $\mathbb{R}$ -linear automorphism of $\mathbf{No}$ with strong inverse ([theorem 23.2](#)): all of them extend to strong automorphisms, preservation of $\mathbf{Oz}$ forces strongness, there is no nonstrong example, 09's characterization is vacuous, and 10's question has the answer yes. The analogous question for $\mathbf{Oz}[i]$ has a positive answer under conjugation or valuation compatibility and is open otherwise ([theorems 30.1](#) and [30.2](#) and [question 32.1](#)); for general $(k, \mathfrak{o})$ see [question 32.13](#).

Question 11.2 (Admissible logarithmic characters [04]). Give testable necessary and sufficient conditions on an additive $h : \Gamma \rightarrow \mathfrak{m}$ for $t^g \mapsto t^g \exp h(g)$ to define a strong bijective field map. Within that class the integer-part condition is exactly $a_\delta(C_\delta) = 0$.

Common positive support is one sufficient regime; the convex condition alone is not asserted to solve the global problem. This is the problem the surcomplex report describes under “what monomial prescriptions do not automatically give”. Sources 12 and 13 ask it again; it stays open ([section 32.1](#)). Source 14 asks it again as a parameter space for the whole leading-term-fixing factor ([section 37.1](#)).

Question 11.3 (Full relative fixed fields [04]). For a set A of parameters, determine $\text{Fix}((\mathcal{U}_{\mathbf{Oz}})_A)$. It is contained in $\mathbb{R}((t^{V_A}))$, with equality for monomial parameters ([corollary 7.5](#)); which conditions on A force equality?

Source 10 asks the definable-closure analogue: when is the bound of [proposition 7.8](#) sharp, and which support and coefficient conditions characterize definable closure in $\mathfrak{S}^\dagger$? Its upper bounds ([proposition 7.8](#) and [corollary 9.11](#)) are obstructions, not converse criteria; a full answer would have to separate algebraic support generation from the information contributed by the order and by $\mathbf{Oz}$ [10]. *Related* (batch 33): source 18's automorphisms fixing A lie in the monomial factor, not in $\mathcal{U}_{\mathbf{Oz}}$, so they do not bear on this question; a single one of them can have fixed field exactly $K_{\Sigma_2(A)} \supseteq \mathbb{R}((t^{V_A}))$ ([remark 63.5](#)). *Related* (batch 34): source 20's relative action, with the larger proper-class fixed field $K_{\mathbf{No} \setminus \mathbf{J}}$, also lies in the monomial factor and does not bear on it ([remark 77.6](#)). Open.

Question 11.4 (Generation and exhaustion [04, 08, 10, 15]). How much of $\mathcal{U}_{\mathbf{Oz}}$ is generated by elementary shears and their support-controlled limits, and can an intrinsic summation convention make this precise without one common shift set for all inputs? Can all strong leading-term-fixing $\mathbf{Oz}$ -preserving automorphisms be described by a necessary and sufficient uniform support condition on their monomial phases?

No density or generation theorem is claimed; fine-topology convergence is not a substitute for a strong summation rule; the finite-dimensional phase condition of [theorem 4.17](#) is sufficient only. Source 10 asks the same for set-sized divisible Γ with its twists as generators, and for compositions indexed by a well-ordered set: [example 6.13](#) shows that one additive family is not enough, and a transfinite composition is not justified merely because each factor is strong; the accumulated changes must be controlled and a genuine limit map with an inverse proved [10]. Source 15 asks it again for every leading-term-fixing omnific automorphism, now that all are strong, asking for an admissibility condition that gives both global summability and an inverse (its Question 10); a topology realizing all Hahn sums offers no separated topological-group shortcut, since on $\mathbf{No}$ every one is indiscrete ([corollary 39.9](#)) [15]. *Status* (batch 32): open.

Question 11.5 (Weaker normal-form data [08]). Which normal-form data weaker than the omega-map remove the monomial nondefinability? Set-sized collections of constants do not ([theorem 9.1](#)).

Nor do every real constant, the zero-cut projections and the floor map, and simplicity and the Gonshor exponential are not definable either ([theorem 9.10](#)) [10].

Question 11.6 (Non-divisible exponent groups [merge]). For non-divisible Γ and $\mathfrak{o} = \mathbb{Z}$, is a strong k -linear 1-automorphism preserving $\mathcal{R}_{\mathbb{Z}}$ characterized by a convex-annihilator condition, as single flows are ([theorem 4.10](#))?

This records the scope correction of [remark 4.11](#). Source 10 assumes divisibility throughout, and its example with $\Gamma = \mathbb{Z}$ is stated for $\mathfrak{o} = k$ ([remark 4.23](#)); the question stays open.

Question 11.7 (Other integer parts [10]). Which parts of the support-cut criterion survive for noncanonical integer parts of a real closed Hahn field, or for subrings whose positive-support condition is replaced by a more general support family?

The proof uses closure under every positive rational power of a monomial and the exact boundary at exponent zero. An arbitrary integer part need not present that boundary termwise, and the criterion should not be transferred to it without a replacement for the least-forbidden-shift argument [10]. Source 10 also records that its construction does not answer the question of [15] on nontrivial omega-map-preserving automorphisms, since its witnesses do not preserve the omega-map, nor factorization problems for omnific integers, nor the order-unit question of the holonomic-rigidity report ([docs/surcomplex/holonomic-rigidity-for-entire-hahn-functions](#)). Source 12 asks the analogous question for automatic strongness ([question 32.13](#)).

Part II

Algebraic-parameter rigidity of the omnific integers

This part is manuscript 02. Its question is different from that of [part I](#): not which automorphisms preserve a support condition, but whether **Oz** admits nonconstant *algebraic* families of embeddings. It does not: the decisive property is that every nonzero element divides an element with roots of every order. It does admit formal families of all orders. The relation to [part I](#) is recorded in [section 19](#).

12 Conventions and the root-covered property

12.1 Conventions of this part

Ring homomorphisms are unital, domains are commutative integral domains, and $\mathbb{N} = \{0, 1, 2, \dots\}$; degrees of polynomials are ordinary integers. This part keeps its source's *large-monomial* convention, which is that of the other omnific reports: for a characteristic-zero field k , a unital subring $\mathfrak{o} \subseteq k$ and a nonzero divisible ordered abelian group Γ ,

$$k((X^\Gamma)) = \left\{ \sum_{g \in S} a_g X^g : S \subseteq \Gamma \text{ reverse well ordered} \right\}, \quad X^g = t^{-g},$$

so the leading exponent $\ell(x) = \max \text{supp } x$ is the largest, and $\ell(xy) = \ell(x) + \ell(y)$. The ideal $\Pi_k(\Gamma)$ consists of the series supported in $\Gamma^{>0}$ (the same set as Π in [part I](#)), and $\mathcal{R}_\mathfrak{o}(k, \Gamma) = \mathfrak{o} + \Pi_k(\Gamma)$ is the ring $\mathcal{R}_\mathfrak{o}$ of [part I](#), the Diophantine report's $\mathcal{R}_\mathfrak{o}(\mathbb{k}, \Gamma)$. For surreals $X = \omega$, so **Oz** = $\mathcal{R}_\mathbb{Z}(\mathbb{R}, \mathbf{No})$ and **Oz**[i] = $\mathcal{R}_{\mathbb{Z}[i]}(\mathbb{C}, \mathbf{No})$. Renamings from the source: its $R_\Gamma(D, k)$ and $\mathcal{P}_\Gamma(k)$ are $\mathcal{R}_\mathfrak{o}(k, \Gamma)$ and $\Pi_k(\Gamma)$; its principal convex subgroup $H(a)$ is $C(a)$, the transcendence report's notation (C_δ in [part I](#); the quotient report's H_a means the opposite, the elements infinitesimal relative to a); its covering element c is z ; its bounded-scale field K_b is K_{bd} and its stages H_j are $\Gamma_{\leq j}$. Elsewhere in this part the letters R, B, A, K, F, L denote rings and fields local to each statement.

The general theorems are proved for sets. Their conclusions for **Oz**, **Oz**[i], **No** and **No**(i) use class functions and set-supported normal forms. A statement about one polynomial contains finitely many coefficients and variables, so the degree arguments apply to a class-sized coefficient domain unchanged. No proper-class scheme $\text{Spec}(\mathbf{Oz})$ and no class compactness or Noetherian theorem is used; the finite-type theorem is a theorem about sets, applied to the set-sized models of [section 18](#). Expressions involving $\text{Aut}(\mathbf{Oz})$ or ML of a class ring abbreviate statements about each automorphism or derivation; $(b, x) \mapsto D_b(x)$ is one definable class operation. For cancellation against a class-sized B , the transcendence-degree arguments involve only finite tuples of indeterminates inside one rational-function field [02].

Definition 12.1 ([02, 16]). A domain R is *root-covered* if for every $0 \neq a \in R$ there are $b \in R$ and $z = ab \neq 0$ such that for every $n \geq 1$ some $z_n \in R$ has $z_n^n = z$. (No compatibility between the z_n is required; the terminology is local to this report.)

Fields are root-covered. The interesting examples have few units: $\mathbf{Oz}^\times = \{\pm 1\}$ and $\mathbf{Oz}[i]^\times = \{\pm 1, \pm i\}$, and in general $\mathcal{R}_\mathfrak{o}(k, \Gamma)^\times = \mathfrak{o}^\times$ by additivity of leading exponents (the Diophantine report's `odg:prop:units`, `odg:def:lem:units`).

Lemma 12.2 (Localization [02]). *If R is root-covered and $S \subseteq R \setminus \{0\}$ is multiplicative, then $S^{-1}R$ is root-covered.*

Proof. $(a/s)(sb) = z$, and the roots z_n remain. $\square$

Root-covered is sufficient for the conclusions below and is not asserted necessary; failure of the monomial criterion is not used to claim failure of cancellation or stable rigidity [02].

13 The monomial-divisibility criterion

For $a > 0$ in Γ let $C(a) = \{g : |g| \leq na \text{ for some } n \geq 1\}$; a is an *order unit* if $C(a) = \Gamma$.

Theorem 13.1 (Monomial divisibility [02, 16]; after `bst:thm:fieldcriterion`). *For $\Gamma \neq 0$ divisible and $R = \mathcal{R}_o(k, \Gamma)$ the following are equivalent: (i) Γ has no order unit; (ii) for every $0 \neq x \in R$ some monomial X^c with $c > 0$ is divisible by x in R . Then the quotient can be chosen in $\Pi_k(\Gamma)$, and R is root-covered.*

Proof. Assume (i). The support of x lies in $[0, a]$ for some $a > 0$. The inverse of x in the Hahn field has support in the subgroup generated by $\text{supp } x$, hence in $C(a)$ (write $x = \alpha X^h(1 + \varepsilon)$ and expand the geometric series). Choose $c > na$ for all n ; then $X^c x^{-1}$ has strictly positive support, lies in $\Pi_k(\Gamma)$, and $x \cdot X^c x^{-1} = X^c$. Divisibility of Γ gives $(X^{c/n})^n = X^c$. Conversely, if a is an order unit and $x = 1 + X^a$, the unique Hahn quotient $X^c/(1 + X^a) = \sum_{n \geq 1} (-1)^{n-1} X^{c-na}$ has a nonzero coefficient at a negative exponent, so it is not in R . $\square$

This is the transcendence report’s field criterion `bst:thm:fieldcriterion` (`docs/surreal/transcendence-over-bounded-support`: the bounded-support ring is a field if and only if Γ has no order unit, by the same two arguments, the inverse support in a proper convex subgroup and the geometric series of $1 - t^u$), restricted to positive-support elements; its “order unit” is defined identically. Source 02’s audit read only the openings of the omnific reports and did not see it. The positive-support phrasing is what the rigidity theorems use.

Example 13.2 ([02]; [19]). In **No**, $\omega > n$ for all ordinary n , so $\omega^\omega/(1 + \omega) = \omega^{\omega-1} - \omega^{\omega-2} + \omega^{\omega-3} - \dots$ is an omnific integer: $1 + \omega$ divides ω^ω . With exponent group $\mathbb{Q}$ no exponent can absorb the tail. The example is L’Innocente and Mantova’s; the distinction is a larger Archimedean scale, not convergence.

Corollary 13.3 (Omnific root coverage [02, 16]). **Oz** and **Oz**[i] are root-covered: every nonzero element divides a positive real monomial ω^c , with quotient of strictly positive support.

Proof. This is immediate from the global monomial clearing `odg:thm:fractions` of the Diophantine report, applied to x^{-1} : some ω^c makes $\omega^c x^{-1}$ purely infinite. Source 02 gives the same argument through [theorem 13.1](#), using a strict upper bound of the set $\{na : n \geq 1\}$; the coefficients’ order is not used. $\square$

14 Finite-type domains carry no varying embedding

The results of this section concern set-sized fields and domains.

Lemma 14.1 (Unbounded roots [02]). *Let F/K be finitely generated and $u \in F^\times$. If u has n th roots in F for an unbounded set of n , then u is algebraic over K .*

Proof. Otherwise extend u to a transcendence basis $u, t_2, \dots, t_r$, with $[F : E] = d < \infty$ for $E = K(u, t_2, \dots, t_r)$. $Z^n - u$ is Eisenstein at the prime u of $K(t_2, \dots, t_r)[u]$, hence irreducible over E by Gauss's lemma, so an n th root has degree $n \leq d$. $\square$

Lemma 14.2 (Algebraic constants [02]). *If F/K is finitely generated, the relative algebraic closure K' of K in F is finite over K ; if A is an integral K -algebra with $\text{Frac } A = F$, then $C_A = A \cap K'$ is a field finite over K .*

Proof. With a transcendence basis $\mathbf{t}$ and $[F : K(\mathbf{t})] = d$, every finitely generated $L \subseteq K'$ has $[L : K] = [L(\mathbf{t}) : K(\mathbf{t})] \leq d$ (adjoining algebraically independent indeterminates preserves linear independence; $\mathbf{t}$ stays independent over L). An L of maximal degree contains every element of K' . For $0 \neq a \in C_A$ the minimal polynomial has nonzero constant term, which expresses a^{-1} as a polynomial in a over K . $\square$

Lemma 14.3 (Finitely many unit translates [02]). *Let A be a finitely generated integral K -algebra and $f \in A$ transcendental over K . Only finitely many $\lambda \in K$ make $f - \lambda$ a unit of A .*

Proof. Extend f to a transcendence basis $f, t_2, \dots, t_r$ of $\text{Frac } A$, and put $B_0 = K[f, t_2, \dots, t_r]$. Generators of A are algebraic over $\text{Frac } B_0$; clearing denominators gives $0 \neq d \in B_0$ with A integral over $B = B_0[1/d]$ in the common overring. If $(f - \lambda)^{-1} \in A$, it lies in $\text{Frac } B_0$ and is integral over the integrally closed B , so $f - \lambda$ divides a power of d in the UFD B_0 , hence divides d ; the primes $f - \lambda$ are pairwise nonassociate. The argument does not assume $B \subseteq A$. $\square$

Theorem 14.4 (Finite-type embedding rigidity [02]). *Let R be a characteristic-zero root-covered domain, K a characteristic-zero field, and A a finitely generated integral K -algebra. Every injective homomorphism $\phi : R \rightarrow A$ has image in the field $C_A = \{x \in A : x \text{ algebraic over } K\}$, finite over K , and extends uniquely to $\text{Frac } R \rightarrow C_A$. If K is relatively algebraically closed in $\text{Frac } A$, then $C_A = K$. Neither K -linearity of ϕ nor a map $K \rightarrow R$ is assumed.*

Proof. For $a \neq 0$ take $ab = z$ and $z_n^n = z$. Then $\phi(z) \neq 0$ has roots of all orders, so it is algebraic (lemma 14.1) and a unit of A (lemma 14.2); hence $\phi(a)$ is a unit. If $f = \phi(a)$ were transcendental, every $f - m = \phi(a - m)$, $m \in \mathbb{Z}$, would be a unit (as $a - m \neq 0$), contradicting lemma 14.3. So $\phi(R) \subseteq C_A$, and nonzero images are invertible there. $\square$

The theorem is nontrivial only for set-sized R , such as the Hahn models of section 18; it is not a disguised statement that a proper class cannot inject into a set. One must not replace C_A by the whole relative algebraic closure: $A = \mathbb{Q}[x, \sqrt{2}x]$ has $\text{Frac } A = \mathbb{Q}(\sqrt{2})(x)$ but $C_A = \mathbb{Q}$, since elements of A have rational constant term; no normality of A is assumed [02]. If K is relatively algebraically closed in $\text{Frac } A$, every embedding $R \rightarrow A$ has image in K , and none exists if $\text{Frac } R$ does not embed in K .

Theorem 14.5 (Pointed parameter spaces [02]). *Let R be a set-sized characteristic-zero root-covered domain, $F = \text{Frac } R$, A a finite-type integral F -algebra with an F -algebra map $\varepsilon : A \rightarrow F$. If a ring homomorphism $\psi : R \rightarrow A$ has $\varepsilon\psi$ equal to the inclusion, then $\psi(x) = x$ for all x .*

Proof. ψ is injective, so $\psi(R) \subseteq C_A$. The restriction of ε to the field C_A is injective and fixes F ; for $a \in C_A$, $a - \varepsilon(a) \in C_A$ has image 0, so $C_A = F$. $\square$

Examples are $A = F[s_1, \dots, s_r, 1/h]$ with an F -point where $h \neq 0$; relations, singularities and localizations are allowed, and relative algebraic closedness is not required. The hypothesis is one common finite-type algebra containing the images of all of R , not separate formulas in unrelated algebras. For a finite extension L/F , $L[s]$ contains the root-covered L as constants and has no F -valued augmentation when $L \neq F$ [02].

15 Polynomial and Laurent coefficient rigidity

These proofs work over arbitrary coefficient domains, need no finite generation and assume no preservation of infinite sums [02].

Theorem 15.1 (Polynomial coefficient rigidity [02]). *If R is root-covered and B a domain, every injective $\phi : R \rightarrow B[T_1, \dots, T_n]$ has image in B .*

Proof. $\deg \phi(z) = n \deg \phi(z_n)$ for all n forces $\deg \phi(z) = 0$, and $\deg \phi(a) + \deg \phi(b) = 0$. $\square$

Lemma 15.2 (Laurent units [02]). *For a domain B , the units of $B[U_1^{\pm 1}, \dots, U_q^{\pm 1}]$ are the $b\mathbf{U}^{\mathbf{e}}$ with $b \in B^\times$ and $\mathbf{e} \in \mathbb{Z}^q$.*

Proof. The widths $M_j - m_j$ of the U_j -exponents are nonnegative and additive, so a unit has a single exponent vector. $\square$

Theorem 15.3 (Laurent coefficient rigidity [02]). *If R is root-covered and B a domain, every injective $\phi : R \rightarrow B[\mathbf{U}^{\pm 1}]$ has image in B ; hence so does every injective $\phi : R \rightarrow B[\mathbf{T}, \mathbf{U}^{\pm 1}]$.*

Proof. The largest and smallest U_j -exponents of $\phi(z)$ are divisible by every n , so $\phi(z)$ is a nonzero constant, a unit over $\text{Frac } B$, and $\phi(a)$ is a Laurent monomial by lemma 15.2. If $a \neq -1$ the same holds for $\phi(a) + 1$, which forces $\phi(a)$ to be constant. For mixed targets apply theorem 15.1 over $B[\mathbf{U}^{\pm 1}]$ first. $\square$

Injectivity and a domain target are needed: $x \mapsto x + sD(x)$ into $R[s]/(s^2)$ can be nonconstant with identity specialization (section 17). No noninjective analogue for all targets is claimed [02].

Theorem 15.4 (Strong coefficient invariance [02]). *Let R be root-covered, B any commutative unital ring, and $\Phi : R[X_1, \dots, X_p, U_1^{\pm 1}, \dots, U_q^{\pm 1}] \rightarrow B[Y_1, \dots, Y_{p'}, V_1^{\pm 1}, \dots, V_{q'}^{\pm 1}]$ an isomorphism with $p + q = p' + q' < \infty$. Then $\Phi(R) = B$, $q = q'$ and $p = p'$. In particular, for $R = \mathbf{Oz}$ or $\mathbf{Oz}[i]$, every isomorphism $R[\mathbf{X}_n] \cong B[\mathbf{Y}_n]$ carries R onto B : R is strongly invariant and cancels against an arbitrary coefficient ring with the same number of variables. If B is also root-covered, the equality of total variable counts can be dropped.*

Proof. The target is a domain, so B is, and theorem 15.3 gives $\Phi(R) \subseteq B$. With $F = \text{Frac } \Phi(R) \subseteq L = \text{Frac } B \subseteq E$, $\text{trdeg}_F E = p + q = p' + q' = \text{trdeg}_L E$, so L/F is algebraic. E is a rational function field over F , which has no new algebraic constants (a nonconstant rational function is transcendental, one variable at a time), so $B \subseteq F$, and comparing coefficients gives $B \subseteq \Phi(R)$. Polynomial variables add no units, and the quotient unit groups give $\mathbb{Z}^q \cong \mathbb{Z}^{q'}$. If both rings are root-covered, apply the first step to Φ and Φ^{-1} . $\square$

Equality of total variable counts cannot be dropped for arbitrary B : $R[X_1, X_2] = B[X_2]$ with $B = R[X_1]$. The general statements do not solve the unrestricted cancellation problem for finitely generated algebras; the strong-invariance terminology and history are older [14] [02].

Theorem 15.5 (Automorphisms of polynomial and torus rings [02]). *For a root-covered R , every automorphism of $R[\mathbf{T}]$ or $R[\mathbf{U}^{\pm 1}]$ preserves R ; for polynomial extensions, restriction to R is split surjective onto $\text{Aut } R$, with kernel the R -algebra automorphisms. Every automorphism of $R[T]$ is $T \mapsto uT + b$ with $u \in R^\times$, $b \in R$, over a unique $\sigma \in \text{Aut } R$; every automorphism of $R[\mathbf{U}^{\pm 1}]$ is $U_i \mapsto u_i \prod_j U_j^{m_{ji}}$ over σ , with $u_i \in R^\times$ and $(m_{ji}) \in \text{GL}_q(\mathbb{Z})$. For $\mathbf{Oz}$, $u \in \{\pm 1\}$; for $\mathbf{Oz}[i]$, $u \in \{\pm 1, \pm i\}$.*

Proof. Apply [theorem 15.3](#) to an automorphism and its inverse; coefficientwise extension is the section. In one variable, $\deg g \deg f = 1$; units give the torus formula via [lemma 15.2](#). $\square$

The affine and torus formulas are standard once coefficients are preserved; the new omnific conclusion is that *every* ring automorphism preserves $\mathbf{Oz}$. No classification of multivariable polynomial automorphisms over R and no claim about the Jacobian conjecture follows [02].

Batch 32. Source 16 carries this section to monomial parameters of arbitrary rank ([part VI](#)): [theorems 15.1](#) and [15.3](#) hold for every cancellative torsion-free monoid algebra $S[U^M]$ ([theorem 47.1](#)), the torus part of [theorem 15.5](#) for every torsion-free exponent group ([theorem 48.1](#)), and [theorem 15.4](#) at every finite rational rank, with exact failure for $\mathbb{Q}^{(\kappa)}$, κ infinite ([theorems 50.1](#) and [50.4](#)).

16 Locally nilpotent derivations, coactions and gradings

A derivation D of a domain is *locally nilpotent* if every element is killed by some iterate, and $\text{ML}(A) = \bigcap_{D \in \text{LND}(A)} \ker D$. Derivations are ring derivations, hence $\mathbb{Z}$ -linear; A need not contain $\mathbb{Q}$.

Lemma 16.1 ([02]). *For a locally nilpotent D of a characteristic-zero domain, $\deg_D(a) = \max\{j : D^j a \neq 0\}$ satisfies $\deg_D(ab) = \deg_D a + \deg_D b$.*

Proof. $D^{r+s}(ab) = \binom{r+s}{r} D^r a D^s b \neq 0$, and higher iterates vanish. $\square$

Theorem 16.2 (Stable locally nilpotent rigidity [02]). *Let R be a characteristic-zero root-covered domain. Every locally nilpotent derivation of any domain containing R kills R , and $\text{ML}(R[T_1, \dots, T_n]) = R$ for every finite n . In particular $\text{ML}(\mathbf{Oz}[\mathbf{T}]) = \mathbf{Oz}$, $\text{ML}(\mathbf{Oz}[i][\mathbf{T}]) = \mathbf{Oz}[i]$, and $\mathbf{Oz}$ has no nonzero locally nilpotent derivation.*

Proof. $\deg_D(z) = m \deg_D(z_m)$ for all m forces $\deg_D z = 0$, hence $\deg_D a = 0$. The partial derivatives $\partial/\partial T_j$ are locally nilpotent with common kernel R . $\square$

Corollary 16.3 (No coactions or gradings [02]). *If R is root-covered, a homomorphism $\psi : R \rightarrow R[s_1, \dots, s_r]$ with identity specialization at $\mathbf{s} = 0$, or $\theta : R \rightarrow R[u_1^{\pm 1}, \dots, u_r^{\pm 1}]$ with identity specialization at $\mathbf{u} = 1$, is the constant inclusion; so there are no nontrivial additive-group or split-torus coactions of finite rank, and every $\mathbb{Z}^r$ -grading of R is concentrated in degree zero.*

Proof. Identity specialization makes the map injective; apply [theorems 15.1](#) and [15.3](#). A grading gives $\sum x_{\mathbf{m}} \mapsto \sum x_{\mathbf{m}} \mathbf{u}^{\mathbf{m}}$. $\square$

A Hahn expansion $\sum a_g X^g$ is not a grading: it has infinitely many terms, and the no-grading result concerns finite homogeneous decompositions. [Theorem 16.2](#) does not say that $R[\mathbf{T}]$ has no locally nilpotent derivation, and it is not a solution of a general stable-rigidity conjecture [6] [02].

17 Euler derivations and formal flows

The parameter s in this section is a new formal indeterminate, not a surreal and not a Hahn monomial.

Proposition 17.1 (Support-preserving Euler derivations [02]). *For additive $\chi : \Gamma \rightarrow (k, +)$, $D_\chi(\sum a_g X^g) = \sum \chi(g) a_g X^g$ is a k -linear derivation of $k((X^\Gamma))$ that preserves $\mathcal{R}_0(k, \Gamma)$, kills $\mathfrak{o}$ and maps $\mathcal{R}_0(k, \Gamma)$ into $\Pi_k(\Gamma)$; and $D_\chi D_\psi = D_\psi D_\chi$ for additive χ, ψ .*

Proof. The coefficient of X^h in xy is a finite sum of $a_g b_{h-g}$, and $\chi(h) = \chi(g) + \chi(h-g)$ splits it; $\chi(0) = 0$. $\square$

In the notation of [part I](#), D_χ is the shift-zero derivation $t^\gamma \mapsto \chi(-\gamma)t^\gamma$. Division by $n!$ is allowed on $\Pi_k(\Gamma)$, and $D_\chi^n(x) \in \Pi_k(\Gamma)$ for $n \geq 1$, so

$$\mathcal{E}_{\chi,s}(x) = \sum_{n \geq 0} \frac{D_\chi^n(x)}{n!} s^n \quad (17.1)$$

has coefficients in $\mathcal{R}_0(k, \Gamma)$; no division of an integer constant occurs.

Theorem 17.2 (Compatible formal flows [02]). *$\mathcal{E}_{\chi,s}$ is a unital ring homomorphism $\mathcal{R}_0(k, \Gamma) \rightarrow \mathcal{R}_0(k, \Gamma)[[s]]$ with identity specialization, satisfying $\mathcal{E}_{\chi,s} \mathcal{E}_{\chi,t} = \mathcal{E}_{\chi,s+t}$; its coefficientwise extension fixing s is an automorphism of $\mathcal{R}_0(k, \Gamma)[[s]]$ with inverse from $-s$, and it induces compatible automorphisms of $\mathcal{R}_0(k, \Gamma)[s]/(s^{N+1})$ for every N .*

Proof. $D_\chi^n(xy)/n! = \sum_r D_\chi^r(x)/r! \cdot D_\chi^{n-r}(y)/(n-r)!$, first in the Hahn field; the coefficient of $s^r t^m$ in both sides of the flow law is $D_\chi^{r+m}(x)/(r!m!)$; each coefficient uses finitely many inputs. Substituting $t = -s$ is valid since both parameters have positive order. $\square$

Theorem 17.3 (Nonalgebraization at the first jet [02]). *If Γ has no order unit and $\chi \neq 0$, the reduction of $\mathcal{E}_{\chi,s}$ modulo s^2 is nonconstant and is induced by no homomorphism $\psi : \mathcal{R}_0(k, \Gamma) \rightarrow \mathcal{R}_0(k, \Gamma)[s]$ with $\text{ev}_{s=0} \psi = \text{id}$, and D_χ is not locally nilpotent.*

Proof. Some $g > 0$ has $\chi(g) \neq 0$; then $D_\chi^n(X^g) = \chi(g)^n X^g \neq 0$. By [theorem 13.1](#) the ring is root-covered, and by [corollary 16.3](#) every such ψ is constant. $\square$

Formal automorphisms exist to *all* orders; the obstruction appears only when a single polynomial family must realize even the first-order part. A formal power series ring is not of finite type, and $R[s]/(s^2)$ is not a domain, so there is no conflict with [theorem 14.4](#) [02]. The flows involve no analytic convergence, no fine-topology limit and no identification of s with a surreal infinitesimal, and they are not an exponentiation of a derivation *inside* the Hahn field [02].

17.1 The monomial orbit field

Let $R = \mathcal{R}_0(k, \Gamma)$ be set sized, $F = \text{Frac } R$ (which contains k : $a = aX^g/X^g$) and $W = \chi(\Gamma)$, a $\mathbb{Q}$ -subspace of k . In $F((s))$ put

$$L_\chi = F\left(\frac{\mathcal{E}_{\chi,s}(X^g)}{X^g} : g > 0\right) = F(e^{ws} : w \in W), \quad e^{ws} = \sum_n w^n s^n / n!. \quad (17.2)$$

It uses only transformed monomials; it is not claimed to contain the image of every Hahn series [02].

Lemma 17.4 (Formal exponential independence [02]). *For distinct $\lambda_j \in F$ the $e^{\lambda_j s}$ are F -linearly independent; for $\mathbb{Q}$ -independent $w_1, \dots, w_r$ the $e^{w_j s}$ are algebraically independent over F .*

Proof. Comparing coefficients of s^n , $n < m$, gives a Vandermonde system. A polynomial relation is a linear relation among $e^{(\alpha \cdot \mathbf{w})s}$ with distinct exponents. $\square$

Theorem 17.5 (Monomial orbit field [02]). $\text{trdeg}_F L_\chi = \dim_{\mathbb{Q}} W$. *If $\chi \neq 0$, L_χ lies in no finitely generated extension of F inside $F((s))$, so $\mathcal{E}_{\chi,s} : R \rightarrow F[[s]]$ factors through no finite-type F -algebra with an F -algebra second map.*

Proof. A $\mathbb{Q}$ -basis (w_j) of W gives algebraically independent $e^{w_j s}$, and each $e^{w s}$ has a power equal to a monomial in finitely many of them, so L_χ is algebraic over $F(e^{w_j s})$. For $0 \neq w \in W$, $e^{w s}$ is transcendental and has roots $e^{(w/n)s}$ in L_χ , so lemma 14.1 excludes a finitely generated overfield, even when $\dim_{\mathbb{Q}} W = 1$. A factorization would put the image of R in a finite-type domain $B \subseteq F[[s]]$ with $\text{Frac } B \supseteq L_\chi$. $\square$

17.2 An omnific family of commuting directions

Theorem 17.6 (Commuting omnific derivations [02]; after `osq:prop:classder`). *For $b \in \mathbf{No}$, $D_b(\sum a_g \omega^g) = \sum q_b(g) a_g \omega^g$ is an $\mathbb{R}$ -linear derivation of $\mathbf{No}$ and, with complex coefficients, a $\mathbb{C}$ -linear derivation of $\mathbf{No}(i)$ commuting with conjugation; it preserves $\mathbf{Oz}$ and $\mathbf{Oz}[i]$. The D_b commute, and every finite family with distinct indices is linearly independent over $\mathbf{Oz}$, respectively $\mathbf{Oz}[i]$.*

Proof. Proposition 17.1 with $\chi = q_b$. For $m_c = \omega^{\omega^c} \in \mathbf{Oz}$, $D_b(m_c) = m_c$ if $b = c$ and 0 otherwise; applying $\sum_j a_j D_{b_j} = 0$ to m_{b_ℓ} gives $a_\ell m_{b_\ell} = 0$ in a domain. $\square$

These derivations, with the test elements m_c , are the quotient report's class-valued derivations `osq:prop:classder` (`docs/surreal/set-sized-quotients-of-omnific-integers`), which combine that report's sources 06, 08 and 10 and state independence over $\mathbb{R}$; the same evaluation gives independence over the omnific ring, as 02 states, and 02 adds their commutation and flows. In the t -convention of part I, $D_b(t^\gamma) = -q_b(\gamma)t^\gamma$.

Corollary 17.7 (Multi-parameter flows [02]). *For distinct $b_1, \dots, b_r$ and $R = \mathbf{Oz}$ or $\mathbf{Oz}[i]$, the map*

$$x \mapsto \sum_{\alpha \in \mathbb{N}^r} \frac{D_{b_1}^{\alpha_1} \dots D_{b_r}^{\alpha_r}(x)}{\alpha_1! \dots \alpha_r!} s_1^{\alpha_1} \dots s_r^{\alpha_r} \quad \text{into } R[[s_1, \dots, s_r]]$$

obeys the additive formal group law; its coordinate directions are independent, and no nonzero coordinate direction is induced by a polynomial family on R .

Proof. Compose one-parameter flows, which commute; independence is theorem 17.6, and the last claim is theorem 17.3 in one parameter. $\square$

Example 17.8 ([02]). For $b = 0$, $q_0(g)$ is the real constant coefficient of the exponent g : $D_0(\omega^r) = r\omega^r$ for real r , $D_0(\omega^\omega) = 0$, and $\mathcal{E}_{0,s}(\omega) = \omega + \omega s + \frac{\omega}{2}s^2 + \frac{\omega}{6}s^3 + \dots$, every coefficient in $\mathbf{Oz}$ although $\frac{1}{2}, \frac{1}{6} \notin \mathbf{Oz}$; “ ωe^s ” abbreviates this series and is not the surreal exponential. Also $D_0(\omega^\omega/(1+\omega)) = -\omega^{\omega+1}/(1+\omega)^2$, again omnific, supported at $\omega - n$ for positive ordinary n (checked coefficientwise to 29 terms by the placement dossier).

Corollary 17.9 (A continuum of independent orbit elements [02]). *For $R = \mathbf{Oz}$, $F = \mathbf{No}$ (or $R = \mathbf{Oz}[i]$, $F = \mathbf{No}(i)$), the monomial orbit field $F(e^{rs} : r \in \mathbb{R})$ of D_0 has transcendence degree $2^{\aleph_0}$ over F , in the sense of a set-sized transcendence basis with finite algebraic relations.*

Proof. $q_0(r) = r$, and $e^{rs} = \mathcal{E}_{0,s}(\omega^r)/\omega^r$ for $r > 0$. A $\mathbb{Q}$ -basis of $\mathbb{R}$, made positive, has $2^{\aleph_0}$ elements; lemma 17.4 holds over the class field F since each relation uses finitely many coefficients. $\square$

Only a lower bound is asserted for the field generated by the formal images of all Hahn series [02]. *Status* (batch 33): for D_0 this field is described exactly by source 17 (part VII): it is linearly disjoint from $\mathbf{No}(s)$ over $\ker D_0$, which is much larger than $\mathbb{R}$; it contains explicit omnific independent families of every set cardinality, so it has no set-sized transcendence basis; and s is transcendental over it (theorems 53.1 and 53.2). For every χ the full image descends to F/F^{D_χ} (corollary 58.7); its size for other characters is open (questions 59.1 and 59.2). The monomial statements above stand.

Remark 17.10 (Actual automorphisms remain abundant [02]). For additive $\chi : \mathbf{No} \rightarrow \mathbb{R}$ and ordinary $t \in \mathbb{R}$, $\tau_{\chi,t}(\sum a_g \omega^g) = \sum e^{t\chi(g)} a_g \omega^g$ is a support-preserving automorphism of $\mathbf{No}$ and $\mathbf{No}(i)$ preserving $\mathbf{Oz}$ and $\mathbf{Oz}[i]$, with $\tau_{\chi,t}\tau_{\chi,u} = \tau_{\chi,t+u}$ and positive multipliers. These character twists are established [15] and not claimed new (remark 9.5); they prevent reading the rigidity theorems as $\text{Aut}(\mathbf{Oz}) = 1$. What fails is realization of a family by a homomorphism into a polynomial or finite-type parameter algebra: with $\chi = q_0$ the twists send ω^r to $e^{tr}\omega^r$ for all real r at once, which no single polynomial parameter algebra can carry.

18 Set-sized workspaces and a fraction-field warning

Let $\Gamma = \bigoplus_{j \geq 0} \mathbb{Q}e_j$, ordered by the sign of the coefficient at the largest index, so $e_{j+1} > ne_j$ for all n : Γ is divisible with no order unit. For $(\mathfrak{o}, k) = (\mathbb{Z}, \mathbb{R})$ or $(\mathbb{Z}[i], \mathbb{C})$, $R = \mathcal{R}_{\mathfrak{o}}(k, \Gamma)$ is a root-covered set (theorem 13.1), so all the results of this part apply to it in ordinary set theory. The order embedding $e_j \mapsto \omega^j$ of exponent groups induces $X^{e_j} \mapsto \omega^{\omega^j}$ and embeds R in $\mathbf{Oz}$, respectively $\mathbf{Oz}[i]$; the coordinate projections give commuting derivations already here, and replacing $\mathbb{Q}e_j$ by $\mathbb{R}e_j$ gives a set-sized realization of corollary 17.9 [02].

Let $\Gamma_{\leq j} = \bigoplus_{\ell \leq j} \mathbb{Q}e_\ell$ (convex) and $K_{\text{bd}} = \bigcup_j k((X^{\Gamma_{\leq j}}))$.

Proposition 18.1 (Bounded-scale fraction field [02]; after `bst:prop:localdensity`). *For this R , $\text{Frac } R = K_{\text{bd}} \subsetneq k((X^\Gamma))$.*

Proof. An element of R with positive leading exponent in $\Gamma_{\leq j}$ has all its exponents in $\Gamma_{\leq j}$ by convexity, so quotients lie in K_{bd} . Conversely $y \in k((X^{\Gamma_{\leq j}}))$ gives $X^{e_{j+1}}y \in R$. Finally $\sum_{j \geq 0} X^{-e_j}$ is a Hahn series in no $k((X^{\Gamma_{\leq j}}))$. $\square$

This is the transcendence report's localization proposition `bst:prop:localdensity` together with its field criterion (condition (2) of `bst:thm:fieldcriterion`) for this group. For the full class ring, $\text{Frac } \mathbf{Oz} = \mathbf{No}$ and $\text{Frac } \mathbf{Oz}[i] = \mathbf{No}(i)$ (`odg:thm:fractions`); that fact does not justify replacing K_{bd} by the full Hahn field, since a fixed set-sized group may lack an upper bound for a set of its scales even without an order unit [02].

19 Relation to Part I and the questions of this part

The two parts are complementary. Part I concerns derivations with *positive* shifts and their exponentials inside the Hahn field; this part concerns shift-zero (support-preserving) derivations and needs an external formal parameter.

- 02's question on formal directions below is partly answered by part I: the positive-shift derivations preserving $\mathbf{Oz}$ are classified exactly, by $c_\delta(C_\delta) = 0$ (theorem 3.4) and, on

common positive shifts, by $a_\delta(C_\delta) = 0$ (theorem 4.7). Those with a common well-ordered set of shifts integrate *inside* **No** to automorphisms of **Oz** (theorem 4.8), so they are integrable by an honest automorphism rather than only by a formal flow; for an arbitrary contracting strong one this holds granted cited theorem 1.1 for $\Gamma = \mathbf{No}$ (remark 1.2). The shift-zero derivations, the Hasse–Schmidt extensions of other derivations of **Oz**, and derivations with no common well-ordered shift set are not covered. *Status* (batch 33): the Hasse–Schmidt clause is settled for every derivation by source 19: each derivation of **Oz** maps it into Π and has the unique iterative integral extension $\delta^n/n!$ (theorem 67.1); integration *inside* **No** beyond the common-shift class is question 59.6.

- There is no conflict with theorem 16.2: the shift derivations $D_{\delta,\theta}$ restrict to derivations of **Oz** but are not locally nilpotent, since $D_{\delta,\theta}^n(t^\gamma) = \theta(\gamma)^n t^{\gamma+n\delta} \neq 0$ when $\theta(\gamma) \neq 0$; and the flows Φ_s are not polynomial in s , since $\Phi_s(t^{-\eta})$ involves every power of s .
- The character twists of remark 17.10 are the factor $M_{\chi,\text{id}}$ of theorem 5.1.

Theorem 15.5 concerns $\text{Aut}(\mathbf{Oz}[T])$, not $\text{Aut}(\mathbf{Oz})$, and does not bear on `osq:q:invisible`. The rational self-maps of **Oz** studied in the quotient report are maps of the ring given by rational functions, not ring automorphisms of **Oz**[T]; the objects differ. Derivations into set-sized modules vanish (`osq:thm:derivations`), which does not conflict with theorem 17.6, whose values lie in the class module Π .

Question 19.1 (Beyond the scale criterion [02]). For a nonzero divisible Γ *with* an order unit, which rings $\mathcal{R}_o(k, \Gamma)$ remain strongly invariant or stably rigid?

Theorem 13.1 shows that their elements need not divide monomials; it does not show that cancellation or locally nilpotent rigidity fails.

Question 19.2 (Formal directions [02]; re-scoped). Which derivations of **Oz** preserve its purely infinite ideal, and which admit an iterative integral Hasse–Schmidt extension $\mathbf{Oz} \rightarrow \mathbf{Oz}[[s]]$? How much larger is this class than the Euler family of theorem 17.6?

Status. Partly answered for positive shifts by part I, as recorded above; open otherwise. The factorial argument of theorem 17.2 is a sufficient condition for the displayed family and is not asserted to classify all integrable derivations or all formal automorphisms tangent to the identity. *Status* (batch 33): *answered* by source 19 (part IX). Every ring derivation of **Oz** or **Oz**[i] maps the ring into its purely infinite ideal and preserves it, because that ideal is $\bigcap_n nR$ (lemma 2.3), and $\delta^n/n!$ is its unique iterative integral Hasse–Schmidt extension (theorem 67.1 and corollary 68.5); no strongness, $\mathbb{R}$ -linearity or common shift set is needed. So both classes are the whole class of derivations, strictly larger than the Euler family: it contains non- $\mathbb{R}$ -linear derivations such as ωd_* (example 71.2). Every formal automorphism tangent to the identity is the exponential of an integral Lie series (theorem 69.3). What remains is a structural description of all derivations (question 75.1) and evaluation of the flows inside **No** (question 59.6).

Question 19.3 (Reduced parameter algebras [02, 16, 19]). How far does coefficient rigidity extend to reduced parameter rings with several irreducible components when a prescribed specialization stays injective on the source?

The present proofs apply to domain quotients on which the map stays injective; a component can kill root-cover elements, and the dual-number example shows that nilpotent parameter rings need a different statement. Source 16 asks it again (its Question 12.8): its arbitrary-rank proofs (part VI) are for domains and do not address components on which the source map has a kernel; it suggests finding conditions under which enough root covers survive in every relevant

domain quotient and then comparing the coefficient values across components [16]. *Status* (batch 32): open. Source 19 refines it (its Question 13.9): for a reduced finite-type parameter algebra, which components force a pointed omnific homomorphism to be constant, and what happens when some component maps cease to be injective? The integral case ([theorem 74.2](#)) and the unobstructed nilpotent jets ([theorem 70.1](#)) are two endpoints; an answer should separate what is due to zero divisors, to nilpotents and to loss of injectivity on a component [19]. *Status* (batch 33): open.

Part III

Automatic strongness and proper embeddings of the omnific pair

This part is built from three manuscripts of batch 30, numbered 11, 12 and 13 here. It has two themes. The first is automatic strongness: every ring automorphism of $\mathbf{Oz}$ extends to a strong automorphism of $\mathbf{No}$, which answers [question 11.1](#) and extends [theorem 5.1](#) and [corollary 5.2](#) to all automorphisms. The second is embeddings: proper self-embeddings of $\mathbf{No}$ can preserve and reflect $\mathbf{Oz}$ and still move a real number, and the strong exact-monomial ones are classified by a convex subgroup of the target value group. Its labels carry the sub-prefix `opa:as:`.

20 Sources, results and conventions of this part

20.1 Three sources and where the merge chose

The three manuscripts were placed together in commit 21375f8 as items 03, 04 and 05 of batch 30. They are numbered 11, 12 and 13 here, after their file prefixes:

- **11** *Coefficient Gaps and Proper Self-Embeddings of the Surreal Field: omnific preservation, topological dichotomies, and nonconjugate copies*, pinned to `cf56b89`; files prefixed `11-coefficient-gaps-`.
- **12** *Automatic Strongness and Proper Embeddings of the Omnific Integers: constant-term duality, full stabilizers, and the surreal-surcomplex boundary*, pinned to `a6c68ac`; files prefixed `12-automatic-strongness-`.
- **13** *Automatic Hahn Linearity of Omnific Isomorphisms: residue duality, full stabilizers, and coefficient-drifting self-embeddings*, pinned to `0865f04`; files prefixed `13-omnific-isomorphisms-`.

The three pins are commits of one line that forked at `c6359e4`, the placement of source 10. None contains `b3fa9e2`, the batch-28 write of source 10 into this report. So no author saw [section 3.3](#), [remark 3.9](#) or the paragraph of partial information after [question 11.1](#); their comparisons were with the report of batch 26, and they cite its strong factorization ([theorem 5.1](#)) and its strongness-free [corollary 5.2](#) accurately.

Sources 12 and 13 prove the same main theorems, in nearly the same order: scalar detection of summability, automatic strongness of constant-term-covariant isomorphisms, automatic strongness of $\text{Aut}(\mathbf{Oz})$, the factorization of the whole stabilizer, scalar adjoints, a proper-class obstruction, a proper embedding moving a real, and surcomplex versions. They are printed once, with both sources named. The merge made these choices:

- *Base of the automatic-strongness sections: 12.* Its abstract theorem assumes only a coefficient isomorphism α and derives $F|_k = \alpha$, where 13 assumes $\sigma(k) = \ell$ ([theorem 22.1](#)); it also proves the factorization for set-sized groups, the Archimedean corollary, the regular-cardinal case and retractions for arbitrary strong embeddings. 13's genuinely different material is added with its proofs: the countable binary witness and its finite-row lemma, kept as a second route to detection ([lemma 21.3](#) and [theorem 21.4](#)); isomorphisms between different, possibly non-divisible groups ([proposition 23.1](#) and [corollary 23.4](#)); the coefficient-matrix formula ([proposition 22.3](#)); the example that constant-term preservation does not give preservation of $\mathbf{Oz}$ ([example 23.6](#)); the small-integer-part fraction example ([example 31.2](#)); and the failure of homogeneity ([corollary 27.5](#)).
- *Base of the embedding sections: 11.* Its bottom-gap classification ([theorem 26.3](#)) contains the embeddings of 12 and 13 as instances ([theorem 27.3](#)). The three explicit maps are the same construction with different order isomorphisms $\mathbf{No} \rightarrow \mathbf{No}_{>0}$ or $\mathbf{No} \rightarrow (1, \infty)$; they are printed once. 12's image intersection and annihilator and 13's non-homogeneity are kept.
- *Printed once, not reprinted.* The reconstruction ingredients that all three reprove (the ideal $\Pi = \bigcap_n n\mathbf{Oz}$, the existential test, the multiplier ring, its units, $\text{Frac } \mathbf{Oz} = \mathbf{No}$, fixing of $\mathbb{R}$) are cited from the Diophantine report and [section 8](#); the convex-support criterion, which 12 and 13 reprove, is [theorems 3.3](#) and [3.8](#); the Taylor coefficient automorphism, which 11 reprove, is `saut:thm:coeffflow` of the surcomplex automorphism report.
- *Credit added.* 12 and 13 do not credit `saut:thm:coeffflow`, which already moves a real b to $b + t^\delta$ by a strong automorphism of $\mathbf{No}$; 11 does. The Taylor motion in every embedding below is that theorem's ([section 26.1](#)).

No mathematical error was found in the three manuscripts. They agree with one another and with the rest of the report: automorphisms preserving $\mathbf{Oz}$ fix $\mathbb{R}$ ([corollary 5.2](#)), embeddings need not. Two statements in their delivered audits are wrong and are recorded in [section A.3](#); neither affects the mathematics.

Batch 31. Source 14, written independently of 11–13, re-derives the automatic-strongness results of this part; where it does, 14 is added to the source tag, and its own material and its notation are in [part IV](#).

20.2 What this part settles

- [Question 11.1](#) is *answered* in all its clauses ([theorem 23.2](#)): every ring automorphism of $\mathbf{Oz}$ extends uniquely to a strongly $\mathbb{R}$ -linear automorphism of $\mathbf{No}$ whose inverse is strong, so preservation of $\mathbf{Oz}$ forces strongness. There is no nonstrong $\mathbf{Oz}$ -preserving automorphism, so the clause of 09 (characterize the nonstrong automorphisms of $(\mathbf{No}, \mathbf{Oz})$ that commute with normal-form sums) is vacuous, and 10's variant (is every $\mathbf{Oz}$ -preserving 1-automorphism strong?) has the answer yes.
- [Theorem 5.1](#) and [corollary 5.2](#) hold for all automorphisms of $(\mathbf{No}, \mathbf{Oz})$ and of the full real Hahn pairs with nonzero divisible set-sized exponent group ([theorem 24.1](#)). For Archimedean groups every stabilizer element is a monomial map ([corollary 24.3](#)), which upgrades [theorem 4.13](#) from 1-automorphisms to the whole stabilizer.
- The second clause of `osq:q:invisible` (how much can an automorphism of $\mathbf{Oz}$ do inside Π while fixing ct ?) is answered ([corollary 24.4](#)); its first clause is untouched.
- Scope. The answers concern $(k, \mathfrak{o}) = (\mathbb{R}, \mathbb{Z})$: $\mathbf{Oz}$ and the full real Hahn pairs. For $(\mathbb{C}, \mathbb{Z}[i])$ they hold when conjugation or the natural valuation is preserved ([section 30](#));

whether every automorphism of $\mathbf{Oz}[i]$ is strong stays open ([question 32.1](#)). For a general characteristic-zero k and unital $\mathfrak{o} \subseteq k$, as in [theorem 3.3](#), nothing here removes the strongness hypothesis ([question 32.13](#)).

- **Embeddings:** a strong field embedding of the full real Hahn pairs sending monomials to monomials and preserving the integer parts is classified by a coefficient section into the Hahn field on the *bottom gap*; real numbers can move exactly when the positive part of the value-group image is not coinital ([theorem 26.3](#) and [corollary 26.7](#)). Whether every embedding of $(\mathbf{No}, \mathbf{Oz})$ reflecting $\mathbf{Oz}$ is strong has a negative answer if there is a measurable cardinal ([question 32.2](#)).

20.3 Conventions and notation of this part

All three sources use the small- t convention (1.1) of [part I](#), $t^\gamma = \omega^{-\gamma}$, so the purely infinite series have *negative* t -exponents. For a field k and an ordered abelian group Γ put $K_\Gamma = k((t^\Gamma))$, Π_Γ for the series supported in $\Gamma^{<0}$ and $\mathfrak{m}_\Gamma$ for those supported in $\Gamma^{>0}$. For $k = \mathbb{R}$ put

$$\mathcal{R}_\Gamma = \mathbb{Z} \oplus \Pi_\Gamma, \quad \mathcal{A}_\Gamma = \mathbb{R} \oplus \Pi_\Gamma,$$

as in [theorem 4.10](#); for $\Gamma = \mathbf{No}$ these are $\mathbf{Oz}$ and $\mathcal{A}_k$. For $k = \mathbb{C}$ write $\Pi_\Gamma^\mathbb{C}, \mathcal{R}_\Gamma^\mathbb{C} = \mathbb{Z}[i] \oplus \Pi_\Gamma^\mathbb{C}$ and $K_\Gamma^\mathbb{C} = \mathbb{C}((t^\Gamma))$; for $\Gamma = \mathbf{No}$ these are $\Pi_\mathbb{C}, \mathbf{Oz}[i]$ and $\mathbf{No}(i)$. The constant term $\text{ct}(x) = [t^0]x$ is the projection P_0 on all of K_Γ ; it is k -linear and not multiplicative, since $\text{ct}(t^\gamma) \text{ct}(t^{-\gamma}) = 0 \neq \text{ct}(1)$ for $\gamma \neq 0$. The *constant-term pairing* is

$$\langle x, y \rangle_0 = \text{ct}(xy), \quad [t^\gamma]x = \langle x, t^{-\gamma} \rangle_0. \quad (20.1)$$

It is symmetric, k -bilinear and nondegenerate, and not positive definite: it is not an inner product, a trace or a residue integral [12, 13].

Strong maps. An additive map is *strongly additive* if it sends every Hahn summable family to a summable family and commutes with its sum; this is one-directional. A strongly additive map need not be $\mathbb{R}$ -linear: the Taylor maps below move real constants. As in [section 1.5](#), a *strong automorphism* is one whose inverse is also strongly additive; every theorem below states both directions where it asserts them. For scalar families in k “summable” means “only finitely many nonzero terms”; this is not convergence in $\mathbb{R}$ [13].

Exponent maps. Here τ is an *ordered additive embedding* of exponent groups; when it is onto it is the order automorphism τ of (5.1). The monomial map $M_{\chi, \tau}$ of (5.1) is defined by the same formula for an embedding τ and then is a strong field embedding. A field embedding is *exact-monomial* if it sends every t^γ to a monomial $t^{\tau(\gamma)}$ with coefficient one (11 says “monomial-preserving”). This does *not* mean commuting with the omega-map $x \mapsto \omega^x$, and it excludes characters and principal-unit corrections [11].

Notation	Meaning in part III , and the sources’ symbols
$\mathcal{R}_\Gamma, \mathcal{A}_\Gamma$	$\mathbb{Z} \oplus \Pi_\Gamma$ and $\mathbb{R} \oplus \Pi_\Gamma$. Sources: I_Γ, A_Γ (11); $A_G, B_\mathbb{R}$ (12); $\mathcal{J}_\Gamma, \mathcal{A}$ (13).
$\langle x, y \rangle_0$	$\text{ct}(xy)$. Sources: $\langle x, y \rangle_0$ (12, 13); 11 writes $[x]_0$ for $\text{ct}x$.
$h_\sigma(g), a_{\sigma, \delta}$	The logarithmic character (3.1). Sources: ℓ_g, a_δ (12); L_γ, a_δ (13).
G_τ	The <i>bottom gap</i> of $\tau : \Gamma \hookrightarrow \Delta$ (definition 26.1). Source: C_τ (11).

Notation	Meaning in part III , and the sources' symbols
ρ	An ordered field embedding $\mathbb{R} \rightarrow \mathbb{R}((t^{\mathbb{G}_\tau}))$, a <i>coefficient section</i> (11). 13's $\rho = \sigma _k$ is α here, 12's letter.
$J_{\tau,\rho}$	The classified embedding (26.1). Source: $\Phi_{\tau,\rho}$ (11).
d	A $\mathbb{Q}$ -derivation of $\mathbb{R}$. Sources: d (11 and the surcomplex report), D (12, 13); D stays reserved for derivations of Hahn fields.
$\rho_{d,\delta}, \mathcal{T}_{d,\delta}$	The Taylor section (26.3) and the ambient Taylor automorphism (26.4), which is Φ_d of <code>saut:thm:coeffflow</code> . Sources: $\rho_{d,\delta}, E_{d,\delta}$ (11); ρ_D (12); E_D (13).
ι_μ	The inner lift (27.1) of an order embedding $\mu : \mathbf{No} \hookrightarrow \mathbf{No}$. Sources: T_h (11); 12's θ is $\iota_{\mu_{12}}$ and 13's second τ is $\iota_{\mu_{13}}$.
$\mu_{11}, \mu_{12}, \mu_{13}$	The order isomorphisms (27.2). Sources: p (11), ψ (12), p (13).
$J_{\mu,d}$	The explicit embedding (27.3). Sources: Φ (11), J_D (12), F_D (13).
$T^\dagger, e_\phi, \text{ret}_\phi$	Constant-term adjoint, $\phi^\dagger(1)$ and the linear retraction of a strong embedding ϕ (section 25). Sources: $T^\dagger, d = j^\dagger(1), E_j$ (12); T^*, A (13).
$\lambda_{\mathbf{On}}$	The class functional (25.2). Sources: Λ (12), λ (13).
$\mathcal{V}_\lambda$	The valuation ball $\{x : v(x) > \lambda\}$. Source: $\mathcal{B}_\lambda$ (11).

The collisions are resolved in the new material only, and the tempting false readings are wrong:

- 11's C_τ is *not* a principal convex subgroup C_δ . It is the largest convex subgroup of the target group meeting $\tau(\Gamma)$ only in 0; in the notation of [definition 2.4](#), $\mathbb{G}_\tau = \bigcap_{h \in \tau(\Gamma) > 0} H_h$. For the explicit maps below it happens to be a principal convex subgroup, C_1 or C_ω ([lemma 27.2](#)). 11's $H = \tau(\Gamma)$ is written $\tau(\Gamma)$, leaving H_g to [definition 2.4](#); 11's class $\mathcal{F}$ of finite surreals is C_1 .
- 12's θ is an *exponent embedding*, not an additive profile as in $D_{\delta,\theta}$; it is written $\iota_{\mu_{12}}$. 13 uses τ both for the order automorphism of $M_{\chi,\tau}$ and for its exponent embedding; the second is $\iota_{\mu_{13}}$.
- 11's and 13's order isomorphisms are both called p , and 11's ρ and 13's ρ are different maps; the renamings above separate them.
- 13 writes the factorization as $\sigma = U \circ M$, 12 and [theorem 5.1](#) as $\sigma = M \circ u$; the two agree with $U = MuM^{-1}$, since the monomial maps normalize $\mathcal{U}_{\mathbf{Oz}}$.
- 11's target group Λ is Δ here, 13's letter, so that λ and Λ stay free; 11's Gaussian ring J_Γ and field L_Γ are $\mathcal{R}_\Gamma^\mathbb{C}$ and $K_\Gamma^\mathbb{C}$.
- The large-cardinal report's sign-sequence embedding, called J there and in its dossier, is written $\hat{j}$ here, to keep J for the strong embeddings of this part.

21 Constant-term tests detect Hahn summability

This section is 12's Section 3 and 13's Section 3, printed once. Let k be any field and $K = k((t^\Gamma))$ a full Hahn field; Γ may be a set or $\mathbf{No}$, with set-sized supports. The point is that one scalar functional, ct , together with multiplication, recognizes Hahn summability.

Lemma 21.1 (Finite intersection [12, 13, 14, 15]). *If $S, T \subseteq \Gamma$ are well ordered, then $S \cap (-T)$ is finite.*

Proof. An infinite subset of the well-ordered S contains a strictly increasing sequence $s_0 < s_1 < \dots$, and then $-s_0 > -s_1 > \dots$ would be a descending sequence in T . $\square$

Lemma 21.2 (Triangular detector [12]). *Let $(x_j)_{j \in I}$ be a set-indexed family in K in which each exponent occurs in only finitely many supports. If $\bigcup_j \text{supp } x_j$ is not well ordered, there is $y \in K$ with countable support and all coefficients in $\{0, 1\}$ such that $\langle x_j, y \rangle_0 \neq 0$ for infinitely many j .*

Proof. Choose a strictly decreasing sequence in the union of the supports. Each support meets it in finitely many places, since an infinite subsequence would descend inside a well order. Select recursively a decreasing subsequence $g_0 > g_1 > \dots$ and distinct indices $j_0, j_1, \dots$ with

$$[t^{g_n}]x_{j_n} \neq 0, \quad [t^{g_m}]x_{j_n} = 0 \quad (m > n) : \quad (21.1)$$

at each stage discard the finitely many places lying in the previously selected supports, and pick a later place and an index whose support contains it. Choose $c_n \in \{0, 1\}$ recursively with $\sum_{m \leq n} c_m [t^{g_m}]x_{j_m} \neq 0$; the new variable c_n has the nonzero coefficient $[t^{g_n}]x_{j_n}$, so at most one of the two choices is forbidden, over every field. Put $y = \sum_n c_n t^{-g_n}$, a Hahn series because the exponents $-g_n$ increase. By (21.1), $\langle x_{j_n}, y \rangle_0$ is exactly that finite sum, nonzero for every n . $\square$

Lemma 21.3 (Binary finite-row lemma [13]). *Let $(L_n)_{n \in \mathbb{N}}$ be nonzero linear forms on k^J , each using only finitely many coordinates. There is $a \in \{0, 1\}^J$ with $L_n(a) \neq 0$ for infinitely many n .*

Proof. Let $E_n \neq \emptyset$ be the finite support of L_n . If $E = \bigcup_n E_n$ is finite, some coordinate j has a nonzero coefficient in infinitely many rows, and $a = e_j$ works. If E is infinite, select rows $n_0 < n_1 < \dots$ inductively. At stage r finitely many coordinates have been assigned, and the rows after n_{r-1} still use infinitely many coordinates; choose n_r whose support contains an unassigned coordinate j_r , set the other new coordinates of that row to 0, and choose $a_{j_r} \in \{0, 1\}$ so that $L_{n_r}(a) \neq 0$ (the two choices change $L_{n_r}(a)$ by a nonzero field element). Mark all of E_{n_r} as assigned; later choices do not change this row. Finally set the remaining coordinates to 0. $\square$

The lemma holds in every characteristic, including over $\mathbb{F}_2$, and uses no uncountability, genericity or analytic argument. Asking for *every* row to be nonzero would be false: over $\mathbb{F}_2$ the forms $x, y, x + y$ cannot all be nonzero [13].

Theorem 21.4 (Scalar detection of Hahn summability [12, 13, 14, 15]). *For a set-indexed family $(x_j)_{j \in I}$ in the full Hahn field K the following are equivalent:*

- (i) (x_j) is Hahn summable;
- (ii) for every $y \in K$, only finitely many j have $\langle x_j, y \rangle_0 \neq 0$;
- (iii) the same holds for every y with countable support and coefficients in $\{0, 1\}$.

If they hold, then for every y

$$\left\langle \sum_j x_j, y \right\rangle_0 = \sum_j \langle x_j, y \rangle_0, \quad (21.2)$$

a finite scalar sum, and $\sum_j x_j$ is the unique element with this property.

Proof. (i) $\Rightarrow$ (ii): with $S = \bigcup_j \text{supp } x_j$ and $T = \text{supp } y$, only exponents in the finite set $S \cap (-T)$ contribute to a constant term (lemma 21.1), and each occurs in finitely many x_j ; all rearrangements in (21.2) are finite. (ii) $\Rightarrow$ (iii) is trivial. (iii) $\Rightarrow$ (i): the tests $y = t^{-g}$ give pointwise finiteness, and if the union of the supports were not well ordered, lemma 21.2 would give a countable binary y violating (iii). Uniqueness follows from the tests t^{-g} and nondegeneracy. $\square$

Second route to (iii) $\Rightarrow$ (i) [13]. Assume (iii); monomial tests give pointwise finiteness. If the union S is not well ordered, choose $\gamma_0 > \gamma_1 > \dots$ in S . Each support meets $\{\gamma_n\}$ finitely, so infinitely many distinct indices $j_0, j_1, \dots$ meet it. The forms $L_r(a) = \sum_{n: \gamma_n \in \text{supp } x_{j_r}} a_n [t^{\gamma_n}] x_{j_r}$ on $k^{\mathbb{N}}$ are nonzero and use finitely many coordinates, so lemma 21.3 gives binary a_n with infinitely many $L_r(a) \neq 0$. For $y = \sum_n a_n t^{-\gamma_n}$, a Hahn series, one has $\text{ct}(yx_{j_r}) = L_r(a)$, contradicting (iii). $\square$

The two routes differ in how cancellation is prevented: 12 first makes the family triangular and then chooses coefficients; 13 keeps the family and chooses coefficients row by row. 12's witness is also countable and binary, although 12 does not state it; 13 does. Every witness has countable set support, so the theorem holds for $\Gamma = \mathbf{No}$ [12, 13]. A third route, 14's induced matching, whose witness has all coefficients 1 and involves no choice of coefficients, is printed in section 34, and 15's disjoint-row variant of it in section 40.

Corollary 21.5 (Countable certificates [13]). *Every nonsummable set-indexed family has a countable subfamily and a countably supported binary y for which infinitely many products $x_j y$ have nonzero constant term.*

Proof. An exponent occurring in infinitely many supports gives a monomial witness; otherwise use lemma 21.2. $\square$

This does not say that summable families are countable: a Hahn sum may have an arbitrarily large set-sized support. It says that a failure of summability has a countable certificate.

Example 21.6 (Monomial tests are not enough [13, 15]). The family $x_n = t^{-n}$, $n \geq 1$, has each exponent in one member, but its supports descend. Each monomial test detects at most one member, whereas the Hahn series $y = \sum_{n \geq 1} t^n$ has $\text{ct}(yx_n) = 1$ for every n .

Remark 21.7 (Finiteness spaces [12, 13, 14]). Lemma 21.1 is finite-intersection orthogonality, and its linearization is the finiteness-space duality of Ehrhard, applied to generalized power series by Blute, Cockett, Jacqmin and Scott [4]. Neither support duality nor the notion of a strong linear map is new; what the sources prove directly is the scalar criterion with its cancellation argument and set supports over the class $\mathbf{No}$. It is an external characterization of summation, not first-order definability of summation, and not convergence in the valuation topology: a Hahn-summable family need not converge in that topology [13].

22 Constant-term-covariant isomorphisms are strong

Theorem 22.1 (Automatic strongness from constant-term covariance [12, 13, 14, 15]). *Let $K = k((t^\Gamma))$ and $L = l((t^\Delta))$ be full Hahn fields, the exponent groups sets or **No**, with set supports. Let $F : K \rightarrow L$ be a field isomorphism and $\alpha : k \rightarrow l$ a field isomorphism with*

$$\text{ct}(Fx) = \alpha(\text{ct } x) \quad (x \in K). \quad (22.1)$$

Then $F(r) = \alpha(r)$ for all $r \in k$, F is α -semilinear, and F and F^{-1} are strongly additive.

Proof. For $r \in k$ and $x \in K$, multiplicativity and (22.1) give $\text{ct}((F(r) - \alpha(r))F(x)) = \alpha(\text{ct}(rx)) - \alpha(r)\alpha(\text{ct } x) = 0$. As F is onto, $F(r) - \alpha(r)$ pairs to zero with all of L , so $F(r) = \alpha(r)$; semilinearity follows. Let (x_j) be summable with sum s . For $y \in L$ write $y = F(z)$; then $\langle F(x_j), y \rangle_0 = \alpha(\langle x_j, z \rangle_0)$ is nonzero for finitely many j (theorem 21.4 in K), so $(F(x_j))$ is summable (theorem 21.4 in L). With $w = \sum_j F(x_j)$, $\langle F(s), y \rangle_0 = \alpha(\sum_j \langle x_j, z \rangle_0) = \sum_j \langle F(x_j), y \rangle_0 = \langle w, y \rangle_0$ for all y , so $F(s) = w$. The inverse satisfies the hypotheses with α^{-1} . $\square$

13, 14 and 15 prove this with the extra hypothesis $F(k) = l$ and $\alpha = F|_k$; 12 derives it. The proof detects summability of the image family before any target sum is used, and it does not represent arbitrary strong functionals, which would fail for **No** (theorem 25.6). Surjectivity enters twice: it supplies a preimage of every target test y , and it identifies the image of k . An embedding supplies only the tests in its image; nondegeneracy of the pairing on the image does not detect all target support obstructions. This is a real boundary, not a removable word: the embeddings of section 27 satisfy (22.1) with $\alpha = \text{id}$ and move real coefficients; they are strongly additive, so they contradict coefficient rigidity without surjectivity, not strongness [12, 13].

Corollary 22.2 (The summation operation is recovered [12, 14]). *An isomorphism of full Hahn fields that carries the coefficient field onto the coefficient field and commutes with the constant term up to that coefficient isomorphism respects the whole set-indexed Hahn summation. It need not preserve the monomial sections.*

Proposition 22.3 (Coefficient matrix [13]). *If in theorem 22.1 $k = l$ and $\alpha = \text{id}$, then for all $f \in K$ and $\delta \in \Delta$*

$$[t^\delta]F(f) = \text{ct}(F^{-1}(t^{-\delta})f) = \sum_{\gamma \in \text{supp } f} [t^\gamma]f [t^{-\gamma}]F^{-1}(t^{-\delta}), \quad [t^\delta]F(t^\gamma) = [t^{-\gamma}]F^{-1}(t^{-\delta}), \quad (22.2)$$

with finite sums. So the coefficient matrix of F is the inverse transpose of that of F^{-1} , with exponent reversal from the pairing.

Proof. $[t^\delta]F(f) = \text{ct}(F(f)t^{-\delta}) = \text{ct}(F(fF^{-1}(t^{-\delta}))) = \text{ct}(fF^{-1}(t^{-\delta}))$ by (22.1); expand the last constant term, which has finitely many contributions by lemma 21.1. $\square$

The formula presupposes the global well-ordering of the image, which theorem 22.1 proves; a term-by-term formal argument alone would leave both the coefficient and the well-ordering gaps open [13].

23 Automatic strongness of the omnific stabilizer

23.1 What the integer part reconstructs

All three sources reprove the reconstruction of the coefficients from the integer part: $\Pi = \bigcap_n n\mathbf{Oz}$ (lemma 2.3), the existential test $\exists y (x^2 = 2y^2)$ (odg:def:thm:ideal), the multiplier ring $(\Pi : \Pi) = \mathcal{A}_k$ with units $\mathbb{R}^\times$ (odg:def:thm:multiplier), $\text{Frac } \mathbf{Oz} = \mathbf{No}$ (odg:thm:fractions; also [19, Proposition 2.4.5], which records it as [7, Theorem 32]), and the fixing of $\mathbb{R}$ by every automorphism (odg:def:thm:autreal, corollary 5.2). They credit these as prior work, and they are not reprinted. The one form needed below that is not already stated in the report is 13's version for isomorphisms between different, possibly non-divisible groups.

Proposition 23.1 (Pair isomorphisms preserve the constant term [12, 13, 14, 15]). *Let Γ, Δ be nonzero ordered abelian groups, sets or $\mathbf{No}$, and let $F : K_\Gamma \rightarrow K_\Delta$ be an ordered field isomorphism of the real Hahn fields with $F(\mathcal{R}_\Gamma) = \mathcal{R}_\Delta$. Then F fixes $\mathbb{R}$ pointwise, maps $\Pi_\Gamma, \mathbb{R}$ and $\mathfrak{m}_\Gamma$ onto $\Pi_\Delta, \mathbb{R}$ and $\mathfrak{m}_\Delta$, and $\text{ct}(Fx) = \text{ct}(x)$ for all x . If Γ and Δ are divisible, “ordered” may be omitted.*

Proof. By lemma 2.3, $\Pi_\Gamma = \bigcap_n n\mathcal{R}_\Gamma$ for every ordered abelian group, so $F(\Pi_\Gamma) = \Pi_\Delta$. The multiplier description $\{x : x\Pi_\Gamma \subseteq \Pi_\Gamma\} = \mathcal{A}_\Gamma$ holds for every nonzero Γ : a nonzero coefficient of x at $\delta > 0$ gives $\text{ct}(xt^{-\delta}) \neq 0$. As F is onto, $F(\mathcal{A}_\Gamma) = \mathcal{A}_\Delta$, and the units $\mathbb{R}^\times$ correspond, so F maps $\mathbb{R}$ onto $\mathbb{R}$; a field isomorphism $\mathbb{R} \rightarrow \mathbb{R}$ is the identity. The order is preserved by hypothesis, or because divisible groups give real closed Hahn fields whose order is defined by squares. The infinitesimals $\{x : |x| < 1/n \text{ for all } n\}$ are $\mathfrak{m}_\Gamma$, so $F(\mathfrak{m}_\Gamma) = \mathfrak{m}_\Delta$, and the decomposition $K = \Pi \oplus \mathbb{R} \oplus \mathfrak{m}$ gives the rest. $\square$

For $\Gamma = \Delta$ divisible this is also a consequence of the parameter-free definitions of theorem 8.2; for $\Gamma = \Delta = \mathbf{No}$ it is part of corollary 5.2. The coefficient field is reconstructed as a set and then fixed pointwise; an ordered embedding $\mathbb{R} \hookrightarrow \mathbf{No}$ need not have image $\mathbb{R}$, and section 27 gives embeddings that move reals by infinitesimals [13].

23.2 The main theorem

Theorem 23.2 (Automatic strongness of omnific automorphisms [12, 13, 14, 15]). *Every unital ring automorphism σ of $\mathbf{Oz}$ extends uniquely to a field automorphism $\tilde{\sigma}$ of $\mathbf{No}$. The extension preserves the order and fixes $\mathbb{R}$ pointwise, and $\tilde{\sigma}$ and $\tilde{\sigma}^{-1}$ are strongly $\mathbb{R}$ -linear: for every Hahn-summable family (f_j) the family $(\tilde{\sigma}(f_j))$ is summable and $\tilde{\sigma}(\sum_j f_j) = \sum_j \tilde{\sigma}(f_j)$. Consequently*

$$\text{Aut}(\mathbf{Oz}) \cong \text{Aut}(\mathbf{No}; \mathbf{Oz}) = \text{Aut}_{\mathbb{R}}^{\text{str}}(\mathbf{No}; \mathbf{Oz}), \quad (23.1)$$

by restriction, with extension to fractions as inverse; here $\text{Aut}(\mathbf{No}; \mathbf{Oz})$ denotes the field automorphisms of $\mathbf{No}$ preserving $\mathbf{Oz}$ setwise and the right-hand side those among them that are strongly $\mathbb{R}$ -linear with strong inverse.

Proof. Extension, uniqueness, order and fixing of $\mathbb{R}$ are odg:def:thm:autreal. By corollary 5.2 (or proposition 23.1) the extension commutes with $P_0 = \text{ct}$, without any strongness assumption. Theorem 22.1 with $k = l = \mathbb{R}$ and $\alpha = \text{id}$ gives strongness of $\tilde{\sigma}$ and its inverse. $\square$

The dependencies are not circular [13, 14, 15]:

$$\mathbf{Oz} \longrightarrow \Pi \longrightarrow \mathcal{A}_k \longrightarrow \mathbb{R} \longrightarrow (\Pi, \mathbb{R}, \mathfrak{m}) \longrightarrow \text{ct} \longrightarrow \text{Hahn summation}.$$

The first five steps use only algebra, the order and finite decompositions; the residue criterion is used only after the constant term has been recovered. The group notation in (23.1) is schematic, in the sense of section 1.4: it concerns individual class maps and their compositions, and forms no class of all of them.

Remark 23.3 (What is and is not asserted [12, 13, 14, 15]). The theorem does not say that every automorphism of the field **No** is strong: Kaplan, Krapp and Serra construct automorphisms of **No** that are not strongly additive [15, Example 4.4], and the theorem shows that none of them preserves **Oz**. Nor does it say that the automorphisms of **(No, Oz)** are few: part I constructs large families of them, and they are strong by construction. The theorem is a regularity statement. It identifies the summation operation as preserved by these maps; invariance under automorphisms is not first-order definability of summation. It also gives an intrinsic certificate for a transported sum: a proposed value g of $\tilde{\sigma}(\sum_j f_j)$ is correct exactly when $\text{ct}(yg) = \sum_j \text{ct}(y\tilde{\sigma}(f_j))$ for every $y \in \mathbf{No}$, each a finite equation [13]. With theorems 3.3 and 5.1 the theorem answers question 11.1; see section 32.1.

Related (batch 32). The Kaplan–Krapp–Serra automorphism already fails a countable sum: it moves the sum of an ω -indexed family of monomials while fixing every summand (the large-cardinal report’s `lce:mf:rem:unrestricted`). By that report’s `lce:mf:thm:spectrum`, a countably strong k -linear map between full Hahn fields over $\mathbb{R}$ or $\mathbb{C}$ that is not strong first fails on a family of measurable cardinality.

Corollary 23.4 (Full real Hahn pairs [12, 13, 14, 15]). *Every isomorphism F in proposition 23.1 is strongly $\mathbb{R}$ -linear with strongly $\mathbb{R}$ -linear inverse. In particular, for a nonzero divisible set-sized Γ , every field automorphism of K_Γ stabilizing $\mathcal{R}_\Gamma$ is strongly $\mathbb{R}$ -linear.*

Proof. Proposition 23.1 and theorem 22.1 with $\alpha = \text{id}$. □

The corollary assumes an isomorphism of the *ambient* Hahn fields. It does not assert that an abstract isomorphism $\mathcal{R}_\Gamma \cong \mathcal{R}_\Delta$, or an automorphism of $\mathcal{R}_\Gamma$, extends to the Hahn fields when $\text{Frac } \mathcal{R}_\Gamma$ is smaller than K_Γ , which can happen (example 31.2, [19, Proposition 2.4.5]); 12 states the automorphism case, 13 the isomorphism case [12, 13].

Corollary 23.5 (Determination by monomials [12, 13, 14, 15]). *Two automorphisms of **Oz** whose extensions agree on every Conway monomial are equal. An **Oz**-preserving field automorphism of **No** fixing every monomial is the identity.*

Proof. Both extensions fix $\mathbb{R}$ and commute with the normal-form sum of every element. □

Example 23.6 (Constant-term preservation does not give preservation of **Oz** [13]). Choose a $\mathbb{Q}$ -linear $q : \mathbb{R} \rightarrow \mathbb{R}$ with $q(1) = 0$ and $q(\sqrt{2}) = 1$, by extending $\{1, \sqrt{2}\}$ to a Hamel basis, and put $\theta(\gamma) = q(\text{ct } \gamma)$, the constant term of the exponent’s own normal form. Then θ is additive, $\theta(1) = 0$ and $\theta(\sqrt{2}) = 1$, and $\Psi(t^\gamma) = t^\gamma \exp(\theta(\gamma)t)$ defines a strong $\mathbb{R}$ -linear 1-automorphism of **No** with inverse given by $-\theta$: it is the flow T^1 of proposition 4.9 with $\delta = 1$, which needs only $\theta(1) = 0$, not $\theta(C_1) = 0$. A term at exponent 0 in $\Psi(t^\gamma)$ requires $\gamma = -n$ with $n \in \mathbb{N}$, and $\theta(-n) = 0$; so $\text{ct} \circ \Psi = \text{ct}$ on monomials and, by strongness, everywhere. But

$$\Psi(t^{-\sqrt{2}}) = t^{-\sqrt{2}} \exp(-t)$$

has coefficient $\frac{1}{2}$ at the positive exponent $2 - \sqrt{2}$, so $\Psi(\mathbf{Oz}) \not\subseteq \mathbf{Oz}$. Constant-term preservation suffices for automatic strongness of isomorphisms, but not for preservation of **Oz**. The merge notes the agreement with theorem 4.1: here $\theta(C_1) \neq 0$, since $\sqrt{2} \in C_1$.

24 The whole stabilizer

Theorem 24.1 (Factorization of every pair automorphism [12, 13, 14, 15]). *Let $\Gamma = \mathbf{No}$, or a nonzero divisible set-sized ordered abelian group. Every field automorphism σ of $K_\Gamma = \mathbb{R}((t^\Gamma))$ with $\sigma(\mathcal{R}_\Gamma) = \mathcal{R}_\Gamma$ factors uniquely as*

$$\sigma = M_{\chi, \tau} \circ u, \quad u \in \mathcal{U}_{\mathcal{R}_\Gamma}(\Gamma),$$

where $\tau(\gamma) = v(\sigma(t^\gamma))$ is an order automorphism of Γ and $\chi(\gamma) = \text{lc}(\sigma(t^\gamma))$ is a positive character; u satisfies the equivalent conditions of [theorem 3.3](#), in particular $a_{u, \delta}(C_\delta) = 0$ for every $\delta > 0$. Conversely every such composition preserves $\mathcal{R}_\Gamma$. So the stabilizer is $\{M_{\chi, \tau}\} \ltimes \mathcal{U}_{\mathcal{R}_\Gamma}(\Gamma)$, and $\text{Aut}(\mathbf{Oz}) \cong \mathcal{U}_{\mathbf{Oz}} \rtimes \{M_{\chi, \tau}\}$.

Proof. By [theorem 23.2](#) or [corollary 23.4](#), σ is strong and fixes $\mathbb{R}$. It preserves the order, hence Archimedean equivalence, which is expressed with ordinary integers; so it maps the valuation ring onto itself and is valued in the sense of [section 5](#). Every character into $\mathbb{R}^\times$ of a divisible group is positive. [Theorem 5.1](#) now applies, with $k = \mathbb{R}$ and $\mathfrak{o} = \mathbb{Z}$. $\square$

This is [theorem 5.1](#) and the second sentence of [corollary 5.2](#) without the hypothesis of strongness. 12 proves the factorization for $\mathbf{No}$ and set-sized groups, 13 for $\mathbf{No}$, in the order $\sigma = U \circ M_{\chi, \tau}$ ([section 20.3](#)); 14 proves it for both, in the order $M \circ u$, reading the leading data from order and the fixed reals, so the existence of the factorization needs no strongness, and the composition law it states is the one after (5.1). All three, and 15, reprove the convex-support criterion: 12 by the least forbidden shift, which is source 10's route ([theorem 3.8](#)), and 13, 14 and 15 by a polynomial in the rational parameter q with infinitely many zeros, which is 04's route ([theorem 3.3](#)); those proofs are printed there. 13's nontrivial example, the shear $t^\gamma \mapsto t^\gamma \exp(q_1(\gamma)t)$ with $\omega^\omega \mapsto \omega^\omega \exp(-\omega^{-1})$, is [example 4.3](#).

Remark 24.2 (Admissibility is retained [12, 13, 14, 15]). [Theorem 24.1](#) classifies *existing* automorphisms. An arbitrary array of additive a_δ with $a_\delta(C_\delta) = 0$ is not thereby shown to define one: the monomial substitution must be summable on every well-ordered support, multiplicative and invertible with the same admissibility. Local convex vanishing is not proved sufficient, and no class-wide exponential–logarithm correspondence is used or established ([question 11.2](#)). The theorem removes a hypothesis on actual automorphisms; it licenses no unchecked infinite substitution.

Corollary 24.3 (Archimedean groups [12, 14, 15]). *For a nonzero divisible Archimedean set-sized Γ , every field automorphism of $\mathbb{R}((t^\Gamma))$ stabilizing $\mathcal{R}_\Gamma$ is a monomial map $M_{\chi, \tau}$.*

Proof. In [theorem 24.1](#), $u \in \mathcal{U}_{\mathcal{R}_\Gamma}(\Gamma) = \{\text{id}\}$ by [theorem 4.13](#). $\square$

[Theorem 4.13](#) gave this among strong maps; automatic strongness removes the qualification. No claim is made that omnific-preserving automorphisms are trivial: the monomial factor, and in higher rank the factor $\mathcal{U}$, can be large [12]. For $\Gamma = \mathbb{Q}$ this is the form $t^q \mapsto c^q t^{\lambda q}$ after [theorem 5.1](#), and for $\Gamma = \mathbb{R}$ the character may be discontinuous ([remark 36.3](#)) [14].

Corollary 24.4 (What an automorphism of $\mathbf{Oz}$ does inside Π [merge]). *Every ring automorphism σ of $\mathbf{Oz}$ satisfies $\text{ct} \circ \sigma = \text{ct}$; its extension fixes $\mathbb{R}$ and is $M_{\chi, \tau} \circ u$, where τ is an order automorphism of $\mathbf{No}$, χ a positive character and $u \in \mathcal{U}_{\mathbf{Oz}}$. Thus on Π it acts first by u , then by the exponent automorphism and positive monomial rescaling $M_{\chi, \tau}$. Here u is a strong leading-term-fixing automorphism whose logarithmic character has, for every $g \neq 0$, support in H_g , the elements infinitesimal relative to g ([theorem 3.3](#), condition (v)); every such composition occurs, and every monomial image stays in the signed Archimedean block of its exponent up to the action of τ ([proposition 3.10](#)).*

Proof. [Theorem 24.1](#) with $\Gamma = \mathbf{No}$, and [theorem 3.3](#) and [proposition 3.10](#) for u . $\square$

With [theorems 7.6, 7.7](#) and [9.1](#) this describes the automorphisms of $\mathbf{Oz}$ inside Π as fully as [part I](#) describes the strong ones; the admissibility of logarithmic data remains [question 11.2](#). Source 10's [remark 3.9](#) is subsumed: its monomial condition now holds for every automorphism because every automorphism is strong.

25 Constant-term adjoints and a proper-class obstruction

In this section Γ, Δ are *sets*, $K = k((t^\Gamma))$ and $L = k((t^\Delta))$ share the coefficient field k , and linearity is over k , not over K . This is the scalar k -linear dual of a full Hahn field; the three-duals report (`docs/surcomplex/three-duals-of-hahn-vector-spaces`, `duals:`) studies K -linear maps on vector-valued Hahn spaces and topological duals, a different convention, and no identification with its functionals is asserted [12, 13]. A k -linear functional $\lambda : K \rightarrow k$ is *strong* if it sends every summable family to a finitely supported scalar family and commutes with the finite sum. T denotes a k -linear map $K \rightarrow L$ in this section only.

Proposition 25.1 (Scalar representability at set size [12, 13, 14, 15]). *Every strong k -linear functional λ on K has a unique representation $\lambda(x) = \text{ct}(ax)$ with $a \in K$; conversely every such functional is strong.*

Proof. Put $a_g = \lambda(t^{-g})$, a function on the set Γ . If its support contained $g_0 > g_1 > \dots$, the monomials t^{-g_n} , with increasing exponents, would form a summable family with all images nonzero; so $a = \sum a_g t^g \in K$. Strongness on the normal-form family of x gives $\lambda(x) = \sum_g [t^g]x a_{-g} = \text{ct}(ax)$, a finite sum; uniqueness is nondegeneracy, and the converse is (21.2). 13 argues on the set $\{\gamma : \lambda(t^\gamma) \neq 0\}$, which meets every well-ordered set finitely and is therefore reverse well ordered, by dependent choice. $\square$

Theorem 25.2 (Adjoint characterization [12, 13, 14, 15]). *A k -linear map $T : K \rightarrow L$ is strongly k -linear if and only if it has a constant-term adjoint $T^\dagger : L \rightarrow K$,*

$$\langle Tx, y \rangle_0 = \langle x, T^\dagger y \rangle_0 \quad (x \in K, y \in L). \quad (25.1)$$

The adjoint is unique, k -linear and strong; $T^{\dagger\dagger} = T$, $(ST)^\dagger = T^\dagger S^\dagger$ and $(T + T')^\dagger = T^\dagger + T'^\dagger$. Multiplication by a fixed series is self-adjoint, and a coefficient-fixing, constant-term-preserving field isomorphism has adjoint its inverse.

Proof. If T is strong, $x \mapsto \text{ct}(T(x)y)$ is a strong functional, represented by a unique $T^\dagger y$ ([proposition 25.1](#)); uniqueness gives linearity. Conversely, given $T^\dagger$ and a summable (x_j) , the scalars $\langle Tx_j, y \rangle_0 = \langle x_j, T^\dagger y \rangle_0$ vanish for almost all j , so (Tx_j) is summable by [theorem 21.4](#), and (21.2) on both sides of (25.1) gives $T(\sum x_j) = \sum Tx_j$. Symmetry of the pairing makes T the adjoint of $T^\dagger$, so $T^\dagger$ is strong. The identities are substitutions; for an isomorphism F with $\text{ct} \circ F = \text{ct}$, $\text{ct}(F(x)y) = \text{ct}(xF^{-1}(y))$. $\square$

The direction “adjoint implies strong” uses only [theorem 21.4](#) and holds for $\mathbf{No}$ as well; the converse genuinely needs set-sized groups ([theorem 25.6](#)) [13]. The theorem can be read as a concrete transpose statement for the linearized finiteness space; transpose duality in finiteness spaces is not claimed new [12].

Theorem 25.3 (A canonical linear retraction [12]). *Let $\phi : K \hookrightarrow L$ be a strong k -linear field embedding and $e_\phi = \phi^\dagger(1)$. Then $e_\phi \neq 0$ and $\phi^\dagger(\phi(a)y) = a\phi^\dagger(y)$ for $a \in K$, $y \in L$. The map*

$$\text{ret}_\phi(y) = e_\phi^{-1}\phi^\dagger(y)$$

is a strong K -linear retraction (L a K -space through ϕ), with $\text{ret}_\phi \circ \phi = \text{id}_K$. If $\text{ct} \circ \phi = \text{ct}$ then $e_\phi = 1$ and $\text{ret}_\phi = \phi^\dagger$. If ϕ is not onto, ret_ϕ is not multiplicative.

Proof. For $b \in K$, $\text{ct}(b\phi^\dagger(\phi(a)y)) = \text{ct}(\phi(ba)y) = \text{ct}(ba\phi^\dagger(y))$, which gives the projection formula by nondegeneracy; $\text{ct}(e_\phi) = \text{ct}(\phi(1)) = 1$. With $y = 1$, $\phi^\dagger(\phi(a)) = ae_\phi$. Multiplication by a fixed series is strong, so ret_ϕ is strong and K -linear. If $\text{ct} \circ \phi = \text{ct}$, then $\text{ct}(xe_\phi) = \text{ct}(x)$ for all x and $e_\phi = 1$. A multiplicative unital retraction would be an injective field homomorphism with $\text{ret}_\phi(y - \phi(\text{ret}_\phi y)) = 0$, forcing $y = \phi(\text{ret}_\phi y)$ for all y . $\square$

13 proves the case $\text{ct} \circ \phi = \text{ct}$ with ϕ coefficient-fixing (its residue splitting); it adds that $\ker \phi^\dagger$ is the annihilator of $\phi(K)$ for the pairing, and that $L = \phi(K) \oplus \ker \phi^\dagger$ as $\phi(K)$ -spaces. More generally a proper field extension admits no unital ring retraction fixing the subfield [13].

Corollary 25.4 (Tower law and complement [12]). *For composable constant-term-preserving strong k -linear field embeddings $K \xrightarrow{\phi} L \xrightarrow{\psi} P$, $\text{ret}_{\psi\phi} = \text{ret}_\phi \circ \text{ret}_\psi$. For one such embedding $L = \phi(K) \oplus \ker \text{ret}_\phi$ as K -spaces, and $\phi \circ \text{ret}_\phi$ is a strong projection onto $\phi(K)$.*

Proof. Adjoints reverse composition and all normalizing elements are 1; the complement follows from $\text{ret}_\phi \circ \phi = \text{id}$. $\square$

ret_ϕ is not a field retraction and not a positive conditional expectation; neither multiplicativity nor positivity is part of its conclusion, and it is canonical only relative to the two pairings and to ϕ [12].

Example 25.5 (Monomial embeddings [12, 13]). For an ordered group embedding $\tau : \Gamma \hookrightarrow \Delta$ and a character $\chi : \Gamma \rightarrow k^\times$, the strong embedding $M_{\chi,\tau}$ preserves constant terms and

$$M_{\chi,\tau}^\dagger \left(\sum_{\eta} y_{\eta} t^{\eta} \right) = \sum_{g \in \Gamma} \chi(g)^{-1} y_{\tau(g)} t^g,$$

supported in the preimage under an order embedding of a well-ordered set; compare constant coefficients, using $\chi(-g) = \chi(g)^{-1}$. The same formula works for class groups with set supports, by replacement on the target support [13]. If $\eta \notin \tau(\Gamma)$, then $M_{\chi,\tau}^\dagger(t^\eta) = M_{\chi,\tau}^\dagger(t^{-\eta}) = 0$ while $M_{\chi,\tau}^\dagger(1) = 1$, so the retraction is not multiplicative on two monomials. For instance $\Gamma = \Delta = \mathbb{Z}$, $\tau(g) = 2g$, $\chi(g) = 2^g$ over $k = \mathbb{Q}$ give a weighted even-coefficient extraction [12].

For a constant-term-compatible embedding ϕ the source tests show only that $\text{ct}(\phi(h)\phi(x_j))$ is finitely supported in j for each $h \in K$; detection in L needs all $y \in L$. An adjoint supplies a source representative of every target test, so the question is whether all target residue functionals can be pulled back; this locates the remaining embedding questions without answering them [13] (questions 32.2 and 32.3).

Theorem 25.6 (A strong functional with no representing surreal [12, 13, 14, 15]). *For $x \in \mathbf{No}$ put*

$$\lambda_{\mathbf{On}}(x) = \sum_{\alpha \in \mathbf{On}} [t^{-\alpha}]x. \quad (25.2)$$

For each x only finitely many terms are nonzero. $\lambda_{\mathbf{On}}$ is a strongly $\mathbb{R}$ -linear functional, but no $a \in \mathbf{No}$ satisfies $\lambda_{\mathbf{On}}(x) = \text{ct}(ax)$ for all x . Hence the strong $\mathbb{R}$ -linear map $x \mapsto \lambda_{\mathbf{On}}(x) \cdot 1$ of $\mathbf{No}$ has no constant-term adjoint.

Proof. A well-ordered set of exponents meets $-\mathbf{On}$ finitely, since an infinite intersection would give a strictly decreasing sequence of ordinals. The same applies to the total support of a summable family, and only finitely many members contribute at each of those exponents; this gives strongness. As $\lambda_{\mathbf{On}}(t^{-\alpha}) = 1$ for every ordinal α , a representing a would have $[t^\alpha]a = 1$ for all α , a proper-class support. An adjoint evaluated at 1 would be such an a . $\square$

The definition is not a proper-class Hahn sum: for each input one first finds the finite subset of its support lying in $-\mathbf{On}$ and adds finitely many reals; the class only selects coefficients [12]. There is no conflict with [theorem 22.1](#): there the inverse of an isomorphism supplies each needed test element, and no arbitrary strong functional is represented. So the isomorphism theorem extends to $\mathbf{No}$ while the self-duality of [proposition 25.1](#) and [theorem 25.2](#) does not [12, 13].

26 The bottom-gap classification of exact-monomial embeddings

This section and the next three are source 11's, with the embeddings of 12 and 13 as instances. Automorphisms preserving $\mathbf{Oz}$ fix $\mathbb{R}$ because $\mathbb{R}$ is parameter-free definable in the pair (`odg:def:cor:internal`, [theorem 8.2](#)); the definition uses a universally quantified multiplier condition, which a proper embedding need not preserve [11, 12], as the Diophantine report already cautions after `odg:def:thm:autreal`. The existential part survives: by `odg:def:thm:ideal` and `odg:def:thm:homct`, every unital ring homomorphism $\mathcal{R}_\Gamma \rightarrow \mathcal{R}_\Delta$ maps Π_Γ into Π_Δ and preserves the constant term [11]. Throughout, Γ and Δ are nonzero divisible ordered abelian groups (sets, or $\mathbf{No}$ with set supports), so K_Γ and K_Δ are real closed, and $\tau : \Gamma \hookrightarrow \Delta$ is an ordered additive embedding.

Definition 26.1 (Bottom gap [11]). The *bottom gap* of τ is

$$\mathbf{G}_\tau = \{c \in \Delta : n|c| < h \text{ for every } h \in \tau(\Gamma)^{>0} \text{ and every } n \geq 1\}.$$

Since $\tau(\Gamma)$ is divisible, this is $\{c : |c| < h \text{ for every } h \in \tau(\Gamma)^{>0}\}$.

The word “gap” refers to a convex subgroup of the value group, not to an omitted cut of the field [11].

Lemma 26.2 (Intrinsic description [11]). $\mathbf{G}_\tau$ is the largest convex subgroup of Δ meeting $\tau(\Gamma)$ only in 0; it is divisible, and $\mathbf{G}_\tau = 0$ if and only if $\tau(\Gamma)^{>0}$ is coinitial in $\Delta^{>0}$. The sum $\tau(\Gamma) + \mathbf{G}_\tau$ is direct and ordered lexicographically with $\tau(\Gamma)$ outer: $h < h'$ in $\tau(\Gamma)$ and $c, c' \in \mathbf{G}_\tau$ give $h + c < h' + c'$. Hence a well-ordered family of well-ordered blocks lying in distinct cosets $h + \mathbf{G}_\tau$ has well-ordered union.

Proof. For $c, c' \in \mathbf{G}_\tau$ and $0 < h \in \tau(\Gamma)$, $|c + c'| < h/2 + h/2$; symmetry and convexity are clear, and $0 < c \in \tau(\Gamma)$ fails the condition at $h = c$. If D is convex with $D \cap \tau(\Gamma) = 0$, then $h \leq n|c|$ with $c \in D$ would put $h \in D$, so $D \subseteq \mathbf{G}_\tau$. Convex subgroups of divisible groups are divisible. If $\tau(\Gamma)^{>0}$ is coinitial no $c > 0$ lies in $\mathbf{G}_\tau$; otherwise some $c > 0$ lies below every $h \in \tau(\Gamma)^{>0}$, hence below every h/n . For the blocks, $h' - h > |c - c'|$; a nonempty subset of a union of blocks has a least outer coordinate and a least element in that block. $\square$

Theorem 26.3 (Bottom-gap classification [11]). *There is a bijection between*

- (i) *strongly additive field embeddings $J : K_\Gamma \hookrightarrow K_\Delta$ with $J(t^\gamma) = t^{\tau(\gamma)}$ for all γ and $J(\mathcal{R}_\Gamma) \subseteq \mathcal{R}_\Delta$; and*
- (ii) *ordered field embeddings $\rho : \mathbb{R} \hookrightarrow \mathbb{R}((t^{\mathbf{G}_\tau}))$,*

given by $\rho = J|_{\mathbb{R}}$ and

$$J_{\tau, \rho} \left(\sum_{\gamma} r_{\gamma} t^{\gamma} \right) = \sum_{\gamma} \rho(r_{\gamma}) t^{\tau(\gamma)}. \quad (26.1)$$

Every $J = J_{\tau, \rho}$ satisfies

$$\begin{aligned} v(Jx) &= \tau(v(x)), & \text{lc}(Jx) &= \text{lc}(x), & \text{ct}(Jx) &= \text{ct}(x), \\ J^{-1}(\mathcal{R}_\Delta) &= \mathcal{R}_\Gamma, & J^{-1}(\Pi_\Delta) &= \Pi_\Gamma. \end{aligned} \quad (26.2)$$

Without strongness, every field embedding with $J(t^\gamma) = t^{\tau(\gamma)}$ and $J(\mathcal{R}_\Gamma) \subseteq \mathcal{R}_\Delta$ still has $J(\mathbb{R}) \subseteq \mathbb{R}((t^{\mathbb{G}_\tau}))$.

Here $\text{ct}(Jx) = \text{ct}(x)$ is an equality of coefficient extractions, not multiplicativity of ct .

Proof. Step 1: residues. An ordered embedding of $\mathbb{R}$ into a real Hahn field fixes $\mathbb{Q}$; a rational bound on $|r|$ bounds $|J(r)|$, so $J(r)$ has nonnegative support, and its constant term lies between all rational bounds of r . So $J(r) = r + (\text{a series with positive support})$, with first exponent 0 and first coefficient r when $r \neq 0$. (An embedding of a real closed field into an ordered field preserves positivity, positive elements being squares; so every field embedding here is ordered.)

Step 2: the gap is necessary. For $\gamma > 0$ and $r \in \mathbb{R}$, $rt^{-\gamma} \in \Pi_\Gamma$, so $J(r)t^{-\tau(\gamma)} \in \Pi_\Delta$ by the existential test. Hence every $c \in \text{supp } J(r)$ has $c < \tau(\gamma)$ for all $\gamma > 0$; with $c \geq 0$ and divisibility, $c \in \mathbb{G}_\tau$. No strongness is used. If J is strong, expanding x into its summable family of terms gives (26.1), hence uniqueness.

Step 3: every ρ is admissible. For a support S , the blocks $\tau(\gamma) + \text{supp } \rho(r_\gamma)$, $\gamma \in S$, lie in distinct cosets of $\mathbb{G}_\tau$ in the order of S (lemma 26.2), so (26.1) is a Hahn series. For a summable family (x_j) , the outer exponents form a well-ordered set, finitely many j contribute at each outer exponent, and each block is a finite union of well-ordered supports; so the images are summable, and additivity of ρ gives strong additivity.

Step 4: field operations. $J(1) = 1$. Products of terms map to $\rho(r)t^{\tau(\gamma)}\rho(s)t^{\tau(\eta)} = \rho(rs)t^{\tau(\gamma+\eta)}$, and the product family is summable (the support principle of section 2); regrouping by Step 3 gives multiplicativity. The first block of $J(x)$ starts at $\tau(v(x))$ with coefficient $\text{lc}(x)$, which gives injectivity, the order and the valuation formula.

Step 5: preservation and reflection. For $\gamma < 0$ the block at $\tau(\gamma)$ is negative, for $\gamma > 0$ positive; the zero block is $\rho(r_0)$ with constant term r_0 ; blocks cannot cancel. So $\text{ct}(Jx) = \text{ct}(x)$. If $x \in \mathcal{R}_\Gamma$ its zero block is the integer r_0 and all other blocks are negative. Conversely, if $J(x) \in \mathcal{R}_\Delta$, every positive block is absent and $r_0 \in \mathbb{Z}$, so $x \in \mathcal{R}_\Gamma$; for Π also $r_0 = 0$. $\square$

So these embeddings automatically *reflect* the integer part and Π . The classification is inside the exact-monomial category. It does not cover maps $t^\gamma \mapsto c_\gamma t^{\tau(\gamma)}(1 + u_\gamma)$ with characters or unit corrections, which already occur among the automorphisms of part I, and it does not assert that omnific-preserving embeddings are strong; beyond the category it gives only the necessary coefficient restriction of Step 2 [11] (questions 32.2 and 32.5).

Corollary 26.4 (Surjectivity [11]). *$J_{\tau, \rho}$ is onto if and only if τ is onto. Then $\mathbb{G}_\tau = 0$, $\rho = \text{id}_\mathbb{R}$ and $J = M_{1, \tau}$.*

Proof. The valuation formula forces τ onto. Then $\mathbb{G}_\tau = 0$ by lemma 26.2, and an ordered embedding $\mathbb{R} \rightarrow \mathbb{R}$ is the identity; reindexing by τ^{-1} inverts $M_{1, \tau}$. $\square$

This agrees with theorem 24.1: an automorphism of the pair that is exact-monomial is $M_{1, \tau}$.

26.1 Taylor sections and the rigidity boundary

A $\mathbb{Q}$ -derivation $d : \mathbb{R} \rightarrow \mathbb{R}$ is additive with $d(rs) = r d(s) + s d(r)$; it need not be continuous. Given a transcendence basis B of $\mathbb{R}/\mathbb{Q}$, arbitrary values $d(b)$, $b \in B$, extend uniquely to $\mathbb{Q}(B)$ and then through the algebraic extension $\mathbb{R}/\mathbb{Q}(B)$: if $p(X) = \sum_j c_j X^j$ is the minimal polynomial of a , then $d(a) = -\sum_j d(c_j) a^j / p'(a)$, with $p'(a) \neq 0$ in characteristic zero, and the extensions agree on finite subextensions. So for a transcendental b there is d with $d(b) = 1$, and every such d vanishes on the real algebraic numbers; the choice uses a transcendence basis of the set $\mathbb{R}/\mathbb{Q}$ and is not computable [11, 12, 13]. This is also the derivation used in `saut:thm:coeffflow`.

Lemma 26.5 (Taylor section [11, 12, 13]). *For $\delta > 0$ in an ordered group and a $\mathbb{Q}$ -derivation d of $\mathbb{R}$,*

$$\rho_{d,\delta}(r) = \sum_{n \geq 0} \frac{d^n(r)}{n!} t^{n\delta} \quad (26.3)$$

is an ordered field embedding of $\mathbb{R}$ into $\mathbb{R}((t^{\mathbb{Z}\delta}))$, with constant term r ; if $d(b) = 1$ then $\rho_{d,\delta}(b) = b + t^\delta$.

Proof. The iterated Leibniz rule $d^n(rs) = \sum_j \binom{n}{j} d^j(r) d^{n-j}(s)$ gives multiplicativity coefficient-wise; the constant term r gives injectivity and the sign; $d^2(b) = d(1) = 0$. $\square$

The ambient version is `saut:thm:coeffflow` of the surcomplex automorphism report, restricted to real coefficients (a derivation of $\mathbb{R}$ extends to $\mathbb{C}$ with $d(i) = 0$, and then the map preserves **No**):

$$\mathcal{T}_{d,\delta} \left(\sum_{\lambda} r_{\lambda} t^{\lambda} \right) = \sum_{\lambda, n \geq 0} \frac{d^n(r_{\lambda})}{n!} t^{\lambda + n\delta} \quad (26.4)$$

is a strong automorphism of K_{Δ} fixing every monomial, preserving valuations and leading coefficients, with inverse $\mathcal{T}_{-d,\delta}$ and $v(\mathcal{T}_{d,\delta}(x) - x) \geq v(x) + \delta$. 11 reproves it and credits the surcomplex report; 12 and 13 use its Taylor ingredient without that credit. On its own it never preserves **Oz** (the remark after [example 4.12](#)): with $\delta = 1$, $\Delta = \mathbf{No}$ and $d(b) = 1$ it sends $\frac{b}{2}\omega = \frac{b}{2}t^{-1}$ to $\frac{b}{2}\omega + \frac{1}{2} \notin \mathbf{Oz}$ [11]. What changes below is its interaction with the image of a proper exponent embedding.

Theorem 26.6 (Exact Taylor admissibility [11]). *Let $\delta > 0$ and let $d \neq 0$ be a derivation of $\mathbb{R}$. The strong exact-monomial embedding $\mathcal{T}_{d,\delta} \circ M_{1,\tau}$ preserves $\mathcal{R}_{\Gamma}$ if and only if $\delta \in \mathbf{G}_{\tau}$, that is $n\delta < \tau(\gamma)$ for all $\gamma > 0$ and $n \geq 1$. It is then $J_{\tau,\rho_{d,\delta}}$ and also reflects $\mathcal{R}_{\Gamma}$.*

Proof. $\mathcal{T}_{d,\delta} \circ M_{1,\tau}$ is $\sum r_{\gamma} t^{\gamma} \mapsto \sum_{\gamma} \rho_{d,\delta}(r_{\gamma}) t^{\tau(\gamma)}$. If $\delta \in \mathbf{G}_{\tau}$, then $\rho_{d,\delta}$ takes values in $\mathbb{R}((t^{\mathbf{G}_{\tau}}))$ and [theorem 26.3](#) applies. Conversely, for r with $d(r) \neq 0$ the coefficient of $\rho_{d,\delta}(r)$ at δ is nonzero, and Step 2 of that theorem forces $\delta \in \mathbf{G}_{\tau}$. $\square$

Corollary 26.7 (Coinitiality is exactly coefficient rigidity [11]). *For fixed τ , every embedding in [theorem 26.3](#) fixes $\mathbb{R}$ pointwise if and only if $\tau(\Gamma)^{>0}$ is coinitial in $\Delta^{>0}$. Otherwise some such embedding sends a chosen transcendental real b to $b + t^{\delta}$ with $\delta > 0$.*

Proof. Coinitial means $\mathbf{G}_{\tau} = 0$, and an ordered embedding $\mathbb{R} \rightarrow \mathbb{R}$ is the identity. Otherwise take $0 < \delta \in \mathbf{G}_{\tau}$ and $d(b) = 1$ in [theorem 26.6](#). $\square$

Example 26.8 (A set-sized model [11]). Let $\Gamma = \mathbb{Q}$, $\Delta = \mathbb{Q} \oplus \mathbb{Q}$ ordered lexicographically with the first coordinate dominant, and $\tau(q) = (q, 0)$. Then $\mathbf{G}_{\tau} = \{0\} \oplus \mathbb{Q}$ and $\delta = (0, 1)$ is admissible; for $q > 0$, $\rho_{d,\delta}(r) t^{(-q,0)} = \sum_n \frac{d^n(r)}{n!} t^{(-q,n)}$ has only negative exponents. No proper class is involved.

27 Explicit proper embeddings of $(\mathbf{No}, \mathbf{Oz})$

27.1 Two levels of normal-form substitution

At the *inner* level write an exponent $g \in \mathbf{No}$ in its large- ω Conway normal form $g = \sum_{a \in S} r_a \omega^a$, S reverse well ordered. For an order embedding $\mu : \mathbf{No} \hookrightarrow \mathbf{No}$ put

$$\iota_\mu(g) = \sum_{a \in S} r_a \omega^{\mu(a)}. \quad (27.1)$$

Lemma 27.1 (Inner lift [11]). *ι_μ is an $\mathbb{R}$ -linear ordered embedding of $(\mathbf{No}, +)$ into itself, with the same leading coefficient; its image is the class of normal forms supported in $\mu(\mathbf{No})$, and $\iota_\mu \iota_{\mu'} = \iota_{\mu\mu'}$.*

Proof. An order embedding preserves reverse well-ordering and identifies no exponents; addition is coefficientwise. Reindex a normal form supported in $\mu(\mathbf{No})$ by μ^{-1} for the image. $\square$

At the *outer* level ι_μ is an exponent embedding, and M_{1, ι_μ} is a strong exact-monomial field embedding fixing $\mathbb{R}$ (theorem 26.3 with $\rho = \text{id}$). ι_μ is only additive: it is not a field map, and μ need not be additive. The three sources use three order isomorphisms:

$$\mu_{11}(a) = \begin{cases} (1-a)^{-1}, & a < 0, \\ a+1, & a \geq 0, \end{cases} \quad \mu_{13}(a) = a + \sqrt{a^2 + 1}, \quad \mu_{12}(a) = 1 + a + \sqrt{a^2 + 1}. \quad (27.2)$$

μ_{11} maps $\mathbf{No}$ onto $\mathbf{No}^{>0}$, its branches covering $(0, 1)$ and $[1, \infty)$. μ_{13} maps $\mathbf{No}$ onto $\mathbf{No}^{>0}$ with inverse $y \mapsto (y - y^{-1})/2$, since $(\sqrt{a^2 + 1} + a)(\sqrt{a^2 + 1} - a) = 1$ and $y \mapsto (y - y^{-1})/2$ is strictly increasing. So $\mu_{12} = 1 + \mu_{13}$ maps $\mathbf{No}$ onto $(1, \infty)$. They use only the ordered-field operations of $\mathbf{No}$ [11, 12, 13].

Lemma 27.2 (The three value images and their gaps). *The images $\iota_{\mu_{11}}(\mathbf{No}) = \iota_{\mu_{13}}(\mathbf{No})$ are the additive group of purely infinite surreals, the normal forms with all exponents positive; $\iota_{\mu_{12}}(\mathbf{No})$ consists of the normal forms with all exponents greater than 1. The gaps are*

$$\mathbf{G}_{\iota_{\mu_{11}}} = \mathbf{G}_{\iota_{\mu_{13}}} = C_1 \text{ [11, 13]}, \quad \mathbf{G}_{\iota_{\mu_{12}}} = C_\omega \text{ [merge]},$$

the finite surreals and the surreals bounded by an ordinary multiple of ω . All three images are cofinal in $\mathbf{No}$ and none of their positive parts is coinital.

Proof. The images follow from lemma 27.1. A positive purely infinite surreal exceeds every real, hence every ordinary multiple of a finite surreal; so $C_1 \subseteq \mathbf{G}$. An infinite c has leading exponent $a > 0$, and $\omega^{a/2}$ is a positive image element below $|c|$; so $c \notin \mathbf{G}$. For μ_{12} : a positive image element has leading term $r\omega^a$ with $r > 0$, $a > 1$, so it exceeds $n\omega$ for every n , and $C_\omega \subseteq \mathbf{G}$. If $c \notin C_\omega$, its leading exponent a exceeds 1, and $\omega^{(1+a)/2}$ is an image element below $|c|$. Arbitrarily large exponents give cofinality; no positive image element is at most 1. $\square$

The purely infinite surreals here form a subgroup of the *value group*; it is the same set as the ideal Π of $\mathbf{No}$, but it is used additively and is not a convex subgroup [11, 13].

27.2 One map, three instances

Theorem 27.3 (Proper embeddings moving a real [11, 12, 13]). *Let b be a transcendental real, d a derivation of $\mathbb{R}$ with $d(b) = 1$, and $\mu \in \{\mu_{11}, \mu_{12}, \mu_{13}\}$. Then*

$$J_{\mu, d}\left(\sum_{\gamma} r_{\gamma} t^{\gamma}\right) = \sum_{\gamma} \sum_{n \geq 0} \frac{d^n(r_{\gamma})}{n!} t^{\iota_{\mu}(\gamma) + n} \quad (27.3)$$

is $J_{\iota_\mu, \rho_{d,1}} = \mathcal{T}_{d,1} \circ M_{1, \iota_\mu}$. It is a proper, strongly additive, exact-monomial ordered field embedding $\mathbf{No} \hookrightarrow \mathbf{No}$ with

$$v(J_{\mu,d}x) = \iota_\mu(v(x)), \quad \text{ct} \circ J_{\mu,d} = \text{ct}, \quad J_{\mu,d}^{-1}(\mathbf{Oz}) = \mathbf{Oz}, \quad J_{\mu,d}(b) = b + \omega^{-1},$$

and it fixes every real algebraic number. Moreover $J_{\mu,d}(\omega) = \omega^\omega$ for $\mu = \mu_{11}, \mu_{13}$ and $J_{\mu_{12},d}(\omega) = \omega^{\omega^2}$. Its restriction is a proper ring embedding $\mathbf{Oz} \hookrightarrow \mathbf{Oz}$; it is not $\mathbb{R}$ -linear, and no automorphism of $(\mathbf{No}, \mathbf{Oz})$ agrees with it on b .

Proof. $1 \in \mathbf{G}_{\iota_\mu}$ by lemma 27.2, so theorems 26.3 and 26.6 give the embedding, strongness, valuation, constant term and reflection. $\rho_{d,1}(b) = b + t$ and $t = \omega^{-1}$; d kills the real algebraic numbers. $\iota_\mu(1) = \omega^{\mu(0)}$ with $\mu_{11}(0) = \mu_{13}(0) = 1$ and $\mu_{12}(0) = 2$, and $J(\omega) = J(t^{-1}) = t^{-\iota_\mu(1)}$. The value $1 = v(t)$ is not in $\iota_\mu(\mathbf{No})$, so $t \notin J(\mathbf{No})$; if the restriction to $\mathbf{Oz}$ were onto, extension to fractions (odg:thm:fractions) would make J onto. Every $\mathbf{Oz}$ -preserving automorphism fixes b (corollary 5.2), and the restriction to $\mathbf{Oz}$ determines the fraction-field map. $\square$

So the rigidity of automorphisms does not pass to embeddings: $(\mathbf{No}, +, \cdot, \mathbf{Oz})$ and $\mathbf{Oz}$ have proper self-embeddings, and an ordered-field embedding of $\mathbf{No}$ preserving and reflecting $\mathbf{Oz}$ need not preserve $\mathbb{R}$ setwise, even when it is strongly additive and preserves constant terms exactly [12]. 11 proves the theorem for μ_{11} through its classification; 12 (for μ_{12}) and 13 (for μ_{13}) give the same block argument directly. The three maps differ only in the order isomorphism: 11's and 13's have the same value image and send $\omega \mapsto \omega^\omega$. For μ_{12} , $\mathbf{G} = C_\omega$, so even $\delta = \omega$ would be admissible (lemma 27.2). For example

$$J_{\mu,d}(bt^g) = bt^{\iota_\mu(g)} + t^{\iota_\mu(g)+1} \quad (g < 0), \quad J_{\mu_{11},d}(b\omega) = b\omega^\omega + \omega^{\omega-1} \in \mathbf{Oz},$$

both terms purely infinite although $J(b)$ is not real; and $J(b^2) = b^2 + 2bt + t^2$, $J(b^{-1}) = \sum_n (-1)^n b^{-n-1} t^n$, $J(t^g) = t^{\iota_\mu(g)}$ [11, 12]. The ordinary Taylor section alone is not a map on arbitrary normal forms, since supports could meet its integer shifts; the proper exponent embedding separates the scales [12, 13].

Corollary 27.4 (Image, reals and a missing test [12]; for all three [merge]). *For each $J = J_{\mu,d}: J(\mathbf{No}) \cap \mathbb{R} = \ker d$, and $\text{ct}(tJ(x)) = 0$ for every $x \in \mathbf{No}$, although $t \neq 0$. So the image does not supply all constant-term tests.*

Proof. If $J(x)$ is constant, every block at $g \neq 0$ vanishes, so $x = r$ is real, and $\rho_{d,1}(r)$ is constant exactly when $d(r) = 0$ (see also proposition 29.2). An exponent $\iota_\mu(g) + n = -1$ would make $\iota_\mu(g)$ a negative integer; but $\iota_\mu(g)$ is 0 or of absolute value above every integer (lemma 27.2). So no image has a term at -1 . $\square$

12 proves the corollary for μ_{12} ; the merge notes that its proof uses only that nonzero values of ι_μ are infinite, true for all three. The annihilator is the missing detector in the isomorphism proof: tests from the image cannot distinguish t from 0, whereas the ambient test t^{-1} does [12].

Corollary 27.5 (Failure of pair homogeneity [13]). *The ordered embedding $\rho_{d,1}: \mathbb{R} \rightarrow \mathbf{No}$ is an embedding of the induced pairs $(\mathbb{R}, \mathbb{Z}) \rightarrow (\mathbf{No}, \mathbf{Oz})$ that extends to the proper pair embedding $J_{\mu,d}$ but to no automorphism of $(\mathbf{No}, \mathbf{Oz})$. So $(\mathbf{No}, \mathbf{Oz})$ is not strongly homogeneous for embeddings of set-sized induced substructures.*

Proof. Restrict the reflection property of theorem 27.3 to $\mathbb{R}$; every $\mathbf{Oz}$ -preserving automorphism fixes b , whereas $\rho_{d,1}(b) = b + t$. $\square$

“Strongly homogeneous” means extension of isomorphisms between set-sized induced substructures, not extension of elementary partial maps in a saturated expansion. The pure ordered field $\mathbf{No}$ has this extension property for set-sized ordered subfields [15, Remark 2.10], so $\rho_{d,1}$ does extend to an ordered-field automorphism of $\mathbf{No}$, and every such extension fails to preserve $\mathbf{Oz}$; the obstruction lies in the integer part [13]. The map also satisfies $\text{ct} \circ J = \text{ct}$ but not $J \circ \text{ct} = \text{ct}$ nor $J \circ \text{ct} = \text{ct} \circ J$: for b these compare b with $b + t$. It violates no hypothesis of [theorem 22.1](#), since it is not onto [13]. It does not commute with the omega-map: $J(\omega^1) = \omega^\omega$ for μ_{11} whereas $\omega^{J(1)} = \omega$; no question of [15] on the omega-map is addressed [11].

28 Closed images and the two ends of the value group

Use the valuation topology with basic neighbourhoods $\mathcal{V}_\lambda = \{x : v(x) > \lambda\}$. On a real Hahn field with nonzero value group it is the order topology; for $\mathbf{No}$ it is the fine topology. No countable basis is assumed, and for proper classes the topological statements are local conditions on balls; no object containing all open subclasses is formed [11].

Theorem 28.1 (Closed images [11]). *Every image $F = J_{\tau,\rho}(K_\Gamma)$ of [theorem 26.3](#) is closed in K_Δ ; if J is proper, F is nowhere dense.*

Proof. Put $\mathbf{G} = \mathbf{G}_\tau$ and $D = \tau(\Gamma) + \mathbf{G}$; image series are supported in D . If x has a nonzero coefficient at $\lambda \notin D$, then $v(x - y) \leq \lambda$ for all $y \in F$. If $\text{supp } x \subseteq D$, split $x = \sum_{h \in S} t^h x_h$ with $x_h \in \mathbb{R}((t^\mathbf{G}))$ by the unique decomposition; the projection $D \rightarrow \tau(\Gamma)$ is nondecreasing, so S is well ordered. Put $r_h = \text{ct}(x_h)$. If $x_h = \rho(r_h)$ for all h , then $x = J_{\tau,\rho}(\sum_h r_h t^{\tau^{-1}(h)}) \in F$. Otherwise pick h and $c \in \mathbf{G}$ where $x_h - \rho(r_h)$ has a nonzero coefficient, and put $L = \max\{h, h + c\}$. The block of any $y \in F$ at h is $\rho(s)$ for one $s \in \mathbb{R}$ (possibly 0); if $s \neq r_h$ then $x - y$ has a nonzero coefficient at h , and if $s = r_h$ at $h + c$. So $x + \mathcal{V}_L$ misses F , and F is closed. A subfield containing a ball $\mathcal{V}_L$ contains every $z \neq 0$, as $z = (zt^\gamma)/t^\gamma$ with both in $\mathcal{V}_L$ for large γ ; so a proper subfield has empty interior, and a closed one is nowhere dense. $\square$

Lemma 28.2 (Induced value embedding [11]). *Every field embedding $f : \mathbf{No} \hookrightarrow \mathbf{No}$ is ordered and induces an ordered additive embedding τ_f of the natural value group with $v(f(x)) = \tau_f(v(x))$. No monomial or strongness hypothesis is needed.*

Proof. Positivity is being a nonzero square. Archimedean equivalence and strict domination are expressed by inequalities with ordinary integers, which f preserves and reflects; multiplicativity gives additivity. $\square$

Theorem 28.3 (Continuity or external discreteness [11]). *Let $f : \mathbf{No} \hookrightarrow \mathbf{No}$ be any field embedding. Exactly one of the following holds.*

- (i) $\tau_f(\mathbf{No})$ is cofinal in $\mathbf{No}$. Then f is continuous, a homeomorphism onto its image, and the image is nondiscrete.
- (ii) $\tau_f(\mathbf{No})$ is not cofinal. Then there is L with $v(y - z) < L$ for all distinct $y, z \in f(\mathbf{No})$; the image is closed and uniformly discrete, and f is nowhere continuous.

Proof. (i) Given L , choose γ with $\tau_f(\gamma) > L$; then $f(\mathcal{V}_\gamma) = f(\mathbf{No}) \cap \mathcal{V}_{\tau_f(\gamma)} \subseteq \mathcal{V}_L$. (ii) Take L above $\tau_f(\mathbf{No})$: nonzero differences in the image have value below L , so $f^{-1}(\mathcal{V}_L) = \{0\}$, not open, and additivity gives discontinuity everywhere. A ball of radius L contains at most one image point, which gives closedness and uniform discreteness. $\square$

There is no contradiction in an order-isomorphic copy of the nondiscrete **No** being discrete: the intrinsic order topology of a non-cofinal ordered subfield can be coarser than its subspace topology; the inverse from the image stays continuous [11].

Theorem 28.4 (The two ends are independent [11]). *For the order embeddings*

$$\mu^{++}(a) = \begin{cases} a, & a \leq 0, \\ a + 1, & a > 0, \end{cases} \quad \mu^{+-} = \mu_{11},$$

$$\mu^{-+}(a) = -\mu_{11}(-a), \quad \mu^{--}(a) = \frac{\mu_{11}(a)}{1 + \mu_{11}(a)},$$

with ranges $(-\infty, 0] \cup (1, \infty)$, $(0, \infty)$, $(-\infty, 0)$ and $(0, 1)$, the value group $\iota_\mu(\mathbf{No})$ is respectively cofinal and coinital, cofinal and not coinital, coinital and not cofinal, neither; here “coinital” refers to the positive part. In each case there is a proper strong exact-monomial self-embedding preserving and reflecting **Oz**: M_{1,ι_μ} in the coinital cases, where coefficient motion is impossible, and $\mathcal{T}_{d,1} \circ M_{1,\iota_\mu}$, moving b to $b + \omega^{-1}$, in the others. The embeddings are continuous with closed nondiscrete image in the cofinal cases and nowhere continuous with discrete image in the others; all four images are nowhere dense.

Proof. By lemma 27.1 the value image is the class of normal forms supported in the range. The range of μ^{++} has arbitrarily high and low levels and misses $(0, 1]$, so the image is proper, cofinal and coinital. μ^{+-} is lemma 27.2. For μ^{-+} all values are infinitesimal, so the group is bounded by 1, but monomials at arbitrarily negative levels make it coinital. For μ^{--} all nonzero values are purely infinite and below ω in absolute value, so the group is neither; its gap is C_1 , since a finite value is dominated by every positive image value and an infinite value with leading level $a > 0$ dominates ω^c for some $0 < c < \min\{a, 1\}$. Apply corollary 26.7 and theorems 28.1 and 28.3. $\square$

So cofinality must not replace coinitality in the rigidity criterion: $J_{\mu_{11},d}$ is continuous with cofinal values and still moves $\mathbb{R}$ [11].

29 Constants, elementarity, conjugacy and parameters

29.1 Elementary reduct, nonelementary expansion

$\mathbb{R}$ is parameter-free definable in $(K_\Gamma, \mathcal{R}_\Gamma)$ by the existential test, the multiplier condition and its units (`odg:def:thm:ideal`, `odg:def:thm:multiplier`; the internal form is `odg:def:cor:internal`); 11 reproves this with explicit formulas D , M , Const , and derives the rigidity of automorphisms (corollary 5.2).

Theorem 29.1 (Logical separation [11]; [12] for its example). *Every field embedding $\mathbf{No} \hookrightarrow \mathbf{No}$ is elementary as an embedding of ordered fields. An embedding of theorem 26.3 (with $\Gamma = \Delta = \mathbf{No}$) that moves some real is not elementary as an embedding of $(\mathbf{No}, \mathbf{Oz})$, although it preserves and reflects **Oz**, and its restriction to **Oz** is not elementary as an embedding of rings.*

Proof. The first assertion is quantifier elimination for real closed ordered fields [21], read formula by formula for the class structure. If $\Phi(b) \notin \mathbb{R}$, the formula $\text{Const}(x)$ defining $\mathbb{R}$ holds of b and fails of $\Phi(b)$. For $\Phi(b) = b + t$ the new target witness is visible: $\omega \in \Pi$ but $(b + t)\omega = b\omega + 1 \notin \Pi$; the universal multiplier condition quantifies over more elements in the target. For the ring: $\text{Frac } \mathbf{Oz} = \mathbf{No}$ (`odg:thm:fractions`), the fraction field is uniformly

interpretable in the ring by pairs modulo $ad = bc$, with the ring as the class of fractions $(z, 1)$, so an elementary ring embedding would induce an elementary embedding of the interpreted pair; represent b as $(b\omega)/\omega$. $\square$

The first assertion is classical. The new point is that a strong, exact-monomial, predicate-reflecting embedding can lose elementarity in the omnific expansion; coefficient fixing is not proved sufficient for elementarity ([question 32.6](#)) [11]. 12 records the same failure: the multiplier definition of $\mathcal{A}_k$ quantifies over all of Π , which an embedding need not preserve. This sharpens the Diophantine report's caution after `odg:def:thm:autreal` that its argument does not apply unchanged to endomorphisms: there are proper ring embeddings $\mathbf{Oz} \hookrightarrow \mathbf{Oz}$ whose fraction-field extensions move a real.

Proposition 29.2 (Ordinary overlap [11]; [12] for Taylor sections). *For $F = J_{\tau,\rho}(K_\Gamma)$: $F \cap \mathbb{R}((t^{\mathbb{G}_\tau})) = \rho(\mathbb{R})$ and $F \cap \mathbb{R} = \{r : \rho(r) = r\}$, a real closed subfield of $\mathbb{R}$; for $\rho = \rho_{d,\delta}$ it is $\ker d$.*

Proof. A nonzero block at $h \neq 0$ lies in $h + \mathbb{G}_\tau$, disjoint from $\mathbb{G}_\tau$, so an image in $\mathbb{R}((t^{\mathbb{G}_\tau}))$ is its zero block $\rho(r)$; if that is a real constant, its standard part gives r . If r is algebraic over the fixed field k , then $\rho(r)$ is a root of the same polynomial, all real roots lie in $\mathbb{R}$, and the residue singles out r ; so k is relatively algebraically closed in the real closed $\mathbb{R}$, hence real closed. $\rho_{d,\delta}(r) = r$ exactly when $d(r) = 0$. $\square$

29.2 Continuum many copies separated by the omnific integers

Lemma 29.3 (One-sided infinitesimal neighbourhoods [11]). *Let $F = J_{\tau,\rho}(K_\Gamma)$ and $\rho(r) \neq r$. Every $y \in F$ with $\text{st}(y) = r$ has $\text{sgn}(y - r) = \text{sgn}(\rho(r) - r)$, and $\rho(r)$ is such a y . For $\rho = \rho_{d,\delta}$ with $d(r) \neq 0$ the sign is that of $d(r)$.*

Proof. $y = J(x)$ with $x = r + \varepsilon$, $v(\varepsilon) > 0$ (by the valuation and constant-term formulas). The nonzero $\rho(r) - r$ has value in $\mathbb{G}_\tau^{>0}$, while $v(J\varepsilon) \in \tau(\Gamma)^{>0}$ is larger; so $y - r = (\rho(r) - r) + J(\varepsilon)$ has the leading term of $\rho(r) - r$, which is $d(r)t^\delta$ for a Taylor section. $\square$

The sign is invariant under every field automorphism of $\mathbf{No}$ fixing $\mathbb{R}$ pointwise, in particular under every $\mathbf{Oz}$ -preserving one ([corollary 5.2](#)), since these preserve order and standard part.

Theorem 29.4 (One field-conjugacy class, continuum many pair classes [11]). *There are proper strong exact-monomial embeddings $J_a : \mathbf{No} \hookrightarrow \mathbf{No}$, $a \in \mathbb{R}$, with images F_a , such that*

- (i) $J_a^{-1}(\mathbf{Oz}) = \mathbf{Oz}$;
- (ii) all F_a have the cofinal value image $\iota_{\mu_{11}}(\mathbf{No})$ and are closed, nondiscrete and nowhere dense;
- (iii) all F_a are conjugate under strong valuation-fixing field automorphisms of $\mathbf{No}$;
- (iv) for $a \neq a'$ no field automorphism of $\mathbf{No}$ fixing $\mathbb{R}$ pointwise, in particular none preserving $\mathbf{Oz}$, maps F_a onto $F_{a'}$.

So the pairs $(F_a, F_a \cap \mathbf{Oz})$ are all isomorphic to $(\mathbf{No}, \mathbf{Oz})$ but lie in pairwise inequivalent positions.

Proof. Choose distinct b_1, b_2 in a transcendence basis of $\mathbb{R}/\mathbb{Q}$ and let d_a be the derivation with $d_a(b_1) = 1$, $d_a(b_2) = a$ and $d_a = 0$ on the other basis elements; no independence of named constants is assumed. Put $J_a = \mathcal{T}_{d_a,1} \circ M_{1,\iota_{\mu_{11}}}$. (i) and (ii) are [theorems 27.3, 28.1](#)

and 28.3. For (iii), $\mathcal{T}_{d_{a'},1}\mathcal{T}_{d_a,1}^{-1}$ maps F_a onto $F_{a'}$; the derivations need not commute, and only the individual inverses are used. For (iv) let $a < q < a'$ with $q \in \mathbb{Q}$ and $r = b_2 - qb_1$: $d_a(r) = a - q < 0 < a' - q = d_{a'}(r)$, so by lemma 29.3 the points of F_a with standard part r lie below r and those of $F_{a'}$ above, and both sets are nonempty. Finally $F_a \cap \mathbf{Oz} = J_a(\mathbf{Oz})$ by reflection. $\square$

This is a lower bound of continuum on the pair-conjugacy types inside one field-conjugacy class, not a classification or an upper bound; the images have identical value data, so no classification can use the value image and its cuts alone [11]. It partly answers 13's question on the conjugacy of its drift images (question 32.8).

29.3 Proper copies fixing prescribed parameters

Lemma 29.5 (Two-level support control [11]). *For a set $A \subseteq \mathbf{No}$ let B be the union of the inner supports of the outer exponents of the members of A , a set. Every order embedding μ fixing B pointwise has $M_{1,\iota_\mu}(a) = a$ for all $a \in A$.*

Proof. ι_μ fixes every outer exponent of every member of A ; this is the inner/outer distinction of lemma 7.1. $\square$

Theorem 29.6 (Parameter-fixed discrete and nondiscrete copies [11]). *For every set $A \subseteq \mathbf{No}$ there are proper strong exact-monomial embeddings $f_A, g_A : \mathbf{No} \hookrightarrow \mathbf{No}$ fixing $A \cup \mathbb{R}$ pointwise with $f_A^{-1}(\mathbf{Oz}) = g_A^{-1}(\mathbf{Oz}) = \mathbf{Oz}$. The image of f_A is closed and uniformly discrete and f_A is nowhere continuous; the image of g_A is closed, nondiscrete and nowhere dense and g_A is a homeomorphism onto it; no field automorphism of $\mathbf{No}$ maps one image onto the other. There is an explicit $\Omega > 1$ with $\Omega^{-1} < |y| < \Omega$ for every $0 \neq y \in f_A(\mathbf{No})$.*

Proof. Take B as in lemma 29.5 and u above B . The map $\mu^{\text{cmp}}(x) = x$ for $x \leq u$, $u + \frac{x-u}{1+x-u}$ for $x > u$, is an order isomorphism onto $(-\infty, u+1)$ fixing B , so every value h of $\iota_{\mu^{\text{cmp}}}$ has $|h| < \beta = \omega^{u+1}$; put $f_A = M_{1,\iota_{\mu^{\text{cmp}}}}$. The map $\mu^{\text{imp}}(x) = x$ for $x \leq u$, $x+1$ for $x > u$, fixes B , omits $(u, u+1]$ and has cofinal range; put $g_A = M_{1,\iota_{\mu^{\text{imp}}}}$. Theorems 26.3, 28.1 and 28.3 and lemma 29.5 give the properties. A field automorphism of $\mathbf{No}$ is a homeomorphism (its value map is onto; lemma 28.2 and theorem 28.3) and cannot exchange a discrete and a nondiscrete subspace. For $\Omega = \omega^\beta$: a nonzero $y \in f_A(\mathbf{No})$ has natural value h with $|h| < \beta$, so its leading monomial has exponent below β and $|y| < \Omega$; apply this to $1/y$. $\square$

The annulus bounds are external: Ω is not in the image, which is internally isomorphic to $\mathbf{No}$ and not Archimedean [11].

Extension elsewhere (batch 31). The independent-copies report docs/surreal/independent-surreal-copies extends the fixing clause, with $\rho = \text{id}$, from two maps to an $\mathbf{On}$ -indexed family with a prescribed intersection (its `isc:cor:parameters`, from `isc:main:copies`). Let H be the $\mathbb{Q}$ -span of the supports of the members of A . There are proper strong exact-monomial embeddings Φ_α ($\alpha \in \mathbf{On}$) preserving and reflecting $\mathbf{Oz}$ and fixing the full Hahn field $\mathbb{R}((t^H)) \supseteq A \cup \mathbb{R}$ pointwise; any two images meet exactly in $\mathbb{R}((t^H))$, and every finite subfamily is jointly linearly disjoint over it. That field is the full support closure of A , not the field generated by A or its real closure. In the terms of theorem 26.3 these maps are $J_{h_\alpha, \text{id}}$ for exponent embeddings h_α fixing H with ranges independent over H (`isc:thm:group`), obtained by class bookkeeping. For $H = 0$ that report also gives explicit maps (`isc:thm:explicit`): $M_{1,\iota_{\sigma_\alpha}}$ with σ_α an order isomorphism of $\mathbf{No}$ onto the interval $(3\alpha, 3\alpha+1)$. These intervals are bounded and pairwise disjoint, and the bottom gap is nonzero. That report makes no topological assertion. [merge]: the exponent map of each explicit copy has bounded range, so

by [theorem 28.3\(ii\)](#) its image is closed and uniformly discrete, as for f_A ; for the general family the alternative is not decided.

Theorem 29.7 (Relative coefficient motion [11]). *Let $A \subseteq \mathbf{No}$ be a set and k_A the field generated over $\mathbb{Q}$ by the real coefficients of the members of A . If $b \in \mathbb{R}$ is transcendental over k_A , there is a proper strong exact-monomial embedding fixing A pointwise, preserving and reflecting $\mathbf{Oz}$, with cofinal value image and $b \mapsto b + t^\delta$ for some $\delta > 0$.*

Proof. Choose d zero on k_A with $d(b) = 1$, through a transcendence basis of $\mathbb{R}/k_A$ containing b . With B as in [lemma 29.5](#), choose v below B and $c < v$, and let $\mu^{\text{rel}}(x) = v - (v - c) \frac{v-x}{1+v-x}$ for $x < v$ and x for $x \geq v$, an order isomorphism onto (c, ∞) fixing B . Positive values of $\iota_{\mu^{\text{rel}}}$ have leading exponent above c , so $\delta = \omega^c$ lies in the gap. Put $\mathcal{T}_{d,\delta} \circ M_{1,\iota_{\mu^{\text{rel}}}}$; it fixes A since the exponent lift fixes each member and d kills its coefficients. Apply [theorem 26.6](#). $\square$

An ordered embedding fixing a real field fixes the reals algebraic over it, so one cannot fix every real and move another; the transcendence hypothesis is needed [11].

30 Surcomplex and Gaussian versions

Identify $\mathbf{No}(i) = \mathbb{C}((t^{\mathbf{No}}))$ coefficientwise with conjugation $\overline{x+iy} = x-iy$; the union of the real and imaginary supports is well ordered, and a family of surcomplex series is summable if and only if its real and imaginary families are. The general Gaussian problem is kept separate from two restricted ones: with conjugation named, and with the natural valuation assumed preserved. They must not be merged into an unconditional classification [12, 13].

Theorem 30.1 (Conjugation-compatible automorphisms [12, 13, 14, 15]). *Every field automorphism F of $\mathbf{No}(i)$ with $F(\mathbf{Oz}[i]) = \mathbf{Oz}[i]$ that commutes with conjugation has the form $F(x+iy) = \sigma(x) + \varepsilon i \sigma(y)$ with σ an $\mathbf{Oz}$ -preserving automorphism of $\mathbf{No}$ and $\varepsilon = \pm 1$, and F and F^{-1} are strongly additive. Conversely every such map has these properties, so this stabilizer is $\text{Aut}(\mathbf{Oz}) \times C_2$; with $F(i) = i$ imposed, only $\varepsilon = 1$ occurs and F is strongly $\mathbb{C}$ -linear.*

Proof. Commuting with conjugation preserves its fixed field $\mathbf{No}$, and $\mathbf{Oz}[i] \cap \mathbf{No} = \mathbf{Oz}$. $F(i)$ is a root of $X^2 + 1$. [Theorem 23.2](#) makes σ strong, and summability is coordinatewise. Real lifts commute with conjugation, which makes the product direct. $\square$

Theorem 30.2 (Valuation-compatible Gaussian automorphisms [12, 13, 14, 15]). *Let F be a field automorphism of $\mathbf{No}(i)$ with $F(\mathbf{Oz}[i]) = \mathbf{Oz}[i]$ that maps the Hahn valuation ring onto itself (equivalently $v(Fx) = \tau(v(x))$) for an ordered automorphism τ of $\mathbf{No}$; values need not be fixed). Then $F(\mathbb{C}) = \mathbb{C}$; with $\alpha = F|_{\mathbb{C}}$, F is α -semilinear, $\text{ct}(Fx) = \alpha(\text{ct } x)$, and F and F^{-1} are strongly additive. The same holds for the fraction-field extension of a ring automorphism of $\mathbf{Oz}[i]$ whenever that extension preserves the valuation, and for valuation-compatible isomorphisms between full complex Hahn fields with nonzero exponent groups preserving their Gaussian integer parts [13].*

Proof. $\Pi_{\mathbb{C}} = \bigcap_n n\mathbf{Oz}[i]$ ([lemma 2.3](#)); its multiplier ring is $\mathbb{C} \oplus \Pi_{\mathbb{C}}$ with units $\mathbb{C}^\times$ (`odg:def:thm:multiplier`), so $F(\mathbb{C}) = \mathbb{C}$. Valuation compatibility preserves $\mathfrak{m}_{\mathbb{C}}$, and $\mathbf{No}(i) = \Pi_{\mathbb{C}} \oplus \mathbb{C} \oplus \mathfrak{m}_{\mathbb{C}}$ gives $\text{ct}(Fx) = \alpha(\text{ct } x)$. Apply [theorem 22.1](#). The fraction-field statement uses $\text{Frac } \mathbf{Oz}[i] = \mathbf{No}(i)$ (`odg:thm:fractions`); the isomorphism statement uses only the ambient fields. $\square$

α may be a wild automorphism of $\mathbb{C}$: coefficientwise application of any automorphism of $\mathbb{C}$ is strong and preserves $\mathbf{Oz}[i]$, since it fixes $\mathbb{Q}$ and sends i to $\pm i$ [13]. This is consistent with the wild lift of [proposition 10.3](#) and with `odg:def:prop:coeffaut`, which are strong.

Neither theorem shows that every automorphism of $\mathbf{Oz}[i]$ has a valuation-compatible extension; conjugation is not assumed intrinsic, and maps moving the real axis are already in the collection ([theorem 10.2](#), `odg:def:lem:twist`). Since $\text{Frac } \mathbf{Oz}[i] = \mathbf{No}(i)$, the open issue is compatibility, not existence of the extension [12, 13] ([question 32.1](#)).

Theorem 30.3 (Conjugation-compatible embeddings [11]). *The strong field embeddings $\Psi : K_\Gamma^\mathbb{C} \hookrightarrow K_\Delta^\mathbb{C}$ that commute with conjugation, satisfy $\Psi(t^\gamma) = t^{\tau(\gamma)}$ and preserve $\mathcal{R}_\Gamma^\mathbb{C}$ are exactly $\Psi(x + iy) = J_{\tau,\rho}(x) + \varepsilon i J_{\tau,\rho}(y)$, $\varepsilon = \pm 1$, with (τ, ρ) as in [theorem 26.3](#); each satisfies $\Psi^{-1}(\mathcal{R}_\Delta^\mathbb{C}) = \mathcal{R}_\Gamma^\mathbb{C}$, and $\Psi(i) = i$ forces $\varepsilon = 1$.*

Proof. Commuting with conjugation maps K_Γ into K_Δ , and $\mathcal{R}_\Gamma^\mathbb{C} \cap K_\Gamma = \mathcal{R}_\Gamma$, so the restriction is classified; $\Psi(i) = \pm i$. The converse is coordinatewise, and $\mathcal{R}^\mathbb{C}$ -membership is membership of both coordinates. $\square$

12's and 13's surcomplex embeddings $x + iy \mapsto J(x) + iJ(y)$ are the case $\varepsilon = 1$ of [theorem 27.3](#); they are proper and strong, commute with conjugation, reflect $\mathbf{Oz}[i]$, move b to $b + t$, and their image meets $\mathbb{C}$ in $\ker d + i \ker d$ [12, 13].

Proposition 30.4 (Complex constants and nonelementarity [11]). *$\mathbb{C}$ is parameter-free definable in $(K_\Gamma^\mathbb{C}, \mathcal{R}_\Gamma^\mathbb{C})$. An embedding of [theorem 30.3](#) with nontrivial ρ is elementary as an embedding of algebraically closed fields, but not in the Gaussian expansion; for $\Gamma = \Delta = \mathbf{No}$ its restriction to $\mathbf{Oz}[i]$ is not elementary as a ring embedding.*

Proof. For $z \in \mathcal{R}_\Gamma^\mathbb{C}$, $z \in \Pi_\Gamma^\mathbb{C}$ if and only if $z^2 = 2w^2$ for some $w \in \mathcal{R}_\Gamma^\mathbb{C}$, since $\sqrt{2} \notin \mathbb{Q}(i)$; the multiplier ring and its units define $\mathbb{C}$ as in the real case (`odg:def:cor:internal`). A moved real b has $\rho(b) \notin \mathbb{C}$. Pure-field elementarity is quantifier elimination for algebraically closed fields [21]; for the ring use $\text{Frac } \mathbf{Oz}[i] = \mathbf{No}(i)$ and the interpreted fraction field as in [theorem 29.1](#). $\square$

Corollary 30.5 (Transferred copies [11]). *The explicit coefficient-moving embedding, the four topological cases and the parameter-fixed copies extend to $\mathbf{No}(i)$, preserving and reflecting $\mathbf{Oz}[i]$; parameters in a set $A \subseteq \mathbf{No}(i)$ are handled through their real and imaginary parts. The family of [theorem 29.4](#) gives field-conjugate surcomplex images that are pairwise nonconjugate under automorphisms preserving both conjugation and $\mathbf{Oz}[i]$.*

Proof. Use $\varepsilon = 1$; the valuation of $x + iy$ is the minimum of the coordinate valuations, so the topology is the product topology. An automorphism preserving conjugation and $\mathbf{Oz}[i]$ restricts to an $\mathbf{Oz}$ -preserving automorphism of $\mathbf{No}$, and each extended image meets $\mathbf{No}$ in the real image; apply [theorem 29.4](#). $\square$

No nonconjugacy is claimed for automorphisms preserving $\mathbf{Oz}[i]$ but not conjugation, which need not fix the real axis pointwise, and non-conjugation-compatible surcomplex embeddings are not classified [11]. *Related* (batch 32). For automorphisms rather than images, the surcomplex automorphism report's finite-symmetry sections give nonconjugacy without conjugation compatibility: `saut:fs:thm:large-family` and `saut:fs:thm:main-count` give $2^{2^{\aleph_0}}$ strong valued involutions of $\mathbf{No}(i)$ preserving $\mathbf{Oz}[i]$, with pairwise nonisomorphic fixed fields, hence pairwise nonconjugate even under arbitrary field automorphisms of $\mathbf{No}(i)$. They do not concern the images of [corollary 30.5](#), and the sentence above stands.

31 Boundaries of the method

Proposition 31.1 (Regular cardinal support bounds [12]). *Let κ be a regular uncountable cardinal, Γ a set, and $K_{<\kappa}$ the field of series in $k((t^\Gamma))$ with support of size $< \kappa$, where summability also requires total support of size $< \kappa$. A family in $K_{<\kappa}$ is summable there if and only if fewer than κ members are nonzero and for every $y \in K_{<\kappa}$ only finitely many $\text{ct}(x_j y)$ are nonzero. Consequently a constant-term-covariant field isomorphism between two such fields is strong for this restricted summation.*

Proof. A summable family has fewer than κ nonzero members, since each contributes at an exponent of a total support of size $< \kappa$ and each exponent receives finitely many; pairing finiteness is [theorem 21.4](#). Conversely, regularity bounds the union of fewer than κ supports of size $< \kappa$; monomials test pointwise finiteness, and the countable witness of [lemma 21.2](#) lies in $K_{<\kappa}$ because κ is uncountable. An isomorphism preserves the number of nonzero members, and the proof of [theorem 22.1](#) carries over. $\square$

The cardinal condition cannot be dropped: a strictly increasing $(g_\alpha)_{\alpha < \kappa}$ in Γ gives (t^{g_α}) with finite pairings against every $y \in K_{<\kappa}$ ([lemma 21.1](#)) but total support of size κ [12]. For singular κ the union argument fails, and no extension is claimed; the proposition identifies no birthday cutoff with a support-bounded Hahn field [12]. 13 records the regular case only inside its singular-cardinal question ([question 32.12](#)). *Related* (batch 32). The first- κ coefficient report (`docs/surcomplex/first-kappa-coefficients`) shows that for singular κ the characterization itself fails (`fkcsb:thm:sums`): a series with κ terms and reverse well-ordered support splits along a cofinal sequence of length $\text{cf } \kappa < \kappa$ into pieces with fewer than κ terms and pairwise disjoint supports, a Hahn-summable family with fewer than κ nonzero members whose sum has support of size κ . That report proves this for its class fields of surreals with support of size $< \kappa$ and notes that the construction works in any set-sized $K_{<\kappa}$ whose exponent group contains a strictly monotone sequence of length κ . It is a statement about summability, not about automorphisms.

Example 31.2 (Small integer parts [13, 14]). The proof of $\text{Frac } \mathbf{Oz} = \mathbf{No}$ uses an upper bound in $\mathbf{No}$ for every set of exponents, and does not transfer to set-sized groups. For $\Gamma = \mathbb{Z}$ and $u = t^{-1}$, $\mathcal{R}_{\mathbb{Z}} = \mathbb{Z} \oplus u\mathbb{R}[u]$ and $\text{Frac } \mathcal{R}_{\mathbb{Z}} = \mathbb{R}(u) \neq \mathbb{R}((t))$: $ru/u = r$ gives $\mathbb{R}(u)$, and $f = \sum_{n \geq 1} t^{n!}$ is not rational, since a rational power series has coefficients obeying an eventual linear recurrence, and arbitrarily long zero blocks followed by a 1 contradict one.

This is the phenomenon of [proposition 18.1](#) and of [19, Proposition 2.4.5]; it is why [corollary 23.4](#) assumes an ambient isomorphism [11, 12, 13].

Remark 31.3 (Distinctions kept [12, 13]). The detector must lie in the field tested: automatic for full Hahn fields and for the countable witness in $K_{<\kappa}$, not for arbitrary subfields (a sufficient condition is 14's countable detector property, [proposition 35.2](#), and [example 35.3](#) is a subfield where detection fails). Strongness is not $\mathbb{R}$ -linearity: $J_{\mu,d}$ is strong but not $\mathbb{R}$ -linear, so the adjoint theorem, which is for k -linear maps, does not apply to it by treating its coefficient lift as scalar multiplication. The retraction ret_ϕ is linear, never a field retraction for a proper extension. The adjoint theorem needs a set of exponents; automatic strongness of isomorphisms does not. No result of this part concerns the Gonshor exponential, logarithm, the Berarducci–Mantova derivation, the omega-map, birthdays or simplicity; the square roots in (27.2) are ordered-field operations reindexing an additive group. No proper-class Zorn lemma and no basis of the class $\mathbf{No}$ is used; the only bases chosen are a transcendence basis and a Hamel basis of the set $\mathbb{R}$ over $\mathbb{Q}$ [13].

32 Status of the questions, and questions of this part

32.1 Answered and re-scoped questions

- [Question 11.1](#) is *answered*. Every ring automorphism of $\mathbf{Oz}$ extends to a strong automorphism of $\mathbf{No}$ ([theorem 23.2](#)), so preservation of $\mathbf{Oz}$ forces strongness and there is no nonstrong $\mathbf{Oz}$ -preserving example; 09’s characterization of the nonstrong pair automorphisms is vacuous, and 10’s question whether every $\mathbf{Oz}$ -preserving 1-automorphism is strong has the answer yes. The analogous question for $\mathbf{Oz}[i]$ has a positive answer under conjugation or valuation compatibility and is open otherwise ([question 32.1](#)).
- `osq:q:invisible` of `docs/surreal/set-sized-quotients-of-omnific-integers`: its second clause is *answered* by [corollary 24.4](#); the nonstrong case left open in [section 11.1](#) does not exist. Its first clause, on class quotient domains, is untouched. The quotient report is not edited by this merge; since batch 31 it records the question as partly answered, with this corollary.
- [Question 11.2](#): unchanged. 12 and 13 ask it again, in the forms “for which support-controlled families of logarithmic characters does $a_\delta(C_\delta) = 0$ imply two-sided substitution admissibility?” and “give necessary and sufficient support conditions on additive $\gamma \mapsto L_\gamma \in \mathfrak{m}$ for $t^\gamma \mapsto t^\gamma \exp L_\gamma$ to extend to a strong automorphism, assuming $a_\delta(C_\delta) = 0$ ”, suggesting common-shift and finite-rank triangular classes. Automatic strongness settles the recognition of existing automorphisms, not their construction.
- [Question 11.4](#) now concerns every automorphism of $(\mathbf{No}, \mathbf{Oz})$ with trivial monomial factor, not only the strong ones; it is otherwise unchanged.
- [Question 11.7](#) is related to 12’s question on more intrinsic scalar tests, merged into [question 32.13](#).
- Scope of the answers: $\mathbf{Oz}$, the full real Hahn pairs, and $\mathbf{Oz}[i]$ with conjugation or valuation data. For general characteristic-zero k and unital $\mathfrak{o}$ the non-claims of 04, 09 and 10 that strongness is not shown ([section B](#)) still apply ([question 32.13](#)).

The novelty statements of 11, 12 and 13 were checked against the tree at the merge. No report proves automatic strongness, the scalar detection of summability, the adjoint theorems, or coefficient-moving embeddings of $(\mathbf{No}, \mathbf{Oz})$; their statements about this report and the Diophantine report were accurate at their pins. What they reprove (reconstruction, the criterion, the strong factorization, the Taylor automorphism) is credited above. Priority remains provisional, as each source says. Source 14’s novelty statement (batch 31), accurate at its pin but stale at the merge, is recorded in [section 37.2](#).

32.2 Questions

Each question is a proposed research direction, not a published open problem [11, 12, 13]. Duplicates across the three sources are merged, with every source named.

Question 32.1 (The unrestricted Gaussian stabilizer [12, 13, 14, 15]). Does every ring automorphism of $\mathbf{Oz}[i]$ have a valuation-compatible extension to $\mathbf{No}(i)$, that is, is $\mathfrak{m}_{\mathbb{C}}$ invariant? Can the positive-support component be recovered from $\mathbf{Oz}[i]$ and its fraction field without naming conjugation or a valuation?

$\mathbb{C}$ and $\Pi_{\mathbb{C}}$ are already intrinsic, so the missing object is $\mathfrak{m}_{\mathbb{C}}$; a positive answer with [theorem 30.2](#) would give automatic strongness for all automorphisms of $\mathbf{Oz}[i]$. A negative

answer should exhibit an automorphism of the ring violating a valuation inequality, not merely moving the real axis or being an unrelated automorphism of the algebraically closed fraction field [12, 13]. 14 asks the same, adding whether equivariance of the full constant term at least follows, and proposes first targets: the full field $\mathbb{C}((t^{\mathbb{Q}}))$, then a rank-two lexicographic exponent group. A counterexample must do more than move the real axis, since the strong phase twists of [remark 36.6](#) do that [14]. *Status* (batch 31): open. 15 asks it again (its Question 4) and notes that its topology does not help: a compatible ring topology cannot reconstruct the missing valuation, since on $\mathbf{No}(i)$ every one is indiscrete ([corollary 39.9](#)); 16 recovers $\mathbf{No}(i)$ as a coefficient field but not its conjugation ([question 52.9](#)). *Status* (batch 32): open. The surcomplex automorphism report’s finite-symmetry sections reprove [theorem 30.2](#) for valued automorphisms by the same argument (`saut:fs:rem:gaussianstrong`) and classify the valued involutions preserving $\mathbf{Oz}[i]$ up to conjugacy (`saut:fs:thm:main-involutions`); they assume the valuation throughout, so they do not address this question.

Question 32.2 (Strongness of omnific embeddings [11, 12, 13, 14]). Let $f : \mathbf{No} \hookrightarrow \mathbf{No}$ be a field embedding with $f^{-1}(\mathbf{Oz}) = \mathbf{Oz}$. Must f be strongly additive? What changes if also $\text{ct} \circ f = \text{ct}$? Without strongness, which coefficient and image restrictions survive; can such an f have a proper dense image, or one that is neither closed nor discrete?

The embeddings of [section 27](#) are strong and do not answer the first clause; a counterexample needs a summable family whose image is not summable or has the wrong sum, and the missing step in the isomorphism proof is access to all target tests ([corollary 27.4](#)) [12, 13]. *Status*. The first two clauses have a negative answer if there is a measurable cardinal, more generally an elementary embedding $j : V \rightarrow M$ into a transitive class with a critical point, available as a class. The large-cardinal report (`docs/foundations-and-computation/large-cardinal-embeddings-and-normal-forms`) shows that the map $\hat{j}$ obtained by applying j to sign sequences is an ordered field embedding of $\mathbf{No}$ that fixes $\mathbb{R}$, preserves and reflects $\mathbf{Oz}$ and preserves constant terms, while it sends a summable family to a summable family whose sum is not the image of the sum; so it is not strongly additive. It is not onto, so it does not conflict with [theorem 23.2](#). Consequently a positive answer is not provable in the base theory unless that theory with a measurable cardinal is inconsistent; whether a counterexample exists without large cardinals stays open. *Status* (batch 32). The large-cardinal report sharpens this. A countably strong k -linear map between full Hahn fields over $\mathbb{R}$ or $\mathbb{C}$ that is not strong first fails on a family whose cardinality is measurable (`lce:mf:thm:spectrum`), and the absence of measurable cardinals is equivalent to strongness of every countably strong coefficient-fixing field embedding of $\mathbf{No}$ preserving $\mathbf{Oz}$ (`lce:mf:thm:equivalence`). So in a universe without measurable cardinals a coefficient-fixing counterexample to the first clause must already fail on a countable family; whether such an embedding must be countably strong is `lce:mf:q:countable`, open. The necessary gap restriction of [theorem 26.3](#) and the dichotomy of [theorem 28.3](#) remain valid without strongness, whereas the closed-image proof uses the strong expansion formula. The questions on dense images and on images neither closed nor discrete are not settled here [11].

Question 32.3 (Coefficient-fixing embeddings and target tests [13, 14]). For $\mathbb{R}$ -linear embeddings preserving the omnific predicate, can every target test $\text{ct}(y \cdot)$ be represented, after pullback, by a source multiplier?

Such adjoints suffice for strongness and are necessary at set size ([theorem 25.2](#)); the functional $\lambda_{\mathbf{On}}$ shows that a proof for $\mathbf{No}$ must use the multiplicativity of the embedding, not a theorem about all strong linear operators [13]. 14 asks which additional hypotheses make a constant-term-compatible embedding of full Hahn fields strong, in particular for embeddings of $\mathbf{No}$ preserving the integer part. The isomorphism proof fails exactly because a target detector

need not lie in the image; at set size an adjoint of the underlying semilinear map suffices, and 14 proposes seeking a criterion that the image supplies all detectors needed for its families, weaker than surjectivity. No embedding theorem is deduced [14]. *Status* (batch 32): *negative under a measurable cardinal*. The large-cardinal report's embedding (its J , written $\hat{j}$ here; `lce:mf:prop:J`) is coefficient-fixing and preserves and reflects $\mathbf{Oz}$, and the pulled-back test of ω^κ has no source multiplier representative (`lce:mf:thm:target`). This already followed from [theorem 25.2](#) and the non-strongness of $\hat{j}$ ([question 32.2](#)); that report adds the explicit test. The version for strongly $\mathbb{R}$ -linear embeddings is its `lce:mf:q:targetstrong`, open.

Question 32.4 (Coefficient-moving embeddings [12]; re-scoped). Which subfields $C_0 \subseteq \mathbb{R}$ occur as $f(\mathbf{No}) \cap \mathbb{R}$ for strongly additive embeddings f with $f^{-1}(\mathbf{Oz}) = \mathbf{Oz}$? Which arise from $\ker d$, and are higher-derivation coefficient lifts needed?

Partly answered by 11: in the exact-monomial strong class the intersections are the fixed fields of the coefficient sections ρ , which are real closed, and Taylor sections give $\ker d$ ([proposition 29.2](#)). Which real closed subfields occur, and the classes with characters, unit corrections or no monomial condition, stay open. A classification should separate the coefficient homomorphism, the value-group embedding and the admissibility of their combination; block separation is sufficient, not necessary [12].

Question 32.5 (Removing exact monomial preservation [11]). Classify the strong embeddings $f : \mathbf{No} \hookrightarrow \mathbf{No}$ preserving $\mathbf{Oz}$ with $f(t^\gamma) = c_\gamma t^{\tau(\gamma)}(1 + u_\gamma)$, $c_\gamma > 0$, $v(u_\gamma) > 0$. Which compatibility and summability conditions are necessary and sufficient, and can every such embedding be normalized by an $\mathbf{Oz}$ -preserving automorphism?

The criterion of [part I](#) is a starting point, but surjectivity is unavailable: corrections on the image need not extend to all target scales. A first step would keep the corrections on one common positive shift set [11].

Question 32.6 (Elementarity of coefficient-fixing pair embeddings [11]). For an ordered additive $\tau : \mathbf{No} \hookrightarrow \mathbf{No}$, when is $M_{1,\tau} : (\mathbf{No}, \mathbf{Oz}) \hookrightarrow (\mathbf{No}, \mathbf{Oz})$ elementary? Is fixing $\mathbb{R}$ sufficient in the exact-monomial strong category?

[Theorem 29.1](#) gives the necessary condition $\rho = \text{id}$; quantifier elimination for real closed fields cannot decide the expanded structure [11].

Question 32.7 (Coefficient sections [11]). Describe the ordered sections $\rho : \mathbb{R} \hookrightarrow \mathbb{R}((t^C))$ with residue the identity. Which come from one Taylor derivation, compositions of such, or summable families of higher derivations, and what distinguishes sections with the same fixed subfield?

The classification reduces the coefficient part of the embedding problem exactly to this question; the one-sided neighbourhood invariant is only a first invariant [11].

Question 32.8 (Conjugacy of proper copies [11, 13]). Classify the images $J_{\tau,\rho}(\mathbf{No})$ up to $\mathbf{Oz}$ -preserving automorphisms of $\mathbf{No}$. How much do the value image, its cuts, the fixed coefficient field and the one-sided neighbourhood data determine? When are two images $J_{\mu,d}(\mathbf{No})$ conjugate, and which scale embeddings admit Taylor deformations with several independently prescribed motions?

Partly answered by [theorem 29.4](#): the value image alone is insufficient, and continuum many types occur in one field-conjugacy class. 13 notes that since every omnific automorphism fixes $\mathbb{R}$, the positions of the deformed coefficient fields may give finer invariants; a restricted target is

$\mathcal{T}_{d,\delta} \circ M_{1,\tau}$ for fixed τ and one shift [11, 13]. *Related* (batch 31): the independent-copies report asks the pair version at a fixed Hahn core: which invariants decide whether an **Oz**-preserving automorphism of **No** fixing $\mathbb{R}((t^H))$ carries one independent pair of copies with that common field to another (its second research question, “Relative classification of pairs of copies”, unlabelled). It proves no conjugacy result, and the present question stays open.

Question 32.9 (Homogeneity of the pair [12, 13]). Which isomorphisms between set-sized substructures of $(\mathbf{No}, \mathbf{Oz})$ extend to automorphisms of the pair?

Fixing the reals of the domain is necessary (corollary 27.5); no sufficiency theorem is proved. The question differs from extension of elementary partial maps in a first-order language. A first target is a back-and-forth theorem for truncation-closed Hahn subfields with prescribed real coefficients and compatible integer parts [12, 13].

Question 32.10 (Surcomplex embeddings without a real form [11]). Classify the embeddings of $(\mathbf{No}(i), \mathbf{Oz}[i])$ without assuming commutation with conjugation, and decide which of the nonconjugate copies of corollary 30.5 stay nonconjugate under the larger group.

$\mathbf{Oz}[i]$ reconstructs $\mathbb{C}$ but gives no pointwise real rigidity for the sign invariant [11].

Question 32.11 (The proper-class strong dual [12, 14, 15]). Can the strong $\mathbb{R}$ -linear dual of **No** be described by class coefficient functionals meeting every well-ordered set finitely, and which maps have transposes that stay set-supported on every argument?

Such a description must avoid unlicensed collections of classes; the issue is foundational as well as algebraic [12]. 14 asks for a natural enlargement of surreal kernels representing every strong $\mathbb{R}$ -linear functional on **No**: formal kernels supported on a proper class whose pairing with every set-sized normal form is finite, which recover the set-sized duality on restriction and explain which strong class operators have surreal-valued adjoints. Such kernels are not surreal numbers, and merely writing a proper-class sum is not a definition [14]. *Status* (batch 32): the first clause is *answered* by 15’s proposition 40.5. The strong $\mathbb{R}$ -linear functionals on **No** are exactly the λ_a of class functions a whose support contains no strictly increasing sequence, equivalently meets every well-ordered set finitely; these are 14’s kernels, and they restrict to the set-sized duality. The second clause, which strong operators have transposes that stay set-supported, is 15’s Question 9, which asks whether the obstruction is a uniform bound on the supports of the adjoint coefficient classes; it stays open.

Question 32.12 (Singular support bounds [12, 13, 15]). For singular uncountable κ , when does a coefficient- and constant-term-compatible automorphism of $K_{<\kappa}$ preserve summability with total support of size $< \kappa$?

Countable binary witnesses lie in $K_{<\kappa}$ and detect well-ordering and local finiteness, but not the cardinal bound on unions; a sharp theorem may need uniform bounds on supports of monomial images, and a counterexample must preserve the restricted fields themselves [12, 13]. *Partial information* (batch 31): by 14’s detection in subfields (proposition 35.2 and remark 35.4) such an automorphism sends every family summable in $K_{<\kappa}$ to a Hahn-summable family with sum the image of the sum; only the size of the union of the image supports remains. The question stays open. 15 asks it again for support-bounded fields (its Question 6), asking that any large-cardinal dependence be kept apart from its ZFC topological obstruction, which uses only countable sequences (part V). *Status* (batch 32): open. The first- κ coefficient report’s `fkcsb:thm:sums` bears on it (note after proposition 31.1): for singular κ , summability in $K_{<\kappa}$ is not decided by the number of nonzero members and ambient summability. It does not answer the question, which asks about automorphisms.

Question 32.13 (Other rings and coefficient fields [12, 14]; [merge]). For which integer parts, truncation subrings or distinguished subrings of a real closed Hahn field can one reconstruct a scalar pairing that forces automatic strongness of their stabilizers? In particular, for which characteristic-zero k and unital $\mathfrak{o} \subseteq k$ does $\sigma(\mathcal{R}_{\mathfrak{o}}) = \mathcal{R}_{\mathfrak{o}}$ force a field automorphism of $k((t^{\Gamma}))$ to be strong?

The omnific proof uses a divisible ideal, its multiplier ring and units to recover coefficients, then the order to recover the infinitesimals; it would be useful to find the weakest hypotheses under which this works and examples of failure [12]. The merge adds the second sentence: for $\mathfrak{o} = k$ the divisible-ideal step fails, and for non-real k without valuation data the order step is missing; compare [question 11.7](#). 14 asks the same for integer parts A of a real closed Hahn field, and whether the full constant term is reconstructible from A by the same chain; it suggests the checkable sufficient condition that the divisibility core of A recover a negative-support complement whose multiplier units form the coefficient field, and asks when that condition is necessary and stable under conjugation [14]. *Partial information* (batch 32): for $\Gamma = \mathbf{No}$, a set-sized characteristic-zero k and $\mathfrak{o} \in \{\mathbb{Z}, \mathbb{Z}[i]\}$ (with $i \in k$ in the second case), the surcomplex automorphism report shows that every automorphism of $\mathcal{R}_{\mathfrak{o}}$ extends to $k((t^{\mathbf{No}}))$ and preserves k and Π (`saut:fs:prop:reconstruction`), and that such an automorphism is strong and semilinear once it is also valued (`saut:fs:rem:gaussianstrong`). The question does not assume the valuation and stays open.

Question 32.14 (Isomorphisms of small integer parts [13, 14]). For which ordered groups Γ, Δ does every ring isomorphism $\mathcal{R}_{\Gamma} \cong \mathcal{R}_{\Delta}$ extend to an isomorphism of the full real Hahn fields preserving order and integer parts?

[Example 31.2](#) shows the fraction field is not the Hahn field in general. One route is an intrinsic summation completion compatible with the pairing; another is to find two Hahn fields with indistinguishable small integer parts [13]. 14 asks the automorphism case for divisible Γ , with uniqueness: first determine $\text{Frac } \mathcal{R}_{\Gamma}$ and its position in K_{Γ} , since extension through the remaining series is a separate problem [14]. *Status* (batch 31): an extension, if it exists, is unique ([proposition 36.5](#), [merge]); existence stays open.

Question 32.15 (Exponential, omega-map and differential structure [11, 12, 13, 18, 20]). Which proper embeddings preserving $\mathbf{Oz}$, and which gap sections and value embeddings, are compatible with the surreal exponential, the omega-map, a surreal derivation or a simplicity relation? Intersect the full omnific stabilizer with the automorphisms preserving such a structure.

The Taylor block embeddings give flexibility only in the algebraic pair and fail omega-map compatibility at 1; the questions must be compared with those of [15], not inferred from the construction, and the exponential rigidity already in the collection is not to be re-solved [11, 12, 13]. Source 18 asks it for group realizations (its Question 11.5): which left-orderable groups remain realizable when the Gonshor exponential, the omega-map or a specified surreal derivation must be preserved, and can an exact fixed-field theorem survive? Its construction does not answer this: compatibility $M_{1,\iota_{\varphi}}(\omega^a) = \omega^{M_{1,\iota_{\varphi}}(a)}$ already ties the two lifting levels together, which a general order action does not ([remark 61.3](#)); relevant rigidity results and questions are in [15] [18]. *Status* (batch 33): open. Source 20 asks it again (its Question 1): which left-orderable groups admit faithful omnific-preserving actions commuting with the omega-map or the Gonshor exponential, and can part of the universality survive either requirement? Its obstruction $\sigma_1^{\text{tr}}(\omega) \neq \omega^{\sigma_1^{\text{tr}}(1)}$ shows that its actions do not answer it; a first step is to translate the compatibility into a condition coupling the index map and both lifts rather than treating the lifts independently [20]. *Status* (batch 34): open.

Question 32.16 (Restricted workspaces [11]). For a support-restricted subfield of a full Hahn field, when does the bottom-gap extension stay inside it, and which restrictions make the canonical integer part have the whole workspace as fraction field?

The block construction preserves well-ordering but not bounds on support cardinality, order type or boundedness; this matters for birthday cutoffs, computable presentations and transseries subfields [11].

Question 32.17 (Compositions [11]). Describe composition and image inclusion of classified maps in terms of the data (τ, ρ) ; the composite of (τ_1, ρ_1) and (τ_2, ρ_2) has exponent map $\tau_2\tau_1$ and section $J_{\tau_2, \rho_2} \circ \rho_1$. Can image inclusion and ambient extendability be decided from smaller invariants?

Status (batch 31): image inclusion is settled in a special case. Take $\rho = \text{id}$ and the exponent ranges $G_i = h_i(\mathbf{No})$, $i \in I$, of the family in the independent-copies report, which fix a common set-sized divisible group H and are independent over it (`isc:thm:group`). Put $E_J = \mathbb{R}((t^{G_J}))$ with $G_J = H + \sum_{j \in J} G_j$, so that $E_{\{i\}} = J_{h_i, \text{id}}(\mathbf{No})$. That report proves $E_J \cap E_T = E_{J \cap T}$, and $E_J \subseteq E_T$ exactly when $J \subseteq T$ (`isc:thm:boolean`). This decides inclusion and intersection among these images and their full Hahn joins. Composition is not treated there, and the general question stays open.

Question 32.18 (Formalization [11, 12, 13, 14, 15]). Formalize the finite-intersection lemma, the triangular or binary detector, the detection theorem and the constant-term isomorphism theorem for arbitrary full Hahn fields; then the multiplier reconstruction and the omnific specialization; and, separately, the bottom-gap classification for set-sized Hahn fields, the closed-image proof and the two-level index lift, isolating the class-size assumptions for $\mathbf{No}$.

The sources propose this order; finite checks remain a separate test layer. Nothing in this part is formalized in Lean [11, 12, 13]. 14 proposes the same order, beginning with an abstract matrix lemma with finite row and column supports ([lemma 34.1](#)), and asks that the full-surreal corollary go through a stated universe-relative or class-coded normal-form interface, not by treating $\mathbf{No}$ as a set, and that the failure of class duality ([theorem 25.6](#)) be formalized separately; its proposed module names are plans only [14]. 15 asks that the topological collapse and the cyclic boundary be formalized first over abstract ordered monomial systems (the dense-interval and cyclic-cover lemmas, the closure-of-zero facts and the partial-sum argument), then instantiated in set-sized Hahn rings and in the repository's surreal field, with no opaque imported obstruction and a minimal interface for Hahn-summable families (its Question 12) [15]. 16's formalization question is [question 52.10](#).

Part IV

A fourth derivation: induced matchings, detector subfields and formal evaluation

This part records manuscript 14, the addition of batch 31. It proves again the main theorems of [part III](#) for the real pair, in less generality; those results are printed once, in [part III](#), with 14 added to their sources, and [section 33.2](#) lists them. What is printed here is what 14 adds: a third proof of scalar detection, by induced matchings, whose witness has all coefficients equal to 1 and needs no choice of coefficients ([section 34](#)); detection inside subfields that contain enough such witnesses ([section 35](#)); and equivariance of formal power-series evaluation,

the general adjoint-matrix formula, the rank-one group $\Gamma = \mathbb{R}$ and the source’s shear and truncation examples (section 36). Its labels carry the sub-prefix `opa:cm:`.

33 Source 14, what it re-derives, and its notation

33.1 The manuscript and its pin

Manuscript 14 is *Automatic Summability from Omnific Arithmetic: constant-term duality, the full omnific stabilizer, and a sharp set-class distinction*, dated 23 September 2026, 24 pages. It is item 07 of batch 31, delivered as the archive `omnific_automatic_summability` in commit `39fe674` and placed as an addition in commit `9d28e28`; it is numbered 14 here after its file prefix, continuing this report’s local sequence. Its finite checks, their recorded run and its source statement are shipped as `code/14-automatic-summability-verify.py`, `data/14-automatic-summability-verification.json` and `14-automatic-summability-SOURCES.md`; its `article.tex`, PDF, README and checksum list are not shipped.

It pins repository commit `3d40856`. That commit contains this report’s `article.tex` exactly as written in batch 26 (`9b80a30`), that is [parts I](#) and [II](#) without the material of source 10, together with the placement `c6359e4` of source 10’s files. It contains neither `b3fa9e2`, the batch-28 write of source 10, nor `21375f8` and `20c4c9f`, the placement and the write of sources 11–13. So 14’s author saw [question 11.1](#) open and saw none of [part III](#); the three batch-30 authors and 14 worked independently of one another. 14 describes the batch-26 report accurately: its criterion concerns strong coefficient-fixing 1-automorphisms, and 14 credits its reconstruction and its convex-scale criterion as antecedents.

33.2 Printed once

The table maps 14’s statements to the results where they are printed. “Same” means the same statement with at most notational differences; otherwise the difference is named. 14’s theorem numbers are those of its own PDF.

14	Printed as	Comparison
Thm. 1.1, Thm. 5.3, Cor. 5.5	theorem 23.2 and corollary 23.4	Less general: 14 treats $\Gamma = \Delta$ divisible, while proposition 23.1 allows ordered isomorphisms between different, possibly non-divisible groups (13).
Lem. 2.2; the Neumann lemma	lemma 2.1 and the support principle of section 2 ; lemma 21.1 is its case of exponent 0	Same.
Lem. 3.1, Thm. 3.3, Cor. 3.4	theorem 21.4	Same statement; 14’s proof is the third route of section 34 .
Thm. 4.1, Cor. 4.2	theorem 22.1 and corollary 22.2	14 assumes $\sigma(k) = l$ with $\alpha = \sigma _k$, as 13 does; 12 derives it.
Props. 5.1, 5.2	The reconstruction cited before proposition 23.1 , and that proposition	Re-derivation.

14	Printed as	Comparison
Lem. 5.4 and the remark after Cor. 5.5	<code>odg:thm:fractions;</code> example 31.2	Same, including the example $\Gamma = \mathbb{Z}$.
Cor. 5.6	corollary 23.5	Same.
The route (5.4)	The dependency chain after theorem 23.2	Same chain.
Thms. 6.1, 6.2, Cor. 6.3	proposition 25.1 and theorem 25.2	Same; the general matrix formula (6.2) is printed as proposition 36.2 .
Thm. 7.1 and the operator after it	theorem 25.6	Same.
Thm. 8.1 and the composition law	theorem 24.1 ; the law after (5.1)	Same.
Thm. 8.2	theorem 3.3 , by 04's polynomial route (lemma 2.2)	Re-proof for $(\mathbb{R}, \mathbb{Z})$.
Cor. 8.3	corollary 24.3 ; the form for $\Gamma = \mathbb{Q}$ after theorem 5.1	Same; the case $\Gamma = \mathbb{R}$ is remark 36.3 .
Prop. 9.1 and the truncation example	example 4.19 , (4.5)	The inverse of that example with the two letters swapped; remark 36.4 .
Prop. 10.1, Thm. 10.2	theorem 30.2 and its proof	Same.
Thm. 10.3	theorem 30.1	Equivalent hypothesis (remark 36.6).
Ex. 10.4	The phase twists recorded after theorem 10.2	An instance.
Questions 12.1–12.8 and the question of its Section 10	section 37	One new question; the others are merged.

So 14 is a third independent derivation of the automatic-strongness theorem, after 12 and 13, and adds nothing to its generality. Its route $\mathcal{R}_\Gamma \rightarrow \Pi \rightarrow \mathcal{A}_\Gamma \rightarrow \mathbb{R} \rightarrow \mathfrak{m} \rightarrow \text{ct} \rightarrow \text{summation}$ is the chain printed after [theorem 23.2](#). Where 14's proof of a shared result differs only in presentation it is not reprinted. Its genuinely different proof, the induced matching, is kept as a marked third route.

33.3 Conventions and notation of this part

This part uses the conventions of [section 20.3](#): the small- t convention $t^\gamma = \omega^{-\gamma}$, $K_\Gamma = k((t^\Gamma))$, Π_Γ , $\mathcal{R}_\Gamma = \mathbb{Z} \oplus \Pi_\Gamma$, $\mathcal{A}_\Gamma = \mathbb{R} \oplus \Pi_\Gamma$ and the pairing $\langle x, y \rangle_0 = \text{ct}(xy)$. Manuscript 14 also writes in the t -convention. Its symbols are renamed as follows.

14's symbol	Here
$A_\Gamma = \mathbb{Z} \oplus \Pi$, and A	$\mathcal{R}_\Gamma$; for $\Gamma = \mathbf{No}, \mathbf{Oz}$.
$B = k \oplus \Pi$	$\mathcal{A}_k$; for $k = \mathbb{R}, \mathcal{A}_\Gamma$.
$\langle x, y \rangle$	$\langle x, y \rangle_0$.
$\tau : k \rightarrow l$, the coefficient isomorphism	α , as in theorem 22.1 .
ϕ , the exponent automorphism; $M_{\chi, \phi}$	τ ; $M_{\chi, \tau}$ of (5.1).
ρ , a ring automorphism of $\mathbf{Oz}$; $\tilde{\rho}$	σ and $\tilde{\sigma}$, as in theorem 23.2 .
ν , a normalized 1-automorphism; $\mathcal{U}, \mathcal{G}, \mathcal{M}$	u ; $\mathcal{U}_{\mathcal{R}_\mathbb{Z}}(\Gamma)$ (and $\mathcal{U}_{\mathbf{Oz}}$); the stabilizer of $\mathcal{R}_\Gamma$; the monomial maps.
$L_\nu(g)$, $a_{\nu, \delta}$	$h_u(g)$, $a_{u, \delta}$ of (3.1).
T^*	$T^\dagger$; the matrix $T_{h, g} = [t^h]T(t^g)$ keeps its name.
Λ	$\lambda_{\mathbf{On}}$ of (25.2); 14's $[\omega^\alpha]x$ is $[t^{-\alpha}]x$.
F , a power series	$\mathbf{F}$.
F , a subfield	K' .
$P_{<\gamma}$, truncation	$T_{<\gamma}$, as in theorem 8.2 .
e_1, e_2 of its shear (e_1 the larger scale); $s = t^{e_2}$	e_2, e_1 of example 4.16 ($e_1 \ll e_2$); s is written t^{e_1} .
h , an additive map	θ .
$u_1, \dots, u_m, u$, infinitesimals	$z_1, \dots, z_m, z$; u is the leading-term-fixing factor.
$d > 0$ in its cancellation example	$\delta > 0$; d is a derivation of $\mathbb{R}$ in part III .
$\mathbf{No}[i], \mathbf{Oz}[i]; A_\mathbb{C}, K_\mathbb{C}, \Pi_\mathbb{C}, B_\mathbb{C}$	$\mathbf{No}(i), \mathbf{Oz}[i]; \mathcal{R}_\Gamma^\mathbb{C}, K_\Gamma^\mathbb{C}, \Pi_\Gamma^\mathbb{C}, \mathbb{C} \oplus \Pi_\Gamma^\mathbb{C}$.
$L = l((u^\Delta))$	$L = l((t^\Delta))$, with t for the monomials of both fields.
$a \ll b$	$a \ll b$, the same relation.

The collisions are resolved here only, and the tempting false readings are wrong:

- 14's A_Γ is *not* $\mathcal{A}_\Gamma$: 14's $A_\Gamma = \mathbb{Z} \oplus \Pi$ is $\mathcal{R}_\Gamma$, and $\mathcal{A}_\Gamma = \mathbb{R} \oplus \Pi$ is 14's B .
- 14's τ is a coefficient isomorphism; τ here is the exponent automorphism, 14's ϕ . 14's ρ is a ring automorphism of $\mathbf{Oz}$; ρ in [part III](#) is a coefficient section.
- 14's F has two meanings, neither of which is the isomorphism F of [theorem 22.1](#) or the profile F of (4.4).
- In 14's shear e_1 is the *larger* scale; in [example 4.16](#) it is the smaller. The letters are swapped here.
- 14's truncation $P_{<\gamma}$ is not one of the zero-cut projections P_-, P_0, P_+ .
- 14's additive h is not the logarithmic character h_σ .

14 calls an additive map *strong* when it sends summable families to summable families with the same sum; this is “strongly additive” here, and 14 states both directions where it asserts them. Its “normalized 1-automorphism” is a 1-automorphism in the sense of [section 1.5](#); 14 does not assume it strong, and by [corollary 23.4](#) every one preserving $\mathcal{R}_\Gamma$ is.

34 Induced matchings: a third route to scalar detection

Lemma 34.1 (An infinite induced matching [14]). *Let $(a_{n,j})_{n \in \mathbb{N}, j \in I}$ be a matrix over a field, I a set. Suppose that every row is nonzero and has finite support, and that every column has finite support. Then there are row indices $n_0 < n_1 < \dots$ and distinct column indices $j_0, j_1, \dots$ with*

$$a_{n_r, j_r} \neq 0, \quad a_{n_r, j_s} = 0 \quad (r \neq s).$$

Proof. Choose a nonzero entry in row $n_0 = 0$, in column j_0 . After r choices let E be the union of the supports of the chosen rows, a finite set of columns. Only finitely many rows meet E , since each column is finite. Choose a row n_r beyond all earlier chosen rows that does not meet E , and a nonzero entry in it, in column j_r . Then $j_r \notin E$, so column j_r vanishes on the earlier chosen rows, and row n_r vanishes on the earlier chosen columns, which lie in E . Induction gives the matching. The selection uses only the pattern of nonzero entries, not the characteristic or the values of the coefficients. $\square$

Third route to (iii) $\Rightarrow$ (i) in [theorem 21.4](#) [14]. Assume (iii). The tests $y = t^{-g}$ give pointwise finiteness. Suppose the union S of the supports is not well ordered, and choose $g_0 > g_1 > \dots$ in S . Put $a_{n,j} = [t^{g_n}]x_j$. Row n is nonzero because $g_n \in S$, and finite by pointwise finiteness. Column j is finite, since infinitely many g_n in $\text{supp } x_j$ would descend in a well-ordered set. [Lemma 34.1](#) gives n_r and j_r ; put

$$y = \sum_{r \geq 0} t^{-g_{n_r}}, \quad (34.1)$$

a Hahn series, since its exponents increase, with countable support and every coefficient equal to 1. Then $\langle x_{j_r}, y \rangle_0 = \sum_{s \geq 0} [t^{g_{n_s}}]x_{j_r} = a_{n_r, j_r} \neq 0$ for every r : the sum is finite, and only its term $s = r$ is nonzero. The distinct j_r contradict (iii). $\square$

The three routes differ in how cancellation is prevented. 12’s triangular detector ([lemma 21.2](#)) makes the family triangular and then chooses coefficients in $\{0, 1\}$ one at a time; 13’s binary lemma ([lemma 21.3](#)) keeps the family and chooses coefficients row by row; 14 passes to an induced matching, a diagonal submatrix, so each selected pairing is one matrix entry and nothing is chosen. 14’s witness is therefore a special case of the tests in (iii): countable support and all coefficients 1. Every construction uses a countable subset of one set-sized support, so the route works for $\Gamma = \mathbf{No}$ and for $\mathbf{No}(i)$ [14].

Corollary 34.2 (Monomial and all-ones tests suffice [14]). *A set-indexed family (x_j) in a full Hahn field $K = k((t^\Gamma))$, Γ a set or $\mathbf{No}$, is Hahn summable if and only if every exponent lies in only finitely many supports and, for every strictly increasing sequence $(h_n)_{n \in \mathbb{N}}$ in Γ , only finitely many j have $\langle x_j, \sum_n t^{h_n} \rangle_0 \neq 0$.*

Proof. Necessity is (i) $\Rightarrow$ (ii) of [theorem 21.4](#). For sufficiency the third route uses only monomial tests and the series (34.1), whose exponents $-g_{n_r}$ increase strictly. $\square$

14 states this through the proof of its detection theorem and its Proposition 11.3 ([proposition 35.2](#)).

Example 34.3 (A nonexistent sum disguised as zero [14]). Let $\delta > 0$ in Γ and, for $n \geq 1$, $x_{n,+} = t^{-n\delta}$ and $x_{n,-} = -t^{-n\delta}$. Each exponent lies in exactly two members, and grouping the members in pairs would cancel every coefficient. The family is not Hahn summable, since $\{-n\delta : n \geq 1\}$ descends. The all-ones series $y = \sum_{n \geq 1} t^{n\delta}$ gives $\langle x_{n,+}, y \rangle_0 = 1$ and $\langle x_{n,-}, y \rangle_0 = -1$ for every n : the test fails before any cancellation can hide the infinitely many contributions.

Here each exponent occurs twice with cancelling coefficients, whereas in [example 21.6](#) each exponent occurs once. The detection theorem recognizes admissibility of the family, not a value assigned by some other summation convention; neither conditional convergence nor an ordering of the index set is used [14].

Remark 34.4 (Summability is not convergence [14]). The family $(t^{n/(n+1)})_{n \geq 1}$ is Hahn summable over $\mathbb{Q}$, since its exponents increase, but its terms do not tend to zero in the valuation topology. A valuation-preserving automorphism is continuous, but continuity speaks only about families that converge in that topology, and the detector also handles bounded increasing supports; for **No**, set-indexed topological convergence meets further size obstructions. The detection theorem reduces summability to scalar product tests, without completeness or net arguments. This agrees with [remark 21.7](#). 14 identifies the dual supports, the sets meeting every well-ordered set finitely, with the reverse-well-ordered sets, as the total-order case of [4, Theorem 3 and Lemmas 4.1–4.2], and claims no origin of that duality. *Batch 32*: for ring topologies with jointly continuous multiplication the obstruction on **No** is complete and is one of density, not of size: only the indiscrete topology realizes every Hahn sum as a limit ([corollary 39.9](#)).

35 Detection in subfields with enough detectors

Definition 35.1 (Countable detector property [14]). Let K' be a subfield of $K = k((t^\Gamma))$, Γ a set or **No**, containing k and every monomial. K' has the *countable detector property* if $\sum_{n \geq 0} t^{h_n} \in K'$ for every strictly increasing sequence $(h_n)_{n \in \mathbb{N}}$ in Γ .

Proposition 35.2 (Detection in a subfield [14]). *Let K' have the countable detector property.*

- (a) *A set-indexed family in K' is Hahn summable in K if and only if, for every $y \in K'$, only finitely many $\langle x_j, y \rangle_0$ are nonzero.*
- (b) *Let σ be a field automorphism of K' with $\sigma(k) = k$ and $\text{ct}(\sigma x) = \alpha(\text{ct } x)$ for $x \in K'$, where $\alpha = \sigma|_k$. If (x_j) in K' is Hahn summable in K , so is $(\sigma(x_j))$. If moreover $s = \sum_j x_j$ lies in K' , then $\sum_j \sigma(x_j) = \sigma(s)$; in particular this sum lies in K' .*

Proof. (a) Necessity holds because $K' \subseteq K$ ([theorem 21.4](#)). The tests of [corollary 34.2](#) are monomials and series (34.1), which lie in K' ; this gives sufficiency. (b) For $y \in K'$ write $y = \sigma(z)$ with $z \in K'$. Then $\langle \sigma x_j, y \rangle_0 = \text{ct}(\sigma(x_j z)) = \alpha(\langle x_j, z \rangle_0)$, nonzero for finitely many j by (a), so $(\sigma(x_j))$ is summable in K by (a). Let s' be its sum in K . By (21.2) in K , for every $y \in K'$,

$$\langle \sigma(s), y \rangle_0 = \alpha(\langle s, z \rangle_0) = \sum_j \alpha(\langle x_j, z \rangle_0) = \sum_j \langle \sigma x_j, y \rangle_0 = \langle s', y \rangle_0,$$

with finite sums. So $\sigma(s) - s'$ pairs to zero with every $y \in K'$, in particular with every monomial, and $\sigma(s) = s'$. $\square$

The inverse satisfies the same hypotheses with α^{-1} . As in the proof of [theorem 22.1](#), the hypothesis $\sigma(k) = k$ may be replaced by covariance $\text{ct}(\sigma x) = \alpha(\text{ct } x)$ for some field automorphism α of k : since K' contains every monomial the pairing separates K' , and

$\sigma(r) = \alpha(r)$ follows for $r \in k$ [merge]. The proposition says nothing about sums that leave K' , and it concerns the summation of the ambient full Hahn field, not a summation intrinsic to K' .

Example 35.3 (Puisseux series have too few detectors [merge]). Let $K = \mathbb{R}((t^{\mathbb{Q}}))$ and let $K' = \bigcup_{N \geq 1} \mathbb{R}((t^{\frac{1}{N}\mathbb{Z}}))$ be the field of Puiseux series; it contains $\mathbb{R}$ and every monomial. The family $x_n = t^{-n/(n+1)}$, $n \geq 1$, is not Hahn summable in K , since its exponents descend. For $y \in K'$ with $\text{supp } y \subseteq \frac{1}{N}\mathbb{Z}$, $\langle x_n, y \rangle_0 = [t^{n/(n+1)}]y$ is nonzero only if $Nn/(n+1) \in \mathbb{Z}$, that is only if $n+1$ divides N , since n and $n+1$ are coprime. So every test in K' gives finitely many nonzero pairings, and (a) of [proposition 35.2](#) fails for K' . Accordingly $\sum_n t^{n/(n+1)}$, the sum of the summable family of [remark 34.4](#), is not a Puiseux series.

Remark 35.4 (Relation to support-bounded fields [merge]). For every uncountable cardinal κ , regular or singular, the field $K_{<\kappa}$ of [proposition 31.1](#) has the countable detector property. So [proposition 35.2](#) gives for all such κ what [proposition 31.1](#) gives for regular κ as far as ambient summability goes: a coefficient- and constant-term-compatible automorphism of $K_{<\kappa}$ sends every family that is summable in the sense of $K_{<\kappa}$ to a family that is Hahn summable in K , with fewer than κ nonzero members and with sum the image of the sum. For singular κ what [question 32.12](#) still asks is only whether the union of the image supports has size $< \kappa$. The warning of [remark 31.3](#), that the detector must lie in the field tested and that this fails for arbitrary subfields, stands: [example 35.3](#) is such a failure, and the countable detector property is a sufficient condition, not a characterization ([question 37.1](#)). 14 does not claim the property for transseries, Puiseux or left-finite subfields; it notes that a series on an increasing sequence of exponents bounded above can fail a left-finiteness restriction [14]. [Question 32.16](#) (11) asks the analogous question for the bottom-gap embeddings.

36 Further consequences

Theorem 36.1 (Formal evaluation is transported [14]). *Let Γ be **No** or a nonzero divisible set-sized ordered abelian group, and let σ be a field automorphism of $K_\Gamma = \mathbb{R}((t^\Gamma))$ with $\sigma(\mathcal{R}_\Gamma) = \mathcal{R}_\Gamma$. For $F \in \mathbb{R}[[X_1, \dots, X_m]]$ and $z_1, \dots, z_m \in \mathfrak{m}_\Gamma$,*

$$\sigma(F(z_1, \dots, z_m)) = F(\sigma(z_1), \dots, \sigma(z_m)). \quad (36.1)$$

In particular $\sigma(\exp z) = \exp(\sigma z)$ and $\sigma(\log(1+z)) = \log(1+\sigma z)$ for $z \in \mathfrak{m}_\Gamma$.

Proof. Write $F = \sum_\nu c_\nu X^\nu$ over $\nu \in \mathbb{N}^m$. The union of the supports of the z_j is a well-ordered subset of $\Gamma^{>0}$, so the Neumann support principle ([section 2](#)) makes $(c_\nu z^\nu)_\nu$ Hahn summable; its sum is the formal evaluation. By [proposition 23.1](#) and [corollary 23.4](#) (and [theorem 23.2](#) for **No**), σ fixes $\mathbb{R}$, maps $\mathfrak{m}_\Gamma$ onto itself and is strongly $\mathbb{R}$ -linear, so $\sigma(\sum_\nu c_\nu z^\nu) = \sum_\nu c_\nu \sigma(z)^\nu$, and the $\sigma(z_j)$ are again infinitesimal. $\square$

No radius of convergence enters: a formal series with arbitrarily fast-growing real coefficients is evaluated at Hahn infinitesimals through support finiteness. For a valuation-compatible Gaussian automorphism as in [theorem 30.2](#), with $\alpha = \sigma|_{\mathbb{C}}$, the same proof gives $\sigma(F(z)) = (\alpha F)(\sigma(z))$, where α acts on the coefficients; for the conjugation-compatible automorphisms of [theorem 30.1](#), α is the identity or complex conjugation [14]. The exponential here is the local one at infinitesimals ([section 2](#)); nothing is claimed about the Gonshor exponential of infinite arguments, whose compatibility is a further condition studied in [15].

Proposition 36.2 (The adjoint as a sign-reversed transpose [14]). *In the setting of [section 25](#), let $T : K \rightarrow L$ be strongly k -linear and put $T_{\eta,\gamma} = [t^\eta]T(t^\gamma)$ for $\gamma \in \Gamma$, $\eta \in \Delta$. Then for every*

$y = \sum_{\eta} y_{\eta} t^{\eta} \in L$ and $\gamma \in \Gamma$

$$[t^{\gamma}]T^{\dagger}(y) = \sum_{\eta \in \Delta} T_{-\eta, -\gamma} y_{\eta}, \quad \text{so} \quad (T^{\dagger})_{\gamma, \eta} = T_{-\eta, -\gamma}, \quad (36.2)$$

with finitely many nonzero terms.

Proof. $[t^{\gamma}]T^{\dagger}(y) = \langle t^{-\gamma}, T^{\dagger}y \rangle_0 = \langle T(t^{-\gamma}), y \rangle_0$ by (25.1), and the last constant term is $\sum_{\eta} [t^{-\eta}]T(t^{-\gamma}) y_{\eta}$, finite by lemma 21.1. $\square$

For a coefficient-fixing isomorphism F with $\text{ct} \circ F = \text{ct}$ one has $F^{\dagger} = F^{-1}$ (theorem 25.2), and (36.2) becomes the coefficient-matrix formula (22.2) of 13. The general formula is 14's; it presupposes the strongness of T , which theorem 25.2 supplies, and 14's suite checks it only on finite matrices (section C).

Remark 36.3 (Rank one with $\Gamma = \mathbb{R}$ [14, 15]). By corollary 24.3 every field automorphism of $\mathbb{R}((t^{\mathbb{R}}))$ preserving $\mathcal{R}_{\mathbb{R}}$ is a monomial map $M_{\chi, \tau}$. Here $\tau(\gamma) = \lambda\gamma$ with $\lambda > 0$, since an order-preserving additive bijection of $\mathbb{R}$ is monotone and additive, hence $\mathbb{R}$ -linear; and $\chi(\gamma) = \exp(\theta(\gamma))$ with $\theta: \mathbb{R} \rightarrow \mathbb{R}$ additive, since χ is positive. θ need not be continuous: any $\mathbb{Q}$ -linear map defined on a Hamel basis will do, and every such pair (λ, θ) gives an automorphism by (5.1). So rank one removes the factor u , not all automorphisms and not all nonclassical coefficient scalings. For $\Gamma = \mathbb{Q}$ the character is $q \mapsto q^c$ (the paragraph after theorem 5.1).

Remark 36.4 (The shear of source 14 and truncation [14, 15]). Let $\Gamma = \mathbb{Q}e_1 \oplus \mathbb{Q}e_2$ with $e_1 \ll e_2$ as in example 4.16, and let θ be the e_2 -coordinate. 14's automorphism, in these letters,

$$t^{ae_2+be_1} \mapsto t^{ae_2+be_1}(1+t^{e_1})^a \quad (a, b \in \mathbb{Q}),$$

is the binomial twist $\sigma_{e_1, \theta}^{1+\xi}$ of (4.5), so it is a strong $\mathbb{R}$ -linear 1-automorphism preserving $\mathcal{R}_{\Gamma}$ (theorem 4.18, since $\theta(C_{e_1}) = 0$); it is the inverse of example 4.19. It sends t^{-e_2} to $\sum_{n \geq 0} (-1)^n t^{-e_2+ne_1}$, a monomial to an infinite series. With $\gamma = -e_2 + \frac{1}{2}e_1$ and $T_{<\gamma}$ the truncation below γ ,

$$T_{<\gamma}(\sigma(t^{-e_2})) = t^{-e_2}, \quad \sigma(T_{<\gamma}(t^{-e_2})) = \sigma(t^{-e_2}) \neq t^{-e_2}.$$

So automatic strongness does not give commutation with truncation at a monomial, even when the exponent automorphism is the identity. This does not contradict theorem 8.2: the definable truncation $T_{<\gamma}(f) = mP_{-}(f/m)$ uses the parameter $m = t^{\gamma}$, and σ carries it to $f \mapsto \sigma(m)P_{-}(f/\sigma(m))$, truncation relative to $\sigma(m) = t^{\gamma}(1+t^{e_1})^{-1}$, which is not a monomial. The tempting reading “every **Oz**-preserving automorphism commutes with truncation at every monomial” is false; the true statements are commutation with P_{-}, P_0, P_{+} (corollary 5.2) and with the projection onto any union of signed Archimedean blocks (proposition 3.10), in agreement with the paragraph after that proposition. 14 relates naming truncations to the complexity studied by Camacho [5].

Proposition 36.5 (Extensions from $\mathcal{R}_{\Gamma}$ are unique [merge]). *Let Γ be a nonzero divisible set-sized ordered abelian group. A ring automorphism σ of $\mathcal{R}_{\Gamma}$ has at most one extension to a field automorphism of K_{Γ} ; an extension, if one exists, fixes $\mathbb{R}$ and is strongly $\mathbb{R}$ -linear with strongly $\mathbb{R}$ -linear inverse.*

Proof. An extension $\tilde{\sigma}$ satisfies $\tilde{\sigma}(\mathcal{R}_{\Gamma}) = \sigma(\mathcal{R}_{\Gamma}) = \mathcal{R}_{\Gamma}$, so by corollary 23.4 it fixes $\mathbb{R}$ and is strongly $\mathbb{R}$ -linear with strong inverse. Its values on monomials are forced: $\tilde{\sigma}(t^{\gamma}) = \sigma(t^{\gamma})$ for $\gamma < 0$, since $t^{\gamma} \in \Pi_{\Gamma}$, and $\tilde{\sigma}(t^{\gamma}) = \sigma(t^{-\gamma})^{-1}$ for $\gamma > 0$. Applied to the normal-form family $(a_{\gamma}t^{\gamma})$ of any f , strongness gives $\tilde{\sigma}(f) = \sum_{\gamma} a_{\gamma}\tilde{\sigma}(t^{\gamma})$, which depends only on σ . $\square$

This answers the uniqueness clause of 14's Question 12.7 without assuming $\text{Frac } \mathcal{R}_\Gamma = K_\Gamma$, which fails in general (example 31.2); 14 notes only that uniqueness on the fraction field is automatic. Existence is question 32.14.

Remark 36.6 (Gaussian statements of source 14 [14]). 14's real-structure theorem assumes $\sigma(\mathbf{No}) = \mathbf{No}$, where theorem 30.1 assumes commutation with conjugation. The hypotheses are equivalent: conjugation fixes exactly $\mathbf{No}$, and conversely $\sigma(i) = \pm i$ gives $\sigma(x + iy) = \sigma(x) \pm i\sigma(y)$, which commutes with conjugation when $\sigma(\mathbf{No}) = \mathbf{No}$. 14's example over $\Gamma = \mathbb{Q}$, the strong map $\sum a_q t^q \mapsto \sum a_q e^{i\pi q/2} t^q$, preserves $\mathcal{R}_{\mathbb{Q}}^{\mathbb{C}}$, fixes $\mathbb{C}$ and sends t to it ; on $\mathbf{No}(i)$ it uses $\chi(\gamma) = \exp(i\pi\theta(\gamma)/2)$ with $\theta(\gamma) = \text{ct}(\gamma)$. It is a phase twist of the kind recorded after theorem 10.2 (odg: def: lem: twist, saut: thm: phase). Since it is strong, it does not bear on question 32.1; it only shows that preservation of $\mathbf{Oz}[i]$ alone does not give the hypothesis of theorem 30.1.

37 Questions of source 14, and stale statements

37.1 Status of the source's questions

14 poses eight questions (its 12.1–12.8), proposed as research problems, not as published open problems, and repeats its Gaussian question at the end of its Section 10. Seven are merged into questions already in the report, with 14 added to their sources:

- 12.1 and the Section-10 question (must every automorphism of $\mathbf{No}(i)$ preserving $\mathbf{Oz}[i]$ be strong; does $\mathbf{Oz}[i]$ determine the valuation ring, or at least ct -equivariance?): question 32.1, open; 14's suggested first targets are recorded there.
- 12.2 (which hypotheses make a constant-term-compatible embedding strong; embeddings of $\mathbf{No}$ preserving $\mathbf{Oz}$): questions 32.2 and 32.3. The first clauses of the former have a negative answer given a measurable cardinal, as recorded there.
- 12.4 (class-supported kernels for strong functionals on $\mathbf{No}$): question 32.11.
- 12.5 (other integer parts): question 32.13, and question 11.7 for the criterion.
- 12.6 (a parameter space for the normalized factor): question 11.2, with question 11.4; open. 14 adds that the convex criterion recognizes existing automorphisms and does not construct them, and that a parameterization must distinguish pointwise positive support from uniform control on summable families.
- 12.7 (does every automorphism of $\mathcal{R}_\Gamma$ extend to K_Γ , uniquely?): the automorphism case of question 32.14. Uniqueness is answered by proposition 36.5; existence stays open.
- 12.8 (a kernel-checked automatic-strongness theorem): question 32.18.

One question is new:

Question 37.1 (Minimal closure for algebraic detection [14, 15]). Can the countable detector property be weakened to a necessary and sufficient closure condition on a subfield K' for the product detection of proposition 35.2(a)? Which natural computable or transseries subfields satisfy that condition?

The induced-matching proof needs one all-ones detector for a subsequence of each bad configuration, not every countable series with arbitrary coefficients, so a closure principle could quantify over bad support configurations rather than over all increasing sequences. The question has algebraic and computability aspects: the existence of a detector need not give an

algorithm that recognizes or constructs it from a chosen representation [14]. [Example 35.3](#) shows that some closure condition is needed [merge]. 15 asks the same (its Question 5): which proper subfields admit the detector, whether closure under the countable sparse detectors is equivalent to a natural support condition, and with what effect on constant-term-preserving embeddings and isomorphisms; it asks for necessary and sufficient support conditions rather than sufficient completeness hypotheses [15]. *Status* (batch 32): partly addressed by the sufficient condition of [definition 35.1](#) and [proposition 35.2](#); open.

37.2 Stale and checked statements

- *Stale novelty.* 14’s abstract says that “the strong-additivity hypothesis can be removed” from this report’s real stabilizer classification, its introduction that it “supplies a missing implication”, and its questions section that its main result “closes the automatic-strongness question for the real omnific pair”. At its pin 3d40856 this was accurate: [question 11.1](#) was open. At the merge it is stale: sources 12 and 13 answered [question 11.1](#) in 20c4c9f ([theorem 23.2](#)). 14 is recorded as an independent third derivation, and its priority statement is not printed as current; its own caveat that global priority is not certified is kept.
- *Checked true.* 14’s description of the pinned report (the criterion concerns strong coefficient-fixing 1-automorphisms; reconstruction and criterion are credited), the example $\text{Frac}(\mathbb{Z} \oplus t^{-1}\mathbb{R}[t^{-1}]) = \mathbb{R}(t)$ for $\Gamma = \mathbb{Z}$, and the finite checks of its suite, rerun for the merge ([section C](#)). No mathematical error was found in 14.
- *Not rechecked.* 14’s pinpoint citations of [17] (an example of a non-strongly additive automorphism) and of [4] (Theorem 3, Lemmas 4.1–4.2) are reported as 14 gives them.

14’s limitations are listed with the other sources’ in [section B](#).

Part V

Topological collapse of Hahn summation

This part records manuscript 15, an addition of batch 32. Its new subject is topological: which ring topologies, with jointly continuous multiplication, realize every Hahn sum as the limit of its net of finite partial sums? For a densely ordered exponent group every such topology on $\mathfrak{o} \oplus \Pi$ is pulled back from the constants along ct , so none is Hausdorff; on $\mathbf{No}$ and $\mathbf{No}(i)$ only the indiscrete topology qualifies; and a set-sized full Hahn field has a Hausdorff one exactly when its exponent group is 0 or order-isomorphic to $\mathbb{Z}$ ([section 39](#)). The algebraic half of 15, a fourth derivation of the automatic strongness of [part III](#), is printed once there with 15 added to the sources; what it adds to that half is in [section 40](#), and its pairing topology, which realizes exactly the Hahn sums but has only separately continuous multiplication, in [section 41](#). Its labels carry the sub-prefix `opa:tc:.`

38 Source 15, what it re-derives, and its notation

38.1 The manuscript and what it read

Manuscript 15 is *Algebraic Rigidity and Topological Collapse of Surreal Arithmetic: Hahn summation, omnific integer topologies, and the discrete-exponent boundary*, dated 23 September

2026, 29 pages. It is item 03 of batch 32, delivered as the archive `Surreal_Arithmetic_Topological_Collapse` in commit `aa268a4` and placed as an addition in commit `7d04483`; it is numbered 15 here, continuing this report’s local sequence. Its finite checks, their recorded run, its build record, its `Makefile` and its source audit are shipped as `code/15-topological-collapse-verify.py`, `data/15-topological-collapse-verification.json`, `data/15-topological-collapse-BUILD_AUDIT.json`, `code/15-topological-collapse-Makefile` and `15-topological-collapse-SOURCE_AUDIT.md`; its `.tex`, PDF and README are not shipped.

15 names no repository commit. Its pins are two blob identifiers of this report’s `article.tex`, read through a connector: `059939b5`, the file as written in `b3fa9e2` (batch 28: [parts I](#) and [II](#) with source 10’s material), in which [question 11.1](#) was still open, and `60f3b172`, the file as written in `20c4c9f` (batch 30, with [part III](#)), which it found in a later directory read together with the audits of 11, 12 and 13 (blobs `050b52f`, `0e4e3a1`, `c4d3939`, as placed in `21375f8`). The merge checked all five identifiers against the history. So 15 saw [part III](#) but neither [part IV](#) (`d4d72d7`) nor the batch-31 reciprocal notes (`ab18444`). Its account of this report is accurate for what it read: it credits automatic strongness, scalar detection, constant-term covariance, adjoints, the class-size failure of duality and the qualified Gaussian versions to 12 and 13, and the criterion, the reconstruction, the strong factorization and the shear examples to this report; it claims as new only its Section 3, the topology. Its reading was incomplete, as its audit says: it does not mention 14’s induced-matching route or detection in subfields, and it does not mention the congruence topologies of the quotient and Diophantine reports, which are instances of its classification ([remark 39.6](#)).

38.2 Printed once

The table maps 15’s statements to the results where they are printed; 15’s numbers are those of its own PDF. “Same” means the same statement up to notation.

15	Printed as	Comparison
Lem. 2.2	lemma 2.1 , the support principle of section 2 ; lemma 21.1	Same.
Thm. 4.2	theorem 21.4	Same, except that 15’s (iii) allows all countably supported tests, without the restriction to coefficients in $\{0, 1\}$; its disjoint-row proof is a variant of the third route (lemma 40.1).
Rem. 4.3	example 40.3	New example, beside example 34.3 .
Cor. 4.4	corollary 40.2	New.
Prop. 4.5	example 21.6	The same family; 15 states it for all finite-support tests.
Thm. 5.1, Rem. 5.2	theorem 22.1 and the paragraph after it	15 assumes $\sigma(k) = \ell$ and $\alpha = \sigma _k$, as 13 and 14 do.
Prop. 5.3	“Adjoint implies strong” in theorem 25.2	Same, for full Hahn fields of any size, as 13 notes.

15	Printed as	Comparison
Lems. 6.1, 6.2, Rem. 6.3	lemma 2.3 , <code>odg: def: thm: multiplier</code> , <code>odg: def: thm: autreal</code> ; proposition 23.1	Re-derivation.
Thm. 6.4	proposition 23.1 and corollary 23.4	Same, for one group.
Lem. 7.1	<code>odg: thm: fractions</code>	Same.
Thm. 1.2, its proof in Section 7, Cors. 7.2, 7.4	theorem 23.2 , (23.1)	Same; 15 credits 12 and 13.
Cor. 7.3; Rem. 7.5	corollary 23.5 ; remark 23.3	Same.
Thm. 8.1	theorem 24.1	Same.
Prop. 8.2	theorem 3.3 , conditions (i), (iv), (v)	Re-proof for $(\mathbb{R}, \mathbb{Z})$ by 04's polynomial route; 15 credits this report.
Rem. 8.3	remark 24.2	Same.
Cor. 8.4, the forms for $\mathbb{Q}$ and $\mathbb{R}$	corollary 24.3 , the paragraph after theorem 5.1 , remark 36.3	Same.
Ex. 8.5	remark 36.4	The same map, the inverse of example 4.19 .
Lem. 9.1	lemma 2.3 , <code>odg: def: thm: multiplier</code>	Re-derivation.
Thms. 9.2, 9.3	theorems 30.1 and 30.2	Same.
Cor. 9.4	corollary 40.4	New, the ring form.
Thm. 10.2, Cor. 10.3	proposition 25.1 and theorem 25.2	Same.
Prop. 10.4, Rem. 10.6	theorem 25.6 and the paragraph after it	Same.
Prop. 10.5	proposition 40.5	New.
Section 3; Section 11	section 39 ; section 41	New.
Questions 1–12	section 42	Six new, six merged.

So 15 is a fourth independent derivation of the automatic strongness of the real pair, after 12, 13 and 14, and it says so: it keeps its proof “for the comparison with the topological collapse theorem, not as a new resolution”. Where its proofs of shared results differ only in presentation they are not reprinted.

38.3 Conventions and notation of this part

This part uses the small- t convention ([1.1](#)) and the notation of [section 20.3](#): $K_\Gamma = k((t^\Gamma))$, Π_Γ , $\mathfrak{m}_\Gamma$, and for $k = \mathbb{R}$ $\mathcal{R}_\Gamma = \mathbb{Z} \oplus \Pi_\Gamma$ and $\mathcal{A}_\Gamma = \mathbb{R} \oplus \Pi_\Gamma$, with the pairing $\langle x, y \rangle_0 = \text{ct}(xy)$. For a field k and a unital subring $\mathfrak{o} \subseteq k$, $\mathcal{R}_\mathfrak{o} = \mathfrak{o} \oplus \Pi$ is the ring of [section 1.5](#), but in [section 39](#) the exponent group is only assumed *densely ordered* (nonzero, with no least positive element), not

divisible; divisible groups are densely ordered, and $\mathcal{R}_\mathfrak{o}$ is then 15's "support ring". $\text{ct} : \mathcal{R}_\mathfrak{o} \rightarrow \mathfrak{o}$ is a ring homomorphism with kernel Π . 15's symbols are renamed as follows.

15's symbol	Here
$\mathcal{T}, \mathcal{T}_0, \mathcal{T}_\beta$, topologies	$\mathfrak{T}, \mathfrak{T}_0, \mathfrak{T}_\beta$.
$R_{\mathfrak{o}, \Gamma} = \mathfrak{o} \oplus \Pi_k$	$\mathcal{R}_\mathfrak{o} = \mathfrak{o} \oplus \Pi$, with Γ densely ordered.
$N = \overline{\{0\}}$	$\overline{\{0\}}$, the closure of zero; $\mathbb{N}$ stays the natural numbers.
D, E (sets of exponents); D_U, D_V	$A, B; A_U, A_V$.
$R_\Gamma, A, \mathfrak{m}; \Pi_\mathbb{C}, A_\mathbb{C}$	$\mathcal{R}_\Gamma, \mathcal{A}_\Gamma, \mathfrak{m}_\Gamma; \Pi_\mathbb{C}, \mathbb{C} \oplus \Pi_\mathbb{C}$.
$\rho = \sigma _k$	α , as in theorem 22.1 .
$\mathcal{U}_{R_\Gamma}; h(g), a_\delta$	$\mathcal{U}_{\mathcal{R}_\mathbb{Z}}(\Gamma); h_u(g), a_{u,\delta}$ of (3.1) .
$\langle f, h \rangle$; summation functional; K^Σ	$\langle f, h \rangle_0$; strong k -linear functional (section 25); the space of these.
$\Lambda; D = \{g : \lambda(t^g) \neq 0\};$ D_a	$\lambda_{\mathbf{On}}$ of (25.2) ; $S_\lambda; S_a$.
$\sigma_{c,r}$	$M_{\chi,\tau}$ with $\chi(q) = c^q, \tau(q) = rq$.
$\mathbf{No}[i], \mathbf{Oz}[i]; c(z)$	$\mathbf{No}(i), \mathbf{Oz}[i]; \bar{z}$.
$\text{Fin}(J)$	$\text{Fin}(J)$, the finite subsets of J directed by inclusion.

The collisions are resolved here only, and the tempting false readings are wrong:

- 15's $\mathcal{T}$ is a topology. In [part III](#) $\mathcal{T}_{d,\delta}$ is the Taylor automorphism, T a linear map with adjoint $T^\dagger$, $T_{<\gamma}$ a truncation and T^1 a flow; none of them is a topology, and the topologies of this part are written $\mathfrak{T}$.
- 15's N is the closure of zero, not the natural numbers.
- 15's sets D and E are not derivations and not the flow $\mathcal{E}$; the letters A, B used instead are sets of exponents, not the rings $\mathcal{A}_k, \mathcal{A}_\Gamma$.
- The neighbourhoods U, V, W of zero are local to this part: V is not the span V_A of [theorem 7.4](#) or a valuation ball $\mathcal{V}_\lambda$, and U is neither $\mathcal{U}$ nor a monomial U^g of [part VI](#).
- 15's "Automorphisms invisible to every compatible omnific topology" is [corollary 39.10](#). It is *not* [corollary 24.4](#), which describes algebraically what an automorphism of $\mathbf{Oz}$ does inside Π ; the topological statement says that no compatible topology can see that action at all.
- 15's $\mathbf{No}[i]$ is the field $\mathbf{No}(i) = \mathbf{No}(i)$ of this report, the same field. Its Hahn-field macro prints $k((t^\Gamma))$, as here; it has no symbol for the relative Hahn hull $\mathcal{H}(A)$ of [theorem 7.4](#).

For proper-class rings the topological statements are read with class-coded neighbourhood systems: a class of indices and one class relation describing membership in each basic neighbourhood. No class whose members are arbitrary proper-class open sets is formed; proofs use individual neighbourhoods, countable sequences and set-supported normal forms [15], in the sense of [section 1.4](#).

39 Hahn-compatible ring topologies

Definition 39.1 (Hahn-compatible ring topology [15]). A *Hahn-compatible ring topology* on a Hahn field, or on one of its subrings A , is a topology $\mathfrak{T}$ making addition, additive inversion and multiplication continuous, such that for every Hahn-summable family $(f_j)_{j \in J}$ of elements of A whose sum lies in A ,

$$\sum_{j \in F} f_j \longrightarrow \sum_{j \in J} f_j \quad (F \in \text{Fin}(J)).$$

Hausdorffness is not assumed; multiplication is required to be *jointly* continuous; the coefficients need not carry their usual topology, and no continuity of ct is assumed.

A much weaker consequence drives the negative results. If $g_0 < g_1 < \dots$ and the monomials at^{g_n} and their Hahn sum lie in A , then

$$at^{g_n} \longrightarrow 0. \quad (39.1)$$

Indeed the partial sums over $\{0, \dots, n\}$ form a cofinal subnet and converge to the sum; their consecutive differences are the terms [15]. So the obstructions below are already forced by countable monomial series.

Lemma 39.2 (Two opposite well orders cannot cover a dense interval [15]). *Let I be a nonempty open interval of a densely ordered set. There are no A, B with $I = A \cup B$ such that A contains no strictly increasing sequence and B contains no strictly decreasing sequence.*

Proof. If B met every nonempty open subinterval of I , density would give recursively a strictly decreasing sequence in B : after $b_n \in B \cap I$, the interval between the left end of I and b_n is nonempty and open. So some nonempty open subinterval $I' \subseteq I$ misses B , hence lies in A ; but I' contains a strictly increasing sequence. $\square$

Lemma 39.3 (Monomial collapse [15]). *Let k be any field, $\mathfrak{o} \subseteq k$ a unital subring and Γ a densely ordered abelian group. If a ring topology on $\mathcal{R}_{\mathfrak{o}}$ satisfies (39.1) for every $a \in k$ and every strictly increasing sequence of negative exponents, then every at^γ with $\gamma < 0$ lies in the closure of zero.*

Proof. Let $a \neq 0$ and $\gamma < 0$, and suppose a neighbourhood W of 0 misses at^γ . Joint continuity gives neighbourhoods U, V of 0 with $UV \subseteq W$. On the interval $I = (\gamma, 0)$ put

$$A = \{x \in I : at^x \notin U\}, \quad B = \{x \in I : t^{\gamma-x} \notin V\}.$$

By (39.1) A contains no strictly increasing sequence, and B none strictly decreasing, since $x \mapsto \gamma - x$ reverses order and maps I into the negative exponents. For $x \in I$ both monomials lie in $\mathcal{R}_{\mathfrak{o}}$ and $(at^x)t^{\gamma-x} = at^\gamma \notin W$, so $x \in A \cup B$, contradicting lemma 39.2. Hence every neighbourhood of 0 contains at^γ , which in a topological group is membership in $\overline{\{0\}}$. $\square$

Only countable monomial null sequences are used: no metric, first countability or description of the neighbourhoods [15].

Theorem 39.4 (Topological collapse onto the constants [15]). *Let k be a field, $\mathfrak{o} \subseteq k$ a unital subring and Γ a densely ordered abelian group. The Hahn-compatible ring topologies on $\mathcal{R}_{\mathfrak{o}} = \mathfrak{o} \oplus \Pi$ are exactly the pullbacks*

$$\mathfrak{T} = \text{ct}^{-1}(\mathfrak{T}_0) = \{\text{ct}^{-1}(O) : O \in \mathfrak{T}_0\} \quad (39.2)$$

of the ring topologies $\mathfrak{T}_0$ on $\mathfrak{o}$, and $\mathfrak{T}_0$ is determined by $\mathfrak{T}$. In particular:

- (i) every such topology has $\Pi \subseteq \overline{\{0\}}$;
- (ii) none is Hausdorff when $\Pi \neq 0$;
- (iii) the finest one is the pullback of the discrete topology of $\mathfrak{o}$;
- (iv) $\overline{\{0\}}^{\mathfrak{T}} = \text{ct}^{-1}(\overline{\{0\}}^{\mathfrak{T}_0})$, and the separated quotient of $(\mathcal{R}_{\mathfrak{o}}, \mathfrak{T})$ is that of $(\mathfrak{o}, \mathfrak{T}_0)$.

Proof. Let $\mathfrak{T}$ be compatible. The closure $\overline{\{0\}}$ is a closed ideal: a closed subgroup, and multiplication by a fixed element is continuous and fixes 0. By [lemma 39.3](#) every negative monomial lies in $\overline{\{0\}}$. Every $p \in \Pi$ is the Hahn sum of its terms; the finite partial sums lie in $\overline{\{0\}}$ and converge to p , so $p \in \overline{\{0\}}$. Thus $\Pi \subseteq \overline{\{0\}}$.

In a topological group every open set O is invariant under translation by $\overline{\{0\}}$: for $x \in O$ choose a neighbourhood V of 0 with $x + V \subseteq O$; as $\overline{\{0\}} \subseteq V$, $x + \overline{\{0\}} \subseteq O$. So every open set is a union of Π -cosets, that is $O = \text{ct}^{-1}(O_0)$ for a unique $O_0 \subseteq \mathfrak{o}$. The sets O_0 form the quotient ring topology on $\mathcal{R}_{\mathfrak{o}}/\Pi \cong \mathfrak{o}$, which is also the subspace topology on the constants $\mathfrak{o} \subseteq \mathcal{R}_{\mathfrak{o}}$ (since $O \cap \mathfrak{o} = O_0$); this gives joint continuity of the induced operations, uniqueness and [\(39.2\)](#).

Conversely, the pullback of a ring topology along the ring homomorphism ct is a ring topology. In a Hahn-summable family only finitely many members have a nonzero constant coefficient, so the constant terms of the finite partial sums are eventually equal to the constant term of the sum; hence the partial sums converge in every pullback topology. (i)–(iv) follow from [\(39.2\)](#). $\square$

The theorem already applies to the set-sized ring $\mathbb{Z} \oplus \Pi$ with $\Gamma = \mathbb{Q}$. So its force is not a proper-class cardinality obstruction of the kind that governs set-sized quotients (`osq:thm:universal` of the quotient report); it comes from a dense exponent interval, formal summation and jointly continuous multiplication together [\[15\]](#).

Corollary 39.5 (A universal separated recipient [\[15\]](#)). *Let B be a Hausdorff topological ring and $\varphi : \mathcal{R}_{\mathfrak{o}} \rightarrow B$ a ring homomorphism taking every Hahn sum to the limit of the corresponding finite partial sums. Then φ factors uniquely through $\text{ct} : \mathcal{R}_{\mathfrak{o}} \rightarrow \mathfrak{o}$.*

Proof. The initial topology of φ is a Hahn-compatible ring topology, and its closure of zero is $\ker \varphi$ because B is Hausdorff. By [theorem 39.4](#), $\Pi \subseteq \ker \varphi$. $\square$

The hypothesis is on the treatment of sums. Arbitrary ring maps from a set-sized support ring need not factor through ct : the identity does not [\[15\]](#).

Remark 39.6 (The congruence topologies are instances [\[merge\]](#)). The collection already has compatible topologies on **Oz**. The finite-congruence topology and, for an ordinary prime p , the p -adic topology are pulled back along ct from the profinite and the p -adic topologies of $\mathbb{Z}$ (`odg:eq:profinite` of the Diophantine report and the paragraph after `osq:thm:completions` of the quotient report, both recording that they are not Hausdorff and that $\Pi = \overline{\{0\}}$). The Lean modules `Surreal/Foundations/OmnificCongruenceTopology.lean` and `Surreal/Foundations/OmnificSeparationTopology.lean` prove that Π is the closure of zero, that neither topology is T_2 , and that ct induces each topology from the corresponding topology of $\mathbb{Z}$ (`isInducing_omnificConstantCoeff_congruence` and its p -adic analogue; see the rows of `docs/FORMALIZATION.md` following `odg:eq:profinite`). By the converse half of [theorem 39.4](#) these topologies are Hahn-compatible, and by [theorem 39.7](#) below their failure to be Hausdorff is forced: every Hahn-compatible ring topology on **Oz** is such a pullback. The Lean results concern those two topologies; nothing of this part is formalized.

Theorem 39.7 (Omnific topologies [15]). *Every Hahn-compatible ring topology on $\mathbf{Oz}$ is the pullback along $\text{ct} : \mathbf{Oz} \rightarrow \mathbb{Z}$ of a unique ring topology on $\mathbb{Z}$, and every Hahn-compatible ring topology on $\mathbf{Oz}[i]$ is the pullback of a unique ring topology on $\mathbb{Z}[i]$. No such topology on either ring is Hausdorff.*

Proof. $\mathbf{No}$ is divisible, hence densely ordered. The proof of [theorem 39.4](#) is elementwise, with $(k, \mathfrak{o}) = (\mathbb{R}, \mathbb{Z})$ or $(\mathbb{C}, \mathbb{Z}[i])$; the descended topology is a topology on the set $\mathbb{Z}$ or $\mathbb{Z}[i]$, and no proper-class sum is formed. $\square$

Proposition 39.8 (Countable monomial tests suffice for the full omnific rings [15]). *In [theorem 39.7](#) it suffices to require (39.1) for countable strictly increasing sequences of negative exponents and fixed coefficients: every ring topology on $\mathbf{Oz}$ or $\mathbf{Oz}[i]$ with this property is a pullback along ct , hence Hahn-compatible.*

Proof. [Lemma 39.3](#) still puts every negative monomial into the closed ideal $\overline{\{0\}}$. For $p \in \Pi$, the support of p is a set of negative surreals; choose a positive surreal δ below every $-g$, $g \in \text{supp } p$, by the cut $\{0\} \mid \{-g : g \in \text{supp } p\}$. Then $t^\delta p \in \Pi$ and $p = t^{-\delta}(t^\delta p) \in \overline{\{0\}}$, an ideal. This replaces the convergence of a possibly uncountable normal form; the second paragraph of the proof of [theorem 39.4](#) finishes. $\square$

The positive buffer δ is a property of the class $\mathbf{No}$. For $\Gamma = \mathbb{Q}$ the support $\{-1/n : n \geq 1\}$ has none, so the proof does not transfer unchanged to set-sized groups [15]; whether the conclusion does is part of [question 42.2](#).

Corollary 39.9 (The surreal and surcomplex fields collapse [15]). *The only Hahn-compatible ring topology on $\mathbf{No}$ or on $\mathbf{No}(i)$ is the indiscrete topology; already the countable condition (39.1) forces it. In particular no Hausdorff topological-field structure on either field realizes every Hahn sum as a topological limit.*

Proof. [Lemma 39.3](#) applies inside either field and puts $t^{-1} \neq 0$ into the ideal $\overline{\{0\}}$, which is therefore the whole field; a topological group in which 0 is dense is indiscrete. The indiscrete topology is trivially compatible. $\square$

Corollary 39.10 (Automorphisms are topologically invisible [15]). *For every ring automorphism σ of $\mathbf{Oz}$ and every Hahn-compatible ring topology on $\mathbf{Oz}$, the elements x and $\sigma(x)$ have the same neighbourhoods. So σ is a homeomorphism and acts trivially on the separated quotient.*

Proof. σ fixes $\mathbb{Z}$ and preserves $\Pi = \bigcap_n n\mathbf{Oz}$ ([lemma 2.3](#)), so $\text{ct} \circ \sigma = \text{ct}$ on $\mathbf{Oz}$; by [theorem 39.7](#) every open set is $\text{ct}^{-1}(O_0)$. $\square$

Algebraically the same automorphisms are large and act on every Archimedean scale ([corollary 24.4](#)); every compatible topology is blind to all of it.

Lemma 39.11 (Covers of an ordered group by opposite well orders [15]). *Let $\Gamma \neq 0$ be a set-sized ordered abelian group with $\Gamma = A \cup B$, where A is reverse well ordered and B is well ordered. Then $\Gamma \cong \mathbb{Z}$ as an ordered group.*

Proof. $A \neq \emptyset$, since Γ is not well ordered ($-n\delta$, $n \geq 1$, descends for $\delta > 0$); so A has a greatest element d , and $(d, \infty) \subseteq B$ is well ordered. Translating $\Gamma^{\geq 0}$ into (d, ∞) by an element above d shows that $\Gamma^{\geq 0}$ is well ordered. Let ϵ be the least positive element. For $g > 0$ some n has $g < n\epsilon$, since otherwise $g - n\epsilon$ would descend in $\Gamma^{\geq 0}$; for the least such n , $0 \leq g - (n-1)\epsilon < \epsilon$ forces $g = (n-1)\epsilon$. So $\Gamma = \mathbb{Z}\epsilon$. $\square$

Theorem 39.12 (Exact boundary for full Hahn fields [15]). *Let Γ be a set-sized ordered abelian group and k a field. The full Hahn field $k((t^\Gamma))$ admits a Hausdorff Hahn-compatible ring topology if and only if $\Gamma = 0$ or $\Gamma \cong \mathbb{Z}$ as an ordered group. More strongly, if Γ is neither, every ring topology on $k((t^\Gamma))$ in which every strictly increasing sequence of monomials t^{g_n} tends to 0 is indiscrete.*

Proof. Suppose such a topology is not indiscrete. Its closure of zero is a proper ideal of a field, hence 0. Choose a neighbourhood W of 0 with $1 \notin W$ and neighbourhoods U, V of 0 with $UV \subseteq W$, and put $A_U = \{g : t^g \notin U\}$, $A_V = \{g : t^g \notin V\}$. Neither contains a strictly increasing sequence, so both are reverse well ordered. Since $t^g t^{-g} = 1 \notin W$, $\Gamma = A_U \cup (-A_V)$, and $-A_V$ is well ordered; lemma 39.11 gives $\Gamma \cong \mathbb{Z}$ if $\Gamma \neq 0$.

Conversely, for $\Gamma = 0$ take the discrete topology of k . For $\Gamma = \mathbb{Z}\epsilon$ put $s = t^\epsilon$ and take the s -adic valuation topology of $k((s))$, a Hausdorff field topology. A well-ordered subset of $\mathbb{Z}$ is bounded below and has finitely many elements below any integer; with pointwise finiteness, for each N only finitely many members of a summable family have a term below N , so the net of finite partial sums converges to the Hahn sum. $\square$

Remark 39.13 (Scope [15]). The cyclic half is an existence statement: the compatible topologies of $k((s))$ are not classified. The theorem separates cyclic from merely discrete groups: lexicographic $\mathbb{Z}^2$ has a least positive element and is still excluded. The field theorem needs neither density nor divisibility but uses both t^g and t^{-g} , so it does not settle the negative-support ring for non-dense groups (question 42.1); for $\Gamma = \mathbb{Z}$ that ring is $\mathfrak{o} + t^{-1}k[t^{-1}]$, whose summable families are finite up to zero members, so its discrete topology is compatible and Hausdorff.

Nothing here conflicts with the order and valuation topologies, with formal Hahn calculus or with strong automorphisms: familiar topologies do not realize every Hahn sum. In $\mathbb{R}((t^\mathbb{Q}))$ the monomials $t^{-1/n}$ have a Hahn sum but do not tend to 0 in the valuation topology; the foundations report (docs/foundations-and-computation/foundations, found:sub:tsum) records such failures for particular topologies, including the rank-one found:ex:boundedrankone over $\mathbb{C}((t^\mathbb{Q}))$, and so do remarks 21.7 and 34.4 here. Theorem 39.12 is the general form [merge]: outside $\Gamma \in \{0, \mathbb{Z}\}$ no Hausdorff ring topology at all does better. Nor is there a conflict with finiteness spaces [4]: a monoid object for a summation-sensitive tensor product does not have jointly continuous multiplication for a Cartesian product topology. And the theorems do not say that these rings lack Hausdorff ring topologies: the discrete topology always exists; it fails to realize the nontrivial Hahn sums [15].

As a consequence, a unital homomorphism φ from **No** (or **No**(i)) into a nonzero Hausdorff topological ring cannot take every Hahn sum to the corresponding limit: the initial topology of φ would be compatible and would separate 1 from 0, against corollary 39.9. For **Oz** and **Oz**[i] every ring topology on $\mathbb{Z}$ or $\mathbb{Z}[i]$ gives a compatible topology by pullback, which accounts at once for the existence of non-Hausdorff compatible topologies and for their loss of all infinite-scale information. Topology remains usable if one keeps a valuation topology with a smaller class of convergent sums, keeps all sums with only separately continuous multiplication (section 41), or uses a summation-aware tensor or bornological framework; these choices preserve different structures and should be stated [15].

Remark 39.14 (Two proof chains [15]). The topological theorems do not use the automorphism classifications of this report: convergence of monomial sums gives excluded exponent sets with opposite well-order properties, then the dense-interval or cyclic-group obstruction, then the collapse. For the omnific rings one uses only the support description of **Oz** and its constant-term quotient; the countable strengthening uses the positive buffer of **No**. The algebraic chain (well-ordered supports, then scalar detection, then covariance implies strongness, with the reconstruction of Π , $\mathcal{A}_k$, $\mathbb{R}$ and $\mathfrak{m}$) is that of part III. Together: the omnific ring algebraically

forces preservation of all Hahn sums (theorem 23.2), while every jointly continuous ring topology realizing those sums forgets Π .

40 The algebraic half: a disjoint-row variant and four additions

15's detection theorem is theorem 21.4. Its proof selects disjoint rows of an incidence system; this is a variant of 14's induced matching (lemma 34.1) that selects family members rather than exponents, and it yields prescribed pairings.

Lemma 40.1 (Disjoint rows [15]). *Let $(F_i)_{i \in I}$ be infinitely many nonempty finite subsets of $\mathbb{N}$ such that each $n \in \mathbb{N}$ lies in only finitely many F_i . Then there are distinct $i_0, i_1, \dots$ with the F_{i_r} pairwise disjoint.*

Proof. Choose one row. Its finitely many elements meet only finitely many other rows; discard those and repeat. At each stage finitely many rows have been discarded and infinitely many remain. $\square$

Variant of the third route in theorem 21.4 [15]. We show that countably supported tests suffice. Assume that every y with countable support has $\langle x_j, y \rangle_0 \neq 0$ for only finitely many j . Monomial tests give pointwise finiteness. If the union S of the supports is not well ordered, choose $\gamma_0 > \gamma_1 > \dots$ in S and put $F_j = \{n : [t^{\gamma_n}]x_j \neq 0\}$. Each F_j is finite (a well-ordered support meets a descending sequence finitely), each n lies in finitely many F_j , and infinitely many F_j are nonempty. Lemma 40.1 gives pairwise disjoint nonempty F_{j_r} ; pick $n_r \in F_{j_r}$ and put

$$y = \sum_{r \geq 0} ([t^{\gamma_{n_r}}]x_{j_r})^{-1} t^{-\gamma_{n_r}}, \quad (40.1)$$

a Hahn series with countable support. By disjointness, $[t^{\gamma_s}]x_{j_r} = 0$ for $s \neq r$, so $\langle x_{j_r}, y \rangle_0 = 1$ for every r , contradicting the assumption for this y . $\square$

With $a_{n,j} = [t^{\gamma_n}]x_j$, disjoint rows give $a_{n_r, j_r} \neq 0$ and $a_{n_s, j_r} = 0$ for $s \neq r$: an induced matching in the sense of lemma 34.1, found by selecting the members j , with the stronger property that their supports along the sequence are disjoint, rather than the exponents n . With coefficients 1 instead of the inverses in (40.1) each pairing would be the single entry $a_{n_r, j_r} \neq 0$, 14's all-ones witness. 15's normalization to 1 is what gives interpolation. The detector (40.1) has countable support but not coefficients in $\{0, 1\}$, so on its own the variant shows that countably supported tests with arbitrary coefficients suffice; the binary form of (iii) is 12's, 13's and 14's. The route works over every field, including finite fields and positive characteristic [15].

Corollary 40.2 (Sparse interpolation inside a bad family [15]). *Let (x_j) be pointwise finite with a union of supports that is not well ordered. There are distinct j_r and a countable strictly increasing $\Lambda \subseteq \Gamma$ such that for every sequence $(c_r) \in k^{\mathbb{N}}$ some y supported in Λ has $\langle x_{j_r}, y \rangle_0 = c_r$ for all r .*

Proof. Use the rows and exponents of (40.1), with numerators c_r in place of 1; the set $\Lambda = \{-\gamma_{n_r}\}$ does not depend on (c_r) . $\square$

Example 40.3 (The naive detector fails [15]). Putting coefficient 1 at every opposite exponent of a descending sequence does not detect non-summability. For $x_n = t^{-n} - t^{-(n+1)}$, $n \geq 1$, the family is not summable (its supports descend), but $y = \sum_{m \geq 1} t^m$ gives $\langle x_n, y \rangle_0 = 1 - 1 = 0$ for every n . A selection is needed: the members x_n with n odd have pairwise disjoint supports, and selecting them, as in lemma 40.1 or 14's matching, gives $y = \sum_{r \geq 0} t^{2r+1}$ with $\langle x_{2r+1}, y \rangle_0 = 1$.

In [example 34.3](#) the cancellation is between members sharing an exponent, and the all-ones test on all exponents already works; here it is between adjacent exponents of one member, and the all-ones test on all exponents fails. 15's Proposition 4.5 is the family of [example 21.6](#): $t^{-n\delta}$ is not summable, a finite-support test pairs nonzero with finitely many members, and $\sum_{n \geq 1} t^{n\delta}$ pairs to 1 with all. So finite-support tests do not suffice and countable ones do. This concerns the support of a test, not whether countable families determine every uncountable sum [15].

[Proposition 23.1](#) transports the *full* constant term. On bounded elements `ct` is the standard part, but on $p + r + \varepsilon$ with $p \in \Pi$ the standard part is undefined while $\text{ct} = r$, so the invariant Π is essential; invariance of the residue on bounded elements would not suffice [15].

Corollary 40.4 (The Gaussian ring with its involution [15]). *Every unital ring automorphism of $\mathbf{Oz}[i]$ that commutes with conjugation extends uniquely to a field automorphism of $\mathbf{No}(i)$ commuting with conjugation, and the extension is strongly additive with strongly additive inverse.*

Proof. For $x + iy \in \mathbf{No}(i)$, `odg:thm:fractions` gives one $d \in \mathbf{Oz} \setminus \{0\}$ with $dx, dy \in \mathbf{Oz}$, so $\text{Frac } \mathbf{Oz}[i] = \mathbf{No}(i)$ and the automorphism extends uniquely to fractions; commutation with conjugation passes from numerators and denominators, and the extension preserves $\mathbf{Oz}[i]$. [Theorem 30.1](#) gives the form and strongness. $\square$

For the bare pair $(\mathbf{No}(i), \mathbf{Oz}[i])$ strongness is not concluded. The proved alternatives are preservation of the valuation ring, commutation with conjugation, or covariance of the full constant term; no inference from “the constants are invariant” to “`ct` is invariant” is made ([question 32.1](#)) [15].

Proposition 40.5 (Class coefficient description of the strong dual [15]). *An $\mathbb{R}$ -linear strong functional λ on $\mathbf{No}$ is determined by the class function $a(g) = \lambda(t^g)$, whose support $S_a = \{g : a(g) \neq 0\}$ contains no strictly increasing sequence, and*

$$\lambda\left(\sum_g c_g t^g\right) = \sum_g c_g a(g), \quad (40.2)$$

a finite sum. Conversely every class function $a : \mathbf{No} \rightarrow \mathbb{R}$ whose support contains no strictly increasing sequence defines by (40.2) a strong $\mathbb{R}$ -linear functional λ_a . It is $\text{ct}(h \cdot)$ for a surreal h exactly when S_a is a set, and then $h = \sum_g a(g)t^{-g}$.

Proof. A strictly increasing (g_n) in S_a would make (t^{g_n}) a summable family with infinitely many nonzero values of λ ; strongness on the normal-form family of an element gives (40.2). Conversely, a set with no strictly increasing sequence meets every well-ordered set finitely, since an infinite subset of a well-ordered set contains one; so the sum in (40.2) is finite, and for a summable family the union of the supports meets S_a in finitely many exponents, each in finitely many members, which gives the finiteness and the interchange of the two sums. If S_a is a set it is reverse well ordered, h is a surreal, and $\text{ct}(hf) = \lambda_a(f)$; a representing h has support exactly $-S_a$, which must be a set. $\square$

$\lambda_{\mathbf{On}}$ of [theorem 25.6](#) is λ_a for the indicator a of $-\mathbf{On}$. A support “with no strictly increasing sequence” is the same as one meeting every well-ordered set finitely, so the proposition describes the strong $\mathbb{R}$ -linear dual of $\mathbf{No}$ exactly by class coefficient functionals meeting every well-ordered set finitely, each an individual class function in the sense of [section 1.4](#); no class of such functionals is formed. This answers the first clause of [question 32.11](#); the second, which strong operators have transposes that stay set-supported, is 15's Question 9 and stays open

(section 42.1). The proof of theorem 21.4 uses only countably supported tests and never represents a class functional, so this enlarged dual places no restriction on theorem 23.2; it only blocks the overgeneralization of proposition 25.1 and theorem 25.2 to No [15].

41 A pairing topology that realizes exactly the Hahn sums

In this section Γ is a set, $K = k((t^\Gamma))$, and k carries the *discrete* topology. Let $\mathfrak{T}_\beta$ be the weak topology generated by the functionals $f \mapsto \langle f, h \rangle_0$, $h \in K$; a basis of neighbourhoods of 0 is

$$U(h_1, \dots, h_m) = \{f : \langle f, h_1 \rangle_0 = \dots = \langle f, h_m \rangle_0 = 0\}. \quad (41.1)$$

Nondegeneracy of the pairing makes $(K, \mathfrak{T}_\beta)$ a Hausdorff topological vector space over the discrete field k . Over $\mathbb{R}$ or $\mathbb{C}$ the usual topology of the coefficients is not used, and $\mathfrak{T}_\beta$ is not identified with the Hahn-field topologies of Flynn and Shamseddine [9] [15].

Theorem 41.1 (Unordered sums in the pairing topology [15]). *For a family $(f_j)_{j \in J}$ in K , the net $s_F = \sum_{j \in F} f_j$, $F \in \text{Fin}(J)$, converges to f in $\mathfrak{T}_\beta$ if and only if (f_j) is Hahn summable with sum f .*

Proof. In a discrete field a net of finite partial sums of scalars converges if and only if all but finitely many summands vanish: beyond some F_0 the net is constant, and comparing F_0 with $F_0 \cup \{j\}$ kills every summand outside F_0 . Applied to $\langle f_j, h \rangle_0$ for each h , convergence of s_F gives the finiteness condition of theorem 21.4, hence summability, and the limit identity with nondegeneracy identifies the sum. Conversely a Hahn sum gives eventually constant scalar nets with the right limits (21.2), which is convergence in the initial topology. $\square$

Corollary 41.2 (Continuous linear maps [15]). *For set-sized full Hahn fields K, L over k , a k -linear map $K \rightarrow L$ is continuous for the pairing topologies if and only if it is strongly k -linear.*

Proof. A continuous map preserves convergent nets of finite partial sums (theorem 41.1). A strong map has a constant-term adjoint (theorem 25.2), so each target test functional composed with it is a source test functional. $\square$

In particular multiplication by a fixed a is continuous, as $\langle af, h \rangle_0 = \langle f, ah \rangle_0$, and the isomorphisms of theorem 22.1 are homeomorphisms, the coefficient isomorphism being one of discrete fields [15].

Theorem 41.3 (Multiplication is not jointly continuous [15]). *If $\Gamma \neq 0$, multiplication $K \times K \rightarrow K$ is not jointly continuous for $\mathfrak{T}_\beta$; so $\mathfrak{T}_\beta$ is not a ring topology, although it realizes exactly the Hahn sums and multiplication is separately continuous.*

Proof. $\text{ct} = \langle \cdot, 1 \rangle_0$ is continuous and $\text{ct}^{-1}(0)$ is open. Joint continuity at $(0, 0)$ would give basic neighbourhoods U, V of the form (41.1) with $\text{ct}(uv) = 0$ for $u \in U, v \in V$, so $U \subseteq V^\perp = \{x : \langle x, v \rangle_0 = 0 \text{ for all } v \in V\}$. By nondegeneracy $x \mapsto \langle x, \cdot \rangle_0$ injects $V^\perp$ into $(K/V)^*$, which is finite dimensional since V has finite codimension. But U has finite codimension in K , which is infinite dimensional ($t^{n\delta}$, $n \geq 0$, are independent for $\delta > 0$), so U is infinite dimensional, a contradiction. $\square$

This particular obstruction is weaker than theorem 39.12, which excludes every Hausdorff compatible ring topology outside the cyclic case; it shows that joint continuity cannot be weakened to separate continuity in that theorem. A summation-detecting topology does not turn Hahn constructions into topological-field constructions [15].

42 Questions of source 15, and stale statements

42.1 Status of the source's questions

15 poses twelve questions, proposed as research directions, not as established historical conjectures. Six are merged into questions of this report, with 15 added to their sources:

- Question 4 (Gaussian automatic strongness without extra structure): [question 32.1](#), open. 15 adds that the topological collapse does not settle it: a compatible ring topology cannot reconstruct the missing valuation, since on $\mathbf{No}(i)$ every one is indiscrete ([corollary 39.9](#)).
- Question 5 (proper Hahn subfields and detector closure): [question 37.1](#). It is *partly addressed* by 14's countable detector property ([definition 35.1](#) and [proposition 35.2](#)), a sufficient condition, and [example 35.3](#) shows that some condition is needed; the necessary-and-sufficient condition and its effect on constant-term-compatible embeddings stay open.
- Question 6 (countable versus unrestricted preservation in support-bounded fields, singular bounds): [question 32.12](#), open.
- Question 9 (class summation duals and adjoints): [question 32.11](#). Its first clause is answered by [proposition 40.5](#); 15's question is the second clause, open.
- Question 10 (generation of the full stabilizer): [question 11.4](#), open.
- Question 12 (a verified core and transfer layer): [question 32.18](#).

Six are new.

Question 42.1 (Nondense truncated support rings [15]). For which nonzero ordered abelian groups Γ does $\mathfrak{o} \oplus \Pi$ admit a Hausdorff Hahn-compatible ring topology? Dense groups are excluded ([theorem 39.4](#)) and $\Gamma = \mathbb{Z}$ admits the discrete topology ([remark 39.13](#)); the first test is lexicographic $\mathbb{Z}^2$.

The full-field obstruction does not settle it, because inverse monomials leave the negative-support ring. An answer should identify an intrinsic order property of the negative cone; the proof of monomial collapse uses bounded intervals, not the whole group [15].

Question 42.2 (Minimal forbidden summation families [15]). For a fixed dense exponent group, what is the smallest specified collection of countable monomial sums whose convergence already forces the constant-term collapse? Does an explicitly listed countable collection suffice for $\mathbb{Z} \oplus \Pi$ with $\Gamma = \mathbb{Q}$, or is more needed when nothing countable is assumed of the topology?

The present proof chooses its sequences after examining the neighbourhoods; removing that dependence would make the obstruction effective and show how much summation data must be named [15]. [Proposition 39.8](#) answers the question for the full omnific rings, where countable monomial sums suffice; its buffer argument fails for $\Gamma = \mathbb{Q}$ [merge].

Question 42.3 (Maximal topologically realizable summation laws [15]). Fix a Hausdorff valuation topology on a Hahn field. Classify the largest subfamilies of the Hahn sums that converge in it and are closed under reindexing, subfamilies, finite products and multiplication by fixed elements. How do they change when the valuation is coarsened?

The criterion that only finitely many summands contribute below each valuation threshold is a starting point, not a classification of alternative topologies and compatible summation structures [15].

Question 42.4 (A non-Cartesian multiplication framework [15]). Can the algebra of surreal Hahn sums be organized in a summation-sensitive tensor category in which multiplication is a continuous morphism, scalar duality has the correct class-size qualification, and the strong automorphisms are exactly the invertible morphisms? Which completion or boundedness conditions are needed?

Finiteness spaces and formal summation algebras are precedents [4, 3]; the problem is a precise full-surreal implementation that does not mistake tensor continuity for joint continuity in a product topology [15].

Question 42.5 (Separating a chosen family of scales [15]). Given a set $E \subseteq \Pi$, which parts of the Hahn summation law must be discarded to obtain a Hausdorff ring topology separating every nonzero member of E from 0? Is the trade-off governed by a support invariant of E ?

The unrestricted law cannot separate even one negative monomial; a sharp restricted result would serve asymptotic models that keep selected scales [15].

Question 42.6 (Definability beyond automorphism invariance [15]). In a language with sorts for admissible set-indexed families, how much of Hahn summation can be recovered from the field and the omnific predicate? Is there a useful definability theorem strictly stronger than automorphism invariance (theorem 23.2) and weaker than naming the Conway monomial section or full set theory?

The topological quotient and the nondefinability of monomials (theorem 9.1) exclude several naive proposals; an answer must fix the language, the size of its sorts and the interpretation of family parameters [15]. Compare question 11.5 and `dsn:q:languages` [merge].

42.2 Stale and checked statements

- *Novelty, checked.* 15 first answered question 11.1 itself, found 12 and 13 in a later read, and changed its claim: automatic strongness is credited to them and not claimed. Its remaining claim, the topological classification, is new at the merge: no report of the collection classifies Hahn-compatible ring topologies; the collection had particular topologies only (remarks 39.6 and 39.13, section 28). Its statement that the inspected audits do not contain the classification is true, and the merge found no other report containing it.
- *Checked true.* The two article blobs and the three audit blobs it cites (section 38.1); its description of question 11.1 as open in the first blob and answered in the second; its credits to the criterion, theorem 5.1 and the shear examples; the finite checks of its suite, rerun for the merge (section C). No mathematical error was found in 15.
- *Not rechecked.* 15's descriptions of [9] (whose topologies it does not identify with $\mathfrak{T}_\beta$) and of [4, 3] are reported as 15 gives them.

15's limitations are listed with the other sources' in section B.

Part VI

Coefficient recovery in monomial extensions of arbitrary rank

This part records manuscript 16, the second addition of batch 32. Its question is that of [part II](#), asked at arbitrary rank: after adjoining external monomial parameters, of any rank and not necessarily finitely generated, can the ring or field still recover its surreal or omnific coefficients? It can. One existential quartic equation defines the coefficient field in the fraction field of a group algebra ([section 45](#)); a sparse power bound survives every monomial refinement ([section 46](#)); coefficient rigidity, automorphism formulas and finite-rank cancellation follow at arbitrary rank, cancellation failing exactly at infinite rank ([sections 47, 48 and 50](#)); and, with a published defect theorem, every field automorphism is monomial exactly when $\text{Hom}(G, \mathbb{Z}) = 0$ ([section 49](#)). The parameters are finite-support external monomials, not Hahn sums. Its labels carry the sub-prefix `opa:cr:`.

43 Source 16, its relation to Part II, and its notation

43.1 The manuscript and its pin

Manuscript 16 is *Recovering Surreal Coefficients from Monomial Extensions: quartic definitions, arbitrary-rank rigidity, and sharp cancellation*, dated 23 September 2026, 24 pages. It is item 08 of batch 32, delivered as the archive `surreal_coefficient_rigidity` in commit `aa268a4` and placed as an addition in commit `7d04483`; it is numbered 16 here. Its checks, their recorded run, its build script, its hash-bearing build record and its source audit are shipped as `code/16-coefficient-recovery-verify.py`, `data/16-coefficient-recovery-verification.json`, `code/16-coefficient-recovery-build.py`, `data/16-coefficient-recovery-build_report.json` and `16-coefficient-recovery-SOURCE_AUDIT.md`; its `article.tex`, PDF, README and checksum list are not shipped (the checksum list and the hashes of the build record were verified at placement).

It pins `1ab41af`, the last commit it observed, which contains [parts I to III](#) of this report (`20c4c9f`) but not the placement `9d28e28` or the write `d4d72d7` of source 14. It read the catalogue, the audits of sources 11 and 12 and a few guides, not this article. Its “prior manuscript” is source 02 itself, the base of [part II](#), retrieved separately with 02’s pin `fb5c4b5`. 16 describes 02 accurately and credits it for the root-covered framework, the monomial-divisibility criterion, polynomial and Laurent coefficient rigidity, strong cancellation, the finite-type obstructions and the formal flows; it claims none of these.

43.2 What it generalizes, and what is printed once

16’s numbers are those of its own PDF; its main theorems are lettered A–D.

16	Printed as	Relation to part II
Main Thm. A(iii), Thm. 6.4	theorems 45.4 and 47.1	Generalize theorems 15.1 and 15.3 from $B[\mathbf{T}, \mathbf{U}^{\pm 1}]$ with finitely many variables to $S[U^{\mathbf{M}}]$ for every cancellative torsion-free monoid, of any rank.
Thm. 7.1	theorem 48.1	Generalizes the torus part of theorem 15.5 , with $\mathrm{GL}_q(\mathbb{Z}) = \mathrm{Aut}(\mathbb{Z}^q)$, to $\mathrm{Aut}(G)$ for every torsion-free G .
Thm. 9.1	theorem 50.1	Generalizes theorem 15.4 from $\mathbb{N}^p \times \mathbb{Z}^q$ to monoids of finite rational rank, and to coefficient rings with closed fraction field.
Thm. 9.3, Main Thm. D	theorems 50.3 and 50.4	Exact failure at infinite rank, parallel to the example $R[X_1, X_2] = B[X_2]$ after theorem 15.4 .
Prop. 6.2	theorem 13.1 and corollary 13.3	Reproved and credited; printed once, 16 added.
Def. 6.1; Ex. 6.3	definition 12.1 ; the workspace of section 18	Same, with the same fraction-field warning.
Prop. 4.1	<code>odg:thm:fractions</code> , <code>odg:prop:units</code> , <code>odg:def:lem:units</code>	Reproved (“classical”); cited.
Question 12.8	question 19.3	Merged.
All else	This part	New.

The proofs of [part II](#) use degrees in finitely many variables. That argument does not reach arbitrary rank: an element and its roots can lie in different finitely generated exponent lattices, and no finitely generated target need contain all the witnesses. 16’s substitute is the sparse bound of [theorem 46.2](#), which is uniform under changes of lattice [16].

43.3 Conventions and notation of this part

Rings are commutative and unital, homomorphisms unital, embeddings injective, and domains have characteristic zero unless a counterexample says otherwise. G is a set-sized torsion-free abelian group and $\mathbf{M}$, $\mathbf{N}$ set-sized cancellative torsion-free commutative monoids, written additively, with group completion $\mathrm{gp}(\mathbf{M})$ and rational rank $\mathrm{rk}_{\mathbb{Q}} \mathbf{M} = \dim_{\mathbb{Q}}(\mathbb{Q} \otimes \mathrm{gp} \mathbf{M})$. The *external monomials* are written U^g :

$$k[U^G] = \left\{ \sum_{g \in S} a_g U^g : S \subseteq G \text{ finite} \right\}, \quad U^g U^h = U^{g+h}, \quad k(U^G) = \mathrm{Frac} k[U^G],$$

and $R[U^{\mathbf{M}}] \subseteq R[U^{\mathrm{gp} \mathbf{M}}]$ is the monoid algebra. A *monomial* is cU^g with $c \neq 0$; $\mathrm{wt} f$ is the number of nonzero coefficients of $f \in k[U^G]$. So $B[\mathbf{U}^{\pm 1}] = B[U^{\mathbb{Z}^q}]$ and $B[\mathbf{T}] = B[U^{\mathbb{N}^p}]$ in the notation of [theorems 15.1 and 15.3](#). An element of $k(U^G)$ is a quotient of two finite sums, not a Hahn series [16].

16's symbol	Here
$X^g, k[G],$ $K_k(G) = \text{Frac } k[G]$	$U^g, k[U^G], k(U^G).$
M, N (monoids); H (a second group)	$\mathbf{M}, \mathbf{N}; G'.$
$v(f), d(f),$ $w(f) = d(f) - v(f)$	$\min \text{supp } f, \max \text{supp } f, \text{wd}(f),$ for an auxiliary order on G .
ℓ , the specializing homomorphism	$\lambda.$
$D = T d/dT$	$\vartheta.$
$\mathcal{U}_\infty(B); D(G) = \bigcap_n nG$	$\text{Rt}_\infty(B); G_{\text{div}}.$
$\mathcal{D}(B), \mathcal{C}(B)$	$\mathfrak{D}(B), \mathfrak{C}(B).$
$\Delta_k(G) =$ $K_k(G)^\times / (k^\times X^G)$	$\text{Def}_k(G) = k(U^G)^\times / (k^\times U^G).$
$\sigma \in \text{Aut}(F), \tau \in \text{Aut}(G),$ $\chi \in \text{Hom}(G, F^\times)$	$\alpha, \tau, \chi.$
$T = \phi^{-1}(S)$ in the cancellation proof	$R_1.$
$R_\Gamma(D, k), \mathcal{P}_\Gamma(k)$	$\mathcal{R}_\circ(k, \Gamma), \Pi_k(\Gamma),$ as in section 12.1 .
$\mathbf{Oz}[i]; \mathbf{No}(i)$	$\mathbf{Oz}[i]; \mathbf{No}(i),$ the same.

The collisions are resolved here only, and the tempting false readings are wrong:

- 16's X^g is an *external indeterminate*. In [part II](#) $X^g = t^{-g} = \omega^g$ is a Hahn monomial of $k((X^\Gamma))$. So U^g here is not a surreal monomial: in $\mathbf{No}(U^G)$ with $g \neq 0$, U^g is transcendental over $\mathbf{No}$, while $\omega^g \in \mathbf{No}$ is a coefficient ([example 51.1](#)).
- 16's $K_k(G)$ is the rational group field $k(U^G)$, not the Hahn field $K_\Gamma = k((t^\Gamma))$ of [part III](#); no Hahn summation is involved.
- 16's $\Delta_k(G)$ is a defect group, not the target group Δ of [part III](#); 16's $D(G)$ is not a derivation; 16's $\mathcal{D}(B)$ and $\mathcal{C}(B)$ are sets of ring elements, not the convex subgroups C_δ and $C(a)$ or the operator coefficients c_δ ; 16's $\mathcal{U}_\infty$ is not the stabilizer $\mathcal{U}$.
- 16's monoid M is not a monomial map $M_{\chi, \tau}$. Its $\tau \in \text{Aut}(G)$ is a group automorphism, with no order, and its χ need not be positive; the automorphisms ([48.1](#)) are the finite-support analogue of $M_{\chi, \tau}$ composed with a coefficient automorphism α , 16's σ .
- 16's v is the minimum of a finite support for an auxiliary order, not the Hahn valuation, and its d is not a derivation of $\mathbb{R}$; its ℓ is not the leading exponent $\ell(x)$ of [part II](#).

Coefficient rings may be the classes $\mathbf{No}, \mathbf{No}(i), \mathbf{Oz}, \mathbf{Oz}[i]$; the exponent groups and monoids are sets. An element of a monoid algebra is a finite list of exponents and coefficients; fractions are coded by the sets of minimum-rank representatives (Scott's trick), not by proper-class equivalence classes. No proper-class spectrum, Hahn completion or set of all class automorphisms is formed, and a displayed automorphism group is schematic: each class automorphism is specified by the displayed data, and each admissible choice gives one. The algebraic arguments involve finitely many supports and coefficients at a time; the one step that uses a published set-theoretic theorem for all $g \in G$ at once descends to a set-sized coefficient subfield ([section 49](#)) [16].

44 Finite-support monomial extensions

A torsion-free abelian group admits a compatible total order (embed it in $\mathbb{Q} \otimes G$ and order a basis lexicographically). Fix one; for $0 \neq f \in S[U^G]$ the least and greatest support exponents and the width $\text{wd}(f) = \max \text{supp } f - \min \text{supp } f$ are additive under products in a domain S , since the extreme coefficients of a product are products of extreme coefficients [16].

Lemma 44.1 (Monomial factors and units [16]). *Let S be a domain. If fg is a nonzero monomial in $S[U^G]$, then f and g are monomials. So $S[U^G]^\times = S^\times U^G$, and $S[U^M]^\times = S^\times U^{M^\times}$ with $M^\times = M \cap (-M)$ in $\text{gp } M$.*

Proof. A monomial has width 0, widths are nonnegative and additive, and a finite support of width 0 is a point. If $fg = 1$ the coefficients are units and the exponents cancel. $\square$

Lemma 44.2 (No new algebraic constants [16]). *For every field k , k is relatively algebraically closed in $k(U^G)$. If $k \subseteq L$ and $G \subseteq G'$, then $k(U^G) \cap L^\times U^{G'} = k^\times U^G$ inside $L(U^{G'})$.*

Proof. $v_0(P/Q) = \min \text{supp } P - \min \text{supp } Q$ is a valuation of $k(U^G)$ with residue field k (16's v ; it is not the Hahn valuation). Let $z \neq 0$ be algebraic over k , with minimal polynomial of nonzero constant term. If $v_0(z) > 0$ the constant term is the unique term of least valuation, and if $v_0(z) < 0$ the term of highest degree is; either contradicts the equation, so $v_0(z) = 0$. Let $c \in k$ be the residue of z ; if $z \neq c$, then $z - c$ is algebraic with $v_0(z - c) > 0$, which is impossible. For the second claim, $P/Q = cU^h$ gives $\text{supp } P = h + \text{supp } Q$, so $h \in G$, and a coefficient comparison gives $c \in k$. No closedness of k is used. $\square$

The valuation is auxiliary; no automorphism below is assumed to preserve it.

Lemma 44.3 (One-variable specialization [16]). *Given finitely many elements of $k[U^G]$ and a finite set of exponent combinations, there is a homomorphism λ from the subgroup generated by their supports to $\mathbb{Z}$ that is injective on the prescribed set, so the substitution $U^g \mapsto T^{\lambda(g)}$ keeps prescribed denominators nonzero. Moreover, for $P, Q \in k[U^G] \setminus \{0\}$:*

- (i) *if $P/Q \notin k$, λ can be chosen so that the specialized fraction is not constant;*
- (ii) *if P/Q is not a monomial, λ can be chosen so that the specialized fraction is not a monomial and $\text{wt } P, \text{wt } Q$ are unchanged.*

Proof. The subgroup is finitely generated and torsion-free, so $\cong \mathbb{Z}^d$, and an integer functional outside finitely many hyperplanes separates a finite set: try $(1, m, \dots, m^{d-1})$, each coincidence excluding finitely many integers m . Include the supports of all denominators. For (i) require injectivity on $\text{supp } P \cup \text{supp } Q$: proportional specializations then force $P = cQ$. For (ii) require injectivity on $\text{supp } P - \text{supp } Q$, which also separates $\text{supp } P$ and $\text{supp } Q$; a specialized quotient cT^j would give $\lambda(\text{supp } P) = j + \lambda(\text{supp } Q)$, every difference $a - b$ in the resulting bijection would map to j and so be one fixed g , and the coefficients would give $P = cU^gQ$. $\square$

The specialization is applied to one identity with its denominators cleared, and chosen to keep them nonzero; it is never a field homomorphism on all of $k(U^G)$ [16].

45 A quartic equation recovers the coefficient field

Lemma 45.1 (Mason–Stothers in characteristic zero [16]). *If nonzero pairwise coprime $A, B, C \in k[T]$, not all constant, satisfy $A + B = C$, then $\max(\deg A, \deg B, \deg C) \leq \deg \text{rad}(ABC) - 1$, where rad is the product of the distinct monic irreducible factors.*

Proof. $W = A'B - AB'$ is nonzero: otherwise $(A/B)' = 0$, so $A/B \in k$ in characteristic zero and coprimality makes all three constant. Since $C = A + B$, also $W = A'C - AC' = C'B - CB'$. A factor of multiplicity m of A , B or C divides W to multiplicity at least $m - 1$, so by coprimality $ABC/\text{rad}(ABC)$ divides W , and $\deg W \leq \deg A + \deg B - 1$ gives the bound for $\deg C$; permute for the others. This is the standard Wronskian proof [1]. $\square$

Lemma 45.2 (The quartic has only constant rational points [16]). *If $a, b \in k(T)$ satisfy $b^2 = 1 + a^4$, then $a, b \in k$.*

Proof. Write $a = P/Q$ in lowest terms. Then $(bQ^2)^2 = P^4 + Q^4$, and a rational function whose square is a polynomial is a polynomial, so $V_0 = bQ^2 \in k[T]$ with $V_0^2 = P^4 + Q^4$. If $P = 0$ we are done; if $V_0 = 0$, coprimality makes P, Q constant. Otherwise P, Q, V_0 are pairwise coprime; if $d = \max(\deg P, \deg Q) > 0$, then $\deg V_0 \leq 2d$ and lemma 45.1 for $P^4 + Q^4 = V_0^2$ gives $4d \leq \deg P + \deg Q + \deg V_0 - 1 \leq 4d - 1$. So $a \in k$, and $b \in k$ by lemma 44.2. $\square$

Theorem 45.3 (Quartic rigidity in every rational group field [16]). *For every characteristic-zero field k and torsion-free abelian group G , every solution of $b^2 = 1 + a^4$ in $k(U^G)$ has $a, b \in k$.*

Proof. If $a \notin k$, choose λ by lemma 44.3(i) keeping a nonconstant and the denominators of a, b nonzero; the identity with cleared denominators specializes to a nonconstant solution in $k(T)$, against lemma 45.2. Then $b \in k$ by lemma 44.2. $\square$

Theorem 45.4 (Uniform coefficient definitions [16]; 16's Main Theorem A). *Let G be a set-sized torsion-free abelian group and M a set-sized cancellative torsion-free monoid.*

(i) *If F is real closed, or algebraically closed of characteristic zero, then in $F(U^G)$*

$$x \in F \iff \exists y (y^2 = 1 + x^4). \quad (45.1)$$

(ii) *If R is a domain whose fraction field has one of these properties, then in $R[U^M]$*

$$x \in R \iff \exists u \exists v (v \neq 0 \wedge u^2 = (1 + x^4)v^2). \quad (45.2)$$

(iii) *(i) applies to $F = \mathbf{No}, \mathbf{No}(i)$ and (ii) to $R = \mathbf{Oz}, \mathbf{Oz}[i]$. Every embedding of $\mathbf{No}$ or $\mathbf{No}(i)$ into a field $k(U^G)$ lands in k , and every embedding of $\mathbf{Oz}$ or $\mathbf{Oz}[i]$ into $S[U^M]$, S a domain, lands in S .*

Proof of (i). In a real closed F , $1 + x^4 > 0$ has a square root; in an algebraically closed one every element does. The converse is theorem 45.3. More generally, if $\phi : F \hookrightarrow k(U^G)$ is any field embedding, choose $b \in F$ with $b^2 = 1 + a^4$ for each $a \in F$; the images satisfy the same equation, so $\phi(a) \in k$. No order, valuation, linearity or summability of ϕ is assumed. $\square$

Corollary 45.5 (Coefficient recovery under isomorphism [16]). *If F, E are real closed or algebraically closed of characteristic zero, every field isomorphism $F(U^G) \cong E(U^{G'})$ restricts to an isomorphism $F \cong E$. For set-sized structures, an elementary embedding of the ambient fields restricts to an elementary embedding of the coefficient fields.*

Proof. The same parameter-free formula defines both coefficient fields; for elementarity relativize every quantifier to (45.1). $\square$

Remark 45.6 (Why a quartic [16]). The conic $Y^2 = 1 + X^2$ fails: $X = (T - T^{-1})/2$, $Y = (T + T^{-1})/2$ is a nonconstant point in $k(T)$. The quartic has positive genus and $1 + x^4$ has a square root at every real closed coefficient, not only on a bounded interval; this gives one formula for the real and the complex cases.

The omnific rings enter through $\text{Frac } \mathbf{Oz} = \mathbf{No}$ and $\text{Frac } \mathbf{Oz}[i] = \mathbf{No}(i)$ with units $\{\pm 1\}$ and $\{\pm 1, \pm i\}$ (`odg:thm:fractions`, `odg:prop:units`), which 16 reproves by the usual monomial clearing (its Proposition 4.1, credited to Conway through [19, Proposition 2.4.5]).

Theorem 45.7 (Existential coefficient recovery for rings [16]). *Let R be a domain whose fraction field F is real closed or algebraically closed of characteristic zero. Then (45.2) defines R in $R[U^M]$, and every embedding $R \hookrightarrow S[U^M]$, S a domain, has image in S .*

Proof. $\text{Frac}(R[U^M]) = F(U^G)$ with $G = \text{gp } M$, since fractions give F and the differences of exponents, and $F \cap R[U^M] = R$ by comparing coefficients. For $x \in R$ choose $b \in F$ with $b^2 = 1 + x^4$ and write $b = u/v$ with $v \neq 0$. Conversely, witnesses give $(u/v)^2 = 1 + x^4$ in $F(U^G)$, so $x \in F$ by theorem 45.3 and $x \in R$. An embedding $R \hookrightarrow S[U^M]$ extends to $F \hookrightarrow \text{Frac}(S)(U^G)$, whose image lies in $\text{Frac } S$ by (i), and $\text{Frac}(S) \cap S[U^M] = S$. $\square$

This completes the proof of theorem 45.4: the same two-witness formula works for every $\mathbf{Oz}[U^M]$ and $\mathbf{Oz}[i][U^M]$, whatever the rank, presentation or divisibility of M .

Corollary 45.8 (Uniform interpretation [16]). *Among monoid algebras whose coefficient rings have real closed or algebraically closed fraction fields of characteristic zero, the coefficient ring is uniformly parameter-free existentially definable. Isomorphisms of such algebras preserve the coefficient rings, even when the monoids have different or infinite ranks; for set-sized models, elementary embeddings restrict to elementary embeddings of the coefficient rings.*

Proof. Use the same formula on both sides, and relativize quantifiers as in corollary 45.5. $\square$

The ring formula contains an inequation, so it is existential but not one polynomial equation; deleting $v \neq 0$ admits the witnesses $u = v = 0$ for every x . The field formula is one equation with one witness. Neither formula defines $\mathbf{Oz}$ inside the pure field $\mathbf{No}$, or $\mathbb{R}$ inside $\mathbf{No}$: the ambient structures are the monoid algebra and its fraction field, and with trivial parameters the ring formula only restates $x \in R$. No algorithm recognizing surreal coefficients is inferred [16].

46 Sparse powers under arbitrary monomial extensions

Lemma 46.1 (Sparse multiplicity bound [16]). *A nonzero Laurent polynomial $f = \sum_{j=1}^s c_j T^{m_j}$ over a characteristic-zero field, with distinct m_j and nonzero c_j , has no nonzero root of multiplicity $\geq s$.*

Proof. With $\vartheta = T d/dT$, a root $\alpha \neq 0$ of multiplicity $\geq s$ gives $\vartheta^i f(\alpha) = \sum_j m_j^i c_j \alpha^{m_j} = 0$ for $0 \leq i < s$, since ϑ^i is a combination of derivatives of order $\leq i$. The Vandermonde determinant $\prod_{j < l} (m_l - m_j)$ is nonzero in characteristic zero, so every $c_j \alpha^{m_j} = 0$. This classical bound is discussed by Mattarei [22]. $\square$

Theorem 46.2 (Uniform sparse power obstruction [16]; 16's Main Theorem B). *Let $k \subseteq L$ be fields of characteristic zero, $G \hookrightarrow G'$ an embedding of torsion-free abelian groups, and $0 \neq r = P/Q \in k(U^G)$ with $P, Q \in k[U^G] \setminus \{0\}$. If r is not a monomial and $r = h^n$ for some $h \in L(U^{G'})$ and $n \geq 1$, then*

$$n \leq \max\{\text{wt } P, \text{wt } Q\} - 1. \quad (46.1)$$

P and Q need not be coprime. The bound is sharp.

Proof. Write $h = A/B$. Only finitely many exponents occur in P, Q, A, B ; by [lemma 44.3\(ii\)](#) specialize them to one variable keeping P, Q, A, B nonzero, the weights of P, Q , and r non-monomial. Then $r_0 = h_0^n$ in $L(T)$. A nonmonomial rational function of one variable has a zero or pole at some $\alpha \neq 0$ in an algebraic closure of L (otherwise numerator and denominator are, after cancellation, powers of T up to scalars). Its order $e \neq 0$ there is a multiple of n , and by [lemma 46.1](#) $|e| = |\text{ord}_\alpha P_0 - \text{ord}_\alpha Q_0| \leq \max\{\text{wt } P, \text{wt } Q\} - 1$; multiplying by powers of T changes no order at $\alpha \neq 0$, and common factors cancel in the difference. So $n \leq |e|$. Sharpness: $r = (1 + T)^n$, or $((1 + T)/(1 - T))^n$, whose numerator and denominator have $n + 1$ terms each. $\square$

The root h may use new exponents, with arbitrarily fine denominators, and new coefficients; the bound does not depend on the rank of G' [16].

Corollary 46.3 (Unbounded roots force a monomial [16]). *If $0 \neq r \in k(U^G)$ has roots of unbounded orders in $L(U^{G'})$, then r is a monomial; in particular if the roots lie in $k(U^G)$ or $k[U^G]$.*

Proof. Fix one presentation P/Q and take a root order above (46.1). $\square$

Lemma 46.4 (Roots of monomials [16]). *If $h^n = cU^g$ in $k(U^G)$ with $n \geq 1$, $c \neq 0$, then $h = dU^a$ with $d \in k^\times$, $a \in G$, $d^n = c$ and $na = g$.*

Proof. In the Laurent polynomial ring of a finite free subgroup containing g and the supports of a numerator and denominator of h , a factorial domain, write $h = P/Q$ coprime; $P^n = cU^gQ^n$ forces P and Q to be units, and [lemma 44.1](#) applies. $\square$

For a domain B put $\text{Rt}_\infty(B) = \{b \neq 0 : \text{for every } n \geq 1 \text{ some } c \in B \text{ has } c^n = b\}$, the *all-order root elements* (the roots need not be compatible), and for a torsion-free group $G_{\text{div}} = \bigcap_{n \geq 1} nG$.

Proposition 46.5 (The all-order root locus [16]). *For a characteristic-zero field k , $\text{Rt}_\infty(k(U^G)) = \{cU^g : c \in \text{Rt}_\infty(k), g \in G_{\text{div}}\}$, and G_{div} is the largest divisible subgroup of G . For a domain S the same holds in $S[U^G]$ with $\text{Rt}_\infty(S)$.*

Proof. [Corollary 46.3](#) makes such an element a monomial and [lemma 46.4](#) identifies its roots; in $S[U^G]$ the roots are monomials with coefficients in S ([lemma 44.1](#)). The converse takes roots of the coefficient and solutions of $na = g$. If $g = nh \in G_{\text{div}}$ and $g = nmu$, torsion-freeness gives $h = mu$; so G_{div} is divisible, and it contains every divisible subgroup. $\square$

Remark 46.6 (Two detectors [16]). For k real closed or algebraically closed, k is also the set of x such that x^2 and $1 + x^2$ both lie in Rt_∞ or vanish: they are then monomials differing by 1, so $x^2 \in k$, and [lemma 44.2](#) applies. This description is infinitary; the quartic replaces the countable family of root requirements by one equation.

47 Root-covered rings at arbitrary rank

A domain R is root-covered ([definition 12.1](#)) exactly when every nonzero element divides an element of $\text{Rt}_\infty(R)$; 16 uses this form. That $\mathbf{Oz}$ and $\mathbf{Oz}[i]$ are root-covered, and that $\mathcal{R}_o(k, \Gamma)$ is so exactly when the divisible $\Gamma \neq 0$ has no order unit, are [corollary 13.3](#) and [theorem 13.1](#), which 16 reproves with the same geometric-series argument and credits to source 02 (its Proposition 6.2), together with the workspace of [section 18](#) and its warning that the fraction field is not the full Hahn field (its Example 6.3).

Theorem 47.1 (Arbitrary-monoid coefficient rigidity [16]). *Let R be a root-covered domain of characteristic zero and S a domain. Every embedding $\phi : R \hookrightarrow S[U^M]$, for a cancellative torsion-free monoid M of any rank, has image in S .*

Proof. For $a \neq 0$ choose $ab = w \in \text{Rt}_\infty(R)$. Then $\phi(w) \neq 0$ has roots of all orders in $S[U^M] \subseteq S[U^{\text{gp}M}]$, so it is a monomial (proposition 46.5), and so is $\phi(a)$ (lemma 44.1). For $a \neq 0, -1$ the same holds for $\phi(a) + 1 = \phi(a + 1)$; two monomials differing by 1 both have exponent 0, since otherwise the finite supports disagree at the constant term. So $\phi(a) \in S$; and $\phi(0) = 0, \phi(-1) = -1$. $\square$

With $M = \mathbb{N}^p \times \mathbb{Z}^q$ this is theorems 15.1 and 15.3; the new cases are the monoids that are not finitely generated, such as $\mathbb{Q}^d$, $\mathbb{Q}_{\geq 0}^d$ or $\mathbb{Q}^{(\kappa)}$, where the degree argument of part II does not apply (section 43.2) [16].

Proposition 47.2 (An intrinsic root-divisor description [16]). *For a domain B let $\mathfrak{D}(B)$ be the set of nonzero a dividing some element of $\text{Rt}_\infty(B)$, and*

$$\mathfrak{C}(B) = \{0, -1\} \cup \{a \neq 0, -1 : a \in \mathfrak{D}(B), a + 1 \in \mathfrak{D}(B)\}.$$

If R is root-covered of characteristic zero, then $\mathfrak{C}(R[U^M]) = R$ for every cancellative torsion-free M . So isomorphisms between such monoid algebras preserve the coefficient rings.

Proof. As in theorem 47.1, $a \in \mathfrak{D}(R[U^M])$ is a monomial, and a and $a + 1$ both monomials force $a \in R$. Root-coveredness gives the reverse inclusion; the definition is intrinsic to the ring. $\square$

$\mathfrak{C}$ is not claimed first order, and a finite truncation is vacuous: for orders $n_1, \dots, n_s$ and $N = \text{lcm}(n_j)$, every nonzero a divides a^N , which has the roots a^{N/n_j} . For $\mathbf{Oz}$ and $\mathbf{Oz}[i]$ the finite alternative is (45.2) [16].

48 Automorphisms of group and monoid algebras

Theorem 48.1 (All group-algebra automorphisms [16]). *Let R be a characteristic-zero domain that is root-covered or has a real closed or algebraically closed fraction field, and G torsion-free abelian. Every automorphism of $R[U^G]$ is uniquely*

$$\phi\left(\sum_g a_g U^g\right) = \sum_g \alpha(a_g) \chi(g) U^{\tau(g)}, \quad \alpha \in \text{Aut } R, \tau \in \text{Aut } G, \chi \in \text{Hom}(G, R^\times), \quad (48.1)$$

and every such triple gives an automorphism.

Proof. Coefficient rigidity (theorem 47.1 or theorem 45.7) for $\phi|_R$ and $\phi^{-1}|_R$ gives $\phi(R) = R$. By lemma 44.1, $R[U^G]^\times / R^\times \cong G$, so ϕ induces $\tau \in \text{Aut } G$ and $\phi(U^g) = \chi(g) U^{\tau(g)}$ with $\chi(g) \in R^\times$; multiplicativity makes χ a character, additivity gives (48.1), and the monomial basis gives uniqueness. Conversely the formula is multiplicative, with inverse from α^{-1}, τ^{-1} and the inverse unit factors. $\square$

Corollary 48.2 (Reconstruction of the exponent group [16]). *If R and S each satisfy a hypothesis of theorem 48.1, an isomorphism $R[U^G] \cong S[U^{G'}]$ forces $R \cong S$ and $G \cong G'$, and these suffice.*

Proof. Apply coefficient rigidity in both directions, then compare unit groups modulo the coefficient units. $\square$

For the omnific rings the characters take values in $\{\pm 1\}$ ($R = \mathbf{Oz}$) or $\{\pm 1, \pm i\}$ ($R = \mathbf{Oz}[i]$), which are trivial on divisible G . So, schematically in the sense of [section 43.3](#),

$$\mathrm{Aut}(R[U^G]) \cong \mathrm{Aut}(R) \times \mathrm{Aut}(G) \quad (R \in \{\mathbf{Oz}, \mathbf{Oz}[i]\}, G \text{ divisible}), \quad (48.2)$$

with $\mathrm{Aut}(\mathbb{Q}^d) = \mathrm{GL}_d(\mathbb{Q})$. The coefficient automorphisms $\mathrm{Aut}(R)$ are not classified here; for $\mathbf{Oz}$ they are the subject of [parts I and III](#) [16].

Theorem 48.3 (Divisible monoids [16]). *Let R, S be as in [corollary 48.2](#) and $\mathbf{M}, \mathbf{N}$ divisible cancellative torsion-free monoids. Every isomorphism $\phi : R[U^{\mathbf{M}}] \rightarrow S[U^{\mathbf{N}}]$ is uniquely $\phi(\sum_m a_m U^m) = \sum_m \alpha(a_m) \chi(m) U^{f(m)}$ with a ring isomorphism $\alpha : R \rightarrow S$, a monoid isomorphism $f : \mathbf{M} \rightarrow \mathbf{N}$ and a character $\chi : \mathbf{M} \rightarrow S^\times$; conversely such data give an isomorphism. For the omnific and Gaussian omnific rings every such character is trivial.*

Proof. Coefficient rigidity gives α . As $\mathbf{M}$ is divisible, U^m has roots $U^{m/n}$ of every order, so its image is a monomial $c_m U^{f(m)}$ with $c_m \in S$ ([corollary 46.3](#)); the same holds for ϕ^{-1} on each U^n , and composing shows that the exponent maps are inverse and the c_m are units. Products make f additive and c_m a character. A character of a divisible monoid into a finite group of exponent e is trivial: $\chi(m) = \chi(m/e)^e = 1$. $\square$

Example 48.4 (Rational orthants [16]). For $\mathbf{M} = \mathbb{Q}_{\geq 0}^d$, a monoid automorphism permutes the extreme rays (a ray is extreme when every decomposition of one of its elements into two elements of $\mathbf{M}$ stays on it) and scales each by a positive rational; it extends to a $\mathbb{Q}$ -linear bijection of $\mathbb{Q}^d$. So $\mathrm{Aut}(\mathbf{M}) \cong (\mathbb{Q}_{>0})^d \rtimes \mathrm{Sym}(d)$ and, for $R = \mathbf{Oz}$ or $\mathbf{Oz}[i]$, $\mathrm{Aut}(R[U^{\mathbf{M}}]) \cong \mathrm{Aut}(R) \times ((\mathbb{Q}_{>0})^d \rtimes \mathrm{Sym}(d))$, in contrast with the large nonlinear automorphism groups of polynomial rings in several variables. (16 writes $\mathfrak{S}_d$ for the symmetric group; $\mathfrak{S}^\dagger$ is source 10's structure here.)

49 Rational group fields: an exact automorphism criterion

In $F[U^{\pm 1}]$ the units force monomial images, but $F(U)$ has $U \mapsto U + 1$. Divisibility of the exponent group can prevent such a change from extending to all monomial roots [16]. The general criterion uses one external theorem.

Cited theorem 49.1 ([20, Theorem 4.5]). *For a set-sized field k of characteristic zero and a torsion-free abelian group G , the defect group $\mathrm{Def}_k(G) = k(U^G)^\times / (k^\times U^G)$ is free abelian.*

[Cited theorem 49.1](#) is from an unrefereed preprint of August 2026. 16 uses it as an identified input and does not reprove it; neither does this report, and no peer-review status is inferred. The results of [sections 44 to 48](#) do not depend on it. For a class field k and a homomorphism $g \mapsto r_g \in k(U^{G'})^\times$ from a set group G , choose numerator and denominator codes for the set of values; their coefficients generate a set subfield k_0 , all r_g lie in $k_0(U^{G'})$, and by [lemma 44.2](#) being a monomial over k is being one over k_0 . So the kernel of $G \rightarrow \mathrm{Def}_{k_0}(G')$ is the monomial kernel, and no proper-class version of [cited theorem 49.1](#) is used [16].

Theorem 49.2 (Exact monomial-automorphism criterion [16]; 16's Main Theorem C). *Let F be real closed or algebraically closed of characteristic zero and G a set-sized torsion-free abelian group. The following are equivalent:*

- (i) $\mathrm{Hom}(G, \mathbb{Z}) = 0$;
- (ii) every field automorphism of $F(U^G)$ is uniquely of the form [\(48.1\)](#), extended to quotients, with $\alpha \in \mathrm{Aut} F$, $\tau \in \mathrm{Aut} G$ and $\chi \in \mathrm{Hom}(G, F^\times)$.

The implication (i) $\Rightarrow$ (ii) uses [cited theorem 49.1](#); for groups generated by elements with unbounded root orders it has the independent proof of [proposition 49.3](#).

Proof. An automorphism ϕ preserves F ([corollary 45.5](#)). Assume (i). By [cited theorem 49.1](#), with the descent above if F is a class, $g \mapsto [\phi(U^g)]$ is a homomorphism from G into a free abelian group, hence zero, since each coordinate projection to $\mathbb{Z}$ vanishes. So every $\phi(U^g)$ is a monomial; the same for ϕ^{-1} gives inverse exponent maps, and coefficient comparison gives (48.1) as in [theorem 48.1](#), extended uniquely to fractions. Conversely, if $\text{Hom}(G, \mathbb{Z}) \neq 0$, a nonzero map has image $d\mathbb{Z} \cong \mathbb{Z}$ and splits, so $G \cong G_0 \oplus \mathbb{Z}$ and $F(U^G) \cong F(U^{G_0})(T)$; the automorphism fixing $F(U^{G_0})$ with $T \mapsto T + 1$ is not monomial. $\square$

The condition says that G has no direct summand $\cong \mathbb{Z}$; it does not say that G is divisible [16].

Proposition 49.3 (An elementary range [16]). *If G is generated by elements g for which $\{n \geq 1 : g \in nG\}$ is unbounded, the description (ii) of [theorem 49.2](#) holds for F real closed or algebraically closed of characteristic zero, without [cited theorem 49.1](#).*

Proof. For such a generator U^g has roots of unbounded orders, and so has its image, which is a monomial by [corollary 46.3](#). Products give every g , and ϕ^{-1} gives the inverse exponent map; F is preserved by the quartic formula. $\square$

Examples are the divisible groups and the $\mathbb{Z}[1/p]$ -modules, and direct sums of groups of this kind, even though sums of generators may have only bounded root orders; so divisibility of every element is not needed [16]. For divisible G and $F = \mathbf{No}$, every character in (48.1) is positive, as $\chi(g) = \chi(g/2)^2$, and for $G = \mathbb{Q}^d$ it is determined by d positive surreal values. Over $\mathbf{No}(i)$ compatible roots matter: $U^q \mapsto e^{2\pi i q} U^q$, $q \in \mathbb{Q}$, is a nontrivial automorphism fixing U^1 ; its coefficients are ordinary roots of unity, and no surcomplex exponential is used. Triples compose as

$$(\alpha, \tau, \chi) \circ (\alpha', \tau', \chi') = (\alpha\alpha', \tau\tau', g \mapsto \alpha(\chi'(g))\chi(\tau'(g))).$$

Lemma 49.4 (Relative rank is relative transcendence degree [16]). *If $A \leq G$ are torsion-free abelian groups, $\text{trdeg}_{k(U^A)} k(U^G) = \dim_{\mathbb{Q}}((G/A) \otimes \mathbb{Q})$; in particular $\text{trdeg}_k \text{Frac}(k[U^M]) = \text{rk}_{\mathbb{Q}} M$.*

Proof. If the images of g_j form a basis of $(G/A) \otimes \mathbb{Q}$, the U^{g_j} are algebraically independent over $k(U^A)$ (clear denominators and compare the distinct cosets of exponents), and every U^g is algebraic over them and $k(U^A)$, since $ng = a + \sum_j m_j g_j$ for some $n > 0$ and $a \in A$. The monoid case uses the group completion. This computation is also used in [20]. $\square$

Theorem 49.5 (Coefficient field and exponent group together [16]). *Let F, E be real closed or algebraically closed of characteristic zero and G, G' set-sized torsion-free abelian groups. Then $F(U^G) \cong E(U^{G'})$ if and only if $F \cong E$ and $G \cong G'$. The forward implication uses [cited theorem 49.1](#) and the two-frame argument of Lin and Wang; it does not say that every isomorphism is monomial.*

Proof. An isomorphism ϕ maps F onto E . Put $A = \{g : \phi(U^g) \in E^\times U^{G'}\}$ and $A' = \{h : \phi^{-1}(U^h) \in F^\times U^G\}$. Monomial comparison in both directions gives $A \cong A'$ and $\phi(F(U^A)) = E(U^{A'})$. The defect maps embed G/A and G'/A' into free abelian groups ([cited theorem 49.1](#)), so both are free and the sequences split; by [lemma 49.4](#) their ranks agree. Hence $G \cong A \oplus \mathbb{Z}^{(I)} \cong A' \oplus \mathbb{Z}^{(I)} \cong G'$. Class coefficients use the descent above; the converse is the coefficientwise monomial construction. $\square$

So rational monomial extensions of $\mathbf{No}$ or $\mathbf{No}(i)$ determine their parameter groups up to isomorphism. This extends a credited reconstruction argument and is not a new proof of Lin and Wang's theorem; it leaves $\text{Aut}(\mathbf{No})$ and $\text{Aut}(\mathbf{No}(i))$ as coefficient data [16].

50 Finite-rank cancellation and the exact infinite-rank failure

Theorem 50.1 (Finite-rank coefficient invariance [16]). *Let R be a characteristic-zero domain that is root-covered or has a real closed or algebraically closed fraction field, let $\mathbf{M}, \mathbf{N}$ be cancellative torsion-free monoids of the same finite rational rank d , and S any domain. Every isomorphism $\phi : R[U^{\mathbf{M}}] \rightarrow S[U^{\mathbf{N}}]$ has $\phi(R) = S$; in particular $R \cong S$.*

Proof. Coefficient rigidity gives $\phi(R) \subseteq S$. Put $R_1 = \phi^{-1}(S)$, so $R \subseteq R_1 \subseteq R[U^{\mathbf{M}}]$, and let $F = \text{Frac } R$, $E = \text{Frac } R_1$. Pulling the $\mathbf{N}$ -presentation back along ϕ gives $\text{trdeg}_E \text{Frac}(R[U^{\mathbf{M}}]) = d$, and $\text{trdeg}_F \text{Frac}(R[U^{\mathbf{M}}]) = d$ by lemma 49.4. Additivity gives $d = \text{trdeg}_F E + d$, so E/F is algebraic as d is finite, and $E = F$ by lemma 44.2. So $R_1 \subseteq F \cap R[U^{\mathbf{M}}] = R$. $\square$

Corollary 50.2 (Finite-rank field coefficient invariance [16]). *Let F be real closed or algebraically closed of characteristic zero, G, G' of the same finite rational rank and E any field. Every isomorphism $F(U^G) \cong E(U^{G'})$ maps F onto E .*

Proof. The field clause of theorem 45.4 puts the image of F inside E ; pull E back to an intermediate field and argue as above. $\square$

These are coefficient-invariance statements, stronger than an isomorphism of coefficient rings. Finite generation is not needed, only finite rank; this is the first difference from the polynomial-cylinder statements of theorem 15.4, which for finitely many polynomial and Laurent variables also give the equality of the numbers of variables of each kind. Laurent cancellation has a large literature [10] [16].

Theorem 50.3 (Infinite-rank counterexamples [16]). *Let κ be an infinite cardinal and $G_\kappa = \mathbb{Q}^{(\kappa)}$.*

- (i) *If R is a root-covered domain of characteristic zero and $S = R[U^{\mathbb{Q}}]$, then $S \not\cong R$ but $R[U^{G_\kappa}] \cong S[U^{G_\kappa}]$.*
- (ii) *If F is real closed or algebraically closed of characteristic zero and $E = F(U^{\mathbb{Q}})$, then $E \not\cong F$ but $F(U^{G_\kappa}) \cong E(U^{G_\kappa})$.*

Proof. $\mathbb{Q} \oplus G_\kappa \cong G_\kappa$ gives both isomorphisms. In (i), $1 + U$ is not a monomial, so by proposition 46.5 and lemma 44.1 it divides no element of $\text{Rt}_\infty(S)$; so S is not root-covered, while R is, and root-coveredness is an isomorphism invariant. In (ii), F satisfies $\forall x \exists y (y^2 = 1 + x^4)$ and E does not, at $x = U^1$ (theorem 45.3). $\square$

Theorem 50.4 (A sharp rank threshold [16]; 16's Main Theorem D). *For $R = \mathbf{Oz}$ or $\mathbf{Oz}[i]$, the property “for every domain S , every isomorphism $R[U^{G_\kappa}] \cong S[U^{G_\kappa}]$ maps R onto S ” holds exactly when κ is finite. For $F = \mathbf{No}$ or $\mathbf{No}(i)$ and arbitrary comparison fields E , the analogous property of $F(U^{G_\kappa})$ has the same threshold. For infinite κ the witnesses are $S = R[U^{\mathbb{Q}}]$ and $E = F(U^{\mathbb{Q}})$.*

Proof. Theorem 50.1 and corollary 50.2 for finite κ ($\text{rk}_{\mathbb{Q}} \mathbb{Q}^\kappa = \kappa$); theorem 50.3 for infinite κ , where no isomorphism can map R onto $S \not\cong R$. $\square$

In the counterexamples the enlarged comparison ring is not root-covered and the enlarged comparison field is neither real closed nor algebraically closed, so [corollary 48.2](#) and [theorem 49.5](#) do not apply to that side; within the original family reconstruction holds at every rank. Cardinal arithmetic, $\kappa + 1 = \kappa$, not a defect of the finite-rank proof, allows the absorption [16].

51 Boundaries of the method

Example 51.1 (Hahn completion destroys the detector [16]). Let k be real closed or algebraically closed of characteristic zero. In the Hahn field $k((t^{\mathbb{Q}}))$,

$$b = \sum_{n \geq 0} \binom{1/2}{n} t^{4n} \quad \text{satisfies} \quad b^2 = 1 + t^4,$$

coefficientwise, so the nonconstant t satisfies the quartic detector there, while no nonconstant solution exists in $k(U^{\mathbb{Q}})$. The results of this part must not be transferred to Hahn fields. There is no conflict with $\sqrt{1 + \omega^4} \in \mathbf{No}$: in $\mathbf{No}(U^G)$ the element ω is a coefficient, and an external U^g with $g \neq 0$ is not a surreal magnitude. Adjoining every rational power of a parameter can increase rigidity, whereas allowing arbitrary Hahn sums destroys the detector [16].

One missing divisibility hypothesis allows translations [16]. For $G = \mathbb{Z}$, $F(U^G) = F(T)$ has $T \mapsto T + 1$. For $G = \mathbb{Q}$ [theorem 49.2](#) excludes it; elementarily, T has every root $T^{1/n}$, while $T + 1$ has two terms and is no n th power in $F(U^{\mathbb{Q}})$ for $n > 1$ by [\(46.1\)](#). At the ring level even $R[T^{\pm 1}]$ has no translation, since $T + 1$ is not a unit; field and ring hypotheses must not be interchanged.

Example 51.2 (Positive characteristic [16]). In characteristic $p > 0$, $(1 + T)^{p^r} = 1 + T^{p^r}$ has two terms for every r , so [\(46.1\)](#) fails. For $G = \mathbb{Z}[1/p]$, $\text{Hom}(G, \mathbb{Z}) = 0$, but $T^{1/p^n} \mapsto T^{1/p^n} + 1$ ($n \geq 0$) are compatible under p th powers and define a nonmonomial automorphism of $k(U^G)$, with inverse from -1 ; so characteristic zero is substantive in [theorem 49.2](#). In characteristic 2, $1 + x^4 = (1 + x^2)^2$ for every x , and the quartic formula fails completely.

Torsion, zero divisors and injectivity [16]. Torsion-free cancellative monoids give domains and separate finite supports. A monoid with a nontrivial idempotent e has $U^e(1 - U^e) = 0$; the two-element idempotent monoid gives $R[Y]/(Y^2 - Y) \cong R \times R$, whose independent factor automorphisms need not preserve the diagonal copy of R . Such examples are outside the theorems. Injectivity keeps the root element nonzero in the target; no classification of noninjective maps into reduced parameter rings is given, and [question 19.3](#) stays open.

What the results say about surreal automorphisms [16]. They classify automorphisms of the constructed extensions relative to the remaining coefficient automorphism. They do not compute $\text{Aut}(\mathbf{No})$ or $\text{Aut}(\mathbf{No}(i))$, and they do not decide whether an automorphism of $\mathbf{No}(i)$ preserving $\mathbf{Oz}[i]$ preserves conjugation or the natural valuation ([question 32.1](#)). Hahn summation is not part of $F[U^G]$ or $F(U^G)$; every displayed sum is finite.

Remark 51.3 (Dependencies and proof obligations [16]). The quartic path uses field algebra and [lemma 45.1](#); the sparse path uses finite specialization and [lemma 46.1](#); both are proved here. Arbitrary-monoid rigidity rests on 02's root-cover framework and the sparse bound; the group-algebra and divisible-monoid formulas on coefficient recovery, units and roots; the general field criterion and [theorem 49.5](#) on [cited theorem 49.1](#); the divisible and prime-local criterion on the sparse bound alone; the cancellation threshold on embedding rigidity, relative transcendence degree and group absorption. 16's checklist of proof obligations is honoured above: monoids are cancellative and torsion-free before completion; specialization acts on a

finite identity with nonzero denominators, injective on $\text{supp } P - \text{supp } Q$ where needed; a root order is compared with a fixed presentation, without a single lattice containing all roots; zero cases precede Mason–Stothers; ring witnesses satisfy $v \neq 0$ and $\text{Frac}(R[U^M]) = \text{Frac}(R)(U^{\text{gp } M})$ is used explicitly; set-sized Hahn root-covered rings are not identified with **No** or with their Hahn fraction field; the field criterion recovers coefficients first and then uses the set-subfield form of the defect theorem; the infinite-rank counterexamples leave the recovered family; characters of the omnific *rings* lie in a finite unit group and vanish on divisible groups, those of the surcomplex *field* need not; all-order root membership is not treated as first order.

52 Questions of source 16, relations, and stale statements

52.1 Status of the source’s questions

16 poses eleven questions (its 12.1–12.11), proposed as research problems, not as published open problems. Question 12.8 (reduced parameter rings and kernels) is [question 19.3](#), with 16 added to its sources; 16’s domain proofs do not reach it. The other ten are new.

Question 52.1 (Beyond rational monomial fields [16]). Which intermediate fields $k(U^G) \subseteq L \subseteq k((t^G))$ (for an ordered G , with $U^g = t^g$) still satisfy quartic rigidity, so that $\exists y (y^2 = 1 + x^4)$ detects only k ?

A first task is to find the smallest algebraic or support-closed extensions with a nonconstant point of the quartic; adjoining the square root of [example 51.1](#) already breaks the detector, so inclusion in a Hahn field is no useful sufficient condition [16]. *Related* (batch 33): in the formal completions the ring formula holds of every element of **Oz**[[s]], **Oz**[i][[s]] and **No**[[s]], and in **No**(i)[[s]] exactly up to an order parity ([theorem 72.7](#) and [proposition 72.8](#)), so the s -adic completion is the extreme case; this does not answer the question.

Question 52.2 (Single-equation ring definitions [16, 19]). For $R = \mathbf{Oz}$ or $\mathbf{Oz}[i]$, can the inequation in [\(45.2\)](#) be removed using further polynomial witnesses, uniformly in M and without naming a nonunit parameter?

A purely equational definition would be a different Diophantine result; deleting $v \neq 0$ is invalid [16]. Source 19, which found the same formula independently ([corollary 72.2](#)), asks the same for $\mathbf{Oz}[U]$ and $\mathbf{Oz}[i][U]$ (its Question 13.4): a positive-existential definition, or a single polynomial equation with further witnesses, or any lower-complexity uniform substitute. Removing $b \neq 0$ is not a formal rewriting, since zero witnesses destroy the test; genus one suffices for the chosen strategy, and no optimality over all formulas is proved [19]. *Status* (batch 33): open.

Question 52.3 (Optimal root certificates [16]). Can [\(46.1\)](#) be refined through the Newton polytopes or coefficient relations of P, Q , keeping invariance under arbitrary torsion-free monomial extensions?

The sharp worst case uses weights only; a refinement would rule out a given root order without multivariate factoring, with finite separating functionals as witnesses, over effective coefficient fields and exponent presentations, not as an algorithm for all surreals [16].

Question 52.4 (Positive characteristic [16]). What replaces $\text{Hom}(G, \mathbb{Z}) = 0$ for monomial automorphism rigidity in characteristic p , and which prime-to- p root tests remain valid?

[Example 51.2](#) rules out the characteristic-zero criterion; Frobenius-divisible monomial and nonmonomial directions should be separated first [16].

Question 52.5 (Cancellation for nonabsorbing infinite groups [16]). For which infinite-rank groups G does coefficient invariance against arbitrary comparison rings still hold for $R[U^G]$?

$\mathbb{Q}^{(\kappa)}$ fails because it absorbs a $\mathbb{Q}$ summand; the finite-rank proof cannot cancel infinite cardinals, but that does not prove failure for every infinite-rank group [16].

Question 52.6 (Monoids without divisibility [16]). How much of a cancellative torsion-free monoid M can be recovered from $R[U^M]$, once the coefficient ring is recovered, without assuming M divisible?

[Theorem 48.3](#) needs all-order roots; nonlinear polynomial automorphisms obstruct the same formula otherwise, and the monoid-ring isomorphism literature should be compared [16].

Question 52.7 (Root-covered coefficient definability [16]). Which root-covered domains admit a uniform first-order definition of the coefficient ring in every $R[U^M]$, beyond the closed-fraction-field case?

$\mathfrak{C}$ is infinitary and its finite truncations are vacuous; the quartic settles the omnific examples, not every root-covered domain [16].

Question 52.8 (Intrinsic monomial sets [16]). Under $\text{Hom}(G, \mathbb{Z}) = 0$, is the monomial subgroup $F^{\times}U^G$ of $F(U^G)$ first-order definable for interesting classes of groups, rather than only invariant under all automorphisms?

For divisible G its all-order-root description is explicit but not first order; invariance never establishes definability [16].

Question 52.9 (Surcomplex real-form recovery [16]). Does additional Gaussian omnific structure in larger extensions recover a preferred conjugation or real form of $\mathbf{No}(i)$, and under exactly which valuation or summation hypotheses?

The present results recover $\mathbf{No}(i)$ as a coefficient field, not its copy of $\mathbf{No}$ or its conjugation, so they do not answer [question 32.1](#) [16]. The question is a relative of `odg:def:q:realform` of the Diophantine report and of [question 32.1](#) [merge].

Question 52.10 (Formalization in stages [16]). Can the specialization lemma, the sparse power bound, the quartic detector and monoid reconstruction be formalized over set-sized coefficient fields and monoids, and then connected to the repository's surreal construction?

16 proposes the order: sparse support separation, the Mason–Stothers quartic corollary, fraction and monoid compatibility, coefficient interpretation, finite-rank cancellation, and only then class descent, keeping the general criterion a separate dependency until defect freeness is formally available. The Lean formalization of Mason–Stothers by Baek and Lee [1] is a possible starting point, not a certificate for 16 [16].

52.2 Relations and stale statements

- *Not answered.* [Question 19.1](#) (groups with an order unit) is not addressed. 16 does not answer `odg:def:q:coeffpe` of the Diophantine report: that question asks for a lower-complexity recovery of $\mathbb{R}$ or $\mathbb{C}$ from the integer-part ring alone, whereas [\(45.2\)](#) recovers R in $R[U^M]$; with trivial M it is vacuous, and it defines neither $\mathbf{Oz}$ nor $\mathbb{R}$ in $\mathbf{No}$ [merge]. [Question 19.3](#) and [question 19.2](#) stay as they were. *Status* (batch 33): [question 19.2](#) is answered by source 19 ([part IX](#)); [question 19.3](#) is refined by 19 and stays open.

- *Checked true.* 16’s description of source 02 and of its pin; the pin `1ab41af` and its tree `fba47f2`, which 16 correctly distinguishes; the SHA-256 hashes in its build record, which match the delivered files; its statement that the reducible-parameter question remains open; its suite, rerun for the merge ([section C](#)). No mathematical error was found in 16.
- *Not rechecked.* [cited theorem 49.1](#) and the two-frame method are Lin and Wang’s; 16 read the preprint and states Theorem 4.5 as displayed, and the merge did not recheck the preprint. The pinpoint descriptions of [\[22\]](#), [\[1\]](#) and [\[10\]](#) are reported as 16 gives them.

16’s limitations are listed with the other sources’ in [section B](#).

Part VII

Full formal orbit fields of the Euler flow

This part records manuscript 17, the first addition of batch 33. Its question is the one left open after [corollary 17.9](#): [theorem 17.5](#) computes the field generated by the transformed *monomials* of a formal Euler flow, and only a lower bound was asserted for the field generated by the formal images of all Hahn series. For the constant-term Euler derivation D_0 of [theorem 17.6](#) ($b = 0$), 17 describes the full image exactly. It is linearly disjoint from $\mathbf{No}(s)$ over the constant field $\ker D_0$, which is much larger than $\mathbb{R}$; it contains explicit omnific independent families of every set cardinality, so it has no set-sized transcendence basis; and s is transcendental over it ([sections 55](#) and [56](#)). After the formal change of parameter $s = \log(1 + v)$ the rational and algebraic trajectories are classified exactly by $C_0(\omega)$ and its relative algebraic closure, and the parameter $u = 1 + v$ becomes recoverable ([section 57](#)). The part treats $\mathbf{No}(i)$ and $\mathbf{Oz}[i]$ as well. Its labels carry the sub-prefix `opa:of:`.

53 Source 17, what it re-derives, and its notation

53.1 The manuscript and its pin

Manuscript 17 is *Full Formal Orbit Fields of Surreal Arithmetic: proper-class independence, exact algebraic trajectories, and the effect of changing the formal parameter*, dated 23 September 2026, 25 pages. It is manuscript 03 of batch 33, delivered as the archive `Surreal_Formal_Orbit_Fields` in commit `73043eb` and placed as an addition in commit `aa9c891`; it is numbered 17 here, continuing this report’s local sequence. Its checks, their recorded run, its requirements file, its build record, its Makefile and its proof and source audits are shipped as `code/17-formal-orbit-fields-verify.py`, `data/17-formal-orbit-fields-verification.json`, `data/17-formal-orbit-fields-requirements.txt`, `data/17-formal-orbit-fields-BUILD_REPORT.json`, `code/17-formal-orbit-fields-Makefile`, `17-formal-orbit-fields-PROOF_AUDIT.md` and `17-formal-orbit-fields-SOURCE_AUDIT.md`; its `article.tex`, PDF, README and checksum list are not shipped. The SHA-256 hashes that its build record gives for `article.tex`, `article.pdf`, `verify.py` and `verification.json` match the delivered files.

It pins `efc5446`, the eight-source text of this report: [parts I](#) to [III](#) (`20c4c9f`) and the placement `9d28e28` of source 14, but not the write `d4d72d7` of [part IV](#) nor [parts V](#) and [VI](#). It read this report’s guide, searched this article around its formal-flow material, and read source 02’s audit and the guides of the Diophantine, quotient, independent-copies and holonomic reports; some connector responses were truncated and only searched (its source audit). Its

account of this report is accurate: it credits [theorem 17.5](#) and [corollary 17.9](#), quotes the non-claim that only a lower bound holds for the full image, and claims none of [part II](#)’s rigidity results.

53.2 Printed once

17’s numbers are those of its own PDF. “Same” means the same statement up to notation.

17	Printed as	Comparison
Lem. 2.1	<code>odg:thm:fractions</code>	Same: one monomial clears a set of surreals, so $\text{Frac } \mathbf{Oz} = \mathbf{No}$ and $\text{Frac } \mathbf{Oz}[i] = \mathbf{No}(i)$. Cited, not reprinted.
Prop. 2.2	proposition 54.1 ; proposition 17.1 and theorem 17.6	Its derivation is D_0 , the case $b = 0$ of theorem 17.6 ; the constant field and $\text{ct}(D_0x) = 0$ are added and printed.
Prop. 4.1, (4.2)	proposition 55.1 , (55.2); theorem 17.2	The ring statements and the flow law are theorem 17.2 for $\chi = q_0$; the field statement and the monomial formula are printed.
Section 4.4	lemma 17.4 , theorem 17.5 , and corollary 17.9	The same Vandermonde proof of the monomial calculation; not reprinted (section 55.1).
Thms. 1.1–1.3, Cor. 1.4	theorems 53.1 to 53.3 and corollary 53.4	Summaries, proved in sections 54 to 58 .
Thm. B.1	theorem 58.6	Printed; corollary 58.7 applies it to every Euler derivation of part II [merge].
Questions 1–10	section 59.3	Eight new; its question on internal evaluation is merged with source 19’s (question 59.6), and its formalization question with those of 18 and 19 (question 59.10).
All else	This part	New.

53.3 Conventions and notation of this part

This part keeps the large-monomial convention of [part II](#) ([section 12.1](#)), which is 17’s: a surreal is $x = \sum_{g \in S} r_g \omega^g$ with S a reverse well-ordered set and $r_g \in \mathbb{R}$, so $\omega^g = X^g = t^{-g}$. For an exponent $g \in \mathbf{No}$, $q_0(g)$ is the real constant coefficient of the normal form of g itself, as in [section 7](#) (the inner normal form), not the coefficient r_g of x . The derivation is

$$D_0\left(\sum_g r_g \omega^g\right) = \sum_g q_0(g) r_g \omega^g, \quad C_0 = \ker D_0, \quad (53.1)$$

the Euler derivation D_b of [theorem 17.6](#) with $b = 0$ and [proposition 17.1](#)’s D_χ with $\chi = q_0$. Its formal flow is [theorem 17.2](#)’s $\mathcal{E}_{0,s}$, written by the same formula ([17.1](#)) on all of $\mathbf{No}$. The

indeterminates s and v are external: no surreal is substituted for them, and $\mathbf{No}[[s]]$, $\mathbf{No}((s))$ are formal power and Laurent series (finitely many negative powers). Field generation means closure under *finite* field operations, not a Hahn closure or closure under arbitrary formal sums; this distinction is essential to the results and their proofs [17]. The symbols are renamed as follows.

17's symbol	Here
$D; h(g) = \text{ct}(g)$	$D_0; q_0(g)$.
$H = \ker h; C = \ker D$	$\ker q_0; \mathbf{C}_0$.
$\mathcal{T}_s$ (Taylor map); $\mathcal{U}_v = (1 + v)^D$	$\mathcal{E}_{0,s}; \Upsilon_v$.
$L_{\text{add}}, L_{\text{mult}}$	$\widehat{L}_0, \widehat{L}_0^\times$ (the full orbit fields).
$F = \mathbf{No}, M = F((s)),$ $A = C(\omega), A' = C(\omega u)$	Written out: $\mathbf{No}, \mathbf{No}((s)), \mathbf{C}_0(\omega), \mathbf{C}_0(\omega u)$.
$\mathfrak{P}$	Π .
$\text{Const}(K, D), C_K$	K^∂ , the constants of a differential field (K, ∂) .
$\delta = D - \partial_s; \delta_m = D - u\partial_u$	$\nabla_s = D_0 - d/ds; \nabla^\times = D_0 - u d/du$.
$\partial = -\omega D$	$\widetilde{D}_0 = -\omega D_0$.
$A_\alpha, B_\alpha, a_\alpha, M_\alpha, E_\alpha, X_\alpha$	$\mathfrak{a}_\alpha, \mathfrak{b}_\alpha, \zeta_\alpha, \omega^{\mathfrak{b}_\alpha}, \exp(\zeta_\alpha z), \Xi_\alpha$.
$\mathbf{No}(i), C(i), \mathbf{Oz}[i]; F_{\mathbb{C}}, C_{\mathbb{C}}$	$\mathbf{No}(i), \mathbf{C}_0(i), \mathbf{Oz}[i]$.
t with $Dt = 1$ (example 54.8)	y_1 .

The collisions are resolved here only, and the tempting false readings are wrong:

- 17's C is not $\mathbb{C}$, not a principal convex subgroup C_δ or $C(a)$, and not C_1 . It is the field $\mathbf{C}_0$ of normal forms supported in $\ker q_0$, a real closed field containing ω^ω and $\omega^{-\omega}$; “constant” means D_0 -constant, not “ordinary real”.
- $\ker q_0$ (17's H) is a subgroup of $\mathbf{No}$, not convex, not the convex subgroup H_g of [definition 2.4](#), and not the hull $\mathcal{H}(A)$ of [theorem 7.4](#); no valuation on a quotient by it is used. 17's h is an exponent character, not a logarithmic character h_σ .
- 17's $\mathcal{T}_s$ is a formal flow into $\mathbf{No}[[s]]$, not the Taylor automorphism $\mathcal{T}_{d,\delta}$ of [part III](#) and not a topology of [part V](#). 17's $\mathcal{U}_v$ is not the stabilizer $\mathcal{U}$; its $u = 1 + v$ is not the normalized factor u of [theorem 24.1](#) or the witness u of [\(45.2\)](#). In this part v is a formal indeterminate: no valuation $v(x)$ occurs here.
- $\widehat{L}_0$ is not [theorem 17.5](#)'s L_χ . For $\chi = q_0$, $R = \mathbf{Oz}$ and $F = \mathbf{No}$, $L_{q_0} = \mathbf{No}(e^{rs} : r \in \mathbb{R})$ is the monomial orbit field, a subfield of $\widehat{L}_0$.
- $\mathfrak{a}_\alpha, \mathfrak{b}_\alpha$ are exponents, not the shears A_s, B_u of [section 6](#); $\exp(\zeta_\alpha z)$ is one explicit Hahn series (the notation $\exp(u)$ of [section 1.5](#) for an infinitesimal u), not a global surreal or surcomplex exponential, not the group $\mathcal{E}_A$ of [theorem 7.4](#) and not the set E of [lemma 7.1](#). Ξ_α is not a Hahn monomial X^g . $z = \omega^{-1}$ is local to [section 56](#), not the covering element z of [definition 12.1](#).
- 17's letters F, A, M, L are fields, not Part III's isomorphism F , the rings $\mathcal{A}_k, \mathcal{A}_\Gamma$, or the monomial maps $M_{\chi,\tau}$. In [section 54.2](#) K, E, M, A, B are fields local to each statement and ∂ is an arbitrary derivation, as in [section 12.1](#).

Class statements are read in NBG with global choice, as in [section 1.4](#). No class-sized sum is formed. An **On**-indexed independent family means that every finite subfamily satisfies no nonzero polynomial relation over the stated field; it yields a set-indexed family of every set cardinality. Tensor products and polynomial ideals below are shorthand for finite-relation statements, and every calculation may be read in a set-sized differential subfield containing its data [17].

53.4 Principal results of source 17

Write $\widehat{L}_0 = \mathbf{No}(\mathcal{E}_{0,s}(x) : x \in \mathbf{No}) \subseteq \mathbf{No}((s))$ for the *full additive orbit field*.

Theorem 53.1 (Exact additive-flow structure [17]; 17's Theorem 1.1). *The fields $\mathbf{No}(s)$ and $\mathcal{E}_{0,s}(\mathbf{No})$ inside $\mathbf{No}((s))$ are linearly disjoint over $\mathbb{C}_0$, and so are $\mathbf{No}$ and $\mathcal{E}_{0,s}(\mathbf{No})$. Consequently:*

- (i) *for every finite tuple $\mathbf{x}$ from $\mathbf{No}$, $\text{trdeg}_{\mathbf{No}(s)} \mathbf{No}(s, \mathcal{E}_{0,s}(\mathbf{x})) = \text{trdeg}_{\mathbb{C}_0} \mathbb{C}_0(\mathbf{x}) = \text{trdeg}_{\mathbf{No}} \mathbf{No}(\mathcal{E}_{0,s}(\mathbf{x}))$;*
- (ii) *$\mathcal{E}_{0,s}(x)$ is algebraic over $\mathbf{No}(s)$ if and only if $x \in \mathbb{C}_0$;*
- (iii) *s is transcendental over $\widehat{L}_0$;*
- (iv) *all finite polynomial relations in a transformed tuple descend, by scalar extension, from its relations over $\mathbb{C}_0$.*

Proof. Linear disjointness is [theorem 55.2](#); (i) and (iv) follow by [lemma 54.11](#) and [\(55.4\)](#); (ii) is [corollary 55.3](#); (iii) is [theorem 55.4](#). $\square$

The constant field here is $\mathbb{C}_0$, not $\mathbb{R}$. The equality of transcendence degrees is an equality of finite integers; the corresponding independence criterion applies to arbitrary set-indexed families [17].

Theorem 53.2 (Explicit omnific witnesses at every cardinal [17]; 17's Theorem 1.2). *For each ordinal α put $\mathbf{a}_\alpha = \omega^{\alpha+1}$, $\mathbf{b}_\alpha = \omega^{\alpha+2}$ and*

$$\Xi_\alpha = \sum_{n \geq 0} \frac{\omega^{\mathbf{b}_\alpha - n(\mathbf{a}_\alpha + 1)}}{n!}. \quad (53.2)$$

Each Ξ_α is a countably supported, purely infinite omnific integer. Every finite subfamily of $(\Xi_\alpha)_{\alpha \in \mathbf{On}}$ is algebraically independent over $\mathbb{C}_0(\omega)$, and $(\mathcal{E}_{0,s}(\Xi_\alpha))_{\alpha \in \mathbf{On}}$ is algebraically independent over $\mathbf{No}(s)$ in the finite-relation sense. In particular $\widehat{L}_0/\mathbf{No}$ has algebraically independent families of every set cardinality; neither $\widehat{L}_0/\mathbf{No}$ nor $\widehat{L}_0(s)/\mathbf{No}(s)$ has a set-sized transcendence basis, and neither is algebraic over a field generated over its base by a set of elements of the ambient formal Laurent field.

Proof. [Lemma 56.4](#), [theorem 56.5](#), and [corollary 56.6](#). $\square$

The final assertion includes sets of generators outside the orbit field: if the whole orbit field were algebraic over such a set-generated field in the same ambient field, its independent families would be bounded by the size of that set [17].

Theorem 53.3 (Exact multiplicative-flow classification [17]; 17's Theorem 1.3). *Put $u = 1 + v$ and $\Upsilon_v(x) = \sum_{n \geq 0} \binom{D_0}{n}(x) v^n$, where $\binom{D_0}{n} = D_0(D_0 - 1) \cdots (D_0 - n + 1)/n!$. Then Υ_v*

is a field embedding of $\mathbf{No}$ into $\mathbf{No}((v))$; its image and $\mathbf{No}(u)$ are linearly disjoint over $\mathbf{C}_0(\omega u) = \Upsilon_v(\mathbf{C}_0(\omega))$. For $x \in \mathbf{No}$,

$$\begin{aligned}\Upsilon_v(x) \in \mathbf{No}(u) &\iff x \in \mathbf{C}_0(\omega), \\ \Upsilon_v(x) \text{ algebraic over } \mathbf{No}(u) &\iff x \text{ algebraic over } \mathbf{C}_0(\omega),\end{aligned}$$

and when these algebraic conditions hold the two finite degrees are equal. Moreover $u = \Upsilon_v(\omega)/\omega$ lies in $\hat{L}_0^\times = \mathbf{No}(\Upsilon_v(x) : x \in \mathbf{No})$, and $(\Upsilon_v(\Xi_\alpha))_{\alpha \in \mathbf{On}}$ is algebraically independent over $\mathbf{No}(u)$ in the finite-relation sense.

Proof. Proposition 57.1, theorem 57.3, and corollaries 57.4 and 57.5. $\square$

Corollary 53.4 (Arithmetic and complex forms [17]; 17's Corollary 1.4). *The two flows restrict to ring embeddings $\mathbf{Oz} \rightarrow \mathbf{Oz}[[s]]$ and $\mathbf{Oz} \rightarrow \mathbf{Oz}[[v]]$, which preserve the integer constant term, read coefficientwise. Replacing $\mathbf{No}, \mathbf{C}_0, \mathbf{Oz}$ by $\mathbf{No}(i), \mathbf{C}_0(i), \mathbf{Oz}[i]$ gives theorems 53.1 to 53.3 with the same witnesses.*

Proof. Propositions 55.1 and 57.1 and theorem 58.4. $\square$

What 17 proposes as new. Taylor morphisms, Euler derivations, independence of exponential solutions and the constants lemma are not new constructions. Taylor morphisms have an extensive differential-algebraic setting [25]; formal summability, automorphisms and derivations in Hahn-type algebras are studied in [3, 15]; composita of a field with its Taylor image belong to the background of Umemura's Galois-hull approach [13], and the field studied here is not identified with a full Galois hull with all its additional derivations or functorial structures. 17 proposes the linked surreal-specific conclusion: exact full-image descent, explicit omnific independent families of unbounded set cardinality, exclusion of the additive parameter, and the exact contrasting classification after multiplicative reparametrization; these go beyond the monomial calculation of part II. Priority has not been established by an exhaustive review; complete proofs are supplied so that correctness can be judged apart from novelty [17]. At the merge this is accurate: part II contains only the monomial field and a lower bound.

54 The Euler derivation D_0 and differential-field lemmas

54.1 The derivation and its constant field

That every surreal is a quotient of two omnific integers, with one monomial denominator for a whole set, is `odg:thm:fractions`; 17 reproves it (its Lemma 2.1, by choosing $\beta > 0$ above the negatives of all exponents in the union of the supports) so as to depend on no later repository claim, and applies it to real and imaginary parts for $\text{Frac } \mathbf{Oz}[i] = \mathbf{No}(i)$ [17]. Normal forms, real closedness and omnific arithmetic are classical [7, 11, 19]. Arbitrary real coefficients are allowed at strictly positive exponents of an omnific integer, so the coefficients $1/n!$ in (53.2) are compatible with integrality; $\text{ct} : \mathbf{Oz} \rightarrow \mathbb{Z}$ is a ring retraction with kernel Π .

Proposition 54.1 (The derivation D_0 and its constants [17]; [02]). *Formula (53.1) defines a strong $\mathbb{R}$ -linear derivation of $\mathbf{No}$ with*

$$D_0\omega = \omega, \quad D_0(\omega^g) = q_0(g)\omega^g, \quad \text{ct}(D_0x) = 0 \quad (x \in \mathbf{No}). \quad (54.1)$$

It preserves Π , sends $\mathbf{Oz}$ into Π , and its constant field is $\mathbf{C}_0 = \{\sum_g r_g \omega^g : \text{supp} \subseteq \ker q_0\}$.

Proof. Extraction of the constant coefficient is additive on normal forms, so $q_0(g + g') = q_0(g) + q_0(g')$. D_0 neither enlarges nor reorders a support, so additivity and $\mathbb{R}$ -linearity hold coefficientwise, and D_0 commutes with every strong sum. Each coefficient of a product is a finite sum, and $q_0(g + g') = q_0(g) + q_0(g')$ splits it into the two Leibniz terms. At exponent 0 the multiplier is $q_0(0) = 0$, so $\text{ct}(D_0x) = 0$; positive supports stay positive and integer constants are killed. By uniqueness of normal forms $D_0x = 0$ exactly when every exponent of x lies in $\ker q_0$. The derivation and its preservation of $\mathbf{Oz}$ are [theorem 17.6](#) with $b = 0$ [02]. $\square$

The field $\mathbf{C}_0$ is much larger than $\mathbb{R}$: $\omega^\omega, \omega^{-\omega} \in \mathbf{C}_0$ while $\omega \notin \mathbf{C}_0$.

Remark 54.2 (Two roles of the constant coefficient [17]). ct is not a ring homomorphism on $\mathbf{No}$: $\text{ct}(\omega \cdot \omega^{-1}) = 1$ while both factors have constant term 0 (as for P_0 in [section 2](#)). Its additivity suffices to make q_0 an additive exponent character; its multiplicativity is used only on $\mathbf{Oz}$, or another ring supported on nonnegative exponents. Confusing the two roles would invalidate the proof of [proposition 54.1](#).

Corollary 54.3 (No additive slice [17]). *There is no $y \in \mathbf{No}$ with $D_0y = 1$.*

Proof. $\text{ct}(D_0y) = 0$, whereas $\text{ct}(1) = 1$. $\square$

Lemma 54.4 (Constants are relatively algebraically closed [17]). *In a differential field (K, ∂) of characteristic zero, K^∂ is relatively algebraically closed in K .*

Proof. If a is algebraic over K^∂ with minimal polynomial P , differentiating $P(a) = 0$ gives $P'(a)\partial a = 0$; P is separable, so $P'(a) \neq 0$ and $\partial a = 0$. $\square$

Corollary 54.5 ([17]). *$\mathbf{C}_0$ is real closed and ω is transcendental over it. The constant field of the complexified derivation on $\mathbf{No}(i)$ is $\mathbf{C}_0(i)$, which is algebraically closed.*

Proof. A positive element of $\mathbf{C}_0$ has a square root in $\mathbf{No}$, algebraic over $\mathbf{C}_0$ and hence in $\mathbf{C}_0$ by [lemma 54.4](#); an odd-degree polynomial over $\mathbf{C}_0$ has a root in $\mathbf{No}$, which lies in $\mathbf{C}_0$ for the same reason. If ω were algebraic over $\mathbf{C}_0$, then $D_0\omega = 0$, contrary to [\(54.1\)](#). On $\mathbf{No}(i)$, $D_0(a + ib) = D_0a + iD_0b$, so both components vanish separately; $\mathbf{C}_0(i)$ is algebraically closed since $\mathbf{C}_0$ is real closed. $\square$

Nothing here asserts that D_0 is the Berarducci–Mantova derivation or that it differentiates a surreal exponential by a prescribed chain rule; it is the normal-form Euler derivation of [part II](#) [17].

54.2 Differential-field lemmas

These lemmas are background differential algebra, not independent novelty claims (compare [16, 25]); 17 proves them to make their role and their standard character visible. The proofs use finite relations only and apply to the class fields in the interpretation of [section 53.3](#) [17].

Lemma 54.6 (Constants are linearly disjoint [17]). *Let (E, ∂) be a differential subfield of (M, ∂) . Then E and M^∂ are linearly disjoint over E^∂ inside M ; so are E and every intermediate field $E^\partial \subseteq B \subseteq M^\partial$.*

Proof. Suppose $b_1, \dots, b_r \in M^\partial$ are linearly independent over E^∂ but dependent over E . Take a dependence of minimal length, normalized as $b_1 + a_2b_2 + \dots + a_rb_r = 0$ with $a_j \in E$. Differentiating gives $\sum_{j \geq 2} \partial(a_j)b_j = 0$, a shorter relation unless every $\partial a_j = 0$; so all $a_j \in E^\partial$, contradicting independence over E^∂ . Thus every finite E^∂ -independent family in M^∂ stays independent over E , which is linear disjointness. $\square$

Equivalently the multiplication map $E \otimes_{E^\partial} B \rightarrow M$ is injective; this is used only as shorthand for the finite statement just proved, so no class-sized tensor product is formed [17].

Lemma 54.7 (No-slice rational constants [17]). *Let (K, ∂) have characteristic zero and no y with $\partial y = 1$. On $K(s)$ put $\nabla_s = \partial - d/ds$, with ∂ acting on coefficients. Then $K(s)^{\nabla_s} = K^\partial$.*

Proof. Let $0 \neq f \in K(s)$ with $\nabla_s f = 0$, and expand it at infinity, $f = a_m s^m + a_{m-1} s^{m-1} + \dots$ with $m \in \mathbb{Z}$, $a_m \neq 0$. The coefficient of s^m gives $\partial a_m = 0$. If $m \neq 0$, the coefficient of s^{m-1} gives $\partial a_{m-1} = m a_m$, a nonzero constant, so $\partial(a_{m-1}/(m a_m)) = 1$, a contradiction. So $m = 0$ and $a_0 \in K^\partial$. If $f \neq a_0$, the same argument applies to the constant $f - a_0$, whose leading degree is negative, which is impossible. So $f = a_0$. $\square$

The expansion at infinity is an ordinary formal Laurent expansion in s^{-1} and needs no valuation on K ; a pole at infinity and a strictly proper rational function are covered by the same argument [17].

Example 54.8 (Why the missing slice matters [17]). On $K = K_0(y_1)$ with $\partial y_1 = 1$ and $\partial K_0 = 0$, the element $y_1 + s$ is a nonconstant ∇_s -constant, so the conclusion of lemma 54.7 fails. Indeed the Taylor image of y_1 is $y_1 + s$, and the parameter $s = (y_1 + s) - y_1$ lies in the orbit field. The no-slice hypothesis is structural, not cosmetic.

Lemma 54.9 (Rational constants with a fixed variable [17]). *If w is an indeterminate over K and $\partial w = 0$, then $K(w)^\partial = K^\partial(w)$.*

Proof. Write a nonzero constant as $P(w)/Q(w)$ with $P, Q \in K[w]$ coprime and Q monic. Then $(\partial P)Q = P(\partial Q)$, so Q divides ∂Q . The leading coefficient of Q is 1, so $\deg \partial Q < \deg Q$ unless $\partial Q = 0$; hence $\partial Q = 0$ and then $\partial P = 0$, and all coefficients lie in K^∂ . The converse inclusion is immediate. $\square$

For a characteristic-zero (K, ∂) put $\mathcal{E}_s^\partial(x) = \sum_{n \geq 0} \partial^n(x) s^n / n!$ and $\Upsilon_v^\partial(x) = \sum_{n \geq 0} \binom{\partial}{n}(x) v^n$; for $(\mathbf{No}, D_0)$ these are $\mathcal{E}_{0,s}$ and Υ_v .

Lemma 54.10 (Horizontal Laurent series are exactly Taylor images [17]). *For a characteristic-zero differential field (K, ∂) ,*

$$K((s))^{\partial - d/ds} = \mathcal{E}_s^\partial(K), \quad (54.2)$$

$$K((v))^{\partial - (1+v)d/dv} = \Upsilon_v^\partial(K). \quad (54.3)$$

Proof. Let $f = \sum_{n \geq N} b_n s^n$ be a nonzero horizontal Laurent series with $b_N \neq 0$. If $N < 0$, the coefficient of s^{N-1} in $(\partial - d/ds)f$ is $-N b_N \neq 0$. So $N \geq 0$, and comparing coefficients gives $\partial b_n = (n+1)b_{n+1}$, hence $b_n = \partial^n(b_0)/n!$ and $f = \mathcal{E}_s^\partial(b_0)$; the converse is the same recurrence. In the second case a negative lowest power is impossible for the same reason, since $1+v$ has constant term 1, and the recurrence is $(n+1)b_{n+1} = (\partial - n)b_n$, which gives $b_n = \binom{\partial}{n}(b_0)$. The recurrences also cover the zero series and prove uniqueness from the initial coefficient. $\square$

This identifies the whole constant field of the Laurent field, not just a subfield of it [17].

Lemma 54.11 (Relations and degrees under linear disjointness [17]). *Let fields E and B in a common field be linearly disjoint over A . For $\mathbf{b} \in B^r$ let $I_A(\mathbf{b}) = \{P \in A[Y_1, \dots, Y_r] : P(\mathbf{b}) = 0\}$. Then*

$$I_E(\mathbf{b}) = E[\mathbf{Y}] I_A(\mathbf{b}). \quad (54.4)$$

In particular algebraic independence and all finite degrees of subextensions of B/A are preserved after adjoining E .

Proof. $A[\mathbf{b}]$ is the image of $A[\mathbf{Y}]$ with kernel $I_A(\mathbf{b})$; tensoring this presentation with the field E is exact, and $E \otimes_A A[\mathbf{b}] \rightarrow EB$ is injective by linear disjointness, which is (54.4). Alternatively, collect the finitely many evaluated monomials of a proposed relation into an A -basis: it stays independent over E , so every relation is generated by A -relations. An A -basis of a finite extension inside B stays independent over E and spans the compositum, which gives equality of degrees. $\square$

For transcendence degree take a transcendence basis of the finitely generated $A(\mathbf{b})$: its independence survives and the algebraic remainder stays algebraic. This justifies the finite-tuple equalities of this part [17].

55 The full additive orbit field

Proposition 55.1 (The formal Taylor embedding [17]; [02]). $\mathcal{E}_{0,s} : \mathbf{No} \rightarrow \mathbf{No}[[s]]$, $x \mapsto \sum_{n \geq 0} D_0^n(x) s^n / n!$, is a unital field embedding, its image a subfield of $\mathbf{No}((s))$. It fixes $\mathbf{C}_0$, satisfies $\mathcal{E}_{0,s}(\omega^g) = \omega^g e^{q_0(g)s}$ with $e^{rs} = \sum_n r^n s^n / n!$, and restricts to a unital embedding $\mathbf{Oz} \rightarrow \mathbf{Oz}[[s]]$ with

$$\text{ct}_*(\mathcal{E}_{0,s}(a)) = \text{ct}(a) \in \mathbb{Z} \subseteq \mathbb{Z}[[s]] \quad (a \in \mathbf{Oz}), \quad (55.1)$$

where ct_* extracts the constant term of each coefficient.

Proof. The definition is coefficientwise and needs no convergence. The iterated Leibniz rule $D_0^n(xy) = \sum_j \binom{n}{j} D_0^j(x) D_0^{n-j}(y)$ gives multiplicativity at each power of s ; additivity and $\mathcal{E}_{0,s}(1) = 1$ are immediate. The coefficient of s^0 is x , so the map is injective, and for $x \neq 0$ multiplicativity shows that $\mathcal{E}_{0,s}(x^{-1})$ is the inverse of $\mathcal{E}_{0,s}(x)$; so the image is a field. The monomial formula follows from $D_0^n(\omega^g) = q_0(g)^n \omega^g$. For $a \in \mathbf{Oz}$ and $n \geq 1$, $D_0^n(a) \in \Pi$, and division by $n!$ only rescales real coefficients at positive exponents; so all coefficients lie in $\mathbf{Oz}$ and only the coefficient of s^0 has a nonzero constant term. The ring statements are theorem 17.2 for $\chi = q_0$ [02]. $\square$

Equation (55.1) concerns the ordinary integer retraction, not a topological or analytic standard-part map. The formal group identity

$$\mathcal{E}_{0,s}(\mathcal{E}_{0,t'}(x)) = \mathcal{E}_{0,s+t'}(x) \quad (55.2)$$

in $\mathbf{No}[[s, t']]$, with $\mathcal{E}_{0,s}$ acting on the coefficients of the t' -series and fixing t' , is the flow law of theorem 17.2 ($D_0^j D_0^k = D_0^{j+k}$ and the binomial theorem at each bidegree). It is the usual iterativity of a Taylor morphism and gives no evaluation at a surreal value of s or t' [17].

Theorem 55.2 (Full additive-image descent [17]). Extend D_0 coefficientwise to $\mathbf{No}((s))$ and put $\nabla_s = D_0 - d/ds$. Then

$$\mathbf{No}^{D_0} = \mathbf{C}_0, \quad \mathbf{No}(s)^{\nabla_s} = \mathbf{C}_0, \quad \mathbf{No}((s))^{\nabla_s} = \mathcal{E}_{0,s}(\mathbf{No}). \quad (55.3)$$

Hence the multiplication maps $\mathbf{No} \otimes_{\mathbf{C}_0} \mathcal{E}_{0,s}(\mathbf{No}) \rightarrow \mathbf{No}((s))$ and $\mathbf{No}(s) \otimes_{\mathbf{C}_0} \mathcal{E}_{0,s}(\mathbf{No}) \rightarrow \mathbf{No}((s))$ are injective in the finite-relation sense; the fraction fields of their images are $\widehat{L}_0$ and $\widehat{L}_0(s)$.

Proof. The first equality is the definition of $\mathbf{C}_0$, the second is corollary 54.3 and lemma 54.7, and the third is lemma 54.10. Apply lemma 54.6 to the differential subfields $\mathbf{No}$ and $\mathbf{No}(s)$ of $\mathbf{No}((s))$, both with constant field $\mathbf{C}_0$. The images generate the stated composita under finite field operations. $\square$

For a finite tuple $\mathbf{x}$ let $I_{C_0}(\mathbf{x})$ be its polynomial relations over C_0 . Since $\mathcal{E}_{0,s}$ fixes C_0 and is injective, $I_{C_0}(\mathcal{E}_{0,s}(\mathbf{x})) = I_{C_0}(\mathbf{x})$, so by [lemma 54.11](#)

$$I_{\mathbf{No}(s)}(\mathcal{E}_{0,s}(\mathbf{x})) = \mathbf{No}(s)[Y] I_{C_0}(\mathbf{x}), \quad (55.4)$$

and the same with $\mathbf{No}$ in place of $\mathbf{No}(s)$. This is the precise sense in which the full-image calculation reduces to $\mathbf{No}/C_0$; it does not assert that deciding polynomial relations over C_0 is algorithmically easy [17].

Corollary 55.3 (Individual trajectories and intersections [17]). *For $x \in \mathbf{No}$ the following are equivalent: (i) $\mathcal{E}_{0,s}(x)$ is algebraic over $\mathbf{No}(s)$; (ii) $\mathcal{E}_{0,s}(x)$ is rational over $\mathbf{No}(s)$; (iii) $x \in C_0$. Moreover $\mathbf{No}(s) \cap \mathcal{E}_{0,s}(\mathbf{No}) = \mathbf{No} \cap \mathcal{E}_{0,s}(\mathbf{No}) = C_0$.*

Proof. By [lemma 54.11](#) and [theorem 55.2](#), $\mathcal{E}_{0,s}(x)$ is algebraic over $\mathbf{No}(s)$ exactly when x is algebraic over C_0 , that is $x \in C_0$ ([lemma 54.4](#)); such an x is fixed by $\mathcal{E}_{0,s}$ and lies in $\mathbf{No}$. An element of the image lying in $\mathbf{No}(s)$ is in particular algebraic over $\mathbf{No}(s)$. $\square$

This excludes nonconstant algebraic Taylor trajectories for highly infinite elements, not only for monomials. Conversely C_0 contains many infinite and infinitesimal elements: “constant trajectory” must not be confused with “ordinary real input” [17].

Theorem 55.4 (Additive parameter exclusion [17]). *s is transcendental over $\widehat{L}_0 = \mathbf{No}\mathcal{E}_{0,s}(\mathbf{No})$.*

Proof. A polynomial relation for s over $\widehat{L}_0$ has finitely many coefficients in the compositum; clearing a common denominator, and using that products of Taylor images are Taylor images, writes it as $\sum_{j=1}^r p_j(s) \mathcal{E}_{0,s}(y_j) = 0$ with $p_j \in \mathbf{No}[s]$, where each coefficient of the original polynomial becomes a subsum. Choose a C_0 -basis $b_1, \dots, b_m$ of the C_0 -span of the y_j ; $\mathcal{E}_{0,s}$ is C_0 -linear, so the relation becomes $\sum_k q_k(s) \mathcal{E}_{0,s}(b_k) = 0$ with $q_k \in \mathbf{No}[s]$. By [theorem 55.2](#) the $\mathcal{E}_{0,s}(b_k)$ are independent over $\mathbf{No}(s)$, so every $q_k = 0$; comparing coefficients of the powers of s shows that every coefficient of the original polynomial vanished. Equivalently, the injective base change identifies $\widehat{L}_0(s)$ with a rational function field over $\widehat{L}_0$. $\square$

For example $e^s = \mathcal{E}_{0,s}(\omega)/\omega \in \widehat{L}_0$, but its formal logarithm s is transcendental over $\widehat{L}_0$. This is compatible with $\log(e^s) = s$ in $\mathbf{No}[[s]]$: field generation is not closure under logarithmic power series, and a large orbit field need not capture every natural formal operation on its elements [17].

55.1 The monomial subfield and the lower bound of Part II

17 recalls the proof of the monomial calculation so that the improvement is visible. It is [lemma 17.4](#) and [theorem 17.5](#) with $\chi = q_0$ and $W = q_0(\mathbf{No}) = \mathbb{R}$: distinct formal exponentials $e^{r_j s}$ are linearly independent over $\mathbf{No}$ by a Vandermonde determinant $\prod_{i < j} (r_j - r_i) \neq 0$, $\mathbb{Q}$ -independent weights give algebraically independent exponentials, and every other one is algebraic over those of a $\mathbb{Q}$ -basis of its weight space; so $L_{q_0} = \mathbf{No}(e^{rs} : r \in \mathbb{R})$ has transcendence degree $\dim_{\mathbb{Q}} \mathbb{R} = 2^{\aleph_0}$ ([corollary 17.9](#)). These are printed once there. That computation is exact for monomials but involves only a set of real weights. [Section 56](#) supplies a class-indexed family of independent inputs whose supports combine a changing integer q_0 -weight with new directions in $\ker q_0$; the additional transcendence is not another choice of basis of $\mathbb{R}$ [17].

56 An explicit independent family of omnific integers

Lemma 56.1 (A rational logarithmic-derivative obstruction [17]). *Let k have characteristic zero and y be an indeterminate. No nonzero $r \in k(y)$ satisfies $r'/r = a$ for a nonzero $a \in k$, where $' = d/dy$.*

Proof. With $r = P/Q$ the identity reads $P'Q - PQ' = aPQ$. The left side has degree less than $\deg P + \deg Q$ unless it vanishes, while the right side is nonzero of exactly that degree. $\square$

Lemma 56.2 (Independent exponential solutions [17]). *Let a characteristic-zero differential field contain $k(z)$ with $\partial z = 1$ and $\partial k = 0$, z transcendental over k . If $a_1, \dots, a_r \in k$ are $\mathbb{Q}$ -linearly independent and nonzero $y_1, \dots, y_r$ satisfy $\partial y_j = a_j y_j$, then $y_1, \dots, y_r$ are algebraically independent over $k(z)$.*

Proof. A relation, after combining equal monomials, is $\sum_{\nu \in S} c_\nu(z) \mathbf{y}^\nu = 0$ with $S \subseteq \mathbb{N}^r$ finite and all $c_\nu \neq 0$. Take one with $|S|$ minimal and normalize the coefficient at some ν_0 to 1; a one-term relation is impossible since the y_j are nonzero. Put $\lambda_\nu = \sum_j \nu_j a_j$. Differentiating and subtracting λ_{ν_0} times the relation removes the ν_0 term, so by minimality $c'_\nu + (\lambda_\nu - \lambda_{\nu_0})c_\nu = 0$ for every other ν . The weights are distinct by $\mathbb{Q}$ -independence, so c'_ν/c_ν is a nonzero constant, contrary to lemma 56.1. $\square$

The proof is elementary differential algebra: no analytic exponentiation, no transcendence theorem for complex numbers and no conjecture on values of an exponential function is used. It works when k is the class field $\mathbf{C}_0$, since each relation uses finitely many of its elements [17].

56.1 Hidden constant parameters

Put $z = \omega^{-1}$ and $\tilde{D}_0 = -\omega D_0 = -z^{-1} D_0$. A nonzero multiple of a derivation is a derivation with the same constant field; $D_0 z = -z$, so $\tilde{D}_0 z = 1$, and z is transcendental over $\mathbf{C}_0$ (corollary 54.5), so $(\mathbf{C}_0(z), \tilde{D}_0)$ is an ordinary rational differential field. For each ordinal α put

$$\mathbf{a}_\alpha = \omega^{\alpha+1}, \quad \zeta_\alpha = \omega^{-\mathbf{a}_\alpha}, \quad \mathbf{b}_\alpha = \omega^{\alpha+2}. \quad (56.1)$$

The exponents $-\mathbf{a}_\alpha$ and $\mathbf{b}_\alpha$ are positive or negative infinite ordinal monomials with constant term 0, so $\zeta_\alpha, \omega^{\mathbf{b}_\alpha} \in \mathbf{C}_0$. The ζ_α are $\mathbb{Q}$ -linearly independent: they are distinct monomials, and a finite nonzero linear combination of distinct monomials is nonzero by uniqueness of normal forms. The infinitesimal $\zeta_\alpha z = \omega^{-(\mathbf{a}_\alpha+1)}$ has the Hahn exponential

$$\exp(\zeta_\alpha z) = \sum_{n \geq 0} \frac{(\zeta_\alpha z)^n}{n!} = \sum_{n \geq 0} \frac{\omega^{-n(\mathbf{a}_\alpha+1)}}{n!},$$

whose support, listed in decreasing order, is a strictly decreasing $\mathbb{N}$ -indexed sequence, hence an admissible countable support. It is this one sum, not an appeal to a global exponential on $\mathbf{No}$ or $\mathbf{No}(i)$ [17].

Lemma 56.3 ([17]). $D_0 \exp(\zeta_\alpha z) = -\zeta_\alpha z \exp(\zeta_\alpha z)$ and $\tilde{D}_0 \exp(\zeta_\alpha z) = \zeta_\alpha \exp(\zeta_\alpha z)$.

Proof. The exponent $-n(\mathbf{a}_\alpha+1)$ has constant term $-n$, so $D_0 \exp(\zeta_\alpha z) = \sum_{n \geq 1} \frac{-n}{n!} \omega^{-n(\mathbf{a}_\alpha+1)} = -\zeta_\alpha z \sum_{m \geq 0} \frac{\omega^{-m(\mathbf{a}_\alpha+1)}}{m!}$; reindexing a set-sized reverse well-ordered support is legitimate. Multiply by $-z^{-1}$. $\square$

Lemma 56.4 (Support positivity [17]). *For every ordinal α and every $n \in \mathbb{N}$, $\mathbf{b}_\alpha - n(\mathbf{a}_\alpha+1) > 0$, and these exponents decrease strictly in n . So $\Xi_\alpha = \omega^{\mathbf{b}_\alpha} \exp(\zeta_\alpha z)$ is exactly the normal form (53.2), and $\Xi_\alpha \in \Pi \subseteq \mathbf{Oz}$.*

Proof. For $n = 0$ this is clear. For $n > 0$ the normal forms of $\mathbf{b}_\alpha = \omega^{\alpha+2}$ and $n\mathbf{a}_\alpha + n = n\omega^{\alpha+1} + n$ have different leading exponents, and the former is larger, so the difference is positive; subtracting $\mathbf{a}_\alpha + 1 > 0$ at each step makes the sequence decrease. Translating the support of $\exp(\zeta_\alpha z)$ by $\mathbf{b}_\alpha$ preserves reverse well-ordering, all translated exponents are positive, and there is no term at exponent 0. $\square$

A sequence of positive exponents need not approach zero: here the higher scale $\mathbf{b}_\alpha$ dominates every finite multiple of $\mathbf{a}_\alpha + 1$. The construction needs this non-Archimedean gap; an Archimedean exponent group would not permit the same omnific shift for every n [17].

Theorem 56.5 (Independence before and after the flow [17]). *Every finite subfamily of $(\Xi_\alpha)_{\alpha \in \mathbf{On}}$ is algebraically independent over $\mathbf{C}_0(z) = \mathbf{C}_0(\omega)$. Its Taylor images are algebraically independent over $\mathbf{No}(s)$ and over $\mathbf{No}$.*

Proof. $\omega^{\mathbf{b}_\alpha} \in \mathbf{C}_0$, so lemma 56.3 gives $\tilde{D}_0 \Xi_\alpha = \zeta_\alpha \Xi_\alpha$; each Ξ_α is nonzero and the ζ_α of a finite subfamily are $\mathbb{Q}$ -independent constants. Lemma 56.2 with $k = \mathbf{C}_0$ gives independence over $\mathbf{C}_0(z)$, hence over $\mathbf{C}_0$, and (55.4) transfers it to the Taylor images over $\mathbf{No}(s)$; the base $\mathbf{No}$ is weaker. $\square$

Corollary 56.6 (No set bounds the orbit transcendence [17]). *For every set cardinal κ the first κ witnesses give an algebraically independent family of cardinality κ in $\hat{L}_0$ over $\mathbf{No}$. So $\hat{L}_0/\mathbf{No}$ has no set-sized transcendence basis, and $\hat{L}_0$ is algebraic over no set-generated extension of $\mathbf{No}$ inside $\mathbf{No}((s))$; the same holds over $\mathbf{No}(s)$ after adjoining s .*

Proof. Restrict theorem 56.5 to the ordinals below the initial ordinal of κ . If $\hat{L}_0$ were algebraic over a field generated by a set S , every independent family over $\mathbf{No}$ in $\hat{L}_0$ would have cardinality at most $\max(|S|, \aleph_0)$; a larger κ contradicts the first assertion. Over $\mathbf{No}(s)$ the argument is identical, since the independence theorem is proved over that base. $\square$

This replaces writing a nonexistent “proper-class cardinal” as a transcendence degree: no common numerical transcendence degree of proper classes is claimed, and the no-set-basis assertion is a consequence of the finite algebraic independence of an explicit family [17].

56.2 Coefficients of the witness trajectories

For reproduction and formalization the coefficients can be written without nested exponential notation:

$$[s^k] \mathcal{E}_{0,s}(\Xi_\alpha) = \frac{1}{k!} \sum_{n \geq 0} \frac{(-n)^k}{n!} \omega^{\mathbf{b}_\alpha - n(\mathbf{a}_\alpha + 1)}, \quad (56.2)$$

$$[v^k] \Upsilon_v(\Xi_\alpha) = \sum_{n \geq 0} \frac{1}{n!} \binom{-n}{k} \omega^{\mathbf{b}_\alpha - n(\mathbf{a}_\alpha + 1)}, \quad (56.3)$$

with $0^0 = 1$ in (56.2). Every inner sum is a Hahn sum supported in the original countable support, so every coefficient is omnific. It would be wrong to justify these formulas by the s -adic convergence of $\sum_n \omega^{\mathbf{b}_\alpha - n(\mathbf{a}_\alpha + 1)} e^{-ns} / n!$: its summands have nonzero constant coefficients. The valid reading is the coefficientwise strong summation in (56.2), while the outer series is formal in s [17].

57 Multiplicative time and the exact algebraic core

57.1 Changing the parameter, not the derivation

Let v be a new formal indeterminate and $u = 1 + v$. The substitution

$$s = \log(1 + v) = v - \frac{v^2}{2} + \frac{v^3}{3} - \cdots \quad (57.1)$$

is invertible with inverse $v = e^s - 1$; both series have zero constant term and nonzero linear term, so composition is defined in formal power series over any characteristic-zero field. Put $\Upsilon_v(x) = \mathcal{E}_{0, \log(1+v)}(x)$. The identity $e^{q \log(1+v)} = (1 + v)^q$ is, coefficient by coefficient, an identity of polynomials in q ; substituting D_0 gives

$$\Upsilon_v(x) = \sum_{n \geq 0} \binom{D_0}{n} (x) v^n, \quad (57.2)$$

which is also characterized by the recurrence of [lemma 54.10](#). Every coefficient is a finite polynomial expression in D_0 [17].

Proposition 57.1 ([17]). Υ_v is a unital field embedding of $\mathbf{No}$ into $\mathbf{No}((v))$ with

$$\Upsilon_v(\omega^g) = \omega^g u^{q_0(g)}, \quad u^r := \sum_{n \geq 0} \binom{r}{n} v^n. \quad (57.3)$$

It fixes $\mathbf{C}_0$, sends ω to ωu , and maps $\mathbf{Oz}$ into $\mathbf{Oz}[[v]]$ with coefficientwise constant term $\text{ct}(a)$. Its group law is multiplicative: $\Upsilon_v(\Upsilon_{v'}(x)) = \Upsilon_{v+v'+vv'}(x)$ for a further indeterminate v' .

Proof. Formal substitution in [proposition 55.1](#) gives the field statement and the monomial formula. For $n \geq 1$, $\binom{D_0}{n}$ is a rational polynomial in D_0 with zero constant term, and each positive iterate of D_0 maps $\mathbf{Oz}$ into Π , which is closed under rational scalars. Finally $\log((1 + v)(1 + v')) = \log(1 + v) + \log(1 + v')$ formally; combine with (55.2), or compare binomial coefficients. $\square$

The field isomorphism given by (57.1) sends the additive orbit field to the multiplicative one, but it does *not* send $\mathbf{No}(s)$ to $\mathbf{No}(u)$: it sends it to $\mathbf{No}(\log u)$. So different answers to algebraicity over the two rational parameter fields are consistent [17].

57.2 The invariant rational field

Theorem 57.2 (Multiplicative rational constants [17]). *On $\mathbf{No}(u)$ let $\nabla^\times = D_0 - u d/du = D_0 - (1 + v) d/dv$, so that $\nabla^\times u = -u$. Then $\mathbf{No}(u)^{\nabla^\times} = \mathbf{C}_0(\omega u)$, and on $\mathbf{No}((v))$ the full constant field is $\Upsilon_v(\mathbf{No})$.*

Proof. Put $w = \omega u$. Since $D_0 \omega = \omega$, $\nabla^\times w = \omega u - \omega u = 0$; w is transcendental over $\mathbf{No}$ and $\mathbf{No}(u) = \mathbf{No}(w)$. In these coordinates $\nabla^\times$ acts as D_0 on coefficients and kills w , so its constant field is $\mathbf{C}_0(w)$ by [lemma 54.9](#). The Laurent statement is (54.3). $\square$

The extra constant ωu is the algebraic reason for the larger multiplicative core; the additive case has no counterpart $y + s$, because D_0 has no slice [17].

57.3 Exact descent over the enlarged core

Since ω is transcendental over $\mathbf{C}_0$, a nonzero denominator from $\mathbf{C}_0[W]$ vanishes neither at ω nor at ωu , so the restriction of Υ_v is the isomorphism of rational function fields

$$\phi : \mathbf{C}_0(\omega) \xrightarrow{\sim} \mathbf{C}_0(\omega u), \quad R(\omega) \mapsto R(\omega u) \quad (R \in \mathbf{C}_0(W)). \quad (57.4)$$

Theorem 57.3 (Full multiplicative-image descent [17]). *$\mathbf{No}(u)$ and $\Upsilon_v(\mathbf{No})$ are linearly disjoint over $\mathbf{C}_0(\omega u)$. For a finite tuple $\mathbf{x}$ from $\mathbf{No}$,*

$$I_{\mathbf{No}(u)}(\Upsilon_v(\mathbf{x})) = \mathbf{No}(u)[\mathbf{Y}] \phi(I_{\mathbf{C}_0(\omega)}(\mathbf{x})), \quad (57.5)$$

with ϕ acting on coefficients. In particular $\text{trdeg}_{\mathbf{No}(u)} \mathbf{No}(u, \Upsilon_v(\mathbf{x})) = \text{trdeg}_{\mathbf{C}_0(\omega)} \mathbf{C}_0(\omega, \mathbf{x})$, and if $\mathbf{C}_0(\omega, \mathbf{x})/\mathbf{C}_0(\omega)$ is finite then $[\mathbf{No}(u, \Upsilon_v(\mathbf{x})) : \mathbf{No}(u)] = [\mathbf{C}_0(\omega, \mathbf{x}) : \mathbf{C}_0(\omega)]$.

Proof. By [theorem 57.2](#), $\mathbf{No}(u)$ has constant field $\mathbf{C}_0(\omega u)$ and the ambient Laurent field has constant field $\Upsilon_v(\mathbf{No})$; apply [lemma 54.6](#). Since Υ_v is a field embedding restricting to ϕ , relations over $\mathbf{C}_0(\omega)$ correspond exactly to relations over $\mathbf{C}_0(\omega u)$ in the image; apply [lemma 54.11](#). $\square$

Corollary 57.4 (Rational and algebraic trajectories [17]). *For $x \in \mathbf{No}$: $\Upsilon_v(x) \in \mathbf{No}(u)$ if and only if $x \in \mathbf{C}_0(\omega)$; $\Upsilon_v(x)$ is algebraic over $\mathbf{No}(u)$ if and only if x lies in the relative algebraic closure $\mathbf{C}_0(\omega)_{\mathbf{No}}^{\text{alg}}$ of $\mathbf{C}_0(\omega)$ in $\mathbf{No}$, and then $[\mathbf{No}(u, \Upsilon_v(x)) : \mathbf{No}(u)] = [\mathbf{C}_0(\omega, x) : \mathbf{C}_0(\omega)]$. The field $\mathbf{C}_0(\omega)_{\mathbf{No}}^{\text{alg}}$ is a real closure of $\mathbf{C}_0(\omega)$ for its induced ordering.*

Proof. The algebraic equivalence and the degrees follow from [theorem 57.3](#). Rationality forces degree one, and conversely $\mathbf{C}_0(\omega)$ maps into $\mathbf{C}_0(\omega u) \subseteq \mathbf{No}(u)$. Positive square roots and roots of odd-degree polynomials exist in the real closed field $\mathbf{No}$ and remain algebraic over $\mathbf{C}_0(\omega)$, so the relative closure is real closed with the induced order. $\square$

This is an exact description, not merely a necessary differential equation, and it classifies arbitrary infinite normal forms once their membership in the relative algebraic closure is known. It supplies no uniform decision procedure for arbitrary surreal presentations [17].

57.4 Parameter recovery and independence together

Corollary 57.5 ([17]). *$\widehat{L}_0^\times$ contains u and independent families of every set cardinality over $\mathbf{No}(u)$; it is algebraic over no set-generated extension of $\mathbf{No}(u)$ inside $\mathbf{No}((v))$.*

Proof. $u = \Upsilon_v(\omega)/\omega$. The Ξ_α are independent over $\mathbf{C}_0(\omega)$ ([theorem 56.5](#)); transfer by (57.5) and repeat the cardinal argument of [corollary 56.6](#). $\square$

Formal parameter	Rational trajectory core in $\mathbf{No}$	The parameter and the full orbit field
s	$\mathbf{C}_0$	s transcendental over $\widehat{L}_0$
$u = 1 + v$	$\mathbf{C}_0(\omega)$	$u \in \widehat{L}_0^\times$

In either row the full orbit field contains the same scale of independent families after the formal change of variable; size and algebraizability are separate invariants of this calculation [17].

58 Worked consequences, the surcomplex case and the abstract theorem

Proposition 58.1 (The monomial test [17]). *Let $g \in \mathbf{No}$ and $r = q_0(g)$. Then $\Upsilon_v(\omega^g)$ is rational over $\mathbf{No}(u)$ if and only if $r \in \mathbb{Z}$, and algebraic over $\mathbf{No}(u)$ if and only if $r \in \mathbb{Q}$; if $r = p/q$ in lowest terms with $q \geq 1$, its degree is q . In additive time, $\mathcal{E}_{0,s}(\omega^g)$ is algebraic over $\mathbf{No}(s)$ exactly when $r = 0$.*

Proof. $q_0(g - r) = 0$, so $\omega^g = c\omega^r$ with $c = \omega^{g-r} \in \mathbf{C}_0^\times$, and it suffices to classify ω^r over $\mathbf{C}_0(\omega)$. If $r = p/q$, put $\xi = \omega^{1/q}$, transcendental over $\mathbf{C}_0$ since $\omega = \xi^q$ is. $Y^q - W$ is Eisenstein at W in $\mathbf{C}_0[W][Y]$, so $[\mathbf{C}_0(\xi) : \mathbf{C}_0(\omega)] = q$; with $ap + bq = 1$, $\xi = (\xi^p)^a(\xi^q)^b$, so $\mathbf{C}_0(\omega, \omega^{p/q}) = \mathbf{C}_0(\xi)$ has degree q (for $p = 0$, $q = 1$). If r is irrational and ω^r satisfies a relation over $\mathbf{C}_0(\omega)$, clearing denominators gives a finite sum of terms $c_{ij}\omega^{i+rj}$ with $c_{ij} \in \mathbf{C}_0$. Each nonzero term is supported in the coset $\ker q_0 + i + rj$, and different pairs (i, j) give different cosets, since $i + rj = i' + rj'$ would make r rational unless both indices agree. The supports are disjoint and nothing cancels, a contradiction. Now apply [corollary 57.4](#); the additive clause is [corollary 55.3](#) with $D_0(\omega^g) = r\omega^g$. $\square$

The cosets of $\ker q_0$ are used for additive support separation only; no quotient order is imposed. A coefficient $c_{ij} \in \mathbf{C}_0$ may have infinite support, but it lies in $\ker q_0$, so the finite-coset argument holds [17].

Example 58.2 (An algebraic trajectory after reparametrization [17]). For $x = \omega^{1/2}$, $\mathcal{E}_{0,s}(x) = \omega^{1/2}e^{s/2}$ is transcendental over $\mathbf{No}(s)$, whereas $\Upsilon_v(x) = \omega^{1/2}(1+v)^{1/2}$ has degree two over $\mathbf{No}(u)$ and satisfies $Y^2 - \omega u = 0$. The input and every coefficient of both series are omnific; the contrast comes only from the different rational parameter fields.

Example 58.3 (Infinite constants and nonconstant infinite series [17]). The infinite monomial ω^ω lies in $\mathbf{C}_0$ and has a fixed trajectory under both flows. In contrast $\Xi_0 = \sum_{n \geq 0} \omega^{\omega^2 - n(\omega+1)}/n!$ is a purely infinite omnific integer transcendental over $\mathbf{C}_0(\omega)$, and its trajectory is transcendental in both parametrizations. The classification does not reduce to deciding whether an input is finite, infinite, a monomial or an infinite sum.

Finite systems and degrees [17]. If $x_1, \dots, x_m$ generate a finite extension of $\mathbf{C}_0(\omega)$, a basis of it maps to a basis after adjoining $\mathbf{No}(u)$ ([theorem 57.3](#)); every polynomial system describing the extension is transformed coefficientwise by $\omega \mapsto \omega u$, and its degree does not collapse under the much larger coefficient field $\mathbf{No}$. For positive integers $q_1, \dots, q_m$ with $q = \text{lcm}(q_1, \dots, q_m)$,

$$[\mathbf{No}(u, u^{1/q_1}, \dots, u^{1/q_m}) : \mathbf{No}(u)] = q : \quad (58.1)$$

the subgroup of $\mathbb{Q}$ generated by the $1/q_j$ is $\frac{1}{q}\mathbb{Z}$, so the monomials generate $\mathbf{C}_0(\omega^{1/q})$ over $\mathbf{C}_0(\omega)$, of degree q by the proof above, and exact descent applies; the factors ω^{1/q_j} of the transformed trajectories lie in $\mathbf{No}^\times$ and do not change the field. This does not reclassify all algebraic extensions of $\mathbf{No}(u)$; it concerns extensions arising from this flow, with an explicit source field.

Theorem 58.4 (The complexified package [17]). *Extend D_0 to $\mathbf{No}(i)$ by $D_0 i = 0$. All additive results hold with $\mathbf{No}(i)$ and $\mathbf{C}_0(i)$ in place of $\mathbf{No}$ and $\mathbf{C}_0$; all multiplicative results hold with the core $\mathbf{C}_0(i)(\omega)$, whose relative algebraic closure in $\mathbf{No}(i)$ is an algebraic closure of it. Both flows preserve $\mathbf{Oz}[i]$, and the real witnesses Ξ_α give independent image families of every set cardinality over $\mathbf{No}(i)(s)$ and over $\mathbf{No}(i)(u)$.*

Proof. The complex constant term still annihilates every derivative, so $D_0 y = 1$ has no solution in $\mathbf{No}(i)$, and the differential-field lemmas apply verbatim; the constant field is $\mathbf{C}_0(i)$ by

corollary 54.5. $\mathbf{No}(i)$ is algebraically closed, so the relative algebraic closure of $\mathbf{C}_0(i)(\omega)$ in it is algebraically closed. The witnesses satisfy $\tilde{D}_0\Xi_\alpha = \zeta_\alpha\Xi_\alpha$ with $\mathbb{Q}$ -independent constants ζ_α , so **lemma 56.2** applies over $\mathbf{C}_0(i)(z)$ directly. Preservation of $\mathbf{Oz}[i]$ follows componentwise from the real statement and $D_0i = 0$. $\square$

Complex conjugation commutes with D_0 , $\mathcal{E}_{0,s}$ and Υ_v , so the extension is compatible with the standard real form; it asserts nothing about other real forms or arbitrary surcomplex field automorphisms, and introduces no global surcomplex exponential [17].

Proposition 58.5 (Omnific generators [17]). $\hat{L}_0 = \mathbf{No}(\mathcal{E}_{0,s}(a) : a \in \mathbf{Oz})$ and $\hat{L}_0^\times = \mathbf{No}(\Upsilon_v(a) : a \in \mathbf{Oz})$; the analogues hold for $\mathbf{Oz}[i]$ and $\mathbf{No}(i)$.

Proof. Write $x = a/b$ with $a, b \in \mathbf{Oz}$, $b \neq 0$ (`odg:thm:fractions`); the image of x is the quotient of the images. The other inclusion is immediate; for $\mathbf{Oz}[i]$ use real and imaginary parts. $\square$

So the large field-theoretic phenomenon is not produced by allowing arbitrary input denominators: it is carried by $\mathbf{Oz}$, and the witnesses lie in Π [17].

Theorem 58.6 (The abstract full Taylor-orbit theorem [17]; 17's Theorem B.1). *Let (K, ∂) be a characteristic-zero differential field. Then K and $\mathcal{E}_s^\partial(K)$ are linearly disjoint over K^∂ . If K has no additive slice ($\partial y \neq 1$ for all y), the same holds for $K(s)$ and $\mathcal{E}_s^\partial(K)$, and s is transcendental over $K\mathcal{E}_s^\partial(K)$. If moreover $b \in K^\times$ satisfies $\partial b = b$, then b is transcendental over K^∂ ; $K(u)$ and $\Upsilon_v^\partial(K)$ are linearly disjoint over $K^\partial(bu) = \Upsilon_v^\partial(K^\partial(b))$, and the rational and algebraic trajectories are classified by $K^\partial(b)$ and its relative algebraic closure in K , with finite degrees preserved.*

Proof. The horizontal identities are **lemma 54.10**; apply **lemma 54.6** to K , and **lemma 54.7** when adjoining s ; the proof of **theorem 55.4** uses only these facts. If b were algebraic over K^∂ , then $\partial b = 0$ by **lemma 54.4**. In the multiplicative case $w = bu$ is a fixed transcendental variable for $\partial - u d/du$, so the constant field is $K^\partial(w)$ by **lemma 54.9**; the rest is **lemma 54.11** after identifying the base fields by $b \mapsto bu$. $\square$

The theorem marks the novelty boundary: its proof combines standard differential-field arguments, and the surreal result needs the separate construction of **section 56**, which shows that K/K^∂ has independent families beyond every set bound, chosen in $\mathbf{Oz}$ with explicit countable normal forms [17].

17 observes, after its first question, that for every nonzero additive exponent character χ the no-slice and full-image descent lemmas continue to hold, because $\chi(0) = 0$. Stated for the Euler derivations of **part II**:

Corollary 58.7 (Every Euler flow [17]; [merge]). *Let $R = \mathcal{R}_o(k, \Gamma)$ be set sized, or $R = \mathbf{Oz}, \mathbf{Oz}[i]$, with $F = \text{Frac } R$, and let $\chi : \Gamma \rightarrow k$ be additive. Then F and $F(s)$ are each linearly disjoint from $\mathcal{E}_{\chi,s}(F)$ over $F^{D_\chi} = \{x \in F : \text{supp } x \subseteq \ker \chi\}$, s is transcendental over $F\mathcal{E}_{\chi,s}(F)$, and for every finite tuple $\mathbf{x}$ from F , $\text{trdeg}_{F(s)} F(s, \mathcal{E}_{\chi,s}(\mathbf{x})) = \text{trdeg}_{F^{D_\chi}} F^{D_\chi}(\mathbf{x})$.*

Proof. D_χ is a derivation of R (**proposition 17.1**), hence of F , and F^{D_χ} is as stated by uniqueness of Hahn expansions. The coefficient of X^0 in $D_\chi y$ is $\chi(0)y_0 = 0$, so $D_\chi y \neq 1$: there is no slice. Apply **theorem 58.6** and **lemma 54.11**. $\square$

So for every χ the full formal image is determined by the extension F/F^{D_χ} , of which **theorem 17.5** measures only the monomial part; its size for characters other than q_0 is **questions 59.1** and **59.2**.

59 Scope, relations and questions of source 17

59.1 Dependency chain and boundary checks

17's dependency chain is short: (1) normal-form arithmetic constructs D_0 and C_0 and gives $\text{ct}(D_0x) = 0$, hence no slice; (2) the rational constants lemma and the linear-disjointness lemma give exact additive descent; (3) rescaling to $\tilde{D}_0$, the constants ζ_α , the witnesses Ξ_α and the logarithmic-derivative obstruction give independent omnific inputs; (4) descent transfers them to image families of every set cardinality; (5) the fixed coordinate ωu and the constants lemma for a fixed variable give multiplicative descent and exact degrees. The two roles of the constant coefficient ([remark 54.2](#)), the absence of class-sized sums, the positivity of every witness exponent and the change of rational base under reparametrization are the critical boundary checks [17]. Its limitations are listed in [section B](#).

59.2 Relations and status

- *The gap in [part II](#).* After [corollary 17.9](#) only a lower bound was asserted for the field generated by the formal images of all Hahn series, and 02's sixth non-claim ([section B](#)) says the same. For D_0 this is now answered exactly ([theorems 53.1](#) and [53.2](#)); for every Euler flow the descent holds ([corollary 58.7](#)), and the size for other characters is open. The monomial statements of [theorem 17.5](#) and [corollary 17.9](#) stay true. Status notes were added after [corollary 17.9](#) and in [section B](#).
- *Not addressed.* [Question 19.2](#) (answered by source 19 in [part IX](#)), [question 19.3](#) and [question 19.1](#). The formal flows are not internal automorphisms of $\mathbf{No}$; evaluation is [question 59.6](#).
- *Independent copies.* The independent-copies report ([docs/surreal/independent-surreal-copies](#)) concerns embeddings *inside* $\mathbf{No}$ with prescribed common Hahn cores; the embeddings here land in an external formal Laurent field, with different base fields and hypotheses, and neither construction substitutes for the other [17].
- *Checked true.* 17's pin, its description of [theorem 17.5](#) and [corollary 17.9](#) and of 02's non-claim, its statement that this report is pending formalization, the four SHA-256 hashes of its build record, and its suite, rerun for the merge ([section C](#)). No mathematical error was found in 17.

59.3 Questions of source 17

17 poses ten questions, proposed as research directions, not as published open problems. Eight are new; its question on internal evaluation is merged with source 19's ([question 59.6](#)), and its formalization question with those of sources 18 and 19 ([question 59.10](#)).

Question 59.1 (Other exponent characters [17]). For a nonzero additive $\chi : \mathbf{No} \rightarrow \mathbb{R}$ and $D_\chi(\sum r_g \omega^g) = \sum \chi(g) r_g \omega^g$, which χ make $\mathbf{No}/\ker D_\chi$ admit independent families of every set cardinality?

The no-slice and descent lemmas continue to hold ([corollary 58.7](#)); what needs new work is the supply of independent inputs. A sufficient criterion would produce many constant parameters and a common coordinate z with $\partial z = 1$ such that the required Hahn exponentials exist and can all be shifted into the omnific region; q_0 satisfies it by construction, and arbitrary characters need not preserve the same scale pattern [17].

Question 59.2 (Set-sized Hahn fields and exact cardinal bounds [17]). For a set-sized Hahn field $k((X^\Gamma))$ with a specified exponent character, determine $\text{trdeg}_{\ker D_\chi} k((X^\Gamma))$, and hence the transcendence degree of its full Taylor orbit field.

Descent reduces the formal problem to an ordinary field extension ([corollary 58.7](#)). Bounds should depend on the kernel of the character, the available support types and the cardinality of the coefficient field; a first test case is a divisible group of ordered rank two with a character reading its lower coordinate. The question is more informative than the dimension of the character's image, which controls monomials but not arbitrary series [17].

Question 59.3 (The algebraic core by supports [17]). Give a normal-form criterion for membership in the relative algebraic closure of $C_0(\omega)$ in $\mathbf{No}$.

[Corollary 57.4](#) classifies multiplicative trajectories through this field but does not describe its supports. Puiseux-type expansions are a natural start, with the topology and the large coefficient field fixed carefully. The induced order of $C_0(\omega)$ does not make ω larger than every element of C_0 ($\omega^\omega \in C_0$ is much larger), so Archimedean intuition does not settle the support question [17].

Question 59.4 (Several commuting Euler directions [17]). For additive characters $\chi_1, \dots, \chi_m$, compute the full image of the joint Taylor map $\exp(\sum_j s_j D_{\chi_j})$ and its rational trajectory core.

The derivations commute ([proposition 17.1](#) and [corollary 17.7](#)); the new issue is the constant field of $K(s_1, \dots, s_m)$ for the mixed operators. Pairwise absence of a slice is not a proof that there are no joint rational first integrals; a multivariable analogue of the expansion at infinity, with independence hypotheses on the characters, would be a concrete advance [17]. Source 19's [question 75.4](#) asks about the jet groups of several parameters, a different invariant.

Question 59.5 (Formal clocks with algebraic trajectories [17]). For an invertible parameter change $s = f(v)$ with $f \in v\mathbb{R}[[v]]$, characterize the x for which $\mathcal{E}_{0,f(v)}(x)$ is algebraic over $\mathbf{No}(v)$.

The two clocks treated are $f(v) = v$ and $f(v) = \log(1 + v)$, with rational horizontal fields C_0 and $C_0(\omega(1 + v))$. In general the mixed derivation is $D_0 - (f'(v))^{-1}d/dv$, which need not preserve the rational parameter field unless $(f')^{-1}$ is rational; that case should be classified first, rather than applying the constants lemma to a field the derivation does not preserve [17].

Question 59.6 (Internal evaluation of formal flows [17, 19]). For a derivation δ of $\mathbf{Oz}$ (17: $\delta = D_0$) and a surreal infinitesimal ε , when is $\sum_n \delta^n(x)\varepsilon^n/n!$ strongly summable, and when does it define an automorphism on all inputs of a prescribed subring? Give a necessary and sufficient support criterion beyond the common-shift setting (19), and determine what algebraic independence survives after evaluation over a specified smaller coefficient field (17).

17: the witness formulas ([56.2](#)) are test cases, and a theorem must control a two-index Hahn family after evaluation. Every evaluated value lies in $\mathbf{No}$, so independence over $\mathbf{No}$ itself cannot survive; a meaningful statement needs a proper smaller base, and even then formal independence cannot imply independence at every specialization (an ordinary nonzero polynomial may vanish at a chosen value). 19: the question is now separate from integrability, since all coefficients exist before evaluation ([theorem 67.1](#)); a criterion must control both well-ordering of supports and finite coefficient multiplicities, not merely each term. The common-shift class of [theorem 4.8](#) is integrated inside $\mathbf{No}$; the merge records the two sources' questions as one.

Question 59.7 (A proper-class differential Galois object [17]). Can the full orbit-field calculation be organized as a class-sized analogue of a Taylor Galois hull with a suitable symmetry functor, and what part is captured by set-sized differential subfields?

The horizontal identities and exact scalar extension are natural starting data. A construction should specify its test algebras and descent to sets and distinguish the compositum studied here from differential closures under further derivations; existing Galois-hull theory [13] is the comparison point, not a framework to be renamed [17].

Question 59.8 (Sharper arithmetic restrictions on witnesses [17]). Can equally large independent families be realized with further constraints, such as a prescribed leading monomial, a fixed finite set of real coefficients, or membership in a specified initial subring of $\mathbf{Oz}$?

The coefficients $1/n!$ and the scale-dependent leading monomial are chosen for the first-order equation; a fixed leading monomial cannot dominate arbitrary new scales in the same way, which suggests a genuine obstruction or a different construction [17].

Question 59.9 (Uniform finite certificates for relation descent [17]). Give a proof-producing reduction that translates polynomial relations among the trajectories of a finite explicitly presented tuple into relations over $\mathbf{C}_0$ or $\mathbf{C}_0(\omega)$.

The equality of relation ideals is proved; an algorithm needs representations of coefficients, equality tests and a way to find a finite basis of evaluated monomials. The natural first domain is finite sums of monomials and finite algebraic extensions of $\mathbf{C}_0(\omega)$ with certified coefficients, not arbitrary normal forms [17].

Question 59.10 (Formalization of Parts VII–IX and X [17, 18, 19, 20]). Formalize in Lean the constants lemmas and the exact trajectory classification (17), the free cut-insertion lemma, the two-level fixed-support theorem and the fraction-field transfer (18), and the divisible-ideal integration theorem (19), first for set-sized structures, then for the repository’s surreal field.

17 proposes the order: a generic characteristic-zero differential field; the rational constants lemmas; the minimal-relation proof of linear disjointness; the Taylor and binomial recurrences; finite relation descent; the support and differential identities of Ξ_α ; and only then cardinal-indexed independence. The generic lemmas are set sized and should be kept apart from universe management; no class-sized basis is needed to formalize “for every set cardinal there is an independent family of that size” [17]. 18 proposes finite ordered actions, Hahn-support relabelling, ordered-group fibers over cut orbits, set-sized bounded recursions and only then a class-sensitive or universe-indexed formulation; its finite checks are no substitute [18]. 19 asks for the divisible-ideal integration theorem for a set-sized characteristic-zero domain first, instantiated in the omnific ring, then finite logarithms and the nonsplit second-jet extension before infinite operator series; the first milestone needs only divisibility, torsion-freeness, iterated Leibniz and unique division in the ideal, while the genus-one definition and the class-size nondefinability need separate geometric and foundational infrastructure [19]. 20 (its Question 12, [part X](#)) asks to formalize the downward-orbit lemma and the weighted Green operator in a Hahn-series library, then the two lifts and the arithmetic restrictions, and whether the class construction can be proved under weaker choice while keeping the explicit translation family without it; the support lemma is a first milestone with no surreal-specific recursion [20]. Nothing in these parts is formalized.

Part VIII

Left-orderable symmetry groups with prescribed fixed fields

This part records manuscript 18, the second addition of batch 33. Its question is not which automorphisms preserve **Oz** but which set-sized groups act faithfully by them, and how precisely their fixed fields can be prescribed. The answer is exact: every nontrivial left-orderable set group acts faithfully on **No** by strongly additive, real-fixing, **Oz**-preserving automorphisms all of whose nonidentity elements have one prescribed fixed field, a two-level support field containing any given set of parameters; conversely every set group acting faithfully on **No**, or on **Oz**, is left-orderable ([theorem 60.1](#)). The surcomplex version classifies the conjugation-compatible groups ([theorem 60.4](#)). The automorphisms are double Hahn lifts: in this report's notation they are the monomial maps M_{1,ι_φ} of [section 27](#), so they lie in the monomial factor of [theorem 24.1](#), not in $\mathcal{U}_{\mathbf{Oz}}$. The new ingredient is an equivariant filling of all set cuts of an ordered class by free ordered fibers. Its labels carry the sub-prefix `opa:es:`.

60 Source 18, its relation to Parts I and III, and its notation

60.1 The manuscript and its pin

Manuscript 18 is *Relative Universality and Exact Fixed Fields of Strong Omnific Symmetries: group realization, two-level Hahn supports, and conjugation on the surcomplex field*, dated 23 September 2026, 24 pages. It is manuscript 05 of batch 33, delivered as the archive `Surreal_Exact_Symmetries_Research` in commit 73043eb and placed as an addition in commit aa9c891; it is numbered 18 here. Its checks, their recorded run, its build script, its `Makefile`, its build record and its proof and source audits are shipped as `code/18-exact-symmetries-verify.py`, `data/18-exact-symmetries-verification.json`, `code/18-exact-symmetries-build.sh`, `code/18-exact-symmetries-Makefile`, `data/18-exact-symmetries-BUILD_REPORT.json`, `18-exact-symmetries-PROOF_AUDIT.md` and `18-exact-symmetries-SOURCE_AUDIT.md`; its `article.tex`, PDF, README and checksum list are not shipped (the checksum list was verified at placement).

It pins `efc5446`, which its audit correctly identifies as a commit with tree `aba6982`; that commit contains [parts I to III](#) of this report but not [parts IV to VI](#). It read the root and catalogue guides, and the guides of this report, of the quotient report and of the holonomic report, not this article; some connector responses were truncated. Its account is accurate for what it read: it attributes to this report the fixed fields and the relative Hahn hull, large parameter stabilizers, nonabelian actions with an explicit wreath product, support-cut twists and parameterwise nondefinability, and claims none of them; the guide it read describes the eight-source report. Its search for “left-orderable” on the default branch found nothing, and at the merge the phrase occurs in no report of the collection outside this part and its staged files.

60.2 What it re-derives, and what is printed once

The key identification is [merge]: 18's first lift T_φ is the inner lift ι_φ of [\(27.1\)](#), for an order automorphism φ of **No**, and its second lift F_φ is the monomial map M_{1,ι_φ} of [\(5.1\)](#), the construction of [lemma 27.1](#) and the paragraph after it. So all of 18's real actions lie in the factor $\{M_{\chi,\tau}\}$ of [theorem 24.1](#) and [corollary 24.4](#), with $\chi = 1$ and $u = \text{id}$; a nonidentity $M_{1,\tau}$

moves the valuation of some monomial, so the realized groups meet $\mathcal{U}_{\mathbf{Oz}}$ trivially. 18's numbers are those of its own PDF.

18	Printed as	Comparison
Lem. 2.1	<code>odg:thm:fractions</code>	Same ($\text{Frac } \mathbf{Oz} = \mathbf{No}$, $\text{Frac } \mathbf{Oz}[i] = \mathbf{No}(i)$); cited.
Lem. 2.2	corollary 5.2 ; <code>odg:thm:floor</code>	Every automorphism preserving $\mathbf{Oz}$ commutes with the floor, with or without order; 18's direct argument is recorded after proposition 61.2 .
Def. 3.1, Prop. 3.2	definition 61.1 and proposition 61.2 ; lemma 27.1	The inner lift and M_{1,ℓ_μ} are in section 27 for order embeddings; 18's proposition adds functoriality, faithfulness on $\mathbf{Oz}$ and preservation of ct and the floor, and is printed.
Lem. 8.2	<code>saut:thm:axis</code>	The same decomposition $\text{Aut}(\mathbf{No}(i); \text{conj}) \simeq \text{Aut}(\mathbf{No}) \times C_2$; printed as lemma 65.2 with 18's ring clause.
Cor. 6.2, last sentence	<code>saut:thm:torsion</code>	Nonidentity order-preserving bijections have infinite order; the same argument.
Prop. 9.1	<code>saut:thm:phase</code> , remark 36.6 , <code>odg:def:lem:twist</code>	The twist is a phase twist; the fixed fields of its powers are new (proposition 65.5).
Cors. 1.2, 1.3	corollaries 60.2 and 60.3	New; they extend theorems 7.2 and 7.6 and contrast with theorem 7.4 (remark 63.5).
Questions	section 66.3	Eight new, two merged.
All else	This part	New.

18's own novelty statement is the combined theorem ([60.2](#)) for *every* left-orderable set group, its free-outside-constants specialization, the field–ring equivalence and the conjugation-compatible classification with nonreal parameters. The double lift, and the existence of nontrivial and nonabelian omnific-preserving automorphisms, are credited to [\[15\]](#) (its Construction 3.10), [\[17\]](#) and this report. This is accurate at the merge.

60.3 Conventions and notation of this part

This part uses the large-monomial (growth) convention of 18 and of [part II](#): $x = \sum_{a \in \text{supp } x} x_a \omega^a$ with $x_a \in \mathbb{R} \setminus \{0\}$ and $\text{supp } x$ a reverse well-ordered set; in the t -convention of [part I](#) $\omega^a = t^{-a}$. The valuation is $v(x) = -\max \text{supp } x$, which is v of [section 1.5](#). For a subclass $B \subseteq \mathbf{No}$ and a set $A \subseteq \mathbf{No}$ put

$$\text{NF}(B) = \{x \in \mathbf{No} : \text{supp } x \subseteq B\}, \quad \mathbf{K}_B = \text{NF}(\text{NF}(B)), \quad \Sigma_2(A) = \bigcup_{x \in A} \bigcup_{a \in \text{supp } x} \text{supp } a. \quad (60.1)$$

$\text{NF}(B)$ is an ordered real vector subspace of $(\mathbf{No}, +)$, so $\mathbf{K}_B$ is a Hahn subfield; for a set B both are sets (their members are coded by functions from subsets of B to $\mathbb{R}$), a set-size statement, not a birthday bound. For example $\mathbf{K}_\emptyset = \mathbb{R}$ and $\mathbf{K}_{\{0\}} = \mathbb{R}((\omega^\mathbb{R}))$, written with growth exponents. $\Sigma_2(A)$ is the set T of lemma 7.1 (inner supports of outer exponents) for real parameters. A map is *strongly additive* if it preserves every set-indexed Hahn-summable family and its sum (section 20.3); here both a map and its inverse are. A group is *left-orderable* if it has a linear order $<_G$ with $g <_G h \Rightarrow kg <_G kh$; right invariance is not required.

18's symbol	Here
$\mathcal{H}(B), \mathbf{K}_B = \mathcal{H}(\mathcal{H}(B));$ $\sigma_2(A)$	$\text{NF}(B), \mathbf{K}_B; \Sigma_2(A).$
T_φ, F_φ	$\iota_\varphi, M_{1, \iota_\varphi}.$
$\mathbf{No}_\mathbb{C}, \mathbf{Oz}_\mathbb{C}; c$	$\mathbf{No}(i), \mathbf{Oz}[i];$ complex conjugation $z \mapsto \bar{z}$, as a group element $\text{cj}.$
Cut D ; its stabilizer H ; components C_D	$\mathbf{c}; G_\mathbf{c}; I_\mathbf{c}.$
The factor H in $H \times C_2$	$G_0.$
Ordered classes X, X_α	$\mathcal{X}, \mathcal{X}_\alpha.$
The twist U , its character χ , the sets E_n	$\Phi_{\text{ph}}, \chi_{\text{ph}}, E_n.$
Homeomorphism h (remark 64.4); $h(t) = t/(2-t)$	$\psi; h_{\text{KB}}.$

The collisions are resolved here only, and the tempting false readings are wrong:

- 18's $\mathcal{H}(B)$ uses the same letter as this report's relative Hahn hull $\mathcal{H}(A) = k((t^{V_A}))$ of theorem 7.4, with a different meaning: $\text{NF}(B)$ keeps exactly the exponents in B and takes no rational span, while $\mathcal{H}(A)$ is the Hahn field on the rational span of the outer supports of a parameter set. Reading 18's " $\mathcal{H}(\mathcal{H}(S))$ " with Part I's hull is wrong.
- 18's T_φ is not the Taylor automorphism $\mathcal{T}_{d,\delta}$, a truncation $T_{<\gamma}$, a linear map T of section 25 or 19's formal variable; it is the additive, not multiplicative, inner lift ι_φ , and it need not fix 1: $\iota_\varphi(1) = \omega^{\varphi(0)}$. 18's F_φ is not Part III's isomorphism F .
- 18's U is not the stabilizer $\mathcal{U}$; its χ is a multiplicative character, as the χ of $M_{\chi,\tau}$, not an additive character χ of D_χ in part II. $\mathbf{K}_B$ is not the surcomplex field of the surcomplex report.
- The cut $\mathbf{c}$, its stabilizer $G_\mathbf{c}$ and the components $I_\mathbf{c}$ replace 18's D, H, C_D , which would read as a derivation, a convex subgroup H_g and a principal convex subgroup C_δ . S is a set of fixed *indices*, not a support.

Class conventions are those of section 1.4, in NBG with global choice: a class action $G \curvearrowright \mathbf{No}$ is one class relation coding $(g, x) \mapsto \sigma_g(x)$; no class of all proper-class automorphisms is formed; recursions through $\mathbf{On}$ have set values, limit stages are set unions, and a fixed global set-like well-order supplies uniform least choices, so no stage chooses among proper-class objects and no support is indexed by all ordinals. Normal forms, real closedness and the set-cut property of $\mathbf{No}$ are classical inputs [7, 11] [18].

60.4 Principal results of source 18

Theorem 60.1 (Exact relative group realization [18]; 18’s Theorem 1.1). *Let $S \subseteq \mathbf{No}$ be a set and G a nontrivial set-sized left-orderable group. There is a faithful class action $g \mapsto \sigma_g$ of G on $\mathbf{No}$ such that:*

- (i) *every σ_g is a strongly additive field automorphism fixing $\mathbb{R}$ pointwise and permuting the coefficient-one Conway monomials;*
- (ii) *every σ_g preserves $\mathbf{Oz}$, the constant coefficient ct and the omnific floor map;*
- (iii) *for every $g \neq 1$,*

$$\text{Fix}_{\mathbf{No}}(\sigma_g) = K_S, \quad \text{Fix}_{\mathbf{Oz}}(\sigma_g) = \mathbf{Oz} \cap K_S; \quad (60.2)$$

- (iv) *the restriction to $\mathbf{Oz}$ is faithful.*

In particular every nontrivial subgroup has common fixed field K_S . Conversely, a set-sized group acting faithfully by field automorphisms of $\mathbf{No}$, or by unital ring automorphisms of $\mathbf{Oz}$, is left-orderable.

The proof is in [section 63](#). $\Sigma_2(A)$ measures the index information that the double lift sees; the first-level support alone is not the right parameter [18].

Corollary 60.2 (Arbitrary set parameters [18]). *For every set $A \subseteq \mathbf{No}$ and every nontrivial set-sized left-orderable G , the action can be chosen with $\text{Fix}(\sigma_g) = K_{\Sigma_2(A)} \supseteq A$ for $g \neq 1$. This fixed field is set sized and real closed, and a single automorphism of infinite order realizes it.*

Corollary 60.3 (Free action outside the ordinary constants [18]). *Every nontrivial set-sized left-orderable group has a strongly additive $\mathbf{Oz}$ -preserving action with $\text{Fix}_{\mathbf{No}}(\sigma_g) = \mathbb{R}$ and $\text{Fix}_{\mathbf{Oz}}(\sigma_g) = \mathbb{Z}$ for $g \neq 1$; equivalently its actions on $\mathbf{No} \setminus \mathbb{R}$ and on $\mathbf{Oz} \setminus \mathbb{Z}$ are free. (“Free” describes the action, not the group.)*

This prescribes the fixed field of *each* nonidentity element, including all nonidentity words of a nonabelian group, which is more than a large parameter stabilizer or one nontrivial automorphism [18].

Theorem 60.4 (Conjugation-compatible group classification [18]; 18’s Theorem 1.4). *A set-sized group G has a faithful action on $\mathbf{No}(i)$ by field automorphisms commuting with conjugation if and only if G is left-orderable or $G \cong G_0 \times C_2$ with G_0 left-orderable; the same holds for faithful unital ring actions on $\mathbf{Oz}[i]$ commuting with conjugation. The sufficient constructions can be strongly additive, $\mathbf{Oz}[i]$ -preserving, and fix any prescribed set of real parameters. If a prescribed parameter set contains a nonreal element, faithful conjugation-compatible actions fixing it exist exactly for the left-orderable groups.*

This concerns actions commuting with a *named* conjugation; it is not a classification of arbitrary surcomplex automorphism groups [18]. The proof is in [section 65](#).

61 Double lifts, orbit supports and two-level hulls

$\text{Frac } \mathbf{Oz} = \mathbf{No}$ and $\text{Frac } \mathbf{Oz}[i] = \mathbf{No}(i)$ are `odg:thm:fractions`; 18 reproves them by choosing b above 0 and the negatives of all exponents, a step that uses the set-cut property of the *full* exponent order and cannot be transferred to a set-sized Hahn subfield [18]. Since a nonzero omnific integer with a positive exponent is infinite, $\mathbf{Oz} \cap \mathbb{R} = \mathbb{Z}$.

Definition 61.1 (Two lifts [18]). For an order automorphism φ of $(\mathbf{No}, <)$ put

$$\iota_\varphi(x) = \sum_{a \in \text{supp } x} x_a \omega^{\varphi(a)}, \quad M_{1, \iota_\varphi}(x) = \sum_{a \in \text{supp } x} x_a \omega^{\iota_\varphi(a)}. \quad (61.1)$$

The first is an additive lift, (27.1) with $\mu = \varphi$; the second is a field lift, the monomial map of (5.1) with $\chi = 1$ and $\tau = \iota_\varphi$: since ι_φ is additive, $\iota_\varphi(-a) = -\iota_\varphi(a)$, so $t^\gamma \mapsto t^{\iota_\varphi(\gamma)}$ is $\omega^a \mapsto \omega^{\iota_\varphi(a)}$.

An arbitrary order automorphism φ need not be additive or fix 0; nevertheless ι_φ is additive and fixes the zero series, though not the number 1 [18].

Proposition 61.2 (Properties of the lifts [18]; [11]). ι_φ is an order-preserving $\mathbb{R}$ -linear automorphism of $(\mathbf{No}, +)$, and M_{1, ι_φ} is an order-preserving field automorphism fixing $\mathbb{R}$ pointwise; both are strongly additive, and

$$\iota_{\varphi\psi} = \iota_\varphi \iota_\psi, \quad M_{1, \iota_{\varphi\psi}} = M_{1, \iota_\varphi} M_{1, \iota_\psi}. \quad (61.2)$$

$\varphi \mapsto M_{1, \iota_\varphi}$ is faithful, and M_{1, ι_φ} preserves $\mathbf{Oz}$, ct and the omnific floor.

Proof. An order isomorphism carries a set-sized reverse well-ordered support to another; coefficients are unchanged and no exponents merge, so addition and real scalars commute with ι_φ , whose inverse is $\iota_{\varphi^{-1}}$. The sign of a series is that of its leading coefficient, which is unchanged, so ι_φ preserves order. Relabelling the exponent union of a summable family by φ preserves reverse well-ordering and the finite incidence at each exponent, which is strong additivity. Now ι_φ is an ordered additive automorphism, and relabelling outer exponents by it gives the same assertions for M_{1, ι_φ} ; constants are fixed since $\iota_\varphi(0) = 0$. For multiplication, $\iota_\varphi(a + b) = \iota_\varphi(a) + \iota_\varphi(b)$ identifies the finite convolution at an exponent d bijectively with the convolution at $\iota_\varphi(d)$. The inverse is $M_{1, \iota_{\varphi^{-1}}}$. Equation (61.2) follows by applying the two relabellings successively, and faithfulness from

$$M_{1, \iota_\varphi}(\omega^{\omega^a}) = \omega^{\omega^{\varphi(a)}}, \quad (61.3)$$

which differs from the input when $\varphi(a) \neq a$. Finally ι_φ preserves the nonnegative cone and fixes 0, so the normal-form description of $\mathbf{Oz}$ (support in $\mathbf{No}^{\geq 0}$, integer coefficient at 0) is preserved in both directions, and the coefficient at 0 is unchanged, since no nonzero exponent is mapped to 0. The floor follows from corollary 5.2. $\square$

For order embeddings μ the first sentence is lemma 27.1, and M_{1, ι_μ} is the strong exact-monomial field embedding of the paragraph after it (11's). 18 proves the floor clause directly: the omnific floor of x is the unique $z \in \mathbf{Oz}$ with $z \leq x < z + 1$ (write $x = x_+ + r + \varepsilon$ with $x_+ \in \Pi$, $r \in \mathbb{R}$, ε infinitesimal and take $x_+ + \lfloor r \rfloor$, or $x_+ + r - 1$ when $r \in \mathbb{Z}$ and $\varepsilon < 0$; distinct omnific integers differ by at least 1), and an order-preserving $\mathbf{Oz}$ -preserving automorphism carries these inequalities to the image [18]. Corollary 5.2 gives it for every automorphism preserving $\mathbf{Oz}$, order being automatic.

Remark 61.3 (What is not preserved [18]). These automorphisms preserve the class of monomials, not the omega-map: $M_{1, \iota_\varphi}(\omega^a) = \omega^{\iota_\varphi(a)}$, not $\omega^{M_{1, \iota_\varphi}(a)}$. They are not asserted to preserve the Gonshor exponential, simplicity, birthdays, or the valuation pointwise; the correct equivariance is $v(M_{1, \iota_\varphi}(x)) = \iota_\varphi(v(x))$. This prevents importing the conclusions into more rigid expansions of $\mathbf{No}$.

Lemma 61.4 (An invariant well-based orbit is a point [18]). *If a group acts by increasing bijections on a linearly ordered class and one orbit is contained in a reverse well-ordered set, that orbit is a singleton.*

Proof. The orbit has a greatest element m ; every group element permutes the orbit increasingly, so it fixes m , and every member of the orbit is an image of m . $\square$

This rules out a tempting but invalid construction: a constant coefficient on a whole nontrivial orbit is not an invariant Hahn series, since such an orbit is not an admissible support [18].

Theorem 61.5 (Exact two-level fixed support [18]). *Let a set-sized group G act on $(\mathbf{No}, <)$ by $g \mapsto \varphi_g$, and let B be the class of common fixed points. Then*

$$\text{Fix}_{\mathbf{No}}(\iota_{\varphi_g} : g \in G) = \text{NF}(B), \quad \text{Fix}_{\mathbf{No}}(M_{1, \iota_{\varphi_g}} : g \in G) = K_B, \quad (61.4)$$

and for a single g the same holds with $B = \text{Fix}(\varphi_g)$.

Proof. If x is fixed by all additive lifts, uniqueness of normal forms makes $\text{supp } x$ invariant; each orbit of a point of $\text{supp } x$ lies in $\text{supp } x$, so it is a point by lemma 61.4, and $\text{supp } x \subseteq B$. Conversely a series supported in B is fixed. Apply the same argument to the outer action $g \mapsto \iota_{\varphi_g}$, whose common fixed class is $\text{NF}(B)$ by the first part. For a single element use its cyclic subgroup, whose orbits are still sets. $\square$

Proposition 61.6 (The two-level support hull [18]). *For sets $A, S, T \subseteq \mathbf{No}$: $A \subseteq K_S$ if and only if $\Sigma_2(A) \subseteq S$; $K_S \cap K_T = K_{S \cap T}$; and $\Sigma_2(K_S) = S$. So $S \mapsto K_S$ is monotone and injective, and $K_{\Sigma_2(A)}$ is the smallest field of this form containing A .*

Proof. Membership in K_S means that every inner exponent of every outer exponent lies in S , which gives the first claim and the intersection formula. For the third, one inclusion is automatic, and $\omega^{\omega^s} \in K_S$ has second-level support $\{s\}$. $\square$

“Of this form” is essential: the proposition does not identify algebraic closure, definable closure, or the smallest surreal subfield containing A [18].

Lemma 61.7 (Ordered fixed fields are real closed [18]). *The common fixed field of any group of field automorphisms of a real closed field is real closed.*

Proof. The automorphisms preserve order, since positive elements are the nonzero squares. The positive square root of a positive fixed element is unique, hence fixed. An odd-degree polynomial with fixed coefficients has a root, and its finite set of roots is permuted increasingly by every automorphism, so each root is fixed. $\square$

62 Free actions on the surreal order

A free action of G on a linear order is one in which no nonidentity element fixes a point; it need not be transitive. The action is first built on an order with the set-cut property of $\mathbf{No}$ and then transported [18].

Definition 62.1 ([18]). A nonempty ordered class $\mathcal{X}$ has the *set-cut property* if for all sets $L, R \subseteq \mathcal{X}$ with $L < R$ there is $x \in \mathcal{X}$ with $L < x < R$; either side may be empty.

Such an order is a proper class (take $L = \mathcal{X}$, $R = \emptyset$ if it were a set), and $\mathbf{No}$ has the property by its cut construction.

Lemma 62.2 (Class-order uniqueness [18]). *In NBG with global choice any two ordered classes with the set-cut property are order isomorphic, and a prescribed set-sized partial order isomorphism extends to an isomorphism.*

Proof. Enumerate both classes by **On** with the fixed global well-order and start from the prescribed partial isomorphism. At each stage perform a forth and a back step: for a new point x of the first order, its comparisons with the current set-sized domain give two sets of assigned images, and the set-cut property gives a point strictly between them, outside the current range; take the least eligible point in the global enumeration, and symmetrically for the back step. At limits take unions, still set-sized partial isomorphisms. The union over all ordinals is total and onto; every stage carries a set, so this is an ordinary set-valued class recursion. $\square$

The same algorithm, with class parameters coding the orders, makes these isomorphisms uniform in a set-indexed family; this is used for the complementary intervals of a parameter set [18].

Lemma 62.3 (Free insertion at a cut orbit [18]). *Let a left-orderable set group G , with a fixed left-invariant order $<_G$, act freely and increasingly on a set-sized linear order $\mathcal{X}$; a cut is a downward-closed $\mathfrak{c} \subseteq \mathcal{X}$ ($\emptyset$ and $\mathcal{X}$ allowed), and G acts on cuts. For each orbit $\mathcal{O}$ of cuts one can insert new points at every cut in $\mathcal{O}$ so that the action extends to a free increasing action on the enlarged order; all cut orbits can be treated at once.*

Proof. Fix a representative $\mathfrak{c} \in \mathcal{O}$ and its stabilizer $G_{\mathfrak{c}}$, which inherits a left order. Take a tagged copy of the set G with the left regular action, and place the point labelled g at the cut $g\mathfrak{c}$: old points of $g\mathfrak{c}$ below it, the other old points above it. For each $\mathfrak{c}' \in \mathcal{O}$ choose $r(\mathfrak{c}')$ with $r(\mathfrak{c}')\mathfrak{c} = \mathfrak{c}'$; the labels over $\mathfrak{c}'$ are the $r(\mathfrak{c}')h$ with $h \in G_{\mathfrak{c}}$, ordered by h , and different cuts are ordered by inclusion. For $k \in G$ write $kr(\mathfrak{c}') = r(k\mathfrak{c}')h_0$ with $h_0 \in G_{\mathfrak{c}}$; then k sends the fiber coordinate h to h_0h , which is increasing because the order of $G_{\mathfrak{c}}$ is left invariant. Comparisons between cuts and with old points are preserved, and the action on each tagged copy is free, being left regular. Distinct cut orbits get disjoint tags and never supply points to the same cut; cuts of a set form a set, so only set-many points and choices are used. $\square$

No right-invariant order was used; replacing “left-orderable” by “bi-orderable” would weaken the theorem [18].

Theorem 62.4 (Free ordered-class realization [18]). *Every set-sized left-orderable group has an increasing free class action on an ordered class with the set-cut property, hence on $(\mathbf{No}, <)$.*

Proof. Let $\mathcal{X}_0$ be a tagged copy of G with its left order and the left regular action, and build an increasing chain $(\mathcal{X}_\alpha)_{\alpha \in \mathbf{On}}$ of set-sized ordered G -sets: at $\alpha + 1$ apply lemma 62.3 to all cuts of $\mathcal{X}_\alpha$, at limits take unions, with fresh ordinal tags and the global well-order for representatives, so the recursion is uniform and set-valued. $\mathcal{X} = \bigcup_\alpha \mathcal{X}_\alpha$ is a proper class, since every successor stage inserts a point above all earlier ones; the action on it is increasing and free, since each point lies in a stage with those properties. If $L < R$ are sets in $\mathcal{X}$, their stages of appearance form a set, so $L \cup R \subseteq \mathcal{X}_\alpha$ for some α ; with $\mathfrak{c} = \{x \in \mathcal{X}_\alpha : x \leq l \text{ for some } l \in L\}$ (empty if L is) the next stage has a point above L and below R . So $\mathcal{X}$ has the set-cut property; transport the action to **No** by lemma 62.2. $\square$

Extending an action to a Dedekind completion would not suffice, since invariant cuts could become fixed points; the inserted free ordered fibers prevent this [18].

Lemma 62.5 (Complementary components of a set [18]). *For a set $S \subseteq \mathbf{No}$ and a downward-closed $\mathfrak{c} \subseteq S$ put $I_{\mathfrak{c}} = \{x \in \mathbf{No} : \mathfrak{c} < x < S \setminus \mathfrak{c}\}$. These classes partition $\mathbf{No} \setminus S$; each is nonempty, convex, and has the set-cut property.*

Proof. For $x \notin S$ the index is $\mathbf{c} = \{s \in S : s < x\}$, and the strict inequalities exclude S . The set-cut property of **No** realizes each cut, including the two unbounded ones; convexity and disjointness follow from the inequalities. For sets $L < R$ in $I_{\mathbf{c}}$, apply the set-cut property of **No** to $\mathbf{c} \cup L$ and $(S \setminus \mathbf{c}) \cup R$. $\square$

Theorem 62.6 (Uniform exact fixed sets on the index order [18]). *For every set $S \subseteq \mathbf{No}$ and every nontrivial set-sized left-orderable G there is a faithful increasing class action $g \mapsto \varphi_g$ on **No** with $\text{Fix}(\varphi_g) = S$ for every $g \neq 1$.*

Proof. On every component $I_{\mathbf{c}}$ install a transported copy of the free action of [theorem 62.4](#), through an order isomorphism $\mathbf{No} \rightarrow I_{\mathbf{c}}$, and let every group element fix S . There are set-many codes $\mathbf{c}$, and the back-and-forth algorithm makes the isomorphisms uniform, so the result is one class action. Each component is invariant with a free increasing action, and the placement of the components and of S is preserved, so the action is increasing and its nonidentity elements fix exactly S ; the complement is nonempty, so it is faithful. $\square$

63 Realization, the converse and exact arithmetic fixed rings

Proof of the sufficient direction of [theorem 60.1](#). Take φ_g from [theorem 62.6](#) and $\sigma_g = M_{1, \iota \varphi_g}$; [\(61.2\)](#) makes this an action, and [proposition 61.2](#) gives (i) and (ii). For $g \neq 1$, [theorem 61.5](#) with $\text{Fix}(\varphi_g) = S$ gives $\text{Fix}(\sigma_g) = K_S$, and intersecting with the invariant ring **Oz** gives the second formula. For faithfulness on **Oz** choose $a \notin S$; then $\varphi_g(a) \neq a$ for every $g \neq 1$, so the witness [\(61.3\)](#) is moved, and it lies in **Oz** because its outer exponent ω^a is positive whatever the sign of a . A nontrivial subgroup H' contains some $h \neq 1$, and its common fixed field lies between K_S and $\text{Fix}(\sigma_h) = K_S$. $\square$

Proof of [corollaries 60.2](#) and [60.3](#). Take $S = \Sigma_2(A)$ and use [proposition 61.6](#); the fixed field is a set by [section 60.3](#) and real closed by [lemma 61.7](#); $G = \mathbb{Z}$ gives a single automorphism of infinite order. For the second corollary take $S = \emptyset$: $K_{\emptyset} = \mathbb{R}$ and $\mathbf{Oz} \cap \mathbb{R} = \mathbb{Z}$. $\square$

Lemma 63.1 (A faithful ordered action gives a left order [18]). *If a set-sized group G acts faithfully by increasing bijections on a linearly ordered class, G is left-orderable.*

Proof. For each $u \neq 1$ choose a point x_u moved by u , by the global well-order; by replacement these witnesses form a set, well-ordered without repetitions as $(x_{\alpha})_{\alpha < \kappa}$. Distinct g, h have distinct image lists, since a witness moved by $h^{-1}g$ separates them. Order g, h by their images at the first index where they differ. This lexicographic order is linear; left multiplication by k preserves the first differing index, k being injective, and the comparison there, k being increasing. $\square$

Proof of the necessary direction of [theorem 60.1](#). Every field automorphism of the real closed field **No** preserves order, positivity being squareness; apply [lemma 63.1](#). This does not assume that the automorphisms fix $\mathbb{R}$ or preserve **Oz**. A unital ring automorphism of **Oz** extends uniquely to $\text{Frac } \mathbf{Oz} = \mathbf{No}$ (`odg:thm:fractions`); extensions respect composition and faithfulness, so the same argument applies. $\square$

The argument needs no strongness; for **Oz** every automorphism is in fact strong ([theorem 23.2](#)), which the proof does not use.

Corollary 63.2 (Admissible set groups [18]). *For a set-sized group G the following are equivalent: (a) G is left-orderable; (b) G has a faithful action by field automorphisms of **No**; (c) G has a faithful action by unital ring automorphisms of **Oz**; (d) G has a faithful strongly*

additive, real-fixing, $\mathbf{Oz}$ -preserving action on $\mathbf{No}$. Every nonidentity ring automorphism of $\mathbf{Oz}$ has infinite order; in particular no nontrivial finite group acts faithfully.

Proof. The equivalences have been proved (the trivial group acts trivially). A left-ordered group is torsion-free: a positive nonidentity element has strictly increasing positive powers, and a negative one has a positive inverse. Apply this to the cyclic group of a finite-order automorphism. $\square$

Torsion-freeness alone is not the criterion: the construction needs an actual left order, and the converse produces one [18]. The infinite order of every nonidentity order-preserving bijection is the last step of `saut:thm:torsion` of the surcomplex automorphism report; the set groups of this report, such as the faithful $\mathbb{Z} \wr \mathbb{Z}$ of [theorem 6.10](#) and the nonsolvable stabilizers of [theorem 7.7](#), are therefore left-orderable [merge].

Corollary 63.3 (An arithmetic fixed-fraction gap [18]). *For each nontrivial group of [corollary 60.3](#) and each $g \neq 1$,*

$$\text{Frac}(\text{Fix}_{\mathbf{Oz}}(\sigma_g)) = \mathbb{Q} \subsetneq \text{Fix}_{\text{Frac } \mathbf{Oz}}(\sigma_g) = \mathbb{R}, \quad (63.1)$$

and the same holds for the common fixed rings and fields of every nontrivial subgroup.

Proof. The fixed ring is $\mathbb{Z}$, the ambient fraction field is $\mathbf{No}$, and the fixed field is $\mathbb{R}$. $\square$

There is no conflict with finite-group invariant theory: every nontrivial group used here is infinite. An invariant real need not be a quotient of two *invariant* omnific integers, although every surreal is a quotient of omnific integers [18].

Corollary 63.4 (Unrestricted set products [18]). *Let $(G_j)_{j \in J}$ be a set-indexed family of left-orderable set groups and $A \subseteq \mathbf{No}$ a set. If $\prod_j G_j$ is nontrivial, it has an action as in [corollary 60.2](#), every nonidentity element having fixed field $K_{\Sigma_2(A)}$.*

Proof. Well-order J , choose left orders on the factors and compare distinct elements at the first differing coordinate; this lexicographic order is left invariant, and the product is a set, so [theorem 60.1](#) applies. $\square$

No infinite composition of individually supported automorphisms, and so no unproved summability of one, is needed; the conclusion is for the full product, not the direct sum [18].

Remark 63.5 (Relation to [part I](#) [merge]). [Theorem 7.6](#) gives faithful actions of $\mathbb{R}^\kappa$ fixing a set P , by 1-automorphisms in $\mathcal{U}_{\mathbf{Oz}}$; [corollary 63.4](#) gives faithful actions of every left-orderable set group, $\mathbb{R}^\kappa$ with its lexicographic order included, fixing P and with exact fixed field $K_{\Sigma_2(P)}$ for every nonidentity element, but in the monomial factor $\{M_{\chi, \tau}\}$ rather than in $\mathcal{U}_{\mathbf{Oz}}$. [Theorem 7.2](#) says that the *common* fixed field of all automorphisms preserving $\mathbf{Oz}$ is $\mathbb{R}$ and the common fixed subring $\mathbb{Z}$; [corollary 60.3](#) gives single automorphisms, and whole groups, with exactly that fixed field and fixed ring. [Theorem 7.4](#) computes the common fixed field of the elementary shears fixing A as $\mathcal{H}(A) = \mathbb{R}((t^{V_A}))$; since every exponent in the rational span V_A has inner support in $\Sigma_2(A)$, $\mathcal{H}(A) \subseteq K_{\Sigma_2(A)}$, usually strictly (for $A = \{\omega\}$ the first is $\mathbb{R}((t^{\mathbb{Q}}))$ and the second is $\mathbb{R}((\omega^{\mathbb{R}}))$), so a single monomial-factor automorphism fixing A fixes more than the shears do in common. [Question 11.3](#) concerns the stabilizer of A in $\mathcal{U}_{\mathbf{Oz}}$ and is not addressed by 18.

64 Explicit automorphisms and examples

Example 64.1 (Translation at the index level [18]). For $\varphi(a) = a + 1$, which has no fixed index, $\iota_\varphi(x) = \omega x$ and $M_{1,\iota_\varphi}(\sum_a x_a \omega^a) = \sum_a x_a \omega^{\omega a}$. The fixed field is $\mathbb{R}$ and the fixed omnific ring is $\mathbb{Z}$; every nonzero power has the same fixed field, since $\varphi^n(a) = a + n$ has no fixed points. This holds for all set-supported normal forms, not only finite sums.

So the fixed field of a surreal automorphism can be a set; [theorem 60.1](#) replaces this cyclic example by an arbitrary left-orderable group without adding fixed elements [18].

Example 64.2 (Dilation at the index level [18]). For $\varphi(a) = 2a$, $\text{Fix}(\varphi) = \{0\}$, so $\text{Fix}(M_{1,\iota_\varphi}) = \mathbb{K}_{\{0\}} = \text{NF}(\mathbb{R}) = \mathbb{R}((\omega^\mathbb{R}))$. Here $\iota_\varphi(\sum r_a \omega^a) = \sum r_a \omega^{2a}$, which differs from the first lift of [example 64.1](#): multiplying by ω and doubling all exponents are different operations.

Example 64.3 (An exact two-point index set [18]). The formula $\varphi(a) = 2a$ for $a < 0$; $\varphi(a) = a$ for $a \in \{0, 1\}$; $\varphi(a) = a/(2 - a)$ for $0 < a < 1$; $\varphi(a) = 2a - 1$ for $a > 1$ defines an increasing bijection of $\mathbf{No}$. The middle branch maps $(0, 1)$ onto itself with inverse $u \mapsto 2u/(1 + u)$, and its only possible fixed points are the endpoints; the outer branches have no fixed points. So $\text{Fix}(\varphi) = \{0, 1\}$ and $\text{Fix}(M_{1,\iota_\varphi}) = \text{NF}(\mathbb{R} + \mathbb{R}\omega)$, a two-level support field realized without a class back-and-forth choice.

Remark 64.4 (A concrete countable route [18]). A countable left-orderable group acts faithfully on $\mathbb{R}$ by increasing homeomorphisms [26, Proposition 1.1]. An individual such homeomorphism ψ extends to an increasing bijection $\widehat{\psi}$ of $\mathbf{No}$ by

$$\widehat{\psi}(r + \varepsilon) = \psi(r) + \varepsilon \quad (r \in \mathbb{R}, \varepsilon \text{ infinitesimal}), \quad (64.1)$$

fixing the positive and negative infinite surreals. The decomposition of a finite surreal into standard part and infinitesimal is unique; inside a real monad the map is a translation, between monads the standard parts decide, and infinite points stay on their side, so $\widehat{\psi}$ is increasing and bijective with inverse $\widehat{\psi}^{-1}$, and $\widehat{\psi_1 \psi_2} = \widehat{\psi_1} \widehat{\psi_2}$. It can be localized to a bounded interval (a, b) by conjugating with the order isomorphism

$$y \mapsto \frac{a+b}{2} + \frac{b-a}{2} \frac{y}{1+|y|}, \quad (64.2)$$

whose core $y \mapsto y/(1 + |y|)$ has inverse $w \mapsto w/(1 - |w|)$ on $(-1, 1)$, extending by the identity off the interval and applying the double lift. This gives explicit faithful parameter-stabilizing actions, but usually with many fixed indices; it must not replace the free cut filling when exact fixed fields are claimed.

Example 64.5 (A left-orderable group that is not bi-orderable [18]). Let $h_{\text{KB}}(t) = t/(2 - t)$ on $[0, 1]$ and define on $\mathbb{R}$

$$a(x) = x + 1, \quad b(n + t) = n + h_{\text{KB}}^{(-1)^n}(t) \quad (n \in \mathbb{Z}, 0 \leq t < 1). \quad (64.3)$$

Both are increasing homeomorphisms (the pieces of b agree at integers), and $aba^{-1} = b^{-1}$, since translation by one changes the parity of the interval index. The action realizes the Klein bottle group $G_{\text{KB}} = \langle a, b : aba^{-1} = b^{-1} \rangle$ faithfully: every word has a normal form $b^m a^n$, with $(b^m a^n)(b^r a^s) = b^{m+(-1)^n r} a^{n+s}$; if $b^m a^n$ acts trivially, its value at an integer gives $n = 0$, and on $(0, 1)$ the map b is h_{KB} , whose nonzero powers are not the identity, so $m = 0$. This also shows that the normal forms describe the presented group. By [lemma 63.1](#) G_{KB} is left-orderable; it is not bi-orderable, since conjugation preserves positivity in a bi-order while $aba^{-1} = b^{-1}$. [Corollary 60.3](#) realizes it on $\mathbf{No}$ freely outside $\mathbb{R}$ and on $\mathbf{Oz}$ freely outside $\mathbb{Z}$. That conclusion uses the general theorem, not the explicit action, in which b fixes every integer: a faithful action is not a free one.

65 Surcomplex fields and Gaussian omnific integers

For a field automorphism σ of $\mathbf{No}$ put $\tilde{\sigma}(x + iy) = \sigma(x) + i\sigma(y)$, a field automorphism of $\mathbf{No}(i)$ commuting with conjugation and fixing i ; for the actions above it fixes $\mathbb{C}$, is strongly additive on complex normal forms and preserves $\mathbf{Oz}[i]$, componentwise [18].

Proposition 65.1 (Complex fixed fields of the realized groups [18]). *For the action of theorem 60.1 and $g \neq 1$, $\text{Fix}_{\mathbf{No}(i)}(\tilde{\sigma}_g) = K_S(i)$ and $\text{Fix}_{\mathbf{Oz}[i]}(\tilde{\sigma}_g) = (\mathbf{Oz} \cap K_S) + i(\mathbf{Oz} \cap K_S)$; for $S = \emptyset$ these are $\mathbb{C}$ and $\mathbb{Z}[i]$. The field $K_S(i)$ is algebraically closed.*

Proof. An element is fixed exactly when both components lie in K_S ; intersect for the ring. K_S is real closed (lemma 61.7), so $K_S(i)$ is algebraically closed. $\square$

Lemma 65.2 (Centralizer of conjugation [18]; `saut:thm:axis`). *Every field automorphism τ of $\mathbf{No}(i)$ commuting with conjugation has the unique form $\tau(x + iy) = \sigma(x) + \epsilon i\sigma(y)$ with $\sigma \in \text{Aut}(\mathbf{No})$ and $\epsilon \in \{\pm 1\}$, and restriction and sign give $\text{Aut}(\mathbf{No}(i); \text{cj}) \simeq \text{Aut}(\mathbf{No}) \times C_2$ (schematically, in the sense of section 60.3). The same holds for $\mathbf{Oz}[i]$ and $\mathbf{Oz}$ with unital ring automorphisms.*

Proof. The field statement is `saut:thm:axis` of the surcomplex automorphism report with $F = \mathbf{No}$: commutation preserves $\text{Fix}(\text{cj}) = \mathbf{No}$, and $\tau(i) = \pm i$. For the ring statement, the fixed ring of conjugation on $\mathbf{Oz}[i]$ is $\mathbf{Oz}$; a unital ring automorphism of $\mathbf{Oz}[i]$ extends to $\mathbf{No}(i) = \text{Frac } \mathbf{Oz}[i]$ and the same argument applies, and conversely the extension of any automorphism of $\mathbf{Oz}$, with either sign, preserves $\mathbf{Oz}[i]$ [18]. $\square$

Proof of theorem 60.4 without parameters. Let G act faithfully and compatibly with conjugation, and restrict to $\mathbf{No}$; by lemma 65.2 the kernel of the restriction lies in C_2 . If it is trivial, G acts faithfully on $\mathbf{No}$ and is left-orderable. Otherwise it is generated by an involution z acting as conjugation, which is central, and the sign homomorphism $\epsilon : G \rightarrow C_2$ is an isomorphism on $\langle z \rangle$; so $G = \ker \epsilon \times \langle z \rangle$, and the restriction is faithful on $\ker \epsilon$, which is therefore left-orderable. Conversely realize a left-orderable G_0 by theorem 60.1 (trivially if $G_0 = 1$) and extend coefficientwise; for $G_0 \times C_2$ add conjugation, which commutes with the extension and meets it trivially, so the product action is faithful, strongly additive and $\mathbf{Oz}[i]$ -preserving. The ring case follows from the same construction and restriction, or from the fraction field. $\square$

Lemma 65.3 (A fixed nonreal point forces positive sign [18]). *If τ commutes with conjugation and fixes $x + iy$ with $y \neq 0$, its sign ϵ is $+1$.*

Proof. With $\epsilon = -1$, comparing imaginary parts gives $\sigma(y) = -y$; but σ preserves the order of $\mathbf{No}$, hence the sign of y . $\square$

Completion of theorem 60.4. If all parameters are real, the real realization fixing them and conjugation both fix them, so both kinds of group occur. If a parameter set $A \subseteq \mathbf{No}(i)$ contains a nonreal element, lemma 65.3 forces every sign to be positive, the restriction is faithful and G is left-orderable; conversely, for left-orderable G apply the real construction to $\{\text{Re } z, \text{Im } z : z \in A\}$ and extend coefficientwise. The argument is the same for parameters in $\mathbf{Oz}[i]$. $\square$

So one nonreal point removes the sign alternative from the classification, not merely shrinking the stabilizer; the only possible nontrivial finite conjugation-compatible group is C_2 , and it cannot fix a nonreal point [18].

Proposition 65.4 (A parity-controlled fixed-field tower [18]). *Let σ generate the cyclic realization with fixed field K_S , so every nonzero power of σ has that fixed field, and put $\tau = \text{cj} \circ \tilde{\sigma}$. For $n \neq 0$, $\text{Fix}_{\mathbf{No}(i)}(\tau^n)$ is K_S for odd n and $K_S(i)$ for even n ; in $\mathbf{Oz}[i]$ these are $\mathbf{Oz} \cap K_S$ and $(\mathbf{Oz} \cap K_S) + i(\mathbf{Oz} \cap K_S)$.*

Proof. cj commutes with $\tilde{\sigma}$, so $\tau^n = \text{cj}^n \tilde{\sigma}^n$. For even n use [proposition 65.1](#). For odd n a fixed $x + iy$ has $\sigma^n(x) = x$ and $\sigma^n(y) = -y$; order preservation forces $y = 0$, and $x \in K_S$. Negative powers are the same. $\square$

65.1 Why conjugation and the real coefficients matter

The complex fixed fields above are real closed or algebraically closed because they come from real ordered actions with a controlled sign. Strong complex monomial automorphisms outside this category behave differently. The following standard character twist marks the boundary of the theorems; it is not a novelty claim [18]. For an exponent a put $\chi_{\text{ph}}(a) = \exp(2\pi i q_0(a))$, the ordinary complex exponential at an ordinary real; q_0 is additive on $\mathbf{No}$, so χ_{ph} is a character.

Proposition 65.5 (A Gaussian-omnific character twist [18]). *The map*

$$\Phi_{\text{ph}}\left(\sum_a z_a \omega^a\right) = \sum_a z_a \chi_{\text{ph}}(a) \omega^a \quad (z_a \in \mathbb{C}) \quad (65.1)$$

is a strongly additive $\mathbb{C}$ -fixing field automorphism of $\mathbf{No}(i)$ preserving $\mathbf{Oz}[i]$, not commuting with conjugation. For $n \neq 0$,

$$\text{Fix}(\Phi_{\text{ph}}^n) = \mathbb{C}((\omega^{E_n})), \quad E_n = \{a \in \mathbf{No} : q_0(a) \in n^{-1}\mathbb{Z}\}, \quad (65.2)$$

and none of these fixed fields is algebraically closed.

Proof. Supports are unchanged and all multipliers are nonzero; the character identity gives multiplicativity by finite convolution; the inverse uses χ_{ph}^{-1} . Since $\chi_{\text{ph}}(0) = 1$, constants and Gaussian-integer constant terms are fixed and nonnegative supports stay nonnegative; strong additivity follows from unchanged incidences. $\omega^{1/4}$ is real while $\Phi_{\text{ph}}(\omega^{1/4}) = i\omega^{1/4}$ is not, so Φ_{ph} does not commute with conjugation. A series is fixed by Φ_{ph}^n exactly when every occupied exponent has $\chi_{\text{ph}}(a)^n = 1$, that is $q_0(a) \in n^{-1}\mathbb{Z}$. With $N = |n|$, $\omega^{1/N}$ is fixed; an m th root with $m \geq 2$ in the fixed field would have leading exponent $1/(Nm) \notin E_n$, so $X^m - \omega^{1/N}$ has no root there. $\square$

Φ_{ph} is the phase twist $P_{-2\pi}$ of `saut:thm:phase` (in the t -convention its character is $e^{-2\pi i q_0(\gamma)}$), of the kind recorded after [theorem 10.2](#) and in [remark 36.6](#) (`odg:def:lem:twist` [merge]); the fixed fields of its powers are 18's. There is no conflict with [proposition 65.1](#), since Φ_{ph} is not the complexification of a real automorphism and does not centralize conjugation, nor with [corollary 63.2](#), since it acts on $\mathbf{Oz}[i]$, not on $\mathbf{Oz}$ [18].

66 Consequences, limits and questions of source 18

66.1 What the theorems establish, and the proof obligations

The field and ring realization problems have the same group-theoretic answer, although their fixed objects differ: every left-orderable set group can act with fixed field exactly $\mathbb{R}$ and fixed omnific ring exactly $\mathbb{Z}$ for each nonidentity element, and inserting a set S at the index level replaces these by K_S and $\mathbf{Oz} \cap K_S$. The fixed-field assignment is deliberately far from

injective on subgroups (every nontrivial subgroup has the same fixed field), an infinite-action phenomenon: no finite Galois correspondence is to be expected from real or algebraic closedness alone. And no set of parameters makes the structure of the field with **Oz**, the named ordinary reals and Hahn summation rigid, in the sense that the constructed automorphisms preserve these data; Hahn summation is not claimed to be a first-order operation of finite arity [18]. What is not established is listed in [section B](#).

18's dependency structure is acyclic: classical inputs (normal forms, real closedness, the set-cut property of **No**); the Hahn layer ([proposition 61.2](#) and [theorem 61.5](#)); the ordered-action layer ([lemma 62.3](#) and [theorems 62.4](#) and [62.6](#)); the synthesis [theorem 60.1](#), whose converse uses only [lemma 63.1](#) and the fraction field; and the complex layer (restriction to the conjugation-fixed field, the sign decomposition, order preservation). No unreviewed theorem of a companion report is a premise; this report's automatic strongness, coefficient reconstruction and exponential rigidity are not imported [18].

Potential failure	How the proof prevents it [18]
An invariant orbit sum is not a Hahn series	Lemma 61.4 makes every orbit in a reverse well-ordered invariant support a singleton.
A filled invariant cut creates a fixed point	Lemma 62.3 inserts a free ordered fiber, not one point.
The fiber order silently needs a bi-order	The coordinate change is $h \mapsto h_0 h$, so left invariance suffices.
A construction chooses sets of class maps	One global choice rule and one set-valued recursion code the action and all component identifications.
The two Hahn levels are confused	The maps are named ι_φ and M_{1,ι_φ} ; parameters are controlled by Σ_2 , not one support.
Faithfulness is lost on Oz	The moved witness ω^{ω^a} has positive outer exponent and lies in Oz .
A complex sign hides in the imaginary part	An ordered real automorphism cannot send $y \neq 0$ to $-y$.

66.2 Status and relations

- *No named question is answered.* [Question 11.3](#) concerns the stabilizer of a parameter set in $\mathcal{U}_{\mathbf{Oz}}$; 18's automorphisms lie in the monomial factor ([section 60.2](#)), so they do not bear on it; a status note records this after the question.
- *Merged questions.* 18's question on preserving additional transcendental structure is merged into [question 32.15](#), and its formalization question into [question 59.10](#).
- *Related.* 18's question on surcomplex actions without a named conjugation ([question 66.7](#)) is a relative of [question 32.1](#); its conjugacy question concerns group actions, whereas [question 32.8](#) concerns proper copies; its birthday question is a relative of [question 32.16](#).
- *Checked true.* The pin [efc5446](#) and its tree [aba6982](#); the catalogue count of 61 reports at the pin; the description of this report's guide as the eight-source report; the absence of "left-orderable" from the other reports of the collection; and its suite, rerun for the merge ([section C](#)). No mathematical error was found in 18.

66.3 Questions of source 18

18 poses ten questions (its 11.1–11.10), proposed as research directions, not as published open problems. Eight are new; 11.5 is merged into [question 32.15](#) and 11.10 into [question 59.10](#).

Question 66.1 (Which real closed fields are exact fixed fields? [18]). Characterize the set-sized real closed K with $\mathbb{R} \subseteq K \subseteq \mathbf{No}$ for which some strongly additive **Oz**-preserving automorphism has $\text{Fix}(\sigma) = K$. [Theorem 60.1](#) settles $K = K_S$. Must every realizable K have a recognizable truncation or support-closure property?

A first target is a Hahn field $\text{NF}(E)$ with E divisible but not of the form $\text{NF}(S)$; the outer fixed-support argument reduces it to realizing E as the fixed subgroup of an ordered additive automorphism of $\mathbf{No}$ [18].

Question 66.2 (Beyond the two-level family [18]). For which divisible ordered subgroups $E \subseteq \mathbf{No}$ can every left-orderable set group be realized by strong **Oz**-preserving actions with $\text{Fix}(\sigma_g) = \text{NF}(E)$ for every $g \neq 1$?

This is not the problem of realizing one automorphism: all group relations and a common exact fixed subgroup must survive the additive-group construction [18]. *Status* (batch 34): partly answered. Source 20's relative action ([theorem 77.5](#)) realizes $E = \text{NF}(\mathbf{No} \setminus J)$ for every nonempty bounded open interval J and every left-orderable set group; this E is a proper class and not of the form $\text{NF}(S)$ for a set S ([remark 77.6](#)). Other E stay open (compare [question 82.2](#)).

Question 66.3 (Conjugacy of free-complement actions [18, 20]). When are two realizations with the same G and S conjugate by a strong **Oz**-preserving automorphism fixing K_S ? Which invariants of the chosen left order survive the cut filling and the double lift?

Different actions can have the same fixed fields for all nontrivial subgroups, so fixed fields do not classify them; centralizers, invariant convex equivalence relations and stabilizers of cuts are candidate finer invariants [18]. Source 20 asks the same (its Question 3): when do two left orders or two equivariant cut constructions give conjugate actions on $(\mathbf{No}, \mathbf{Oz})$, and can the order on a regular orbit, the arrangement of index orbits or a support invariant distinguish them? An embedding classification does not imply uniqueness, and a uniform fixed field cannot distinguish them [20].

Question 66.4 (Centralizers and normalizers [18]). Compute the centralizer and normalizer of an exact realization, first for the translation lift of [example 64.1](#), then for a free nonabelian realization. Can a prescribed extension of G be realized as the normalizer quotient while keeping the exact fixed field?

Maps commuting with translation at the index level are a natural start; not every field centralizer element need come from an index permutation, so the constructed subgroup and the ambient automorphism structure must be distinguished [18].

Question 66.5 (Birthday-controlled realization [18]). For a specified set-sized initial surreal field, under which cardinal and closure hypotheses can the equivariant cut filling be carried out inside it, and what least height realizes a given left-ordered group and parameter set?

A solution must bound separately the number of cuts filled, the supports generated and the lengths of the sign sequences; truncating the class recursion does not settle them [18]. Compare [question 32.16](#).

Question 66.6 (Arithmetic invariant-fraction fields [18]). Determine $\text{Frac}(\mathbf{Oz} \cap \mathbf{K}_S)$ inside $\mathbf{K}_S$ as a function of S . When is it $\mathbf{K}_S$, and how large can the gap be?

[Equation \(63.1\)](#) is the extreme case $S = \emptyset$, with $\mathbb{Q} \subsetneq \mathbb{R}$. Support bounds inside the set-sized exponent group $\text{NF}(S)$, rather than bounds available only in $\mathbf{No}$, should control further examples [18].

Question 66.7 (Surcomplex actions without named conjugation [18, 20]). Characterize the set groups with faithful $\mathbf{Oz}[i]$ -preserving strong actions on $\mathbf{No}(i)$ when commutation with conjugation is dropped. Which fixed-field geometries arise from combining exponent actions with character twists such as [\(65.1\)](#)?

[Theorem 60.4](#) is not asserted in this larger category; the non-algebraically-closed fixed fields of [proposition 65.5](#) are a test family [18]. It is a relative of [question 32.1](#), which asks whether such automorphisms are strong at all. Source 20 asks it too (its Question 5), for the set groups acting faithfully on $(\mathbf{No}(i), \mathbf{Oz}[i])$ without conjugation, and which arithmetic or support data force preservation of the real form: without the real form the direct-product argument no longer applies [20].

Question 66.8 (Fixed fields and orbit types together [18]). Can the fixed fields of selected subgroups be prescribed together with the stabilizers of individual nonfixed surreals? Which subgroup-lattice data are compatible with strong Hahn summation?

The present action realizes one pattern: every nonconstant element has trivial stabilizer. More general cut fibers could encode nontrivial stabilizers, but the action must stay increasing and the orbit-support obstruction remains [18].

Part IX

All derivations of the omnific integers integrate

This part records manuscript 19, the third addition of batch 33. It answers [question 19.2](#): every ring derivation of $\mathbf{Oz}$ or $\mathbf{Oz}[i]$ maps the ring into its purely infinite ideal and preserves it, and has a unique iterative integral Hasse–Schmidt extension, $\delta^n/n!$ ([theorem 67.1](#)). The reason is not Hahn summability but the identity $\Pi = \bigcap_n n\mathbf{Oz}$ of [lemma 2.3](#) and the unique divisibility of Π . 19 develops this into an integral exponential–logarithm description of all formal automorphisms tangent to the identity, a jet tower that lifts at every stage but never splits, and a classification of the sections of the residue map ([sections 69 to 71](#)). A second, logical, half contrasts polynomial coefficient recovery, which is [part VI](#)’s quartic formula re-derived, with the formal completion, in which that formula becomes universal and no set of parameters defines the canonical coefficient copy ([sections 72 and 73](#)). Its labels carry the sub-prefix `opa:hs:`.

67 Source 19, the question it answers, and its notation

67.1 The manuscript and its pin

Manuscript 19 is *All Derivations Integrate: Formal Symmetry and Coefficient Definability of Omnific Arithmetic*, dated 23 September 2026, 25 pages. It is manuscript 08 of batch 33,

delivered as the archive `omnific_formal_symmetry` in commit `73043eb` and placed as an addition in commit `aa9c891`; it is numbered 19 here. Its checks, their recorded run, its build script, its build record and its source audit are shipped as `code/19-formal-symmetry-verify.py`, `data/19-formal-symmetry-verification.json`, `code/19-formal-symmetry-build.sh`, `data/19-formal-symmetry-BUILD_REPORT.json` and `19-formal-symmetry-SOURCE_AUDIT.md`; its `article.tex`, PDF and README are not shipped.

It pins `889b87c`, which contains [parts I to IV](#) of this report (`d4d72d7`) but neither the batch-31 reciprocal notes (`ab18444`) nor [parts V and VI](#) (placed in `7d04483`, written in `c315ac9`). So 19 did not see [theorem 45.4](#), and its quartic formula is an independent re-derivation. It read the catalogue, this report's guide (truncated) and ranges of this article around [part II](#), several of them truncated, and used only returned text (its source audit). Its account is accurate: it quotes [question 19.2](#), notes that the text establishes extensions for Euler derivations and discusses positive shifts but leaves the general case open, and credits [theorems 14.5, 15.4, 17.2](#) and [17.6](#), [lemma 14.1](#), and [proposition 18.1](#).

67.2 What it answers, re-derives and generalizes

19 answers [question 19.2](#), all three clauses: the derivations preserving Π and those with an iterative integral Hasse–Schmidt extension are both the whole class of ring derivations, which is strictly larger than the Euler family: it contains non- $\mathbb{R}$ -linear derivations such as ωd_* ([example 71.2](#)). A structural description of all derivations is not given ([question 75.1](#)). 19's numbers are those of its own PDF.

19	Printed as	Comparison
Lem. 3.1	lemma 2.3 ; <code>odg:eq:divkernel</code>	Same, with the remark that Π_R is an $\mathbb{R}$ - or $\mathbb{C}$ -vector space; cited.
Lem. 3.2	<code>odg:thm:fractions</code>	Same (one monomial clears a set, complex coefficients allowed); cited.
Lem. 6.2, first part	theorem 17.6 (D_b with $b = \eta$), <code>osq:prop:classder</code>	Same derivations; the Witt relation is new (lemma 70.2).
Lem. 8.1	theorem 45.3 for $G = \mathbb{Z}^n$	Same statement; 19's genus-one proof is kept as a second route (lemma 72.1).
Thm. 8.2, Cor. 8.3	theorem 45.4(ii) , (iii), theorem 45.7	Same formula (19's a, b are the u, v of (45.2)); 16's is for every monoid algebra of any rank, 19's for intermediate rings of finite rank; printed once in the joint generality (corollary 72.2 , [merge]).
Warnings after Prop. 8.5	remark 45.6 and example 51.2	The conic and characteristic-2 failures are the same; the genus-boundary example is new.
Lem. 10.1	The fresh-scale choice of lemma 7.1 and theorem 9.1	A new application of the same technique.
Lem. 11.1	lemma 14.1	Same; cited.
Thm. 11.3	theorem 14.5	19's class-size version for the four rings, called background by 19; printed (theorem 74.2).

19	Printed as	Comparison
Thm. 9.2	example 51.1	Strengthens that example from one nonconstant solution to all of $R[[s]]$.
Questions	section 75.3	Seven new, four merged.
All else	This part	New.

19's novelty statement lists the removal of all derivation restrictions from the formal-directions question, the integral formal correspondence and nonsplit jet tower, the section classification and the existential/nondefinability boundary; formal Lie theory in characteristic zero [12, 24, 2], the positive-genus argument [27], strong invariance [14] and the results of [part II](#) are credited. At the merge this is accurate except that the existential formula is now also 16's ([part VI](#)), which 19 could not see.

67.3 Conventions and notation of this part

This part uses the growth convention of [part II](#), $x = \sum_{\gamma} c_{\gamma} \omega^{\gamma}$ with reverse well-ordered set support, and R denotes one of $\mathbf{Oz}$, $\mathbf{Oz}[i]$, $\mathbf{No}$, $\mathbf{No}(i)$, with $K = \text{Frac } R$ (so $K = \mathbf{No}$ or $\mathbf{No}(i)$). For the arithmetic rings Π_R is the purely infinite ideal: $\Pi_{\mathbf{Oz}} = \Pi$ and $\Pi_{\mathbf{Oz}[i]} = \Pi_{\mathbb{C}}$, so $\mathbf{Oz} = \mathbb{Z} \oplus \Pi$ and $\mathbf{Oz}[i] = \mathbb{Z}[i] \oplus \Pi_{\mathbb{C}}$; put $\Pi_R^{\sharp} = \Pi_R$ for the rings and $\Pi_R^{\sharp} = R$ for the fields. The formal variable is Part II's s ; $R[[s]]$ is a ring of formal power series, whose multiplication uses finite sums in each degree; no substitution of a surreal for s and no summability of the coefficient family is implied, and every exponential and logarithm below is an s -adic operator series [19]. A ring homomorphism fixing s preserves every ideal (s^n), hence is s -adically continuous and determined by its values on coefficients. $\text{Der}(R)$ is the collection of ring derivations (additive maps with the Leibniz rule; not analytic derivatives).

19's symbol	Here
T (formal variable)	s .
$\mathfrak{I}_R; J$	$\Pi_R; \Pi_R^{\sharp}$.
H_n (Hasse–Schmidt components)	$\mathbf{H}_n$.
$L = \sum_j T^j \delta_j$ (Lie series); $\Delta = \Phi - \text{id}$	$\mathbf{L} = \sum_j s^j \delta_j$; written $\Phi - \text{id}$.
$G(R), G_N(R); S = R[[T]],$ S_N	$\mathcal{G}(R), \mathcal{G}_N(R)$; written out.
$\ell_{\eta}(\gamma); D_{\eta}; E_m = \omega^m D_0$	$q_{\eta}(\gamma); D_{\eta}$ (theorem 17.6); written $\omega^m D_0$.
C_R	$\mathbf{C}_R$.
$\Theta(x)$	$\mathbf{Q}(x)$.
Coefficient section $s : R \rightarrow R[[T]]$	Residue section ς .
D_d	d_* , the coefficientwise extension of a derivation d of $\mathbb{R}$ or $\mathbb{C}$.
The curve C in Prop. 8.5	$\mathbf{X}$.

19's symbol	Here
$\mathbf{No}[i]$, $\mathbf{Oz}[i]$	$\mathbf{No}(i)$, $\mathbf{Oz}[i]$.

The collisions are resolved here only, and the tempting false readings are wrong:

- 19's T is the formal parameter s of [section 17](#): not a linear map T of [section 25](#), not 10's variable (written ξ in [part I](#)) and not 18's lift.
- 19's Θ is a formula, written $\mathbf{Q}$; it is not source 10's additive profile Θ of [section 1.5](#).
- 19's “coefficient section” is a unital homomorphism $\varsigma : R \rightarrow R[[s]]$ with $\varsigma \bmod s = \text{id}$, a section of the residue map; it is *not* Part III's coefficient section $\rho : \mathbb{R} \rightarrow \mathbb{R}((t^{\mathbb{G}_\tau}))$ ([section 20.3](#) and [question 32.7](#)), a field embedding into the Hahn field on a bottom gap. The same phrase names two different notions; here the first is called a *residue section*.
- 19's H_n are not the convex subgroups H_g or H_δ ; 19's E_m are not $\mathcal{E}_A$, E or 18's $\mathbf{E}_n$; 19's C_R is not C_δ , $C(a)$, $\mathbb{C}$, nor $\mathbf{C}_0$ of [part VII](#) (the constants of the single derivation D_0); $\mathbf{C}_R$ is the field or ring of common constants of *all* derivations. 19's L is not L_χ or $\hat{L}_0$; its J is not an embedding $J_{\tau,\rho}$; its $G(R)$ is not a group G of [parts VI](#) and [VIII](#); its Δ is not the target group Δ .
- 19's ℓ_η is q_η of [section 7](#), the coefficient of ω^η in the exponent, not the leading exponent $\ell(x)$ of [section 12.1](#). Its D_d is not $D_{\delta,\theta}$; d is a derivation of $\mathbb{R}$ as in [part III](#).
- In [\(67.1\)](#) the witnesses a, b are the u, v of [\(45.2\)](#).

Foundations: all algebra is first proved for set-sized rings; for the class rings the collections $\text{Der}(R)$ and $\mathcal{G}(R)$ are external collections of admissible class maps, and a displayed correspondence is a theorem scheme about given maps. Normal forms, formal series and coefficient families used in a computation are sets; recursions have finite or $\mathbb{N}$ -length; no proper-class summation or large-cardinal assumption occurs, and a proposed rational-function solution of a curve equation involves finitely many coefficients, which lie in a set-sized subfield, so no proper-class scheme is formed [19].

67.4 Principal results of source 19

Theorem 67.1 (Automatic formal integration [19]; 19's Theorem 1.1). *Let $R = \mathbf{Oz}$ or $\mathbf{Oz}[i]$. Every ring derivation δ of R satisfies $\delta(R) \subseteq \Pi_R$ and $\delta(\Pi_R) \subseteq \Pi_R$. There is exactly one iterative Hasse–Schmidt derivation with first component δ , namely $\mathbf{H}_n = \delta^n/n!$, and $x \mapsto \sum_n \mathbf{H}_n(x)s^n$ extends to an s -fixing automorphism of $R[[s]]$, inverse to the one obtained from $-\delta$.*

Theorem 67.2 (Complete formal symmetry [19]; 19's Theorem 1.2). *Let R be one of $\mathbf{Oz}$, $\mathbf{Oz}[i]$, $\mathbf{No}$, $\mathbf{No}(i)$, and let $\mathcal{G}(R)$ consist of the s -fixing automorphisms of $R[[s]]$ that induce the identity modulo s . Exponential and logarithm are mutually inverse correspondences between $s\text{Der}(R)[[s]]$, the Lie series acting coefficientwise as continuous derivations, and $\mathcal{G}(R)$; all factorials and logarithmic denominators are legitimate even for the two rings not containing $\mathbb{Q}$. Every automorphism of $R[s]/(s^{N+1})$ fixing s and inducing the identity modulo s lifts to $\mathcal{G}(R)$; the jet group $\mathcal{G}_N(R)$ has nilpotency class exactly N , and for $N \geq 2$ its central extension over $\mathcal{G}_{N-1}(R)$ does not split.*

Theorem 67.3 (A definability boundary [19]; 19's Theorem 1.3). *Put*

$$Q(x) : \Longleftrightarrow \exists a \exists b (b \neq 0 \wedge a^2 = (x^4 + 1)b^2). \quad (67.1)$$

For $R \in \{\mathbf{Oz}, \mathbf{Oz}[i], \mathbf{No}, \mathbf{No}(i)\}$ this parameter-free formula defines exactly R in every finite polynomial or Laurent polynomial ring over R ; for $R = \mathbf{Oz}$ or $\mathbf{Oz}[i]$ it holds for every element of $R[[s]]$. In all four cases the canonical copy $R \subseteq R[[s]]$ is not definable in the pure ring language with parameters from any set, even with s and every ideal (s^n) named, although $R[[s]]/(s)$ is interpretable.

The proofs are in [sections 68 to 70, 72 and 73](#).

68 Divisibility and automatic integration

For $R = \mathbf{Oz}$ or $\mathbf{Oz}[i]$, $\Pi_R = \bigcap_{n \geq 1} nR$, and Π_R is uniquely divisible by every positive integer; this is [lemma 2.3](#) (with $\mathfrak{o} = \mathbb{Z}, \mathbb{Z}[i]$) and `odg:eq:divkernel`, which 19 reproves as the sole structural input of automatic integration (a purely infinite normal form divided by n keeps its positive support; the constant term of an element of nR lies in $n\mathbb{Z}$ or $n\mathbb{Z}[i]$, and only 0 lies in all of these). Moreover Π_R is a vector space over $\mathbb{R}$, respectively $\mathbb{C}$, since multiplying coefficients by an ordinary scalar keeps positive support [19]. That one monomial $\omega^\beta \in \Pi_R$, $\beta > 0$, satisfies $\omega^\beta A \subseteq \Pi_R$ for any set $A \subseteq K$, so that $\text{Frac } R = K$, is `odg:thm:fractions` (complex coefficients make no difference) [19].

Proposition 68.1 (Formal fraction fields [19]). $\text{Frac}(\mathbf{Oz}[[s]]) = \mathbf{No}((s))$ and $\text{Frac}(\mathbf{Oz}[i][[s]]) = \mathbf{No}(i)((s))$. More generally every set-indexed array of elements of K has one monomial denominator making all entries purely infinite.

Proof. A quotient of nonzero power series over R is a Laurent series over K . Conversely, for $f \in K((s))$ some $s^q f$ has nonnegative degrees; apply `odg:thm:fractions` to its countable set of coefficients to get m with $ms^q f \in \Pi_R[[s]]$, so $f = (ms^q f)/(ms^q)$. The array statement is the same support-union argument, with no multiplication in the index set. $\square$

Remark 68.2 (Why the class scope matters [19]). The common denominator uses a bound for a set of surreal exponents; it cannot be imported into a fixed set-sized exponent group, where even a divisible group without an order unit may have sets of scales with no bound. This is the fraction-field warning of [proposition 18.1](#).

Definition 68.3 ([19]). A Hasse–Schmidt derivation of R is a sequence of additive maps $(H_n)_{n \geq 0}$ with $H_0 = \text{id}$ and

$$H_n(xy) = \sum_{r+q=n} H_r(x) H_q(y); \quad (68.1)$$

it is *iterative* if

$$H_r \circ H_q = \binom{r+q}{r} H_{r+q} \quad (r, q \geq 0). \quad (68.2)$$

Integral means that all H_n take values in R , not in its rationalization or fraction field; it does not refer to integration of functions.

Lemma 68.4 (Automatic invariance [19]). Every derivation δ of $R = \mathbf{Oz}$ or $\mathbf{Oz}[i]$ preserves Π_R and maps R into Π_R ; so $\delta^j(R) \subseteq \Pi_R$ for $j \geq 1$.

Proof. If $x \in \Pi_R$, write $x = ny_n$ for every $n \geq 1$. A derivation kills 1, hence every integer, and $\delta(x) = n\delta(y_n) \in nR$ for every n , so $\delta(x) \in \Pi_R$. For $\mathbf{Oz}$ write $x = z + u$ with $z \in \mathbb{Z}$, $u \in \Pi$; then $\delta(x) = \delta(u) \in \Pi$. For $\mathbf{Oz}[i]$, $i^2 = -1$ gives $2i\delta(i) = 0$, so $\delta(i) = 0$ in the characteristic-zero domain, δ kills $\mathbb{Z}[i]$, and the same decomposition applies. $\square$

Proof of theorem 67.1. For $n \geq 1$, $\delta^n(x)$ lies in the uniquely divisible Π_R , so $H_n(x) = \delta^n(x)/n!$ is defined in R ; δ commutes with division by positive integers on Π_R , since applying it to $ny = x$ gives $n\delta(y) = \delta(x)$, whose solution in the ideal is unique. The iterated Leibniz identity $\delta^n(xy) = \sum_r \binom{n}{r} \delta^r(x) \delta^{n-r}(y)$, divided in the fraction field with every term already in R , gives (68.1), and $H_r H_q(x) = \delta^{r+q}(x)/(r!q!) = \binom{r+q}{r} H_{r+q}(x)$ gives iterativity. For uniqueness, (68.2) with $r = 1$ gives $\delta H_n = (n+1)H_{n+1}$; from $H_0 = \text{id}$, $H_1 = \delta$ and torsion-freeness, $(n+1)!H_{n+1} = \delta^{n+1}$. Extend the generating map to $R[[s]]$ fixing s : each output coefficient uses finitely many inputs, and $\exp(s\delta)\exp(-s\delta) = \text{id}$ is a finite binomial cancellation in each degree, with all divided terms defined as above. $\square$

Corollary 68.5 (Answer to question 19.2 [19]). *Both classes of derivations asked for in question 19.2 are the whole class of ring derivations of $\mathbf{Oz}$; no hypothesis of $\mathbb{R}$ -linearity, strong Hahn summability, a common set of shifts or a support description is needed. The same holds for $\mathbf{Oz}[i]$.*

This answers formal integration, not the different question of which formal flows can be evaluated inside $\mathbf{No}$ or $\mathbf{No}(i)$ (question 59.6), and it gives no support classification of all derivations (question 75.1); these remain meaningful after the integrability obstruction has disappeared [19]. For the positive-shift derivations of part I the Hasse–Schmidt flow $\exp(s\partial)$ is formal, while theorem 4.8 integrates those with a common shift set by honest automorphisms of $\mathbf{No}$ [merge].

Proposition 68.6 (Divisible-ideal integration [19]). *Let A be a characteristic-zero domain in which $J_A = \bigcap_{n \geq 1} nA$ is a uniquely divisible ideal, and suppose $A = A_0 \oplus J_A$ additively for a subring A_0 all of whose elements are algebraic over $\mathbb{Q}$ in $\text{Frac } A$. Then every derivation of A takes values in J_A and has the unique iterative extension $H_n = \delta^n/n!$.*

Proof. A derivation extends to $\text{Frac } A$ by the quotient rule. For a algebraic over $\mathbb{Q}$ with minimal polynomial p , $p'(a)\delta(a) = 0$ and p is separable, so $\delta(a) = 0$; thus δ kills A_0 . The argument with ny_n preserves J_A , and the proof of theorem 67.1 applies. $\square$

In particular this holds for $\mathbb{Z} \oplus \Pi_{\mathbb{R}}(\Gamma)$ and $\mathbb{Z}[i] \oplus \Pi_{\mathbb{C}}(\Gamma)$ for every set-sized ordered abelian exponent group Γ , with reverse well-ordered positive supports: no divisibility, Archimedean or no-order-unit hypothesis is needed, so the formal result is independent of the scale criterion theorem 13.1 used for polynomial rigidity [19].

Proposition 68.7 (Formal additive actions [19]). *For commuting derivations $\delta_1, \dots, \delta_r$ of $\mathbf{Oz}$ or $\mathbf{Oz}[i]$, $H_{\alpha} = \delta_1^{\alpha_1} \cdots \delta_r^{\alpha_r} / (\alpha_1! \cdots \alpha_r!)$ is the unique iterative r -parameter Hasse–Schmidt extension with these first components. Conversely the first components of an iterative additive r -parameter action commute.*

Proof. Every nonempty composition has image in the uniquely divisible ideal; the commuting one-parameter exponentials multiply, and their additive laws give the multivariate iterative law; uniqueness is by induction on the multi-index. Conversely the coefficient of $s_i s'_j$ in the two orders of composing an additive action gives $\delta_i \delta_j = \delta_j \delta_i$. $\square$

This extends corollary 17.7 from the Euler directions D_b to all commuting derivations [merge].

69 Formal automorphisms and their integral logarithms

In this section R is one of the four rings. For derivations $\delta_1, \delta_2, \dots$ of R put

$$\mathbf{L} = \sum_{j \geq 1} s^j \delta_j, \quad \mathbf{L} \left(\sum_{n \geq 0} a_n s^n \right) = \sum_{m \geq 1} \left(\sum_{j=1}^m \delta_j(a_{m-j}) \right) s^m, \quad (69.1)$$

a continuous derivation of $R[[s]]$ fixing s , raising s -order, with image in $s\Pi_R^\sharp[[s]]$ [19].

Lemma 69.1 (Divisibility for formal coefficient maps [19]). *Let $R = \mathbf{Oz}$ or $\mathbf{Oz}[i]$ and let Φ be a ring endomorphism of $R[[s]]$ fixing s and inducing the identity modulo s . Then $\Phi(\Pi_R[[s]]) \subseteq \Pi_R[[s]]$ and $(\Phi - \text{id})(R[[s]]) \subseteq s\Pi_R[[s]]$; the same holds in every quotient $R[s]/(s^{N+1})$.*

Proof. Coefficientwise, $\Pi_R[[s]] = \bigcap_n nR[[s]]$, and a ring homomorphism preserves each $nR[[s]]$. For $\mathbf{Oz}$, Φ fixes the integers and s , hence, by continuity, $\mathbb{Z}[[s]]$; decompose each series into an integer series plus one in $\Pi[[s]]$. For $\mathbf{Oz}[i]$, Φ also fixes i : if $\Phi(i) = i + \sum_{n \geq 1} a_n s^n$ with a first nonzero a_n , the coefficient of s^n in $\Phi(i)^2 = -1$ gives $2ia_n = 0$; the same first-degree argument works in a truncation. So $\mathbb{Z}[i][[s]]$ is fixed and the decomposition works, and identity reduction supplies the factor s . $\square$

Lemma 69.2 (Invertibility from the first jet [19]). *Every ring endomorphism Φ of $R[[s]]$ fixing s and congruent to the identity modulo s is an automorphism.*

Proof. $\Phi - \text{id}$ is s -linear with image in $sR[[s]]$, so it maps $s^m R[[s]]$ into $s^{m+1} R[[s]]$, and $\sum_{n \geq 0} (\text{id} - \Phi)^n$ is defined coefficientwise and is a two-sided additive inverse; a bijective ring homomorphism has a ring-homomorphic inverse. No rational denominator is used. $\square$

Theorem 69.3 (Exponential–logarithm correspondence [19]). *The maps*

$$\exp \mathbf{L} = \text{id} + \sum_{n \geq 1} \frac{\mathbf{L}^n}{n!}, \quad \log \Phi = \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} (\Phi - \text{id})^n \quad (69.2)$$

are mutually inverse between the Lie series (69.1) and $\mathcal{G}(R)$. Every logarithm has exactly one expansion (69.1) with each δ_j a ring derivation of R , and there is no further integrability condition on (δ_j) .

Proof. *The divided operators exist.* For the fields division is ordinary. For the rings, every positive power of $\mathbf{L}$ takes values in $\Pi_R[[s]]$, and lemma 69.1 gives the same for positive powers of $\Phi - \text{id}$; these ideals are uniquely divisible coefficientwise, and terms of order above m do not affect the coefficient of s^m . *Exponentials are automorphisms.* Iterated Leibniz in each degree makes $\exp \mathbf{L}$ multiplicative; it fixes s , reduces to the identity, and has inverse $\exp(-\mathbf{L})$ by the binomial identity for powers of one operator. *Logarithms are derivations.* Φ extends to an s -fixing automorphism of $K[[s]]$ by $a/b \mapsto \Phi(a)/\Phi(b)$, the denominator having nonzero constant term, so one may work over $\mathbb{Q}$. Modulo s^{M+1} put $\Phi^{[X]} = \sum_{n=0}^M \binom{X}{n} (\Phi - \text{id})^n$; for every $r \in \mathbb{N}$ this is Φ^r , hence multiplicative, so each coefficient of $\Phi^{[X]}(fg) - \Phi^{[X]}(f)\Phi^{[X]}(g)$ is a polynomial in X over a characteristic-zero field vanishing at every $r \geq 0$, hence zero. Differentiating at $X = 0$, with $\frac{d}{dX} \binom{X}{n} \Big|_{X=0} = (-1)^{n-1}/n$ for $n \geq 1$, gives the Leibniz rule for $\log \Phi$ modulo s^{M+1} , for every M , hence in $K[[s]]$ and in $R[[s]]$, where the logarithm already takes its values. *Coefficients and inverses.* Writing $\log \Phi(a) = \sum_j s^j \delta_j(a)$ for $a \in R$, additivity and Leibniz in each degree make each δ_j a derivation, and s -linearity and continuity give (69.1). The identities $\log(\exp X) = X$ and $\exp(\log(1 + X)) = 1 + X$ hold after every finite truncation for a nilpotent operator X ; apply them in $K[[s]]$, all operators restricting to $R[[s]]$ by the first step. Uniqueness of coefficients is immediate. $\square$

Remark 69.4 (The role of classical theory [19]). The logarithm proof is the classical formal Lie argument, reproduced to make the denominators explicit; [12, 2] give much broader settings. The arithmetic point is that all positive divided terms stay in $\Pi_R[[s]]$, although $\frac{1}{2} \notin \mathbf{Oz}, \mathbf{Oz}[i]$.

Every Hasse–Schmidt sequence with $H_0 = \text{id}$ defines a homomorphism $R \rightarrow R[[s]]$, hence an element of $\mathcal{G}(R)$ by lemma 69.2, so theorem 69.3 describes *all* of them, iterative or not. If $\Phi|_R = \text{id} + sH_1 + s^2H_2 + s^3H_3 + \dots$, then $\delta_1 = H_1$, $\delta_2 = H_2 - \frac{1}{2}H_1^2$, $\delta_3 = H_3 - \frac{1}{2}(H_1H_2 + H_2H_1) + \frac{1}{6}H_1^3$, and conversely $H_1 = \delta_1$, $H_2 = \delta_2 + \frac{1}{2}\delta_1^2$, $H_3 = \delta_3 + \frac{1}{2}(\delta_1\delta_2 + \delta_2\delta_1) + \frac{1}{6}\delta_1^3$, compositions in their actual order (the operators need not commute) and every divided expression acting through the divisible ideal. An iterative one-parameter sequence is exactly the case $\log \Phi = s\delta$ [19].

Corollary 69.5 (Unique roots and stabilizers [19]). *$\mathcal{G}(R)$ is uniquely divisible: $\Phi^{1/n} = \exp((\log \Phi)/n)$ is its unique n th root, so it is torsion-free. For a set $A \subseteq R[[s]]$ the logarithm identifies the pointwise stabilizer of A with the Lie series L with $L(a) = 0$ for all $a \in A$; for constant series $a \in R$ this means $\delta_j(a) = 0$ for all j .*

Proof. $(\exp L)^n = \exp(nL)$, and the logarithm gives uniqueness. If Φ fixes a , every positive power of $\Phi - \text{id}$ kills a , so $\log \Phi$ does; conversely $L(a) = 0$ gives $\exp L(a) = a$. Compare coefficients for the last clause. $\square$

70 An unobstructed but nonsplit jet tower

Let $\mathcal{G}_N(R)$ be the group of automorphisms of $R[s]/(s^{N+1})$ fixing s and acting as the identity modulo s ; always $N \geq 1$.

Theorem 70.1 (Finite logarithms and lifting [19]). *Exponential and logarithm identify $\mathcal{G}_N(R)$ with $\{\sum_{j=1}^N s^j \delta_j : \delta_j \in \text{Der}(R)\}$ under the truncated Baker–Campbell–Hausdorff operation. Every finite jet lifts to $\mathcal{G}(R)$, and for $N \geq 2$ there is a central exact sequence*

$$0 \longrightarrow (\text{Der}(R), +) \xrightarrow{\delta \mapsto \text{id} + s^N \delta} \mathcal{G}_N(R) \longrightarrow \mathcal{G}_{N-1}(R) \longrightarrow 1. \quad (70.1)$$

Proof. The proof of theorem 69.3 works modulo s^{N+1} with finite series, and exponentiating the same finite polynomial in derivations on $R[[s]]$ gives a lift. A kernel element has logarithm $s^N \delta$ and exponential $\text{id} + s^N \delta$, since $2N \geq N + 1$; composition in the kernel is addition, and the commutator of an operator of order N with one of positive order has order at least $N + 1$, which gives centrality. $\square$

Surjectivity of all lifting maps must not be confused with splitting of the extensions, which fails here [19].

Lemma 70.2 (Euler directions and a Witt relation [19]). *For $\eta \in \mathbf{No}$ the derivation $D_\eta(\sum c_\gamma \omega^\gamma) = \sum c_\gamma q_\eta(\gamma) \omega^\gamma$ of theorem 17.6 is a derivation of each of the four rings, and for ordinary integers $m, n \geq 0$*

$$[\omega^m D_0, \omega^n D_0] = (n - m) \omega^{m+n} D_0, \quad [D_0, \omega D_0] = \omega D_0 \neq 0. \quad (70.2)$$

Proof. q_η is additive, the output support lies in the input support, each exponent of a product receives finitely many contributions, and $q_\eta(\alpha + \beta) = q_\eta(\alpha) + q_\eta(\beta)$ gives Leibniz; the exponent 0 is killed, so the arithmetic rings are preserved. These are the derivations of theorem 17.6 and `osq:prop:classder`. A ring element times a derivation is a derivation, and $\omega^m D_0$ maps $\mathbf{Oz}$ into $\omega^m \Pi \subseteq \Pi$. Since $D_0(\omega^n) = n\omega^n$, the second-derivative terms cancel in the commutator, which gives (70.2); and $(\omega D_0)(\omega) = \omega^2 \neq 0$. $\square$

Lemma 70.3 (Leading commutator [19]). *If $L = s^a\delta + O(s^{a+1})$ and $M = s^b\epsilon + O(s^{b+1})$ with $a, b \geq 1$, then $[\exp L, \exp M] = \text{id} + s^{a+b}[\delta, \epsilon] + O(s^{a+b+1})$, for the group commutator $ghg^{-1}h^{-1}$.*

Proof. Multiply the four operator series through degree $a + b$: terms involving only L or only M cancel, the first mixed terms are $LM - ML$, and every other mixed term has order at least $a + b + 1$. $\square$

Theorem 70.4 (Exact jet complexity [19]). *For each of the four rings $\mathcal{G}_N(R)$ has nilpotency class exactly N , and for every $N \geq 2$ the sequence (70.1) has no homomorphic section.*

Proof. With $\mathcal{G}^{(a)}$ the elements congruent to the identity modulo s^a , $[\mathcal{G}^{(a)}, \mathcal{G}^{(b)}] \subseteq \mathcal{G}^{(a+b)}$, so every commutator of length $N + 1$ is trivial in $\mathcal{G}_N(R)$. Commuting $\exp(s\omega D_0)$ successively $N - 1$ times with $\exp(sD_0)$ gives, by lemmas 70.2 and 70.3, $\text{id} + s^N\omega D_0$ modulo s^{N+1} up to the orientation of the commutators, nontrivial on ω ; for $N = 1$ the nonzero group $(\text{Der}(R), +)$ is abelian. For nonsplitting, $g = \exp(sD_0)$ and $h = \exp(s^{N-1}\omega D_0)$ commute in $\mathcal{G}_{N-1}(R)$, their commutator starting in degree N ; lifts to $\mathcal{G}_N(R)$ differ from these by central kernel elements, so their commutator is still $\text{id} + s^N\omega D_0 \neq \text{id}$ by lemma 70.3. A homomorphic section would send commuting elements to commuting elements, also for $N = 2$. $\square$

Every finite formal symmetry extends, but compatible extensions cannot be chosen by a group homomorphism at any nontrivial stage: absence of individual lifting obstructions is not absence of extension classes [19].

Remark 70.5 (Low-order identities [19]). For $L = s\delta + s^2\epsilon$, modulo s^4 , $\exp L = \text{id} + s\delta + s^2(\epsilon + \frac{\delta^2}{2}) + s^3(\frac{\delta\epsilon + \epsilon\delta}{2} + \frac{\delta^3}{6})$; for $\Phi = \text{id} + sA + s^2B + s^3C$, $\log \Phi = sA + s^2(B - \frac{A^2}{2}) + s^3(C - \frac{AB + BA}{2} + \frac{A^3}{3})$. These are operator identities and fail if $AB + BA$ is replaced by $2AB$. The order-two commutator is $\exp(s\delta)\exp(s\epsilon)\exp(-s\delta)\exp(-s\epsilon) = \text{id} + s^2[\delta, \epsilon]$ modulo s^3 ; for $\delta = X d/dX$ and $\epsilon = X^2 d/dX$ it sends X to $X + s^2X^2$ modulo s^3 , so two commuting first jets cannot be lifted to commuting second jets; the higher-order version replaces the second s by s^{N-1} . These are 19's independent checks (its Appendix B).

71 Residue sections and their common core

A *residue section* (19's "coefficient section") is a unital ring homomorphism $\varsigma : R \rightarrow R[[s]]$ with $\varsigma \bmod s = \text{id}_R$; its image is a coefficient copy onto which reduction modulo s is an isomorphism. The notion is marked: the residue ring is fixed, not allowed to change by an automorphism [19].

Theorem 71.1 (All residue sections [19]). *Restriction to R is a bijection between $\mathcal{G}(R)$ and the residue sections; so $\mathcal{G}(R)$ acts simply transitively on them, and each is given by a unique Lie series $\sum_j s^j \delta_j$ through theorem 69.3. No smoothness assumption is needed.*

Proof. A section ς extends to $\Phi_\varsigma(\sum a_n s^n) = \sum \varsigma(a_n) s^n$, each coefficient a finite sum; this is a ring endomorphism with identity reduction, hence an automorphism (lemma 69.2), and the only continuous s -fixing extension of ς . Conversely every element of $\mathcal{G}(R)$ restricts to a section, and $\Phi_{\varsigma_2} \Phi_{\varsigma_1}^{-1}$ is the unique automorphism carrying ς_1 to ς_2 . $\square$

Example 71.2 (Directions beyond the Euler family [19]). Let $b \in \mathbb{R}$ be transcendental over $\mathbb{Q}$. A derivation $d : \mathbb{R} \rightarrow \mathbb{R}$ with $d(b) = 1$ exists: take a transcendence basis containing b , differentiate that variable in its rational function field, and extend through the algebraic extension, $d(a) = -(dp)(a)/p'(a)$ for a algebraic with separable polynomial p (over $\mathbb{C}$ use a transcendence basis over $\overline{\mathbb{Q}}$). Its coefficientwise extension $d_*(\sum c_\gamma \omega^\gamma) = \sum d(c_\gamma) \omega^\gamma$ is a

derivation of **No** (output support inside input support, finite sums in products), kills the integer constant term and preserves **Oz**; so ωd_* is a derivation of **Oz**, with $(\omega d_*)(b) = \omega$ and $(\omega d_*)(\omega) = 0$ on the ambient field. It is not $\mathbb{R}$ -linear and not an Euler direction, yet [theorem 67.1](#) integrates it; it is not claimed to exhaust the non-Euler directions. For ωD_0 one orbit is simple: $(\omega D_0)^n(\omega) = n! \omega^{n+1}$ and $\exp(s \omega D_0)(\omega) = \omega/(1 - \omega s)$, whereas $\exp(s D_0)(\omega) = \omega e^s$ ([example 17.8](#)). A rational formula for one orbit is not a finite-type formula for the action on every element.

The derivations d are those of [part III](#), where they drive the Taylor sections $\rho_{d,\delta}$; they are choice-dependent and not computable ([section B](#)).

Put $\mathbf{C}_R = \mathbb{Z}, \mathbb{Z}[i], \mathbb{R}_{\text{alg}}, \overline{\mathbb{Q}}$ for $R = \mathbf{Oz}, \mathbf{Oz}[i], \mathbf{No}, \mathbf{No}(i)$, where $\mathbb{R}_{\text{alg}}$ is the field of real algebraic numbers and $\overline{\mathbb{Q}} \subseteq \mathbb{C}$ its algebraic closure.

Lemma 71.3 (Constants of all derivations [19]). *The elements of R killed by every derivation of R are exactly those of $\mathbf{C}_R$.*

Proof. Every derivation kills the elements algebraic over $\mathbb{Q}$ ([proposition 68.6](#)), which covers $\mathbf{C}_R$. If x has a nonzero exponent γ , choose η in the normal form of γ ; then $q_\eta(\gamma) \neq 0$ and $D_\eta(x)$ has a nonzero coefficient at γ . So a common constant is an ordinary real or complex number; in the rings these are $\mathbb{Z}$ or $\mathbb{Z}[i]$, and in the fields a coefficient transcendental over $\mathbb{Q}$ is moved by some d_* as in [example 71.2](#). $\square$

Theorem 71.4 (Formal invariants and the intersection of all coefficient copies [19]). *For each of the four rings, $\text{Fix}(\mathcal{G}(R), R[[s]]) = \mathbf{C}_R[[s]]$ and $\bigcap_{\varsigma} \varsigma(R) = \mathbf{C}_R$, the intersection over all residue sections, with $\mathbf{C}_R$ embedded as constant series.*

Proof. Every derivation kills $\mathbf{C}_R$, so every Lie series and its exponential fix $\mathbf{C}_R[[s]]$. Conversely, if $f = \sum a_n s^n$ is fixed by every $\exp(s\delta)$, the degree-one coefficient gives $\delta(a_0) = 0$, and inductively, once $\delta(a_0) = \dots = \delta(a_{m-1}) = 0$, the coefficient of s^{m+1} in $(\exp(s\delta) - \text{id})f$ is $\delta(a_m)$; so every coefficient lies in $\mathbf{C}_R$ by [lemma 71.3](#). The intersection lies in the standard copy R ; for $x \in R$ in every $\Phi(R)$, $\Phi^{-1}(x) \in R$ is a constant with residue x , so it is x , and x is fixed by every Φ . $\square$

All derivations are needed in the field cases: the Euler family fixes every ordinary coefficient, and the coefficient derivations cut the fixed fields down to their algebraic parts; for the rings the Euler family already reduces to $\mathbb{Z}$ or $\mathbb{Z}[i]$ [19]. Compare [theorem 7.2](#), where the automorphisms of **No** preserving **Oz** fix $\mathbb{R}$: the formal automorphisms here need not fix the reals.

72 The quartic formula before and after completion

Lemma 72.1 (A genus-one rational-function test [19]; second route). *Let F have characteristic zero and let $x, y \in F(U_1, \dots, U_n)$ satisfy $y^2 = x^4 + 1$. Then $x, y \in F$.*

Proof (19's route). For $n = 1$: over an algebraic closure of F , $X^4 + 1$ has four distinct roots, and the smooth projective curve with function field $F(X, Y)$, $Y^2 = X^4 + 1$, is a double cover of the X -line branched exactly there (the two points over infinity are unramified); Riemann–Hurwitz gives $2g - 2 = 2(-2) + 4 = 0$, so $g = 1$. A nonconstant solution would give a nonconstant morphism from $\mathbb{P}^1$ to this curve, the map extending over the smooth projective source, and Riemann–Hurwitz would give $-2 = \deg f \cdot 0 + \deg \text{Ram}(f) \geq 0$ [27]. So x is constant, y is algebraic over F , and the algebraic constants of $F(U)$ are F . For n variables treat U_1 as the variable over $F(U_2, \dots, U_n)$ and repeat. $\square$

This is [theorem 45.3](#) for $G = \mathbb{Z}^n$; 16's proof, by Mason–Stothers ([lemmas 45.1](#) and [45.2](#)) and specialization, works for every torsion-free G . The merge keeps 19's genus-one argument as a second route.

Corollary 72.2 (Existential coefficient recovery in intermediate rings [16, 19]; [merge]). *Let A be a characteristic-zero domain whose fraction field K is real closed or algebraically closed, G a torsion-free abelian group, and B a ring with $A \subseteq B \subseteq K(U^G)$ and $B \cap K = A$. Then Q of (67.1) defines exactly A in B . This covers every finite polynomial, Laurent or mixed polynomial–Laurent extension of A (19's Theorem 8.2) and every monoid algebra $A[U^M]$ ([theorem 45.7](#)).*

Proof. For $x \in A$, $x^4 + 1$ has a square root in K (positive in a real closed field; automatic in an algebraically closed one), written a/b with $a, b \in A$, $b \neq 0$. Conversely witnesses $a, b \in B$, $b \neq 0$, give $(a/b)^2 = x^4 + 1$ in $K(U^G)$, so $x \in K$ by [theorem 45.3](#) (or, for $G = \mathbb{Z}^n$, by [lemma 72.1](#)), and $x \in B \cap K = A$. $\square$

Remark 72.3 (The formula for the four rings [19]). For $R \in \{\mathbf{Oz}, \mathbf{Oz}[i], \mathbf{No}, \mathbf{No}(i)\}$, $\text{Frac } R$ is real or algebraically closed (`odg:thm:fractions`), so Q defines the coefficient copy in every finite polynomial or Laurent ring over R ; the class-field case of [lemma 72.1](#) reduces to a set-sized field containing the finitely many coefficients of a proposed solution. The formula has two existential witnesses, no parameters and the inequation $b \neq 0$, and after clearing denominators its equation has total degree six, not four. The inequation must not be suppressed ($a = b = 0$ would witness every element): this is an *existential*, not a positive-existential or single-equation, definition ([question 52.2](#)) [19].

Proposition 72.4 (Definable polynomial evaluation [19]). *In $A[U]$ with A as in [corollary 72.2](#), after naming U , the graph of evaluation at zero into the embedded coefficient ring is defined by $Q(c) \wedge \exists q (f - c = Uq)$. For each fixed n the first $n+1$ coefficients are defined by $Q(c_0), \dots, Q(c_n)$ and $f - \sum_{j \leq n} c_j U^j = U^{n+1}q$.*

Proof. The formula restricts candidates to constants, and comparison of polynomial coefficients gives existence and uniqueness; no variable ranges over exponents or degrees. $\square$

Proposition 72.5 (Curve parameter fields [19]). *Let K be a set-sized real closed or algebraically closed field of characteristic zero and $L = K(X)$ the function field of a smooth projective geometrically integral curve of genus g . For an integer $d \geq g + 2$, $\exists y (y^2 = x^{2d} + 1)$ defines exactly K in L .*

Proof. $Y^2 = X^{2d} + 1$ has genus $d - 1$ by the same double-cover computation, and a nonconstant solution would give a nonconstant morphism from X to a curve of genus $d - 1 > g$, forbidden by Riemann–Hurwitz in characteristic zero. Every $x \in K$ has a square root of $x^{2d} + 1$ in K , and geometric integrality makes K the algebraic constants of L . $\square$

The genus threshold is sufficient, not an optimal complexity bound: at the boundary $d = g + 1$ one may take X to be the completed curve $Y^2 = X^{2d} + 1$ itself, whose nonconstant coordinate is a counterexample [19]. The quadratic test fails already in a rational function field, $(\frac{U+U^{-1}}{2})^2 - (\frac{U-U^{-1}}{2})^2 = 1$, which is [remark 45.6](#); in characteristic 2 the quartic test is satisfied by $y = x^2 + 1$, as in [example 51.2](#). The characteristic-zero and positive-genus hypotheses are substantive [19].

Lemma 72.6 (Formal square roots of units [19]). *Let K have characteristic zero and $f = f_0 + f_1 s + \dots \in K[[s]]$ with $f_0 \neq 0$. For each c with $c^2 = f_0$ there is exactly one square root of f with constant term c .*

Proof. Write it as $\sum y_n s^n$ with $y_0 = c$ and solve $y_n = (f_n - \sum_{j=1}^{n-1} y_j y_{n-j})/(2c)$ for $n \geq 1$; this is a formal recursion, not a convergence statement. $\square$

Theorem 72.7 (The polynomial test becomes universal [19]). *For $R = \mathbf{Oz}$, $\mathbf{Oz}[i]$ or $\mathbf{No}$, every $f \in R[[s]]$ satisfies $Q(f)$.*

Proof. Let $K = \text{Frac } R$ and f_0 the constant coefficient. In the real cases $f_0^4 + 1 > 0$. In the Gaussian case $f_0^4 + 1 \neq 0$ as well: the roots of $X^4 + 1$ in $\mathbf{No}(i)$ are its four ordinary complex roots, and none lies in $\mathbf{Oz}[i]$, whose ordinary elements are $\mathbb{Z}[i]$, since a Gaussian integer with fourth power -1 would have norm one and be one of $\pm 1, \pm i$, whose fourth powers are 1. So $f^4 + 1$ has a square root $y \in K[[s]]$ by lemma 72.6. For $R = \mathbf{No}$ take $a = y$, $b = 1$. For the rings, `odg:thm:fractions` applied to the countable set of coefficients of y gives $m \in \Pi_R \setminus \{0\}$ with $my \in \Pi_R[[s]]$; then $a = my$ and $b = m$ satisfy $a^2 = (f^4 + 1)b^2$, $b \neq 0$. $\square$

This is stronger than one nonconstant solution: a formula that recognizes polynomial coefficients exactly becomes true of the whole omnific completion. The proof needs formal square roots and a *single* denominator for their infinitely many coefficients [19]. It strengthens example 51.1, where one nonconstant solution in a Hahn field breaks 16's detector, and bears on question 52.1: the s -adic completion is the extreme case, in which no element is excluded; it does not answer that question [merge].

Proposition 72.8 (The surcomplex-field exception is an order parity [19]). *For $f \in \mathbf{No}(i)[[s]]$, $Q(f)$ holds exactly when $f^4 + 1 = 0$ or $\text{ord}_s(f^4 + 1)$ is even; in particular whenever f_0 is not a root of $X^4 + 1$.*

Proof. $\text{Frac}(\mathbf{No}(i)[[s]]) = \mathbf{No}(i)((s))$, and a nonzero Laurent series over an algebraically closed characteristic-zero field is a square exactly when its order is even (factor out the leading power and apply lemma 72.6); if $f^4 + 1 = 0$ take $a = 0$, $b = 1$. If f_0 is not a quartic root, $f^4 + 1$ has order 0. If $f_0 = \xi$ is one and $f \neq \xi$, then $f^4 + 1 = (f - \xi)g$ with $g_0 = 4\xi^3 \neq 0$, so the relevant order is $\text{ord}_s(f - \xi)$. $\square$

In a finite jet ring $R[s]/(s^{N+1})$, $N \geq 1$, the formula is universal for a simpler reason: $b = s^N \neq 0$ has $b^2 = 0$, so $a = 0$ is a witness for every x . This nilpotent phenomenon does not explain theorem 72.7, whose ambient ring is a domain [19].

73 No set of parameters defines the canonical coefficient copy

The following obstruction applies to every first-order formula in the stated language, not only to Q [19].

Lemma 73.1 (A fresh Euler direction [19]). *Let A be a set of elements of $R[[s]]$, R any of the four rings. There is $\eta \in \mathbf{No}$ such that D_η kills every coefficient of every member of A , while $D_\eta(w) = w \neq 0$ for $w = \omega^{\omega^\eta} \in \mathbf{Oz}$.*

Proof. Collect all coefficients of all members of A (a set, by a product with $\mathbb{N}$), then all outer exponents of their normal forms, then all inner exponents of those exponents; let B be the resulting set and choose $\eta \notin B$, for instance above B . Then $q_\eta(\gamma) = 0$ for every collected outer exponent, so D_η kills every coefficient, while $q_\eta(\omega^\eta) = 1$; the outer exponent ω^η is positive, so $w \in \mathbf{Oz}$. $\square$

This is the fresh-scale choice of lemma 7.1, used in theorem 9.1, applied to a formal completion [merge].

Theorem 73.2 (Parameterwise nondefinability [19]). *For every set $A \subseteq R[[s]]$ some $\Phi \in \mathcal{G}(R)$ fixes A pointwise and does not preserve the canonical copy R . So R is not definable in the pure ring language with parameters from any set, even after naming s and every ideal (s^n) .*

Proof. Take D_η from lemma 73.1 and $\Phi = \exp(sD_\eta)$. It kills every coefficient of every member of A , so it fixes A ; it fixes s , acts as the identity on the residue and preserves every (s^n) . And $\Phi(w) = w \sum_n s^n/n! = w + ws + \frac{w}{2}s^2 + \dots$, all coefficients in R ($w/n! \in \Pi$), with a nonzero coefficient of s , so $\Phi(w) \notin R$. A definable class is invariant under every automorphism fixing its parameters and the symbols of the language. $\square$

In the real cases the witness also preserves the lexicographic order of formal series, given by the sign of the first nonzero coefficient, which no member of $\mathcal{G}(R)$ changes; in the complex cases the real-coefficient Euler witnesses commute with coefficientwise conjugation. Adding these structures does not remove the obstruction; naming an actual residue section would remove it by definition and is excluded [19].

Corollary 73.3 (An interpretable quotient without its canonical lift [19]). *With s named, $R[[s]]/(s)$ is an interpretable ring isomorphic to R , but its canonical constant-series section is not definable with any set of parameters, and neither is the embedded evaluation $f \mapsto f_0 \in R \subseteq R[[s]]$.*

Proof. $f \sim g \iff \exists h (f - g = sh)$ and the induced operations interpret the quotient. A definable canonical section, or the embedded evaluation, would have definable image R , against theorem 73.2. $\square$

Corollary 73.4 (No definable residue section [19]). *No residue section has definable image in the pure ring language with s named and any set of parameters.*

Proof. Its image is $\Phi(R)$ for some $\Phi \in \mathcal{G}(R)$ (theorem 71.1); pulling a definition back by Φ would define R with the inverse images of the parameters, again a set. $\square$

Completion does not destroy the residue ring; it destroys a distinguished lift of it into the completed ring, and theorems 71.1 and 71.4 describe the competing lifts and their small common core [19]. Polynomial definability, on the other hand, does not imply uniform recovery in arbitrary algebraic function fields (the boundary example after proposition 72.5).

74 Compatibility with algebraic-parameter rigidity

The formal symmetries do not contradict part II; 19 gives a short argument for the four rings instead of importing the root-covered theory wholesale [19]. Its first lemma, that an element with n th roots for unboundedly many n in a finitely generated extension L/F is algebraic over F (an Eisenstein argument), is lemma 14.1.

Lemma 74.1 (Closed source fields have algebraic image [19]). *If K is real closed or algebraically closed and $\sigma : K \hookrightarrow L$ is a field embedding into a finitely generated extension L/F , every element of $\sigma(K)$ is algebraic over F ; no F -linearity is assumed.*

Proof. In the algebraically closed case every nonzero element has roots of all orders, so lemma 14.1 applies. In the real closed case $a^2 > 0$ has roots of all orders for $a \neq 0$, so $\sigma(a)^2$ is algebraic over F , and so is $\sigma(a)$. $\square$

Theorem 74.2 (No nonconstant pointed finite-type family [19]). *Let R be one of the four rings, $K = \text{Frac } R$, B a finitely generated integral K -algebra and $\varepsilon : B \rightarrow K$ a K -algebra homomorphism. If $\psi : R \rightarrow B$ is a ring homomorphism with $\varepsilon\psi(x) = x$ for all x , then $\psi(x) = x$ for all x .*

Proof. ψ is injective and extends to $K \hookrightarrow \text{Frac } B$, so each $\psi(x)$ is algebraic over K (lemma 74.1). The elements of B algebraic over K form a field C (the minimal polynomial of a nonzero algebraic a expresses a^{-1} as a polynomial in a). $\varepsilon|_C$ is an injective field map fixing K , and for $a \in C$, $a - \varepsilon(a) \in C$ maps to 0, so $a \in K$. Hence $C = K$ and $\psi(x) = \varepsilon\psi(x) = x$. $\square$

The finite-generation and degree arguments use finitely many generators even when K is a class field and may be read as finite-witness field arguments, without a proper-class scheme [19]. Theorem 14.5 proves the pointed statement for every set-sized root-covered domain; 19 calls its theorem background to it, and its contribution is the class-size reading for these four rings, through real or algebraic closedness of K .

Corollary 74.3 (Every nontrivial formal section escapes finite type [19]). *The image of a nonstandard residue section $\varsigma : R \rightarrow R[[s]]$ lies in no finitely generated K -subalgebra of $K[[s]]$. In particular every nonzero derivation yields a formal flow that is not one pointed integral finite-type algebraic parameter family on all of R .*

Proof. Such a subalgebra is an integral K -algebra with evaluation at $s = 0$ into K ; theorem 74.2 forces ς to be standard. For a nonzero derivation the first coefficient already makes the section nonstandard. $\square$

The corollary does not say that every individual orbit is transcendental (the rational orbit of ω under ωD_0 in example 71.2 is a warning), nor that no formal parameter can ever be evaluated in a Hahn field; the issue is one finite-type algebra containing the images of *every* input [19]. It is the analogue for all derivations of theorems 17.3 and 17.5 [merge].

75 Scope, stale statements and questions of source 19

75.1 What the proofs require

Result	Decisive input and boundary [19]
Automatic integration	$\Pi_R = \bigcap nR$, unique divisibility, and derivations killing $\mathbb{Z}$ or $\mathbb{Z}[i]$; no Hahn strongness.
Full logarithm	$\Phi - \text{id}$ has image in $s\Pi_R[[s]]$, so positive divided terms stay integral; the parameter is fixed.
Jet lifting	Truncated logarithms have derivation coefficients, exponentiated in the complete ring; not smoothness of a finite-type scheme.
Nonsplit jet tower	$[D_0, \omega D_0] = \omega D_0$ and the leading commutator; individual lifts always exist.
Residue sections	s -adic completeness and identity reduction, not Noetherianity or Cohen structure theory; only marked sections are classified.
Polynomial formula	Real or algebraic closedness of the fraction field and the genus-one obstruction; the formula has an inequation.

Result	Decisive input and boundary [19]
Universal completed test	A square root with nonzero constant term and one monomial clearing its countably many coefficients, a full-class support argument.
Parameterwise nondefinability	A set of parameters has a set of inner exponents; a fresh Euler coordinate fixes them and moves a new coefficient.
Nonalgebraization	Finite type, an integral parameter algebra and a pointed specialization; disconnected or nonreduced alternatives are not included.

19's dependency map: (1) the normal-form decomposition, $\Pi_R = \bigcap nR$ and unique division; (2) $\delta(\Pi_R) \subseteq \Pi_R$ from $\delta(ny) = n\delta(y)$, and $\delta(R) \subseteq \Pi_R$ from the arithmetic constants; (3) $\delta^n/n!$, iterated Leibniz and iterative uniqueness by torsion-freeness; (4) the formal difference lemma and the logarithm identity modulo every s^M , giving the correspondence and jet lifting; (5) the Euler witnesses, $[D_0, \omega D_0] = \omega D_0$, exact nilpotency and nonsplitting; (6) the complete-ring section argument and the common constants; (7) separately, the genus-one lemma and the fraction-field criterion for $\mathbf{Q}$; (8) uniform denominators with formal square roots, and fresh inner exponents for nondefinability. The three branches (divisible-ideal formal algebra, positive-genus recovery, fresh-support nondefinability) are independent until combined, and no automorphism-classification theorem of the repository is used, so a correction to one would not affect them [19]. Its limitations are listed in [section B](#).

75.2 Status of this report's questions, and stale statements

- *Answered.* [Question 19.2](#), by [theorem 67.1](#) and [corollary 68.5](#); status notes are added after the question, in [section 19](#) and in [section 52.2](#). 02's third non-claim (no classification of all derivations) stays true ([section B](#)).
- *Merged.* 19's question on definitional complexity duplicates [question 52.2](#); its question on reduced parameter spaces refines [question 19.3](#); its question on Hahn evaluation is merged with 17's into [question 59.6](#); its formalization question into [question 59.10](#). 19 is added to their sources.
- *Partly answered by another source.* 19's question on orbit-field transcendence ([question 75.2](#)) is answered for D_0 by [part VII](#).
- *Stale against the tree.* None of 19's mathematical statements is stale; its formula $\mathbf{Q}$ was new at its pin [889b87c](#) and is 16's at the merge ([section 67.1](#)), so it is printed once in [corollary 72.2](#). Its source audit notes truncated connector reads, and that packaging happened after midnight UTC on 24 September although the manuscript is dated 23 September.
- *Checked true.* The pin and its content; 19's reading of [question 19.2](#) and its credits; the 61-report catalogue at the pin; its suite, rerun for the merge ([section C](#)). No mathematical error was found in 19.

75.3 Questions of source 19

19 poses eleven questions (its 13.1–13.11), proposed as continuations, not as established named problems. Seven are new; 13.2 is merged into [question 59.6](#), 13.4 into [question 52.2](#), 13.9 into

question 19.3 and 13.11 into question 59.10.

Question 75.1 (A structural description of all derivations [19]). Can every derivation of $\mathbf{Oz}$ be decomposed into coefficient derivations and support-controlled exponent directions? If not, construct and characterize a derivation that escapes such a decomposition.

Automatic integrability removes one obstacle but imposes neither $\mathbb{R}$ -linearity nor strongness. A first step is to restrict the extended derivation to the ordinary real coefficients and subtract explicitly described coefficient motions; a strongness theorem for the remainder would be substantially stronger than the results here [19]. This is what remains of question 19.2.

Question 75.2 (Orbit-field transcendence [19]). For a nonzero derivation δ , what are the possible transcendence degrees of the field generated by K and $\exp(s\delta)(R)$ inside $K((s))$? Can a nontrivial full omnific section have an orbit field of finite transcendence degree but infinite algebraic degree over a rational function field?

Corollary 74.3 excludes one common finitely generated algebra but not this field-theoretic question; the Euler family already has many independent exponentials (corollary 17.9) [19]. *Status* (batch 33): for $\delta = D_0$ it is answered by part VII: the orbit field has independent families of every set cardinality and no set-sized transcendence basis (theorem 53.2); for every Euler derivation the full image descends to $F/F^{D_\times}$ (corollary 58.7). Other derivations, and the second clause, stay open.

Question 75.3 (Additional structure for a residue section [19]). What is the weakest natural expansion of the formal ring that admits a definable residue section, without naming a section outright?

Corollary 73.4 rules out every section in the pure ring language with s and any set of parameters, so an expansion must add more than constants and the s -adic filtration; differential operators, truncations and an externally specified simplicity relation are candidates whose strength remains to be analyzed [19].

Question 75.4 (Several formal parameters and the jet Lie algebra [19]). Develop the analogue of theorem 70.4 for $R[[s_1, \dots, s_r]]$ with the total-degree filtration: determine the extension classes, centers and lower central series of its jet groups and their dependence on $\text{Der}(R)$.

Integrality extends directly to finitely many parameters; the group invariants are not computed, and the one-parameter witness gives a lower bound by restriction [19]. Compare question 59.4, on orbit fields of several Euler directions.

Question 75.5 (Centralizers and formal conjugacy [19]). Classify formal automorphisms with prescribed logarithmic leading term up to conjugation in $\mathcal{G}(R)$. Which resonances of $\text{ad}(\delta_1)$ prevent the elimination of higher logarithmic coefficients?

The logarithm formulas turn the first stages into linear equations in derivations; a classification could show which residue sections become equivalent under additional differential data [19].

Question 75.6 (Fixed exponent groups and bounded parameters [19]). For set-sized Hahn arithmetic rings, characterize exactly when one monomial clears the coefficients of a given formal series, and which parameter sets admit a fresh-coordinate nondefinability witness.

Integration and the formal Lie correspondence hold for arbitrary ordered exponent groups (proposition 68.6); the common denominator and the all-set-parameter argument need separate cofinality and support hypotheses, and conflating the two would erase the set-class distinction [19].

Question 75.7 (Coefficient intersections in other completions [19]). Does an analogue of [theorem 71.4](#) hold for Hahn-parameter completions, Puiseux-parameter completions or iterated Laurent series? Which completions keep a small ordinary arithmetic intersection of all coefficient copies?

The s -adic order makes each operator coefficient a finite calculation; a generalized parameter support can destroy that finiteness, so replacing s by a monomial is not a proof [19].

Part X

Universal symmetries and exact difference equations

This part records manuscript 20, the addition of batch 34. It proves again, independently, the universality theorem of [part VIII](#): the set groups acting faithfully on **No**, or on **Oz**, are exactly the left-orderable ones, and they can act so that every nonidentity element fixes exactly $\mathbb{R}$ ([theorem 76.1](#)). Its new material is of two kinds. On the symmetry side it adds that the orbit of ω is algebraically independent, and a relative action fixing a given set of parameters whose common fixed field is a proper-class two-level field over which a whole interval of double exponentials is independent ([theorem 77.5](#)). On the equation side it solves difference equations for these automorphisms: additive equations with a weighted Green operator ([theorem 78.5](#)), all constant-coefficient operators ([theorem 79.1](#)), multiplicative equations globally and relative to parameters ([theorems 79.5](#) and [79.7](#)), every scalar first-order equation ([theorem 80.1](#)), and cocycles of groups with a central element or free groups ([theorem 80.2](#) and [proposition 80.5](#)). The difference-equation core was already in the single-dilation report `docs/surcomplex/single-dilation-hahn-support (dsup:)`, for monomial automorphisms of set-sized Hahn fields; the correspondence is recorded in [section 76.2](#). Its labels carry the sub-prefix `opa:us:`.

76 Source 20, what it re-derives, and its notation

76.1 The manuscript and its pin

Manuscript 20 is *Universal Symmetries and Exact Difference Equations over the Surreals: left-orderable groups, omnific stabilizers, fixed fields, and surcomplex descent*, dated 23 September 2026, 25 pages (title page, contents, twelve questions, two appendices and bibliography included). It is manuscript 01 of batch 34, delivered as the archive `Surreal_Universal_Symmetries_and_Difference_Equations` (inner directory `surreal_universal_symmetries`) in commit `bbe23a3` and placed as an addition in commit `a7a435f`; it is numbered 20 here. Its checks, their recorded run, its `Makefile`, its build record and its source and proof audit are shipped as `code/20-universal-symmetries-verify.py`, `data/20-universal-symmetries-verification.json`, `code/20-universal-symmetries-Makefile`, `data/20-universal-symmetries-BUILD_REPORT.json` and `20-universal-symmetries-SOURCE_AND_PROOF_AUDIT.md`; its `article.tex`, PDF and README are not shipped. 20's numbers are those of its own PDF.

It pins `e93a06d` (tree `42aec05`). That commit contains [parts I](#) to [IV](#) of this report (the batch-31 text with the reciprocal notes `ab18444`) and the single-dilation, autonomous-dilation and surcomplex-automorphism reports, but neither the write of [parts V](#) and [VI](#) (`c315ac9`) nor the placement `aa9c891` of sources 17–19. So 20 could not see [part VIII](#): its universality theorem

is an independent re-derivation. It read the root guide, the catalogue and five report guides (this report's, the independent-copies, quotient, surcomplex-automorphism and differential-equations reports), some truncated, and the first hundred lines of the Diophantine article, not this article and not the single-dilation report (its audit, Section 1). Its account of what it read is accurate: it credits the two Hahn lifts to [15] (Definitions 2.5 and 2.8, Construction 3.7 and Proposition 3.9, the first and second lifts) and [18], the monomial maps and arithmetic preservation to this report, and the centralizer factorization to the surcomplex automorphism report (`saut:thm:axis`). Its novelty statement is incomplete against its own pin, which already contained the single-dilation report, and it is stale at the merge ([section 82.2](#)).

76.2 What it re-derives, and what is printed once

20	Printed as	Comparison
Thm. 1.1 (i)–(iii) and the fixed field $\mathbb{R}$	corollaries 60.3 and 63.2 and proposition 65.1	Same theorem; 20 is an independent second derivation. The algebraic independence of the orbit of ω is new (proposition 77.3).
Lem. 2.3	<code>odg:thm:fractions</code>	Same; cited.
Lem. 3.2	lemma 63.1	Same lexicographic argument (20 chooses a witness for each pair, 18 for each nonidentity element).
Def. 3.3 ($\eta_{\mathbf{On}}$), Thm. 3.4, Rem. 3.5	definition 62.1 , theorem 62.4 , and lemma 62.2	Same theorem; 20's cut filling (a whole regular copy of G at <i>every</i> cut) differs from 18's (fibers over cut orbits) and is kept as a second route (proposition 77.1).
Lems. 4.1, 4.2	proposition 61.2 and lemma 27.1 ; (5.1)	Same lifts; 20's complex and conjugation clauses are printed in lemma 77.2 .
Lem. 4.3, Thm. 4.4	lemma 61.4 , theorem 61.5 , and proposition 65.1	Same; 20's proof by forward and backward iterates is recorded after lemma 77.2 .
Cor. 4.5	theorem 60.1 with $S = \emptyset$; proposition 65.1	Same.
Prop. 4.6 ($s = 1$)	example 64.1	Same map; the family, its group law and the choice-free subgroup actions are new (proposition 77.4).
Thm. 5.1	corollary 60.2 (existence)	Different fixed field: 20's is a proper class, 18's is set sized and minimal among the K_S containing the parameters; both printed and compared (remark 77.6). The independence clauses are new.
Lem. 6.2	<code>dsup:lem:orbit</code>	Same lemma, with the chambers swapped between growth and valuation notation; 20 states it for ordered classes without addition, which lemma 79.4 needs.

20	Printed as	Comparison
Thm. 7.1	<code>dsup:thm:resolvent</code> with $\chi = 1$	Identical formula (remark 78.6); dsup's is for set-sized Γ and transfers to classes by its set localization; 20's proof is direct for classes and for Oz -preserving double lifts.
Rem. 7.2	dsup's remark "Not a residue homomorphism"	Same counterexample.
Thm. 8.1, Cor. 8.2	<code>dsup:cor:poly</code> , <code>dsup:cor:fixed</code>	dsup's needs a polynomial split over the coefficient field; 20 treats real non-split polynomials by conjugation.
Thm. 9.2	<code>dsup:thm:multsingle</code> , <code>dsup:thm:surreal</code> (2)	The first specializes dsup's criterion; the second is the same statement for rational dilations.
Thm. 9.4	<code>dsup:thm:multsingle</code>	A specialization, made explicit by lemma 79.4 and the inner projection.
Cor. 11.3	<code>dsup:cor:simultaneous</code>	The global case; 20's one-operator proof is kept as a second route.
Thm. 11.2, H^1	<code>dsup:thm:dilationcohom</code>	dsup has $\mathbb{Z}^d$ acting by rational dilations; 20 has every group with a central element.
Prop. 12.1	lemma 65.2 ; <code>saut:thm:axis</code>	Same; credited by 20.
Thm. 12.2	theorem 60.4	Same classification, same route; 20's finite-subgroup clause is weaker than <code>saut:thm:torsion</code> .
Rem. after Thm. 10.1	<code>adr:sr:cor:free</code> and the non-claims of the autonomous-dilation report	The same remark ($x \mapsto \sigma x - x$ misses 1) for its dilations.
Questions	section 82.3	Eight new, four merged.
All else	This part	New.

Genuinely new in 20, beyond [part VIII](#) and the single-dilation report: the algebraic independence of the orbit of ω ([proposition 77.3](#)) and of $\{\omega^{\omega^a} : a \in J\}$ over the relative fixed field, and that proper-class fixed field itself ([theorem 77.5](#)); the translation family ([proposition 77.4](#)); the inner difference lemma ([lemma 79.4](#)) and with it the explicit image $\ker \varpi_B^{\text{in}}$ in the relative multiplicative criterion ([theorem 79.7](#)); the first-order criterion ([theorem 80.1](#)); the operator field ([corollary 79.3](#)); cocycle normal forms for every group with a central element ([theorem 80.2](#), [remark 80.4](#)) and for free groups ([proposition 80.5](#)); the choice-free set model and the η_κ -orders ([propositions 81.1](#) and [81.2](#)). 20 is an **Oz**-preserving double lift, not a rational dilation: dsup's S_q multiplies exponents by q , while the maps here relabel inner indices. The single-dilation proofs use only a coefficient-fixing monomial automorphism with an ordered additive exponent map, and transfer to them [merge].

76.3 Conventions and notation of this part

This part uses the growth convention of [parts II](#) and [VIII](#): $x = \sum_{\gamma} c_{\gamma} \omega^{\gamma}$ with reverse well-ordered set support; in the t -convention of [part I](#) $\omega^{\gamma} = t^{-\gamma}$. ω^{γ} is always Conway's monomial, not $\exp(\gamma \log \omega)$; a positive exponent, even an infinitesimal one, gives an infinite monomial. Let $k = \mathbb{R}$ or $\mathbb{C}$ and write $\mathbf{No}_{\mathbb{R}} = \mathbf{No}$, $\mathbf{No}_{\mathbb{C}} = \mathbf{No}(i)$ (the surcomplex field, 20's $\mathbf{No}[i]$). Put

$$\mathbf{No}_k = \Pi_k \oplus k \oplus \mathbf{m}_k, \quad \mathbf{V}_k = \ker \text{ct} = \Pi_k \oplus \mathbf{m}_k, \quad (76.1)$$

where Π_k and $\mathbf{m}_k$ are the series supported in $\mathbf{No}^{>0}$ and $\mathbf{No}^{<0}$ ($\Pi_{\mathbb{R}} = \Pi$, the purely infinite part, not necessarily positive; $\mathbf{m}_k$ the infinitesimals), and ct is the coefficient of ω^0 . For $x \neq 0$, $\text{lc}(x)$ is the coefficient of the largest exponent $\gamma(x)$, and $x = \text{lc}(x) \omega^{\gamma(x)} (1 + \varepsilon_x)$ with $\varepsilon_x \in \mathbf{m}_k$ uniquely; $\text{lc}(xy) = \text{lc}(x) \text{lc}(y)$. $\mathbf{Oz} = \mathbb{Z} \oplus \Pi$ and $\mathbf{Oz}[i] = \mathbb{Z}[i] \oplus \Pi_{\mathbb{C}}$; on them ct is a ring homomorphism, since two nonnegative exponents add to 0 only if both are 0. For a subclass $E \subseteq \mathbf{No}$ of exponents, $k((\omega^E))$ is the class of series in $\mathbf{No}_k$ supported in E (a full Hahn subfield when E is an additive subgroup, set sized or not; inversion uses finite sums of elements of E), and ϖ_E is the *outer* support projection $\sum c_{\gamma} \omega^{\gamma} \mapsto \sum_{\gamma \in E} c_{\gamma} \omega^{\gamma}$ on field elements. For a subclass B of *indices* and an exponent $\gamma = \sum_a r_a \omega^a$, $\varpi_B^{\text{in}} \gamma = \sum_{a \in B} r_a \omega^a$ is the *inner* projection. With Part VIII's $\text{NF}(B) = \{\gamma : \text{supp } \gamma \subseteq B\}$ and $\mathbf{K}_B = \text{NF}(\text{NF}(B))$, put $\mathbf{K}_{B,\mathbb{R}} = \mathbf{K}_B$ and $\mathbf{K}_{B,\mathbb{C}} = \mathbb{C}((\omega^{\text{NF}(B)})) = \mathbf{K}_B(i)$. For a *class* B these are classes.

20's symbol	Here
$\mathbf{No}[i]$, $\mathbf{Oz}[i]$; K_k ; $M_k = \ker c_0$	$\mathbf{No}(i)$, $\mathbf{Oz}[i]$; $\mathbf{No}_k$; $\mathbf{V}_k = \ker \text{ct}$.
c_0 (its macro $\backslash \text{ct}$); lc ; Π_k , $\mathbf{m}_k$	ct ; lc ; Π_k , $\mathbf{m}_k$ (unchanged meaning).
p, p_g ; T_p, T_g ; $\sigma_p, \sigma_T, \sigma_g$	φ, φ_g ; $\iota_{\varphi}, \iota_{\varphi_g}$; $M_{1,\iota_{\varphi}}, M_{1,\tau}, \sigma_g = M_{1,\iota_{\varphi_g}}$ (Parts III and VIII).
T (an ordered additive automorphism of the exponent group)	τ .
$F = \text{Fix}(p)$; H_F ; $k((\omega^{H_F}))$	$B = \text{Fix}(\varphi)$; $\text{NF}(B)$; $\mathbf{K}_{B,k}$.
$J = (u, u+1)$; $H_J, F_{J,k}$; B_A ; y_a	$\mathbf{J}$; $\text{NF}(\mathbf{No} \setminus \mathbf{J})$, $\mathbf{K}_{\mathbf{No} \setminus \mathbf{J},k}$; $\Sigma_2(A)$; w_a .
$H = \text{Fix}(T)$, K_H ; P_H ; Q_F	Γ_{τ} , $k((\omega^{\Gamma_{\tau}}))$; $\varpi_{\Gamma_{\tau}}$; ϖ_B^{in} .
f_+, f_- (orbit directions)	$f_{\uparrow}, f_{\downarrow}$.
$\mathcal{G}_c$ (Green operator)	Gr_c .
$\mathcal{R}_F$ (multiplicative retraction)	mr_B .
$S_s, S = S_1$	$\sigma_s^{\text{tr}}, \sigma_1^{\text{tr}}$.
$P \in k[X, X^{-1}]$; $k(X)$	$\mathbf{p} \in k[\mathbf{Y}, \mathbf{Y}^{-1}]$; $k(\mathbf{Y})$.
$h \in \text{Hom}(G, k)$, $\chi \in \text{Hom}(G, k^{\times})$ (cocycle characters)	$\text{ch}^+, \text{ch}^{\times}$.
j ; $\mathcal{A} = \text{Aut}(\mathbf{No}, \mathbf{Oz})$; identity e	cj ; written out; 1.

20's symbol	Here
$\eta_{\mathbf{On}}$ -order	an ordered class with the set-cut property (definition 62.1).

The collisions are resolved here only, and the tempting false readings are wrong:

- 20's S_s is the double lift of the index *translation* $a \mapsto a + s$, so $S_s(\omega^\gamma) = \omega^{\omega^s \gamma}$ and $S_1 \neq \text{id}$; the single-dilation and autonomous-dilation reports' S_q , S_d multiply exponents by a positive number ($S_d(\omega^g) = \omega^{dg}$), so $S_1 = \text{id}$ there. Reading 20's S_1 as a dilation is wrong; it is σ_1^{tr} here, the map of [example 64.1](#).
- 20's $\mathcal{G}_c$ is not Part III's bottom gap $\mathbf{G}$, the surcomplex report's group $\mathcal{G}$ or Part IX's $\mathcal{G}(R)$; 20's $\mathcal{R}_F$ is not Part III's retraction ret_ϕ , dsup's resonance set R_λ or the ring $\mathcal{R}_\circ$; its P_H is not dsup's P_λ in general (it is P_1 for $\chi = 1$) nor Part I's zero-cut projections P_-, P_0, P_+ ($\varpi_{\{0\}} = P_0 = \text{ct}$ only in the global case).
- 20's M_k is not a monomial map $M_{\chi, \tau}$; its H, H_F, H_J, K_H are not the convex subgroups H_g , the hull $\mathcal{H}(A)$ or Part VIII's $\text{NF}(B)$ *as notation* (though $H_F = \text{NF}(F)$); its F is neither Part III's isomorphism F nor 18's F_φ ; its J is not an embedding $J_{\tau, \rho}$.
- $f_\uparrow$ ($\tau\gamma > \gamma$) and $f_\downarrow$ ($\tau\gamma < \gamma$) record orbit direction, not the sign of γ ; they agree with the signs for σ_1^{tr} but not in general. dsup writes the valuation chambers $A_\pm = \{g : \theta g \gtrless g\}$ with $t^g = \omega^{-g}$, so its f_+ is $f_\downarrow$ here.
- 20's j (conjugation) is cj , not the surcomplex report's involution j ; 20's χ in cocycles is a character of the acting group G , not a character of the exponent group as in $M_{\chi, \tau}$; its X is an operator variable, not a monomial X^g of Part II.

Foundations are those of [sections 1.4](#) and [60.3](#): NBG with global choice, a set-sized group acting on a class means one class map $G \times \mathbf{No} \rightarrow \mathbf{No}$ satisfying the action identities, $\text{Aut}(\mathbf{No}, \mathbf{Oz})$ and its “subgroups” are metanotation for joint actions and elementwise factorizations [[15](#), Remark 2.1], class recursions have set-valued stages, every support and every index set of a sum is a set, and no cut with a proper-class side is filled [[20](#)]. *Strong* means strongly additive in both directions ([section 60.3](#)); it is a support condition with a finite coefficient calculation at each exponent, not convergence in the fine order topology.

76.4 Principal results of source 20

Theorem 76.1 (Universality with minimal fixed fields [[20](#); [18](#)]; 20's Theorem 1.1). *For a set-sized group G the following are equivalent: (i) G is left-orderable; (ii) G acts faithfully by field automorphisms on $\mathbf{No}$; (iii) G acts faithfully by ring automorphisms on $\mathbf{Oz}$; (iv) G acts faithfully on $(\mathbf{No}, \mathbf{Oz})$ by strong, $\mathbb{R}$ -linear, exact-monomial field automorphisms σ_g with*

$$\text{Fix}(\sigma_g) = \mathbb{R} \quad (g \neq 1), \quad \{\sigma_g(\omega) : g \in G\} \text{ algebraically independent over } \mathbb{R}. \quad (76.2)$$

The constructed action extends $\mathbb{C}$ -linearly to $\mathbf{No}(i)$, commutes with conjugation, preserves $\mathbf{Oz}[i]$ and has $\text{Fix}(\sigma_g) = \mathbb{C}$ for $g \neq 1$.

The equivalence and the fixed fields are [corollaries 60.3](#) and [63.2](#) and [proposition 65.1](#), proved by [18](#); [20](#) proves them independently, and only the independence clause is new ([proposition 77.3](#)).

Requiring the fixed field of *each* nonidentity element to be $\mathbb{R}$ is more than a common fixed field $\mathbb{R}$; the proof uses a free action on the inner index order, not on the field, whose automorphisms all fix 0 and 1 [20].

Theorem 76.2 (Uniform difference obstructions [20]; 20’s Theorem 1.2). *Let $G \neq 1$ act as in the construction of theorem 76.1, $g \neq 1$ and $\sigma = \sigma_g$. Then*

$$\text{im}(\sigma - 1) = \ker \text{ct}, \quad \ker(\sigma - 1) = k, \quad (76.3)$$

$$\{\sigma x/x : x \in \mathbf{No}_k^\times\} = \text{lc}^{-1}(1), \quad \text{Fix}(\sigma) \cap \mathbf{No}_k^\times = k^\times. \quad (76.4)$$

If $\mathbf{p} \in k[\mathbf{Y}, \mathbf{Y}^{-1}]$ is nonzero, $\mathbf{p}(\sigma)$ is bijective on $\ker \text{ct}$; on $\mathbf{No}_k$ it is bijective if $\mathbf{p}(1) \neq 0$, and if $\mathbf{p}(1) = 0$ its kernel is k and its image $\ker \text{ct}$. Moreover $\text{im}(\sigma - 1 : \mathbf{Oz} \rightarrow \mathbf{Oz}) = \Pi$ and $\ker(\sigma - 1 : \mathbf{Oz} \rightarrow \mathbf{Oz}) = \mathbb{Z}$, and likewise with $\mathbf{Oz}[i]$, $\Pi_{\mathbb{C}}$ and $\mathbb{Z}[i]$.

The proofs are theorems 78.5, 79.1 and 79.5 and corollary 78.7. For rational dilations the additive and multiplicative clauses are `dsup:thm:surreal` (1)–(2), which 20 did not cite.

Example 76.3 (An infinite-support omnific solution [20]). For the explicit automorphism $\sigma_1^{\text{tr}}(\sum_{\gamma} c_{\gamma} \omega^{\gamma}) = \sum_{\gamma} c_{\gamma} \omega^{\omega^{\gamma}}$ of example 64.1,

$$x = \sum_{n \geq 1} \omega^{\omega^{-n}} \in \mathbf{Oz}, \quad \sigma_1^{\text{tr}} x - x = \omega, \quad (76.5)$$

the support of x being strictly decreasing and positive; whereas $\sigma_1^{\text{tr}} x - x = 1$ has no solution in $\mathbf{No}(i)$. The obstruction is a coefficient, not a size bound.

77 Universality, fixed fields and algebraic independence

77.1 What is printed once

Every surreal is a quotient of omnific integers, so a ring automorphism of $\mathbf{Oz}$ extends uniquely to $\mathbf{No}$ (`odg:thm:fractions`); 20 proves it by choosing δ above 0 and the negatives of all exponents, as 18 does (section 61). A set group acting faithfully by increasing bijections on a linearly ordered class is left-orderable (lemma 63.1); 20 chooses a witness $x_{g,h}$ with $gx_{g,h} \neq hx_{g,h}$ for each pair, well-orders the set of witnesses and compares at the first witness that separates, the same lexicographic argument. Every field automorphism of the real closed $\mathbf{No}$ is increasing, which gives necessity in theorem 76.1; this does not assert that every surreal field automorphism fixes $\mathbb{R}$ [20]. An ordered class with the set-cut property (20’s $\eta_{\mathbf{On}}$ -order) is order isomorphic to $\mathbf{No}$ (lemma 62.2), and every set group with a left order acts freely and increasingly on $\mathbf{No}$ (theorem 62.4). 20 builds that action by a different cut filling, kept as a second route:

Proposition 77.1 (Free equivariant cut completion, second route [20]; 20’s Theorem 3.4). *Let G be a set group with a left-invariant order. Put $L_0 = G$ with left translation. Given a set L_{α} with a free increasing G -action, let $\mathcal{D}_{\alpha}$ be the set of downward-closed subsets $C \subseteq L_{\alpha}$, $\emptyset$ and L_{α} included, and insert at every $C \in \mathcal{D}_{\alpha}$ a disjoint copy $\{(\alpha, C, h) : h \in G\}$ of the ordered set G : old points keep their order, an old point x is below (α, C, h) exactly when $x \in C$, copies at distinct cuts are ordered by inclusion of the cuts and each copy by the order of G . Then*

$$g \cdot (\alpha, C, h) = (\alpha, gC, gh) \quad (77.1)$$

extends the action freely and increasingly to the linear order $L_{\alpha+1}$; with unions at limits, $L = \bigcup_{\alpha \in \mathbf{On}} L_{\alpha}$ has the set-cut property, and the action transports to a free increasing action on $\mathbf{No}$.

Proof. Downward-closed subsets of a linear order are linearly ordered by inclusion, so the rules define a linear order. The map (77.1) preserves cut inclusion and the old-point rule, and is increasing on each copy by left invariance; both coordinates respect composition. On new points $gh = h$ forces $g = 1$, and old points stay free. Every stage is a set. If $A < B$ are sets in L , their stages are bounded by some α ; the downward closure of A in L_α (empty if A is) contains A and excludes B , and a point inserted there lies strictly between them. So L has the set-cut property and is a proper class. Both L and $\mathbf{No}$ have set-like enumerations along $\mathbf{On}$ by global choice and their set exhaustions; the back-and-forth of lemma 62.2 gives $L \cong \mathbf{No}$. $\square$

18 inserts, over each cut orbit, fibers indexed by the stabilizer of a representative cut (lemma 62.3); 20 inserts a whole regular copy of G at every cut, including cuts not stabilized by anything, so the new points need no choice of orbit representatives. Inserting one point at a cut stabilized by $g \neq 1$ would create a fixed point: the regular copy is what the fixed-field conclusions need. The construction does not preserve Dedekind completeness and asserts no free action on the real line [20].

Lemma 77.2 (The outer lift in both coefficient cases [20]; [11, 18]). *Let τ be an ordered additive automorphism of $(\mathbf{No}, +)$. The monomial map $M_{1,\tau}(\sum_\gamma c_\gamma \omega^\gamma) = \sum_\gamma c_\gamma \omega^{\tau(\gamma)}$ is a strong k -linear field automorphism of $\mathbf{No}_k$. It fixes k , preserves ct and lc , preserves Π_k and $\mathfrak{m}_k$, preserves $\mathbf{Oz}$, respectively $\mathbf{Oz}[i]$, sends monomials to monomials, and for $k = \mathbb{C}$ commutes with conjugation. For an increasing bijection φ of $\mathbf{No}$, ι_φ is such a τ , $\iota_\varphi(1) = \omega^{\varphi(0)}$, and $\varphi \mapsto M_{1,\iota_\varphi}$ is a faithful action.*

Proof. This is (5.1) with $\chi = 1$ and proposition 61.2; additivity of τ transports every finite convolution, and $\tau(0) = 0$ gives unitality. Order preservation gives $\gamma > 0 \Leftrightarrow \tau\gamma > 0$, hence the support and arithmetic clauses; the leading and zero coefficients are unchanged. Conjugation acts only on coefficients, the map only on exponents [20]. $\square$

The inner lift ι_φ is additive, not a field map, and not unital; its additivity is what the outer level needs. For an increasing bijection q of a linear order and a reverse well-ordered S with $q(S) = S$, q fixes S pointwise: if $q(a) > a$ the forward orbit, if $q(a) < a$ the backward orbit, is an infinite increasing sequence in S [20]. This is the cyclic case of lemma 61.4, and applied at both levels it gives, for $B = \text{Fix}(\varphi)$,

$$\text{Fix}(\iota_\varphi) = \text{NF}(B), \quad \text{Fix}(M_{1,\iota_\varphi}) = K_{B,k}, \quad (77.2)$$

so $\text{Fix}(\iota_\varphi) = \{0\}$ and $\text{Fix}(M_{1,\iota_\varphi}) = k$ when φ has no fixed point (20's Theorem 4.4; theorem 61.5 and proposition 65.1). $\text{NF}(B)$ is an additive group and an $\mathbb{R}$ -vector space by its support description.

77.2 The regular orbit of ω

Proposition 77.3 (Independence of the orbit of ω [20]). *Let $g \mapsto \varphi_g$ be a free increasing action of G on $\mathbf{No}$ and $\sigma_g = M_{1,\varphi_g}$. Then*

$$\sigma_g(\omega) = \omega^{\omega^{\varphi_g(0)}}, \quad (77.3)$$

and $\{\sigma_g(\omega) : g \in G\}$ is algebraically independent over k .

Proof. $\iota_{\varphi_g}(1) = \omega^{\varphi_g(0)}$ gives (77.3). Freeness makes the $\varphi_g(0)$ pairwise distinct, so the inner monomials $\omega^{\varphi_g(0)}$ are linearly independent over $\mathbb{Q}$. A monomial $\prod_g \sigma_g(\omega)^{n_g}$ of a polynomial in finitely many orbit points is $\omega^{\sum n_g \omega^{\varphi_g(0)}}$, and distinct exponent vectors give distinct outer exponents; the nonzero coefficients of distinct monomials cannot cancel. $\square$

Proof of theorem 76.1. Necessity and the equivalences are corollary 63.2. For (iv), choose a left order, take the free action of theorem 62.4 or proposition 77.1, and put $\sigma_g = M_{1, \iota\varphi_g}$; lemma 77.2 and (77.2) give the structure and the fixed fields, faithfulness on $\mathbf{Oz}$ follows from `odg:thm:fractions`, and proposition 77.3 gives independence, with real or complex coefficients. $\square$

For the global action every nontrivial subgroup $U \leq G$ has $\mathbf{No}_k^U = k$ and fixed omnific ring $\mathbb{Z}$ or $\mathbb{Z}[i]$ (20's Corollary 4.5; theorem 60.1 and proposition 65.1). This is not a finite Galois correspondence: $\mathbf{No}_k/k$ is transcendental and a fixed field need not distinguish the infinite subgroups acting on it [20].

77.3 An explicit family

Proposition 77.4 (Translations of the inner index [20]; 20's Proposition 4.6). *For $s \in \mathbf{No}$ put*

$$\sigma_s^{\text{tr}} \left(\sum_{\gamma} c_{\gamma} \omega^{\gamma} \right) = \sum_{\gamma} c_{\gamma} \omega^{\omega^s \gamma}. \quad (77.4)$$

Then $\sigma_s^{\text{tr}} \sigma_t^{\text{tr}} = \sigma_{s+t}^{\text{tr}}$, and for $s \neq 0$ the map σ_s^{tr} has all the fixed-field and difference-equation properties of a nonidentity element of the global construction proved in this part. For every set-sized additive subgroup $U \subseteq \mathbf{No}$, $s \mapsto \sigma_s^{\text{tr}}$ is a faithful joint action of U needing no choice of a cut completion.

Proof. $\sigma_s^{\text{tr}} = M_{1, \iota\varphi_s}$ for the translation $\varphi_s(a) = a + s$, whose inner lift is multiplication by ω^s ; translations compose additively, and φ_s has no fixed point for $s \neq 0$. The results below use only fixed-point-freeness of the inner index action. $\square$

σ_1^{tr} is the map of example 64.1. Exact monomial preservation is not preservation of the omega-map: $\sigma_1^{\text{tr}}(\omega) = \omega^{\omega}$ while $\omega^{\sigma_1^{\text{tr}}(1)} = \omega$ (compare remark 61.3) [20]. It is not a dilation of exponents by a rational number (section 76.3).

77.4 Fixing a set of parameters

A fixed field equal to $\mathbb{R}$ cannot contain a nonreal parameter; 20's relative statement has a larger explicit common fixed field.

Theorem 77.5 (Universality inside every set-parameter stabilizer [20]; 20's Theorem 5.1). *Let $A \subseteq \mathbf{No}(i)$ be a set and G a set-sized left-orderable group. Put*

$$\Sigma_2(A) = \bigcup_{x \in A} \bigcup_{\gamma \in \text{supp } x} \text{supp } \gamma, \quad (77.5)$$

choose $u > \Sigma_2(A)$ and let $J = (u, u + 1)$. There is a faithful strong $\mathbb{C}$ -linear action $g \mapsto \sigma_g$ on $\mathbf{No}(i)$ preserving $\mathbf{No}$, $\mathbf{Oz}[i]$ and conjugation, fixing A pointwise, such that for every $g \neq 1$

$$\text{Fix}(\sigma_g) = K_{\mathbf{No} \setminus J, \mathbb{C}} = \mathbb{C}((\omega^{\text{NF}(\mathbf{No} \setminus J)})). \quad (77.6)$$

The restriction to $\mathbf{No}$ preserves $\mathbf{Oz}$ and has fixed field $K_{\mathbf{No} \setminus J}$. For each $a \in J$ the orbit of $w_a = \omega^{\omega^a}$ is algebraically independent over $K_{\mathbf{No} \setminus J, k}$, and so is $\{w_a : a \in J\}$, in the sense of all finite polynomial tests.

Proof. $\Sigma_2(A)$ is a set (real and imaginary parts taken together), so u exists. A nonempty bounded open interval of $\mathbf{No}$ has the set-cut property (add the endpoints to the appropriate sides of a set cut) and a set-like enumeration, so $J \cong \mathbf{No}$ (lemma 62.2). Transport a free increasing action of G onto J and extend by the identity off J : the φ_g are increasing bijections of $\mathbf{No}$ with $\text{Fix}(\varphi_g) = \mathbf{No} \setminus J$ for $g \neq 1$. Lift twice. Every exponent of an element of A lies in $\text{NF}(\mathbf{No} \setminus J)$, so A is fixed, and (77.2) gives (77.6).

For independence, distinct finite integer combinations of the ω^a , $a \in J$, lie in distinct cosets of $\text{NF}(\mathbf{No} \setminus J)$: a nonzero difference is supported in J . In a polynomial in finitely many w_a with coefficients in $K_{\mathbf{No} \setminus J, k}$, distinct monomials therefore contribute supports in disjoint cosets and cannot cancel. Finally $\sigma_g(w_a) = w_{\varphi_g(a)}$, and freeness on J makes the orbit indices distinct. $\square$

So a set group has a faithful action on $(\mathbf{No}, \mathbf{Oz})$ fixing a given set of surreal parameters if and only if it is left-orderable (20's Corollary 5.2; corollaries 60.2 and 63.2). The fixed field in (77.6) is a two-level support field; the theorem does not identify all possible fixed fields, and does not claim that an arbitrary Hahn subfield can replace it; its independence statement concerns a proper-class family only through finite subsets, since no polynomial has a proper class of variables [20].

Remark 77.6 (Two relative fixed fields [merge]). Corollary 60.2 fixes the same A (real) with $\text{Fix}(\sigma_g) = K_{\Sigma_2(A)}$ for $g \neq 1$, the index action being free off the set $\Sigma_2(A)$ (theorem 62.6); 20's is free on the interval J and the identity on the class $\mathbf{No} \setminus J$. Since $\Sigma_2(A) \subseteq \mathbf{No} \setminus J$ and $S \mapsto K_S$ is monotone (proposition 61.6),

$$A \subseteq K_{\Sigma_2(A)} \subsetneq K_{\mathbf{No} \setminus J} \subsetneq \mathbf{No} :$$

18's fixed field is set sized, real closed and the smallest of the form K_S containing A ; 20's is a proper class (it contains ω^{ω^b} for every $b \notin J$). The two constructions answer different questions: 18 prescribes a minimal fixed field, 20 keeps a whole interval of indices free, which gives the independent family $\{w_a : a \in J\}$ and the uniform obstruction maps of theorem 79.7 and remark 80.4. Both lie in the monomial factor of theorem 24.1, not in $\mathcal{U}_{\mathbf{Oz}}$, so neither bears on question 11.3. The fixed exponent group $\text{NF}(\mathbf{No} \setminus J)$ is divisible and not of the form $\text{NF}(S)$ for a set S , so theorem 77.5 realizes, for every left-orderable set group, a new instance of question 66.2.

78 Downward orbits and the weighted Green operator

The support result behind exact difference solvability applies to arbitrary ordered index sets, with no Archimedean hypothesis and no cofinal orbit [20].

Lemma 78.1 (Sequential test [20]). *In ZFC a linear order is reverse well-ordered if and only if it has no infinite strictly increasing sequence, and every sequence in a reverse well-ordered set has an infinite nonincreasing subsequence.*

Proof. A subset with no greatest element yields a strictly increasing sequence by recursive choice; the values of such a sequence have no greatest element. For the second claim, color a pair $i < j$ by whether $s_i < s_j$; select successively a point and an infinite tail on which its pair color is constant, then infinitely many selected points of one color. This gives a strictly increasing or a nonincreasing subsequence, and the former is impossible. $\square$

Lemma 78.2 (Downward-orbit lemma [20]; `dsup:lem:orbit`). *Let U be an increasing bijection of a linearly ordered set or class E with $U(a) < a$ for all $a \in E$, and $S \subseteq E$ a reverse well-ordered set. Then $\bigcup_{n \geq 0} U^n(S)$ is a reverse well-ordered set, and each point lies in $U^n(S)$*

for only finitely many $n \geq 0$. Hence the Hahn translates of a series supported on S , with arbitrary scalar weights, form a summable family.

Proof. The union is a set by replacement. Let $t_j = U^{n_j}(s_j)$, $s_j \in S$, be strictly increasing. Pass to a subsequence with (n_j) nondecreasing (one value infinitely often, or a strictly increasing subsequence) and then, by lemma 78.1, with (s_j) nonincreasing. For $i < j$, $U^{n_j}(s_j) \leq U^{n_i}(s_j) \leq U^{n_i}(s_i)$, a contradiction. If $a \in U^{n_j}(S)$ for increasing n_j , the points $U^{-n_j}(a) \in S$ are strictly increasing, again a contradiction. Scalar weights only remove support. $\square$

This is dsup 's increasing-orbit hull, with the growth order reversed; 20 states it for a bare ordered class, which the inner level of lemma 79.4 needs (there the index class carries no addition).

Lemma 78.3 (The downward direction on every orbit [20]). *For an increasing bijection q of a linear order E , the complement of $\text{Fix}(q)$ is invariant, and on it $U(a) = \min\{q(a), q^{-1}(a)\}$ and $V(a) = \max\{q(a), q^{-1}(a)\}$ are mutually inverse increasing bijections with $U(a) < a$.*

Proof. Whether $q(a) > a$ is constant along the orbit of a ; on orbits with $q(a) > a$, $(U, V) = (q^{-1}, q)$, otherwise (q, q^{-1}) . If $a < b$, the smaller value at a is below both candidates at b , so U is increasing; likewise V . $\square$

Remark 78.4 (What the lemma does not say [20]). It does not say that $\sum_{n \geq 0} q^n(f)$ is always summable: on an upward orbit that family has increasing support, and the downward direction must be chosen separately on the two invariant orbit regions. Nor does it say that exponents tend cofinally to $-\infty$. Both distinctions matter in the surreal value group (compare the shear remark after `dsup:cor:labeled`).

78.1 The Green operator

Let Γ be an exponent group (a set, or **No**), τ an ordered additive automorphism of Γ , and $\sigma = M_{1,\tau}$ on $k((\omega^\Gamma))$, all series with set support. Put $\Gamma_\tau = \text{Fix}(\tau)$, write $\varpi = \varpi_{\Gamma_\tau}$, and let $f_\uparrow, f_\downarrow$ keep the exponents with $\tau\gamma > \gamma$, respectively $\tau\gamma < \gamma$, so that $f = \varpi f + f_\uparrow + f_\downarrow$.

Theorem 78.5 (Weighted Green operator [20]; `dsup:thm:resolvent`). *For $c \in k^\times$ and $\varpi f = 0$ the series*

$$\text{Gr}_c(f) = \sum_{n \geq 1} c^{n-1} \sigma^{-n}(f_\uparrow) - \sum_{n \geq 0} c^{-n-1} \sigma^n(f_\downarrow) \quad (78.1)$$

is Hahn summable and is the unique solution x of $(\sigma - c)x = f$ with $\varpi x = 0$. Consequently

$$\ker(\sigma - 1) = k((\omega^{\Gamma_\tau})), \quad \text{im}(\sigma - 1) = \ker \varpi, \quad (78.2)$$

$$(\sigma - c)^{-1}f = \frac{\varpi f}{1 - c} + \text{Gr}_c(f - \varpi f) \quad (c \neq 1), \quad (78.3)$$

the last an inverse on all of $k((\omega^\Gamma))$; for $c \in k$ these maps are $k((\omega^{\Gamma_\tau}))$ -linear.

Proof. On the complement of Γ_τ use U of lemma 78.3: the terms on $f_\uparrow$ are downward iterates of σ^{-1} , those on $f_\downarrow$ of σ , and all supports lie in the union of lemma 78.2 for $\text{supp } f$, which gives summability with finite multiplicity; the support avoids Γ_τ . The finite partial sums satisfy

$$(\sigma - c) \sum_{n=1}^N c^{n-1} \sigma^{-n} f_\uparrow = f_\uparrow - c^N \sigma^{-N} f_\uparrow, \quad (78.4)$$

$$(\sigma - c) \left(- \sum_{n=0}^{N-1} c^{-n-1} \sigma^n f_\downarrow \right) = f_\downarrow - c^{-N} \sigma^N f_\downarrow, \quad (78.5)$$

and at each exponent the residual eventually vanishes, by finite multiplicity; strongness justifies the infinite cancellation, so $(\sigma - c)\text{Gr}_c(f) = f$. If $(\sigma - c)x = 0$, then $\text{supp } x$ is τ -invariant, hence inside Γ_τ (the cyclic case of [lemma 61.4](#)), where σ is the identity and $(1 - c)x = 0$. This proves uniqueness on $\ker \varpi$, the kernels and [\(78.3\)](#); for $c = 1$ the Γ_τ -part of every difference vanishes. Multiplying an exponent outside Γ_τ by a Γ_τ -supported series stays in a coset disjoint from Γ_τ , so ϖ is $k((\omega^{\Gamma_\tau}))$ -linear, as is the unique normalized inverse. $\square$

Remark 78.6 (Relation to the single-dilation report [\[merge\]](#)). In the t -convention $\sigma = M_{1,\tau}$ is `dsup:eq:sigma` with $\theta = \tau$ and $\chi = 1$; its chamber $A_+ = \{g : \theta g > g\}$ is the set of t -exponents $g = -\gamma$ with $\tau\gamma < \gamma$, so `dsup`'s f_+ is $f_\downarrow$. With this dictionary its Q_c (the resolvent `dsup:eq:Q`) is the right side of [\(78.3\)](#) for $c \neq 1$, and $Q_1 = \text{Gr}_1$ on $\ker \varpi$; its resonance projection P_1 is ϖ . `dsup` proves it for set-sized Γ and any character; for a class exponent automorphism its proofs localize to a set-sized subgroup containing the input supports and their orbits (`dsup`'s set localization before `dsup:thm:surreal`), which is what 20's direct class argument avoids.

The projection ϖ need not be multiplicative: a monomial outside Γ_τ and its inverse give the counterexample for `ct`, and `dsup`'s remark “Not a residue homomorphism” is the same. `ct` is additive and k -linear but not a ring homomorphism on **No** ($\text{ct}(\omega) = \text{ct}(\omega^{-1}) = 0$, $\text{ct}(\omega\omega^{-1}) = 1$); `lc` is multiplicative, not additive; additive obstructions use `ct`, multiplicative ones `lc`. In the relative construction $\Gamma_\tau = \text{NF}(\mathbf{No} \setminus \mathbf{J})$, and no convexity is assumed [\[20\]](#).

78.2 The global action and omnific solutions

For the global action, $\Gamma_\tau = \{0\}$ and $\varpi = \text{ct}$, so [theorem 78.5](#) gives [\(76.3\)](#). An ordered additive τ preserves the sign of every exponent, and so do its iterates, so [\(78.1\)](#) preserves Π_k and $\mathbf{m}_k$ separately [\[20\]](#).

Corollary 78.7 (Normalized arithmetic primitives [\[20\]](#)). *For $g \neq 1$ in the global action, $\sigma_g - 1$ is bijective on Π_k and on $\mathbf{m}_k$. Each purely infinite series has a unique purely infinite primitive, its primitives in **Oz** differ exactly by integers, and in **Oz** $[i]$ by Gaussian integers.*

Proof. Restrict the normalized inverse to each support region; the constants of **Oz**, **Oz** $[i]$ are $\mathbb{Z}$, $\mathbb{Z}[i]$. $\square$

Example 78.8 (Two exact primitives for σ_1^{tr} [\[20\]](#)). The normalized solutions are

$$(\sigma_1^{\text{tr}} - 1)^{-1}(\omega) = \sum_{n \geq 1} \omega \omega^{-n}, \quad (\sigma_1^{\text{tr}} - 1)^{-1}(\omega^{-1}) = - \sum_{n \geq 0} \omega^{-\omega^n}, \quad (78.6)$$

the first purely infinite and omnific, the second infinitesimal. For $c \in k^\times$, $(\sigma_1^{\text{tr}} - c)^{-1}(\omega) = \sum_{n \geq 1} c^{n-1} \omega \omega^{-n}$, valid also for $c = 1$ on the normalized subspace. For a nonzero ordinary constant b , $(\sigma_1^{\text{tr}} - 1)x = b$ has no solution.

79 Constant-coefficient operators and multiplicative equations

Theorem 79.1 (Scalar operator classification [\[20\]](#); `dsup:cor:poly`). *For a nonidentity σ of the global action and a nonzero $\mathbf{p} \in k[\mathbf{Y}, \mathbf{Y}^{-1}]$, $\mathbf{p}(\sigma)$ restricts to bijections of $\mathbf{V}_k$, Π_k and $\mathbf{m}_k$; on **No** $_k$ its kernel and image are those of [theorem 76.2](#).*

Proof. For $k = \mathbb{C}$ factor $\mathbf{p}$ into a nonzero scalar, a power of $\mathbf{Y}$ and factors $\mathbf{Y} - c$, $c \neq 0$, with multiplicity. σ is invertible, every $\sigma - c$ is invertible on $\mathbf{V}_\mathbb{C}$ by [theorem 78.5](#), and the factors

and their inverses preserve the two sign subspaces. On k , $\mathbf{p}(\sigma)$ is multiplication by $\mathbf{p}(1)$, and $\mathbf{No}_k = k \oplus \mathbf{V}_k$. For $k = \mathbb{R}$ solve over $\mathbb{C}$ and conjugate: $\mathbf{p}$ and the input are real, so the conjugate of the unique normalized solution (or of the unique solution when $\mathbf{p}(1) \neq 0$) is again one, hence equal to it and real; the same uniqueness gives the sign subspaces. $\square$

$\mathbf{dsup}$'s corollary needs $\mathbf{p}$ split over the coefficient field; for $k = \mathbb{R}$ the conjugation step removes that hypothesis here.

Corollary 79.2 (No generalized constant eigenvectors [20]; $\mathbf{dsup:cor:fixed}$). *$\ker(\sigma - 1)^m = k$ for every $m \geq 1$: no nonconstant element satisfies a nonzero homogeneous linear difference equation with coefficients in k .*

Corollary 79.3 (A field of difference operators [20]). *The additive class $\mathbf{V}_k$ is a vector space over $k(\mathbf{Y})$, with $\mathbf{Y}$ acting by σ and $(\mathbf{p}/\mathbf{q}) \cdot f = \mathbf{p}(\sigma)\mathbf{q}(\sigma)^{-1}f$ for $\mathbf{q} \neq 0$; Π_k and $\mathbf{m}_k$ are subspaces.*

Proof. Nonzero polynomial operators are bijective on $\mathbf{V}_k$ and commute, so the localization identities define a field action; a nonzero $\mathbf{p}$ cannot act as zero on the nonzero space $\mathbf{V}_k$; stability of the subspaces is from the proof above. $\square$

$k(\mathbf{Y})$ is a field of operators: $\mathbf{Y}$ is not identified with a surreal scalar, and the action is not multiplication by a new coefficient subfield. The constant line is the only obstruction to extending this structure to $\mathbf{No}_k$ with the same action of $\mathbf{Y}$ [20].

79.1 The inner difference and the global multiplicative criterion

Lemma 79.4 (The exponent difference is onto [20]). *If φ has no fixed point, $\iota_\varphi - 1$ is a bijective $\mathbb{R}$ -linear map of $(\mathbf{No}, +)$. In general $\text{im}(\iota_\varphi - 1) = \ker \varpi_B^{\text{in}}$ and $\ker(\iota_\varphi - 1) = \text{NF}(B)$ for $B = \text{Fix}(\varphi)$, with a unique solution β of $(\iota_\varphi - 1)\beta = \gamma$, $\varpi_B^{\text{in}}\beta = 0$ when $\varpi_B^{\text{in}}\gamma = 0$.*

Proof. Apply the proof of [theorem 78.5](#) with $c = 1$ to the inner normal forms, indexed by the ordered class $\mathbf{No}$ with index permutation φ ; it uses only the order of the index class, not addition of indices ([lemmas 78.2](#) and [78.3](#)). The fixed indices are B , the projection is ϖ_B^{in} , and the downward-orbit sums supply the inverse; for $B = \emptyset$ the projection and the kernel are zero. $\square$

For $\varepsilon, h \in \mathbf{m}_k$ the formal series $\log(1 + \varepsilon) = \sum_{n \geq 1} (-1)^{n+1} \varepsilon^n / n$ and $\exp(h) = \sum_{n \geq 0} h^n / n!$ are summable (finite sums of a reverse well-ordered negative support) and give mutually inverse group isomorphisms $1 + \mathbf{m}_k \leftrightarrow \mathbf{m}_k$, with the usual identities since every rearrangement is finite; a strong coefficient-fixing automorphism commutes with both. Only these local maps are used, not a surcomplex exponential at arbitrary arguments ($\mathbf{dsup}$'s $\mathbf{dsup:eq:explog}$) [20].

Theorem 79.5 (Exact multiplicative obstruction [20]; $\mathbf{dsup:thm:multsingle}$). *Let $\sigma = \sigma_g$, $g \neq 1$, in the global action and $a \in \mathbf{No}_k^\times$. Then $\sigma x / x = a$ has a solution $x \neq 0$ if and only if $\text{lc}(a) = 1$; then exactly one solution has leading coefficient 1, and the solutions are its multiples by $k^\times$.*

Proof. $\text{lc}(\sigma x) = \text{lc}(x)$ and multiplicativity give necessity. Conversely write $a = \omega^\gamma(1 + \varepsilon)$, $\varepsilon \in \mathbf{m}_k$. By [lemma 79.4](#) there is a unique β with $(\iota_{\varphi_g} - 1)\beta = \gamma$, and by [corollary 78.7](#) a unique $h \in \mathbf{m}_k$ with $(\sigma - 1)h = \log(1 + \varepsilon)$. Then $x = \omega^\beta \exp(h)$ has leading coefficient 1 and $\sigma x / x = \omega^{\iota_{\varphi_g}\beta - \beta} \exp(\sigma h - h) = a$. A quotient of two solutions is fixed, hence in $k^\times$. $\square$

In dsup's form a solution exists iff some g satisfies $(\theta - 1)g = \gamma$, $\chi(g) = \text{lc}(a)$ and $P_1 h = 0$; here $\chi = 1$, the first condition always holds by [lemma 79.4](#) and the third by $P_1 = \text{ct}$ on $\mathfrak{m}_k$, which leaves $\text{lc}(a) = 1$ [merge]. Equivalently the sequence of classes and class maps

$$1 \longrightarrow k^\times \longrightarrow \mathbf{No}_k^\times \xrightarrow{x \mapsto \sigma x/x} \mathbf{No}_k^\times \xrightarrow{\text{lc}} k^\times \longrightarrow 1 \quad (79.1)$$

is exact, shorthand for the kernel, image and surjectivity statements, not an assertion that a proper-class quotient is a set [20].

Example 79.6 (Monomial and unit equations [20]). For σ_1^{tr} the normalized solution of $\sigma_1^{\text{tr}} x/x = \omega$ is $x = \omega^{1/(\omega-1)}$, and that of $\sigma_1^{\text{tr}} x/x = 1 + \omega^{-1}$ is $x = \exp(-\sum_{n \geq 0} (\sigma_1^{\text{tr}})^n \log(1 + \omega^{-1}))$, the inner sum summable by [lemma 78.2](#) on negative support. The equations $\sigma_1^{\text{tr}} x/x = 2$ and $\sigma_1^{\text{tr}} x/x = -1$ have no nonzero solution even in $\mathbf{No}(i)$.

79.2 A multiplicative retraction for the relative action

Let φ be any increasing bijection of $\mathbf{No}$, $B = \text{Fix}(\varphi)$, $\sigma = M_{1, \iota_\varphi}$, and $\varpi = \varpi_{\text{NF}(B)}$ (outer) and ϖ_B^{in} (inner): different linear maps on different normal-form levels. For $a = d\omega^\gamma \exp(\ell)$ with $d = \text{lc}(a) \in k^\times$ and $\ell \in \mathfrak{m}_k$ put

$$\text{mr}_B(a) = d\omega^{\varpi_B^{\text{in}} \gamma} \exp(\varpi \ell) \in \mathbf{K}_{B,k}^\times; \quad (79.2)$$

$\varpi \ell$ still has negative support, and $\varpi_B^{\text{in}} \gamma \in \text{NF}(B)$.

Theorem 79.7 (Relative multiplicative retraction [20]; 20's Theorem 9.4). $\text{mr}_B : \mathbf{No}_k^\times \rightarrow \mathbf{K}_{B,k}^\times$ is a σ -invariant multiplicative retraction, and

$$\{\sigma x/x : x \in \mathbf{No}_k^\times\} = \ker \text{mr}_B; \quad (79.3)$$

$x \mapsto \sigma x/x$ is a bijection of $\ker \text{mr}_B$ onto itself. Equivalently $\sigma x/x = a$ is solvable exactly when

$$\text{lc}(a) = 1, \quad \varpi_B^{\text{in}} \gamma(a) = 0, \quad \varpi \log(a/(\text{lc}(a)\omega^{\gamma(a)})) = 0, \quad (79.4)$$

all solutions differ by elements of $\mathbf{K}_{B,k}^\times$, and exactly one has $\text{mr}_B(x) = 1$.

Proof. Leading coefficients multiply, leading exponents and logarithms of normalized units add, and both projections are additive, so mr_B is a homomorphism. On $\mathbf{K}_{B,k}^\times$ the leading exponent lies in $\text{NF}(B)$ and the normalized unit and its logarithm are $\text{NF}(B)$ -supported, so mr_B is the identity there. $\varpi_B^{\text{in}} \iota_\varphi = \varpi_B^{\text{in}}$ and $\varpi \sigma = \varpi$ make it σ -invariant, which gives necessity. If $\text{mr}_B(a) = 1$, uniqueness of the factorization gives (79.4); [lemma 79.4](#) gives β with $(\iota_\varphi - 1)\beta = \gamma(a)$, $\varpi_B^{\text{in}} \beta = 0$, and [theorem 78.5](#) gives $h \in \mathfrak{m}_k$ with $(\sigma - 1)h = \ell$, $\varpi h = 0$. Then $x = \omega^\beta \exp(h)$ has $\text{mr}_B(x) = 1$ and $\sigma x/x = a$. Quotients of solutions lie in $\mathbf{K}_{B,k}^\times$ by (77.2), where mr_B is the identity, so two normalized solutions coincide. $\square$

For the global action $B = \emptyset$, $\varpi_B^{\text{in}} = 0$, $\varpi = \text{ct}$ vanishes on $\mathfrak{m}_k$, and $\text{mr}_B = \text{lc}$. For the relative action of [theorem 77.5](#), $B = \mathbf{No} \setminus \mathbf{J}$ and the retraction is the *same map for every nonidentity element*, so both obstruction spaces are uniform while a prescribed set of parameters stays fixed. mr_B is a homomorphism of multiplicative groups, not a field homomorphism, and is not asserted to preserve $\mathbf{Oz}$: projecting an inner exponent can change its sign, which ι_φ cannot [20]. It specializes `dsup:thm:multsingle` (with $\chi = 1$ its conditions are $\gamma \in \text{im}(\theta - 1)$, $\text{lc}(a) = 1$ and $P_1 h = 0$); what is new is the explicit image $\text{im}(\iota_\varphi - 1) = \ker \varpi_B^{\text{in}}$ of [lemma 79.4](#) and the retraction packaging.

80 First-order equations and cocycles

Theorem 80.1 (Variable-coefficient criterion [20]; 20's Theorem 10.1). *Fix a nonidentity σ of the global action, $a \in \mathbf{No}_k^\times$, $b \in \mathbf{No}_k$, $c = \text{lc}(a)$, and let u be the leading-coefficient-one solution of $\sigma u/u = a/c$. The equation*

$$\sigma x - ax = b \quad (80.1)$$

has a unique solution for every b if $c \neq 1$. If $c = 1$, it is solvable exactly when

$$\text{ct}(b/\sigma u) = 0, \quad (80.2)$$

and then its solutions are $u(y_0 + k)$ with $y_0 = \text{Gr}_1(b/\sigma u)$, the solution with $\text{ct}(y_0) = 0$. For $a = 0$ the unique solution is $\sigma^{-1}b$.

Proof. [Theorem 79.5](#) gives u , since $\text{lc}(a/c) = 1$. With $x = uy$ and $\sigma u = (a/c)u$,

$$\sigma(uy) - a(uy) = \sigma u(\sigma y - cy), \quad (80.3)$$

and [theorem 78.5](#) with $\Gamma_\tau = \{0\}$ finishes. Replacing u by du , $d \in k^\times$, multiplies (80.2) by d^{-1} . $\square$

For $c \neq 1$ an explicit solution is

$$x = u \left[\frac{\text{ct}(b/\sigma u)}{1 - c} + \text{Gr}_c \left(\frac{b}{\sigma u} - \text{ct} \left(\frac{b}{\sigma u} \right) \right) \right]. \quad (80.4)$$

A general coefficient is reduced to its ordinary leading coefficient, but the forcing term must undergo the same gauge change: testing $\text{ct}(b)$ would give the wrong criterion. Variable-coefficient operators of order two or more and matrix equations are not classified, and $(\mathbf{No}(i), \sigma)$ is not difference closed: $\sigma x - x = 1$ has no solution [20]. The autonomous-dilation report records the same fact for its dilations ($S_d x - x = 1$, in its comment after `adr:sr:cor:free`).

80.1 Groups with a central element

An *additive cocycle* is a family $b_g \in \mathbf{No}_k$ with $b_{gh} = b_g + \sigma_g b_h$, a *multiplicative* one a family $a_g \in \mathbf{No}_k^\times$ with $a_{gh} = a_g \sigma_g(a_h)$; coboundaries are $\sigma_g m - m$ and $\sigma_g(u)/u$. Throughout, $G \neq 1$ acts by the global construction [20].

Theorem 80.2 (Central cocycle normal forms [20]; 20's Theorem 11.2). *Suppose the center of G contains $z \neq 1$. Every additive cocycle has a unique expression*

$$b_g = \text{ch}^+(g) + (\sigma_g - 1)m, \quad \text{ch}^+ \in \text{Hom}(G, (k, +)), \quad m \in \mathbf{V}_k, \quad (80.5)$$

and every multiplicative cocycle a unique expression

$$a_g = \text{ch}^\times(g) \sigma_g(u)/u, \quad \text{ch}^\times \in \text{Hom}(G, k^\times), \quad u \in \mathbf{No}_k^\times, \quad \text{lc}(u) = 1. \quad (80.6)$$

Proof. $\text{ch}^+(g) = \text{ct}(b_g)$ is a homomorphism since $\text{ct} \sigma_g = \text{ct}$, and $d_g = b_g - \text{ch}^+(g) \in \mathbf{V}_k$ is a cocycle. Comparing d_{gz} and d_{zg} ,

$$(\sigma_z - 1)d_g = (\sigma_g - 1)d_z. \quad (80.7)$$

Let $m = (\sigma_z - 1)^{-1}d_z \in \mathbf{V}_k$; the inverse commutes with every σ_g , z being central and the inverse on $\mathbf{V}_k$ unique, so (80.7) gives $d_g = (\sigma_g - 1)m$. ct determines ch^+ , and the z -equation

the normalized m . Multiplicatively, $\text{ch}^\times(g) = \text{lc}(a_g)$ is a homomorphism and $d_g = a_g/\text{ch}^\times(g)$ a cocycle with leading coefficient 1. Take the normalized u with $\sigma_z(u)/u = d_z$ and divide by its coboundary: the cocycle r_g has $r_z = 1$ and $\text{lc}(r_g) = 1$, and comparing r_{gz} with r_{zg} gives $\sigma_z(r_g) = r_g$, so $r_g \in k^\times$ and $r_g = 1$. Uniqueness follows from leading coefficients and the normalized z -equation. $\square$

With cohomology read through these normal forms, $H^1(G, (\mathbf{No}_k, +)) = \text{Hom}(G, (k, +))$ and $H^1(G, \mathbf{No}_k^\times) = \text{Hom}(G, k^\times)$; the formulas, not a set-sized quotient, are the assertions [20]. For $G = \mathbb{Z}^d$ acting by rational dilations the same groups appear in `dsup:thm:dilationcohom`, degree 1; there all degrees are computed, here all groups with a central nonidentity element.

Corollary 80.3 (Commuting inhomogeneous equations [20]; `dsup:cor:simultaneous`). *Let $g_1, \dots, g_r$ commute, $g_1 \neq 1$, and $f_i \in \mathbf{No}_k$. Some x satisfies $(\sigma_{g_i} - 1)x = f_i$ for all i if and only if*

$$\text{ct}(f_i) = 0, \quad (\sigma_{g_i} - 1)f_j = (\sigma_{g_j} - 1)f_i \quad (\text{all } i, j), \quad (80.8)$$

and then x is unique modulo k .

Proof (second route, 20). Necessity is immediate. Take $x = (\sigma_{g_1} - 1)^{-1}f_1 \in \mathbf{V}_k$; the error $e_i = (\sigma_{g_i} - 1)x - f_i \in \mathbf{V}_k$ satisfies $(\sigma_{g_1} - 1)e_i = 0$ by compatibility, so $e_i = 0$. Two solutions differ by an element fixed by g_1 , a constant. $\square$

`dsup`'s route solves the system with the explicit Koszul contraction $y = \sum_j Q_j E_{j-1} f_j$ (`dsup:eq:simultaneousolution`); 20 needs only the inverse of one nonidentity element, which exists because the global action is uniform [merge].

Remark 80.4 (The same normal forms relative to parameters [20]). For the relative action of [theorem 77.5](#), with $B = \mathbf{No} \setminus J$ and common fixed field $F_0 = K_{B,k}$, replace in [theorem 80.2](#) k by F_0 , ct by $\varpi_{\text{NF}(B)}$ and lc by mr_B , with the normalizations $\varpi_{\text{NF}(B)}m = 0$ and $\text{mr}_B(u) = 1$. These give valid unique normal forms for groups with a central nonidentity element: the projections are invariant under all group elements and the identity on F_0 , and [theorems 78.5](#) and [79.7](#) make every nonidentity difference bijective on the normalized kernels, so the two proofs apply unchanged. No weighted inverse with coefficients in $F_0 \setminus k$ is needed.

80.2 Free groups

Proposition 80.5 (Free-group comparison [20]; 20's Proposition 11.4). *For the free group F_r on $s_1, \dots, s_r$, $r \geq 2$, every additive cocycle is equivalent modulo a coboundary to exactly one with*

$$b_{s_1} = h_1, \quad b_{s_i} = h_i + v_i \quad (2 \leq i \leq r), \quad h_i \in k, \quad v_i \in \mathbf{V}_k, \quad (80.9)$$

so constants are not the only invariants. Multiplicatively there is a unique generator normal form $a_{s_1} = c_1$, $a_{s_i} = c_i w_i$ with $c_i \in k^\times$ and $\text{lc}(w_i) = 1$.

Proof. A cocycle on a free group is determined by arbitrary generator values, extended along reduced words with $b_{s^{-1}} = -\sigma_{s^{-1}}b_s$, which makes adjacent inverse letters cancel. Subtract $h_i = \text{ct}(b_{s_i})$ and the unique coboundary whose s_1 -value is the remaining part of b_{s_1} , possible since $\sigma_{s_1} - 1$ is bijective on $\mathbf{V}_k$; the other values stay arbitrary in $\mathbf{V}_k$. A coboundary preserving the normalized first value comes from an element fixed by s_1 , a constant, and vanishes. Multiplicatively remove $c_i = \text{lc}(a_{s_i})$, which extend to a character of F_r , normalize the first generator by [theorem 79.5](#), and argue the same way. $\square$

Free groups are left-orderable [23], so [theorem 76.1](#) supplies the action. Uniform one-automorphism solvability does not by itself trivialize the cocycles of noncommutative groups [20].

81 The surcomplex classification and set-sized models

81.1 Named conjugation

Every automorphism of $\mathbf{No}(i)$ preserving $\mathbf{Oz}[i]$ and commuting with cj is uniquely $\tilde{\alpha} \text{cj}^\epsilon$ with α an automorphism of $(\mathbf{No}, \mathbf{Oz})$ and $\epsilon \in \{0, 1\}$, so the centralizer of cj is $\text{Aut}(\mathbf{No}, \mathbf{Oz}) \times C_2$ in the joint-action sense (20's Proposition 12.1). This is the ring clause of [lemma 65.2](#) and, for the field, `saut:thm:axis`, which 20 credits; its proof (commutation preserves $\mathbf{No} = \text{Fix}(\text{cj})$ and $\mathbf{No} \cap \mathbf{Oz}[i] = \mathbf{Oz}$, and $i \mapsto \pm i$) is 18's. No strongness is assumed [20].

20's Theorem 12.2 is [theorem 60.4](#) without parameters: a set group acts faithfully on $(\mathbf{No}(i), \mathbf{Oz}[i], \text{cj})$ if and only if it is left-orderable or $H \times C_2$ with H left-orderable ($G_0 \times C_2$ in Part VIII), by the same route: the kernel of restriction to $\mathbf{No}$ has order at most two; if nontrivial its element acts as cj and is central, the sign $\epsilon : G \rightarrow C_2$ splits it off, and $\ker \epsilon$ acts faithfully on $\mathbf{No}$. Its consequence that finite subgroups are trivial or of order two, from torsion-freeness of left-orderable groups, is weaker than `saut:thm:torsion` (every nontrivial finite subgroup of $\text{Aut}(\mathbf{No}(i))$ has order two). The theorem requires named conjugation: it does not classify automorphisms preserving only $\mathbf{Oz}[i]$, all automorphisms of the pure field or all real forms, and it does not assert that arbitrary real field automorphisms fix $\mathbb{R}$ [20].

81.2 Set-sized models and saturation

Proposition 81.1 (A direct ZFC model for every left-orderable group [20]). *Let G be a set group with a left order, I its underlying ordered set, $\Gamma = \{\sum_{a \in S} r_a e_a : S \subseteq I \text{ reverse well-ordered, } r_a \in \mathbb{R}\}$ the ordered Hahn group (sign of the largest index), ι_g the inner lift of left translation, and $K = k((\omega^\Gamma))$ the set-sized Hahn field. Then the proofs of [sections 78 to 80](#) apply verbatim: the fixed fields are k , $\iota_g - 1$ is bijective for $g \neq 1$, the additive and multiplicative obstructions and the operator classification hold, and $\mathbb{Z} \oplus \Pi_{\mathbb{R}}$, $\mathbb{Z}[i] \oplus \Pi_{\mathbb{C}}$ are preserved.*

Proof. Left translation is a free increasing index action, so every step above goes through with Γ in place of the surreal exponent group; no cardinal arithmetic hypothesis is used. $\square$

What the class completion adds is the identification of the actual inner index order with $\mathbf{No}$, hence realization inside the specified surreal and omnific objects [20].

Proposition 81.2 (Saturated orders at a regular cardinal [20]; 20's Proposition 13.1). *Let κ be an infinite regular cardinal with $\kappa^{<\kappa} = \kappa$ and $|G| \leq \kappa$, G left-ordered. Then G has a free increasing action on an η_κ -order of cardinality κ .*

Proof. Run the construction of [proposition 77.1](#) for κ stages, inserting copies only at downward closures of subsets of size $< \kappa$, a G -invariant family. Each stage has at most $\kappa^{<\kappa} = \kappa$ such cuts and stays of size at most κ . Separated subsets of total size $< \kappa$ lie in one stage by regularity, and the next stage inserts a point between them. At least one point is added at each stage, so the size is exactly κ ; freeness and order preservation are unchanged. $\square$

For $\kappa = \omega$ every countable left-orderable group acts freely on a countable dense order without endpoints. This is not a free extension to the Dedekind completion, and for larger κ

no preservation of a chosen surreal birthday cutoff is asserted [20]. Compare the countable route of [remark 64.4](#), which gives faithful but not free actions.

82 Scope, relations and questions of source 20

82.1 What the theorems determine, and the proof obligations

[Theorem 76.1](#) classifies the set groups that *embed* as actions, not the automorphism object itself: it neither realizes a group as the entire automorphism group of a named structure nor classifies conjugacy classes of the actions. The fixed-field mechanism (an invariant reverse well-ordered support contains no nontrivial ordered orbit), the independence mechanism (separate inner monomials) and the solvability mechanism (summable downward half-orbits) are different consequences of one two-level construction. 20's dependency chain is: left order $\Rightarrow$ free equivariant cut completion $\Rightarrow \varphi_g$ on $\mathbf{No} \Rightarrow \iota_{\varphi_g} \Rightarrow \sigma_g$; invariant support $\Rightarrow \text{Fix}(\sigma_g) = k$; downward orbits $\Rightarrow \text{Gr}_c \Rightarrow$ additive and operator classification; $(\iota_{\varphi_g} - 1)^{-1}$ and Gr_1 on $\mathfrak{m}_k \Rightarrow$ the multiplicative criterion $\Rightarrow$ first-order and cocycle normal forms. Neither automatic strongness nor an exponential decomposition is an input [20].

Proof obligation	Check and boundary [20]
Class sizes	Stages, cuts at a stage, supports, witness collections and orbit unions of a set support are sets; automorphisms are one joint class action.
Freeness	Each new cut receives an entire regular G -orbit; fixed points are excluded for every nonidentity element, not only for the group.
Two lifts	The inner lift is additive, not multiplicative; the outer lift needs and uses that additivity.
Fixed fields	Equality of series gives invariant support; forward and backward orbits exclude a moved exponent.
Green summability	Reverse well-ordering and finite multiplicity are proved separately; scalar weights add no support; no convergence or cofinality.
Arithmetic	Positive support stays positive under forward and backward iterates; the constants of the omnific rings are $\mathbb{Z}$, $\mathbb{Z}[i]$.
Multiplicative step	The global exponent equation uses the free inner index action; the relative theorem uses both projections; logarithms only on $1 + \mathfrak{m}_k$.
Variable coefficients	After $x = uy$ the resonant obstruction is $\text{ct}(b/\sigma u)$, not $\text{ct}(b)$; the weighted formula uses coefficients in k , not in a relative fixed field.
Cocycles	The central element is used explicitly; the free-group normal form shows extra invariants without it.
Complex classification	Conjugation is named; the centralizer factorization is credited as prior.

Proof obligation	Check and boundary [20]
Verification	Finite exact-arithmetic checks test formulas and residuals, not infinite summability, class recursions, priority or formalization.

82.2 Status and relations

- *No named question is answered.* Checked against [questions 11.1, 11.3, 11.4, 32.1 and 32.15](#), the single-dilation report’s questions (`dsup:sec:questions`) and the surcomplex report’s (`saut:sec:questions`). 20’s automorphisms lie in the monomial factor, so [question 11.3](#) (the stabilizer in $\mathcal{U}_{\mathbf{Oz}}$) is not addressed, and none of the questions of [15, Section 5] on the exponential, simplicity or the omega-map is answered.
- *Partly answered here.* [Question 66.2](#): [theorem 77.5](#) realizes $E = \text{NF}(\mathbf{No} \setminus \mathbf{J})$, for every nonempty bounded open interval $\mathbf{J}$ and every left-orderable set group (status note after the question). 20’s own question on fixed sets beyond two-level complements is partly settled by [part VIII \(question 82.2\)](#).
- *Merged.* 20’s questions on additional surreal structure, conjugacy invariants, the Gaussian ring without conjugation and formalization are merged into [questions 32.15, 59.10, 66.3 and 66.7](#), with 20 added.
- *Stale against the tree.* 20’s contribution statement (the “combined structure” of universality, exact fixed fields and difference solvability, its audit’s Section 3) is stale at the merge for the universality and fixed-field half, which is [part VIII](#) (placed after its pin), and was incomplete at its pin for the difference half: the resolvent, the polynomial operators, the multiplicative obstruction, the commuting systems and the dilation cohomology were in the single-dilation report ([section 76.2](#)), which 20 did not read. Both are credited, and only the material listed after the table in [section 76.2](#) is printed as new. Its reading of the collection, and its crediting of the lifts and of the centralizer, are accurate.
- *Checked true.* The pin [e93a06d](#), its tree [42aec05](#) and its content; the 61-report catalogue at the pin; the KKS numbering it cites (the first and second lifts are Construction 3.7 and Proposition 3.9 of [15], 18’s double lift is its Construction 3.10); every proof of [sections 77 to 81](#) and the worked calculations below; its suite, rerun for the merge ([section C](#)). No mathematical error was found in 20.

Example 82.1 (A worked real equation [20]; 20’s Appendix A). For $f = 3\omega^2 - 5\omega^{-1} + 7$, the equation $(\sigma_1^{\text{tr}} - 2)x = f$ has the unique solution

$$x = -7 + 3 \sum_{n \geq 1} 2^{n-1} \omega^{2\omega^{-n}} + 5 \sum_{n \geq 0} 2^{-n-1} \omega^{-\omega^n} : \quad (82.1)$$

the constant part is $7/(1-2)$, the exponent 2 is in the upward region and -1 in the downward one, the two supports are reverse well-ordered and disjoint, and (78.4)–(78.5) give f coefficient-wise. $(\sigma_1^{\text{tr}} - 1)x = f$ has no solution since $\text{ct}(f) = 7$; without the constant it has a unique solution with $\text{ct}(x) = 0$, with the weights at $c = 1$. For a localized action replace 7 by the whole $\text{NF}(B)$ -supported part: a nonconstant series can then be fixed, so testing ct alone is not sufficient.

82.3 Questions of source 20

20 poses twelve questions (its Questions 1–12), proposed as research directions, not as published problems. Eight are new; its Question 1 is merged into [question 32.15](#), Question 3 into [question 66.3](#), Question 5 into [question 66.7](#) and Question 12 into [question 59.10](#).

Question 82.2 (Prescribed fixed sets beyond two-level complements [20]). Which additive subgroups $H \subseteq \mathbf{No}$ occur as $\text{Fix}(\iota_\varphi)$ for an ordered action free on the complement of its fixed indices, every nonidentity element having the same fixed set? Which full Hahn subfields occur as the common individual fixed field of an arbitrary left-orderable group?

$\text{NF}(\mathbf{No} \setminus J)$ is a large explicit family; subgroups not defined by supports are a separate problem [20]. *Status* (batch 34): by (77.2) every such H is $\text{NF}(B)$ with B the common fixed class; [theorem 62.6](#) realizes every set B and [theorem 77.5](#) every complement of a bounded open interval, for every left-orderable set group. The second clause is [questions 66.1](#) and [66.2](#). Open otherwise.

Question 82.3 (Full automorphism-group realization [20]). Can every set-sized left-orderable group be realized as the entire symmetry group of a naturally expanded surreal structure, not merely as a subgroup? What parameter or predicate resources are needed?

Naming a set of ordinary parameters does not eliminate the nontrivial actions ([theorem 77.5](#)) [20].

Question 82.4 (Higher-order variable coefficients [20]). For $L = \sum_{j=0}^n a_j \sigma^j$ with $a_j \in \mathbf{No}_k$, describe kernel and image by finitely or transfinitely structured support invariants. Does a slope decomposition reduce some such operators to [theorem 80.1](#)?

One leading coefficient suffices in order one; no scalar factorization theorem for all operators is proved [20].

Question 82.5 (Matrix difference equations [20]). For $A \in \text{GL}_n(\mathbf{No}_k)$, when does $\sigma Y = AY$ have a fundamental matrix in $\text{GL}_n(\mathbf{No}_k)$? Is there an exact matrix analogue of the leading-coefficient obstruction, possibly with several invariants?

A determinant gives only a necessary rank-one test [20].

Question 82.6 (Relative first-order equations [20]). For the relative action with common fixed field F_0 , when is $\sigma - c$ invertible for $c \in F_0^\times$, not only $c \in k^\times$? Does this solve every scalar first-order equation relative to F_0 ?

[Theorem 79.7](#) gauges a to $\text{mr}_B(a) \in F_0^\times$; the remaining twisted operator has a fixed-field coefficient that may itself have infinite support, and the scalar weights of (78.1) do not settle that [20].

Question 82.7 (Centerless groups and higher cocycles [20]). Extend the cocycle classification beyond groups with a central nonidentity element and free groups. How do relations, amenability or geometric properties influence the extra zero-constant invariants? What happens for higher cocycles?

[Proposition 80.5](#) is a test case against an unjustified general vanishing principle [20]. For $\mathbb{Z}^d$ acting by rational dilations all degrees are in `dsup:thm:dilationcohom`.

Question 82.8 (Small workspaces and birthday bounds [20]). For a group and a set of input equations, find a smallest useful set-sized Hahn workspace stable under the action and all normalized inverses. Can a transparent birthday bound be attached to it?

Countably many orbit translates preserve set support but do not control birthdays of new exponents [20]. Compare dsup’s set localization, which bounds cardinality but not birthdays, and [questions 32.16](#) and [66.5](#).

Question 82.9 (Effective Green operators [20]). For computable descriptions of exponents and index actions, which instances of [\(78.1\)](#) admit algorithms deciding coefficient queries, support membership or leading terms? Which finite certificates verify an exact solution?

The class back-and-forth is not an algorithm; the translation family σ_s^{tr} is a better first target [20].

A Provenance ledger

A.1 Section by section

Section	Sources and choices
1	Structure from 09; imported results from all three automorphism sources; the Fact 4.2 recheck is the merge’s (remark 1.2) . 10’s provenance and symbols were added with it (batch 28); item (8) of section 1.1 and the rows and paragraph for 11, 12 and 13 in section 1.2 were added in batch 30; item (9), the row and paragraph for 14, and the counts on the title page and in the abstract were updated in batch 31; items (10) and (11), the rows and paragraph for 15 and 16, the abstract and the counts in batch 32; items (12)–(14), the rows and paragraph for 17, 18 and 19, the abstract, the notation pointer and the counts in batch 33; item (15), the row and paragraph for 20, the abstract, the notation pointer and the counts in batch 34.
2	04’s one-shift lemma and coefficient polynomial; 09’s and 04’s ring decompositions; 04’s and 08’s divisible part. Units and constant products are cited from the Diophantine report, not reprinted. 10’s one-shift lemma is lemma 2.1 .
3	The criterion is printed once. <i>Choice:</i> 09’s generality (k of characteristic zero, unital $\mathfrak{o}$) with 04’s import-free proof; the first implication is 09’s, which removes 04’s use of $\bigcap n\mathfrak{o} = 0$. 09’s derivation form is printed as theorem 3.4 ; its operator form, which uses cited theorem 1.1 , as corollary 3.5 . Example 3.2 is from the placement dossier’s check. Section 3.3 is 10’s: its support-cut form printed as further equivalent conditions, its least-forbidden-shift proof as a marked second route, necessity without strongness, and Archimedean blocks.

Section	Sources and choices
4	Shears from 04 and 09 (after <code>saut:thm:shiftflow</code>); exact displacement from 04 (08 and 09 have the same computation); 09's uniform shifts; 04's common-shift correspondence as a second, proved route; 08's divisibility-free threshold as a marked case; the translation example from 04, 08, 09 and the surcomplex report; rank dichotomy from 04 and 09 (08's for single flows); bottom subgroup from 04; phase shears from 08. Section 4.7 is 10's: twists with a general profile, the binomial family, their fixed fields, finite orbits and algebraic independence; 10's rank witness and divisibility example are recorded in remark 4.23 .
5	Factorization from 04 and 09 (credited to [17] and <code>saut:thm:decomp</code>); all pair automorphisms from 08 and 09 through <code>odg:def:thm:autreal</code> . 10 re-derives both.
6	Bracket law from 04, 08, 09; filtration, rank bound, balanced brackets, mixed-parameter lemma and sharpness from 09; 04's commutator formula and 08's faithful $\mathbb{Z} \wr \mathbb{Z}$ printed as one group in 04's normalization; the all-times remark is the merge's. Example 6.13 is 10's.
7	Fixed field $\mathbb{R}$ from 04, 08, 09 (explicit choice of 08, 09); separation, hull, monomial parameters and closure from 04; $\mathbb{R}^\kappa$ and nonsolvability from 09. Section 7.4 is 10's.
8	Reconstruction and definitions of $\mathbb{Z}$ and Π from 04, 08, 09 are cited to the Diophantine report; printed extras: 08's order formula, the projections (09), floor (08), truncation and the set-sized version (09).
9	Set-parameter nondefinability from 04, 08, 09 printed once with 04's strengthening; representative rules from 09; twists, one-term class and RV from 04; contrasts added by the merge. Section 9.2 is 10's; the outer bound in corollary 9.11 and the remark on <code>cor:question54</code> are the merge's.
10	Complex criterion and conjugation-compatible witnesses from 09 and 04; real axis inside the kernel from 04; wild lift and Gaussian fixed field from 08 and 09 (09's definition of $\mathbb{C}$ is cited to the Diophantine report); proposition 10.5 from 10.
11	Status records and questions: 04's four, 08's first and third, 09's nonstrong question (merged with 04's first), and the merge's non-divisibility question. 10's four questions: nonstrong maps and generation are merged into questions 11.1 and 11.4 , definable closure is recorded under question 11.3 , and other integer parts is question 11.7 ; the item on <code>dsn:q:languages</code> and the correction of 10's novelty statement are the merge's.
part II	Manuscript 02 throughout, with credits to <code>bst:thm:fieldcriterion</code> , <code>bst:prop:localdensity</code> , <code>osq:prop:classder</code> and <code>odg:thm:fractions</code> ; section 19 is the merge's. The batch-33 status notes after corollary 17.9 and questions 19.2 and 19.3 and in section 19 are the merge's, from 17 and 19.

Section	Sources and choices
Part III	Manuscripts 11, 12 and 13 (batch 30). 20: the merge's. 21–24: 12's spine with 13's binary witness (second route), set-pair isomorphisms, coefficient matrix and example 23.6; the reconstruction is cited, not reprinted; the convex criterion reproved by 12 and 13 is theorems 3.3 and 3.8; corollary 24.4 is the merge's. 25: 12's and 13's adjoint theory, 12's general retraction and tower law, 13's kernel statement. 26–29: 11's, with the embeddings of 12 and 13 as the instances of theorem 27.3, 12's image corollary (extended to all three by the merge), 13's non-homogeneity; the Taylor automorphism is <code>saut:thm:coeffflow</code> (credit added for 12 and 13); the gap C_ω in lemma 27.2 is the merge's. 30: 12 and 13 for automorphisms, 11 for embeddings. 31: 12's regular cardinals, 13's small integer parts, both sources' distinctions. 32: the thirty questions of the three sources (ten each) merged into eighteen, with 12's and 13's admissibility question recorded under question 11.2; the status records and the note on the large-cardinal report are the merge's. Batch 31: 14 added to the sources of the results it re-derives, pointers to part IV, and 14's comments and status notes after questions 32.1, 32.3, 32.11 to 32.14 and 32.18.
part IV	Manuscript 14 (batch 31). 33: the merge's, from 14's comparison with the report. 34: 14's matching lemma and detection proof (third route), its cancellation example and topology remark. 35: 14's definition and proposition; the Puiseux example, the weakened hypothesis after the proposition and remark 35.4 are the merge's. 36: 14's formal evaluation, matrix formula, $\Gamma = \mathbb{R}$ remark, shear and truncation example and Gaussian statements, translated; proposition 36.5 is the merge's. 37: 14's eight questions, one new, seven merged, and the stale-novelty record.
part V	Manuscript 15 (batch 32). 38: the merge's, from 15's comparison and the blob check. 39: 15's Section 3, with its scope and audit remarks; remark 39.6 and the comparison with <code>found:sub:tsum</code> in remark 39.13 are the merge's. 40: 15's disjoint-row lemma (a variant of the third route), interpolation, cancellation example, Gaussian ring corollary and class description; the comparison with 14's matching and the reading of proposition 40.5 as an answer to the first clause of question 32.11 are the merge's. 41: 15's Section 11. 42: 15's twelve questions, six new, six merged. Its algebraic half is printed in part III, with 15 added to the sources.
part VI	Manuscript 16 (batch 32). 43: the merge's, from 16's comparison with source 02. 44–51: 16's Sections 2–10, translated to external monomials U^g ; its Proposition 4.1, Definition 6.1, Proposition 6.2 and Example 6.3 are printed in part II and the Diophantine report. 52: 16's eleven questions, ten new, one merged; the relation to <code>odg:def:q:coeffpe</code> is the merge's. The status notes and pointers in part II are the merge's.

Section	Sources and choices
part VII	Manuscript 17 (batch 33). 53: the merge's, from 17's Section 1 and its comparison with part II; the principal results are 17's Theorems 1.1–1.3 and Corollary 1.4. 54–58: 17's Sections 2–7 and Appendix B, translated to D_0 , C_0 , $\mathcal{E}_{0,s}$, Υ_v ; its Lemma 2.1 is cited to the Diophantine report and its monomial comparison to section 17; corollary 58.7 is the merge's, from 17's remark after its first question. 59: 17's Sections 8–9, its ten questions (eight new, two merged with 18's and 19's).
part VIII	Manuscript 18 (batch 33). 60: the merge's, from 18's Section 1 and Appendix B; the identification of its lifts with ι_φ and M_{1,ι_φ} is the merge's. 61–65: 18's Sections 2–9; its Lemmas 2.1 and 8.2 are cited to the Diophantine and surcomplex reports; remark 63.5 and the remarks on <code>saut:thm:torsion</code> and <code>saut:thm:phase</code> are the merge's. 66: 18's Sections 10–11 and Appendix A, its ten questions (eight new, two merged).
part IX	Manuscript 19 (batch 33). 67: the merge's, from 19's Section 1 and source audit. 68–74: 19's Sections 2–11 and Appendix B, with s for 19's T ; its Lemmas 3.1, 3.2 and 11.1 are cited; its Theorem 8.2 and Corollary 8.3 are printed once with part VI as corollary 72.2 and remark 72.3, its Lemma 8.1 kept as a second route; the relations to example 51.1, theorem 4.8, corollary 17.7, and lemma 7.1 are the merge's. 75: 19's Section 12, Appendix A and its eleven questions (seven new, four merged).
part X	Manuscript 20 (batch 34). 76: the merge's, from 20's Section 1, Appendix B and source audit, with its Theorems 1.1, 1.2 and Example (1.3)–(1.4). 77–81: 20's Sections 2–13; its Lemmas 2.3, 3.2, 4.1, 4.3, Definition 3.3, Theorem 4.4, Corollaries 4.5, 5.2, Proposition 12.1 and Theorem 12.2 are printed once in part VIII (or cited to the Diophantine report); its Theorem 3.4 is kept as a second route (proposition 77.1); its Lemma 6.2, Theorem 7.1, Theorem 8.1, Corollary 8.2, Theorems 9.2, 9.4 and Corollary 11.3 are printed with their correspondence to the single-dilation report; remarks 77.6 and 78.6, the comparisons with <code>dsup:</code> , <code>adr:</code> and <code>saut:thm:torsion</code> , and the relation of theorem 79.5 to <code>dsup:thm:multisingle</code> are the merge's. 82: 20's Section 14, Appendix A and its twelve questions (eight new, four merged).

A.2 Printed once, and where the merge chose

- *The classification* (04 thm. criterion, 08 thm. threshold, 09 thms. derivation and stabilizer): theorem 3.3 with 04's proof, theorem 3.4, corollary 3.5, and 08's theorem 4.10 as the divisibility-free single-flow case. 04's $a_{\sigma,\delta}$ and 09's c_δ are kept as different functionals with equivalent conditions (example 3.2).
- *Rank triviality* (04, 09; 08 for single flows): theorem 4.13. *Common fixed field* $\mathbb{R}$ (04, 08, 09): theorem 7.2. *Set-parameter nondefinability* (04, 08, 09): theorem 9.1.

- *The metabelian non-nilpotent group*: 04's $\langle A_s, B_u \rangle$ and 08's $\langle A_1, B_1 \rangle$ are the same group at $s = u = -1$; printed once as [theorems 6.8](#) and [6.10](#).
- *Strong factorization* (04, 09): [theorem 5.1](#).
- *Not reprinted* (cited to the Diophantine report): units and constant products, Pell rigidity, $\text{Frac } \mathbf{Oz} = \mathbf{No}$, the multiplier ring, the reconstruction of $\mathbb{R}$ and of the order, automorphisms fix $\mathbb{R}$, the split restriction to $\text{Aut } \mathbb{C}$, the first-order and existential definitions of $\mathbb{Z}$, Π and $\mathbb{C}$.
- *Kept as second routes*: 04's common-shift correspondence ([theorem 4.8](#)) beside the imported one; 04's separation lemma beside the explicit choice in [theorem 7.2](#); 09's Pell-divisibility definition of $\Pi_{\mathbb{C}}$ is mentioned beside the Diophantine report's.
- *Added by the merge* (tagged [merge], complete proofs): the combined statement of [theorem 3.3](#); [example 3.2](#); [remark 6.11](#); [question 11.6](#); with source 10, the converse in [theorem 4.18](#) and the outer bound in [corollary 9.11](#).
- *Source 10, printed once* (batch 28): its criterion (10 thm. criterion, cor. logcut) as [theorem 3.3](#) with the further equivalent conditions of [theorem 3.8](#); its rank theorem as [theorem 4.13](#); its stabilizer splitting as [theorem 5.1](#); its fixed field for $F = \exp \xi$ as [theorem 4.2](#); its parameter theorem as [theorem 9.1](#) (with the two-term witness added as [theorem 9.8](#)); monomials and the omega-map in its nondefinability theorem as [theorem 9.1](#) and [corollary 9.2](#); its inner-support bound as a consequence of [theorem 7.4](#); its Gaussian theorem as [theorem 10.1](#) with the extras of [proposition 10.5](#); its nonabelian example beside [theorem 6.8](#); its one-shift lemma as [lemma 2.1](#); its floor, fraction-field and coefficient-rigidity statements as [corollary 5.2](#), `odg:thm:fractions`, `odg:def:thm:multiplier` and `odg:def:thm:autreal`, not reprinted.
- *Source 10, kept as a second route*: its least-forbidden-shift proof of necessity ([theorem 3.8](#)), beside 04's polynomial argument; its direct proof of the inner-support bound ([corollary 9.11](#)).
- *Sources 12 and 13, printed once* (batch 30): scalar detection (12 thm. 3.3, 13 thm. 3.3) as [theorem 21.4](#); the constant-term isomorphism theorem (12 thm. 4.1, 13 thm. 4.1) as [theorem 22.1](#), with 12's weaker hypothesis; automatic strongness of $\text{Aut}(\mathbf{Oz})$ (12 thm. 5.5, 13 thm. 6.1) as [theorem 23.2](#); the set pairs (12 cor. 5.6, 13 prop. 5.5 and cor. 6.2) as [proposition 23.1](#) and [corollary 23.4](#); monomial determination (12 cor. 5.7, 13 cor. 6.3); the factorization (12 thms. 6.1–6.2, 13 thms. 7.1–7.2) as [theorem 24.1](#); scalar representability and adjoints (12 prop. 7.2, thm. 7.3; 13 thms. 8.1, 8.3); the retraction (12 thm. 7.4, 13 prop. 8.4) as [theorem 25.3](#); monomial adjoints (12 ex. 7.5, 13 ex. 8.5); the class functional (12 thm. 8.1, 13 ex. 11.2); the surcomplex automorphism theorems (12 thms. 10.1–10.2, 13 thms. 10.1–10.2). Their reprinted reconstruction (12 lems. 5.1–5.2 and lem. 5.4, prop. 5.3; 13 lems. 5.1–5.2 and prop. 5.3) is cited from the Diophantine report and [corollary 5.2](#), and their reproofs of the criterion are the two routes of [theorems 3.3](#) and [3.8](#).
- *Sources 11, 12 and 13, printed once*: the embedding moving a real (11 thm. 6.3, 12 thm. 9.5, 13 thm. 9.5) as [theorem 27.3](#), an instance of 11's classification; $J(\mathbf{No}) \cap \mathbb{R} = \ker d$ (12, 11 prop. 8.5) as [corollary 27.4](#) and [proposition 29.2](#); nonelementarity (12 rem. 9.8, 11 thm. 8.3) as [theorem 29.1](#); the surcomplex embeddings (12 prop. 10.3, 13 cor. 10.4) as the case $\varepsilon = 1$ of [theorem 30.3](#). 11's reproof of the Taylor automorphism is `saut:thm:coeffflow`; its constants and automorphism rigidity (11 prop. 8.1, cor. 8.2) are `odg:def:thm:multiplier`, `odg:def:cor:internal` and [corollary 5.2](#).

- *Kept as a second route* (batch 30): 13's binary finite-row proof of detection beside 12's triangular detector ([theorem 21.4](#)).
- *Added by the merge* (batch 30): [corollary 24.4](#), the gap C_ω for 12's exponent embedding ([lemma 27.2](#)), the extension of [corollary 27.4](#) to all three maps, and the second sentence of [question 32.13](#).
- *Source 14, printed once* (batch 31; its numbering): Thm. 1.1, Thm. 5.3 and Cor. 5.5 as [theorem 23.2](#) and [corollary 23.4](#); Lem. 2.2 as [lemma 2.1](#) and [lemma 21.1](#); Thm. 3.3 and Cor. 3.4 as [theorem 21.4](#); Thm. 4.1 and Cor. 4.2 as [theorem 22.1](#) and [corollary 22.2](#); Props. 5.1–5.2 as the cited reconstruction and [proposition 23.1](#); Lem. 5.4 as `odg:thm:fractions`; Cor. 5.6 as [corollary 23.5](#); Thms. 6.1–6.2 and Cor. 6.3 as [proposition 25.1](#) and [theorem 25.2](#); Thm. 7.1 as [theorem 25.6](#); Thm. 8.1 as [theorem 24.1](#); Thm. 8.2 as [theorem 3.3](#); Cor. 8.3 as [corollary 24.3](#); Prop. 9.1 as the inverse of [example 4.19](#); Prop. 10.1 and Thms. 10.2–10.3 as [theorems 30.1](#) and [30.2](#); Ex. 10.4 as the phase twists of `odg:def:lem:twist` (table in [section 33.2](#)).
- *Source 14, kept as a third route*: its induced-matching proof of detection ([lemma 34.1](#)), beside 12's triangular and 13's binary detectors.
- *Added by the merge* (batch 31): [example 35.3](#), [remark 35.4](#), and [proposition 36.5](#), the weakened hypothesis after [proposition 35.2](#), and the status notes after [questions 32.12](#) and [32.14](#).
- *Source 15, printed once* (batch 32; its numbering; table in [section 38.2](#)): Lem. 2.2 as [lemma 2.1](#) and [lemma 21.1](#); Thm. 4.2 as [theorem 21.4](#); Prop. 4.5 as [example 21.6](#); Thm. 5.1 as [theorem 22.1](#); Prop. 5.3 as a direction of [theorem 25.2](#); Lems. 6.1–6.2 as the cited reconstruction and [proposition 23.1](#); Thm. 6.4 as [proposition 23.1](#) and [corollary 23.4](#); Lem. 7.1 as `odg:thm:fractions`; Thm. 1.2, Cors. 7.2 and 7.4 as [theorem 23.2](#); Cor. 7.3 as [corollary 23.5](#); Thm. 8.1 as [theorem 24.1](#); Prop. 8.2 as [theorem 3.3](#); Rem. 8.3 as [remark 24.2](#); Cor. 8.4 as [corollary 24.3](#); Ex. 8.5 as [remark 36.4](#); Thms. 9.2–9.3 as [theorems 30.1](#) and [30.2](#); Thm. 10.2, Cor. 10.3 and Prop. 10.4 as [proposition 25.1](#) and [theorems 25.2](#) and [25.6](#).
- *Source 15, kept as a variant of the third route*: its disjoint-row proof of detection ([lemma 40.1](#)), beside 14's induced matching.
- *Source 16, printed once* (batch 32): Def. 6.1, Prop. 6.2 and Ex. 6.3 as [definition 12.1](#), [theorem 13.1](#) and [corollary 13.3](#) and the workspace of [section 18](#); Prop. 4.1 as `odg:thm:fractions`, `odg:prop:units` and `odg:def:lem:units`; Question 12.8 as [question 19.3](#). Its generalizations of [theorems 15.1](#) and [15.3](#) to [15.5](#) are printed as new results in [part VI](#), since they are stronger.
- *Added by the merge* (batch 32): [remark 39.6](#); the comparison with the foundations report in [remark 39.13](#); the reading of [proposition 40.5](#) as an answer to the first clause of [question 32.11](#); the remark after [question 42.2](#); the relation after [question 42.6](#); the relation after [question 52.9](#) and to `odg:def:q:coeffpe` ([section 52.2](#)); the status notes after [questions 11.4](#), [19.3](#), [32.1](#), [32.11](#), [32.12](#), [32.18](#) and [37.1](#); the pointers after [theorem 15.5](#) and in [remark 34.4](#).
- *Source 17, printed once* (batch 33; its numbering; table in [section 53.2](#)): Lemma 2.1 as `odg:thm:fractions`; the derivation of Proposition 2.2 as [theorem 17.6](#) with $b = 0$ (its constant field and $\text{ct}(D_0x) = 0$ printed in [proposition 54.1](#)); the ring and flow statements of Proposition 4.1 and ([55.2](#)) as [theorem 17.2](#); the monomial comparison of its Section 4.4 as [lemma 17.4](#), [theorem 17.5](#), and [corollary 17.9](#).

- *Source 18, printed once* (batch 33; table in [section 60.2](#)): Lemma 2.1 as `odg:thm:fractions`; Lemma 2.2 as [corollary 5.2](#) (its direct floor argument recorded after [proposition 61.2](#)); the lifts of Definition 3.1 as ι_φ and M_{1,ι_φ} ([lemma 27.1](#)); Lemma 8.2 as `saut:thm:axis`, with its ring clause printed; the infinite order in Corollary 6.2 as in `saut:thm:torsion`; the twist of Proposition 9.1 as the phase twist `saut:thm:phase`.
- *Source 19, printed once* (batch 33; table in [section 67.2](#)): Lemma 3.1 as [lemma 2.3](#); Lemma 3.2 as `odg:thm:fractions`; the derivations D_η of Lemma 6.2 as [theorem 17.6](#); Lemma 8.1 as [theorem 45.3](#) for $G = \mathbb{Z}^n$; Theorem 8.2 and Corollary 8.3 as [theorems 45.4](#) and [45.7](#), printed once in the joint generality [corollary 72.2](#); its conic and characteristic-2 warnings as [remark 45.6](#) and [example 51.2](#); Lemma 11.1 as [lemma 14.1](#).
- *Kept as a second route* (batch 33): 19’s genus-one proof of the quartic rigidity ([lemma 72.1](#)), beside 16’s Mason–Stothers and specialization proof.
- *Added by the merge* (batch 33): [corollary 58.7](#); [corollary 72.2](#) in its joint generality; [remark 63.5](#); the identification of 18’s lifts with ι_φ and M_{1,ι_φ} ; the remarks relating 18’s twist to the phase twists and its torsion statement to `saut:thm:torsion`; the relations of 19’s results to [theorem 4.8](#), [corollary 17.7](#), [lemma 7.1](#), [example 51.1](#), and [question 52.1](#); the merged questions [questions 59.6](#) and [59.10](#); the status notes after [corollary 17.9](#) and [questions 11.3](#), [19.2](#), [19.3](#), [32.15](#), [52.2](#) and [75.2](#), in [sections 19](#) and [52.2](#), after [theorem 7.6](#), and in [section B](#).
- *Source 20, printed once* (batch 34; table in [section 76.2](#)): Theorem 1.1 (i)–(iii) and its fixed fields as [corollaries 60.3](#) and [63.2](#) and [proposition 65.1](#); Lemma 2.3 as `odg:thm:fractions`; Lemma 3.2 as [lemma 63.1](#); Definition 3.3 as [definition 62.1](#); Lemmas 4.1–4.3 and Theorem 4.4 as [proposition 61.2](#), [lemma 61.4](#), and [theorem 61.5](#) (with [lemma 77.2](#) for the general outer lift); Corollary 4.5 as [theorem 60.1](#); Corollary 5.2 as [corollary 60.2](#); Proposition 12.1 as [lemma 65.2](#) and `saut:thm:axis`; Theorem 12.2 as [theorem 60.4](#). Its difference core, which re-derives `dsup:lem:orbit`, `dsup:thm:resolvent`, `dsup:cor:fixed`, `dsup:cor:poly`, `dsup:thm:multsingle` and `dsup:cor:simultaneous` for class-sized double lifts, is printed in [part X](#) with each correspondence stated, since the single-dilation report states these only for set-sized fields and, on **No**, for rational dilations.
- *Kept as second routes* (batch 34): 20’s cut filling by a regular copy of G at every cut ([proposition 77.1](#)), beside 18’s fibers over cut orbits; its one-operator proof of [corollary 80.3](#), beside `dsup`’s Koszul contraction.
- *Added by the merge* (batch 34): [remarks 77.6](#) and [78.6](#); the dictionary between 20’s operators and `dsup`’s (chambers, Q_c , P_1) and the specialization remarks after [theorems 79.5](#) and [79.7](#) and [corollary 80.3](#); the status note after [question 82.2](#); the merged questions [questions 32.15](#), [59.10](#), [66.3](#) and [66.7](#); the status notes after [questions 11.3](#) and [66.2](#).

A.3 Corrections and stale statements

1. *Divisibility* (09): its remark that divisibility is essential is correct for $\mathcal{A}_k$ and not for $\mathcal{R}_{\mathbb{Z}}$: the translation does not preserve $\mathbb{Z} + t^{-1}\mathbb{R}[t^{-1}]$ ([remark 4.11](#)).
2. *Fact 4.2 of [15]* (09): the placement dossier doubted that it states the bijection; the merge re-read the PDF and found that it does, with only an assertion for proper classes ([remark 1.2](#)). 09’s claims $\mathcal{U}_{\mathbf{0}\mathbf{z}} = \text{Exp}(\mathcal{L}_A)$ and the upper half of “exactly $r - 1$ ” in its nonsolvability theorem rest on that assertion and are stated so ([corollary 3.5](#) and [theorem 7.7](#)).

3. *Stale novelty* (04, 08, 09): their reconstruction, first-order recovery of $\mathbb{R}$ and definitions of Π were written at pins `fb5c4b53` and `befe739b`, before the Diophantine report's reconstruction section was placed; they are cited, not claimed ([section 8](#)). 04's remark that its reconstruction is not first order is superseded by `odg:def:cor:internal`. 08's definition of Π has the structure of `odg:cor:definablect`, which was in the tree at 08's pin.
4. *08's Question 2* is answered ([section 8](#)) and dropped.
5. *Credits added*: `saut:thm:shiftflow` (08 and 09 did not credit it), `saut:thm:decomp` (04 and 09 credited only [17]), the rigidity report's `thm:no` and `cor:question54` (09 cited only [15, Proposition 5.2]), `bst:thm:fieldcriterion`, `bst:prop:localdensity`, `osq:prop:classder` and `odg:thm:fractions` (for 02); and the surcomplex report's $T_1(t) = t/(1-t)$ is noted not to preserve **Oz** ([example 4.12](#)).
6. *Repository statements checked true*: at both pins the catalogue listed 51 reports (04, 08); the fixed-shift flow is `saut:thm:shiftflow` (04); phase twists preserve **Oz**[i] and move the real axis (04); the exponential-rigidity question is answered in its report (04); Pell rigidity and a definition of $\mathbb{Z}$ were in the Diophantine report (08); $\text{Frac } \mathbf{Oz} = \mathbf{No}$ was there (09, 02); no report contained an **Oz**-automorphism classification (04, 08, 09). 04 reports two date fields for [15] v3 (23 April and 24 August 2026); it is cited as version 3 of 23 April 2026, as elsewhere in the collection.
7. *Stale novelty* (10): at its pin `58cd8e1`, which does not contain this report, 10 stated that no inspected report guide contains the support-cut criterion and proposed the criterion, the rank boundary, the specialized fixed fields and the parameter witnesses as new. At the merge these are [theorems 3.3](#), [4.2](#), [4.13](#) and [9.1](#); 10 is credited as an independent fifth source, and only the material listed in [section 1.2](#) is printed as its contribution ([section 11.1](#)).
8. *Checked true* (10): $\text{Frac } \mathbf{Oz} = \mathbf{No}$ is in the Diophantine report (`odg:thm:fractions`); the multiplier identity, real-coefficient rigidity and the Gaussian phase twists recorded in that report's reconstruction audit are now `odg:def:thm:multiplier`, `odg:def:thm:autreal` and `odg:def:lem:twist`; 10 credits them and its own proofs of them are not claimed as new. No mathematical error was found in 10.
9. *Status notes* (batch 30): [question 11.1](#) is answered by [theorem 23.2](#). The statements that recorded it as open or partly answered now carry status notes: [remark 3.6](#), [remark 3.9](#), the paragraph after [theorem 5.1](#), [corollary 5.2](#), the item on `osq:q:invisible` in [section 11.1](#), and the paragraph after [question 11.1](#). No label, statement number or earlier text was removed; the non-claims of 04, 09 and 10 are kept as source records with a status note ([section B](#)).
10. *A false reason, corrected* (09; merge text of batch 26): after [corollary 10.4](#) the report said that a real automorphism “preserves order and cannot move an ordinary real”. Read literally this is false: the Taylor automorphism `saut:thm:coeffflow` is strong and order preserving and moves a real b to $b + t^\delta$. The correct reason is that an automorphism preserving **Oz** fixes $\mathbb{R}$, which is definable in the pair. The sentence is corrected in place; no result depended on it, so this is a correction, not a retraction.
11. *Credits added* (batch 30): `saut:thm:coeffflow` for the Taylor motion used by 12 and 13, which do not credit it (13's audit calls it an established mechanism); 11 credits it.
12. *Repository statements checked* (batch 30): 12's statement that this report's Theorem 5.1 is the strong factorization and that its Corollary 5.2 fixes $\mathbb{R}$ and commutes with the projections and floor without strongness, and that 10's audit disclaims a classification of

nonstrong automorphisms, is accurate at its pin `a6c68ac`. 13's statements on [theorems 3.3](#) and [5.1](#) and on the Diophantine reconstruction, and 11's on the surcomplex Taylor motions and exponent lifts, the Diophantine ideal test, multiplier and fraction field and the automorphism rigidity of this report, are accurate at their pins. Their statements that no report proves automatic strongness, scalar detection or coefficient-moving embeddings are still true at the merge.

13. *Wrong statements in delivered files* (batch 30), disclosed and not edited, since delivered files are kept byte-identical: 11's audit (`11-coefficient-gaps-SOURCE_AUDIT.md`, Section 2) calls `a6c68ac3826ac337762b71a2ef901997d019260c` "a tree SHA, not a commit SHA"; it is a commit (it is 12's pin), whose tree is `12bfe3`. 11's pin `cf56b89`, its tree `21d04b0` and its four blob hashes are correct. 13's audit (`13-omnific-isomorphisms-SOURCE_AUDIT.md`) gives `71c1710` as the blob of the top-level `README.md` at its pin `0865f04`; that blob is the file at the earlier commit `b895e86`, and at `0865f04` it is `8c636c7`. Neither affects the mathematics.
14. *Stale novelty* (14, batch 31): at its pin `3d40856`, which contains the batch-26 text of this report but not `20c4c9f`, 14 stated that the strongness hypothesis "can be removed" from this report's real stabilizer classification and that it supplies the missing implication. That was accurate at the pin and is stale at the merge: sources 12 and 13 answered [question 11.1](#) in `20c4c9f`. 14 is credited as an independent third derivation ([section 37.2](#)); no mathematical error was found in it.
15. *Delivered-file behaviour* (14, batch 31): 14's script, shipped as `code/14-automatic-summability-verify.py`, always writes `data/verification.json` under the parent of its own directory and has no output option; its delivered `README` says it writes `data/verification.json`. Run in place it would create an unprefixed `data/verification.json` in this report; the file is not edited, and the collection `README` gives rerun instructions on a copy.
16. *Repository statements checked* (15, batch 32): 15 pins no commit but two blobs of this report's source; `059939b5` is the file written in `b3fa9e2` and `60f3b172` the file written in `20c4c9f`, and its three audit blobs are those of `21375f8`. Its statements that [question 11.1](#) was open in the first and answered by 12 and 13 in the second, and that the inspected audits do not contain its topological classification, are accurate. Its reading omitted [part IV](#) and the congruence topologies of the neighbouring reports ([section 38.1](#)). No mathematical error was found in 15 ([section 42.2](#)).
17. *Repository statements checked* (16, batch 32): its pin `1ab41af` and tree `fba47f2` are as stated, its description of source 02 is accurate, and the SHA-256 hashes of its build record match its delivered files. No mathematical error was found in 16 ([section 52.2](#)); its general automorphism criterion and pair reconstruction rest on an unrefereed preprint ([cited theorem 49.1](#)).
18. *Delivered-file behaviour* (15, 16, batch 32): `code/15-topological-collapse-verify.py` writes `verification.json` beside itself unless `--output` is given, so run in place it would create `code/verification.json`; `code/15-topological-collapse-Makefile` names the delivered `Surreal_Arithmetic_Topological_Collapse.tex` and `verify.py`. `code/16-coefficient-recovery-verify.py` likewise writes `verification.json` beside itself unless `--output` is given. `code/16-coefficient-recovery-build.py` creates `build_logs/` beside itself, then runs `verify.py` and `pdflatexarticle.tex` in its own directory, rewriting `article.pdf` and `verification.json` there; in this report those delivered names are absent, so it creates `code/build_logs/` and stops. The files are not edited; the collection `README` gives rerun instructions on a copy.

19. *Repository statements checked* (17, 18, 19, batch 33): 17 and 18 pin `efc5446` (18 correctly gives its tree `aba6982`), the eight-source text with [parts I to III](#); 19 pins `889b87c`, the nine-source text with [part IV](#); none contains [parts V and VI](#). Their descriptions of this report are accurate at their pins: 17's of [theorem 17.5](#) and [corollary 17.9](#) and of the lower-bound non-claim, 18's of the guide's fixed-field, hull, stabilizer, wreath-product and nondefinability results, 19's of [question 19.2](#) as open. The catalogue listed 61 reports at both pins; "left-orderable" occurs in no other report of the collection. 19's formula Q was new at its pin and is 16's at the merge, so it is printed once ([corollary 72.2](#)). The SHA-256 hashes in 17's build record match its delivered files. No mathematical error was found in 17, 18 or 19 ([sections 59.2, 66.2 and 75.2](#)).
20. *Status notes* (batch 33): [question 19.2](#) is answered by [theorem 67.1](#); the passages recording it as open (the status after the question, the item in [section 19](#), the sentence in [section 52.2](#)) carry status notes, as do the lower bound after [corollary 17.9](#) and 02's sixth non-claim. No label, number or earlier sentence was removed.
21. *Delivered-file behaviour* (17, 18, 19, batch 33): `code/17-formal-orbit-fields-verify.py` writes `--output`, by default `verification.json` in the working directory; `code/17-formal-orbit-fields-Makefile` runs `latexmk` on `article.tex` and `pythonverify.py--outputverification.json` in the working directory, so run from this report's root it would rebuild this report in place and then fail. `code/18-exact-symmetries-verify.py` writes `verification.json` beside itself unless `--output` is given (run in place: `code/verification.json`); `code/18-exact-symmetries-build.sh` changes to its own directory and runs `python3verify.py`, absent under that name, so it stops; `code/18-exact-symmetries-Makefile` calls `build.sh` and `verify.py` by their delivered names, and its `clean` target deletes `article.*` auxiliary files in the working directory. `code/19-formal-symmetry-verify.py` writes `data/verification.json` under the parent of its own directory unless `--output` is given, so run in place it would create an unprefix `data/verification.json` in this report; `code/19-formal-symmetry-build.sh` changes to its own directory, creates `code/build/` and `code/data/`, and then fails on the absent `code/code/verify.py`. The build records of 18 and 19 give PDF sizes but no file hashes. The files are not edited; the collection README gives rerun instructions on a copy.
22. *Repository statements checked* (20, batch 34): its pin `e93a06d` (tree `42aec05`) contains [parts I to IV](#) and the single-dilation report, but not the write of [parts V and VI](#) nor [parts VII to IX](#). Its description of the five guides it read, its credits of the lifts to [15] and of the centralizer to `saut:thm:axis`, and its KKS numbering are accurate. No mathematical error was found in 20 ([section 82.2](#)).
23. *Stale novelty* (20, batch 34): its contribution statement is stale at the merge for universality and exact fixed fields, which are [part VIII's](#), and was incomplete at its pin for the difference-equation core, which the single-dilation report already contained; both are credited, and only the material listed in [section 76.2](#) is printed as new.
24. *Delivered-file behaviour* (20, batch 34): `code/20-universal-symmetries-verify.py` writes `verification.json` beside itself unless `--output` is given (run in place: `code/verification.json`); `code/20-universal-symmetries-Makefile` runs `latexmk` on `article.tex` and `python3verify.py` in the working directory, so run from this report's root it would rebuild this report in place and then fail on the absent `verify.py`, and its `clean` target runs `latexmk-c` there. `data/20-universal-symmetries-BUILD_REPORT.json` gives file sizes but no hashes, and its recorded PDF (25 pages) and `README.md` are not shipped. The files are not edited; the collection README gives rerun instructions on a copy.

A.4 Symbols renamed from the sources

$\mathcal{J}$ (08, 09) and $\mathcal{I}$ (04) are Π ; B (09), $\mathcal{A}$ (04), $\mathcal{B}_k$ (08) are $\mathcal{A}_k$; A_D (09), O_D (04), A_k (08), $R_\Gamma(D, k)$ (02) are $\mathcal{R}_o$ or $\mathcal{R}_o(k, \Gamma)$, with the coefficient ring D renamed o because D names derivations; $\mathcal{P}_\Gamma(k)$ (02) is $\Pi_k(\Gamma)$; H_δ (08) and $H(a)$ (02) are C_δ and, in [part II](#), $C(a)$; $\mathcal{U}_{A_D}$ (09) and $\mathcal{U}_D(G)$ (04) are $\mathcal{U}_{\mathcal{R}_o}(\Gamma)$; $D_{\varphi, \delta}$ (08) is $D_{\delta, -\varphi}$; q_η (09), $[\omega^\beta] \cdot$ (04), c_a (08) and χ_b (02) are q_β ; 09's $H = \omega^\eta$ and H_j are $\eta = \omega^\beta$ and η_j ; 08's A_1, B_1, u, E are $A_{-1}, B_{-1}, u_0, \exp$; 02's covering element c is z , its K_b and H_j are K_{bd} and $\Gamma_{\leq j}$. 08's ω -convention statements are translated to the t -convention ([section 1.5](#)); [part II](#) keeps 02's large-monomial convention ([section 12.1](#)). Source 10 (large-monomial convention, $X^g = t^{-g}$, translated): H_h is C_δ and C_g is H_g (letters swapped); $\text{pr}_+, \text{ct}, \text{pr}_-$ are P_-, P_0, P_+ (subscripts reversed); $\mathcal{I}$ is Π ; R_D is $\mathcal{R}_o$; B is $\mathcal{A}_k$; $\mathcal{U}_D$ is $\mathcal{U}_{\mathcal{R}_o}(\Gamma)$; the shift h is δ ; u_g is $1 + \varepsilon_\sigma(-g)$; λ_b is q_b ; P and T are Θ and ξ ; $\sigma_{\lambda, F, h}$ is $\sigma_{h, -\lambda}^F$; $\mathcal{N}^\dagger$ is $\mathfrak{S}^\dagger$ ([section 1.5](#)). Sources 11, 12 and 13 (all in the t -convention; the table of [section 20.3](#)): $I_\Gamma, A_G, \mathcal{J}_\Gamma$ are $\mathcal{R}_\Gamma$; $A_\Gamma, B_\mathbb{R}, \mathcal{A}$ are $\mathcal{A}_\Gamma$; 12's ℓ_g and 13's L_γ are $h_\sigma(g)$; 11's C_τ is $\mathbf{G}_\tau$ and its H is $\tau(\Gamma)$, its $\mathcal{F}$ is C_1 , its Λ is Δ , its $\Phi_{\tau, \rho}$ is $J_{\tau, \rho}$, its $E_{d, \delta}$ is $\mathcal{T}_{d, \delta}$, its S_τ is $M_{1, \tau}$, its T_h is ν_μ , its p is μ_{11} , its h_{jk} are $\mu^{\pm\pm}$, its h_c, h_j, h in the parameter theorems are $\mu^{\text{cmp}}, \mu^{\text{jmp}}, \mu^{\text{rel}}$, its $\mathcal{B}_\lambda$ is $\mathcal{V}_\lambda$, its J_Γ, L_Γ are $\mathcal{R}_\Gamma^\mathbb{C}, K_\Gamma^\mathbb{C}$; 12's θ is $\iota_{\mu_{12}}$, its ψ is μ_{12} , its D, ρ_D and J_D are $d, \rho_{d,1}$ and $J_{\mu_{12}, d}$, its $j^\dagger(1)$ and E_j are e_ϕ and ret_ϕ , its Λ is $\lambda_{\mathbf{On}}$, its $\mathbf{Sc}$ is $\mathbf{No}(i)$; 13's second τ is $\iota_{\mu_{13}}$, its p is μ_{13} , its $\rho = \sigma|_k$ is α , its D, E_D and F_D are $d, \rho_{d,1}$ and $J_{\mu_{13}, d}$, its T^* is $T^\dagger$, its λ is $\lambda_{\mathbf{On}}$, its $\mathcal{P}$ is Π , and its factorization $U \circ M$ is $M \circ u$ with $U = MuM^{-1}$. The large-cardinal report's J is $\hat{j}$. No normalization changes. Source 14 (the table of [section 33.3](#)): its $A_\Gamma = \mathbb{Z} \oplus \Pi$ and A are $\mathcal{R}_\Gamma$; its B is $\mathcal{A}_k$ ($\mathcal{A}_\Gamma$ for $k = \mathbb{R}$); its $\langle x, y \rangle$ is $\langle x, y \rangle_0$; its coefficient isomorphism τ is α and its exponent automorphism ϕ is τ ; its $\rho, \tilde{\rho}$ are $\sigma, \tilde{\sigma}$; its $\nu, \mathcal{U}, \mathcal{G}, \mathcal{M}$ are $u, \mathcal{U}_{\mathcal{R}_\mathbb{Z}}(\Gamma)$, the stabilizer and the monomial maps; its $L_\nu(g), a_{\nu, \delta}$ are $h_u(g), a_{u, \delta}$; its T^* is $T^\dagger$; its Λ is $\lambda_{\mathbf{On}}$; its power series F is $\mathbf{F}$ and its subfield F is K' ; its $P_{<\gamma}$ is $T_{<\gamma}$; its shear letters e_1, e_2 are e_2, e_1 and its s is t^{e_1} ; its additive h is θ ; its infinitesimals u_j are z_j ; its $d > 0$ is δ ; its $\mathbf{No}[i], \mathbf{Oz}[i], A_\mathbb{C}, K_\mathbb{C}, \Pi_\mathbb{C}, B_\mathbb{C}$ are $\mathbf{No}(i), \mathbf{Oz}[i], \mathcal{R}_\Gamma^\mathbb{C}, K_\Gamma^\mathbb{C}, \Pi_\Gamma^\mathbb{C}, \mathbb{C} \oplus \Pi_\Gamma^\mathbb{C}$; its target monomial u is t . No normalization changes. Source 15 (the table of [section 38.3](#); t -convention): its topologies $\mathcal{T}, \mathcal{T}_0, \mathcal{T}_\beta$ are $\mathfrak{T}, \mathfrak{T}_0, \mathfrak{T}_\beta$; its $\mathcal{R}_{o, \Gamma} = o \oplus \Pi_k$ is $\mathcal{R}_o$ with Γ densely ordered; its closure of zero N is $\{0\}$; its exponent sets D, E, D_U, D_V are $\mathbf{A}, \mathbf{B}, \mathbf{A}_U, \mathbf{A}_V$; its $R_\Gamma, A, \mathfrak{m}$ are $\mathcal{R}_\Gamma, \mathcal{A}_\Gamma, \mathfrak{m}_\Gamma$; its $\rho = \sigma|_k$ is α ; its $\mathcal{U}_{R_\Gamma}, h(g), a_\delta$ are $\mathcal{U}_{\mathcal{R}_\mathbb{Z}}(\Gamma), h_u(g), a_{u, \delta}$; its $\langle f, h \rangle$ is $\langle f, h \rangle_0$; its summation functionals are strong functionals; its Λ is $\lambda_{\mathbf{On}}$; its D and D_a (supports of functionals) are S_λ and S_a ; its $\sigma_{c, r}$ is $M_{\chi, \tau}$; its $\mathbf{No}[i], \mathbf{Oz}[i]$ and $c(z)$ are $\mathbf{No}(i), \mathbf{Oz}[i]$ and $\bar{z}$; its invisibility corollary is [corollary 39.10](#), not [corollary 24.4](#). No normalization changes. Source 16 (the table of [section 43.3](#)): its external $X^g, k[G], K_k(G)$ are $U^g, k[U^G], k(U^G)$; its monoids M, N are $\mathbf{M}, \mathbf{N}$ and its second group H is G' ; its v, d, w are $\text{min supp}, \text{max supp}, \text{wd}$; its ℓ is λ and its Td/dT is ϑ ; its $\mathcal{U}_\infty(B)$ and $D(G)$ are $\text{Rt}_\infty(B)$ and G_{div} ; its $\mathcal{D}(B), \mathcal{C}(B)$ are $\mathfrak{D}(B), \mathfrak{C}(B)$; its $\Delta_k(G)$ is $\text{Def}_k(G)$; its coefficient automorphism σ is α ; its $T = \phi^{-1}(S)$ is R_1 ; its $\mathfrak{S}_d$ is $\text{Sym}(d)$; its $R_\Gamma(D, k), \mathcal{P}_\Gamma(k)$ are $\mathcal{R}_o(k, \Gamma), \Pi_k(\Gamma)$; its $\mathbf{Oz}[i]$ is $\mathbf{Oz}[i]$. No normalization changes. Source 17 (the table of [section 53.3](#); large-monomial convention): its D and $h = \text{ct}$ are D_0 and q_0 ; its $H = \ker h$ and $C = \ker D$ are $\ker q_0$ and C_0 ; its $\mathcal{T}_s$ and $\mathcal{U}_v$ are $\mathcal{E}_{0, s}$ and Υ_v ; its $L_{\text{add}}, L_{\text{mult}}$ are $\hat{L}_0, \hat{L}_0^\times$; its F, M, A, A' are written out; its $\mathfrak{P}$ is Π ; its $\text{Const}(K, D)$ and C_K are K^∂ ; its $\delta = D - \partial_s, \delta_m$ and $\partial = -\omega D$ are $\nabla_s, \nabla^\times$ and $\tilde{D}_0$; its $A_\alpha, B_\alpha, a_\alpha, M_\alpha, E_\alpha, X_\alpha$ are $\alpha_\alpha, \mathfrak{b}_\alpha, \zeta_\alpha, \omega^{\mathfrak{b}_\alpha}, \exp(\zeta_\alpha z), \Xi_\alpha$; its $\mathbf{No}(i), C(i), \mathbf{Oz}[i]$ are $\mathbf{No}(i), C_0(i), \mathbf{Oz}[i]$; its t with $Dt = 1$ is y_1 . No normalization changes. Source 18 (the table of [section 60.3](#); growth convention): its $\mathcal{H}(B)$ and $\mathcal{K}_B$ are $\text{NF}(B)$ and $\mathbf{K}_B$ (not the hull $\mathcal{H}(A)$); its σ_2 is Σ_2 ; its T_φ and F_φ are ι_φ and M_{1, ι_φ} ; its $\mathbf{No}_\mathbb{C}, \mathbf{Oz}_\mathbb{C}$ and c are $\mathbf{No}(i), \mathbf{Oz}[i]$ and cj ; its cut D , stabilizer H and components C_D are $\mathbf{c}, G_\mathbf{c}$ and $I_\mathbf{c}$; its factor H in $H \times C_2$ is G_0 ; its classes X, X_α are $\mathcal{X}, \mathcal{X}_\alpha$; its U, χ, E_n are $\Phi_{\text{ph}}, \chi_{\text{ph}}, E_n$; its homeomorphism h is ψ and its $h(t) = t/(2-t)$ is h_{KB} . No normalization changes. Source 19 (the table of [section 67.3](#); growth convention): its formal variable T is s ; its $\mathfrak{J}_R$ and J are Π_R and $\Pi_R^\dagger$; its H_n are $\mathbf{H}_n$; its Lie series L is $\mathbf{L}$; its $G(R), G_N(R)$ are

$\mathcal{G}(R)$, $\mathcal{G}_N(R)$; its ℓ_η is q_η and its E_m is $\omega^m D_0$; its C_R is $\mathbb{C}_R$; its Θ is $\mathbb{Q}$; its coefficient sections s are residue sections ς (not Part III's coefficient sections ρ); its D_d is d_* ; its $\Delta = \Phi - \text{id}$ is written out; its curve C is $\mathbf{X}$; its $\mathbf{No}[i]$, $\mathbf{Oz}[i]$ are $\mathbf{No}(i)$, $\mathbf{Oz}[i]$. No normalization changes. Source 20 (the table of [section 76.3](#); growth convention): its $\mathbf{No}[i]$, $\mathbf{Oz}[i]$, K_k , $M_k = \ker c_0$ are $\mathbf{No}(i)$, $\mathbf{Oz}[i]$, $\mathbf{No}_k$, $V_k = \ker \text{ct}$ (not a monomial map $M_{\chi,\tau}$); its c_0 is ct ; its p , T_p , σ_p , σ_T are φ , ι_φ , M_{1,ι_φ} , $M_{1,\tau}$; its $F = \text{Fix}(p)$, H_F are B , $\text{NF}(B)$; its J , H_J , $F_{J,k}$, B_A , y_a are J , $\text{NF}(\mathbf{No} \setminus J)$, $K_{\mathbf{No} \setminus J,k}$, $\Sigma_2(A)$, w_a ; its $H = \text{Fix}(T)$, K_H , P_H , Q_F are Γ_τ , $k((\omega^{\Gamma_\tau}))$, ϖ_{Γ_τ} , ϖ_B^{in} ; its $f_\pm$ are $f_\uparrow$, $f_\downarrow$; its Green operator $\mathcal{G}_c$ is Gr_c (not $\mathbf{G}$ or $\mathcal{G}(R)$); its retraction $\mathcal{R}_F$ is mr_B (not ret_ϕ); its S_s is σ_s^{tr} (the translation lift, $\sigma_1^{\text{tr}} \neq \text{id}$, not a dilation S_q of the single-dilation report); its $P \in k[X, X^{-1}]$ and $k(X)$ are $\mathfrak{p} \in k[\mathbf{Y}, \mathbf{Y}^{-1}]$ and $k(\mathbf{Y})$; its cocycle characters h , χ are ch^+ , $\text{ch}^\times$; its j is cj and its identity e is 1. No normalization changes.

B Non-claims, by source

No source claims to solve a named conjecture, and none claims Lean verification, peer review or historical priority; the finite suites check identities only. The report keeps every limitation stated by each source.

Source 09.

1. The classification is only within strong, coefficient-fixing, leading-term-preserving automorphisms; it does not assert that an abstract automorphism preserving $\mathcal{R}_o$ is strong, and nonstrong pair automorphisms are not classified (the definability recovery applies to them; the logarithmic classification does not).
2. Coefficient functionals come from an existing strong derivation; arbitrary functions satisfying the vanishing condition are not asserted to assemble into an operator.
3. The shears are plain-field automorphisms and are not claimed to preserve the Gonshor exponential.
4. No set or NBG class of class-function graphs is formed; actions are encoded by single class relations; supports are sets.
5. The exponential–logarithm correspondence and the factorization pattern are imported, not reproved or claimed; the explicit many-shift constructions avoid the class version of the import.
6. No new factorization theorem for omnific integers is claimed; [19] addresses a different problem.
7. $\mathbf{Oz}$ is not identified with a model of full ordinary arithmetic.
8. Truncation at a named monomial is not a uniform definable choice of boundary.
9. Nondefinability is relative to the field-and-integer-part language; Conway normal forms remain set-theoretically defined.
10. Hahn-sum formulas are not claims of convergence in the fine topology.
11. Modulus equivariance is not a claim that moduli are fixed.
12. The repository comparison was bounded and targeted; search coverage is not a proof of novelty.
13. The exact-arithmetic program checks finite identities only; it does not decide well-ordering, prove the correspondence, verify class-theoretic statements or certify novelty; the formalization order is a proposal, not implemented modules.

Source 04.

1. It does not classify abstract (nonstrong) automorphisms of **Oz**; the fraction-field extension of an abstract automorphism is not known to be strong.
2. It imposes no exponential, omega-map, birthday or simplicity structure, and does not revisit the collection's exponential-rigidity question as if unanswered.
3. The criterion classifies existing maps; it is not a construction for arbitrary logarithmic characters, and the convex condition alone is not asserted to solve global admissibility.
4. The relative fixed field is computed for the elementary subgroup only; the full stabilizer only for monomial parameters; the Hahn hull is not asserted to be definable closure.
5. Class maps are treated individually; no class of all automorphisms is formed.
6. Nothing is claimed after naming a proper class of parameters.
7. Bare nondefinability of coefficient-one monomials already follows from character twists; the nondefinability statement alone is not advertised as a new mechanism.
8. $\text{Frac } \mathbf{Oz} = \mathbf{No}$ is a full-class statement; a set-sized Hahn field need not be the fraction field of its integer-part ring.
9. Its intrinsic coefficient invariance was not claimed to be first order (now superseded by the Diophantine report).
10. Real and imaginary shear times preserve different structures; its Gaussian result is not the first surcomplex automorphism moving the real axis.
11. No generation or density theorem; fine-topology convergence is not strong summation.
12. The fixed-shift exponential, the general monomial/kernel decomposition and the exponential-derivation correspondence are prior work.
13. Its finite checks are not proofs and do not implement arbitrary normal forms; no Lean file, no repository checkout or build; historical priority unverified.

Source 08.

1. The threshold covers the elementary family only; rank-one rescalings and characters preserve the integer part; not every leading-term-preserving automorphism is claimed to be the exponential of one fixed-shift derivation.
2. Finite-dimensional phase images are a sufficient condition only.
3. No first-order Gaussian reconstruction was claimed (one now exists, [section 8](#)).
4. The automorphisms are not asserted to commute with the surreal exponential, the Berarducci–Mantova derivation or the omega-map; the wreath product does not contradict rigidity for those structures.
5. Parameter avoidance fails for set-sized fields with the whole field named; a universe-relative collection is not identified with the proper class.
6. No classification of the full stabilizer, and no solution of additional-structure problems.
7. The reconstruction is not a homomorphism $\mathbf{Oz} \rightarrow \mathbb{R}$.
8. The nondefinability is not a statement about adding an arbitrary class predicate, nor about automorphisms required to commute with the omega-map.
9. Strong summation is not topological convergence.

10. Pell rigidity, the existence of nontrivial Hahn automorphisms, the derivation–exponential construction and the monomial-character viewpoint are not claimed new; some elementary lemmas may be known; priority is provisional.
11. Its questions are not named historical problems.
12. No Lean; not refereed; the checks are regression tests, not proofs of well-ordering, parameter avoidance or definability.

Source 02.

1. Root-covered is sufficient, not claimed necessary; failure of the monomial criterion is not used to claim failure of cancellation or rigidity.
2. The finite-type theorems are theorems about sets; no $\text{Spec}(\mathbf{Oz})$ and no class compactness; “ $\text{Aut}(\mathbf{Oz})$ ” and ML of class rings are shorthand; the finite-type theorem is not a statement that a class cannot inject into a set.
3. It does not classify $\text{Aut}(\mathbf{Oz})$, all derivations, or multivariable polynomial automorphisms, and makes no Jacobian-conjecture claim.
4. Not every homomorphism into every target is constant: injectivity and domain targets are required; no noninjective analogue is given.
5. It does not claim that $R[\mathbf{T}]$ has no locally nilpotent derivation; a Hahn expansion is not a grading.
6. The orbit-field formula concerns monomials; only a lower bound holds for the full formal image.
7. Formal flows involve no analytic convergence and no fine-topology limits, s is not a surreal, and they are not exponentiation inside a Hahn field.
8. The cancellation theorem does not solve the unrestricted cancellation problem; $\text{ML}(R[\mathbf{T}]) = R$ is not a solution of a general stable-rigidity conjecture.
9. Not new: character twists [15], strong-invariance terminology [14], the Makar–Limanov framework [6], Euler derivations, Vandermonde arguments, and the example of [19].
10. No named factorization conjecture is claimed solved; the literature search is not a priority certificate.
11. The questions are not presented as published open problems; no Lean, no peer review; the checks are finite identities.

Source 10.

1. The classification is within strong, coefficient-fixing, leading-term-fixing automorphisms; nonstrong and coefficient-moving automorphisms are not classified, and strongness is not a harmless omitted hypothesis.
2. The criterion classifies existing strong maps; an abstract character obeying the cut condition is not asserted to give an automorphism, since summability and an inverse must be proved.
3. The rank theorem concerns the leading-term-fixing factor only; it does not say that all rank-one automorphisms are trivial, and monomial maps such as $t^\gamma \mapsto t^{2\gamma}$ and character rescalings are not counterexamples. Archimedean rank is not the order-unit condition of the holonomic report.

4. The strong valued decomposition, principal-unit characters and their crossed law [17], simplicity rigidity [15], the existence of nontrivial strongly linear 1-automorphisms, Hahn normal forms and the support description of $\mathbf{Oz}$ [19] are prior work; the splitting is not a new decomposition theorem for $\text{Aut}(\mathbf{No})$.
5. Not every truncation is preserved, only signed Archimedean blocks.
6. The fixed field is relatively, not absolutely, algebraically closed; no exact transcendence degree of a class extension is claimed.
7. The nonabelian example gives no presentation or exhaustive description of the stabilizer; no transfinite composition or closure theorem is claimed.
8. No proper-class-indexed sum is formed and no class of automorphisms is formed; class maps are individual, in NBG with global choice.
9. The bounded-birthday automorphism does not preserve birthdays or the simplicity hierarchy globally.
10. The definable-closure statements are upper bounds, not descriptions of definable closure and not quantifier elimination; they do not recover every coefficient from the projections.
11. The formal exponential is not the Gonshor exponential.
12. $\text{Frac } \mathbf{Oz} = \mathbf{No}$, the multiplier identity and coefficient rigidity are prior repository results, credited and not claimed; automatic fixing of coefficients is asserted only for the full surreal field, not for arbitrary $\mathcal{R}_o \subseteq K$.
13. $\mathcal{R}_o$ is called integer-part-shaped, not an integer part in every generality; no ordering of the complex field is asserted and no global surcomplex exponential is introduced.
14. The divisibility example is for $o = k$ and does not assert that translation preserves $\mathbb{Z} + t^{-1}\mathbb{R}[t^{-1}]$; the criterion is not transferred to other integer parts.
15. Initial subdomains, truncation-closed subrings and subrings stabilized by an automorphism are different notions, and none is substituted for another.
16. It does not answer the omega-map question of [15], factorization problems for omnific integers, or the holonomic order-unit question; no named published conjecture is solved, and its questions are research questions.
17. Its finite checks do not prove summability, well-ordering of arbitrary supports, the existence of separating functionals, class-size statements or nondefinability.
18. No Lean, no repository clone or build, no peer review; novelty is provisional, and the repository and literature comparison was targeted, not exhaustive.

Source 11.

1. It is an unrefereed research draft with no Lean verification; priority is provisional and the novelty assessment limited; the existence of proper Hahn-field embeddings is not claimed as new.
2. The classification carries the exact-monomial hypothesis: it is not a classification of all surreal embeddings, does not assert that omnific-preserving embeddings are strong, and does not cover characters or unit corrections.
3. Taylor automorphisms, exponent lifts, the reconstruction of constants and the classical inputs (Hahn fields, normal forms, real and algebraic closedness, quantifier elimination, extension of derivations) are prior work.

4. The derivations of $\mathbb{R}$ are choice-dependent and noncomputable.
5. “Monomial-preserving” does not mean commuting with the omega-map, and “strong” does not include $\mathbb{R}$ -linearity; no omega-map or exponential question of [15] is claimed, and the explicit map fails omega-map compatibility at 1.
6. Elementarity of pure-field embeddings is classical; fixing the coefficients is not proved sufficient for elementarity of the pair.
7. The continuum of conjugacy types is a lower bound, not a classification.
8. No nonconjugacy is claimed for automorphisms preserving $\mathbf{Oz}[i]$ but not conjugation, and non-conjugation-compatible surcomplex embeddings are not classified.
9. $\text{Frac } \mathcal{R}_\Gamma = K_\Gamma$ is not assumed for set-sized Γ .
10. No collection of class automorphisms or of open subclasses is formed; the topological claims are local.
11. The annulus bounds are external.
12. The finite tests prove no infinite, class, conjugacy or model-theoretic statement.
13. No repository was modified or built.

Source 12.

1. It is unrefereed, has no Lean verification, solves no named published conjecture and certifies no priority; the repository comparison was limited to four files and five public sources, the full articles and the ledger were not read, and a local clone failed.
2. Support duality and finiteness spaces are not new; $\text{Frac } \mathbf{Oz} = \mathbf{No}$ is classical; the reconstruction and the convex criterion are prior repository results.
3. Admissibility is retained: arbitrary coefficient arrays are not claimed to define automorphisms.
4. The proper embeddings are not a counterexample to strongness of all embeddings, and are not claimed initial, exponential or omega-compatible; the derivation is not computable.
5. The adjoint theorem is for set-sized groups only, fails for $\mathbf{No}$, and does not apply to non- $\mathbb{R}$ -linear maps.
6. No unrestricted Gaussian valuation-preservation theorem is proved, and preservation of the real axis is not assumed.
7. The support-bound result is for regular uncountable κ only; no singular case, and no identification with a birthday cutoff.
8. ret_ϕ is not a field retraction or a positive conditional expectation.
9. No preservation of the exponential, logarithm, omega-map, birthdays or simplicity is claimed.
10. Invariance under automorphisms is not first-order definability.
11. The set-pair corollary does not assert that automorphisms of $\mathcal{R}_\Gamma$ extend when $\text{Frac } \mathcal{R}_\Gamma$ is smaller.
12. Omnific-preserving automorphisms are not claimed to be trivial.
13. The distinction from the three-duals report is not an exhaustive non-overlap claim.
14. Finite checks are not proofs.

Source 13.

1. It is AI-assisted and unrefereed, not proof-assistant checked, not a resolution of a named conjecture, and makes no priority claim; a failed search is not novelty, and it does not claim to settle every question posed.
2. Earlier repository results are credited: fractions, reconstruction, the strong stabilizer and the shears.
3. The scalar strong-dual representation is set-size only; the automorphism theorem is not.
4. The residue criterion is not valuation-topological convergence and not first-order definability of summation.
5. Not all automorphisms of **No** are strong (the examples of [15] stand), and omnific automorphisms are not trivial.
6. Local convex vanishing is not proved sufficient, and no class-wide operator exponential–logarithm correspondence is used.
7. It does not prove that every automorphism of **Oz**[*i*] is valuation-compatible, and conjugation is not intrinsic; its two Gaussian theorems must not be merged into an unconditional classification.
8. The full-pair theorem assumes an ambient isomorphism; isomorphisms of the small integer parts are not claimed to extend.
9. No Gonshor exponential, Berarducci–Mantova derivation or omega-map is used; its exponent map is not a field map; no proper-class Zorn lemma or basis is used.
10. The drift embedding comes with no uniqueness claim; the failure of homogeneity concerns set-sized induced substructures, not elementary maps.
11. Finite checks do not verify infinite summability, transcendence bases or surjectivity.
12. No repository build was run, and the formalization ledger was not read.

Source 14.

1. It is an AI-assisted research draft, not refereed and not Lean-verified; “theorem” means that a proof is supplied. Global priority is not certified: the literature search is not a certificate of absence, and the priority of the omnific application, of the detector’s formulation and of the class counterexample needs independent review.
2. Finiteness-space duality [4] and the general decompositions of Hahn-field automorphisms [17, 15] are prior work; no priority is claimed for those frameworks, and the underlying support duality is not claimed to originate there. The reconstruction and the convex-scale criterion are credited to this report and re-proved.
3. The unrestricted Gaussian problem is not solved: neither automatic strongness of every automorphism preserving **Oz**[*i*] nor a nonstrong counterexample is claimed, and the phase example is strong, so it disproves nothing.
4. No embedding theorem: the isomorphism proof uses surjectivity essentially, and no statement about constant-term-compatible embeddings is inferred.
5. No compatibility with the global Gonshor exponential is claimed; the formal exponential and logarithm are evaluated at infinitesimals only. Continuity, the Conway monomial cross-section and arbitrary truncations are neither assumed nor concluded, and commutation with every truncation is false.

6. The representation of strong functionals and the adjoint theory hold for set-sized exponent groups only and fail for **No**.
7. It does not assert that every ring automorphism of every $\mathcal{R}_\Gamma$ extends to its full Hahn field; the fraction-field argument is for **Oz** only.
8. The convex criterion classifies existing normalized automorphisms; it does not construct automorphisms from additive data satisfying the vanishing condition, since joint support conditions and invertibility are needed.
9. Summability, the monomials and a cross-section are not claimed first-order definable; the intersection formula for Π is not a single first-order formula; the conclusion is invariance of an externally quantified operation, not definability.
10. The class-group notation $\text{Aut}(\mathbf{No}, \mathbf{Oz})$ and its factorization are elementwise; no NBG class of proper-class graphs is formed; every summed family is set-indexed and every series has set support, in NBG with global choice.
11. The countable detector property is a sufficient closure condition only; it is not claimed for transseries, Puiseux or left-finite subfields.
12. The adjoints are for a bilinear algebraic pairing, not a Hilbert-space inner product; $\sigma^\dagger = \sigma^{-1}$ is not analytic unitarity, and the pairing is not a Hermitian norm.
13. Formal summability is not convergence of partial sums or in the valuation topology; no ordering of the index set is used; the automorphism–derivation correspondence of [3] is not used.
14. Its finite checks validate finite identities only: they do not establish well-ordering of arbitrary supports, quantify over all series, verify class replacement or prove the main theorem. No Lean compilation was performed, and the proposed Lean modules and names are plans only.
15. Conway’s irreducibility conjectures are not addressed and no famous open conjecture is claimed settled; its questions are proposed research problems, not published open problems.
16. The repository was read, not modified; its comparison was targeted, not an audit of every manuscript; the Wikipedia article it lists was an orientation source only.

Source 15.

1. It is an AI-assisted manuscript, not independently refereed and not checked in a proof assistant; “theorem” means that a written proof is supplied. Historical priority is not certified; the source comparison was targeted, not an exhaustive audit of the repository or the literature, and its reading of this report was incomplete (the updated article was not read in its entirety). No repository build, Lean run or modification was performed; a local clone failed.
2. The topological results are about compatibility with *every* Hahn sum. Ordinary ring topologies without this summation requirement are not excluded, and the rings are not claimed to lack Hausdorff ring topologies: the discrete topology always exists and fails only to realize the nontrivial sums.
3. Joint continuity of multiplication is essential: the pairing topology of [section 41](#) is Hausdorff, realizes exactly the Hahn sums, and has only separately continuous multiplication.

4. The support-ring classification assumes a densely ordered exponent group. For non-dense groups such as lexicographic $\mathbb{Z}^2$ the truncated ring $\mathfrak{o} \oplus \Pi$ is open; the full-field theorem does not settle it.
5. The cyclic case is an existence statement, not a classification of all compatible topologies.
6. The countable-only strengthening uses the positive buffer of the class **No**; it is not asserted for set-sized groups, and its proof fails for $\Gamma = \mathbb{Q}$.
7. Automatic strongness, scalar detection, covariance, adjoints, the class-size failure of duality and the qualified Gaussian results are credited to sources 12 and 13; the disjoint-row detector is an alternative proof, not a priority claim; the convex criterion, the reconstruction, the factorization and the shear examples are credited to this report; its reconstruction proofs are not claimed new.
8. Unrestricted Gaussian automatic strongness is not proved: valuation or conjugation compatibility is assumed explicitly, the bare pair $(\mathbf{No}(i), \mathbf{Oz}[i])$ is not claimed strong, and the collapse theorem does not settle it.
9. Global admissibility of arbitrary monomial prescriptions is not solved; the convex criterion is not an existence theorem.
10. Invariance is not definability: no first-order definition of families or sums and no definable monomial section is claimed.
11. Class statements concern individual class maps and class-coded neighbourhood systems; no class of class objects and no proper-class sum is formed; families are set-indexed and supports are sets.
12. The self-duality and the equivalence of strongness with adjointability need set-sized groups and fail for **No**.
13. The detector needs a full Hahn field or a subfield containing its countable detectors; the isomorphism argument uses surjectivity and is not stated for embeddings.
14. The pairing topology uses the discrete topology of the coefficients, not the usual topology of $\mathbb{R}$ or $\mathbb{C}$, and is not identified with the topologies of [9].
15. Finiteness spaces and formal summation algebras [4, 3] are prior work; being a monoid object for a summation-sensitive tensor product is not joint continuity, and no class operator-exponential theorem is imported.
16. Its finite checks do not test arbitrary topologies, infinite supports, proper classes or the infinite covering lemmas: a finite rational mesh can always be covered by two finite sets.
17. No named historical conjecture is claimed resolved; its questions are proposed directions, not established conjectures.

Source 16.

1. It is a proof-bearing research draft, not peer reviewed and with no Lean verification of its new results; historical priority is not certified, some proposed claims may already follow from published results, and the repository comparison was targeted (catalogue, guides and audits, earlier reads from the then-current **main**), not an exhaustive review; a local clone failed and no repository build or Wolfram run took place.
2. No named Conway factorization conjecture and no unrestricted automorphism problem is claimed solved: $\text{Aut}(\mathbf{No})$ and $\text{Aut}(\mathbf{No}(i))$ are left as coefficient data.

3. The general criterion [theorem 49.2](#) and [theorem 49.5](#) depend on [cited theorem 49.1](#), from an unrefereed preprint; no peer-review status is inferred, the theorem is not reproved, and the two-frame method is credited to Lin and Wang.
4. The field formula does not define $\mathbf{Oz}$ (or $\mathbb{R}$) inside the pure field $\mathbf{No}$; the ring formula is interpreted in $R[U^M]$, uses an inequation, and is not a single Diophantine equation.
5. The root-divisor set $\mathfrak{C}$ is infinitary and not first order; its finite truncations are vacuous; all-order root membership is not treated as a first-order formula; no algorithm recognizing surreal coefficients is inferred.
6. Nothing is claimed about Gaussian stabilizers (whether automorphisms of $\mathbf{No}(i)$ preserving $\mathbf{Oz}[i]$ preserve conjugation or the valuation), Hahn summation or conjugation: the extensions have finite supports and are not Hahn completions, and the results must not be transferred to Hahn fields.
7. Everything is in characteristic zero, for domains and cancellative torsion-free monoids; positive characteristic, torsion and idempotents are outside the theorems.
8. Injectivity is essential; noninjective maps are not classified, and reduced parameter rings are not addressed ([question 19.3](#) stays open).
9. The infinite-rank failure is proved only for $\mathbb{Q}^{(\kappa)}$; other infinite-rank groups are open.
10. The quartic obstruction, the sparse multiplicity bound, the monomial unit theorem, normal forms, the transcendence-rank computation and defect freeness are antecedents, not inventions; the root-cover framework, the scale criterion and finite-polynomial cancellation are source 02's.
11. Class-sized coefficient cases are elementwise, with Scott-trick codes and set-subfield descent; no proper-class spectrum, Hahn completion or set of all class automorphisms is formed, and automorphism-group formulas are schematic.
12. An external monomial is not identified with a surreal magnitude.
13. Its finite checks are finite tests, not proofs: they cannot prove the arbitrary-rank results, quantify over all roots, validate class foundations or establish novelty. The Lean formalization of Mason–Stothers [1] and the repository's Lean code are not verification of 16.
14. Its questions are proposals, not published open problems.

Source 17.

1. Taylor morphisms, Euler derivations, formal binomial operators, the independence of exponential solutions, the constants and linear-disjointness lemmas, the rational logarithmic-derivative obstruction, Vandermonde and Eisenstein arguments, abstract scalar extension, and the idea of combining a field with its Taylor image are not new [[25](#), [16](#), [3](#), [15](#)]; the abstract lemmas may exist in more general forms.
2. The orbit field is not identified with a Galois hull with its additional derivations or functorial structures [[13](#)], and no new general Galois theory is claimed.
3. The theorems are for the one derivation D_0 ; they do not transfer automatically to a derivation with an additive slice ([example 54.8](#)), nor to a changed Hahn presentation without checking how the character and the constant field change.
4. s and v are external formal indeterminates: no convergence theorem permitting substitution of a surreal is proved, the witness formulas are not s -adically convergent sums, and independence over $\mathbf{No}$ cannot survive an internal evaluation into $\mathbf{No}$.

5. No single cardinal is called the transcendence degree of a proper-class extension; the no-set-basis statement is a consequence of set-indexed independence.
6. No algorithm recognizes elements of $C_0(\omega)$ or of its relative algebraic closure from arbitrary presentations; relation descent is not a decision procedure, and (58.1) does not classify all algebraic extensions of $\mathbf{No}(u)$.
7. The formal flows are not claimed to define nontrivial internal automorphisms of $\mathbf{No}$.
8. D_0 is not claimed to be the Berarducci–Mantova derivation or to obey a chain rule for the surreal exponential; the one Hahn exponential used is an explicit sum, and no global surcomplex exponential is introduced.
9. Nothing is claimed about other real forms of $\mathbf{No}(i)$ or arbitrary surcomplex automorphisms.
10. ct is not a ring homomorphism on $\mathbf{No}$; $\ker q_0$ is not asserted convex, and no valuation on a quotient is used.
11. The comparison with the independent-copies report distinguishes internal copies from external formal embeddings; neither is substituted for the other.
12. No named conjecture is claimed solved (unique factorization, Gonshor birthday bounds, exponential automorphisms, the Riemann hypothesis); its questions are research directions, not published open problems.
13. Priority is not established by an exhaustive review; the repository comparison was targeted, some connector reads were truncated and only searched, and the large manuscripts, companions and Lean library were not read completely.
14. The finite checks are not proofs: they do not certify normal-form theorems, Hahn summability, all ordinal supports, class foundations or novelty.
15. No Lean formalization or proof-assistant check, no peer review; the build record concerns PDF production only, and the repository’s Lean builds were not rerun.

Source 18.

1. The double Hahn lift, and the existence of nontrivial or nonabelian omnific-preserving automorphisms, are not new ([15], Construction 3.10; [17]; this report’s fixed fields, hulls, stabilizers, wreath product and nondefinability).
2. No classification of $\text{Aut}(\mathbf{Oz})$; it is not claimed that an arbitrary such automorphism is strongly additive (the converse argument works without strongness).
3. No classification of all fixed subfields of $\mathbf{No}$ or of the conjugacy classes of the realized actions; the minimality of $K_{\Sigma_2(A)}$ holds only within the family K_S , not among all subfields, and is not algebraic or definable closure.
4. The maps preserve the monomial class, not the omega-map; no preservation of the Gonshor exponential, a surreal derivation, simplicity, birthdays or the valuation pointwise is claimed.
5. The order identification and the cut filling use the full class $\mathbf{No}$; no birthday bound and no transfer to a birthday cutoff.
6. It is not claimed that every ordered field automorphism of $\mathbf{No}$ fixes every ordinary real.
7. Theorem 60.4 concerns actions commuting with a named conjugation; complex actions without it are not classified.

8. The countable dynamical realization of [26] is used only for the optional explicit route and must not replace the cut filling when exact fixed fields are claimed; torsion-freeness alone is not the criterion.
9. The rigidity consequence concerns the constructed automorphisms, which preserve the field, $\mathbf{Oz}$, the named reals and Hahn summation; Hahn summation is not claimed to be a first-order operation of finite arity.
10. The character twist is a standard boundary example, not a novelty claim.
11. No class of all proper-class automorphisms is formed; actions are single class relations.
12. No resolution of Conway’s refinement conjecture, the general product-birthday conjecture or an exponential-automorphism classification; its questions are directions, not published problems.
13. The finite checks do not verify infinite summability, class recursion, equivariant cut filling, real closedness or novelty.
14. The source inspection was at guide level (catalogues and guides, not every proof in the 61 reports), some connector responses were truncated, and the search for “left-orderable” covered the default branch only and does not prove absence.
15. No Lean proof; not refereed; “proved here” does not mean “first proved here”, and priority is not certified.

Source 19.

1. No classification of derivations by support; derivations are not claimed $\mathbb{R}$ -linear or strong, and ωd_* is not claimed to exhaust the non-Euler directions.
2. The integration result is not the Hahn-evaluation question: no formal flow is claimed to be evaluable inside $\mathbf{No}$ or $\mathbf{No}(i)$, and no surreal value of s is used.
3. Formal exponentiation, unipotent logarithms and higher derivations [12, 24, 2], the Riemann–Hurwitz and positive-genus arguments [27], the generic complete-ring section argument, strong invariance [14] and the results of part II are not new; no Zariski cancellation or multivariable automorphism classification is claimed.
4. Only marked residue sections are classified.
5. $\mathbf{Q}$ is existential, not positive-existential or a single equation; no optimality of the genus threshold or of the formula’s complexity is claimed; the surcomplex-field completion has a parity exception.
6. In finite jet rings the formula is universal for a nilpotent reason, unrelated to theorem 72.7.
7. Nondefinability of the canonical copy does not forbid interpreting the residue ring; polynomial definability does not imply recovery in arbitrary function fields.
8. Corollary 74.3 does not say that every orbit is transcendental or that no formal parameter can be evaluated.
9. Nothing is proved about the whole automorphism group of $\mathbf{Oz}$ beyond credited facts, and no canonical real form of $\mathbf{No}(i)$ is identified.
10. $\text{Der}(\mathbf{Oz})$ and $\mathcal{G}(\mathbf{Oz})$ are external collections of class maps; displayed correspondences are theorem schemes; no proper-class scheme is formed.
11. The repository’s automatic-strongness theorem is not used, and the repository’s Lean build was not rerun.

12. Its finite checks use the model $\mathbb{Z} + X\mathbb{Q}[X]$ and finite jets only; they prove nothing about surreal supports, all derivations, class maps or genus.
13. The literature search is targeted, not a priority search; the repository comparison is fixed at 889b87c, some connector reads were truncated, and packaging happened after midnight UTC on 24 September.
14. No Lean formalization, no peer review, no certified priority; no independently published long-standing conjecture is claimed settled, and its questions are continuations, not established problems.

Source 20.

1. Manuscript proofs only: not independently refereed, not formalized in Lean, and no repository build or Lean audit was run.
2. The two Hahn lifts [15, 18] and the real-form centralizer factorization (`saut:thm:axis`) are prior constructions; the contribution claimed is their combination with a free ordered action and the fixed-field, inverse-operator, retraction and cocycle results.
3. Priority is not certified: the repository review was targeted (five report guides, some truncated, at the pin e93a06d) and not exhaustive, and the literature search did not establish a priority history.
4. No named published problem is claimed solved; in particular none of the exponential or omega-map questions of [15, Section 5].
5. The maps need not preserve simplicity, birthdays, the omega-map as a function ($\sigma_1^{\text{tr}}(\omega) \neq \omega^{\sigma_1^{\text{tr}}(1)}$), the Gonshor exponential or the Berarducci–Mantova derivation; exact monomial preservation is not commutation with the omega-map.
6. Higher-order variable-coefficient and matrix difference equations are not classified.
7. $(\mathbf{No}(i), \sigma)$ is not difference closed: $\sigma x - x = 1$ has no solution.
8. The leading-coefficient-only multiplicative criterion is proved only for the global free-index action; the relative action has the three obstructions of theorem 79.7 and cannot be reduced to the leading coefficient.
9. mr_B is a multiplicative-group map, not a field homomorphism, and is not asserted to preserve $\mathbf{Oz}$.
10. ϖ_{Γ_τ} (and ct on $\mathbf{No}$) is not multiplicative; ct is a ring homomorphism only on $\mathbf{Oz}$ and $\mathbf{Oz}[i]$.
11. $k(Y)$ acts by operators, not as scalars or a new coefficient subfield.
12. Formal logarithms and exponentials are used only on $1 + \mathfrak{m}_k$ and $\mathfrak{m}_k$; no global surcomplex exponential is introduced.
13. The fixed-field results give no finite Galois correspondence; fixed fields do not distinguish subgroups.
14. Theorem 77.5 does not identify all fixed fields, nor claim that an arbitrary Hahn subfield can be the fixed field; its proper-class independence is only through finite subsets.
15. Theorem 76.1 classifies the set groups that embed as actions, not full automorphism groups of a named structure, and not the conjugacy classes of the actions.
16. The complex classification needs a named standard conjugation; automorphisms preserving only $\mathbf{Oz}[i]$, all automorphisms of the pure field and all real forms are not classified.

17. It is not claimed that every field automorphism of **No** fixes $\mathbb{R}$; that property holds for the constructed maps.
18. The cut completion does not preserve Dedekind completeness and gives no free action on the real line; the η_κ -orders carry no birthday cutoff and are not Dedekind completions.
19. No weighted Green inverse with coefficients in a relative fixed field beyond k is claimed ([question 82.6](#)); the exact sequences and cohomology groups are shorthand for explicit normal forms, not quotients of proper classes; no universal vanishing of cocycles of noncommutative groups is asserted.
20. The downward-orbit lemma does not assert summability of forward orbits or cofinal decrease of exponents; the class back-and-forth is not an algorithm.
21. The finite checks (5,555 assertions) certify no infinite summability, arbitrary support, class recursion, $\eta_{\mathbf{On}}$ isomorphism, global choice argument, originality or priority.
22. Its questions are research directions, not published problems.

Status note (batch 33). Non-claim 6 of 02 (only a lower bound for the full formal image) is kept as its record; for D_0 the full image is now described exactly ([part VII](#)), and for other characters its size is open. Non-claim 3 of 02 (no classification of all derivations) stays true: [part IX](#) shows that all derivations integrate, not what they are. Non-claim 2 of 18 is kept; for **Oz** every automorphism is in fact strong ([theorem 23.2](#)).

Status note (batch 30). Non-claim 1 of 09, non-claim 1 of 04 and non-claim 1 of 10 say that nonstrong automorphisms are not classified and that strongness is not shown. They are kept as the sources' records. For (**No**, **Oz**) and the full real Hahn pairs with divisible exponent group, [theorem 23.2](#) and [corollary 23.4](#) now show that every automorphism is strong, so [theorem 24.1](#) classifies them all; for **Oz** $[i]$ without conjugation or valuation data, and for a general $(k, \mathfrak{o})$, those non-claims still describe the state of knowledge ([questions 32.1](#) and [32.13](#)).

Counts: 13 non-claims from 09, 13 from 04, 12 from 08, 11 from 02, 18 from 10, 13 from 11, 14 from 12, 12 from 13, 16 from 14, 17 from 15, 14 from 16, 15 from 17, 15 from 18, 14 from 19, 22 from 20.

Status note (batch 34). Non-claims 1 and 3 of 18 (no classification of fixed subfields; minimality only within the family K_S) and non-claim 14 of 20 are consistent: 20's relative fixed field $K_{\mathbf{No} \setminus \mathbf{J}}$ is another member of that family, indexed by a class ([remark 77.6](#)).

C Finite checks

Fifteen suites ship with the report, byte-identical to their delivery, under the prefixes of their manuscripts. They check finite identities in exact arithmetic; they do not prove Hahn summability, class-sized statements, definability or novelty. They were rerun on copies for the merges (10's for the batch-28 addition, 11's, 12's and 13's for the batch-30 addition, 14's for the batch-31 addition, 15's and 16's for the batch-32 additions, 17's, 18's and 19's for the batch-33 additions, 20's for the batch-34 addition) with Python 3.14.4 and SymPy 1.14.0, with the recorded results reproduced: 15's rewritten record is identical to the shipped one, 16's differs only in its `python` and `elapsed_seconds` fields, 17's only in its `python_version` field, and 18's, 19's and 20's are identical up to line endings.

Script and record	Content and behaviour
code/09-omnific-preservin g-verify_finite_identie s.py; data/09-omnific-preservin g-verification_output.txt	Standard library. Product rule and bracket on finite Laurent polynomials, balanced brackets for ranks 2 to 8, mixed group-commutator coefficients with square-zero parameters, shear iterates, exponential laws, the rank-one escape, Pell growth. Twelve PASS lines and “ALL CHECKS PASSED”. Prints only.
code/04-preserving-autom orphisms-verification.py; data/04-preserving-autom orphisms-verification_re port.json	SymPy 1.14.0 (data/04-preserving-automorphisms-requirements.txt). 7,062 assertions in 17 groups: exponential group law, bracket formula, coefficient polynomial, first corrections, commutator coefficients in lexicographic $\mathbb{Q}^3$. <i>Writes</i> its report to --output, by default verification_report.json in the working directory.
code/08-integer-part-sym metries-verify.py; data/08-integer-part-sym metries-verification.json	Standard library. 9,469 checks in 15 categories: derivation rule, falling-factorial iterates, flow composition, bracket, Vandermonde determinants, the semidirect-product law, commutator phases. <i>Always writes verification.json</i> beside itself.
code/02-parameter-rigidit y-verify.py; data/02-parameter-rigidit y-verification.json	Standard library. 7,041 assertions in 13 families: Euler product rule, factorial Leibniz, truncated flow composition, commutation, geometric remainders, Vandermonde determinants, binomial identity. Writes verification.json beside itself unless --output is given.
code/10-support-cut-verif y.py; data/10-support-cut -verification.json	Standard library, exact rational arithmetic. 28,668 assertions in 19 categories: binomial coefficients, formal additive composition and inverses by two independent power algorithms, rank-two composition, inverse and multiplicativity, signed support projections, exact first displacement, the least-bad-shift coefficient and the rational root crossing zero, the three-scale commutator. Writes verification.json beside itself unless --output is given.
code/11-coefficient-gap s-verify_finite_models.py; data/11-coefficient-gap s-verification.json	Standard library, exact fractions, seed 20260923. 3,880 assertions in 15 categories: truncated Taylor identities over $\mathbb{Q}[b, b^{-1}]$, multiplication of finite outer Hahn polynomials with lexicographic shifts, Taylor inversion, integer-part preservation and reflection, signs of negative blocks, rational samples of the index maps, parameter fixing. Prints to standard output; writes a file only with --output.
code/12-automatic-strongn ess-verify.py; data/12-automatic-strongn ess-verification.json	Standard library, exact fractions, seed 20260923. 17,774 assertions: 10,800 triangular-detector checks, 4,502 monomial adjoint and transfer checks, 2,472 rank-two Taylor-block and binomial checks. <i>Writes verification.json</i> beside itself unless --output is given.

Script and record	Content and behaviour
code/13-omnific-isomorphisms-verify.py; data/13-omnific-isomorphisms-verification.json	Standard library, exact fractions, seed 20260923. 13,885 assertions in 16 categories: binary finite-row choices and their persistence, descending-support detection, coefficient reconstruction, monomial adjoints and retractions, scale-separated Taylor models. <i>Writes</i> <code>../data/verification.json</code> relative to its own directory unless <code>--output</code> is given.
code/14-automatic-summability-verify.py; data/14-automatic-summability-verification.json	Standard library, exact fractions, seed 20260923 for the adjoint trials. Generalized binomial identities to degree 20 for 31 rational parameters (961 multiplication and 31 inverse cases); the induced matching on a banded 256-row prefix (rows meeting columns $n, n+1, n+3$), with a matching of size 64 and its all-ones detector; the sign-reversed transpose (36.2) in 100 random trials with 13 source and 9 target exponents; the finite prefixes of example 34.3 for 1, 2, 5, 16 and 64 pairs. Prints its report and <i>always writes</i> <code>../data/verification.json</code> relative to its own directory; it has no output option.
code/15-topological-collapse-verify.py; data/15-topological-collapse-verification.json	Standard library, exact fractions, seed 20260923. 27,139 assertions in 14 categories: constant-term pairings and coefficient extraction, disjoint rows and the sparse detector with interpolation, the cancellation boundary of example 40.3, monomial transport, rank-two Vandermonde and inverse identities for the shear, order reflection $x \mapsto \gamma - x$ and fixed products on negative intervals, bounded increasing supports, valuation non-escape and cyclic remainders. It does not test arbitrary topologies or the infinite covering lemmas. <i>Writes</i> <code>verification.json</code> beside itself unless <code>--output</code> is given.
code/16-coefficient-recovery-verify.py; data/16-coefficient-recovery-verification.json	Standard library, exact integers and fractions, seed 20260923. 1,314 checks in 12 categories: polynomial <i>abc</i> instances, quartic finite tests, sparse polynomial and rational powers and sharpness, Vandermonde determinants, support-difference separation, specialization multiplicativity, automorphism composition and inverses, truncated Hahn roots of $1+t^4$, and the boundary examples. <i>Writes</i> <code>verification.json</code> beside itself unless <code>--output</code> is given. 16's <code>code/16-coefficient-recovery-build.py</code> is its delivered build script (section A.3).

Script and record	Content and behaviour
<code>code/17-formal-orbit-fields-verify.py;</code> <code>data/17-formal-orbit-fields-verification.json</code>	SymPy 1.14.0 (<code>data/17-formal-orbit-fields-requirements.txt</code>), exact rationals and symbolic polynomial identities. 30,900 assertions in 24 groups: finite Hahn Leibniz and Taylor and binomial products, exponential and binomial convolution, additive and binomial horizontal recurrences, formal clock substitution and the multiplicative group law, the rational invariant ωu and parameter recovery, witness recurrences and derivatives, finite ordinal-polynomial models of the positivity and decrease of the witness exponents, the rational logarithmic-derivative degree bound, Vandermonde and Kummer certificates. It does not verify Hahn summability, all ordinal supports, class statements or novelty. <i>Writes</i> <code>--output</code> , by default <code>verification.json</code> in the working directory. 17's <code>code/17-formal-orbit-fields-Makefile</code> names its delivered files (section A.3).
<code>code/18-exact-symmetries-verify.py;</code> <code>data/18-exact-symmetries-verification.json</code>	Standard library, exact fractions, seed 20260923, deterministic. 11,330 assertions in 19 categories: finite two-level Hahn polynomials (double-lift addition, multiplication, inverses, functoriality, signs, constant coefficients, omnific membership, the two-level fixed-support criterion for three explicit index maps, hull intersections) and the Klein bottle relation, group law, inverses and normal-form witnesses. It does not check infinite summability, class recursion, cut filling or real closedness. <i>Writes</i> <code>verification.json</code> beside itself unless <code>--output</code> is given; 18's <code>code/18-exact-symmetries-build.sh</code> and <code>code/18-exact-symmetries-Makefile</code> rerun it in place (section A.3).
<code>code/19-formal-symmetry-verify.py;</code> <code>data/19-formal-symmetry-verification.json</code>	Standard library, exact fractions, seed 20260923, deterministic. 1,576 checks in 17 groups in the finite model $\mathbb{Z} + X\mathbb{Q}[X]$ with finite s -jets (operators $X^{m+1}d/dX$): divisible-ideal images, ordinary and Hasse-Schmidt Leibniz, iterativity, integral coefficients, exponential and logarithm inversion to order 5, leading commutators and nonsplit jet witnesses to order 7, the Witt relation, and the formal square-root recursion to order 10. It is not a proof about surreal supports, derivations, class maps or genus. <i>Writes</i> <code>data/verification.json</code> under the parent of its own directory unless <code>--output</code> is given (section A.3).

Script and record	Content and behaviour
<code>code/20-universal-symmetries-verify.py;</code> <code>data/20-universal-symmetries-verification.json</code>	Standard library, exact fractions, seed 20260923, deterministic. 5,555 assertions in 29 groups: finite inner and outer Hahn lifts (linearity, order, composition, addition, multiplication, constant and leading coefficients, fixed projections, positive-support preservation), localized index actions, the weighted Green residuals (78.4)–(78.5) for both directions and constant-plus-tail inputs, distinct outer polynomial exponents and the relative independence and projection tests, the monomial part of the relative retraction, gauge, cocycle and commuting compatibility identities with cleared denominators, and truncated formal logarithm and exponential identities. It does not check infinite summability, arbitrary supports, class recursion, the $\eta_{\mathbf{O}\mathbf{n}}$ isomorphism or global choice. <i>Writes</i> <code>verification.json</code> beside itself unless <code>--output</code> is given; 20's <code>code/20-universal-symmetries-Makefile</code> names its delivered files (section A.3).

The placement dossiers also recorded independent checks (281 for the automorphism cluster, including example 3.2, the commutator formula for symbolic s, u , the leading terms of theorem 6.8 for $n \leq 4$, the conjugation lemma and the failure example of remark 4.11; for 02 the quotient-rule identity of example 17.8 to 29 terms; and 8,724 assertions for sources 11–13, including the Taylor-block embedding in a lexicographic $\mathbb{Z}^2$ model, the failure of the ambient Taylor automorphism to preserve $\mathbf{Oz}$ (section 26.1), the binary finite-row algorithm over $\mathbb{F}_2$ and $\mathbb{Q}$, the triangular detector, the least-forbidden-shift coefficient and example 23.6); these scripts are not shipped.

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Reports of the collection are cited by directory and label: the omnific Diophantine report `docs/surreal/omnific-diophantine-geometry (odg:)`, the quotient report `docs/surreal/set-sized-quotients-of-omnific-integers (osq:)`, the surcomplex automorphism report `docs/surcomplex/surcomplex-field-automorphisms (saut:)`, the rigidity report `docs/surreal/exponential-automorphism-rigidity (bare labels)`, the transcendence report `docs/surreal/transcendence-over-bounded-support (bst:)`, the single-dilation report `docs/surcomplex/single-dilation-hahn-support (dsup:)` and the definable-surreals report `docs/foundations-and-computation/definable-surreals-and-omnific-integers (dsn:)`, the three-duals report `docs/surcomplex/three-duals-of-hahn-vector-spaces (duals:)`, the independent-copies report `docs/surreal/independent-surreal-copies (isc:)`, the large-cardinal report `docs/foundations-and-computation/large-cardinal-embeddings-and-normal-forms`, cited by directory only before batch 32 and by its `lce:mf` labels in the batch-32 notes, and the first- κ coefficient report `docs/surcomplex/first-kappa-coefficients (fk:)`. Part V also cites the foundations report `docs/foundations-and-computation/foundations (found:)`, and Part X the autonomous-dilation report `docs/surcomplex/autonomous-dilation-relations (adr:)`.